

The Physics of Golf

Force, Drift, and Control in the Golf Swing

An AffineDrift Textbook

AffineDrift

<https://affinedrift.com>

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Preface

Every golf swing brings together muscular work, motion already acquired, stored elastic energy, and changing contact geometry. The question running through this book is how those interactions shape club delivery: what the current state makes possible, how an input changes the motion, and what evidence lets us distinguish competing explanations.

The useful starting point is a specified mechanical model. We choose bodies, joints, flexible elements, ground and grip conditions, and the quantities to predict. A simple pendulum makes a few interactions visible. A whole-body model can ask more detailed questions, but it also requires more measurements and assumptions. The goal is to make each step understandable and testable, with golf as the concrete setting for multibody mechanics, control and biological inference.

What Drift and Control Mean

For a model with an affine input map, a local state description takes the form

$$\dot{x} = f(x, t) + G(x, t)u.$$

Here x contains the states retained by the model, t is time, u is its declared input, f is the state derivative at zero input, and $G(x, t)u$ is the input contribution at that same state and time. Position and velocity are typical states; muscle activation, tissue deformation and shaft vibration may require others. Time-dependent loads and changing contact modes must also be specified. A smooth equation within one contact mode does not by itself describe an impact or a change of mode.

We call f **drift**. It includes the kinematic rate of position change as well as the modeled zero-input dynamics. Drift and control are contributions to a state derivative; they are not two fundamentally different kinds of physical force. Whether the input is net joint torque, muscle excitation or another channel changes the meaning of zero input and may change whether the model is affine in that input at all. The relevant equations, assumptions and derivations appear in the chapters that use them.

Setting a modeled torque input to zero is a precisely defined calculation. It does not make a golfer's muscles go limp. The retained state can still contain activation and elastic deformation, and the effective plant can retain passive forces or other specified loads. Net torque alone also cannot identify muscle recruitment or co-contraction. Earlier muscular work helped create the very state from which the calculation starts.

Momentum is a property of motion, not an additional force or energy source. A shaft or tendon can return energy previously stored in it. Ideal stationary constraints can transmit forces and redistribute energy between connected bodies while their total constraint power is zero; moving supports require their own work accounting. These distinctions let us trace work and motion through the swing without treating an accelerating segment as proof of local muscular power or treating continued motion as proof of muscular inactivity.

From a Decomposition to a Golf Explanation

At a fixed state, the affine equation separates the modeled instantaneous input contribution. A forward zero-input rollout asks a different question: what trajectory follows when the declared input remains zero over an interval? The changing state then changes the dynamics. Comparing that rollout with another trajectory requires an explicit initial state, horizon, retained loads, contact rules and output metric. Neither calculation automatically identifies the physiological cause of an observed swing.

Control authority is also a task question. The ability to change clubface orientation, impact location or speed depends on available inputs, their bounds and dynamics, the remaining time, contact constraints and the chosen output. A large drift-to-control norm ratio cannot establish that every relevant correction is unavailable. Nor does it establish a universal phase in which golfers lose control, or prove why elite players perform better. Those are hypotheses requiring appropriately defined calculations and evidence.

The aspects of the model are intertwined. Moving a joint center changes hand paths and force moment arms. Changing the grip changes loop closure and transmitted loads. Ground contact changes feasible body motion and reaction forces. Muscle activation and force-length-velocity behavior constrain how joint moments develop, while shaft deformation changes the state delivered to impact. A change that improves one fitted motion trace may therefore have several explanations. Comparing alternatives and propagating uncertainty are part of the mechanical argument.

How to Read This Book

The chapters build from pendulum examples and joint mechanics to constrained multibody dynamics, energy, flexible structures, muscles, inverse problems, motor control and impact. Worked models establish what follows from declared assumptions. Measurements test whether those assumptions and predictions are adequate for a task. Claims about human strategy, learning, tissue injury or coaching outcomes require evidence beyond a successful mechanical calculation.

Read with three questions in view: What was specified? What was observed? What conclusion follows? The distinctions connect the equations to practical investigation. They also keep a model's limitations visible while allowing it to explain something useful. The book is intended as a technical reference and a guide to reasoning about the swing; it does not prescribe one universal technique or present modeled tissue loads as a clinical diagnosis.

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Nomenclature

Throughout this book, we use the following notation conventions and symbol definitions. Chapter-local definitions specify the model, reference frame, units and any exceptions to these conventions. Reusing a letter does not identify two physical quantities.

General Conventions

- Scalars are generally lowercase Roman or Greek letters: m, t, θ, ω ; energy symbols T, V are exceptions.
- Vectors are bold lowercase: $\mathbf{x}, \mathbf{u}, \mathbf{q}, \mathbf{v}$.
- Matrices are bold uppercase: $\mathbf{A}, \mathbf{B}, \mathbf{M}, \mathbf{J}$.
- Sets and manifolds are script or blackboard bold: $\mathcal{Q}, \mathcal{M}, \mathbb{R}$.
- A dot above a symbol denotes a time derivative: $\dot{\mathbf{q}} = \frac{d\mathbf{q}}{dt}$.
- Subscript “drift” denotes zero-declared-input ($\mathbf{u} = \mathbf{0}$) evolution, retaining gravity, passive forces and internal state; it is autonomous only when the model has no explicit time dependence.
- Subscript “head” refers to the clubhead; “hand” to the grip/handle point.

Coordinates and Kinematics

$\mathbf{q} \in \mathbb{R}^n$ Rigid generalized coordinates in a local independent chart. Flexible coordinates $\boldsymbol{\eta}$ are kept separate here.

$\dot{\mathbf{q}}, \ddot{\mathbf{q}}$ Coordinate velocity and acceleration. Nonminimal representations may use speeds $\mathbf{v}$ satisfying $\dot{\mathbf{q}} = \mathbf{N}(\mathbf{q})\mathbf{v}$, with different coordinate and speed dimensions.

$\mathbf{x} \in \mathbb{R}^{n_x}$ Full state. In local mechanical coordinates, $\mathbf{x} = [\mathbf{q}^T, \dot{\mathbf{q}}^T]^T$; with flexibility, append $\boldsymbol{\eta}, \dot{\boldsymbol{\eta}}$ once. Activation, hysteresis or other internal states $\mathbf{z}$ must also be included when modeled.

n Number of rigid-body degrees of freedom.

m Number of flexible shaft modal coordinates in this coordinate convention; chapter-local m may instead denote mass.

$\theta_1, \theta_2, \dots$ Joint angles (shoulder, elbow, wrist, etc.).

$\mathbf{p}_{\text{head}} \in \mathbb{R}^3$ Position of the clubhead center of mass.

$\mathbf{v}_{\text{head}} \in \mathbb{R}^3$ Velocity of the clubhead.

$\mathbf{p}_{\text{hand}} \in \mathbb{R}^3$ Position of the grip/hand point.

Dynamics and Forces

$\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{n \times n}$ Mass matrix; symmetric positive definite for independent coordinates with a nondegenerate positive kinetic-energy metric.

$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{n \times n}$ A generally nonunique matrix representation of velocity-dependent inertial loads. The generalized load is $\mathbf{C}\dot{\mathbf{q}}$, not $\mathbf{C}$ alone.

$\mathbf{g}(\mathbf{q}) \in \mathbb{R}^n$ Gravitational generalized load on the left-hand side of the declared force balance; units depend on the corresponding coordinates.

$\mathbf{d}(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^n$ Dissipative (damping/friction) forces.

$\mathbf{u} \in \mathbb{R}^p$ Declared actuator input; generalized torques, muscle forces and neural commands are different possible input choices, not interchangeable measurements.

$\mathbf{B}(\mathbf{q}) \in \mathbb{R}^{n \times p}$ Input-to-generalized-force map. It may be a selection matrix; redundant actuation permits $p > n$.

$\boldsymbol{\tau}_{\text{total}}$ Total generalized-force vector in the stated balance, with each component power-conjugate to its coordinate velocity.

$\boldsymbol{\tau}_{\text{drift}}$ Generalized drift term in the stated balance with declared control set to zero ($\mathbf{u} = \mathbf{0}$), retaining explicit time dependence when present.

$\boldsymbol{\tau}_{\text{input}}$ Generalized control force: contribution from the applied control channel ($\mathbf{B}\mathbf{u}$).

$\boldsymbol{\lambda}$ Constraint multipliers; $\mathbf{Q}_c = \mathbf{J}_c^T \boldsymbol{\lambda}$ is the generalized constraint force. Multiplier units depend on constraint scaling.

Affine Control Decomposition

$\mathbf{x}$ The full system state, equivalent to $\mathbf{x}$ (used interchangeably for readability).

$f(\mathbf{x})$ Zero-declared-input vector field of the full effective plant. Use $f(\mathbf{x}, t)$ for explicit time dependence, or include time as an augmented state.

$G(\mathbf{x})$ Input (control) matrix field, mapping control inputs to state derivatives.

$G(\mathbf{x})\mathbf{u}$ Control vector field: state-derivative contribution from the declared control channel.

$\dot{\mathbf{x}} = f(\mathbf{x}) + G(\mathbf{x})\mathbf{u}$ The control-affine system equation.

$\mathbf{x}^{\text{ZTCF}}(t)$ Forward Zero-Torque Counterfactual trajectory (evolution of the declared effective plant with $\mathbf{u} = \mathbf{0}$).

$\mathbf{a}_{\text{ZVCF}}(\mathbf{q}, \mathbf{z})$ Instantaneous Zero-Velocity Counterfactual acceleration with velocity and declared control set to zero.

$\text{DCR}_{W, \mathcal{U}}(\mathbf{x})$ Drift-Control Ratio: $\|\mathbf{W}\mathbf{a}_{\text{drift}}\| / (\sup_{\mathbf{u} \in \mathcal{U}} \|\mathbf{W}\mathbf{B}_a \mathbf{u}\| + \varepsilon)$. Declare the acceleration map $\mathbf{B}_a$ after solving constraints, feasible input set $\mathcal{U}$, weighting $\mathbf{W}$ and positive regularizer ε in the denominator's units. This pointwise norm ratio is not a reachability certificate.

Constraint Mechanics

$\Phi(\mathbf{q}, t) = \mathbf{0}$ Holonomic constraints, allowing explicit time dependence; $\mathbf{J}_c \dot{\mathbf{q}} = -\dot{\Phi}_t$.

$\mathbf{J}_c(\mathbf{q}, t)$ Constraint Jacobian: $\mathbf{J}_c = \frac{\partial \Phi}{\partial \mathbf{q}}$.

$\mathbf{J}_{\text{loop}}$ Loop-closure Jacobian for parallel mechanisms.

$\boldsymbol{\lambda}$ Lagrange multipliers under the convention $\mathbf{Q}_c = \mathbf{J}_c^T \boldsymbol{\lambda}$.

$\mathcal{N}(\mathbf{J}_c)$ Feasible virtual-displacement directions at fixed time, for regular constraints. Actual velocities lie in this null space only when $\dot{\Phi}_t = 0$.

$\mathcal{R}(\mathbf{J}_c^T)$ Range of $\mathbf{J}_c^T$ (directions in which constraints exert force).

Shaft and Flexibility

$\boldsymbol{\eta} \in \mathbb{R}^m$ Shaft modal coordinates (flexible deformation modes).

$\dot{\boldsymbol{\eta}}$ Shaft modal velocities.

K_s Shaft bending stiffness (or modal stiffness matrix).

D_s Shaft damping coefficient (or modal damping matrix).

EI Shaft flexural rigidity (Young's modulus $\times$ second moment of area), with units N m^2 .
Modal stiffness and damping units depend on mode normalization.

Energy and Power

$T(\mathbf{q}, \dot{\mathbf{q}})$ Kinetic energy: $T = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}}$.

$V(\mathbf{q})$ Potential energy (gravitational + elastic).

$P_{\text{drift}}(t)$ Coordinate-balance quantity $\boldsymbol{\tau}_{\text{drift}}^T \dot{\mathbf{q}}$. If drift includes velocity-dependent inertial terms, this is not a separate physical energy source; the kinetic-energy balance also contains $\frac{1}{2} \dot{\mathbf{q}}^T \dot{\mathbf{M}} \dot{\mathbf{q}}$.

$P_{\text{input}}(t)$ Power delivered by the declared control channel: $P_{\text{input}} = (\mathbf{B}\mathbf{u})^T \dot{\mathbf{q}}$.

$P_{\text{constraint}}(t)$ Actual power $\mathbf{Q}_c^T \dot{\mathbf{q}} = -\boldsymbol{\lambda}^T \dot{\Phi}_t$. It vanishes for ideal stationary constraints. Ideal moving constraints can exchange energy even though their virtual work is zero.

W Work (time-integrated power).

Biomechanics and Tissue

$a_i \in [0, 1]$ Muscle activation level for muscle i .

F_i^M Muscle force for muscle i .

ℓ_i^M Muscle fiber length.

v_i^M Muscle fiber contraction velocity.

K_{tissue} Tissue stiffness (for fascia, tendon, ligament).

σ Stress (force per unit area in tissue).

ε Tissue strain (deformation per unit length); distinct from the DCR regularizer.

Pendulum Models

L_1, L_2, L_3 Link lengths (proximal, medial, distal).

m_1, m_2, m_3 Link masses.

I_1, I_2, I_3 Link moments of inertia about their centers of mass.

ℓ_1, ℓ_2, ℓ_3 Distance from joint to center of mass for each link.

τ_1, τ_2, τ_3 Applied torques at each joint.

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Part I

Foundations

Chapter 1

Why Physics Matters in Golf

In Plain Language 1.1. The Central Mystery

A swing can feel easy while transmitting large forces. A large joint moment can occur while the club barely speeds up. Two swings can reach similar positions and then produce different strikes. To understand these possibilities, we must connect the motion we see to the forces, energy and constraints that make it possible.

Our central question is: **how does a golfer organize a coupled mechanical system to deliver the club to a useful impact state?** That question includes speed, orientation, contact location and timing. It also includes the body's ability to produce forces, tolerate loads and respond to disturbances. No single angle, force peak or ratio answers it.

1.1 What We Actually Know About Swings

Video and motion capture describe how the body and club move. Force plates measure external contact forces and moments. Other instruments can observe club delivery, ball launch or muscle electrical activity. Each observes a different part of the process. A useful explanation connects these observations through a model whose assumptions can be inspected and whose predictions can be tested.

For example, an observed increase in clubhead speed establishes a change in motion. It does not identify which muscles supplied the energy. An inverse-dynamics calculation can estimate net joint forces and moments, given suitable kinematics, inertial parameters and external loads. It still does not uniquely determine the forces of individual muscles. Several muscles, tendons and passive tissues can contribute to the same net result.

We will distinguish three kinds of statement throughout this book: an identity derived within a declared model, a prediction that depends on fitted or assumed parameters, and an empirical claim about golfers. The first can be checked mathematically; the second needs numerical and model checks; the third needs relevant measurements. Moving from one category to another requires evidence.

1.2 Two Incomplete Stories

1.2.1 The Kinematic Story

Kinematics describes positions, orientations and their time derivatives. It is indispensable: geometry determines where the club can reach, and velocity determines how it approaches the ball. A pose alone, however, omits velocity and any internal state needed to predict

what follows. A measured hip angle is also meaningless without its reference frame, rotation convention and point in the swing.

Dynamics links the motion to applied forces and moments, mass distribution and constraints. The relation works in both directions: forces change motion, while configuration and velocity change the response to those forces. The same wrist moment can have different effects when the arm–club geometry differs. This dependence is one reason a coaching cue cannot be translated into a universal torque prescription.

1.2.2 The Strength and Flexibility Story

Force capacity, range of motion and coordination restrict the movements a person can produce. They do not select a unique swing. Even an ideal actuator has limits on torque and speed; a muscle’s response also depends on activation and mechanical history. A model that prescribes arbitrary joint torques has abstracted away those physiological restrictions.

A useful analysis therefore asks both whether a motion produces desirable impact conditions and whether the body can generate it under the relevant loading and contact conditions. More available strength need not produce greater clubhead speed in every swing. Less visible motion or less net joint torque need not mean less muscular activity.

1.3 The Missing Story: Forces, Motion and Work

For a point mass in an inertial frame, Newton’s equation is $m\mathbf{a} = \sum \mathbf{F}$. Momentum $\mathbf{p} = m\mathbf{v}$ describes its motion; force is the rate of change of momentum. Momentum is not an additional applied force. In rotating coordinates, centrifugal and Coriolis terms help express the same dynamics, but they must not be added again as physical external forces in an inertial-frame force balance.

Acceleration changes the velocity vector. It can change speed, direction, or both. For a smooth path with nonzero speed v and local curvature radius ρ ,

$$\mathbf{a} = \dot{v}\mathbf{t} + \frac{v^2}{\rho}\mathbf{n}.$$

Here $\mathbf{t}$ is the unit tangent and $\mathbf{n}$ points toward the center of curvature. The normal term v^2/ρ is not the full acceleration unless the tangential term vanishes. A real clubhead does not generally follow a circle about a fixed shoulder, so arm length alone does not determine ρ .

For that point mass, power and kinetic energy obey

$$P = \sum \mathbf{F} \cdot \mathbf{v} = \frac{dK}{dt}, \quad K = \frac{1}{2}mv^2.$$

A force perpendicular to velocity can turn the path without changing kinetic energy. Work is power integrated over time, or force integrated along displacement. Thus a large force, a large acceleration, a large power and a large accumulated work are four different claims.

A rigid club adds rotation. Its kinetic energy includes COM translation and rotation about the COM. A force applied at a moving grip does work at the grip’s velocity, and an applied couple does work at the club’s angular velocity. Clubhead speed by itself is insufficient to compute either total club energy or the work of each applied load.

In Plain Language 1.2. Why Attribution Matters

Consider a manufactured point mass of $m = 0.20$ kg on a massless tether of fixed length $L = 1.25$ m, attached to a fixed pivot. Gravity is $g = 9.81$ m/s²; there is no drag or hinge friction. At the bottom, the mass moves horizontally at $v = 40$ m/s. These values are chosen to expose a mechanical distinction, not to represent a measured golfer.

The required upward normal acceleration is $v^2/L = 1280$ m/s², about $130.48g$. Upward tension T and downward gravity give

$$T - mg = m \frac{v^2}{L}, \quad T = 257.962 \text{ N.}$$

Both forces are perpendicular to the velocity at that instant, so their instantaneous powers are zero. The mass nevertheless has $K = 160$ J of kinetic energy supplied before this instant. Large tension does not establish that the support is currently adding energy, nor does it tell us what a human hand would expend to sustain the load.

1.4 Two Kinds of Terms: A Preview

The book uses a **drift–input decomposition** to ask a precise question: at a given state, what motion rate would a chosen model predict at zero selected input, and how does changing that input alter the prediction? This is a mathematical split of a model’s state derivative. It is not a universal division of physical forces into effortless effects and muscular effort.

1.4.1 Drift: Evolution at the Chosen Input Zero

Drift depends on the model, its state and the definition of input zero. It can include motion resulting from energy supplied earlier. For example, a fast unforced pendulum already contains kinetic energy; inertia shapes its evolution but does not create energy. If elastic tissues or a flexible shaft are modeled, their stored energy and internal state also matter.

A past control changes the state at which future drift is evaluated. Consequently, saying that an instantaneous acceleration is attributed to drift does not mean that earlier muscle action was unimportant. Nor does it mean that the trajectory is unavoidable: a feasible future input can alter it.

1.4.2 Control: The Selected Model Input

In our first rigid-link models, the input is an applied generalized actuator torque. A detailed human model might instead use muscle excitation and include activation and contraction dynamics. These are different input choices. A muscle command is not generally an instantaneous, unconstrained joint-torque command.

Antagonist muscles can exert opposing moments with a small or zero net joint moment. Such co-contraction can change joint stiffness and tissue loads. Therefore **zero net torque is not muscle relaxation**, and net mechanical work is not metabolic expenditure. Claims about effort require measurements or an additional physiological model.

Key Principle 1.1. Drift vs. Control in a Simple Fall

A freely falling point mass, released from rest through a height h in uniform gravity without air resistance, gains mgh in kinetic energy. At $h = 1$ m and $m = 0.20$ kg, this is 1.962 J and the final speed is $\sqrt{2gh} = 4.429$ m/s. To reach 40 m/s from rest using only gravity in this model would require a drop of 81.55 m.

This comparison is an energy check for the specified mass, not a percentage of a golfer's energy budget. A golfer's segments have different height changes, initial velocities, muscular work and elastic storage. A quantitative attribution must identify the system boundary and time interval and balance all relevant energy flows. There is no general rule here that passive dynamics supplies most downswing work or that muscles mainly steer.

1.5 The Central Equation

For independent local coordinates q of an ideal rigid mechanical model, with no additional external loads or unresolved constraints, write

$$M(q)\ddot{q} + c(q, \dot{q}) + g(q) = B(q)u. \quad (1.1)$$

The symmetric positive-definite mass matrix M describes inertia. The vector c is the velocity-dependent inertial bias. We define $g = \nabla_q V$ using gravitational potential V , so the gravitational generalized force is $-g$. The matrix B maps the selected inputs into generalized forces. When each independent angular coordinate has its own conjugate actuator torque, B can be the identity. A factorization $c = C\dot{q}$ is common but the matrix C is not unique. These conventions follow standard multibody mechanics [Tedrake, n.d.b].

Solving for acceleration gives

$$\ddot{q} = \underbrace{-M^{-1}(c + g)}_{a_0} + \underbrace{M^{-1}Bu}_{a_u}.$$

The inverse is analytical notation; numerical implementations solve a linear system. The term $M\ddot{q}$ is part of the force balance, not itself a drift force. For $x = (q, \dot{q})$, the corresponding state equation is

$$\dot{x} = f(x) + G(x)u, \quad f(x) = \begin{bmatrix} \dot{q} \\ a_0 \end{bmatrix}, \quad G(x) = \begin{bmatrix} 0 \\ M^{-1}B \end{bmatrix}. \quad (1.2)$$

Drift includes the kinematic block $\dot{q}$. This state closes the ideal rigid model; it does not claim that pose and rate fully specify a human. Generalized velocities may also differ from coordinate rates in spatial representations, as Chapter 2 explains.

1.5.1 Constraints and Energy Complete the Interpretation

When coordinates retain an ideal, stationary holonomic constraint $h(q) = 0$, its reaction appears as $J^T\lambda$, where $J = \partial h / \partial q$. The dynamics and the differentiated constraint must be solved together:

$$M\ddot{q} + c + g = Bu + J^T\lambda, \quad J\ddot{q} + \dot{J}\dot{q} = 0.$$

For independent constraint rows and nonsingular M , these equations determine acceleration and reaction. Changing u generally changes λ too; one cannot remove an actuator torque while arbitrarily freezing the old reaction. Foot contact additionally requires feasible normal forces and friction. A foot cannot pull on an ordinary floor; a model may need to change contact mode when a constraint ceases to be feasible.

For ideal stationary constraints, $J\dot{q} = 0$ and their total power is $\lambda^T J\dot{q} = 0$. They can nevertheless transmit forces and transfer energy between connected bodies. A moving support can supply power. For the simple rigid system with stationary ideal constraints, no dissipation and no other loads,

$$E = \frac{1}{2}\dot{q}^T M \dot{q} + V(q), \quad \dot{E} = \dot{q}^T B u.$$

Gravity exchanges potential and kinetic energy; internal inertial coupling redistributes motion. Neither creates total mechanical energy. Modeling the club alone places hand forces outside its boundary, whereas modeling body and club together makes their contact internal. This boundary change must be stated when comparing energy accounts.

1.5.2 The Input Baseline Matters

If $u = \bar{u}(x) + w$, then exactly the same dynamics can be written

$$\dot{x} = [f(x) + G(x)\bar{u}(x)] + G(x)w.$$

The new drift contains the selected baseline input. As a scalar illustration, let angular acceleration be $a = -2 + 0.5u$, with a measured in rad/s^2 , torque u in N m and the gain 0.5 in $(\text{rad/s}^2)/(\text{N m})$. Take $u = 6$. The total is $a = 1$. With baseline $\bar{u} = 4$, the zero-residual acceleration is 0 and $w = 2$ supplies the same $a = 1$. A ratio of drift magnitude to input-induced magnitude changes from 2/3 to 0 although the physical acceleration is unchanged. The admissible input set must be shifted with the baseline as well.

A Drift-Control Ratio (DCR) therefore requires a declared input zero, state or task quantity, norm, units and treatment of small denominators. It measures the stated decomposition; it does not alone measure energetic efficiency, tissue load or expertise. Comparing skill groups or prescribing a swing from it requires separate validation. Affine dependence on inputs at one fixed state also does not imply that complete nonlinear swing trajectories can be superposed.

1.6 Why a Physics Textbook for Golfers?

The purpose is to connect useful observations to mechanisms and testable questions. Existing research already combines motion analysis, inverse dynamics, forward simulation and energy analysis. Our contribution should be a clear account of how those perspectives fit together, with enough detail to reproduce an argument and understand its limitations.

For example, Nesbit and Serrano used kinematically driven body and club models to estimate work and power for one selected swing from each of four amateur golfers [Nesbit and Serrano, 2005]. That supports an explicit energy accounting approach. Model-derived joint work in four swings is not a direct measurement of muscle metabolism or a population law relating skill to passivity.

MacKenzie and Sprigings developed a three-dimensional forward model with a flexible shaft and torque generators, fitted aspects of its motion to a golfer and optimized a clubhead-speed objective [MacKenzie and Sprigings, 2009a]. Its methods include actuation and timing; it is not evidence that a universal passive-work percentage applies to golf. A result optimized for one objective within one model must be tested before being used as individual coaching advice.

1.6.1 Connecting the Swing to the Shot

A useful chain of reasoning starts with feasible body and club states, then computes the effects of inputs and contacts over time, then evaluates the delivered club state at impact. Ball launch requires a collision model or measurements of the collision. Flight then depends on the ball's launch velocity, spin and aerodynamic environment. Optimizing one joint moment or maximizing clubhead speed is not equivalent to optimizing the final shot.

Timing is part of the mechanics. Changing a torque earlier in the downswing changes later geometry, velocity and the forces needed to preserve contact. It can also change when and where impact occurs. A comparison at the same clock time may therefore answer a different question from a comparison at each swing's actual impact. This is why later chapters distinguish local acceleration attribution from finite-time trajectory and impact sensitivity.

A coaching cue can be investigated through this chain: measure how the player's motion changes, estimate loads under declared assumptions, predict delivery, and check the result against observed impact and ball flight. The mechanics does not guarantee that the cue produces the intended change, or that the same cue acts identically on another golfer.

1.7 The Double Pendulum: Your Guide

Chapter 3 uses two effective rigid links joined by planar hinges, with a fixed proximal pivot. Its coordinates describe one link's absolute angle and the second link's relative angle. It exposes inertial coupling, energy transfer and the distinction between joint acceleration and endpoint acceleration in a model small enough to derive and check completely.

These are effective links, not a unique anatomical map. An arm-club interpretation and an upper-arm-forearm interpretation are different models. The baseline omits bilateral grasp, torso translation, three-dimensional face orientation, shaft deformation and ball contact. It is a starting point for a model hierarchy, not proof that those omitted effects are small.

Some unforced double pendulums exhibit chaotic dynamics in some regimes. Nonlinearity alone does not establish chaos, and a sensitive impact result over a short interval is not a demonstration of long-time chaotic behavior. Feedback can reduce some disturbances under suitable conditions; its presence alone does not guarantee stability. The relevant question for a swing is how plausible initial-state, input and model uncertainty affects the outcome over the swing interval.

1.7.1 What Happens If You Let Go Mid-Swing?

Three interventions must be kept separate. Setting an ideal hinge's drive torque to zero retains the hinge and its reaction forces. Removing the attachment changes the constraint and hence the equations. Asking a person to relax changes an evolving physiological system;

it is not automatically either of those mathematical operations. The coaching word “release” must therefore be operationally defined before a model can test it.

Return to the manufactured tethered point mass. Choose an inertial plane with horizontal coordinate X and upward Y , with the pivot at the origin. Let θ increase from downward vertical toward positive X . Retaining the hinge with zero drive torque gives

$$\ddot{\theta} = -\frac{g}{L} \sin \theta, \quad (X, Y) = (L \sin \theta, -L \cos \theta).$$

Start at $\theta(0) = 0$ and $\dot{\theta}(0) = 32$ rad/s. Removing the tether without an impulse instead preserves initial position and velocity but gives

$$(X, Y) = (40t, -1.25 - \frac{1}{2}gt^2).$$

The first trajectory curves upward initially because the retained tether supplies tension; the second falls below the horizontal tangent. Both have zero drive torque. Their different accelerations demonstrate why “zero torque” does not specify a trajectory without the retained constraints and initial state.

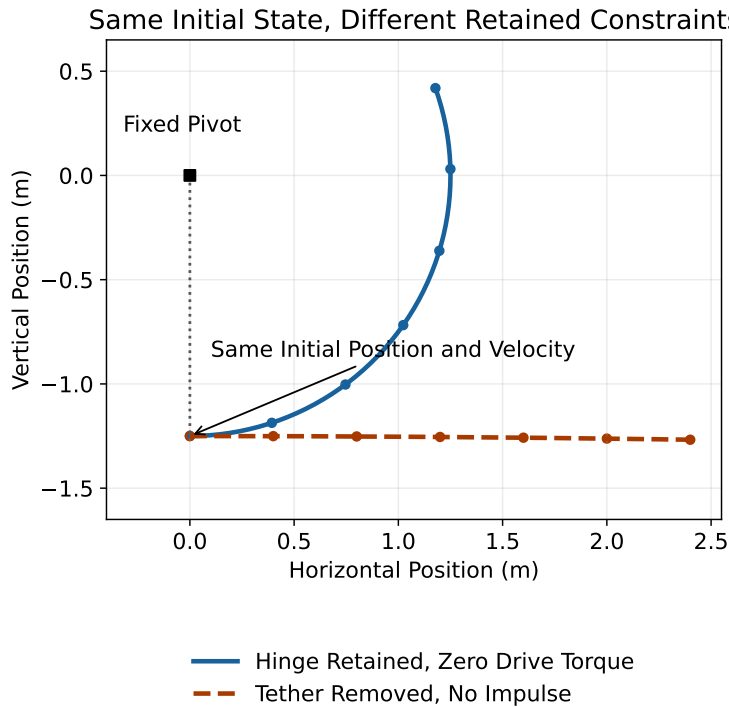


Figure 1.1: Same Initial State, Different Constraints. Manufactured Point-Mass Example With $L = 1.25$ m, Initial Speed 40 m/s and $g = 9.81$ m/s². Markers Are 10 ms Apart Through 60 ms. The Retained Hinge Has Zero Drive Torque; the Released Mass Has No Tether Force. Neither Trajectory Is a Measured Golf Swing.

For a released rigid club under uniform gravity and no aerodynamic force, it is the **center of mass** that follows a ballistic path. The clubhead also moves because the club rotates.

Zero external moment about the COM conserves angular momentum; it does not generally keep a three-dimensional body's angular velocity constant.

The retained example also disproves a supposed need for nonzero drive torque at every instant of a predictable motion. A specified initial state and zero-input dynamics already define a trajectory. Whether that trajectory reaches a particular target, remains feasible or is robust to uncertainty is a further question. Chapter 3 computes such model trajectories with declared, manufactured parameters.

1.8 A Road Map

Chapter 2 defines frames, coordinates, configuration, velocity and predictive state. It explains why a pose is not a complete description of an evolving system.

Chapter 3 derives the two-link equations, checks their energy balance and separates mathematical coupling from claims about anatomy or chaotic behavior.

Chapter 4 examines applied loads, generalized moments and directional acceleration. Chapter 5 develops the chosen-input affine form and its limits. Together these tools support later treatments of constraints, inverse dynamics, control, uncertainty and impact.

Key Principle 1.2. Key Takeaways

1. Motion, forces and constraints must be interpreted together. Geometry changes force response, and forces change future geometry.
2. Normal acceleration can be large while instantaneous power is zero. Momentum and inertia are not new sources of energy.
3. Drift is a model's evolution at a specified input zero. It can reflect earlier work and retained internal state.
4. Net torque, muscle activation, contact force, mechanical work and metabolic effort are distinct quantities.
5. A useful model carries its assumptions through to finite-time impact predictions and is tested against the observations relevant to its claim.

In Plain Language 1.3. On This Site

- **The Affine Nature of the Golf Swing** — Overview of drift-input modeling in the swing.
- **Part 1: Control-Affine Framework** — Derivation of the chosen-input decomposition.
- **Superposition in Control-Affine Systems** — Fixed-state attribution and local perturbations in multibody models.

1.9 Chapter Exercises

1. **Free Fall.** A point mass is released from rest 1 m above a horizontal floor. Ignore air resistance and use $g = 9.81 \text{ m/s}^2$. Find flight time and speed immediately before contact. Does doubling mass change them?
2. **The Spinning Ball.** Contrast a sphere spinning about a vertical axis at an ideal point contact with a sphere initially backspinning about a horizontal axis while its center is stationary. What does tangential contact friction do in each idealization? Why can real balls eventually stop?
3. **The Retracting Tether.** A mass moves in a horizontal plane without gravity or friction. An inward tension acts along a tether through a fixed hole, while tether length r decreases. Is there torque about the hole? Can tension do positive work?
4. **The Mystery of Cues.** Does “rotate your hips faster” specify a unique joint torque? What additional observations and assumptions would an inverse-dynamics calculation require?
5. **Two Throws.** The same ball starts from rest at the same initial height in two trials and leaves the hand at the same final height, at speeds v and $2v$. Neglect air resistance during the throws. Compare the changes in kinetic energy and gravitational work. Can peak muscle force be inferred from those speeds?
6. **Drift and the Baseline.** Recompute the scalar example $a = -2 + 0.5u$ at $u = 6$ using baseline $\bar{u} = 4$. Explain why the changed attribution does not demonstrate improved skill.

1.10 Worked Answers

1. Free Fall

The displacement is $h = gt^2/2$, so $t = \sqrt{2h/g} = 0.4515 \text{ s}$ and the final speed is $gt = 4.429 \text{ m/s}$. Both are independent of mass in this model: doubling gravitational force also doubles inertia. Contact deceleration at the floor is outside this calculation.

2. The Spinning Ball

For pure vertical-axis spin, the ideal bottom point lies on the spin axis and has no tangential slip. A point-contact model with tangential Coulomb friction alone has no torsional resistance about that normal. A real finite contact patch can develop a resisting torsional moment.

For horizontal-axis backspin with a stationary center, the contact point slips. Tangential friction changes both translation and spin and dissipates mechanical energy while slipping. Once ideal rolling without slip is reached on a rigid level surface, rolling can persist without dissipative friction. Real deformation, rolling resistance and air drag can remove energy. Ice is not an exactly frictionless model.

3. The Retracting Tether

The tension has zero moment about the hole because its line of action passes through it. Yet its power is $P = -T\dot{r}$, which is positive during retraction. With no external torque about the hole, angular momentum $mr^2\dot{\theta}$ is conserved even as energy increases through work done by the puller. For $T = 5$ N and $\dot{r} = -0.3$ m/s, power is 1.5 W. “Pulling harder” is therefore not synonymous with applying a larger pivot torque.

4. The Mystery of Cues

The cue names a desired change in motion, not an actuator command. We need a defined coordinate and reference frame, time-resolved motion and its derivatives, inertial properties, external loads and a feasible contact model. Even then, inferred net joint torques do not uniquely identify muscle forces or establish which coaching cue would produce them.

5. Two Throws

The kinetic-energy changes are $mv^2/2$ and $2mv^2$, so the second is four times the first. With the same initial and final heights, gravitational work is identical: $mg(Y_{\text{initial}} - Y_{\text{final}})$. Under the stated assumptions the larger final energy requires more non-gravitational work on the ball, but peak force depends on the path and force-time history. Muscle forces further depend on body mechanics; ball speed alone does not determine them.

6. Drift and the Baseline

The original terms are -2 and 3 , summing to 1 . The shifted terms are $-2 + 0.5(4) = 0$ and $0.5(6 - 4) = 1$. Nothing about the motion or body has changed. Only the definition of the zero-input comparison changed, so a different ratio cannot by itself demonstrate a physiological or coaching improvement.

Chapter 2

The Language of Motion

In Plain Language 2.1. Why We Need a Language

A statement such as “the hips turn early” leaves several questions open: which body rotates relative to which frame, which angle describes that rotation, and early relative to what event? Mathematics helps us answer those questions only after we define the quantities. An equation with an unspecified angle can be just as ambiguous as a coaching phrase.

This chapter builds a description of motion from configuration, velocity and state. We then connect joint motion to the motion of a distal endpoint. These distinctions matter because a useful golf model must explain both how the body moves and what the club delivers at impact. A precise motion description is the beginning of that investigation; it does not by itself identify the forces or the control strategy.

2.1 Coordinates: Measuring Position

2.1.1 Cartesian Coordinates and Body Orientation

A point in space has Cartesian coordinates (x, y, z) relative to a stated origin and three perpendicular axes. These describe that point’s location. They do not describe an entire club: two clubs can put the same point at the same location with different face orientations. A rigid body’s configuration generally needs both position and orientation.

Coordinates are descriptions, not physical causes. Changing from laboratory coordinates to joint coordinates does not change the motion. It changes the numbers and the equations used to describe it. Mixing numbers from different frames or angle conventions does change the calculation, usually incorrectly.

2.1.2 Generalized Coordinates and Degrees of Freedom

A configuration specifies the positions and orientations of the modeled bodies, without their velocities. In a regular local description, **independent generalized coordinates** $\mathbf{q} = (q_1, \dots, q_n)$ parameterize that configuration using n variables. This local number is the configuration’s number of degrees of freedom (DOF).

A single coordinate chart need not cover every configuration without a seam or singularity. An angle and that angle plus 2π describe the same orientation of an ideal revolute hinge. We can unwrap an angle along a motion, but the unwrapped number also records turns. Alternatively, redundant coordinates can be useful if their constraints are enforced; Cartesian coordinates are not an inferior choice.

For an ideal planar pendulum with a fixed pivot and rigid link of length L ,

$$x = L \sin \theta, \quad y = -L \cos \theta, \quad x^2 + y^2 = L^2.$$

One angle suffices even though we wrote two position components. This assumes motion confined to a plane. A spherical pendulum free to swing in space has two configuration DOF. A string gives a fixed-distance constraint only while taut; it cannot supply the compressive force a rigid rod can. These are different physical models.

2.1.3 A Declared Two-Link Model

We will use the same geometric model as Chapter 3: two rigid links in a fixed inertial plane, a fixed base pivot O , a connecting hinge H , and distal endpoint P . There are two ideal revolute joints and no joint limits in this baseline.

The labels “shoulder” and “elbow” can be helpful analogies, but they do not make this an anatomical arm model. Interpreting the hinge as an elbow differs from interpreting it as an arm–club connection. We therefore use link 1, link 2 and hinge throughout. A real swing adds moving supports, three-dimensional orientation, both arms, compliant structures and contact.

Use horizontal x positive to the right and vertical y positive upward. The first angle θ_1 increases counterclockwise from downward vertical. The second angle θ_2 is the counterclockwise **relative** angle from link 1 to link 2. Thus

$$\mathbf{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \quad \alpha_1 = \theta_1, \quad \alpha_2 = \theta_1 + \theta_2.$$

The α_i are absolute link angles in the plane. Earlier notation θ_s, θ_e corresponds to θ_1, θ_2 only under this declared idealization; it is not a pair of measured anatomical rotations. At $\mathbf{q} = (0, 0)$ both links hang downward. Zero does not mean pointing toward a golf target.

2.2 Configuration Space and Constraints

2.2.1 A Map of Configurations

For unrestricted revolute joints, the two angles each lie on a circle. Their product is a torus. A square plot of angles is a useful map of this configuration space, provided opposite edges are identified: crossing an angle seam is not a physical jump.

Joint limits or collision exclusions remove configurations from this ideal space. A rectangular angle window can illustrate chosen independent limits; it is not a universal map of human shoulder and elbow motion. Anatomical limits depend on the joint definition, other joint positions, loading and the individual. We do not infer them from the ideal model.

This distinction between a space and its coordinates is developed in the [Modern Robotics configuration-space treatment](#). The important practical question is what the chosen coordinates represent and where that representation remains valid.

2.2.2 Admissibility Is Not Reachability

A configuration can satisfy geometric constraints yet be unreachable from a specified starting state with the available torques, time and contact conditions. **Admissibility** concerns the

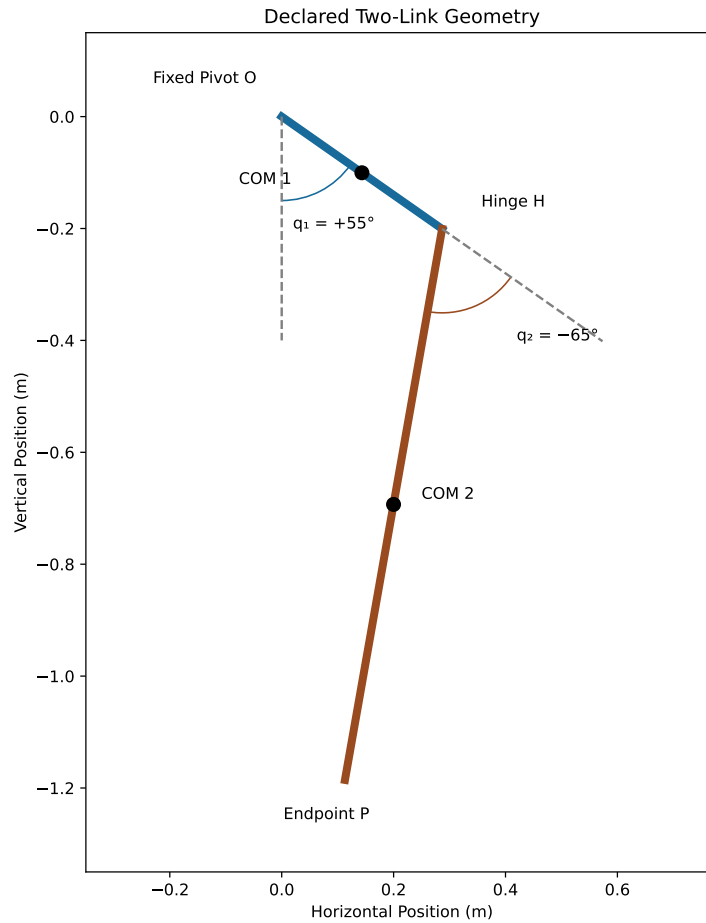


Figure 2.1: Two-Link Coordinate Convention. The Computed Geometry Is Shared With Chapter 3; It Illustrates a Model, Not Measured Golf Anatomy.

declared constraints. **Dynamic reachability** also depends on the initial state, equations, input bounds and elapsed time.

For example, a link arrangement may be geometrically possible while a requested transition to it in 20 ms exceeds the model's actuator bounds. The number 20 ms here defines a hypothetical request, not a measured human capability. Configuration space alone cannot decide whether the transition can be performed.

2.2.3 Position, Velocity and Contact Constraints

A smooth equality constraint $h(\mathbf{q}) = 0$ whose Jacobian has independent rows restricts configuration. Differentiating it gives the compatible velocity condition

$$\mathbf{H}(\mathbf{q})\dot{\mathbf{q}} = 0, \quad \mathbf{H} = \frac{\partial h}{\partial \mathbf{q}}.$$

Some velocity constraints cannot be integrated into position constraints. Rolling without slipping is a standard example. It limits instantaneous motion without necessarily reducing configuration dimension. The [Modern Robotics constraint treatment](#) distinguishes these holonomic and nonholonomic cases.

For the two-link baseline, using hinge angles already enforces rigid attachment; we need not add another equation to hold the hinge together. If instead we modeled each body's free Cartesian pose, attachment would appear as explicit constraints. The same physical reaction can therefore be implicit in one formulation and explicit in another.

Real grip and ground contacts require a declared contact mode. Sticking is possible only when the required contact forces and moments lie within the model's admissible contact set. Feet remaining in place do not freeze every torso coordinate, and a held club does not eliminate wrist motion. Slip, separation and changing contact patches can change the equations.

2.2.4 What a Constraint Reaction Means

In an ideal constrained model, a reaction is solved together with the acceleration to enforce compatible motion. It is not generally a freely chosen actuator input. Nevertheless, changing muscle or actuator forces can change that reaction. Mathematical classification as a reaction does not identify which tissues carry the physical load.

A hard joint stop is one possible idealization. Biological resistance can involve passive tissue deformation, active muscle forces and rate-dependent effects before a nominal range limit is reached. Replacing that response with an impenetrable wall does not predict tissue loading or injury. Chapter 7 develops the force balance and work assumptions more carefully.

2.3 State: What Is Needed to Predict Motion?

2.3.1 Velocity and Its Units

The generalized velocity is the time derivative $\dot{\mathbf{q}}$. For an angle, its units are radians per second. Degrees may be useful for reporting values, but differentiation of sine and cosine in the formulas below assumes radians.

For illustration, a planar link rotating at $500^\circ/\text{s}$ has angular speed $500\pi/180 \approx 8.727$ rad/s. At fixed radius 0.35 m, a point on that link has speed about 3.054 m/s. This is a unit conversion and rigid-link calculation, not a typical measured shoulder rate.

In this planar model, $\omega_1 = \dot{\theta}_1$ and $\omega_2 = \dot{\theta}_1 + \dot{\theta}_2$. In a general three-dimensional model, the derivatives of three orientation coordinates are not simply the three components of angular velocity; a representation-dependent map is required.

2.3.2 A State Belongs to a Specified Model

For a smooth, unconstrained, second-order mechanical model with independent coordinates, a convenient state is

$$\mathbf{x} = \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \end{bmatrix}, \quad \dot{\mathbf{x}} = \begin{bmatrix} \dot{\mathbf{q}} \\ \mathbf{a}(t, \mathbf{q}, \dot{\mathbf{q}}, \mathbf{u}) \end{bmatrix}.$$

The acceleration function $\mathbf{a}$ comes from a declared dynamic model; $\mathbf{u}$ denotes its inputs. An initial state, model parameters and future input history or feedback rule define an initial-value problem. Standard regularity and existence assumptions are also needed for a unique solution. Knowing only the instantaneous forces supplies instantaneous acceleration, not all future commands.

Position alone cannot distinguish passing through a configuration quickly from resting there. Conversely, zero velocity at one instant does not tell us whether motion will reverse, remain at rest or accelerate onward; the model and inputs decide.

The $2n$ state dimension is appropriate to this mechanical model, not to every biological model. Muscle activation, tendon or shaft deformation, controller memory and other internal variables may require additional states. Delay models can require a history rather than a finite list of instantaneous values. Contacts add modes and compatibility conditions. A video-derived joint configuration is therefore not automatically a complete predictive state.

In Plain Language 2.2. A State Is More Than a Pose

Imagine two swings with the same visible link arrangement. Different rates already make their futures different. Even matching those rates may be insufficient if stored elastic deformation, muscle activation or the upcoming control inputs differ. State selection is a modeling commitment about what information is enough to advance the equations.

2.4 Phase Space and Trajectories

For the unconstrained two-link baseline, $\mathbf{x} = (\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2)$ has four components locally. The velocity at a configuration can take many values: configuration does not assign one unique velocity arrow. A tangent vector $(\dot{\theta}_1, \dot{\theta}_2)$ lies in the tangent space at a configuration; the full state-space direction is $(\dot{\mathbf{q}}, \ddot{\mathbf{q}})$.

Here phase space means the position–velocity state space (the tangent bundle of configuration space). Hamiltonian mechanics instead uses positions and conjugate momenta; a nonsingular mechanical Legendre map relates these descriptions. Neither convention adds a new physical motion.

A **trajectory** is a time-parameterized state history. Its configuration projection forgets the velocities. A plot of $(\theta_1, \dot{\theta}_1)$ forgets the second joint, so it is a projection, not a slice fixing the omitted variables. Projected curves can cross without the full states being equal. Even full-state uniqueness concerns the same time-dependent model and input rule; different commands need not produce the same continuation.

For smooth motion, configuration and velocity evolve continuously. An ideal impulsive impact can leave position continuous while resetting velocity. Hybrid state histories are then piecewise smooth with jumps, rather than one everywhere-smooth curve. The moment at which the club reaches a ball also matters: comparing two swings at the same clock time differs from comparing them at their respective impact events.

2.5 From Joint Motion to Endpoint Motion

2.5.1 Positions and Centers of Mass

Write the radial and tangent directions as

$$\mathbf{e}(\alpha) = \begin{bmatrix} \sin \alpha \\ -\cos \alpha \end{bmatrix}, \quad \mathbf{t}(\alpha) = \frac{d\mathbf{e}}{d\alpha} = \begin{bmatrix} \cos \alpha \\ \sin \alpha \end{bmatrix}.$$

With link lengths L_1, L_2 and COM distances r_1, r_2 from their proximal hinges,

$$\begin{aligned} \mathbf{r}_H &= L_1 \mathbf{e}(\theta_1), \\ \mathbf{r}_P &= L_1 \mathbf{e}(\theta_1) + L_2 \mathbf{e}(\theta_1 + \theta_2), \\ \mathbf{r}_{\text{COM},1} &= r_1 \mathbf{e}(\theta_1), \\ \mathbf{r}_{\text{COM},2} &= L_1 \mathbf{e}(\theta_1) + r_2 \mathbf{e}(\theta_1 + \theta_2). \end{aligned}$$

These are planar positions in the declared fixed frame. A specified embedding can place that plane in three-dimensional space; two coordinates still do not reconstruct unrestricted spatial arm or club motion. The COM need not be at half the link length, and the distal endpoint need not be the COM.

For the examples here use $L_1 = 0.35$ m, $L_2 = 1.0$ m, $r_1 = 0.175$ m and $r_2 = 0.5$ m. These are the manufactured geometric parameters of Chapter 3. Kinematics does not require masses. When dynamics is added, that chapter supplies masses 2.5 kg and 0.4 kg and separate centroidal inertias. These numbers do not calibrate a golfer.

2.5.2 The Jacobian Is a Directional Map

Differentiating the endpoint position gives

$$\dot{\mathbf{r}}_P = \mathbf{J}_P(\mathbf{q})\dot{\mathbf{q}},$$

$$\mathbf{J}_P = \begin{bmatrix} L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) & L_2 \cos(\theta_1 + \theta_2) \\ L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2) & L_2 \sin(\theta_1 + \theta_2) \end{bmatrix}.$$

Each column gives endpoint velocity per unit rate of one coordinate while the other rate is zero. The columns add with their signed rates. Describing a matrix as “large” is incomplete: specify the input-rate direction and norm, and the output direction or speed being compared.

At $\mathbf{q} = (0, 0)$,

$$\mathbf{J}_P = \begin{bmatrix} 1.35 & 1.0 \\ 0 & 0 \end{bmatrix}.$$

Rates $(1, -1.35)$ rad/s give zero instantaneous endpoint velocity; $(1, 1.35)$ rad/s give 2.7 m/s to the right. The same geometry can produce cancellation or reinforcement. Neither calculation alone measures mechanical advantage, required torque or available power.

The determinant is $L_1 L_2 \sin \theta_2$. At aligned links the endpoint velocity map loses rank, although both joint coordinates still exist. This task singularity does not by itself make the mass matrix singular. Angular-rate addition is kinematics; inertial coupling depends on the mass distribution and kinetic energy derived in Chapter 3.

2.5.3 Acceleration and a Moving Support

Another derivative introduces a term even when joint accelerations vanish:

$$\ddot{\mathbf{r}}_P = \mathbf{J}_P \ddot{\mathbf{q}} + \dot{\mathbf{J}}_P \dot{\mathbf{q}},$$

$$\dot{\mathbf{J}}_P \dot{\mathbf{q}} = -L_1 \dot{\theta}_1^2 \mathbf{e}(\theta_1) - L_2 (\dot{\theta}_1 + \dot{\theta}_2)^2 \mathbf{e}(\theta_1 + \theta_2).$$

These terms account for the changing velocity directions as links rotate. They are accelerations, not individual muscle forces. The [MIT multibody derivation](#) provides the corresponding relative-angle kinematics and mechanical formulation.

If the pivot translates while the plane's axes remain inertially fixed, add $\dot{\mathbf{r}}_O$ to endpoint velocity and $\ddot{\mathbf{r}}_O$ to endpoint acceleration. If the coordinate frame also rotates, differentiating that frame adds further terms. A fixed-pivot model cannot silently represent the effects of a moving hand path or turning torso.

2.6 A Consistent Manufactured Trajectory

Replace an assumed “typical swing” table with a differentiable example whose signs can be checked. Let $T = 0.5$ s and $s = t/T$ for $0 \leq t \leq T$:

$$\begin{aligned} \theta_1 &= \frac{\pi}{2}(1 - s^2), & \theta_2 &= -\frac{\pi}{4}(1 - s). \\ \dot{\theta}_1 &= -\frac{\pi}{T}s, & \dot{\theta}_2 &= \frac{\pi}{4T}, & \ddot{\theta}_1 &= -\frac{\pi}{T^2}, & \ddot{\theta}_2 &= 0. \end{aligned}$$

t (s)	θ_1 (deg)	θ_2 (deg)	$\dot{\theta}_1$ (deg/s)	$\dot{\theta}_2$ (deg/s)
0	90	-45	0	90
0.25	67.5	-22.5	-180	90
0.5	0	0	-360	90

This is an illustrative segment, not measured golf data or a full address-to-follow-through swing. At its start, the first coordinate has a turning point while the second link is already rotating. At its end, both links point downward, but the state differs from rest. There is no assumed ball collision at that endpoint.

The same values can be reproduced directly:


```

1 # Manufactured Chapter 2 trajectory
2 import numpy as np
3
4 duration_s = 0.5
5 time_s = np.array([0.0, 0.25, 0.5])
6 phase = time_s / duration_s
7 q_deg = np.column_stack((90 * (1 - phase**2), -45 * (1 - phase)))
8 rate_deg_s = np.column_stack((-180 * phase / duration_s, np.full(3, 45 /
9     duration_s)))
10 q_rad = np.deg2rad(q_deg)
11 rate_rad_s = np.deg2rad(rate_deg_s)

```

At $t = T$, substitution into the Jacobian gives

$$\dot{r}_P = \begin{bmatrix} -2.2\pi \\ 0 \end{bmatrix} \text{ m/s} \approx \begin{bmatrix} -6.912 \\ 0 \end{bmatrix} \text{ m/s}.$$

The acceleration is $(-5.4\pi, 3.65\pi^2) \text{ m/s}^2$: leftward tangential and upward normal components. This makes the interplay explicit: configuration sets the velocity map, signed rates set the current motion and curvature terms, and joint accelerations supply the remaining acceleration.

A fully actuated ideal model with invertible inertia and unrestricted input moments can realize this smooth segment by inverse dynamics. That statement supplies no evidence that a human can realize it with physiological limits, contact constraints or the intended impact task. Those are additional questions.

2.7 From Motion to Forces and Golf Outcomes

Forward dynamics uses the model, state and inputs to compute acceleration. Inverse dynamics uses a sufficiently differentiated motion, inertial parameters and external-load assumptions to compute required net generalized forces. It generally does not determine a unique set of individual muscle, grip or ground forces. Multiple load distributions can produce the same generalized effect.

For a useful golf investigation, define the task as well as the state. Clubhead position and speed do not specify face orientation, contact location or ball launch. Earlier forces and energy exchanges influence the state arriving near impact; posture and rates determine the instantaneous kinematic map; inertia and feasible inputs constrain what can change before contact. Ground and grip conditions influence those feasible inputs. A contact model then relates the arriving club and ball states to the collision outcome.

This is why the aspects must be studied together. Kinematics supplies the geometry of motion; dynamics supplies the force and energy balance; control specifies how inputs are chosen; measurements test whether the chosen model and explanations match a golfer. None of these layers can substitute for the others.

Key Principle 2.1. Key Takeaways

1. Declare the body, frame, angle reference and sign before interpreting a coordinate.
2. Configuration dimension, admissible velocity dimension and predictive state dimension are different questions.

3. A complete state is complete for a specified model and future input rule.
4. The Jacobian maps signed rates to task velocity; acceleration also includes changing-geometry terms.
5. Constraint reactions depend on the equations and loading. Their mathematical role does not identify individual tissues or guarantee feasible contact.
6. A consistent synthetic example checks reasoning; it does not establish human performance or a coaching prescription.

2.8 Chapter Exercises With Worked Answers

1. Count the Configuration and Velocity Dimensions

Consider a planar rigid pendulum, a spherical rigid pendulum, an unrestricted planar two-link chain, and an oriented sphere rolling without slipping on a horizontal plane. State the assumptions.

Answer. Their configuration dimensions are 1, 2, 2 and 5 respectively. For the sphere, two numbers locate the center in the plane and three specify orientation. With radius R and angular velocity ω , zero velocity at the contact point gives

$$v_x = R\omega_y, \quad v_y = -R\omega_x.$$

There are three freely selectable instantaneous velocity components, not five: two independent no-slip conditions constrain the five configuration rates. Spin ω_z remains allowed by this ideal point-contact condition. Forbidding spin is an additional assumption. Ignoring orientation changes the question to locating the center rather than describing the full rigid-body configuration.

A human upper-body count cannot be inferred from the words “shoulder, elbow and wrist” alone. A declared open chain with a 3-DOF shoulder, 1-DOF elbow and 2-DOF wrist has six configuration DOF at a fixed base. Adding forearm rotation, a second arm or closed grasp constraints changes that model and requires a new count.

2. Draw the Configuration Map

Draw a square $(-\pi, \pi] \times (-\pi, \pi]$ for the ideal two-link chain and identify opposite edges. Add an illustrative restriction $|\theta_2| \leq \pi/2$.

Answer. The unrestricted square represents a torus, not four physical walls. The added restriction removes configurations with larger relative angles; it does not establish a human range limit or guarantee that every remaining point is reachable within a prescribed time and torque budget.

3. Calculate a Point Speed

For $L_1 = 0.35$ m and $\dot{\theta}_1 = 500^\circ/\text{s}$, find hinge speed with the base fixed.

Answer. Convert to radians first: $|\dot{\mathbf{r}}_H| = 0.35(500\pi/180) \approx 3.054$ m/s. At $\theta_1 = 0$ the positive-rate velocity points right; at $\theta_1 = \pi/2$ it points up. These are signs in the declared frame, not target-relative swing directions.

4. Distinguish Endpoint Velocity From Acceleration

At $\mathbf{q} = (0, 0)$ and rates $(1, 0)$ rad/s, set both joint accelerations to zero.

Answer. Endpoint velocity is $(1.35, 0)$ m/s, but acceleration is $(0, 1.35)$ m/s². Zero joint angular acceleration does not mean zero Cartesian acceleration. The velocity direction is changing.

5. Interpret a Phase Projection

Can two points with identical $(\theta_1, \dot{\theta}_1)$ have different endpoint velocities?

Answer. Yes. With rates $(1, 0)$ rad/s, configurations $(0, 0)$ and $(0, \pi/2)$ give endpoint velocities $(1.35, 0)$ and $(0.35, 1)$ m/s. The projection has discarded the second angle. In the manufactured trajectory, a zero first-coordinate rate at $t = 0$ likewise does not imply a motionless endpoint: its speed is $\pi/2$ m/s.

6. Interpret a Limit Without Inventing Physiology

A model imposes a rigid joint stop. What changes if tissue compliance is modeled instead?

Answer. An ideal reaction enforces the stop's compatibility conditions, potentially with an impact law at engagement. A compliant model allows deformation and requires a force–deformation–rate law and any internal states it uses. Actuation can affect the load in either formulation. Neither formulation alone establishes a person's safe range or predicts injury.

2.9 On This Site

Continue with Chapter 3, Chapter 7, and Chapter 20. For spatial descriptions, see [rotation representations](#); for model size, see [degrees of freedom and dimensionality](#). The site's [notation reference](#) supplies the broader conventions; each model must still declare its particular frame and input definitions.

Chapter 3

The Double Pendulum: Golf's Simplest Useful Model

In Plain Language 3.1. Why This Model?

A two-link model lets us ask a precise question: how do geometry, inertia, gravity and applied joint moments combine to produce motion at a distal endpoint? The answer already contains nonlinear coupling, moving-joint energy transfer and a distinction between joint motion and task motion. These are useful ingredients for understanding the golf swing.

The simplification does not follow from a claim that the wrist, hips or torso contribute little. Here the first link is an effective proximal body and the second is an effective distal body. Interpreting their hinge as an elbow produces a different anatomical model from interpreting it as an arm-club connection. We will call them link 1, link 2 and the connecting hinge throughout. Their coordinates are not measurements of particular anatomical joints.

The model is planar, rigid and supported at a fixed pivot. It omits a moving hand path, bilateral grasp, shaft deformation, spatial clubface orientation and ball contact. Deriving its equations exactly does not validate those simplifications for a human swing. Nor does having explicit differential equations give a general closed-form trajectory: numerical integration is normally needed.

3.1 The Double Pendulum: Physical Setup

The System

Let O be the fixed pivot and H the connecting hinge. Link lengths are L_1, L_2 , masses M_1, M_2 , and COM distances from each proximal hinge are r_1, r_2 . The centroidal planar moments of inertia are I_1, I_2 . They have units kg m^2 and exclude the parallel-axis contribution. Define

$$K_1 = I_1 + M_1 r_1^2, \quad K_2 = I_2 + M_2 r_2^2.$$

Thus K_i is the inertia about the body's own proximal hinge, whereas I_i is about its COM. For a moving hinge, $\frac{1}{2}K_i\omega_i^2$ alone is not the body's full kinetic energy: COM translation must still be accounted for correctly. A point mass at the COM has $I_i = 0$ but generally $K_i > 0$.

The hinges impose position attachment and permit relative rotation. They are frictionless in this baseline; prescribed actuator moments can nevertheless act about them. The applied generalized moments are τ_1 at the base and τ_2 between the bodies. No muscle model

is implied by these inputs.

$$\mathbf{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \quad \alpha_1 = \theta_1, \quad \alpha_2 = \theta_1 + \theta_2. \quad (3.1)$$

Angles increase counterclockwise from downward vertical in an inertial plane with horizontal x and upward y . The first coordinate is absolute; the second is relative. Zero first angle means downward, not toward the target. The distal endpoint P is a geometric point a distance L_2 from H ; it is not necessarily the distal COM.

In Plain Language 3.2. Absolute vs. Relative Angles

Both coordinate conventions describe the same physical motion. Relative coordinates make a hinge actuator's power easy to express; absolute coordinates can make some energy terms simpler. Neither removes inertial coupling. The absolute distal angular velocity is $\dot{\theta}_1 + \dot{\theta}_2$, not $\dot{\theta}_2$.

Writing $\boldsymbol{\alpha} = \mathbf{A}\mathbf{q}$ with $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, virtual work requires

$$\mathbf{Q}_\alpha = \mathbf{A}^{-T} \boldsymbol{\tau} = \begin{bmatrix} \tau_1 - \tau_2 \\ \tau_2 \end{bmatrix}, \quad \mathbf{M}_\alpha = \mathbf{A}^{-T} \mathbf{M}_q \mathbf{A}^{-1}.$$

The hinge moment acts oppositely on the two bodies. Its total power is $\tau_2(\dot{\alpha}_2 - \dot{\alpha}_1) = \tau_2\dot{\theta}_2$. The full actuator power is therefore $\boldsymbol{\tau}^T \dot{\mathbf{q}}$ in relative coordinates. Changing coordinates without transforming forces breaks this equality.

Numerical Parameters for Illustration

The following manufactured parameters match the later affine and constraint-force examples, so their results can be compared directly. This is not a measured golfer or a fitted arm-club model. Link 2 has endpoint length 1.0 m; its COM is at 0.5 m. We use $g = 9.81 \text{ m/s}^2$.

	Link 1	Link 2
L (m)	0.35	1.0
M (kg)	2.5	0.4
r (m)	0.175	0.5
I_{COM} (kg m ²)	0.025	0.03
K_{hinge} (kg m ²)	0.1015625	0.13

Point-mass and rigid-body parameters describe different mass distributions: adding a hinge inertia to an existing COM-translation term double counts its parallel-axis contribution. For compatibility with the expanded notation below, $L_{i,\text{cm}} = r_i$.

In Plain Language 3.3. Roadmap: Deriving the Equations of Motion

First locate both COMs and compute their velocities. Add translational and centroidal rotational kinetic energy, then gravitational potential energy. Apply the forced Euler-Lagrange equations and collect mass, velocity-bias and potential-gradient terms. Finally, independently check energy balance and map the joint mo-

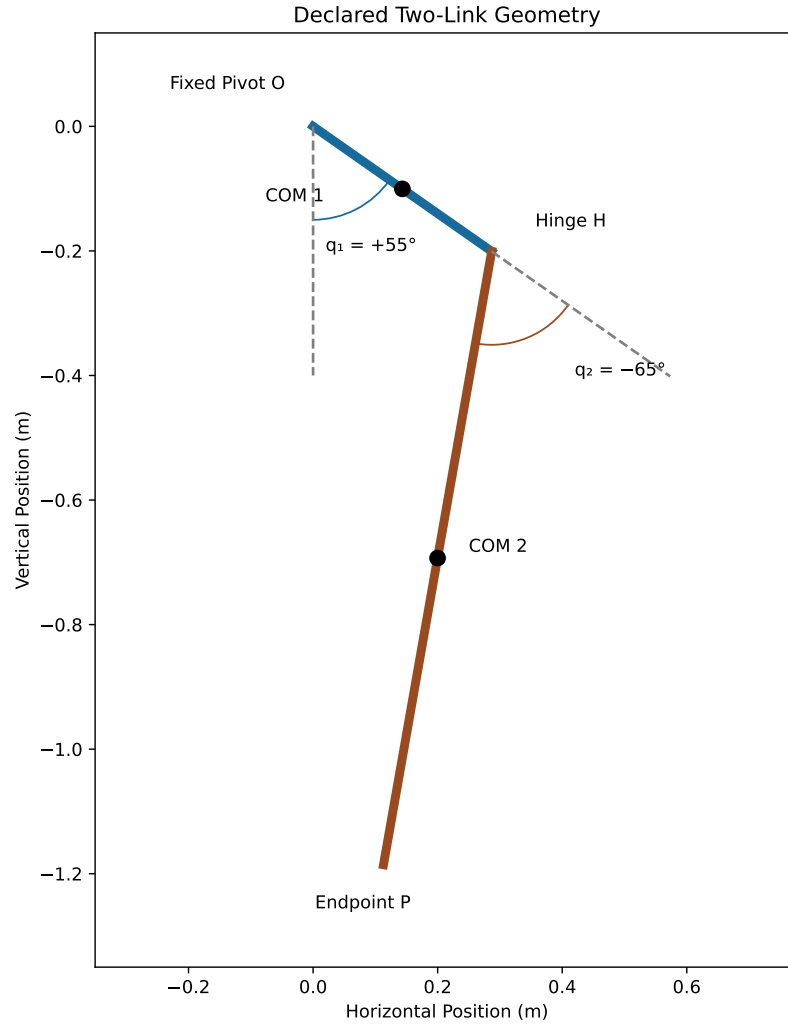


Figure 3.1: Declared Two-Link Geometry. The First Angle Is Measured From Downward Vertical; the Second Is Relative to the First Link. COM Markers and Angle Arcs Are Computed From the Same Geometry. The Distal Endpoint and COM Are Different Points. This Is a Manufactured Planar Model, Not Measured Golf Anatomy.

tion to the endpoint task. These checks connect the algebra to quantities a reader can interpret physically.

3.2 The Kinetic Energy

To derive the equations of motion, we start with energy. Every moving system has kinetic energy.

Kinetic Energy of Link 1

Link 1 rotates about the fixed pivot. Its COM translation and centroidal rotation combine to give:

$$T_1 = \frac{1}{2} K_1 \dot{\theta}_1^2 \quad (3.2)$$

Here $K_1 = I_1 + M_1 L_{1,\text{cm}}^2$. The centroidal inertia is retained; the point-mass limit is obtained by setting $I_1 = 0$, not $K_1 = 0$.

Kinetic Energy of Link 2

Link 2 has a moving proximal hinge. Its absolute angular rate and COM velocity must both be included.

The connecting hinge is at position:

$$\mathbf{r}_H = L_1(\sin \theta_1, -\cos \theta_1) \quad (3.3)$$

Its velocity is:

$$\dot{\mathbf{r}}_H = L_1 \dot{\theta}_1 (\cos \theta_1, \sin \theta_1) \quad (3.4)$$

The center of mass of link 2 is at distance $L_{2,\text{cm}}$ from the connecting hinge, at angle $\theta_1 + \theta_2$:

$$\mathbf{r}_{\text{cm},2} = L_1(\sin \theta_1, -\cos \theta_1) + L_{2,\text{cm}}(\sin(\theta_1 + \theta_2), -\cos(\theta_1 + \theta_2)) \quad (3.5)$$

Its velocity is:

$$\dot{\mathbf{r}}_{\text{cm},2} = L_1 \dot{\theta}_1 (\cos \theta_1, \sin \theta_1) \quad (3.6)$$

$$+ L_{2,\text{cm}}(\dot{\theta}_1 + \dot{\theta}_2)(\cos(\theta_1 + \theta_2), \sin(\theta_1 + \theta_2)) \quad (3.7)$$

The kinetic energy of link 2 is the sum of COM translation and centroidal rotation:

$$T_2 = \frac{1}{2} M_2 |\dot{\mathbf{r}}_{\text{cm},2}|^2 + \frac{1}{2} I_2 (\dot{\theta}_1 + \dot{\theta}_2)^2 \quad (3.8)$$

Expanding $|\dot{\mathbf{r}}_{\text{cm},2}|^2$:

$$|\dot{\mathbf{r}}_{\text{cm},2}|^2 = [L_1 \dot{\theta}_1 \cos \theta_1 + L_{2,\text{cm}}(\dot{\theta}_1 + \dot{\theta}_2) \cos(\theta_1 + \theta_2)]^2 \quad (3.9)$$

$$+ [L_1 \dot{\theta}_1 \sin \theta_1 + L_{2,\text{cm}}(\dot{\theta}_1 + \dot{\theta}_2) \sin(\theta_1 + \theta_2)]^2 \quad (3.10)$$

After expanding and simplifying (using $\cos^2 + \sin^2 = 1$ and $\cos \alpha \cos \beta + \sin \alpha \sin \beta = \cos(\alpha - \beta)$):

$$|\dot{\mathbf{r}}_{\text{cm},2}|^2 = L_1^2 \dot{\theta}_1^2 + L_{2,\text{cm}}^2 (\dot{\theta}_1 + \dot{\theta}_2)^2 \quad (3.11)$$

$$+ 2L_1 L_{2,\text{cm}} \dot{\theta}_1 (\dot{\theta}_1 + \dot{\theta}_2) \cos \theta_2 \quad (3.12)$$

So:

$$T_2 = \frac{1}{2} M_2 [L_1^2 \dot{\theta}_1^2 + L_{2,\text{cm}}^2 (\dot{\theta}_1 + \dot{\theta}_2)^2 + 2L_1 L_{2,\text{cm}} \dot{\theta}_1 (\dot{\theta}_1 + \dot{\theta}_2) \cos \theta_2] \quad (3.13)$$

$$+ \frac{1}{2} I_2 (\dot{\theta}_1 + \dot{\theta}_2)^2 \quad (3.14)$$

Total Kinetic Energy

$$T = T_1 + T_2 \quad (3.15)$$

$$= \frac{1}{2} K_1 \dot{\theta}_1^2 + \frac{1}{2} M_2 [L_1^2 \dot{\theta}_1^2 + L_{2,\text{cm}}^2 (\dot{\theta}_1 + \dot{\theta}_2)^2 \quad (3.16)$$

$$+ 2L_1 L_{2,\text{cm}} \dot{\theta}_1 (\dot{\theta}_1 + \dot{\theta}_2) \cos \theta_2] + \frac{1}{2} I_2 (\dot{\theta}_1 + \dot{\theta}_2)^2 \quad (3.17)$$

Collecting terms:

$$T = \frac{1}{2} (K_1 + M_2 L_1^2) \dot{\theta}_1^2 \quad (3.18)$$

$$+ \frac{1}{2} (M_2 L_{2,\text{cm}}^2 + I_2) (\dot{\theta}_1 + \dot{\theta}_2)^2 \quad (3.19)$$

$$+ M_2 L_1 L_{2,\text{cm}} \dot{\theta}_1 (\dot{\theta}_1 + \dot{\theta}_2) \cos \theta_2 \quad (3.20)$$

In Plain Language 3.4. Why Kinetic Energy Matters

Kinetic energy is energy of motion, not a fuel supply separate from the mechanics. In this model, energy can enter through applied moments and can exchange between kinetic and gravitational potential forms. In a golfer, chemical energy, elastic storage, dissipation, moving support and impact add further terms. There is no reason for all backswing energy to end up in club motion; other segments still move and energy can be stored or dissipated.

The conservative variational statement is stationary action, $\delta \int (T - V) dt = 0$, with fixed endpoint variations. Stationary does not always mean a minimum, and prescribed nonconservative moments require the forced equations below. This principle is not an optimization claim about a golfer's chosen movement.

3.3 The Potential Energy

The work of a uniform gravitational field is represented by gravitational potential energy for the two bodies.

Definition 3.1. Gravitational Potential Energy

The potential energy of a mass at height h is $V = Mgh$ (relative to some reference).

For link 1, the center of mass is at height:

$$h_1 = -L_{1,\text{cm}} \cos \theta_1 \quad (3.21)$$

The pivot is the zero-height reference; downward COM positions have negative height. For link 2, the center of mass is at height:

$$h_2 = -L_1 \cos \theta_1 - L_{2,\text{cm}} \cos(\theta_1 + \theta_2) \quad (3.22)$$

Total potential energy:

$$V = M_1 g h_1 + M_2 g h_2 \quad (3.23)$$

$$= -M_1 g L_{1,\text{cm}} \cos \theta_1 - M_2 g [L_1 \cos \theta_1 + L_{2,\text{cm}} \cos(\theta_1 + \theta_2)] \quad (3.24)$$

In Plain Language 3.5. The Potential Energy Landscape

Gravity does work $W_g = V(q_a) - V(q_b)$ between configurations. A path ending lower can release potential energy, but a downswing label alone does not specify both COM heights. Lifting the system earlier required work, and gravity supplies no net work over a closed configuration cycle. Its contribution to club speed must be assessed together with actuator work and the energy retained in the proximal body.

3.4 The Lagrangian and Euler-Lagrange Equations

Now we use the Lagrangian formalism.

In Plain Language 3.6. Why the Lagrangian Approach?

Generalized coordinates solve the ideal hinge geometry before dynamics are computed. The equal-and-opposite hinge forces cancel in virtual work over compatible motions, so they need not be explicit unknowns in the coordinate equations. They still exist physically and must be recovered from body force balances when load or interface power is the question. Newton–Euler and Lagrange formulations express the same mechanics; recursive force-balance methods are also efficient for large systems.

Definition 3.2. Lagrangian

The Lagrangian is defined as:

$$\mathcal{L} = T - V \quad (3.25)$$

It's the kinetic energy minus the potential energy.

The Euler-Lagrange equations state that the equations of motion are:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = \tau_i \quad (3.26)$$

for each coordinate q_i . Here, τ_i is the generalized applied moment conjugate to q_i . In a biomechanical model, mapping muscle forces to this resultant is a separate modeling step.

This looks abstract. But it's a recipe: compute two partial derivatives of the Lagrangian, take one time derivative, subtract them, and you get the equation of motion for that coordinate.

Let's do it for the double pendulum.

The θ_1 Equation

We need:

- $\frac{\partial \mathcal{L}}{\partial \theta_1}$
- $\frac{\partial \mathcal{L}}{\partial \dot{\theta}_1}$

From T (Equation 3.20), the terms involving $\dot{\theta}_1$ are:

$$\frac{\partial T}{\partial \dot{\theta}_1} = (K_1 + M_2 L_1^2) \dot{\theta}_1 + (M_2 L_{2,\text{cm}}^2 + I_2)(\dot{\theta}_1 + \dot{\theta}_2) \quad (3.27)$$

$$+ M_2 L_1 L_{2,\text{cm}} [(\dot{\theta}_1 + \dot{\theta}_2) + \dot{\theta}_1] \cos \theta_2 \quad (3.28)$$

$$= (K_1 + M_2 L_1^2) \dot{\theta}_1 + (M_2 L_{2,\text{cm}}^2 + I_2)(\dot{\theta}_1 + \dot{\theta}_2) \quad (3.29)$$

$$+ M_2 L_1 L_{2,\text{cm}} (2\dot{\theta}_1 + \dot{\theta}_2) \cos \theta_2 \quad (3.30)$$

Taking the time derivative:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}_1} \right) = (K_1 + M_2 L_1^2) \ddot{\theta}_1 + (M_2 L_{2,\text{cm}}^2 + I_2)(\ddot{\theta}_1 + \ddot{\theta}_2) \quad (3.31)$$

$$+ M_2 L_1 L_{2,\text{cm}} (2\ddot{\theta}_1 + \ddot{\theta}_2) \cos \theta_2 \quad (3.32)$$

$$- M_2 L_1 L_{2,\text{cm}} (2\dot{\theta}_1 + \dot{\theta}_2) \dot{\theta}_2 \sin \theta_2 \quad (3.33)$$

Now the partial of T with respect to θ_1 :

$$\frac{\partial T}{\partial \theta_1} = 0 \quad (3.34)$$

The kinetic energy depends on both rates and the relative angle, but not on an overall rotation of the pair in this fixed inertial plane.

The partial of V with respect to θ_1 :

$$\frac{\partial V}{\partial \theta_1} = M_1 g L_{1,\text{cm}} \sin \theta_1 + M_2 g L_1 \sin \theta_1 + M_2 g L_{2,\text{cm}} \sin(\theta_1 + \theta_2) \quad (3.35)$$

$$= (M_1 L_{1,\text{cm}} + M_2 L_1) g \sin \theta_1 + M_2 g L_{2,\text{cm}} \sin(\theta_1 + \theta_2) \quad (3.36)$$

So:

$$-\frac{\partial \mathcal{L}}{\partial \theta_1} = -\frac{\partial (T - V)}{\partial \theta_1} = \frac{\partial V}{\partial \theta_1} \quad (3.37)$$

The Euler-Lagrange equation becomes:

$$(K_1 + M_2 L_1^2) \ddot{\theta}_1 + (M_2 L_{2,\text{cm}}^2 + I_2)(\ddot{\theta}_1 + \ddot{\theta}_2) \quad (3.38)$$

$$+ M_2 L_1 L_{2,\text{cm}} (2\ddot{\theta}_1 + \ddot{\theta}_2) \cos \theta_2 \quad (3.39)$$

$$- M_2 L_1 L_{2,\text{cm}} (2\dot{\theta}_1 + \dot{\theta}_2) \dot{\theta}_2 \sin \theta_2 \quad (3.40)$$

$$= \tau_1 - (M_1 L_{1,\text{cm}} + M_2 L_1) g \sin \theta_1 - M_2 g L_{2,\text{cm}} \sin(\theta_1 + \theta_2) \quad (3.41)$$

Similarly, for θ_2 :

$$(M_2 L_{2,\text{cm}}^2 + I_2)(\ddot{\theta}_1 + \ddot{\theta}_2) + M_2 L_1 L_{2,\text{cm}} \ddot{\theta}_1 \cos \theta_2 \quad (3.42)$$

$$+ M_2 L_1 L_{2,\text{cm}} \dot{\theta}_1^2 \sin \theta_2 \quad (3.43)$$

$$= \tau_2 - M_2 g L_{2,\text{cm}} \sin(\theta_1 + \theta_2) \quad (3.44)$$

In Plain Language 3.7. Reading the Equations

The acceleration terms express coupled inertia. The remaining quadratic rate terms arise from differentiating kinetic energy; they are generalized moments, not automatically individual hinge forces. Gravity on the right is $-\nabla V$. At a small positive common tilt, its physical generalized moment restores the pair toward downward vertical. Setting only $\theta_1 = 0$ does not make both gravity terms zero when the distal body remains tilted.

The term proportional to $\dot{\theta}_1^2 \sin \theta_2$ can influence relative motion, but its sign in the bias vector is not the sign of the resulting acceleration. The full coupled system must be solved. Calling a term centrifugal does not establish outward inertial acceleration or a universal straightening action.

3.5 Rewriting in Matrix Form

The Euler-Lagrange equations are hard to parse. Let's rewrite them in the standard form:

$$\mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau} \quad (3.45)$$

where:

- $\mathbf{M}(\mathbf{q})$ is the **mass matrix** (moment of inertia matrix).
- $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ contains the Coriolis and centrifugal terms.
- $\mathbf{g}(\mathbf{q})$ is the potential gradient; the physical generalized gravity torque is $-\mathbf{g}$.
- $\boldsymbol{\tau}$ is the applied generalized moment vector.

The Mass Matrix

Collecting all terms with $\ddot{\theta}_1$ and $\ddot{\theta}_2$:

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} M_{11}(\theta_2) & M_{12}(\theta_2) \\ M_{21}(\theta_2) & M_{22} \end{bmatrix} \quad (3.46)$$

where:

$$M_{11}(\theta_2) = K_1 + M_2 L_1^2 + M_2 L_{2,\text{cm}}^2 + I_2 + 2M_2 L_1 L_{2,\text{cm}} \cos \theta_2 \quad (3.47)$$

$$M_{12}(\theta_2) = M_{21}(\theta_2) = M_2 L_{2,\text{cm}}^2 + I_2 + M_2 L_1 L_{2,\text{cm}} \cos \theta_2 \quad (3.48)$$

$$M_{22} = M_2 L_{2,\text{cm}}^2 + I_2 \quad (3.49)$$

The matrix is symmetric and positive definite for the declared positive inertial parameters. It depends on relative configuration, not on velocity. Its off-diagonal terms express kinetic coupling. M_{11} describes the proximal inertia when relative acceleration is held zero; it is not generally the denominator for free-distal response to proximal torque. Eliminating the distal equation at fixed state gives the incremental relation

$$\delta \ddot{\theta}_1 = \frac{\delta \tau_1 - (M_{12}/M_{22})\delta \tau_2}{M_{11} - M_{12}^2/M_{22}}.$$

The positive denominator is the Schur effective inertia. The affine and constraint chapters extend this comparison to input capacity and explicitly imposed locks.

The Coriolis/Centrifugal Matrix

Define $b = M_2 L_1 r_2$. The full velocity-bias vector is unambiguous once the coordinates are specified:

$$\mathbf{c}(\mathbf{q}, \dot{\mathbf{q}}) = b \sin \theta_2 \begin{bmatrix} -2\dot{\theta}_1 \dot{\theta}_2 - \dot{\theta}_2^2 \\ \dot{\theta}_1^2 \end{bmatrix}.$$

A Christoffel factorization satisfying $\mathbf{C}\dot{\mathbf{q}} = \mathbf{c}$ is

$$\mathbf{C} = b \sin \theta_2 \begin{bmatrix} -\dot{\theta}_2 & -(\dot{\theta}_1 + \dot{\theta}_2) \\ \dot{\theta}_1 & 0 \end{bmatrix}. \quad (3.50)$$

$\mathbf{C}$ has units $\text{kg m}^2/\text{s}$ and its product with angular rate has units N m . Putting $\dot{\theta}_1^2$ in C_{21} would introduce an extra rate factor when the matrix multiplies velocity. Other factorizations can produce the same $\mathbf{c}$; matrix entries alone are not unique physical force channels.

This choice also gives $\dot{\mathbf{M}} - 2\mathbf{C}$ skew-symmetric. Consequently

$$\dot{T} = \dot{\mathbf{q}}^T (\mathbf{M}\ddot{\mathbf{q}} + \mathbf{c}), \quad \frac{d}{dt}(T + V) = \boldsymbol{\tau}^T \dot{\mathbf{q}}.$$

This is an independent sign and factor check. It shows why velocity-dependent terms can alter motion and exchange energy between bodies without being an additional energy source. The derivation follows standard manipulator mechanics, with the present COM-inertia and gravity conventions stated explicitly. [Tedrake, n.d.b]

The Gravity Vector

$$\mathbf{g}(\mathbf{q}) = \begin{bmatrix} (M_1 L_{1,\text{cm}} + M_2 L_1)g \sin \theta_1 + M_2 g L_{2,\text{cm}} \sin(\theta_1 + \theta_2) \\ M_2 g L_{2,\text{cm}} \sin(\theta_1 + \theta_2) \end{bmatrix} \quad (3.51)$$

3.6 Numerical Example: A Declared Two-Link State

Before Contact

Consider $\mathbf{q} = (0, -5^\circ)$ and $\dot{\mathbf{q}} = (10, 9)$ rad/s. These values are chosen for arithmetic, not presented as population averages or a measured impact. With the declared parameters,

$$\mathbf{M} \approx \begin{bmatrix} 0.420029758 & 0.199733629 \\ 0.199733629 & 0.130000000 \end{bmatrix} \text{ kg m}^2,$$

$$\mathbf{c} \approx \begin{bmatrix} 1.592335420 \\ -0.610090199 \end{bmatrix} \text{ N m}, \quad \mathbf{g} \approx \begin{bmatrix} -0.170999567 \\ -0.170999567 \end{bmatrix} \text{ N m}.$$

For zero applied moments, $\ddot{\mathbf{q}} = \mathbf{M}^{-1}(-\mathbf{c} - \mathbf{g})$ is

$$\ddot{\mathbf{q}} \approx (-23.166250085, 41.601299717) \text{ rad/s}^2.$$

Applying (5, 0) N m at that same state instead gives

$$\ddot{\mathbf{q}} \approx (21.020338033, -26.287527914) \text{ rad/s}^2.$$

The proximal input changes both accelerations and reverses the distal relative acceleration here. That instantaneous comparison does not predict a terminal club speed without integrating the resulting motion and specifying the task.

In Plain Language 3.8. Interpretation of the Force Components

The Schur effective inertia is 0.113156508 kg m², whereas M_{11} is 0.420029758 kg m². Using the latter to predict free-distal proximal response would be incorrect. The negative values of $\mathbf{g}$ mean the actual generalized gravity moments $-\mathbf{g}$ are positive at this configuration.

The 5 N m proximal input supplies 50 W at $\dot{\theta}_1 = 10$ rad/s. Torque magnitude, acceleration response and power answer different questions. No comparison with an unsupported generic shoulder-torque budget can determine which muscles supplied this power or how efficiently a golfer moved. The total kinetic energy is 44.242514485 J, shared between the two physical bodies.

3.7 The Interaction Torque: How the Links Couple

Definition 3.3. Interaction Torque

In inverse dynamics, one may call the terms involving another coordinate's motion interaction terms. For example, the first equation contains $M_{12}\ddot{\theta}_2$. This is an algebraic decomposition conditional on coordinates and bookkeeping. It is not by itself a separately measured hinge torque or a unique amount of transferred energy.

Example 3.1. The Coupled Interaction of Two Links

For a specified motion, changing distal acceleration changes the proximal moment required to maintain that motion. For specified applied moments, both accelerations instead change through the inverse mass matrix. Those inverse- and forward-dynamics questions must not be interchanged.

The ideal hinge can transmit a force even when its applied rotational moment is zero. If $\mathbf{F}_H$ acts on link 2 at the moving point H , it supplies power $\mathbf{F}_H \cdot \mathbf{v}_H$ to that body and the opposite power to link 1. With τ_2 acting on link 2, the physical body moment powers are $\tau_2 \dot{\alpha}_2$ and $-\tau_2 \dot{\alpha}_1$; their sum is $\tau_2 \dot{\theta}_2$. The base moment adds $\tau_1 \dot{\theta}_1$. Physical body energies use each COM and centroidal inertia, not arbitrary blocks of the generalized mass matrix.

A whip-like speed pattern is therefore a possible motion to investigate, not a proof that an amateur should accelerate the elbow harder or that an expert feels a particular load. A useful comparison specifies input history, initial energy, constraints, endpoint result and segment power balances. The constraint-force chapter develops that accounting explicitly.

From Joint Motion to Endpoint Motion

Let $\mathbf{e}(\alpha) = (\sin \alpha, -\cos \alpha)^T$ and $\mathbf{t}(\alpha) = (\cos \alpha, \sin \alpha)^T$. The endpoint and its Jacobian are

$$\mathbf{p} = L_1 \mathbf{e}(\theta_1) + L_2 \mathbf{e}(\alpha_2), \quad \mathbf{J}_P = [L_1 \mathbf{t}(\theta_1) + L_2 \mathbf{t}(\alpha_2) \quad L_2 \mathbf{t}(\alpha_2)].$$

Then

$$\dot{\mathbf{p}} = \mathbf{J}_P \dot{\mathbf{q}}, \quad \ddot{\mathbf{p}} = \mathbf{J}_P \ddot{\mathbf{q}} - L_1 \dot{\theta}_1^2 \mathbf{e}(\theta_1) - L_2 (\dot{\theta}_1 + \dot{\alpha}_2)^2 \mathbf{e}(\alpha_2).$$

The last two terms are normal accelerations toward the respective moving rotation centers in this decomposition. They remain even at zero joint angular acceleration. Endpoint speed is not simply proximal angular rate times the sum of lengths unless relative rotation and geometry satisfy that special rigid-rotation assumption.

At the numerical state above, endpoint velocity is (22.427699264, -1.655959112) m/s. With zero applied moments, endpoint acceleration is (41.719934305, 393.019565568) m/s². Its magnitude is dominated by the normal component; that does not mean speed is increasing at the same rate. The speed derivative is the projection $\dot{\mathbf{p}} \cdot \dot{\mathbf{p}} / |\dot{\mathbf{p}}|$ when speed is nonzero.

The Jacobian determinant is $L_1 L_2 \sin \theta_2$. At aligned links it loses rank: one instantaneous endpoint acceleration component cannot be chosen freely using joint accelerations. This kinematic task singularity does not make the mass matrix singular. Underactuation, task rank and inertia are different properties.

3.8 Chaotic Dynamics: What Sensitivity Establishes**Definition 3.4. Chaotic Dynamics**

Sensitive finite-time response alone does not diagnose chaos. A chaos claim needs a specified dynamical system and regime, typically sustained sensitive dependence on a bounded invariant set, with an appropriate diagnostic such as an asymptotic

Lyapunov exponent. Large transient amplification, an unstable equilibrium or two different terminal outcomes is not sufficient by itself.

Experiments and simulations on a mechanical double pendulum have demonstrated chaotic motion in specified conditions. That evidence concerns the studied pendulum and initial states, not a controlled human swing. [Shinbrot et al., 1992] The unforced double pendulum can have regular and chaotic regimes. Nonlinearity does not make every initial condition chaotic. Near the stable downward equilibrium, the conservative linearization is

$$\mathbf{M}_0 \delta \ddot{\mathbf{q}} + \mathbf{K}_g \delta \mathbf{q} = 0, \quad \mathbf{K}_g = g \begin{bmatrix} M_1 r_1 + M_2 L_1 + M_2 r_2 & M_2 r_2 \\ M_2 r_2 & M_2 r_2 \end{bmatrix}.$$

The declared model has small-oscillation angular frequencies approximately 3.644068205 and 7.552873204 rad/s, obtained from $\det(\mathbf{K}_g - \omega^2 \mathbf{M}_0) = 0$. Both are real and positive. This local oscillatory approximation supplies a counterexample to inevitable exponential error growth, without proving global regularity at larger amplitudes.

In Plain Language 3.9. Sensitivity to Initial Conditions

For a short controlled movement, a more direct question is how perturbations change the actual endpoint task over its finite duration. With state $x = (q, \dot{q})$ and $\dot{x} = f(x, u, t)$, linearize along a declared nominal motion:

$$\delta \dot{x} = A(t) \delta x + B(t) \delta u, \quad \delta x(T) = \Psi(T, 0) \delta x(0) + \int_0^T \Psi(T, s) B(s) \delta u(s) ds.$$

Here $A = f_x$, $B = f_u$ and Ψ is the state-transition matrix. Map this state perturbation through the task derivative and specify scales before comparing angle, angular-rate and position errors. Feedback changes the linearized system; timing shifts change the input or event description. Effects can amplify, cancel or contract depending on direction and phase.

Thus a release shifted by 0.01 s does not automatically miss the sweet spot. The present endpoint model has no clubface contact location at all. An impact event also changes the evaluation time, which must be included in event sensitivity. The later affine and brain-control chapters connect finite-horizon mechanics to input bounds, observation and neuromuscular dynamics.

3.9 Decomposing the Equation of Motion

The zero-input drift is obtained by solving $\ddot{q} = M^{-1}(-c - g)$; $M\ddot{q}$ on the left is not a separate force that the system supplies by itself. Applied generalized moments may represent actuators, passive tissues or externally imposed loads depending on the declared input convention. A net moment is not a neural command, individual muscle force or metabolic effort.

For golf, the fixed-pivot planar model is a first model-ladder member. Nesbit's three-dimensional study used measured motion to drive a full-body model with a flexible club and emphasized changing hand paths and wrist motion. These features are precisely what

the fixed-pivot two-link reduction omits. Its inverse-dynamics outputs depend on segment, contact and motion-processing assumptions; the study does not calibrate our parameters or establish that wrist effects are negligible. Association with skill does not independently identify an optimal control policy. [Nesbit, 2005]

A useful extension adds only the degrees of freedom or physical effects needed for the target observable, then checks the improvement against relevant measurements. A model predicting endpoint position may still be inadequate for face orientation, shaft strain, bilateral hand loads or tissue stress. The connecting argument is constructive: derive the mechanics, define the task, quantify uncertainty, and test which omitted mechanisms change the conclusion.

Key Principle 3.1. Key Takeaways

Use one coordinate convention and distinguish COM from hinge inertia. Derive M , c and g from the same energy model, verify $\dot{E} = \tau^T \dot{q}$, and solve the coupled equations before interpreting acceleration. Recover segment powers and loads from physical body boundaries. Map joint motion through endpoint kinematics, including normal acceleration. Treat sensitivity, chaos, effort and skill as separate questions requiring their own observables and evidence.

Looking Ahead

Chapter 4 develops forces and moments. Chapter 5 separates the declared drift and input response. The constraint-force chapter explains compatible motion and boundary work; the triple-pendulum chapter adds another independently moving body. These extensions should preserve this chapter's sign, inertia and power conventions while declaring any changed parameters.

Exercises 3.1. Chapter Exercises

1. Distinguish Centroidal and Hinge Inertia

For a separate uniform-rod example with $M = 1.5$ kg and $L = 1$ m, calculate both inertias. The answer is $I_{\text{COM}} = ML^2/12 = 0.125$ kg m² and $K_{\text{end}} = I_{\text{COM}} + M(L/2)^2 = 0.5$ kg m². These are not the numerical link-2 parameters used above. Adding K_{end} to COM translation would double count the parallel-axis term.

2. Compute Gravitational Work

Using the chapter parameters, compare $(150^\circ, -70^\circ)$ with $(0, 0)$. Substitution into V gives a potential drop of 12.192849345 J, so gravity does that positive amount of work from the first configuration to the second. It is a total-system energy contribution, not necessarily club kinetic energy; actuator work and initial kinetic energy must also be included.

3. Identify the Rigid-Rotation Assumption

If the links are aligned, relative angular rate is zero and the pair rotates at 10.47 rad/s, the endpoint's inward normal acceleration is $\omega^2(L_1 + L_2) = 147.988215$ m/s², or 15.085444954 g . This is centripetal inertial acceleration. An outward centrifugal

term belongs to a rotating-frame balance. If the relative rate is nonzero, use the full two-link expression instead.

4. Solve an Underdetermined Task Request

At $(0, 0)$, request 50 m/s^2 horizontal tangential endpoint acceleration. The condition is $1.35\dot{\theta}_1 + \dot{\theta}_2 = 50$: there is a family, not a unique pair. Examples are $(0, 50)$, $(20, 23)$ and $(37.037037037, 0) \text{ rad/s}^2$. With rates $(10, 9) \text{ rad/s}$, all have upward normal acceleration $0.35(10)^2 + 1(9)^2 = 396 \text{ m/s}^2$. Additional torque limits or a declared optimization objective can select among feasible pairs. This smooth pre-contact model cannot infer ball-impact accelerations without a collision model.

5. Keep Gravity Signs and Acceleration Separate

At $(150^\circ, -70^\circ)$, physical generalized gravity moments are

$$-\mathbf{g} \approx (-4.764830311, -1.932192811) \text{ N m}.$$

They tend toward decreasing the declared angles in virtual work; naming a forward or backward golf direction requires a target-frame convention. Actual acceleration is $M^{-1}(-\mathbf{c} - \mathbf{g} + \boldsymbol{\tau})$ and need not have the sign of each isolated moment component.

6. Sensitivity Is Not a Chaos Diagnosis

The small-oscillation calculation gives regular linear modes near the hanging equilibrium, whereas a larger-energy unforced trajectory may require a separate chaos analysis. A controlled golf movement requires its input or feedback law, initial state, model parameters and finite-duration task before sensitivity can be computed. Neither the label double pendulum nor two perturbed trajectories alone establishes that a human swing is chaotic.

Part II

The Affine Framework

Chapter 4

Forces and Torques: Where They Come From

In Plain Language 4.1. Decomposing the Forces

The golf swing combines applied forces, coupled motion and contact. To understand that combination, first specify what is being balanced: a physical body's linear momentum, its angular momentum, a generalized joint equation, or mechanical energy. These are connected descriptions, but their terms are not interchangeable.

This chapter builds on the two-link model of Chapter 3. It explains how to recover physical loads, distinguish them from coordinate-dependent terms, and map the resulting motion to an endpoint task. A large force need not supply large power, a negative torque need not remove energy, and a net joint moment does not identify individual muscle forces. Keeping those distinctions explicit makes the connection to golf more informative.

4.1 Five Parts of a Mechanical Accounting

For the declared fixed-pivot, planar two-link model,

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{c}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau}, \quad \mathbf{c} = \mathbf{C}\dot{\mathbf{q}}, \quad \mathbf{g} = \nabla V. \quad (4.1)$$

The coordinates are $q_1 = \theta_1$, measured counterclockwise from downward vertical, and relative angle $q_2 = \theta_2$. Neither is automatically an anatomical shoulder or elbow angle. The following five parts organize the equation and its physical interpretation:

- **Acceleration terms** $\mathbf{M}\ddot{\mathbf{q}}$ come from differentiating momentum. They are not an additional external agency pushing the body.
- **Velocity-bias terms** $\mathbf{c}$ arise from the same kinetic-energy model in moving generalized coordinates. Their decomposition into individual C_{ij} entries is not unique.
- **Gravity** has physical generalized moment $-\mathbf{g}$. The positive gradient appears on the left because it has been moved there.
- **Applied generalized moments** $\boldsymbol{\tau}$ represent the declared actuators and other loads included in that input convention. In a muscle-driven model, their production has additional geometry and internal dynamics.
- **Constraint reactions** act at physical interfaces. Ideal hinge reactions eliminated by compatible coordinates still exist, but must not be added a second time to the same reduced equation.

These are not five independent physical force sources to add without regard to signs and body boundaries. The kinetic terms together play the role of $m\mathbf{a}$. A more general model can include elastic and dissipative loads, external wrenches, and explicit contact reactions on the right. The standard manipulator formulation provides this organization; the actual terms depend on the declared model. [Tedrake, n.d.b]

4.2 Inertial Terms and Coupled Acceleration

For a body of constant mass, expressed in an inertial frame,

$$\sum \mathbf{f}_{\text{ext}} = m\mathbf{a}_C. \quad (4.2)$$

Here C is its COM. For a rigid body in planar motion, moments about its COM satisfy

$$\sum n_C = I_C \dot{\omega}. \quad (4.3)$$

The corresponding spatial equation in body axes is $\sum \mathbf{n}_C = \mathbf{I}_C \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{I}_C \boldsymbol{\omega})$. A scalar $I\alpha$ formula is therefore a planar or fixed-axis specialization, not a universal three-dimensional joint equation. Choosing a moving point instead of the COM requires the appropriate moment-transport terms.

Example 4.1. Specified Acceleration and Required Moments

Use Chapter 3's manufactured parameters, not a measured golfer: masses (2.5, 0.4) kg, proximal length 0.35 m, distal endpoint length 1 m, COM distances (0.175, 0.5) m, centroidal inertias (0.025, 0.03) kg m², and $g = 9.81$ m/s².

At $\mathbf{q} = (0, -5^\circ)$, with both rates zero, prescribe $\ddot{\mathbf{q}} = (1000\pi/180, 0)$ rad/s². Full inverse dynamics gives

$$\mathbf{M}\ddot{\mathbf{q}} \approx (7.33090223, 3.48600945) \text{ N m}, \quad \boldsymbol{\tau} \approx (7.15990266, 3.31500988) \text{ N m}.$$

The second moment is needed to maintain the prescribed zero relative acceleration. Applying only the first moment with a free distal actuator produces a different motion. For an incremental proximal input at fixed state with distal input unchanged, the acceleration gain is the inverse Schur inertia, approximately 8.83731762 (rad/s²)/(N m), rather than $1/M_{11}$. Constraints change this comparison, as developed in Chapter 7.

Why Mass Distribution Matters

For a fixed axis, $I = \int r_\perp^2 dm$. Adding mass while preserving geometry increases that inertia; moving mass outward can increase it without changing total mass. In a multibody system, off-diagonal inertia, distal freedom and the requested task also matter. A lighter club does not by itself establish greater speed, lower physiological effort or shorter ball flight: those conclusions require a player's input response, impact properties and a flight model. Compare equipment under a declared task and constraints, not from mass alone.

4.3 Velocity-Dependent Terms: Coriolis and Centrifugal

Centrifugal Force and Inertial Acceleration

A particle moving uniformly on a circle has inward acceleration. The force of a taut string on the particle is inward; the particle's outward force on the string acts on a different body. In a frame rotating with the particle, an outward centrifugal inertial term can balance the inward physical force. These are compatible descriptions once the body and frame are named.

Definition 4.1. Circular-Motion Acceleration

For radius r and constant angular rate ω , let $\mathbf{e}_r$ point outward. Then

$$\mathbf{a}_{\text{inertial}} = -\omega^2 r \mathbf{e}_r, \quad \mathbf{f}_{\text{centrifugal}} = +m\omega^2 r \mathbf{e}_r \quad (4.4)$$

in the corresponding rotating-frame force balance. The second expression is an inertial force term, not an extra physical outward force on the particle. Nonuniform circular motion also has tangential acceleration $r\dot{\omega}$.

Example 4.2. A Declared Circular Particle

For a separate 0.2 kg particle, $r = 1.35$ m and $\omega = 30$ rad/s imply speed 40.5 m/s and inward acceleration 1215 m/s². Its required inward resultant is 243 N. This is an ideal circular-particle calculation. It is not a whole-club grip measurement, and it does not establish shaft stress or muscle activity. Gravity and other applied loads must be included when finding a particular support force.

Example 4.3. Curvature in the Two-Link Model

The hinge H at distance 0.35 m from the fixed pivot has normal acceleration $0.35\dot{q}_1^2$ toward that pivot. The distal endpoint has a further normal term about moving H involving $(\dot{q}_1 + \dot{q}_2)^2$. Their vector sum need not point toward the fixed pivot. A velocity-bias term in a generalized equation cannot be renamed a physical elbow reaction or assumed to straighten a human arm; first solve the coupled accelerations and use the actual geometry.

Coriolis Force and the Moving Frame

Let a frame have moving origin O, angular velocity $\mathbf{\Omega}$ and angular acceleration $\dot{\mathbf{\Omega}}$. Express every vector in common axes at the instant of evaluation. For relative position $\mathbf{r}$, velocity $\mathbf{v}_{\text{rel}}$ and acceleration $\mathbf{a}_{\text{rel}}$, transport gives

$$\mathbf{a} = \mathbf{a}_O + \mathbf{a}_{\text{rel}} + \dot{\mathbf{\Omega}} \times \mathbf{r} + \mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{r}) + 2\mathbf{\Omega} \times \mathbf{v}_{\text{rel}}.$$

The Coriolis contribution here has a positive sign. When solving Newton's law for relative acceleration, it moves to the other side:

$$\mathbf{f}_{\text{Cor}} = -2m\mathbf{\Omega} \times \mathbf{v}_{\text{rel}}. \quad (4.5)$$

This is the rotating-frame inertial force. The other transport terms likewise change sides. An inertial observer attributes the particle's acceleration to its physical applied forces; the rotating observer uses the added inertial terms. Omitting the frame or mixing these two equations reverses the interpretation.

Definition 4.2. Generalized Velocity Bias

The two-link expression uses $b = m_2 L_1 r_2 = 0.07 \text{ kg m}^2$:

$$\mathbf{c} = b \sin q_2 \begin{bmatrix} -2\dot{q}_1 \dot{q}_2 - \dot{q}_2^2 \\ \dot{q}_1^2 \end{bmatrix}.$$

The cross-rate product is often called Coriolis coupling and the squared-rate terms centrifugal coupling. These labels organize the generalized equation. They do not identify independent physical contact forces or a unique $\mathbf{C}$ matrix.

Example 4.4. The Sign of a Bias Is Not Assistance

At $\mathbf{q} = (0, -5^\circ)$ and rates $(10, 9) \text{ rad/s}$, the bias is

$$\mathbf{c} \approx (1.59233542, -0.61009020) \text{ N m}.$$

Positive c_1 is on the left of the equation. In inverse dynamics it increases the required first input for a fixed acceleration; in forward dynamics its contribution is $-\mathbf{M}^{-1}\mathbf{c}$. A positive first component therefore cannot be described as a positive applied shoulder torque helping motion. Joint coupling also means an isolated component's sign does not determine the full acceleration.

What Quadratic Scaling Establishes

At fixed configuration, scaling every rate by s gives $\mathbf{c}(\mathbf{q}, s\dot{\mathbf{q}}) = s^2 \mathbf{c}(\mathbf{q}, \dot{\mathbf{q}})$. This statement does not apply to every velocity-dependent physical load: viscous damping is typically linear and aerodynamic laws require their own assumptions. Scaling one rate alone also does not scale all cross terms uniformly.

In Plain Language 4.2. Speed, Power and the Chosen Task

Larger bias terms need not help the desired endpoint motion or reduce required input. Reversing all rates leaves this quadratic bias unchanged but reverses its contraction with velocity. Direction matters as much as magnitude.

The kinetic-energy identity is $\dot{T} = \dot{\mathbf{q}}^T(\mathbf{M}\dot{\mathbf{q}} + \mathbf{c})$, so $\dot{E} = \boldsymbol{\tau}^T \dot{\mathbf{q}}$ when gravitational potential is included. Bias terms can mediate coupling without being a new energy supply. Calling a motion easy, economical or optimal requires a specified effort measure and comparison; the manipulator equation alone supplies none of those conclusions.

4.4 Gravitational Forces and Potential

In a uniform field, each body experiences $m\mathbf{g}_0$, where $\mathbf{g}_0 = (0, -9.81)$ m/s² in this inertial plane. Its magnitude does not depend on joint configuration; its generalized moments do. With COM distances r_1, r_2 ,

$$\mathbf{g}(\mathbf{q}) = g \begin{bmatrix} (m_1 r_1 + m_2 L_1) \sin q_1 + m_2 r_2 \sin(q_1 + q_2) \\ m_2 r_2 \sin(q_1 + q_2) \end{bmatrix}, \quad \mathbf{Q}_g = -\mathbf{g}. \quad (4.6)$$

Reading the Gravity Terms

The first component contains both the motion of H and the distal COM relative to H. Setting $q_1 = 0$ removes only the first sine term; a tilted distal body retains a gravity moment. At $(0, 0)$ both gradients vanish and the configuration is a stable hanging equilibrium for this conservative model. At $(\pi, 0)$ both vanish but the equilibrium is unstable. Zero moment does not imply zero weight or zero joint force.

At $(150^\circ, -70^\circ)$, physical generalized gravity moments are

$$\mathbf{Q}_g \approx (-4.76483031, -1.93219281) \text{ N m}.$$

These signs refer to the declared coordinates. Translating them to forward or backward golf motion requires a target-frame convention and the full coupled dynamics.

Gravity Work and the Complete Cycle

Define $\Delta V = V_{\text{final}} - V_{\text{initial}}$. Then

$$W_g = -\Delta V = V_{\text{initial}} - V_{\text{final}}, \quad P_g = -\mathbf{g}^T \dot{\mathbf{q}}. \quad (4.7)$$

A lower final potential gives positive gravity work. Returning both coordinates to the same configuration gives zero net gravity work, regardless of the final speed. Actuator work, initial kinetic energy, storage and dissipation explain changes in the full energy balance; gravity cannot supply net energy over a closed configuration cycle.

For the declared model, descent from $(150^\circ, -70^\circ)$ to $(0, 0)$ releases 12.192849345 J. The reverse ascent costs that potential increase. Without a trajectory, one cannot determine whether gravity supplies or absorbs power early or late in a downswing. Nor does its work specify how much energy ends in the distal body.

4.5 Muscular Forces and the Input Model

$$\boldsymbol{\tau} = \mathbf{R}(\mathbf{q})\mathbf{F}_T + \boldsymbol{\tau}_{\text{other}}. \quad (4.8)$$

Here $\mathbf{F}_T$ contains nonnegative tendon tensions, $\mathbf{R}$ contains signed moment arms, and $\boldsymbol{\tau}_{\text{other}}$ collects other loads included in the chosen convention. An inverse-dynamics net moment does not uniquely recover the tendon tensions. Equal antagonist moments can cancel while both tensions remain substantial. The result also depends on which passive loads, external wrenches and residual actuators have already been modeled.

Constraints on Muscular Forces

A simple torque box $\tau_i^{\min} \leq \tau_i \leq \tau_i^{\max}$ can be an engineering approximation, but muscle tension being nonnegative does not imply symmetric independent bounds. Moment arms depend on configuration; multiarticular muscles couple joints; force capacity depends on activation, fiber length and shortening velocity, with tendon mechanics introducing further states. Numerical capacity bounds require measurements or an explicitly validated actuator model for the population and state of interest.

Consider a manufactured antagonist pair with signed moment arms $(0.03, -0.03)$ m. Tensions $(100, 100)$ N and $(200, 200)$ N both produce zero net moment. They are not the same internal loading state. Net zero moment or net zero joint power cannot demonstrate muscle inactivity or zero metabolic cost.

From Excitation to Realized Moment

A muscle-driven state description can include excitation $\mathbf{u}$, activation $\mathbf{a}$, fiber/tendon states $\mathbf{z}$, and their dynamics:

$$\dot{\mathbf{a}} = \mathbf{f}_a(\mathbf{a}, \mathbf{u}), \quad \dot{\mathbf{z}} = \mathbf{f}_z(\mathbf{q}, \dot{\mathbf{q}}, \mathbf{a}, \mathbf{z}), \quad \mathbf{F}_T = \mathbf{f}_T(\mathbf{q}, \dot{\mathbf{q}}, \mathbf{a}, \mathbf{z}).$$

This is a model structure, not a universal physiological law. OpenSim's documented activation and forward-dynamics workflows distinguish excitation, activation, fiber state and realized force. They motivate retaining these states rather than assuming an arbitrarily prescribed net torque is an instantaneous neural command. [OpenSim, 2020, Hicks, 2024]

Once inputs, initial states and an admissible contact mode are specified, forward integration gives

$$\dot{\mathbf{q}}(t) = \dot{\mathbf{q}}(0) + \int_0^t \ddot{\mathbf{q}}(s) ds, \quad \mathbf{q}(t) = \mathbf{q}(0) + \int_0^t \dot{\mathbf{q}}(s) ds. \quad (4.9)$$

The acceleration is evaluated along the evolving state, not held at its initial value. Feedback and activation dynamics change that evolution. Chapter 24 develops the relation to observation and motor control.

4.6 Constraint Forces and Contact Feasibility

Definition 4.3. Constraint Reactions

An ideal reaction enforces an assumed compatible motion. A revolute hinge can transmit forces while allowing free relative rotation about its axis; an actuator moment about that axis is a separate load. A rigid attachment has a different constraint set. Real contact must also satisfy unilateral and friction conditions, so an arbitrary computed reaction may be infeasible.

Why Some Reactions Disappear From Reduced Equations

Compatible angular coordinates solve the ideal hinge geometry. Equal-and-opposite interface forces cancel in virtual work over compatible motions, eliminating them from the reduced generalized equations. They can be recovered from physical body force and moment

balances. This elimination does not make them zero, and not every choice of generalized coordinates eliminates every constraint.

For explicit stationary constraints, a common formulation is

$$M\ddot{\mathbf{q}} + \mathbf{c} + \mathbf{g} = \boldsymbol{\tau} + \mathbf{J}_c^T \boldsymbol{\lambda}, \quad \mathbf{J}_c \ddot{\mathbf{q}} + \dot{\mathbf{J}}_c \dot{\mathbf{q}} = 0.$$

The reaction depends on motion, applied loads and the constraint mode. A frictional contact additionally needs compressive normal force and tangential force within its friction limit. If the required load lies outside the admissible set, use a slip/separation model. Simply lowering applied torque does not necessarily cause slip; nor does increasing it monotonically increase every reaction component.

4.7 Putting It Together: Attribution Before Contact

Use $\mathbf{q} = (0, -5^\circ)$, $\dot{\mathbf{q}} = (10, 9)$ rad/s and prescribed $\ddot{\mathbf{q}} = (20, -30)$ rad/s². This is a manufactured state before contact, not a measured impact. It fixes all quantities needed for an instantaneous inverse-dynamics calculation:

$$\begin{aligned} M\ddot{\mathbf{q}} &\approx (2.40858629, 0.09467258) \text{ N m}, \\ \mathbf{c} &\approx (1.59233542, -0.61009020) \text{ N m}, \\ \mathbf{g} &\approx (-0.17099957, -0.17099957) \text{ N m}, \\ \boldsymbol{\tau} &\approx (3.82992214, -0.68641719) \text{ N m}. \end{aligned}$$

Each row is a two-component vector with the same convention. Both off-diagonal inertia contributions are included. The input power is 32.12146671 W. The negative distal input at positive relative rate absorbs 6.17775470 W in that generalized channel; positive proximal power more than offsets it. A negative torque at a negative rate would instead supply positive power.

The Endpoint Acceleration

With $\mathbf{e}(\alpha) = (\sin \alpha, -\cos \alpha)^T$, the endpoint acceleration is

$$\ddot{\mathbf{p}} = \mathbf{J}_P \ddot{\mathbf{q}} - L_1 \dot{q}_1^2 \mathbf{e}(q_1) - L_2 (\dot{q}_1 + \dot{q}_2)^2 \mathbf{e}(q_1 + q_2). \quad (4.10)$$

Here $\mathbf{J}_P$ is the full endpoint Jacobian from Chapter 3. The speed and distance to the base alone do not determine local path curvature; v^2/r requires the appropriate curvature radius, not an assumed shoulder-to-head distance.

$$\ddot{\mathbf{p}} \approx (28.50127615, 395.49784344) \text{ m/s}^2. \quad (4.11)$$

The total includes vector normal and tangential contributions; their magnitudes must not be added as scalars. Smooth swing acceleration and collision acceleration are separate regimes. Predicting the ball-club contact impulse, duration or peak force requires contact mechanics and incident states absent from this two-link model.

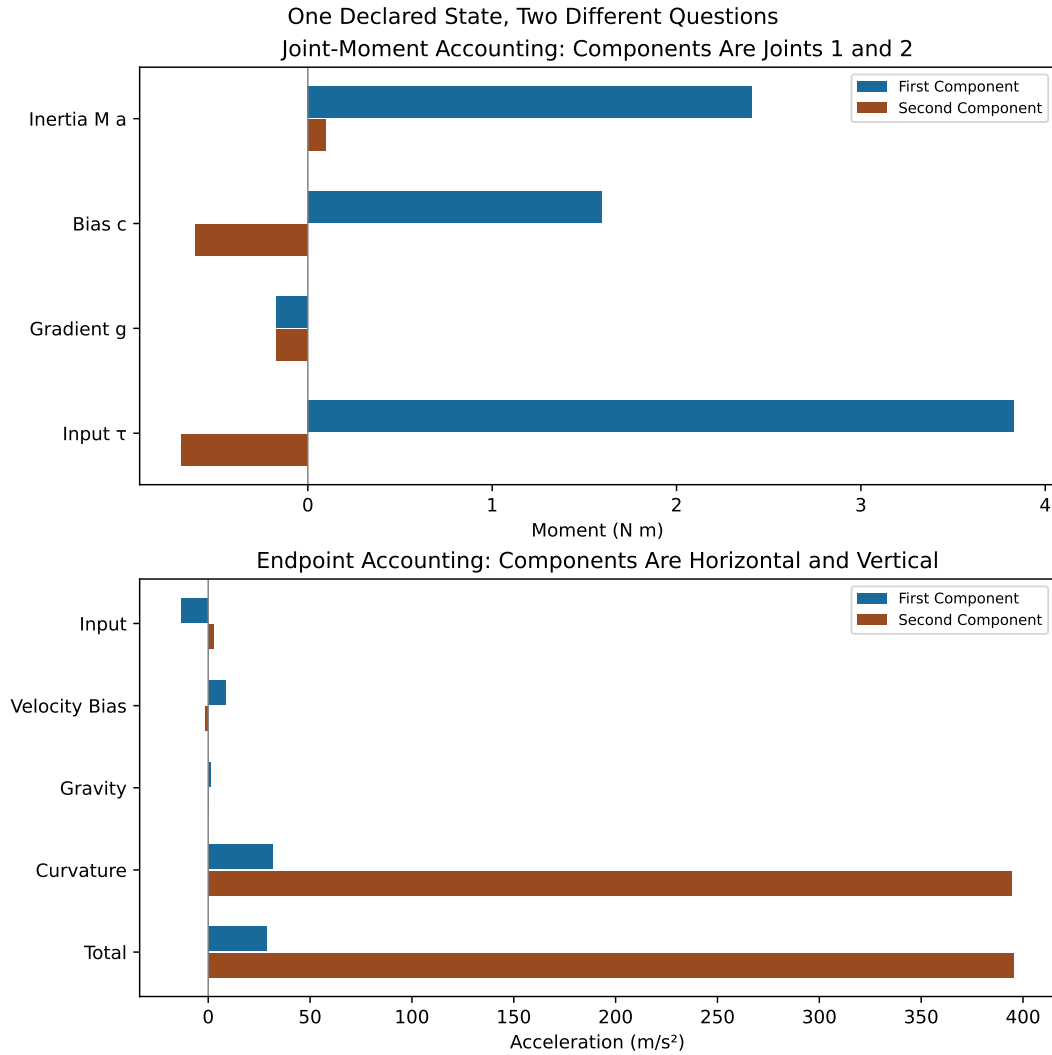


Figure 4.1: One Declared State, Two Different Questions. Top: Inverse-Dynamics Moment Components at the Two Joints. Bottom: Fixed-State Contributions to Horizontal and Vertical Endpoint Acceleration. Input and Bias Contributions Must Be Mapped Through Inverse Inertia and the Endpoint Jacobian; Curvature Remains a Separate Kinematic Term. The State Is Manufactured, Not a Measured Golf Swing.

Where Do These Accelerations Come From?

At this fixed state, an algebraic decomposition of the same endpoint acceleration is

$$\ddot{\mathbf{p}} = \mathbf{J}_P \mathbf{M}^{-1} \boldsymbol{\tau} - \mathbf{J}_P \mathbf{M}^{-1} \mathbf{c} - \mathbf{J}_P \mathbf{M}^{-1} \mathbf{g} + \dot{\mathbf{J}}_P \dot{\mathbf{q}}.$$

The four vectors, in horizontal/vertical m/s², are approximately

$$\begin{aligned} \mathbf{a}_{\text{input}} &\approx (-13.21865815, 2.47827787), \\ \mathbf{a}_{\text{bias}} &\approx (8.79688124, -1.45418002), \\ \mathbf{a}_{\text{gravity}} &\approx (1.45982994, -0.15254042), \\ \mathbf{a}_{\text{curvature}} &\approx (31.46322313, 394.62628601). \end{aligned}$$

They sum to the endpoint value above. The large curvature term does not correspond to an independent applied force or energy source. This is instantaneous attribution with the state held fixed. Removing an input throughout a swing changes the state, Jacobian and every subsequent contribution, so terminal causal comparisons require a new forward solution. Imposed constraints also alter the acceleration map.

Recovering the Distal Body's Interface Load

For an actual club, use the whole-club COM and whole-club mass. Let $\mathbf{f}_H$ be the resultant hand force on the club and $\mathbf{f}_{\text{other}}$ other non-gravitational external forces. Then

$$\mathbf{f}_H = m_c(\mathbf{a}_C - \mathbf{g}_0) - \mathbf{f}_{\text{other}}. \quad (4.12)$$

For a planar rigid club, its required hand moment at H also satisfies

$$\mathbf{n}_H = I_C \dot{\boldsymbol{\omega}} - (\mathbf{r}_H - \mathbf{r}_C) \times \mathbf{f}_H - \mathbf{n}_{\text{other},C}. \quad (4.13)$$

The planar cross product denotes its out-of-plane component. Multiplying clubhead acceleration by a nominal head mass omits shaft/grip mass distribution, COM motion and moment balance. A guessed tangential acceleration cannot establish a numerical grip-force range.

For the declared distal link, $\mathbf{a}_{C2} \approx (17.75063808, 215.24892172)$ m/s², giving $\mathbf{f}_H \approx (7.10025523, 90.02356869)$ N and $\mathbf{n}_H = \boldsymbol{\tau}_2 \approx -0.68641719$ N m. Recovering the proximal body's balance gives base force $(15.85025523, 158.29856869)$ N and the same $\boldsymbol{\tau}_1$ obtained from the generalized equation. Independent Newton–Euler and energy calculations check these results.

In Plain Language 4.3. Resultant Load Is Not Grip Squeeze

A resultant wrench constrains the sum of contact loads, not the distribution of finger pressure or the forces supplied by each hand. Opposing hand forces can change internal loading while preserving the same net force and moment on the club. Grip squeeze can also change frictional capacity and compliance, which a perfectly rigid attachment model omits.

Choi and Park's instrumented-grip study combines measured internal grip loading with spatial upper-limb inverse dynamics, illustrating why separate hand loads need information beyond a net club wrench. Its nine professional participants and modified

551 g club with foam balls limit direct transfer to ordinary driver swings. It does not establish a universal squeeze prescription. [Choi and Park, 2020]
 OpenSim likewise distinguishes net generalized forces from the structural reaction loads produced after actuator forces are specified. Reports must identify the receiving body, frame and included actuators. Neither output alone is tissue stress or proof of muscle inactivity. [Hicks, 2019]

4.8 Wrenches: Forces, Moments and Power

Definition 4.4. A Wrench With a Declared Reference

Use force-first ordering $\mathbf{w}_H = (\mathbf{f}, \mathbf{n}_H)$ and matching twist $\mathbf{V}_H = (\mathbf{v}_H, \boldsymbol{\omega})$, expressed in the same axes. The moment is about H, not an unspecified point. The rigid-body power is

$$P = \mathbf{w}_H^T \mathbf{V}_H = \mathbf{f} \cdot \mathbf{v}_H + \mathbf{n}_H \cdot \boldsymbol{\omega}.$$

For offset $\mathbf{d} = \mathbf{r}_K - \mathbf{r}_H$, shift to K using

$$\mathbf{n}_K = \mathbf{n}_H - \mathbf{d} \times \mathbf{f}, \quad \mathbf{v}_K = \mathbf{v}_H + \boldsymbol{\omega} \times \mathbf{d}.$$

These transformations preserve power. Rotating axes rotates both three-vectors consistently. Other texts use moment-first ordering; the matching twist must change too. A resultant wrench describes the net rigid-body load, not its detailed contact-pressure distribution. [Lynch and Park, n.d.]

At ideal hinge H, force power into the distal body is $\mathbf{f}_H \cdot \mathbf{v}_H$, and the proximal body receives its negative. Here those powers are ± 24.85089331 W. Hinge moment powers are $\tau_2(\dot{q}_1 + \dot{q}_2)$ on body 2 and $-\tau_2\dot{q}_1$ on body 1. Their sum is $\tau_2\dot{q}_2$. The individual body kinetic-energy derivatives are approximately 20.3125 and 15.05795849 W; together they equal input power plus 3.24899178 W of gravity power. A zero net internal force power does not imply zero power into each body.

In Plain Language 4.4. The Full-Body Problem

For the combined golfer–club system, hand–club forces are internal and cancel in the total momentum balance. Ground contact, weight, aerodynamic loads and ball contact are external. The total external force equals total mass times system-COM acceleration; external moments about the COM equal the rate of total angular momentum about it. Dynamic balance does not generally mean that the force and moment sums are zero.

A floating-body model commonly uses generalized velocity $\mathbf{v}$, kinematics $\dot{\mathbf{q}} = \mathbf{N}(\mathbf{q})\mathbf{v}$, actuator map $\mathbf{S}^T$, and contact map $\mathbf{J}_c^T$. Its base equations are not directly actuated by muscle moments. Ground reactions connect internal actuation to motion while satisfying contact feasibility. A moving torso also changes arm-base motion; treating it as fixed changes the question being answered. The physics remains consistent across this hierarchy when frames, loads and omitted mechanisms are declared.

4.9 Numerical Summary: Compare Declared States

Instead of assigning unsupported torques to named swing phases, hold $\mathbf{q} = (0, -5^\circ)$ and $\ddot{\mathbf{q}} = (20, -30)$ fixed while scaling rates (10, 9) by s . Every row is independently reproducible inverse dynamics, not a time history:

s	c_1		c_2	τ_1		τ_2	P_{input}
0	0		0	2.237587	-0.076327		0
1	1.592335	-0.610090		3.829922	-0.686417		32.121467
2	6.369342	-2.440361		8.606928	-2.516688		126.838188

Moments are N m and power is W. The inertia contribution and gravity gradient are unchanged, whereas bias scales by s^2 . In this example greater speed requires greater first input and more positive total input power for the prescribed acceleration. This is a counterexample to inevitable passive assistance, not a claim that faster golf always requires more effort. A phase name alone specifies neither configuration, rate, acceleration nor input.

4.10 The Fundamental Insight

The useful argument connects levels of description: applied loads change body momentum; geometry and inertia couple that change across joints; endpoint kinematics map it to a task; actuator dynamics and contact determine which motions are feasible. Physical power balances then track energy supplied, transferred, stored or dissipated. Measurements test the parts of that model relevant to the intended conclusion.

Key Principle 4.1. Key Takeaways

Name the body and frame before naming a force. Use full coupled dynamics rather than a single diagonal inertia. Distinguish physical gravity from the gradient moved to the left. Keep normal acceleration in the endpoint map. Recover whole-body COM loads and moments before interpreting grip measurements. Evaluate signed power rather than torque sign alone, and distinguish a net wrench from individual contact or muscle forces. Claims about optimal effort, skill or passive dominance require a stated task, admissible inputs and independent evidence.

Looking Ahead

Chapter 5 separates a declared zero-input drift from applied input response. Chapter 7 develops compatible motion and reaction power, and Chapter 24 adds the observation and actuation states needed for motor control. These extensions can test how the mechanisms combine over a swing; they do not follow from a universal table of helpful and opposing forces.

Exercises 4.1. Chapter Exercises

1. Prescribed Inertial Moments

At $\mathbf{q} = (0, 0)$, zero rates, prescribe $\ddot{\mathbf{q}} = (500\pi/180, 0)$ rad/s². The generalized gravity moment vanishes, and inverse dynamics requires approximately (3.67010017, 1.74532925) N m. The distal input is essential to the prescribed zero relative acceleration. One first-coordinate acceleration alone would leave the problem underdetermined.

2. Circular Motion and the Receiving Body

For a separate 1 kg particle in uniform circular motion at 10.47 rad/s and radius 1.35 m, acceleration is 147.988215 m/s² inward and the resultant force is 147.988215 N inward. Its outward reaction on a string acts on the string. The rotating-frame centrifugal term on the particle is outward; do not add it to the inertial-frame force balance.

3. Specified Coriolis Coupling

Specify $q_2 = -5^\circ$ and rates (8, 5) rad/s. Then $c_1 = 0.64059471$ and $c_2 = -0.39045773$ N m. These are left-side bias components, not standalone applied moments. Without configuration, the original rate-only request has no numerical answer. A sudden change in rate additionally needs an impulse or a finite acceleration history.

4. Gravity and Coordinate Signs

At $(150^\circ, -70^\circ)$, $\mathbf{Q}_g \approx (-4.76483031, -1.93219281)$ N m. The gradient is its negative. To infer acceleration, solve with the mass matrix, velocity terms and applied loads. To call it forward or backward, first define the golf target frame.

5. Complete Inverse Dynamics

Using the chapter's declared before-contact state and acceleration, add $\mathbf{M}\ddot{\mathbf{q}}$, $\mathbf{c}$ and $\mathbf{g}$ componentwise to obtain (3.82992214, -0.68641719) N m. The power is 32.12146671 W. Omitting $M_{12}\ddot{q}_2$ would add 5.99200887 N m to the first estimate and solve a different problem.

6. COM Force, Moment and Contact

Use the distal-link COM acceleration from the worked example: subtract gravity and multiply by 0.4 kg to obtain (7.10025523, 90.02356869) N. Balance moments about its COM to recover -0.68641719 N m at H. These determine a resultant in the ideal model, not finger pressure. Assess actual contact using a friction/compliance model and appropriate measurements.

7. Gravity Over a Closed Configuration Cycle

Specify both endpoints as (0, 0) and $(150^\circ, -70^\circ)$. Potential rises 12.192849345 J on ascent. Gravity work is negative that amount on ascent and positive on return; net gravity work is zero. If the final speed differs, explain it using the rest of the energy balance. Specifying only q_1 leaves the distal COM height unknown.

Chapter 5

The Affine Structure: Drift and Control

In Plain Language 5.1. The Big Reveal

A swing combines energy supplied by muscles with motion shaped by inertia, gravity, contact and tissue mechanics. The control-affine description makes one part of this interaction explicit: at a fixed state, how does a chosen input change the state derivative? It is a useful mathematical structure, not proof that skilled movement is predominantly passive.

For an ideal unconstrained rigid model in local coordinates,

$$M(q)\ddot{q} + c(q, \dot{q}) + g(q) = B(q)u. \quad (5.1)$$

The same model can be written

$$\dot{x} = f(x) + G(x)u. \quad (5.2)$$

The meaning of zero input, the admissible inputs and the retained state must accompany this equation. It does not decide which part of the swing supplies energy, which control is optimal, or how the nervous system implements a policy. Those questions require additional calculations and evidence.

5.1 From Equations of Motion to State Space

Defining State

For n independent local joint coordinates, write $v = \dot{q}$ and

$$x = \begin{bmatrix} q \\ v \end{bmatrix} \in \mathbb{R}^{2n}. \quad (5.3)$$

A two-joint planar example has $x = (q_1, q_2, v_1, v_2)^T$. This is a local chart, not a complete human description or a globally unique representation of periodic angles. A general rigid-body implementation may instead use $\dot{q} = N(q)v$, with different position and velocity dimensions. Activation, flexible tissues and shaft deformation require their own state variables.

$$\dot{x} = \begin{bmatrix} v \\ \dot{v} \end{bmatrix}. \quad (5.4)$$

A zero velocity at one instant is not a zero acceleration. The top block describes the rate of position change; the lower block determines how velocity changes.

Solving for Accelerations

Assume M is symmetric positive definite on the independent coordinates. Define c as the complete velocity-dependent inertial bias, and $g = \nabla_q V$ for the gravitational potential. A matrix C may satisfy $c = Cv$, but that factorization is not unique; dropping selected entries without deriving their product changes the dynamics. The standard manipulator formulation and its coordinate limitations are developed in [Tedrake, n.d.b].

$$\dot{v} = M^{-1}(Bu - c - g). \quad (5.5)$$

Numerically, solve the mass-matrix system rather than explicitly forming its inverse. The inverse notation is useful for analysis:

$$\dot{v} = \underbrace{-M^{-1}(c + g)}_{a_0(q,v)} + \underbrace{M^{-1}Bu}_{a_u(q,v,u)}. \quad (5.6)$$

Additional declared passive generalized forces belong in the corresponding bias; a moving base or contact introduces further terms or constraints.

The Control-Affine Form

$$f(x) = \begin{bmatrix} v \\ a_0 \end{bmatrix}, \quad G(x) = \begin{bmatrix} 0_{n \times m} \\ M^{-1}B \end{bmatrix}, \quad \dot{x} = f(x) + G(x)u. \quad (5.7)$$

Drift contains the kinematic block v as well as the uncontrolled acceleration. The zero position block of G means torque changes acceleration immediately in this ideal model; later position changes follow through integration. It does not mean position is uncontrollable over time.

Definition 5.1. Control-Affine Systems

At each fixed state, an affine input map has a constant offset and a linear part:

$$F(x, u) = f(x) + G(x)u. \quad (5.8)$$

Its state dependence may be nonlinear or linear. The linear system $\dot{x} = Ax + Bu$ is also control-affine, with $f = Ax$. An affine combination of two inputs produces the corresponding affine combination of derivatives at the same state; this does not imply superposition of complete nonlinear trajectories. Nonzero f means the input-to-derivative map is generally not a homogeneous linear map.

In Plain Language 5.2. Why Affine Matters

The columns of G describe local directions available through the chosen inputs. The set $f + GU$ describes attainable derivatives when inputs lie in $\mathcal{U}$. This separates the local mechanics from the problem of selecting an input for a particular goal. Torque limits, muscle redundancy and delayed activation make $\mathcal{U}$ and its evolution

as important as the matrix itself.

Here u is generalized actuator torque. It is not automatically muscle excitation. Two muscle patterns can yield the same net joint torque but different stiffness, metabolic cost and future force capacity. Defining zero torque therefore does not identify a relaxed muscle state.

5.2 Explicit Form for the Double Pendulum

Define two rigid links joined by ideal planar hinges, with the proximal pivot fixed. Let q_1 measure the first link from downward vertical and q_2 the second link relative to the first, so its absolute angle is $q_1 + q_2$. Link masses are m_i , proximal-to-COM distances r_i , first-link length l_1 , and COM inertias I_i . Torques u_1, u_2 are conjugate to these relative coordinates; $B = I$. This idealization retains coupling but does not resolve anatomical wrist planes, both arms or a flexible club.

Set

$$A = I_1 + m_1 r_1^2 + m_2 l_1^2, \quad D = I_2 + m_2 r_2^2, \quad b = m_2 l_1 r_2.$$

The COM positions are $p_1 = r_1(\sin q_1, -\cos q_1)^T$ and $p_2 = l_1(\sin q_1, -\cos q_1)^T + r_2(\sin(q_1 + q_2), -\cos(q_1 + q_2))^T$. Their translational energies and the absolute angular rates $v_1, v_1 + v_2$ give

$$M = \begin{bmatrix} A + D + 2b \cos q_2 & D + b \cos q_2 \\ D + b \cos q_2 & D \end{bmatrix}, \quad T = \frac{1}{2} v^T M v.$$

The potential and complete bias are

$$\begin{aligned} V &= -g_0 [(m_1 r_1 + m_2 l_1) \cos q_1 + m_2 r_2 \cos(q_1 + q_2)], \\ c &= b \sin q_2 \begin{bmatrix} -2v_1 v_2 - v_2^2 \\ v_1^2 \end{bmatrix}, \\ g &= g_0 \begin{bmatrix} (m_1 r_1 + m_2 l_1) \sin q_1 + m_2 r_2 \sin(q_1 + q_2) \\ m_2 r_2 \sin(q_1 + q_2) \end{bmatrix}. \end{aligned}$$

Here g_0 is gravitational acceleration, while g is a generalized-force vector. The second bias component can be nonzero when $v_2 = 0$; a rotating first link still affects the second. All inertias are about the COM, avoiding double-counting parallel-axis terms.

The Drift Vector Field

With $h = c + g$ and $K = M^{-1}$,

$$f(x) = \begin{bmatrix} v_1 \\ v_2 \\ -K_{11}h_1 - K_{12}h_2 \\ -K_{21}h_1 - K_{22}h_2 \end{bmatrix}. \quad (5.9)$$

Both bias components enter each acceleration through inverse inertia. Dependence on the current state does not erase the effect of prior control: earlier torques helped create that state.

The Control Matrix

$$G = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix}. \quad (5.10)$$

$$u = (u_1, u_2)^T. \quad (5.11)$$

$$(\dot{q}_1, \dot{q}_2, \dot{v}_1, \dot{v}_2)^T = f(x) + G(x)u. \quad (5.12)$$

The off-diagonal entries express coupled acceleration. They do not assign the net joint torque to a unique muscle or determine the club's Cartesian acceleration without kinematics.

Example 5.1. Drift in the Double Pendulum at Rest

Use manufactured parameters $m_1 = 2.5$ kg, $m_2 = 0.4$ kg, $l_1 = 0.35$ m, $r_1 = 0.175$ m, $r_2 = 0.5$ m, $I_1 = 0.025$ and $I_2 = 0.030$ kg m², and $g_0 = 9.81$ m/s². They are a declared calculation, not fitted golfer data.

At $q = (0, 0)$ and $v = 0$, both gravity torques vanish, so $f = 0$. The vertical downward configuration is a modeled equilibrium, not every golfer's address posture.

At $q = (45^\circ, 0)$ and $v = 0$,

$$M = \begin{bmatrix} 0.4205625 & 0.2 \\ 0.2 & 0.13 \end{bmatrix} \text{ kg m}^2,$$

$$g \simeq \begin{bmatrix} 5.3933 \\ 1.3873 \end{bmatrix} \text{ N m}, \quad a_0 \simeq \begin{bmatrix} -28.8732 \\ 33.7484 \end{bmatrix} \text{ rad/s}^2.$$

The relative second-joint acceleration is positive even though gravity draws the total system toward lower potential. It cannot be calculated from the first gravity component and K_{11} alone. An inverse inertia has units (kg m²)⁻¹; angular units should accompany reported rates.

5.3 Drift as a Vector Field

Definition 5.2. Vector Field

A state vector field assigns a tangent vector to each state. Its integral curves satisfy the differential equation at every time. A drawn path through some arrows is not necessarily such a curve. A projection of a four-dimensional trajectory can be displayed in two coordinates, but its apparent derivative may also depend on the hidden coordinates.

In Plain Language 5.3. What Is a Vector Field?

A two-dimensional slice of a double-pendulum state space is not generally invariant: the trajectory can leave it immediately. To obtain a genuine two-state example, impose the ideal stationary constraint $q_2 = 0$, $v_2 = 0$. Its reaction maintains the lock

and performs zero work. With zero proximal torque, the reduced equation is

$$(A + D + 2b)\ddot{q}_1 + g_0(m_1r_1 + m_2l_1 + m_2r_2)\sin q_1 = 0.$$

The figure integrates this equation from $(q_1, v_1) = (0.9, 0)$. This is drift of a different, constrained effective plant; it is not the free two-joint system with both torques set to zero. Arrow lengths are normalized for visibility after the displayed coordinate scaling; the computed curve retains the dynamics.

Flowing Along the Drift

Setting an actuator torque to zero retains the declared joints and contact. Releasing a mechanical lock changes the constraints; letting go of the club removes grip forces and changes the physical system. Neither operation is represented merely by setting $u = 0$ in the same fixed-pivot model. A muscle's neural drive can also become zero while activation and tendon force persist.

In Plain Language 5.4. Why Understanding Drift Matters

For the ideal free rigid model, $v^T c = \frac{1}{2}v^T \dot{M}v$. Differentiating $E = T + V$ therefore gives

$$\dot{E} = v^T B u.$$

Velocity-dependent inertial terms redistribute motion and kinetic energy within the model; they do not create mechanical energy. Gravity exchanges potential and kinetic energy. A stationary ideal constraint has zero reaction power, while moving support, damping or deformation changes the energy budget.

Acceleration, speed and power answer different questions. A force perpendicular to point velocity can bend a path without doing work. Torque can increase or decrease mechanical energy according to its power $u^T B^T v$. A large acceleration term is not proof that its source supplied most of the energy in the swing.

5.4 Control as a Steering Input

Control can steer, accelerate, brake or maintain a constraint. Which effect matters depends on the input direction, configuration, velocity and task. Calling every muscular contribution “steering” hides its energetic role. The useful question is how an admissible change in input changes the outcome.

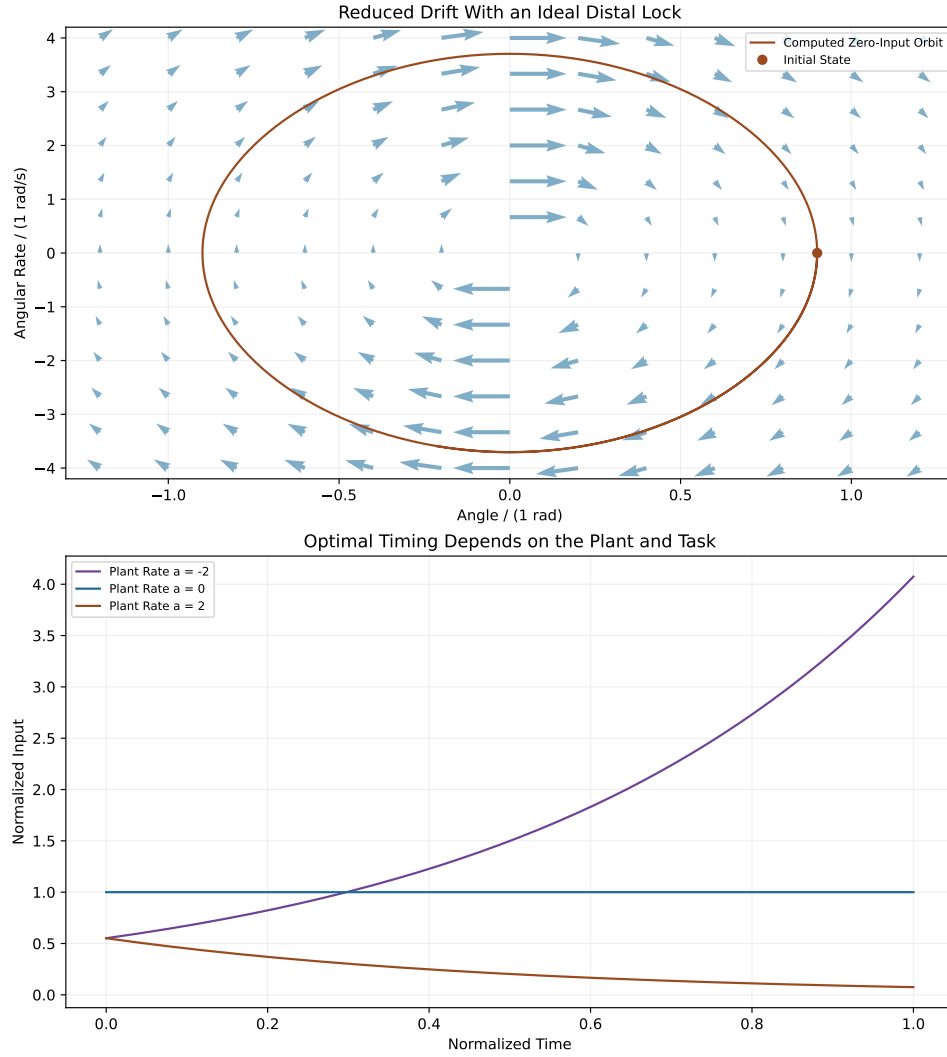


Figure 5.1: Declared Models of Drift and Optimal Timing. The Upper Panel Shows Normalized Directions and a Computed Orbit for the Reduced Plant With an Ideal Distal Lock. The Lower Panel Shows Exact Minimum-Energy Inputs for Three Normalized Scalar Plants With the Same Endpoint Task. Neither Panel Contains Measured Golfer Data.

The Control Matrix as Leverage

Example 5.2. Control Leverage at Different Configurations

For a perturbation $\delta u = (\delta u_1, 0)^T$ with both joints free, $\delta \dot{v}_1 = K_{11}\delta u_1$. In a coupled two-joint system,

$$K_{11} = \frac{1}{M_{11} - M_{12}^2/M_{22}}.$$

The denominator is the Schur complement. If the second coordinate is held fixed instead, the proximal acceleration response is $\delta u_1/M_{11}$ and a constraint reaction supplies the required second torque. These are different experiments.

At $q_2 = 0$ in the manufactured example, $M_{11} = 0.4205625 \text{ kg m}^2$ and the free effective inertia is approximately 0.1128702 kg m^2 . Unit proximal torque produces approximately 8.860 rad/s^2 of free proximal response, versus 2.378 rad/s^2 with the distal coordinate locked. At $q_2 = 90^\circ$, the free effective inertia becomes $A = 0.1505625 \text{ kg m}^2$, so the free response is smaller, approximately 6.642 rad/s^2 , despite a smaller M_{11} . “More folded means more leverage” is not a general rule.

These are incremental control responses at the same state. The total acceleration also includes a_0 , and a prescribed distal torque would add its own off-diagonal contribution. Full club acceleration additionally requires the Jacobian and its velocity-dependent term.

5.5 The Drift-Control Ratio (DCR)

Definition 5.3. Drift-Control Ratio

Specify the input origin, admissible set $\mathcal{U}$, state and an output/scaling map W . Define a capacity and a ratio by

$$C_W(x) = \max_{u \in \mathcal{U}} \|WG(x)u\|_2, \quad D_W(x) = \frac{\|Wf(x)\|_2}{C_W(x)}, \quad C_W > 0. \quad (5.13)$$

Here W may select the acceleration block or normalize all derivative components using declared reference scales. Position rates and angular accelerations cannot simply be combined as unscaled numbers. A ratio using the realized input $\|WGu(t)\|$ answers a different question from a capacity ratio. Report which one is intended.

This ratio is not a work fraction, a measure of skill, or a complete reachability test. If capacity is zero, report that condition explicitly; an arbitrary epsilon changes the diagnostic and must have compatible units. Both numerator and capacity zero is indeterminate. A positive numerator with zero capacity indicates no authority through the selected map, not a finite numerical ratio.

For a centered Euclidean torque ball $\|u\|_2 \leq u_{\max}$, $C_W = u_{\max}\sigma_{\max}(WG)$. A componentwise torque box or muscle-force polytope gives a different optimization. In particular, a 30 N m two-joint Euclidean bound does not mean both joints may independently apply 30 N m .

In Plain Language 5.5. Reading the DCR

A large capacity ratio says the chosen drift norm exceeds the largest chosen control norm at that state. A small ratio says some admissible input has a larger norm; it does not show that the actual input is large or points in the useful direction. Two vectors can reinforce, oppose or be orthogonal, so their norm ratio is not an additive attribution of the observed motion.

For example, in normalized coordinates let $f = (1, 0)^T$, $G = \text{diag}(100, 0.01)$ and $\|u\|_2 \leq 1$. The capacity ratio is 0.01. Yet the available second-component response is at most 0.01: large first-direction authority does not help a task requiring second-direction correction. A directional capacity can instead use $\max_{u \in \mathcal{U}} d^T W G u$ for a declared task direction d .

A state-coordinate change $z = h(x)$ gives $f_z = Dh f$ and $G_z = Dh G$. Euclidean norms generally change. Equivalent physical comparisons require transporting the metric or the specified output; discarding different blocks after a coordinate change can define a different diagnostic. For nonlinear changes of configuration coordinates, acceleration also acquires a velocity-quadratic transformation term.

The split depends on the chosen input origin too. Under $u = u_0(x) + R(x)w$, the same dynamics become $\dot{x} = [f + G u_0] + G R w$. The admissible input set must be transformed with it. The physical attainable-derivative set can remain the same while the designated drift changes. A nominal feedback policy used as a new zero input is not evidence that its energy supply became passive.

Address: A Defined Equilibrium

The manufactured downward equilibrium has zero drift and nonzero torque capacity. Its capacity ratio is zero. It is equally true whether the realized torque is zero or is about to change. This does not establish a universal large torque requirement: the required torque depends on the desired acceleration and available time.

Top of Backswing: A Specified State

For the manufactured state $q = (150^\circ, -70^\circ)$ and $v = 0$, $a_0 \simeq (-16.9466, 5.2045)^T \text{ rad/s}^2$. With the stated 30 N m Euclidean torque bound and an unweighted angular-acceleration norm, capacity is approximately 651.113 rad/s² and the ratio is 0.02723. This is a counterexample calculation, not a measured golfer's top position or a realistic muscle-capacity estimate.

Downswing: Velocity Bias and Coupled Response

At $q = (140^\circ, -60^\circ)$ and $v = (5, 0) \text{ rad/s}$ in the same model, $c = (0, -1.51554)^T \text{ N m}$ and $g \simeq (5.57376, 1.93219)^T \text{ N m}$. The complete drift acceleration is approximately $(-35.7444, 42.1629)^T \text{ rad/s}^2$. Its capacity ratio is 0.07705. A faster state need not cross any universal threshold; geometry, velocity direction, gravity and the admissible input set all matter.

Impact and Beyond: Specify the Event and System

Club-point acceleration is $\ddot{p} = J(q)\dot{v} + \dot{J}(q,v)v$. Its normal and tangential components are not interchangeable with joint torque. Neither 180 m/s^2 nor 180 times gravitational acceleration is a universal impact value; an acceleration alone cannot supply an equivalent torque without geometry and force mapping. The pre-impact smooth model does not represent the collision impulse. Contact and post-impact muscle activity require their own model and evidence.

In Plain Language 5.6. The Three Comparisons

Distinguish three computations: the instantaneous drift/control decomposition, the accumulated mechanical work, and the attainable change in a task before impact. Each can be useful; none substitutes for the others. Increasing a capacity ratio by reducing the torque bound makes the actuator weaker. It cannot by itself improve the best attainable outcome. Universal early/middle/late DCR categories and elite-versus-novice sensations do not follow from the equations.

5.6 Riding the Drift: A Hypothesis to Test

Mechanical structure can reduce the input needed for a specified task. A pendulum descending under gravity is an elementary example of exchanging stored potential for kinetic energy. But an energy-efficient movement may also require opposing drift to stop, change direction, avoid an obstacle or stabilize an unstable state. “Align with drift” is therefore a candidate strategy whose costs and outcomes must be calculated.

A work/power study by Nesbit and Serrano used kinematically driven full-body and club models for four amateur golfers to estimate work and energy transfer [Nesbit and Serrano, 2005]. That scope supports distinguishing modeled joint/club work from instantaneous acceleration; it does not establish a universal DCR phase law, individual muscle activation or a globally optimal neural strategy. Net joint work is not a direct measurement of metabolic expenditure.

5.7 The Geometry of the Golf Swing

Constraint Insight 5.1. The Reachability Problem

Reachability asks which states can be attained from a specified initial state over a specified time under admissible inputs and state constraints. A golf task also maps those states into club pose, velocity, impact conditions and ball outcome. Strength is one constraint, not a complete distance prediction.

If $\mathcal{U}_1 \subseteq \mathcal{U}_2$ with the same plant, initial condition, horizon and other constraints, every trajectory feasible under the smaller set remains feasible under the larger. Increasing available control cannot worsen the best achievable objective under these assumptions. It does not guarantee improvement, tell a golfer which input to use, or rank technique against strength across different people.

Along a nominal trajectory, a smooth local model has $\delta\dot{x} = A(t)\delta x + B_x(t)\delta u$. At a fixed terminal time,

$$\delta x(T) = \Phi(T, 0)\delta x(0) + \int_0^T \Phi(T, s)B_x(s)\delta u(s) ds.$$

A terminal task map projects this result into the outcome. The kernel determines how an input at each time affects the task; its direction and size can change or cancel. Delay and activation restrict which inputs can occur. If impact time changes with the trajectory, event-time sensitivity must also be included. These considerations explain why an instantaneous norm ratio cannot rank early versus late control.

5.8 Optimal Control and the Swing

Specify a terminal cost ϕ , running cost L , horizon and constraints:

$$\min_{u(\cdot)} J = \phi(x(T)) + \int_0^T L(x, u, t) dt, \quad \dot{x} = f(x) + G(x)u, \quad u(t) \in \mathcal{U}.$$

An effort surrogate such as $u^T R u$ has declared scaling and is not mechanical power. Mechanical work integrates $u^T B^T v$ over time; summing several torque values in N m gives neither work nor integrated effort. A model may also constrain torque rate, activation, contact, joint limits, terminal state or uncertainty. An optimizer needs all of them to define what “best” means.

In Plain Language 5.7. Solving the Optimal Control Problem

For a smooth normal fixed-horizon problem, define the control Hamiltonian

$$H = L + p^T(f + Gu), \quad \dot{p} = -\partial_x H, \quad u^*(t) \in \arg \min_{u \in \mathcal{U}} H.$$

Pontryagin’s principle supplies necessary conditions, with the appropriate terminal/transversality conditions; a candidate is not automatically a global optimum. With a free terminal state, $p(T) = \nabla \phi$ in this simple formulation; terminal constraints add multipliers. This Hamiltonian is not generally mechanical energy [Tedrake, n.d.c].

Dynamic programming instead reasons about a value function. Under suitable smoothness its finite-horizon equation is $-\partial_t V = \min_u [L + \nabla_x V^T(f + Gu)]$ with terminal data $V(x, T) = \phi(x)$. Verification needs the stated boundary, admissibility and regularity conditions; nonsmooth values require more general treatment. It is not just another name for solving a single trajectory’s two-point boundary-value equations [Tedrake, n.d.a].

Numerical solutions require feasibility, integration/discretization checks and assessment of local versus global optimality. A successful fit to a golfer’s movement does not establish that the nervous system solved the optimizer or shares its cost function. Learning and feedback need independently testable predictions.

Example 5.3. A Solved Comparison of Control Timing

In normalized variables consider $\dot{z} = az + u$, $z(0) = 0$, $z(1) = 1$, and minimize $\int_0^1 u^2 dt$ with no active input bound. The endpoint condition is $1 = \int_0^1 e^{a(1-t)} u(t) dt$. Cauchy–Schwarz gives

$$J \geq \frac{1}{W_a}, \quad W_a = \int_0^1 e^{2a(1-t)} dt, \quad u^*(t) = \frac{e^{a(1-t)}}{W_a}.$$

Equality proves the global minimum for this specified linear task. For $a = 0$, $W_a = 1$ and the optimal input is constant. For $a < 0$, earlier effects decay and the optimal input grows toward the end. For $a > 0$, earlier effects grow and the optimal input is larger early. The figure compares $a = -2, 0, 2$.

These plants are all control-affine. The timing follows the task sensitivity and cost, not the word “affine” or a universal DCR threshold. Bounds below the displayed input peaks change the solution. A golf model has more states, constraints and uncertainty; the example disproves a universal timing rule rather than prescribing a swing.

5.9 Putting It All Together: The Unified Picture

The System

Define the body, club, supports, contacts and energy boundary before splitting forces. Drift is the zero-input derivative of that declared plant.

The State Space

Include the states needed to predict force and motion: joint configuration and velocity alone may omit activation, material memory and shaft modes. Specify coordinate charts and observation uncertainty.

The Drift

Prior input shaped the current state. Large present drift does not mean little past muscle work, little current energy expenditure or weak future control.

The Control

The full inverse inertia, actuator mapping and admissible set determine instantaneous response. Contact and constraints can change it, while activation and delay change the time at which input is available.

The Skill

A proposed strategy should predict measured task outcomes, energy costs or robustness under specified conditions. None of these is identified by an effortless feeling, a low input in one model or a large ratio alone.

Key Principle 5.1. Key Takeaways

Control-affine form is a precise local input structure. It makes useful calculations possible, but those calculations must still be done: complete mechanics, feasible control, energy accounting, task sensitivity and evidence. The strongest argument for exploiting mechanical structure shows how it improves a defined task under constraints. It does not require declaring muscles irrelevant or DCR the quantity every golfer should maximize.

Exercises 5.1. Chapter Exercises and Worked Answers

1. Compute the Declared Rest State

For the manufactured two-link model at $q = (0, 0)$, $v = 0$, compute f . Does setting torque to zero mean releasing the club?

Worked answer. The bias is zero because velocity is zero, and both potential gradients vanish at downward vertical. Thus $a_0 = 0$ and all four components of f are zero. The ideal hinges and fixed pivot remain. Releasing the club changes contact and requires a different model. Zero net torque also does not specify muscle activation or coactivation.

2. Compute a Raised Rest State

Use $q = (150^\circ, -70^\circ)$ and $v = 0$. Why must both gravity components enter both accelerations?

Worked answer. The defined potential gives $g \simeq (4.76483, 1.93219)^T$ N m, and $M \simeq \begin{bmatrix} 0.328445 & 0.153941 \\ 0.153941 & 0.130000 \end{bmatrix}$ kg m². Solving $Ma_0 = -g$ gives $a_0 \simeq (-16.9466, 5.2045)^T$ rad/s². Inverse-inertia coupling mixes both loads. The first two state-derivative components remain zero at this instant even though velocity begins to change.

3. Separate Velocity Bias From Gravity

For $q = (140^\circ, -60^\circ)$ and $v = (5, 0)$ rad/s, compare the two acceleration contributions.

Worked answer. Although $v_2 = 0$, $c_2 = b \sin(q_2) v_1^2 = -1.51554$ N m. The complete contributions are $-M^{-1}c \simeq (-13.6289, 28.9563)^T$ and $-M^{-1}g \simeq (-22.1155, 13.2066)^T$ rad/s². Their vector sum is $(-35.7444, 42.1629)^T$. Comparing only torque magnitudes or omitting c_2 loses the coupled response. This instantaneous decomposition is not an energy-source fraction.

4. Interpret a Capacity Ratio

Use the acceleration-only norm and 30 N m Euclidean input ball for the four reported states: downward rest, $(45^\circ, 0)$ at rest, raised rest, and the moving state in Exercise 3. Does the ratio identify actual muscular work?

Worked answer. The ratios are approximately 0, 0.04158, 0.02723 and 0.07705. They use the largest singular value of M^{-1} multiplied by 30, rather than an assumed realized torque. They are not monotone in the listed angle changes and do not define golf phases. Work requires a realized torque history and the associated joint rates. Reducing the input bound increases a nonzero capacity ratio while reducing available control.

5. Determine Optimal Timing From a Defined Task

For the normalized scalar endpoint problem, derive the least-energy input and determine whether early input is always preferred.

Worked answer. The terminal solution is $z(1) = \int_0^1 e^{a(1-t)} u(t) dt$. Cauchy–Schwarz bounds its square by $W_a \int_0^1 u^2 dt$. Equality requires u proportional to $e^{a(1-t)}$; enforcing the endpoint gives $u^* = e^{a(1-t)}/W_a$ and cost $1/W_a$. Positive, zero and negative a respectively favor early, constant and late input. These are exact results for the stated normalized, unconstrained task, not a golf timing prescription.

6. Compare Larger and Smaller Input Sets

If otherwise identical models have torque balls of radius 30 and 50 N m, what can be concluded about their best feasible performance and DCR?

Worked answer. Every 30 N m-admissible trajectory is also 50 N m-admissible. The optimum cannot become worse for the same maximization objective, or higher for the same minimization cost, when the feasible set expands. Improvement is not guaranteed. At the same state with nonzero drift, the 30 N m capacity ratio is 50/30 times the 50 N m ratio. A larger ratio therefore cannot be a universal performance objective. Comparing different people additionally changes geometry, force capacity, coordination and uncertainty.

7. Compare Golf With Throwing

What transfers from this chapter to throwing, and what must be modeled again?

Worked answer. A smooth torque-driven multibody phase can share the manipulator and control-affine structure. Segment geometry, actuation, constraints, initial conditions, costs and task outputs differ. Ball release changes contact and introduces event timing; golf impact adds collision mechanics. No DCR values, optimal control schedule or strength-versus-technique ranking transfers without those specifications. The same mathematical form supports different physical predictions.

Closing Thoughts: The Unity of Physics and Swing

The affine decomposition connects a mechanical model to a control question. It is most useful when the reader can follow the entire chain: define the physical system, derive its dynamics, identify feasible inputs, propagate their effects to the task, account for energy and compare predictions with data. A simple model can reveal an important mechanism while remaining incomplete as a description of a golfer.

Induced acceleration analysis applies these mappings to specified generalized-force contributions at a given state. It must retain the same constraints and cannot infer individual muscle forces from net joint torque alone. The following chapters develop these connections; the purpose of the common framework is to make their assumptions and consequences comparable.

Chapter 6

The Zero-Torque Counterfactual

In Plain Language 6.1. The Central Question

A moving club retains momentum, and the forces acting next depend on the current configuration, velocity, loads and internal state. A simulator can copy that state, set specified applied torques to zero, and calculate what follows. This is the forward Zero-Torque Counterfactual (ZTCF): a controlled comparison inside a model.

The useful question is precise: which input was changed, which state was copied, and which laws were retained? A branch that resembles a swing does not establish relaxed muscles. A branch that departs from a reference does not identify a particular muscle or control strategy. The calculation connects mechanics to a stated intervention; measurements and a separate identification argument are needed to connect it to a golfer.

6.1 Formal Definition of the ZTCF

Choose generalized coordinates q , velocities $v = \dot{q}$, internal states z , and parameters p . Internal states can include actuator activation, tendon strain, controller memory or flexible modes when the chosen model represents them. They must either evolve under specified equations or be explicitly held fixed as part of the intervention. The rigid three-link fixture discussed below has no such additional states.

For an unconstrained mechanical model with applied input u , write

$$\begin{aligned}\dot{q} &= v, \\ M(q)\dot{v} + h(q, v, z, t; p) &= B(q, z, t; p)u, \\ \dot{z} &= \psi(q, v, z, u, t; p).\end{aligned}\tag{6.1}$$

Here h is the complete retained bias on the left-hand side, including velocity-product and gravity terms and the signed contributions of other retained loads. This convention makes the zero-input acceleration $-M^{-1}h$. Generalized coordinates need not all be angles; each row has its own force or moment units. The familiar rigid-body structure follows from kinetic and potential energy [Lynch and Park, 2017, Featherstone, 2008].

The full-state equation is control-affine, $\dot{x} = f(x, t) + G(x, t)u$, only when all state equations are affine in the selected input within the chosen mode. Direct generalized torque input commonly has this property; a nonlinear activation or actuator command need not. State the physical input level before using affine superposition.

Four constructions answer different questions. A **pointwise ZTCF sample** evaluates $f(x(t), t)$ at one reference state. A **stitched pointwise ZTCF trace** collects those samples along the reference; it is not an integrated path. A **forward ZTCF trajectory** solves the

zeroed-input initial-value problem. A **branched ZTCF trajectory** starts that solve at a reference state and compares it with a separately specified reference branch. The full-state vector f includes velocity and internal-state derivatives; only its mechanical acceleration block has acceleration units.

Zeroing every declared input gives

$$\dot{x}^0 = f(x^0, t), \quad x^0(t_0) = x_0. \quad (6.2)$$

If other channels remain, their input histories or feedback policies belong to the retained protocol. A time-varying load also remains time-dependent. The zeroed-input plant is autonomous only when the remaining laws have no explicit time dependence, or when the necessary clock/controller states are included. A recorded open-loop torque replay and a feedback controller reacting to the new state define different reference experiments.

6.2 Normative ZTCF Contract

Every executable or quantitative record uses `affinedrift.ztcf-intervention/v1`. The published normative schema and Python golden fixture freeze model/version/revision, inputs, state, units, frame, retained controls, constraints, contact, loads and internal states, solver/version, horizon, preconditions, postconditions, and failure states. The ZTCF is a model-conditioned intervention. The record separates simulated trajectory difference from any contribution measure. It states each causal estimand and physiological interpretation, and records non-identifiability explicitly. Unavailable or engine-unsupported interventions fail closed; no MATLAB, Simulink, or cross-engine parity result is inherited from the Python fixture. The public records are [Normative Schema](#) and [Rigid Three-Link Golden Fixture](#).

The current fixture uses model version 2.0: a rigid three-link chain with consistent rigid inertia and bias forces. It integrates no flexible shaft states or additional shaft inertia. The historical version 1 fixture is preserved as a record of the earlier mixed-inertia calculation and is unsupported by the current adapter. The model change requires a new result; it does not inherit the old numerical or physical interpretation.

The analytical two-rod calculation later in this chapter checks the mechanics underlying a pointwise evaluation. It does not add a registered two-link simulator, publish a forward ZTCF result, or certify an engine. A new executable branch requires its own supported model and intervention record; the existing three-link fixture cannot supply those missing declarations.

6.3 The Decomposition: What the ZTCF Reveals

At the same time and full state, with the same parameters and admissible constraint mode, an affine acceleration solve gives

$$a(x, u, t) = a_0(x, t) + H(x, t)u, \quad a(x, u, t) - a(x, 0, t) = H(x, t)u. \quad (6.3)$$

For the unconstrained model, $a_0 = -M^{-1}h$ and $H = M^{-1}B$. This identity holds at every common state, not just the initial time. A reference branch and a zeroed-input branch normally share their state at the branch instant; later they generally do not. Subtracting their accelerations then includes the change in the drift itself:

$$a(x^r, u, t) - a(x^0, 0, t) = [a_0(x^r, t) - a_0(x^0, t)] + H(x^r, t)u. \quad (6.4)$$

Consequently, integrating the nominal input contribution along the reference is generally different from recomputing the trajectory after input removal. The removed input changes subsequent states, and those states change the forces and accelerations. This is the connection to induced-acceleration analysis: pointwise attribution and trajectory intervention share a dynamics model but answer different questions.

6.3.1 Constraints and Physical Outputs

For equality constraints at a fixed feasible mode, solve accelerations and reactions together:

$$\begin{bmatrix} M & -A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} a \\ \lambda \end{bmatrix} = \begin{bmatrix} Bu - h \\ b \end{bmatrix}. \quad (6.5)$$

The acceleration constraint is $Aa = b$, with b carrying curvature and prescribed support motion. Positive-definite M and independent rows of A give a unique equality-constrained solve. Its zero-input baseline includes the necessary reactions and constraint acceleration; it is generally not $-M^{-1}h$. Contact forces must also satisfy the declared unilateral and friction conditions. A zeroed input can cause separation or a mode transition; retaining an impossible contact mode does not define a valid physical comparison.

For a fixed geometric point $y(q)$, acceleration is $\ddot{y} = Ja + \dot{J}v$. At a common state the transport term cancels in the input increment, giving $\Delta\ddot{y} = JHu$. On separated branches both J and $\dot{J}v$ can differ. A clubhead-speed contribution additionally uses the velocity-direction projection; angular acceleration alone is not clubhead acceleration, speed change or energy transfer.

6.3.2 Force Bookkeeping and Inverse Dynamics

In the unconstrained fully actuated case $B = I$,

$$Q_{\text{inertial}} = Ma, \quad Q_{\text{retained}} = -h, \quad u = Ma + h = M(a - a_0). \quad (6.6)$$

Thus a hypothetical scalar component of inertial generalized load of 100 N m and retained load of 85 N m leaves 15 N m applied input. The first number must mean a component of Ma . If inverse dynamics already reports the required applied input as 100 N m, subtracting the retained load again is double counting. The same inverse-dynamics model can compute all these terms; ZTCF does not reveal information hidden from the equations.

With $B \neq I$, the acceleration difference determines Bu , not necessarily a unique u . With constraints, changes in reaction also enter the force balance; feasible acceleration alone can leave input directions unidentified. The full mass matrix is essential even when there are only two angular coordinates. Replacing it with its diagonal discards the coupling being investigated.

6.4 Computing the ZTCF: Algorithm and Simulation

Copy the complete declared state at t_0 , including any solver-required contact, actuator and controller state. Record the model revision, parameter convention, zeroed channels and their physical level. Specify the reference input policy and every retained load, constraint, mode rule and internal-state law. Use identical initial states for a paired intervention.

An illustrative branch switch applies only to the selected channel:

$$M\dot{v} + h = k_u Bu, \quad k_u \in \{0, 1\}. \quad (6.7)$$

It does not automatically reset activation, elastic strain, velocity or contact forces. A switch diagram is an implementation pattern, not evidence of a supported Simulink or other engine.

Integrate each declared branch to the stated time or event. Record integration method, version, step size or error tolerances, event localization, and failure behavior. Compare both full-state closure and the output relevant to the question. For example, face orientation at a fixed time and face orientation at each branch's own impact event are different estimands. A branch that misses the impact surface has no impact value; it must not inherit the reference event.

Numerical checks should include forward/inverse closure, constraint residuals, physically appropriate energy or power balances, and step refinement in the reported output. An autonomous conservative model with ideal fixed constraints conserves mechanical energy; damping, moving supports or external work change that balance. Energy conservation by itself does not prove trajectory accuracy. Large acceleration and nonlinearity alone do not establish numerical stiffness: fast decaying modes relative to the solution time scale can restrict explicit stability. Contact discontinuities additionally require an event or nonsmooth formulation, not simply a smaller arbitrary step.

6.5 The Zero-Velocity Counterfactual

The **instantaneous Zero-Velocity Counterfactual (ZVCF)** evaluates acceleration at the same configuration with specified velocities reset and declared input zero. For the unconstrained convention above, holding z, t, p fixed gives

$$a_{\text{ZVCF}}(q, z, t) = -M(q)^{-1}h(q, 0, z, t). \quad (6.8)$$

If flexible or actuator velocities are stored in z , say whether they are also reset. Setting rigid joint velocity to zero does not silently reset all internal motion. In a rigid model with gravity as the only retained zero-velocity load this is $-M^{-1}h_g$. Elastic strain, prescribed loads and internal-state forces can survive zero joint velocity, so ZVCF is not generally gravity-only.

The difference

$$a_0(q, v, z, t) = a_{\text{ZVCF}}(q, z, t) + a_{\text{velocity}}(q, v, z, t) \quad (6.9)$$

defines a_{velocity} by subtraction under identical remaining declarations. It includes all effects changed by that velocity reset, not just a selected Coriolis term. For constrained models, recompute the constraint acceleration baseline consistently, and first check that the reset is feasible. A prescribed coordinate $q = t$ requires $\dot{q} = 1$; resetting it to zero violates the velocity constraint. A contact law can also change mode at zero velocity.

This state reset is distinct from removing an input. It changes kinetic energy and may require an impulse if interpreted physically. Integrating afterward would define a separate initial-state intervention. Holding velocity zero for an interval would require a constraint or controller, not free evolution from the instantaneous sample.

6.6 Two Uniform Rods: A Mechanics Check

Use two planar uniform rods with a fixed base, no contact, no damping and uniform gravity. These are manufactured mechanics parameters, not anthropometry or a fitted golf swing. The inertial frame has x right and y up. Angle q_1 is counterclockwise from $+x$; q_2 is relative to rod 1, so rod 2 has absolute angle $q_1 + q_2$. Positive inputs do positive work against positive coordinate velocities.

Take masses $(m_1, m_2) = (2, 0.2)$ kg and lengths $(L_1, L_2) = (0.4, 0.3)$ m. The COM distances are $r_i = L_i/2$, and COM inertias are $J_i = m_i L_i^2/12$. A pivot inertia would instead be $J_i + m_i r_i^2$; do not add the parallel-axis term twice. Set $g = 9.81$ m/s² and $k = m_2 L_1 r_2 = 0.012$ kg m².

The COM positions are

$$\begin{aligned} c_1 &= r_1 (\cos q_1, \sin q_1)^T, \\ c_2 &= L_1 (\cos q_1, \sin q_1)^T \\ &\quad + r_2 (\cos(q_1 + q_2), \sin(q_1 + q_2))^T. \end{aligned} \quad (6.10)$$

Sum translational COM and rotational kinetic energies:

$$T = \frac{1}{2} \sum_i m_i \|\dot{c}_i\|^2 + \frac{1}{2} J_1 v_1^2 + \frac{1}{2} J_2 (v_1 + v_2)^2. \quad (6.11)$$

Collecting the coefficients in $T = \frac{1}{2} v^T M v$ gives

$$\begin{aligned} M_{11} &= J_1 + m_1 r_1^2 + J_2 + m_2 (L_1^2 + r_2^2) + 2k \cos q_2, \\ M_{12} &= M_{21} = J_2 + m_2 r_2^2 + k \cos q_2, \\ M_{22} &= J_2 + m_2 r_2^2. \end{aligned} \quad (6.12)$$

The potential energy is

$$V = g[(m_1 r_1 + m_2 L_1) \sin q_1 + m_2 r_2 \sin(q_1 + q_2)]. \quad (6.13)$$

With $T = \frac{1}{2} v^T M v$, the Euler–Lagrange equations give $Ma + h_v + h_g = u$, where

$$\begin{aligned} h_v &= k \sin q_2 \begin{bmatrix} -(2v_1 v_2 + v_2^2) \\ v_1^2 \end{bmatrix}, \\ h_g &= g \begin{bmatrix} (m_1 r_1 + m_2 L_1) \cos q_1 + m_2 r_2 \cos(q_1 + q_2) \\ m_2 r_2 \cos(q_1 + q_2) \end{bmatrix}. \end{aligned} \quad (6.14)$$

Both velocity-bias components matter. The first contains a factor-two cross term and a squared distal-rate term; the second contains the proximal-rate squared. Their signs follow the stated relative-angle convention. They satisfy $v^T h_v = \frac{1}{2} v^T \dot{M} v$, the identity needed for conservative energy balance.

At $q = (45, 70)$ degrees and $v = (8, 12)$ rad/s, the combined angle has cosine

$$\cos(q_1 + q_2) = \cos 115^\circ = -0.422618. \quad (6.15)$$

All computations use unrounded values; displayed matrices are rounded:

$$M \simeq \begin{bmatrix} 0.152875 & 0.010104 \\ 0.010104 & 0.006000 \end{bmatrix} \text{ kg m}^2. \quad (6.16)$$

$$h_v \simeq \begin{bmatrix} -3.788841 \\ 0.721684 \end{bmatrix}, \quad h_g \simeq \begin{bmatrix} 3.205248 \\ -0.124377 \end{bmatrix} \quad \text{N m.} \quad (6.17)$$

The mobility and zero-input acceleration are

$$M^{-1} \simeq \begin{bmatrix} 7.360562 & -12.395482 \\ -12.395482 & 187.541158 \end{bmatrix} \quad (\text{kg m}^2)^{-1}, \quad (6.18)$$

$$a_0 = -M^{-1}(h_v + h_g) \simeq \begin{bmatrix} 11.699484 \\ -119.253631 \end{bmatrix} \quad \text{rad/s}^2. \quad (6.19)$$

This is an instantaneous mechanics calculation. The first coordinate accelerates positively and the second negatively at this snapshot; the configuration is not labeled an anatomical downswing posture. The gravity-only slice is $(-25.134131, 63.056316)^T \text{ rad/s}^2$, while the velocity-bias contribution is $(36.833615, -182.309947)^T \text{ rad/s}^2$. Their sum recovers a_0 .

For a specified reference acceleration $a^r = (2, 8)^T \text{ rad/s}^2$ at exactly the same state,

$$u = M(a^r - a_0) = Ma^r + h_v + h_g \simeq \begin{bmatrix} -0.197009 \\ 0.665516 \end{bmatrix} \quad \text{N m.} \quad (6.20)$$

Diagonal-only multiplication would give approximately $(-1.482810, 0.763522)^T \text{ N m}$ and would fail forward/inverse closure. The difference is substantial because the reference acceleration at one coordinate also requires generalized input through the off-diagonal inertia. Neither correct net input identifies a muscle, activation or effort.

6.7 The Drift-Control Ratio and Correction Authority

For a stated angular-acceleration projection W , norm and admissible input set $\mathcal{U}$, define

$$\text{DCR}_{W,\mathcal{U}}(x) = \frac{\|W a_0(x)\|}{\sup_{u \in \mathcal{U}(x)} \|WH(x)u\| + \varepsilon}. \quad (6.21)$$

The regularizer ε has the denominator's units and must be reported. A zero control-effect denominator means there is no authority in that projection; regularization does not create it. For physical point acceleration, use the complete baseline $Ja_0 + \dot{J}v$ and operator JH . Coordinate rescaling changes an unqualified component norm, so specify what the projection measures.

For the two-rod snapshot, choose $W = (0, 1)$, absolute value, $B = I$, $\varepsilon = 0$, and the illustrative torque box $|u_1| \leq 2 \text{ N m}$, $|u_2| \leq 0.2 \text{ N m}$. These are chosen limits, not muscular capacity estimates. The exact scalar projection bound is

$$\begin{aligned} s &= \sum_j |(WM^{-1})_j| \bar{u}_j \\ &= 12.395482(2) + 187.541158(0.2) \\ &\simeq 62.299196 \quad \text{rad/s}^2. \end{aligned} \quad (6.22)$$

The bound is attained at a vertex of the torque box. The corresponding ratio is

$$\text{DCR} = 119.253631/62.299196 \simeq 1.914208. \quad (6.23)$$

Using only a single Coriolis force divided by one torque bound would be a different quantity. Here both actuator channels influence the selected angular acceleration.

At fixed configuration and parameters, uniformly scaling both rates by α gives

$$h_v(q, \alpha v) = \alpha^2 h_v(q, v), \quad a_0(q, \alpha v) = -M^{-1}(\alpha^2 h_v + h_g). \quad (6.24)$$

Doubling the rates in this artificial snapshot yields

$$a_{0,2} \simeq -666.183472 \text{ rad/s}^2, \quad \text{DCR} \simeq 10.693292. \quad (6.25)$$

This is a rate-scaling experiment, not a later measured swing state. The total ratio is not exactly four times the first: gravity does not scale with rate, and drift components can cancel. Along a real trajectory, posture, admissible inputs, contact and internal-state forces can also change. Quadratic velocity bias therefore does not establish a universal rising DCR or a late-downswing threshold.

6.7.1 Magnitude Is Not Direction or Reachability

Consider an abstract two-dimensional acceleration drift $(1, 0)^T$ and scalar input $|u| \leq 1$. The input directions $H_1 = (1, 0)^T$ and $H_2 = (0, 1)^T$ both give Euclidean DCR equal to one. The first cancels drift with $u = -1$; the second cannot change the first component. Equal ratios therefore allow different instantaneous cancellation authority.

For the one-dimensional double integrator $\ddot{y} = u$, $|u| \leq a_{\max}$, the displacement change relative to the zero-input branch at time T is

$$\Delta y(T) = \int_0^T (T-s)u(s) ds, \quad |\Delta y(T)| \leq \frac{1}{2}a_{\max}T^2. \quad (6.26)$$

This bound is attained by constant extreme input when no other constraints apply. It exposes the role of remaining time even when instantaneous drift acceleration is zero. A golf impact claim additionally needs the actual input map, state limits, output and event geometry, and uncertainty. DCR and one zero-input branch do not supply that reachable set.

6.8 Why ZTCF Cannot Be Measured Directly

Motion capture or IMUs provide measurements from which pose and motion are estimated under calibration and reconstruction assumptions. Differentiation amplifies noise; filtering, synchronization, uncertainty and sensitivity must be documented. EMG records electrical activity related to muscle excitation, not a direct measurement of muscle force or a uniquely identified activation state. A force plate measures the ground-contact wrench; grip loading requires an instrumented grip/club or a model-based inference. These observations constrain different parts of the model.

The unperformed branch is not observed in the recorded swing. Within an exact fixed deterministic model, a paired branch difference is the consequence of the specified intervention under its retained protocol. Comparing that branch to human data introduces parameter error, state-estimation error, omitted loads, contact mismatch and model discrepancy. Those are different claims and need different uncertainty analyses.

Zero generalized torque can coexist with substantial antagonistic muscle forces. A model may retain activation-dependent stiffness only if its intervention defines how that stiffness and activation evolve. Calling the retained terms passive does not establish a physiologically realizable muscle state. Causal interpretation for a person requires a defensible structural and measurement model; deterministic software replay alone supplies neither muscle identification nor a coaching prescription.

6.9 Summary and Preview

The central chain of reasoning is: define the input and full state; derive the retained dynamics; distinguish a pointwise evaluation from an integrated intervention; project into a named physical output; then test numerical and model uncertainty. The same mass matrix couples torque and acceleration, while state evolution couples present input removal to future drift. ZVCF adds a distinct state reset, and DCR adds a magnitude comparison. None of these steps by itself identifies biological effort or finite-horizon control authority.

The next chapter studies constraint forces. Those forces are essential to a feasible counterfactual whenever supports and contacts matter, even when all declared applied torques are zero.

Exercises 6.1. Chapter Exercises

1. Starting from the two COM positions above, recover all entries of M using the displayed sum of translational and rotational kinetic energies. Explain why a COM inertia and a pivot inertia cannot be interchanged without a parallel-axis correction.
2. Derive h_v from the Euler–Lagrange equations and h_g from ∇V . Verify $v^T h_v = \frac{1}{2} v^T \dot{M} v$, and recover both components of the stated reference input. Explain why zero applied input does not imply constant velocity.
3. For a uniform rod with mass 2 kg and length 0.7 m, let θ measure displacement from vertically downward. At $\theta = 45^\circ$, the pivot inertia is $mL^2/3 = 0.326667 \text{ kg m}^2$ and gravity acceleration is $-3g \sin \theta / (2L) = -14.864395 \text{ rad/s}^2$. Verify these values. An illustrative +10 N m input adds $30.612245 \text{ rad/s}^2$; it is not a measured muscle capacity.
4. Replay the registered three-link Python golden fixture without changing its parameters or source authority. Report its terminal state and units, then assess step refinement separately. Explain why matching the fixture cannot establish a second engine, flexible-shaft validity or an individual golfer’s muscle state.
5. Define a fixed-time and an impact-event branch comparison. State what happens if a branch never reaches impact. For the double-integrator example, derive the displacement bound and explain why adding a final-velocity constraint can change the attainable endpoint.

Part III

Constraints and Mechanisms

Chapter 7

Constraint Forces: Coupling, Work and Control

In Plain Language 7.1. Why Constraint Forces Matter

The hand can accelerate the club by exerting a force and a moment on it. The same interaction loads the hand in the opposite direction. Understanding the swing requires both sides of that interaction: how the bodies remain connected, how their motion determines the required reaction, and how muscles and external supports influence that motion.

The zero-torque counterfactual in the preceding chapter asks what a specified model would do when its declared control input is removed. It does not establish that the human downswing is passive. Here we ask a different question: what forces are necessary to maintain a specified connection, and what work do those forces perform on each part of the system?

A connection can carry considerable load while supplying no net power to the combined bodies. It can nevertheless supply positive power to the club and negative power to the arm. That distinction is central to understanding a kinetic chain. It does not make muscles irrelevant: muscle forces influence both the motion and the reactions, and ideal constraints themselves supply no energy source.

7.1 Holonomic Constraints and Their Jacobians

Definition 7.1. Holonomic Constraint

For local configuration coordinates $\mathbf{q} \in \mathbb{R}^n$, a holonomic constraint is a restriction

$$\Phi(\mathbf{q}, t) = \mathbf{0}. \quad (7.1)$$

The constraint is scleronomic when it has no explicit time dependence. A prescribed moving support generally gives a time-dependent, or rheonomic, constraint. Both descriptions require a choice of system boundary: a hand represented as a prescribed trajectory is an external support for a club-only model; the same hand represented as a moving body belongs inside a coupled hand-club model.

An ideal point attachment satisfies $\mathbf{x}_A(\mathbf{q}) - \mathbf{x}_B(\mathbf{q}) = \mathbf{0}$, where the two material points belong to different bodies. They coincide but can move through space. A no-slip hand-grip point constraint therefore equates hand and grip positions; it does not hold the grip stationary in the world. A rigid grasp additionally constrains relative

orientation. A point attachment alone permits relative rotation and cannot represent all moments that a distributed grasp can transmit.

In Plain Language 7.2. What a Constraint Is

A joint connects specified locations on adjacent bodies. In a rigid-link model, the distance between a body's own joint centers is fixed. The elbow does not keep every point of the forearm at one fixed distance from the shoulder: that distance changes as the elbow bends. A free hinge keeps its two attachment points together while allowing relative rotation. Free rotation and physical separation are different models. The idealization matters. Real cartilage, tendons, skin and grip material deform; contact can slip or separate. A perfectly rigid connection is an approximation whose validity depends on the motion, loads and time scale being studied. Increasing muscular stiffness is not literally the creation of a new exact holonomic equation.

Definition 7.2. The Constraint Jacobian

Define

$$\mathbf{J}_c = \frac{\partial \Phi}{\partial \mathbf{q}}, \quad \mathbf{J}_c \dot{\mathbf{q}} + \Phi_t = \mathbf{0}. \quad (7.2)$$

With m equations, $\mathbf{J}_c$ has m rows and n columns. The derivative Φ_t is taken at fixed configuration. For a stationary constraint, allowed velocities lie in $\ker \mathbf{J}_c$. For a moving constraint they instead satisfy an affine equation, generally an offset of that null space.

These equations use local coordinates with $\dot{\mathbf{q}}$ as the velocity variables. A model with quaternion configurations and a separate generalized velocity $\mathbf{v}$ must include its kinematic map $\dot{\mathbf{q}} = \mathbf{N}(\mathbf{q})\mathbf{v}$; the velocity Jacobian is then $\Phi_q \mathbf{N}$. Quaternion normalization is not an extra physical joint restraint.

Example 7.1. Relative Lock or World-Orientation Guide?

For the two-link model, let q_1 be the first link's angle from downward vertical and q_2 the second link's angle relative to the first. The second body's absolute angle is $\theta_2 = q_1 + q_2$.

A relative joint lock has $q_2 = q_{2*}$ and Jacobian $[0 \ 1]$: both bodies share the same absolute angular velocity. A different constraint is

$$\Phi = q_1 + q_2 - \theta_* = 0, \quad (7.3)$$

with

$$\mathbf{J}_c = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad \dot{q}_2 = -\dot{q}_1. \quad (7.4)$$

This holds the second body at a fixed world orientation. Its relative angle changes as the first link rotates. It is an ideal world-orientation guide, not a locked wrist. In these coordinates a pure external moment on the second body contributes equal generalized moments to q_1 and q_2 , explaining the column $\mathbf{J}_c^T$. By contrast, an internal relative-joint moment contributes to q_2 alone: its opposite effects on the two bodies are already encoded by the coordinate definition.

7.2 The Constrained Equations of Motion

In Plain Language 7.3. What Are Constraint Forces, Really?

A stationary frictionless table can support a book without doing work on it. The normal reaction prevents penetration, while tangential motion remains free. A rough table can exert friction and remove mechanical energy during sliding. A rising table can do positive work on the book. Thus the phrase “constraint force” alone does not determine the power: we need the contact law, the motion of its point of application and the system boundary.

Theorem 7.1. Constrained Lagrangian Dynamics

In a fixed ideal bilateral constraint mode, assume a positive-definite mass matrix, smooth local coordinates and a velocity bias vector $\mathbf{c}$. Write

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{c}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q}) = \mathbf{B}(\mathbf{q})\mathbf{u} + \mathbf{Q}_o + \mathbf{J}_c^T \boldsymbol{\lambda}. \quad (7.5)$$

Here $\mathbf{g} = \nabla V$ is the potential gradient; its force contribution after rearrangement is $-\mathbf{g}$. The vector $\mathbf{Q}_o$ contains other declared loads, such as damping or an applied wrench. The vector $\mathbf{c}$ contains the velocity-dependent inertial terms; these terms are not independent energy sources. Define $\mathbf{r} = \mathbf{B}\mathbf{u} + \mathbf{Q}_o - \mathbf{c} - \mathbf{g}$ so that $\mathbf{M}\ddot{\mathbf{q}} = \mathbf{r} + \mathbf{J}_c^T \boldsymbol{\lambda}$. The multiplier is conjugate to the chosen constraint equation. It is a force in newtons for a suitably normalized positional constraint, or a moment in newton-metres for an angular constraint. Rescaling or mixing equations changes multiplier values. The physically meaningful generalized reaction is $\mathbf{J}_c^T \boldsymbol{\lambda}$, together with an explicitly defined map to a spatial wrench. Multipliers are not universally force magnitudes.

In Plain Language 7.4. Reading the Constrained EOM

The same dynamics determine motion and reaction together. Applied muscle moments enter the load vector; the joint reaction is whatever the specified ideal connection requires in response. A large reaction is therefore not evidence that the load is independent of muscle action. Nor does a large numerical force in newtons establish dominance over a moment in newton-metres: those quantities need a common mechanical mapping before comparison.

An ideal reaction has zero virtual work on admissible virtual displacements. This statement is about a force–displacement pairing. With mixed angular and translational coordinates it is safer than imagining ordinary perpendicular arrows in a Euclidean picture of configuration space.

7.3 Solving for the Constraint Forces

Algorithm 7.1. Computing Lagrange Multipliers

For a stationary constraint, differentiation gives

$$\mathbf{J}_c \ddot{\mathbf{q}} = -\dot{\mathbf{J}}_c \dot{\mathbf{q}}. \quad (7.6)$$

There is no additional unconstrained-acceleration term in this kinematic identity. For the general time-dependent constraint, define the vector

$$\boldsymbol{\gamma} = \boldsymbol{\Phi}_{qq}[\dot{\mathbf{q}}, \dot{\mathbf{q}}] + 2\boldsymbol{\Phi}_{qt}\dot{\mathbf{q}} + \boldsymbol{\Phi}_{tt}, \quad \mathbf{J}_c \ddot{\mathbf{q}} + \boldsymbol{\gamma} = \mathbf{0}.$$

Each component of $\boldsymbol{\Phi}_{qq}[\dot{\mathbf{q}}, \dot{\mathbf{q}}]$ is the corresponding constraint Hessian contracted with velocity twice. For stationary constraints, $\boldsymbol{\gamma} = \dot{\mathbf{J}}_c \dot{\mathbf{q}}$.

Substitution of the dynamics gives

$$\mathbf{J}_c \mathbf{M}^{-1}(\mathbf{r} + \mathbf{J}_c^T \boldsymbol{\lambda}) = -\boldsymbol{\gamma}, \quad (7.7)$$

and hence the Schur-complement system

$$\underbrace{\mathbf{J}_c \mathbf{M}^{-1} \mathbf{J}_c^T}_{\mathbf{S}} \boldsymbol{\lambda} = -\mathbf{J}_c \mathbf{a}_0 - \boldsymbol{\gamma}, \quad \mathbf{a}_0 = \mathbf{M}^{-1} \mathbf{r}. \quad (7.8)$$

Solve for $\boldsymbol{\lambda}$ and recover $\ddot{\mathbf{q}} = \mathbf{a}_0 + \mathbf{M}^{-1} \mathbf{J}_c^T \boldsymbol{\lambda}$. The free acceleration appears once. Subtracting it a second time produces the wrong reaction.

In Plain Language 7.5. Why This Procedure Works

The free dynamics predict an acceleration. The constraint specifies the correction needed to keep the connection compatible. If $\mathbf{M}$ is positive definite and $\mathbf{J}_c$ has full row rank, then $\mathbf{S}$ is positive definite: for nonzero $\mathbf{y}$, $\mathbf{y}^T \mathbf{S} \mathbf{y} = (\mathbf{J}_c^T \mathbf{y})^T \mathbf{M}^{-1} (\mathbf{J}_c^T \mathbf{y}) > 0$. The reaction is then unique for those equations and loads.

With redundant rows, multipliers may be nonunique even when the generalized reaction and acceleration are unique. A minimum-norm multiplier selected by a pseudoinverse depends on constraint scaling; it is not automatically the real distribution of contact forces or muscle loads. Remove redundant equations or state the additional physical model that resolves load sharing. Near rank loss, inspect conditioning and residuals rather than treating a large multiplier as a biological discovery.

The coupled calculation can also be checked without the Schur reduction:

$$\begin{bmatrix} \mathbf{M} & -\mathbf{J}_c^T \\ \mathbf{J}_c & \mathbf{0} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} \mathbf{r} \\ -\boldsymbol{\gamma} \end{bmatrix}.$$

Compare both residuals, $\mathbf{M} \ddot{\mathbf{q}} - \mathbf{r} - \mathbf{J}_c^T \boldsymbol{\lambda}$ and $\mathbf{J}_c \ddot{\mathbf{q}} + \boldsymbol{\gamma}$. Initial position and velocity must also be compatible: acceleration-level enforcement does not repair an already separated joint or an inadmissible incoming velocity. Numerical drift control and physical constraint activation are distinct operations. Standard multibody treatments discuss bilateral constraints, contact

modes and their acceleration/impulse formulations; the equations here are derived with the sign convention above. [Tedrake, n.d.b]

7.4 The Null Space and Range of the Constraint Jacobian

Definition 7.3. Allowed Velocities and Reaction Covectors

For stationary constraints, $\ker \mathbf{J}_c$ is the tangent velocity space. Ideal reaction covectors belong to $\text{range}(\mathbf{J}_c^T)$ and annihilate every tangent velocity:

$$(\mathbf{J}_c^T \boldsymbol{\lambda})^T \mathbf{v} = 0 \quad \text{whenever} \quad \mathbf{J}_c \mathbf{v} = 0.$$

Acceleration is generally not tangent. A point travelling around a circle needs a normal centripetal acceleration even though its velocity is tangent. The curvature term $\boldsymbol{\gamma}$ expresses precisely that requirement.

Example 7.2. Null Space for the Orientation Guide

The guide $q_1 + q_2 = \theta_*$ has tangent basis $\mathbf{n} = (1, -1)^T$ and reaction covectors proportional to $(1, 1)^T$. The relative lock $q_2 = q_{2*}$ instead has tangent basis $(1, 0)^T$ and reaction covectors proportional to $(0, 1)^T$. An ordinary free two-link chain already includes its point attachment in the coordinates and has neither of these extra restrictions. Its joint force must be recovered from body force balance; no extra angular-lock multiplier is needed to make the bodies remain attached.

Constraint Insight 7.1. Constraints Restrict the Effective DOF

At a regular configuration with constant Jacobian rank r , the constraint manifold has local dimension $n - r$. Counting equations gives $n - m$ only when all m equations are independent. Contact activation and singular configurations can change that local description.

The acceleration solves a mass-weighted least-change problem:

$$\min_{\mathbf{a}} \frac{1}{2} (\mathbf{a} - \mathbf{a}_0)^T \mathbf{M} (\mathbf{a} - \mathbf{a}_0) \quad \text{subject to} \quad \mathbf{J}_c \mathbf{a} = -\boldsymbol{\gamma}.$$

Its solution is $\mathbf{a} = \mathbf{a}_0 - \mathbf{M}^{-1} \mathbf{J}_c^T \mathbf{S}^{-1} (\mathbf{J}_c \mathbf{a}_0 + \boldsymbol{\gamma})$. This is a projection onto an affine set of compatible accelerations in the mass metric. It is not an ordinary Euclidean projection of acceleration into the velocity null space. The distinction matters whenever constraints curve or move.

7.5 Energy Transfer via Constraint Forces

Theorem 7.2. Zero Power Constraint Condition

The instantaneous power supplied by an ideal reaction is

$$P_c = (\mathbf{J}_c^T \boldsymbol{\lambda})^T \dot{\mathbf{q}} = \boldsymbol{\lambda}^T \mathbf{J}_c \dot{\mathbf{q}} = -\boldsymbol{\lambda}^T \boldsymbol{\Phi}_t. \quad (7.9)$$

For a compatible stationary ideal constraint,

$$P_c = 0. \quad (7.10)$$

This concerns the sum of reaction powers over the declared system. It does not say that each body receives zero power, or that total energy is constant when other forces do work. For time-independent rigid-body kinetic energy $T = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}}$, conservative potential V , and a consistent velocity bias, the complete balance is

$$\frac{d}{dt}(T + V) = \dot{\mathbf{q}}^T (\mathbf{B}\mathbf{u} + \mathbf{Q}_o) - \boldsymbol{\lambda}^T \boldsymbol{\Phi}_t.$$

The identity uses $\dot{\mathbf{q}}^T \mathbf{c} = \frac{1}{2} \dot{\mathbf{q}}^T \dot{\mathbf{M}} \dot{\mathbf{q}}$. A moving boundary, friction, damping, deformation or impact needs its own work or energy accounting; the stationary ideal theorem does not remove those effects.

In Plain Language 7.6. The Power Paradox

Suppose an ideal point connection moves with common velocity $\mathbf{v}_J$. Let $\mathbf{F}$ be the force on body B from body A. The powers at the connection are $P_B = \mathbf{F} \cdot \mathbf{v}_J$ and $P_A = -\mathbf{F} \cdot \mathbf{v}_J$. Positive power adds mechanical energy to the receiving body; negative power removes it. The sum is zero although either term can be large.

For an entirely illustrative instant, $\mathbf{F} = (100, 0, 0)$ N and $\mathbf{v}_J = (4, 0, 0)$ m/s give $P_B = 400$ W and $P_A = -400$ W. The force is parallel to the common velocity, not perpendicular to it. Cancellation comes from the opposite forces acting at equal point velocities. If those powers persisted for 0.01 s, 4 J would cross the interface; an instantaneous power value alone does not establish that integrated transfer.

The direction can reverse during a swing. A proximal segment slowing down while a distal segment speeds up is not by itself proof of this power pathway: gravity, muscle work, changes in each body's translation and rotation, and other joint loads also contribute.

Definition 7.4. Energy Partition

For each rigid body i , define its physical kinetic energy using its center-of-mass velocity $\mathbf{v}_{G_i}$ and angular velocity $\boldsymbol{\omega}_i$:

$$T_i = \frac{1}{2} m_i \|\mathbf{v}_{G_i}\|^2 + \frac{1}{2} \boldsymbol{\omega}_i^T \mathbf{I}_{G_i} \boldsymbol{\omega}_i, \quad T = \sum_i T_i. \quad (7.11)$$

Express inertia and angular velocity in the same frame. The separate terms $\frac{1}{2}M_{11}\dot{q}_1^2$, $\frac{1}{2}M_{22}\dot{q}_2^2$ and $M_{12}\dot{q}_1\dot{q}_2$ in a coordinate expansion are not generally the energies of body 1, body 2 and a third physical “coupling reservoir.” Each body’s COM velocity may depend on several generalized velocities. Cross terms change with the coordinate choice; physical body energies do not.

For a connection transmitting both a force $\mathbf{F}$ and a moment $\boldsymbol{\mu}$ on B, write both moments about the same joint point. The interface sum is

$$P_A + P_B = \mathbf{F} \cdot (\mathbf{v}_B - \mathbf{v}_A) + \boldsymbol{\mu} \cdot (\boldsymbol{\omega}_B - \boldsymbol{\omega}_A). \quad (7.12)$$

An ideal coincident point joint cancels the force term. A frictionless hinge has no reaction moment along its free rotation axis, so the ideal moment also has zero relative power. A rigid lock cancels both relative velocities. A motor acting across a rotating joint can instead supply power $\boldsymbol{\mu} \cdot (\boldsymbol{\omega}_B - \boldsymbol{\omega}_A)$; a damper can remove it. Those are distinct from the ideal zero-power reaction.

To compute a body’s mechanical-energy change, integrate all of its nonconservative boundary powers and include changes in its potential energy. The club can gain energy from a moving hand even when ideal hand–club interaction contributes zero net power to the combined model. Whole-system and segment statements must never silently exchange boundaries.

7.6 A Compatible Two-Link Constraint Example

Example 7.3. Reaction of a World-Orientation Guide

Use the complete manufactured model from the affine chapter: $m_1 = 2.5$ kg, $m_2 = 0.4$ kg, $l_1 = 0.35$ m, COM distances $r_1 = 0.175$ m and $r_2 = 0.5$ m, centroidal inertias $I_1 = 0.025$ and $I_2 = 0.03$ kg m², and $g = 9.81$ m/s². The first pivot is fixed. The two bodies are attached at a frictionless point hinge; there is no applied joint torque, damping or impact during the following interval. These are illustrative rigid links, not fitted human segments.

For reproducibility, put $a = I_1 + m_1 r_1^2 + m_2 l_1^2$, $d = I_2 + m_2 r_2^2$, $b = m_2 l_1 r_2$, $h_1 = m_1 r_1 + m_2 l_1$, and $h_2 = m_2 r_2$. The operators are

$$\mathbf{M} = \begin{bmatrix} a + d + 2b \cos q_2 & d + b \cos q_2 \\ d + b \cos q_2 & d \end{bmatrix}, \quad \mathbf{c} = b \sin q_2 \begin{bmatrix} -2\dot{q}_1 \dot{q}_2 - \dot{q}_2^2 \\ \dot{q}_1^2 \end{bmatrix},$$

$$V = -g[h_1 \cos q_1 + h_2 \cos(q_1 + q_2)], \quad \mathbf{g} = \nabla V.$$

The companion calculation reuses that chapter’s operator implementation, which is checked against independent recursive Newton–Euler dynamics.

Add the world-orientation guide $q_1 + q_2 = 115^\circ$. Start at $\mathbf{q} = (45^\circ, 70^\circ)$ with $\dot{\mathbf{q}} = (8, -8)$ rad/s. Both position and velocity are compatible, and

$$\mathbf{J}_c = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad \mathbf{J}_c \dot{\mathbf{q}} = 0, \quad \mathbf{J}_c \ddot{\mathbf{q}} = 0. \quad (7.13)$$

At this instant,

$$\mathbf{M} \approx \begin{bmatrix} 0.32844532 & 0.15394141 \\ 0.15394141 & 0.13000000 \end{bmatrix} \text{ kg m}^2,$$

$$\mathbf{c} \approx \begin{bmatrix} 4.20982294 \\ 4.20982294 \end{bmatrix} \text{ N m}, \quad \mathbf{g} \approx \begin{bmatrix} 5.78413025 \\ 1.77817588 \end{bmatrix} \text{ N m}.$$

Without the orientation guide, but retaining the hinge attachment, the free acceleration is

$$\mathbf{a}_0 \approx (-19.86390770, -22.53939122) \text{ rad/s}^2.$$

Solving the constrained system gives

$$\lambda \approx 5.35099959 \text{ N m}, \quad \ddot{\mathbf{q}} \approx (-26.60658776, 26.60658776) \text{ rad/s}^2.$$

The equal generalized moments have powers $\lambda \dot{q}_1 \approx 42.80799674 \text{ W}$ and $\lambda \dot{q}_2 \approx -42.80799674 \text{ W}$. Their sum is zero. These are coordinate powers, not arm and club powers. Physically the guide supplies a moment to the second body, whose absolute angular velocity is zero, so the guide does no work on that body either. The body's hinge-force power can still be nonzero because its COM translates. Calling the two coordinate products a measured transfer from arm to club would repeat the energy-partition error.

In Plain Language 7.7. Why the Hinge and Lock Distinction Matters

Both the free and guided cases above keep the bodies attached. Removing the guide permits the second body's absolute orientation to change; it does not disconnect the club. A true relative lock instead fixes q_2 and starts with $\dot{q}_2 = 0$. It is a third experiment with a different compatible initial velocity. Comparing it with the first two without stating that change would confuse an initial-condition effect with a constraint effect.

The numerical comparison uses exact reduced coordinates for each linear constraint. For the guide, $\mathbf{q} = \mathbf{q}_0 + (1, -1)^T s$ and $\dot{\mathbf{q}} = (1, -1)^T \dot{s}$; the equation is $(\mathbf{n}^T \mathbf{M} \mathbf{n}) \ddot{s} = -\mathbf{n}^T (\mathbf{c} + \mathbf{g})$. This satisfies position and velocity constraints directly. Independently solving the full saddle-point equations checks the multiplier signs.

7.6.1 Engaging and Releasing a Constraint

A new stationary lock cannot be imposed smoothly on a velocity that violates it without specifying an engagement process. In a perfectly inelastic ideal capture at a compatible configuration, let $\mathbf{\Lambda}$ be the constraint impulse and neglect bounded forces during the instantaneous event. Then

$$\mathbf{M}(\dot{\mathbf{q}}^+ - \dot{\mathbf{q}}^-) = \mathbf{J}_c^T \mathbf{\Lambda}, \quad \mathbf{J}_c \dot{\mathbf{q}}^+ = 0,$$

$$\mathbf{\Lambda} = -\mathbf{S}^{-1} \mathbf{J}_c \dot{\mathbf{q}}^-, \quad \dot{\mathbf{q}}^+ = \dot{\mathbf{q}}^- - \mathbf{M}^{-1} \mathbf{J}_c^T \mathbf{S}^{-1} \mathbf{J}_c \dot{\mathbf{q}}^-.$$

Using the unchanged configuration and inertia across the impulse,

$$T^+ - T^- = -\frac{1}{2} (\mathbf{J}_c \dot{\mathbf{q}}^-)^T \mathbf{S}^{-1} (\mathbf{J}_c \dot{\mathbf{q}}^-) \leq 0.$$

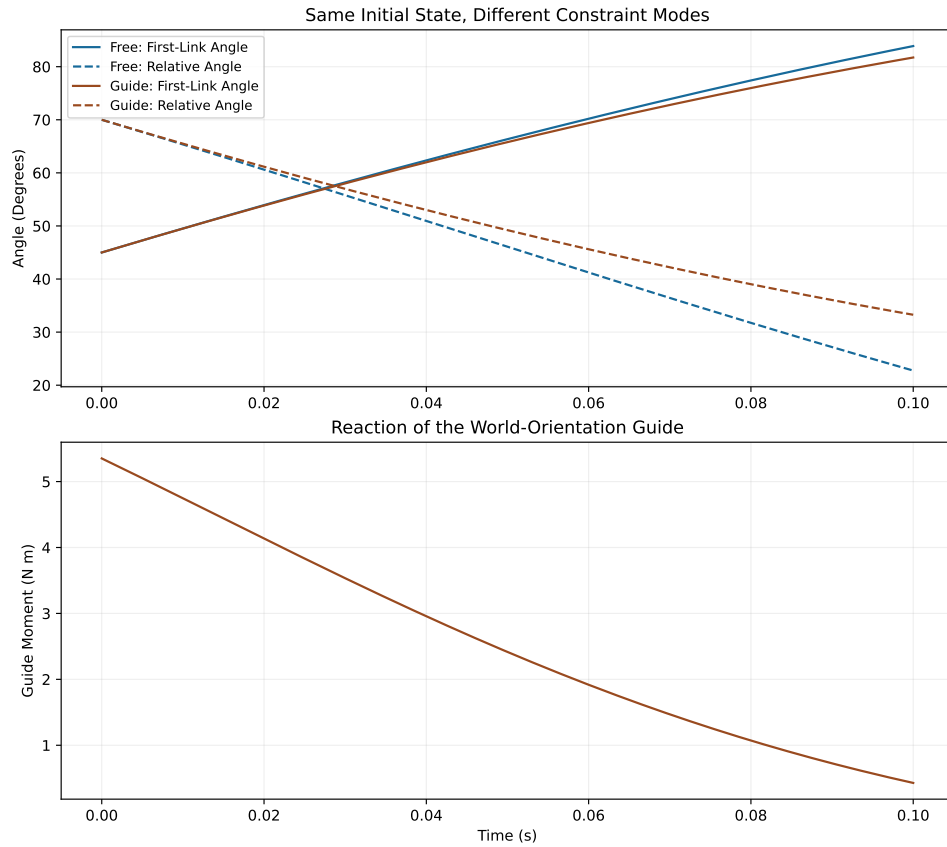


Figure 7.1: Compatible Free and Guided Two-Link Motion. Both Runs Start From the Same Position and Velocity With Zero Applied Joint Torque. The Guide Holds the Second Body at a Fixed World Orientation; It Is Not a Relative Wrist Lock. Its Reaction Moment Changes the Motion While Its Power Remains Zero. The Curves Are Manufactured Model Results, Not Golfer Measurements.

This is a kinetic-energy loss, representing unresolved dissipation in this ideal capture law. It does not contradict zero reaction power during subsequent smooth compatible motion. An elastic, compliant or actively controlled engagement requires a different event model and energy budget.

For the same configuration but the incompatible incoming velocity (8, 12) rad/s, capture by the world-orientation guide gives $\dot{\mathbf{q}}^+ \approx (11.18026202, -11.18026202)$ rad/s and $\Lambda \approx -2.52386004$ N m s. The total kinetic energy falls from 34.64862561 J to 9.41002518 J, a loss of 25.23860043 J. This is not a zero-work redistribution between bodies.

Conversely, releasing an ideal constraint without an impulse preserves position and velocity at that instant. Acceleration may jump because the reaction disappears. No energy is automatically added at release. Real releases can involve stored elastic energy or active work, which must be represented explicitly rather than attributed to removal of an algebraic equation.

7.7 Constraint Forces and Joint Reaction Forces

Key Principle 7.1. Define the Reaction, Frame and System Boundary

A joint reaction is a force and moment transmitted across a specified interface. Whether it is internal depends on the system: hand-club interaction is internal to a model containing both bodies and external to the club alone. Ground contact is external to a golfer-club system that excludes Earth. Muscle forces can be internal to the whole body yet external to an isolated segment. Internal and active are not opposite categories.

To relate a spatial wrench to generalized force, define a twist $\mathbf{t} = [\mathbf{v}_P^T \ \boldsymbol{\omega}^T]^T = \mathbf{J}_P \dot{\mathbf{q}}$ and wrench $\mathbf{w} = [\mathbf{F}^T \ \boldsymbol{\mu}_P^T]^T$ at the same point P , in the same frame and ordering. Power duality gives

$$\mathbf{Q} = \mathbf{J}_P^T \mathbf{w}, \quad \mathbf{Q}^T \dot{\mathbf{q}} = \mathbf{F} \cdot \mathbf{v}_P + \boldsymbol{\mu}_P \cdot \boldsymbol{\omega}. \quad (7.14)$$

For an internal pair, use the relative twist and both bodies' contributions. A generalized moment does not determine a unique three-dimensional joint force by division by one lever arm. Even $\boldsymbol{\mu} = \mathbf{r} \times \mathbf{F}$ leaves the component of force along $\mathbf{r}$ unresolved; a pure couple is another possible source of moment.

If $\mathbf{r}$ points from P to P' , then $\boldsymbol{\mu}_{P'} = \boldsymbol{\mu}_P - \mathbf{r} \times \mathbf{F}$ and $\mathbf{v}_{P'} = \mathbf{v}_P + \boldsymbol{\omega} \times \mathbf{r}$. These transformations preserve wrench power. Mixing a moment about one point with velocity at another does not. Report the receiving body, point, frame, sign and whether the value is a net wrench, ideal reaction or muscle-resolved contact estimate.

OpenSim's Joint Reactions Analysis illustrates the distinction in practice: reported joint forces and moments depend on the loads included in the model and the selected reporting frame/body. They are not the same output as inverse-dynamics generalized joint moments. Muscle forces and even actuator representation affect the interpretation of the reported reaction. A rigid-body balance inferred from motion is therefore not a direct measurement of every anatomical contact load. [Hicks, 2019]

7.8 Coupling and Control in the Kinetic Chain

Key Principle 7.2. The Kinetic Chain Combines Actuation and Coupling

Mechanical coupling lets forces applied in one region influence motion elsewhere. Muscles supply and absorb work, maintain or change posture, alter joint loading, and influence contact conditions. Inertia and geometry determine how those actions propagate. Gravity contributes according to the changing potential energy. These mechanisms coexist in a single equation rather than competing as separate explanations of the swing.

The multiplier calculation makes this interdependence explicit. At fixed configuration, velocity and regular constraint mode, suppose the input enters as $\mathbf{B}\mathbf{u}$ and other terms are independent of $\mathbf{u}$. Then

$$\frac{\partial \lambda}{\partial \mathbf{u}} = -\mathbf{S}^{-1} \mathbf{J}_c \mathbf{M}^{-1} \mathbf{B},$$

$$\frac{\partial \ddot{\mathbf{q}}}{\partial \mathbf{u}} = [\mathbf{I} - \mathbf{M}^{-1} \mathbf{J}_c^T \mathbf{S}^{-1} \mathbf{J}_c] \mathbf{M}^{-1} \mathbf{B}.$$

Actuation changes the reaction and its permitted acceleration effect simultaneously. Part of an input can be balanced by a reaction rather than producing acceleration in a forbidden direction. The constraint-dependent map is the appropriate local input map for a drift/control comparison. Changing a contact mode can change that map abruptly; changing muscular activation can also alter the underlying force law and future state.

These derivatives hold within the stated ideal mode. They do not say that the complete physiological system is affine in neural excitation: activation dynamics, muscle force-length-velocity behavior, compliance and contact transitions require additional states and constitutive laws.

Constraint Insight 7.2. Energy and Momentum Transfer Through Constraints

To substantiate a proposed kinetic-chain mechanism, identify the receiving body and compute interface force and moment powers over an interval. Close its mechanical-energy balance using all other loads and potential changes. Separately check whole-system momentum: linear momentum changes with external force, and angular momentum about a fixed inertial point changes with external moment. Ground and gravity can make those totals nonconserved even when internal reaction powers cancel.

Angular speed is not angular momentum, and neither is kinetic energy. Changes in body configuration and velocity direction can increase one while decreasing another. A sequence of speed peaks does not uniquely identify energy flow, muscular timing or an optimal sequence of joint locks. Likewise, a large ideal reaction can be necessary to turn a trajectory while doing little work. Its magnitude alone is not a measure of useful energy transfer or efficiency.

A useful coaching hypothesis connects an action to a measurable result: for example,

a change in hand trajectory may alter both club loading and interface work. Testing it requires comparable task conditions, a declared system model, uncertainty estimates and a controlled comparison. The equations do not establish a universal instruction to grip harder, relax muscles or unlock joints in a prescribed order.

7.9 Constraint Forces in Real Swings: Grip Forces and Ground Reaction Forces

Example 7.4. Grip Force During a Golf Swing

“Grip force” can mean local normal pressure integrated over contact, a net hand force, a six-axis resultant wrench, or an internal wrench measured between two grip sections. These are different observables. Opposing forces from two hands can partly cancel in their resultant while creating a moment through their separation. Neither a pressure total nor a net wrench directly identifies the individual muscle forces supporting the grasp.

Choi and Park measured an internal six-axis grip wrench in nine professional golfers using a split instrumented grip, then combined it with kinematics and a rigid-body inverse-dynamics model. They observed counteracting hand forces, demonstrating why net club motion alone need not reveal how the two hands share load. The instrumented club differed in mass from a regular club; foam balls, hand overlap and rigid-segment assumptions limit interpretation. Their results support measuring internal load sharing, not a universal grip-force range or a conclusion that muscles are inactive. [Choi and Park, 2020]

In a real grasp, normal pressure helps sustain the frictional tractions needed for the required wrench. A no-slip model is valid only while the demanded loads are feasible for that contact. Greater squeezing force may change friction capacity and tissue stiffness, but does not by itself specify the net club wrench or work. With a given kinematic trajectory and inertial model, inverse dynamics can infer a resultant load; it cannot uniquely recover activation, co-contraction or every local contact traction.

Example 7.5. Ground Reaction Forces and the Closed Kinematic Chain

For a golfer–club system of total mass M_T , the external force balance is

$$\mathbf{F}_{\text{ground}} + M_T \mathbf{g}_0 + \mathbf{F}_{\text{other}} = M_T \mathbf{a}_G,$$

where $\mathbf{g}_0$ is the downward gravitational acceleration vector and $\mathbf{a}_G$ the whole-system COM acceleration. Ground reaction forces are measured in newtons; normalization by body weight must specify whose weight and whether the club is included. As an illustrative SI check, with $M_T = 80$ kg, no other vertical force and upward COM acceleration 2 m/s^2 , the summed vertical reaction is $80(9.81 + 2) = 944.8 \text{ N}$. This is a calculated state, not a typical golfer value. Zero vertical COM acceleration gives 784.8 N for the same system.

Ordinary nonadhesive contact can push but cannot sustain an arbitrary tensile nor-

mal force. A candidate active-contact solution must have nonnegative normal reaction; otherwise that contact mode is inadmissible and separation must be considered. Sticking also requires feasible tangential traction, commonly approximated by $\|\mathbf{F}_t\| \leq \mu F_n$. Sliding requires a friction law, and a finite foot contact patch restricts the admissible center of pressure and moments. Writing only $z_{\text{foot}} = 0$ does not establish no slip, no rotation or feasible support.

If all contacting material points are stationary on rigid ground, ideal contact power is zero. Muscles can still do internal work and change total mechanical energy. COM power $\mathbf{F}_{\text{ground}} \cdot \mathbf{v}_G$ is not the ground's point-of-application power; it appears in the COM translational energy balance and must not be substituted for the full multibody energy budget. Deforming footwear, moving ground and slip modify that budget.

Muscular action changes body motion and contact demand, so ground reaction and muscle action remain coupled. The ground need not keep the entire lower body stationary, and the feet can redistribute pressure or rotate. Without ground contact, a golfer-club system can still reorient its segments subject to external-force and angular-momentum balances; absence of contact does not imply a universal backward slide.

7.10 Summary and Key Insights

Key Principle 7.3. Key Takeaways: Constraint Forces

Define the connection. Attachment, free rotation, a relative lock and a world-orientation guide impose different conditions. Contact additionally needs feasibility and mode rules.

Solve compatible dynamics. Use position, velocity and acceleration consistency, the correct curvature term and one free-acceleration contribution. Interpret rank, conditioning and multiplier units before interpreting magnitudes.

Account for power at a boundary. Stationary ideal reactions have zero net power on their declared system. Individual bodies can exchange energy, moving supports can do work, and constraint activation can dissipate energy through an impulse.

Keep physical and coordinate quantities distinct. Generalized reaction powers are not automatically segment powers. Use COM kinetic energies and wrenches paired with velocities at matching points and frames.

Connect mechanics to control and evidence. Muscles, geometry, inertia and external support jointly determine the motion and reaction. Use measured loads and reproducible balances to test claims about a swing; neither a speed sequence nor a large reaction establishes a passive or optimal mechanism.

The next chapter extends the model to three links. The chapters on parallel mechanisms, induced acceleration, muscle force generation and motor control develop the closed-chain, loading and physiological parts of this argument.

Exercises 7.1. Chapter Exercises: Constraint Forces

1. Explain Interface Power

An ideal connection moves at $(4, 0, 0)$ m/s. Body B receives $(100, 0, 0)$ N from A. Compute the powers and explain their signs.

Worked answer. B receives $+400$ W and A receives -400 W at this interface. The pair sums to zero because the point velocities coincide and the forces oppose each other. Positive work adds mechanical energy to the receiving body. Perpendicularity of force and point velocity is not required. A statement about energy transferred over time requires integrating power, while a statement about either body's total energy change requires its other loads as well.

2. Distinguish the Two Angular Constraints

With q_1 measured from downward vertical and q_2 relative, compare $q_1 + q_2 = 115^\circ$ with $q_2 = 70^\circ$ at $(45^\circ, 70^\circ)$.

Worked answer. Both pass through this configuration. The first has $\mathbf{J}_c = [1 \ 1]$ and tangent direction $(1, -1)$: the second body stays at fixed world orientation as the first moves. The second has $\mathbf{J}_c = [0 \ 1]$ and tangent direction $(1, 0)$: the relative angle stays fixed and both absolute angles turn together. Their generalized reactions are respectively (λ, λ) and $(0, \lambda)$. These columns are coordinate representations, not sketches of two physical forces in space.

3. Compute a Pivot Reaction

Consider a uniform slender rod of mass 2 kg and length 0.7 m rotating about a fixed end in a vertical plane. At the instant it points straight down, its angular speed is 10 rad/s. With no applied moment, find the upward pivot force.

Worked answer. The COM is 0.35 m from the pivot and has upward centripetal acceleration $r_G \omega^2 = 35$ m/s². Gravity has zero moment about the pivot at this instant, so the angular acceleration is zero. Upward force balance gives $F_P - mg = mr_G \omega^2$, hence $F_P = 70 + 19.62 = 89.62$ N. The 70 N term is the required net radial force, not the full reaction. The endpoint acceleration cannot be multiplied by the entire rod mass to obtain its COM force balance. In an inertial frame the acceleration is inward and is called centripetal; an outward centrifugal term belongs to an appropriate rotating-frame description. The consistent centroidal inertia is $mL^2/12 \approx 0.08167$ kg m² and pivot inertia is $mL^2/3 \approx 0.32667$ kg m².

4. Compare Compatible Trajectories

Integrate the chapter's complete two-link model for 0.1 s from $(45^\circ, 70^\circ)$ and $(8, -8)$ rad/s, first with free relative rotation and then with the world-orientation guide. Keep all other assumptions fixed.

Worked answer. The free and guided final states are

$$\begin{aligned} \mathbf{q}_{\text{free}} &\approx (83.88564444^\circ, 22.75882070^\circ), \\ \dot{\mathbf{q}}_{\text{free}} &\approx (5.31225354, -7.54851685) \text{ rad/s}, \\ \mathbf{q}_{\text{guide}} &\approx (81.73079376^\circ, 33.26920624^\circ), \\ \dot{\mathbf{q}}_{\text{guide}} &\approx (4.64868766, -4.64868766) \text{ rad/s}. \end{aligned}$$

Both conserve their initial total mechanical energy to numerical precision. The guide

enforces the 115-degree sum throughout; the free case does not. The reproduced figure plots both angles and the guide moment. A large multiplier means a large corrective moment is needed for that state and mode, not that the guide supplies large work.

Use exact reduced coordinates for the guide, check the full dynamics residual independently, and refine integration tolerance and step size. Here DOP853 with relative tolerances 10^{-10} and 10^{-12} and maximum steps 0.002 and 0.001 s gives state differences below 10^{-9} and energy ranges below 10^{-9} J. These numerical checks verify this manufactured example, not human-swing validity. A separate relative-lock run starts from (8,0) rad/s and ends at $q_1 \approx 85.34277402^\circ$, $q_2 = 70^\circ$, $\dot{q}_1 \approx 6.04969667$ rad/s; its different initial energy prevents a like-for-like comparison with the first two.

5. Interpret a Nearly Constant Wrist Angle

A video shows a nearly constant relative wrist angle during part of the downswing. Does that establish an ideal lock or identify its reaction?

Worked answer. It establishes an approximately constant measured angle within the reconstruction error and time resolution. Similar motion can arise from active stabilization, passive stiffness, an external constraint or a momentary dynamic balance. Frame rate alone does not resolve those alternatives. Define three-dimensional segment frames, estimate orientation and derivative uncertainty, and combine kinematics with external-load measurements and a stated inertial/contact model. Perturbation-response measurements can help distinguish stiffness and active regulation. Even then, net inverse-dynamics moments do not uniquely identify muscle forces or prove a perfect lock.

6. Account for Energy During Capture

Apply the chapter's perfectly inelastic orientation-guide capture to the incompatible incoming velocity (8,12) rad/s. Partition kinetic energy by physical body and determine whether all lost club energy reaches the first link.

Worked answer. Use $\mathbf{v}_{G_1} = r_1 \dot{q}_1 (\cos q_1, \sin q_1)$ and

$$\mathbf{v}_{G_2} = l_1 \dot{q}_1 (\cos q_1, \sin q_1) + r_2 (\dot{q}_1 + \dot{q}_2) (\cos(q_1 + q_2), \sin(q_1 + q_2)).$$

The absolute angular rates are $\dot{q}_1$ and $\dot{q}_1 + \dot{q}_2$. Before capture, $T_1 = 3.25$ J and $T_2 \approx 31.39862561$ J. After capture, $T_1 \approx 6.34756783$ J and $T_2 \approx 3.06245734$ J. Body 1 gains about 3.09756783 J while body 2 loses about 28.33616826 J. The difference, 25.23860043 J, is dissipated by the assumed capture law. Position and potential are unchanged during the impulse. Subsequent smooth guide motion has zero guide power; that theorem must not be applied across the incompatible capture event.

Chapter 8

The Triple Pendulum: Coupling, Release and Control

In Plain Language 8.1. What an Additional Joint Lets Us Investigate

A moving arm and club exchange energy through their connections. Muscles can add or absorb mechanical energy, while joint forces redirect motion and transfer power between bodies. An additional joint gives the system another way to change its shape and redistribute its motion. It does not guarantee a faster club, a particular release sequence or less muscular work.

The useful question is therefore specific: for the same bodies, initial state, admissible inputs and task, how does allowing an additional relative rotation change the attainable club motion? This chapter builds the mechanics needed to answer that question. A two-link arm-club model already includes a wrist hinge. Here the third link resolves an upper arm and forearm separately; it is a choice of resolution, not a universal minimum for explaining golf speed.

Our investigation connects four quantities that are often conflated: joint torque, segment power, angular velocity and clubhead velocity. A large club angular velocity need not imply large contemporaneous wrist power. Conversely, small net wrist work does not establish that the wrist is uncontrolled or mechanically unimportant. The configuration and the history of applied forces determine how those quantities relate.

8.1 The Triple Pendulum Model

Definition 8.1. Three Rigid Links and Three Relative Angles

Use a fixed shoulder pivot and three ideal planar revolute joints. Link 1 represents an upper arm, link 2 a forearm-and-hand aggregate, and link 3 a rigid club. Each link has length L_i , positive mass m_i , COM distance c_i from its proximal joint, and positive planar rotational inertia I_i about its own center of mass. These are independent specified body properties. An inertia measured about a proximal joint must first be converted using $I_{i,\text{joint}} = I_i + m_i c_i^2$; adding that offset again would double-count it. The angle q_1 is measured counterclockwise from downward vertical. The angles q_2 and q_3 are measured from the preceding link. Thus $q_2 = 0$ means aligned links, whereas $q_2 = 90^\circ$ means a right angle. The absolute orientations are

$$\theta_1 = q_1, \quad \theta_2 = q_1 + q_2, \quad \theta_3 = q_1 + q_2 + q_3.$$

The configuration space of these ideal unrestricted hinges is a three-torus; computations use local or unwrapped radian coordinates in $\mathbb{R}^3$. Anatomical limits would restrict this domain. Relative joint rates $v = \dot{q}$ differ from absolute segment rates $\omega_i = \dot{\theta}_i$.

8.1.1 Understanding the Three Segments

The shoulder is fixed in this example. There is no rotating torso, moving pelvis, second arm, shaft bending or ground-contact model hidden inside its parameters. The grip hinge is an ideal planar coordinate, not a measurement of anatomical wrist extension, radial deviation or forearm rotation. In particular, it cannot predict three-dimensional clubface orientation. Those features require additional bodies, coordinates and constraints.

Mass alone does not order rotational inertia: the distance of mass from the reference axis matters. A long light club can have greater inertia about its grip than a heavier compact body about another axis. Nor does an inertia ranking establish which segment must move fastest in a coupled chain. The relevant kinetic energy includes COM motion and all absolute angular velocities.

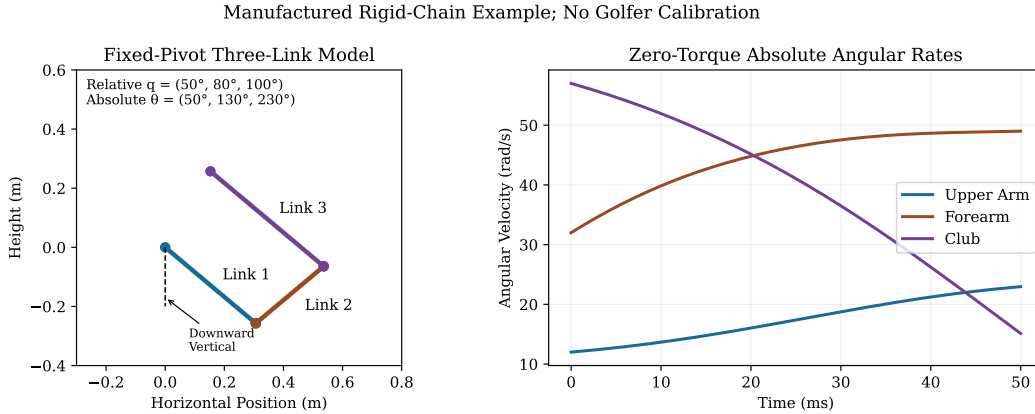


Figure 8.1: Three-Link Geometry and an Unforced Continuation From the Declared Initial State. Absolute Club Angular Velocity Decreases in This Example; an Additional Joint Does Not Guarantee Distal Acceleration. These Are Manufactured Dynamics, Not a Measured Golf Swing.

Write the direction of a link as $e(\theta) = (\sin \theta, -\cos \theta)^T$. The COM position and its Jacobian are

$$r_i = \sum_{k < i} L_k e(\theta_k) + c_i e(\theta_i), \quad J_i = \frac{\partial r_i}{\partial q}.$$

For $j \leq i$, column j of J_i contains $\sum_{k=j}^{i-1} L_k e'(\theta_k) + c_i e'(\theta_i)$; later columns are zero. Let s_i be the vector with its first i entries equal to one and all later entries zero. Then $\dot{r}_i = J_i v$ and $\omega_i = s_i^T v$. This construction fixes the meaning of every subsequent term.

8.2 Equations of Motion for the Triple Pendulum

Definition 8.2. Kinetic Energy, Gravity and Applied Torque

For the rigid, frictionless chain,

$$T = \frac{1}{2} \sum_i (m_i \|J_i v\|^2 + I_i (s_i^T v)^2) = \frac{1}{2} v^T M(q) v, \quad V = \sum_i m_i g_0 (r_i)_y,$$

where $g_0 = 9.81 \text{ m/s}^2$ and upward height is positive. Define $g(q) = \nabla_q V$. Lagrange's equations give

$$M(q)\ddot{q} + c(q, v) + g(q) = \tau. \quad (8.1)$$

The vector c contains the velocity-dependent inertial terms; it is not itself a Coriolis matrix. The input τ_i is the net applied joint couple conjugate to relative rate v_i , so its mechanical power is $\tau_i v_i$. It is not an individual muscle force or neural excitation.

To verify the gravity convention, set $h_1 = m_1 c_1 + (m_2 + m_3)L_1$, $h_2 = m_2 c_2 + m_3 L_2$ and $h_3 = m_3 c_3$. Then $V = -g_0 \sum_i h_i \cos \theta_i$ and

$$g = g_0 \begin{bmatrix} h_1 \sin \theta_1 + h_2 \sin \theta_2 + h_3 \sin \theta_3 \\ h_2 \sin \theta_2 + h_3 \sin \theta_3 \\ h_3 \sin \theta_3 \end{bmatrix}.$$

At rest with every link pointing down, gravity torque is zero. Displacing the chain slightly requires a holding torque with the sign of this potential gradient; the unforced acceleration uses its negative after solving through M .

8.3 The Mass Matrix: Understanding Coupling

Definition 8.3. Complete Three-Link Inertia

The COM construction yields

$$M = \sum_i (m_i J_i^T J_i + I_i s_i s_i^T) = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{12} & M_{22} & M_{23} \\ M_{13} & M_{23} & M_{33} \end{bmatrix}. \quad (8.2)$$

For compact complete entries, define

$$\begin{aligned} a &= I_1 + m_1 c_1^2 + (m_2 + m_3)L_1^2, \\ b &= I_2 + m_2 c_2^2 + m_3 L_2^2, \quad d = I_3 + m_3 c_3^2, \\ \alpha &= m_2 L_1 c_2 + m_3 L_1 L_2, \quad \beta = m_3 L_1 c_3, \quad \gamma = m_3 L_2 c_3. \end{aligned}$$

Every constant has units kg m^2 . The entries are

$$\begin{aligned} M_{11} &= a + b + d + 2\alpha \cos q_2 + 2\beta \cos(q_2 + q_3) + 2\gamma \cos q_3, \\ M_{22} &= b + d + 2\gamma \cos q_3, \quad M_{33} = d, \\ M_{12} &= b + d + \alpha \cos q_2 + \beta \cos(q_2 + q_3) + 2\gamma \cos q_3, \\ M_{13} &= d + \beta \cos(q_2 + q_3) + \gamma \cos q_3, \\ M_{23} &= d + \gamma \cos q_3. \end{aligned}$$

In particular, M_{11} includes all three COM inertias, and both M_{22} and M_{12} include $I_2 + I_3$. Their presence follows from the absolute angular rates. Omitting them requires an explicitly justified point-mass approximation, not merely the observation that a club is light. A club generally does not have $c_3 = L_3/2$; that is only an assumption in the manufactured examples below.

For any nonzero v , the angular contribution $\sum_i I_i (s_i^T v)^2$ is positive because the three s_i span the velocity space. Thus the declared model has a symmetric positive-definite M at every configuration. This property survives a task-Jacobian singularity: the tip may lose an instantaneous motion direction even while the joint-space inertia remains invertible.

8.3.1 What the Mass Matrix Entries Mean

At a fixed state, an incremental applied torque produces $\delta \ddot{q} = M^{-1} \delta \tau$. The inverse, rather than an isolated entry of M , determines acceleration response. Off-diagonal entries describe inertial coupling in the stated coordinates; they do not guarantee simultaneous motion, a universal ordering such as $M_{13} > M_{12}$, or diagonal dominance. Torques or constraints can hold selected coordinates fixed while others move.

The diagonal M_{ii} describes the inertia for accelerating coordinate i while all other accelerations are held zero, with the necessary reaction torques supplied. If the other joints instead respond freely to an incremental torque only at i , the apparent inertia is

$$I_{i,\text{free}} = \frac{1}{(M^{-1})_{ii}} = M_{ii} - M_{ir} M_{rr}^{-1} M_{ri},$$

the Schur complement of the remaining coordinates r . These are different experiments. Neither quantity alone measures anatomical strength, energetic efficiency or control precision.

Example 8.1. A Computed Matrix and Its Inverse

Use $L = (0.4, 0.3, 0.5)$ m, $m = (2, 0.5, 0.2)$ kg, $I = (0.05, 0.01, 0.01)$ kg m^2 and $c_i = L_i/2$. This is a manufactured short-club chain, not calibrated adult-driver anthropometry. At $q = (50, 90, 100)^\circ$,

$$M \simeq \begin{bmatrix} 0.259148 & 0.036844 & 0.000199 \\ 0.036844 & 0.056541 & 0.019895 \\ 0.000199 & 0.019895 & 0.022500 \end{bmatrix} \text{ kg m}^2. \quad (8.3)$$

The unrounded matrix has eigenvalues approximately 0.012310, 0.060183 and

0.265696 kg m². Its computed inverse is

$$M^{-1} \simeq \begin{bmatrix} 4.45187 & -4.19125 & 3.66665 \\ -4.19125 & 29.62083 & -26.15467 \\ 3.66665 & -26.15467 & 67.53886 \end{bmatrix} (\text{kg m}^2)^{-1}. \quad (8.4)$$

An incremental 1 N m at joint 3 therefore changes the acceleration vector by approximately (3.67, -26.15, 67.54) rad/s². The free apparent inertia at that coordinate is 0.01481 kg m², compared with $M_{33} = 0.02250$ when the other accelerations are fixed. A large response can amplify disturbances as well as useful inputs; it does not prove high precision.

8.4 Coriolis Terms in the Triple Pendulum

8.4.1 The Velocity-Dependent Terms and Their Signs

The complete Christoffel expression is

$$c_i = \sum_{j,k} \frac{1}{2} \left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{jk}}{\partial q_i} \right) v_j v_k. \quad (8.5)$$

Keeping only $\sum_{j,k} (\partial M_{ij} / \partial q_k) v_j v_k$ omits the kinetic-energy gradient. A useful independent check is $v^T c = \frac{1}{2} v^T \dot{M} v$. Multiplying the equation of motion by v^T then gives $\dot{T} + \dot{V} = \tau^T v$. Velocity-dependent inertia can redistribute motion while satisfying this energy balance; it is not an additional energy source.

For the distal coordinate the cancellations simplify considerably:

$$c_3 = \beta \sin(q_2 + q_3) v_1^2 + \gamma \sin q_3 (v_1 + v_2)^2. \quad (8.6)$$

There is no $\partial M_{23} / \partial q_2$ contribution because $M_{23} = d + \gamma \cos q_3$ is independent of q_2 . Also, v_3 cancels from this particular generalized component, although it remains in the other components and therefore can affect $\ddot{q}_3$ through the coupled solve. Neither c_3 nor $-c_3 / M_{33}$ alone is the freely responding club acceleration.

Example 8.2. A Signed Inertial Term

At $q = (50, 80, 100)^\circ$ and $v = (12, 20, 25)$ rad/s with the preceding parameters, $\beta = 0.02$ and $\gamma = 0.015$ kg m². Therefore

$$c_3 = 0.02 \sin 180^\circ (12)^2 + 0.015 \sin 100^\circ (32)^2 \simeq 15.127 \text{ N m}. \quad (8.7)$$

This term sits on the left of the equation of motion. It is not a measured wrist-muscle torque or proof of distal acceleration. Reversing both relative configuration angles reverses its sign at fixed rates; at straight alignment it vanishes. In the full unforced example below, the club's absolute angular rate decreases despite its initially large value.

8.5 The Whip Effect: What Constraint Release Changes

Definition 8.4. An Ideal Release Event

Suppose a mechanical lock holds $q_3 = \bar{q}_3$ with $v_3 = 0$. With $A = (0, 0, 1)$, its equations are

$$\begin{bmatrix} M & -A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} \tau - c - g \\ 0 \end{bmatrix}.$$

The multiplier supplies the joint reaction required to maintain the lock. Because $Av = 0$, its power $\lambda^T Av$ is zero. This is a stationary ideal constraint; a driven constraint can do work.

Remove the lock with no impulse and no instantaneous change in potential or body properties. Position and velocity are continuous: $q^+ = q^-$ and $v^+ = v^-$. Mechanical energy is consequently continuous. Acceleration can jump when the reaction is removed. There is no instantaneous inheritance of another body's momentum and no injection of energy by the removal itself.

8.5.1 Why the State at Release Matters

The state reached before release reflects all earlier inputs and constraints. Changing a holding interval can change that state, the subsequent force transmission and the timing of a useful tip trajectory. That provides a mechanism by which release timing can matter. It does not imply that a later release always improves speed or that shoulder, elbow and wrist must unlock at universal times.

For the manufactured chain, use $q = (50, 80, 100)^\circ$ and the compatible locked rate $v = (12, 20, 0)$ rad/s with zero applied joint torques. Immediately before removal,

$$\ddot{q}^- \simeq (199.109, -308.438, 0)^T \text{ rad/s}^2, \quad \lambda \simeq 8.594 \text{ N m}.$$

Immediately afterward the same state gives

$$\ddot{q}^+ \simeq (158.666, -65.564, -596.883)^T \text{ rad/s}^2.$$

The negative relative wrist acceleration is a computed result, not a release rule. The change in acceleration satisfies $M(\ddot{q}^- - \ddot{q}^+) = A^T \lambda$. Both sides of the event have the same energy. In contrast, suddenly imposing a lock on an incompatible nonzero v_3 generally requires an impulse and a separately specified impact law; that is a different event.

8.6 Wrist Cock and Release as a Constraint Mechanism

Definition 8.5. A Pose, a Lock and a Spring Are Different Models

A bent wrist pose by itself specifies neither a force nor stored energy. In these relative coordinates, a fixed wrist angle is $q_3 = \bar{q}_3$. Fixing the sum of the second and third relative angles would instead fix the club orientation relative to the upper arm, a different constraint. Neither condition should be inferred merely from a projected

video angle.

An ideal lock enforces a geometric condition. An active joint controller uses torque to maintain an approximate pose. A passive elastic joint supplies a configuration-dependent torque and stores recoverable energy. A viscoelastic joint also depends on deformation history and dissipates energy. These mechanisms can produce similar-looking motion while having different power and control requirements.

8.6.1 Wrist Cock and Recoverable Energy

For a declared linear torsional spring with natural angle $q_{3,0}$,

$$U_s = \frac{1}{2}k(q_3 - q_{3,0})^2, \quad \tau_s = -k(q_3 - q_{3,0}).$$

With parallel damping $\tau_d = -bv_3$, the rigid-chain balance becomes

$$\frac{d}{dt}(T + V + U_s) = \tau_{\text{active}}^T v - bv_3^2, \quad b \geq 0.$$

Recoil can supply power when U_s decreases. Estimating how much a human wrist supplies requires measured torque–angle behavior, its natural configuration, rate/history effects and the actual multi-axis motion. A cock angle or a generic tendon stiffness is insufficient to establish a universal 5–10 J store or its fraction of club energy.

An actively held posture can have zero joint mechanical power while muscles consume metabolic energy or exchange energy internally. Net joint power also cannot identify co-contraction. This is why mechanical work, motor commands, tissue loading and subjective effort must remain separate in the explanation. The muscle-force and fascia chapters develop the actuator and material states needed to connect them.

8.7 Triple Pendulum ZTCF Family: A Computed Counterfactual

Setting the declared generalized input to zero gives

$$\dot{q} = v, \quad \dot{v} = -M(q)^{-1}(c(q, v) + g(q)). \quad (8.8)$$

Here the model is frictionless and has no spring, muscle-activation or material-memory state. Gravity remains. If those states or passive forces are added, they must remain in the appropriate counterfactual equations; zero neural excitation would not mean zero muscle force. A pointwise zero-torque acceleration and a forward trajectory that keeps torque zero for 50 ms answer different questions. The trajectory evolves through states different from any actively controlled continuation.

Use the preceding manufactured body parameters with $q(0) = (50, 80, 100)^\circ$ and $v(0) = (12, 20, 25)$ rad/s. Convert initial angles to radians before integration. The initial acceleration is approximately $(131.936, 809.896, -1371.119)$ rad/s², computed from the complete equations rather than assigned independently.

Table 8.1: Manufactured Zero-Torque Continuation: Absolute Rates and Tip Speed

t (ms)	ω_1 (rad/s)	ω_2 (rad/s)	ω_3 (rad/s)	Tip (m/s)
0	12.000	32.000	57.000	23.976
25	17.382	46.411	41.051	12.937
50	22.961	49.001	15.120	7.924

The angle vectors at 25 and 50 ms are approximately $(70.582, 117.600, 113.367)^\circ$ and $(99.942, 157.321, 85.308)^\circ$. At 50 ms the relative rates are $(22.961, 26.041, -33.881)$ rad/s; the club's absolute rate is their sum, 15.120 rad/s. Confusing v_3 with ω_3 would reverse the interpretation of its rotation.

The reproducible calculation uses DOP853 with relative tolerance 10^{-10} , absolute tolerance 10^{-12} and maximum step 1 ms, sampled every 0.1 ms. Tightening those tolerances to 10^{-12} and 10^{-13} and halving the maximum step changes each state component by less than 10^{-8} in its own radian or radian-per-second units. The variation of $T + V$ stays below 10^{-7} J. COM-Jacobian inertia is independently compared with the printed closed entries, and recursive Newton–Euler bias with the Christoffel expression. These checks verify this manufactured calculation; they do not validate it as a golfer.

The tip speed falls from 23.976 to 7.924 m/s in this interval. The club kinetic energy falls from 31.241 to 5.419 J, while upper-arm kinetic energy rises from 9.360 to 34.268 J. This is a counterexample to inevitable proximal-to-distal energy concentration. Other initial states and torque histories can produce distal acceleration; the equations specify how to test them. Reversibility of ideal mechanics does not supply a universal preferred direction of energy transfer.

8.7.1 Drift, Available Control and the Task Output

The six-state dynamics are control-affine for this torque-input model:

$$\dot{x} = f(x) + G(x)\tau, \quad f = \begin{bmatrix} v \\ -M^{-1}(c + g) \end{bmatrix}, \quad G = \begin{bmatrix} 0 \\ M^{-1} \end{bmatrix}.$$

Adding a coordinate does not automatically increase a drift-control ratio. The ratio requires a stated output or state metric, input units, admissible torque set and regularization scale. Comparing f with an unspecified control vector conflates available control capacity with the particular torque a golfer happened to apply. A large instantaneous drift contribution cannot establish negligible future control authority or explain neuromuscular effort by itself.

For the rigid tip position $y = r_{\text{tip}}(q)$, where c_3 is replaced by L_3 in the COM-position construction,

$$\dot{y} = J_{\text{tip}}v, \quad \ddot{y} = -J_{\text{tip}}M^{-1}(c + g) + \dot{J}_{\text{tip}}v + J_{\text{tip}}M^{-1}\tau.$$

The geometric acceleration $\dot{J}_{\text{tip}}v$ matters as well as joint acceleration. Speed, direction, strike location and three-dimensional face orientation are distinct task quantities. This planar model represents only the first two and a planar tip location. It cannot justify claims about minimal clubface-control effort.

A useful comparison propagates a bounded torque perturbation over a specified time interval and measures its effect on the chosen output, with identical starting state and

mechanics. If the endpoint is a ball-contact event, changing the trajectory also changes contact time; evaluate the event consistently. Sensitivity to a joint angle may amplify both intended corrections and sensor or execution error. High sensitivity is therefore not synonymous with precision or easy control.

8.8 Comparison: Double vs. Triple Pendulum

The comparison must specify what the two-link model represents. An upper-arm/forearm chain has no club. An arm/club chain has a wrist hinge but lumps arm shape. A model of torso, arm and club has three segments but resolves different anatomy from this chapter. Counting links does not make these models interchangeable.

A controlled reduction of our model locks $q_3 = \bar{q}_3$ and writes

$$q = Nz + \bar{q}, \quad N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \bar{q} = (0, 0, \bar{q}_3)^T.$$

The reduced inertia is $M_r = N^T M N$, with projected bias, gravity and input, and $\dot{q} = N \dot{z}$. This preserves the full mass distribution and its kinetic energy. Deleting the club mass or its rotational inertia is not the same reduction. For a different locked coordinate, use its corresponding embedding.

Allowing the third coordinate enlarges the kinematic possibilities. An optimization over a truly larger feasible set cannot reduce the optimum only if every previous trajectory remains feasible, the objective is unchanged, and sufficient admissible torque can reproduce the old lock. Removing a mechanical lock while keeping tight torque bounds need not satisfy that inclusion. A free joint is not an actuator with unlimited holding capacity.

Comparisons should therefore match initial energy and state mapping, body parameters, input and power limits, duration or contact event, and task objective. Report both attainable speed and strike constraints. A universal percentage improvement, fixed mph table or DCR increase cannot be deduced from link count.

8.9 Energy Redistribution in the Triple Pendulum

Definition 8.6. Segment Energy and Joint Power

For segment i ,

$$T_i = \frac{1}{2} m_i \|J_i v\|^2 + \frac{1}{2} I_i \omega_i^2, \quad \omega_i = \sum_{j \leq i} v_j.$$

Using only $\frac{1}{2} I_i v_i^2$ omits COM translation and substitutes relative for absolute rotation. The correct segment energies sum to T ; coordinate cross terms in $\frac{1}{2} v^T M v$ are not additional stores to assign independently to bodies.

At an ideal joint, the connected bodies share the velocity v_P of the joint point. A reaction force F can supply power $F \cdot v_P$ to one body and $-F \cdot v_P$ to the other. These powers cancel for the complete chain but can substantially change individual segment energies. Equal and opposite joint couples have total power $\tau_i(\omega_i - \omega_{i-1}) = \tau_i v_i$, with the fixed base rate zero. A

joint force can thus transfer substantial power to the club while the local wrist couple also adds or absorbs power. These contributions are compatible, not competing descriptions of a single quantity.

To identify a particular transfer, compute each body's external and intersegment force and couple power and include gravitational and elastic terms in the chosen balance. A sequence of angular-speed peaks alone does not identify the transfer path, amount of work or actuator responsible. Nor does one body's decrease in rotational kinetic energy prove its muscles actively brake to accelerate another body.

8.9.1 When a Distal Segment Can Move Faster

For a single rotation with an identified rotational energy E_{rot} ,

$$|\omega| = \sqrt{\frac{2E_{\text{rot}}}{I}}. \quad (8.9)$$

This explains how a smaller inertia can support a larger rate at the same allocated rotational energy. In a moving chain, however, that allocation is an outcome of dynamics, and total segment energy also includes translation. Neither the equal-energy premise nor monotonic distal acceleration follows from the formula. Clubhead linear speed further depends on the vector sum of all upstream velocity contributions.

8.9.2 What Golf Research Adds

Sprigings and Mackenzie used an optimized planar torso–arm–club model with bounded muscle-torque generators to compare release strategies. Their reported advantage from delaying release was modest, and active wrist torque after release was advantageous. They also reported substantial power transfer into the club through wrist-joint forces. This model result illustrates why transferred power and local actuator power must be assessed together; it does not validate our upper-arm/forearm/club parameters or prescribe an individual golfer's release time. [Sprigings and Mackenzie, 2002]

Nesbit's motion-driven three-dimensional analysis examined 85 amateur golfers using a full-body model and flexible club. The study reported marked individual differences and identified wrist motion, hand trajectory and work as relevant to club motion. Its modeled forces and torques depend on measured kinematics and mechanical assumptions; they are not direct measurements of separate muscle forces or the result of randomly assigning release strategies. Its richer model also exposes quantities that a planar hinge cannot resolve, including clubface orientation. [Nesbit, 2005]

The common lesson is methodological. First define the mechanical system and power ports. Then ask which observed quantities constrain it, which counterfactuals it can support, and which intervention would distinguish alternative explanations. A passive power-transfer pathway does not make the whole golfer passive. Conversely, active control can exploit coupled dynamics without supplying all of a distal segment's instantaneous power locally.

8.10 Summary: The Triple Pendulum and the Kinetic Chain

Three links let us resolve an additional relative motion. A consistent model connects mass distribution to coupled acceleration, joint forces to segment power, torque to mechanical work, and the resulting configuration and rates to the club's task-space motion. Release changes an admissible motion or a force law; ideal impulse-free removal preserves the instantaneous state and energy. Whether subsequent motion helps the strike remains a question for the specified state, inputs and task.

The investigation should track preparation, energy supply, timing, material storage, feedback and strike constraints together. None is reducible to a count of links or an instantaneous drift ratio. The next step toward a golfer can add a moving base, three-dimensional wrist and shoulder motion, both arms, flexible tissues and contact. A spine connecting pelvis to shoulders is a serial connection; closed loops arise when an additional path reconnects bodies, as when both hands couple the arms through a shared club. The topology and constraint forces must be modeled explicitly.

Exercises 8.1. Chapter Exercises and Worked Answers

1. Release Timing and the Comparison Being Made

Explain why the statement “later release gives greater club speed at the same effort” is not a consequence of adding a third link. State what would make a comparison meaningful.

Worked answer. Removing a stationary ideal lock with no impulse preserves q , v and mechanical energy at that instant. Any benefit must come from the state reached before removal and the subsequent dynamics. Equal subjective effort does not specify equal joint work, equal torque bounds or equal initial energy. Compare the same bodies, state mapping, input limits, initial energy and strike objective; report the resulting torque histories, work, duration and club output. Some states benefit from a given timing change and others do not. The chapter's release example initially accelerates the relative wrist coordinate in the negative direction.

2. Complete Inertia With Different Body Parameters

Set $m = (3, 1, 0.2)$ kg, $L = (0.4, 0.3, 0.5)$ m, $I = (0.1, 0.02, 0.01)$ kg m², $c_i = L_i/2$, $q_2 = 90^\circ$ and $q_3 = 100^\circ$. Compute the three diagonal entries and M_{23} . Explain the difference between a mass-matrix entry and an acceleration response.

Worked answer. Substitution gives $M_{11} = 0.450398$, $M_{22} = 0.077791$, $M_{33} = 0.022500$ and $M_{23} = 0.019895$ kg m². The first entry includes $I_1 + I_2 + I_3$, not just I_1 ; the second includes $I_2 + I_3$. At fixed state, freely responding acceleration increments satisfy $\delta\ddot{q} = M^{-1}\delta\tau$. Holding other accelerations fixed instead requires their reaction torques. The size of M_{23} by itself cannot determine the sign or size of wrist acceleration under an arbitrary torque vector.

3. Differentiate the Actual Coupling

For the worked-example parameters, estimate $\partial M_{23}/\partial q_2$ from $q_2 = 89^\circ$ and 91° at fixed $q_3 = 100^\circ$. Compute c_3 at the initial state of the counterfactual example.

Worked answer. Both values of $M_{23} = d + \gamma \cos q_3$ are identical. Their difference divided by $2\pi/180$ radians is zero. Multiplying that derivative by any velocity product still gives zero. The nonzero complete component is instead $c_3 = \beta \sin(q_2 + q_3)v_1^2 + \gamma \sin q_3(v_1 + v_2)^2 = 15.127 \text{ N m}$ at $(50, 80, 100)^\circ$ and $(12, 20, 25) \text{ rad/s}$. To compute acceleration, solve all three coupled equations, including gravity and the other bias components. Comparing this single generalized component with an unsupported “typical wrist torque” would not establish passive dominance.

4. Integrate Before Interpreting the Whip

Reproduce the 50 ms zero-torque continuation. Report relative and absolute club rates and tip speed at the final time. Compare a constant-initial-acceleration approximation with the integration and explain the difference.

Worked answer. The final relative rate is $v_3 = -33.881 \text{ rad/s}$, the absolute club rate is $\omega_3 = 15.120 \text{ rad/s}$, and the tip speed is 7.924 m/s . The initial absolute club rate was 57 rad/s . On the specified 0.1 ms output grid, its maximum is at the initial sample, not at the end. A quadratic Taylor step must use $q(0) + v(0)\Delta t + \frac{1}{2}\ddot{q}(0)\Delta t^2$ entirely in radians. For q_3 it predicts approximately 73.421° , compared with the integrated 85.308° . The changing configuration, velocity-dependent bias and coupled inertia make the initial acceleration a poor constant approximation over this interval. Numerical refinement and energy checks address integration error; they do not remove uncertainty in the physical model.

5. Track Segment Energy and Whole-System Balance

Compute segment kinetic energies at 0, 25 and 50 ms. Does this example concentrate energy in the club? Does the sum of kinetic energies have to stay constant?

Worked answer. In upper-arm, forearm, club order, the kinetic energies in joules are approximately $(9.360, 18.640, 31.241)$, $(19.639, 23.758, 13.886)$ and $(34.268, 15.457, 5.419)$. The club fraction of total kinetic energy decreases, so this example does not show obligatory distal concentration. Kinetic energy totals change from 59.241 to 57.283 to 55.144 J because gravitational potential changes. The sum $T + V$ is conserved within the reported numerical tolerance. Include moving-COM energy and absolute segment rotation; $\frac{1}{2}I_i v_i^2$ does not reproduce either ledger.

6. What a Video Can and Cannot Establish

Design a study of wrist-release timing and club speed. Specify a measurable event, coordinate convention and comparison, and identify what ordinary slow-motion video cannot resolve.

Worked answer. One possible event is the first crossing of a declared three-dimensional arm–club angle during the downswing, with an explicit rule for multiple crossings. This is a kinematic event, not proof that a lock vanished or muscle torque became zero. Estimate it from calibrated synchronized observations, report frame timing and reconstruction uncertainty, and align it to a separately measured strike event. Measure clubhead speed and strike outcomes independently. Compare repeated trials within golfers or a controlled intervention with matched equipment and a defined task; model individual variation rather than inferring a universal rule from one professional and one amateur. A single projected video cannot identify a three-dimensional joint angle, muscle force, elastic energy store or the causal effect of timing. Inverse dynamics additionally needs inertial parameters and external/contact forces; estimating separate muscle contributions needs further assumptions or measurements.

Chapter 9

Parallel Mechanisms and Loop Constraints

The golf swing combines articulated chains, closed contact paths, muscle-generated loads, and elastic deformation. A useful model must say which connections it represents before it counts freedoms or interprets forces. The two-handed grip closes a path through the arms and club. Two supported feet close another through the legs and ground. The spine connects pelvis and thorax along an articulated path; that connection alone does not form a loop.

The central question is how these connections change the motion produced by a specified set of inputs, and what measurements can distinguish different ways of producing the same motion. Geometry defines feasible motion. Inertia and applied loads determine its acceleration. Contact reactions enforce the declared contact model. Muscle and tissue mechanics determine available forces and energy. None of those roles can be inferred merely by calling the golfer a parallel mechanism.

9.1 Draw the Actual Mechanical Graph

Represent each rigid segment as a graph vertex and each joint or maintained contact as an edge. A serial chain has one unbranched path. An open articulated tree can branch without containing a cycle. A closed chain contains a cycle: there must be a distinct return path, not an instruction to go back along the same spine or arm. Serial models can have strongly coupled dynamics even when their joint coordinates are independently selectable.

For a fixed-torso upper-body model, trace the cycle as thorax, left upper arm, left forearm/hand, club, right hand/forearm, right upper arm, thorax. Hand offsets, wrist joints, and shoulder-girdle motion can be resolved further when needed. For the lower body, trace ground, left foot, left leg, pelvis, right leg, right foot, ground. The spine links the pelvis to the thorax between these structures. Removing one hand contact opens the upper loop; removing one foot contact changes the lower contact structure. A raised heel can change the admissible contact wrench without completely detaching the foot.

A pelvis–spine–thorax sequence is an open path. Defining a relative angle between pelvis and thorax does not add a physical return connection. Soft tissue can introduce additional force paths, but a tendon spanning joints is usually represented by a length-dependent force law, not an inextensible closure constraint. Its finite stiffness must not be silently converted into a rigid kinematic restriction.

The distinction agrees with the humanoid examples in the [Modern Robotics constrained-dynamics treatment](#). The same geometry can be represented by independent joint coordinates plus loop constraints or by a larger set of segment poses plus joint constraints. Mixing those two counts subtracts the same restriction twice.

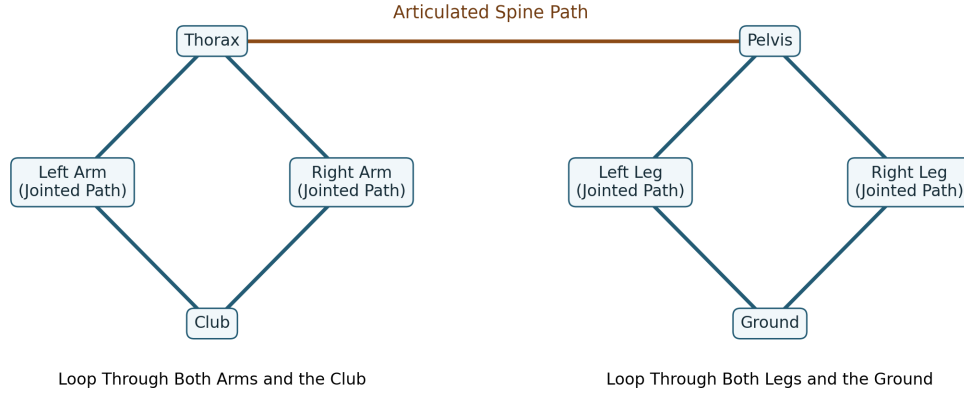


Figure 9.1: Actual Closed Paths and the Connecting Spine Path. Arms and Legs Represent Jointed Paths; Contact at Both Hands or Both Feet Closes Each Cycle.

9.2 Mobility and Independent Constraints

Let $q \in \mathbb{R}^n$ be local coordinates for an open-tree model. Suppose the additional, time-independent holonomic closures are

$$\Phi(q) = 0, \quad J_c(q) = D_q \Phi(q). \quad (9.1)$$

If J_c has constant rank r near a feasible configuration, the local configuration manifold has dimension $n - r$. This is a regular-point statement. At a singular point, the linearized null space can include directions that do not extend to a finite feasible motion; higher-order closure conditions matter. Neither an actuator being passive nor a muscle being inactive adds a kinematic constraint by itself.

For a connected linkage with N bodies including one grounded body, J joints, and f_i freedoms at joint i , the generic Gruebler count is

$$d_{\text{generic}} = s(N - 1) - \sum_{i=1}^J (s - f_i), \quad s = 3 \text{ (planar)}, \quad s = 6 \text{ (spatial)}. \quad (9.2)$$

It assumes that the counted joint restrictions are independent. In a planar linkage made only of one-freedom revolute or prismatic joints this becomes $d_{\text{generic}} = 3(N - 1) - 2J$. A four-bar with four bodies including ground and four revolute joints therefore has one generic freedom. An unactuated revolute joint still has one freedom. A fixed contact removes freedoms; it is not a positive correction called a passive constraint.

A negative generic count is a reason to inspect the model, independence, and geometry. It is not proof that the actual mechanism is immobile or that tissue elasticity is the only reason a human can move. Omitted arm or leg bodies, counting joint freedoms as joints, and mixing planar and spatial restrictions can all produce meaningless results. Special rigid mechanisms can move because some restrictions are dependent.

9.2.1 A Stewart-Platform Count With All the Parts Included

Consider a six-leg UPS platform: each leg has a universal joint with two freedoms, a prismatic joint with one, and a spherical joint with three. Each leg contains two telescoping rigid members. Including base and moving platform, $N = 14$, $J = 18$, and $\sum_i f_i = 6(2 + 1 + 3) = 36$. Thus

$$d_{\text{UPS}} = 6(14 - 1 - 18) + 36 = 6. \quad (9.3)$$

Away from singularities, these six freedoms can describe the moving platform's pose. Locking the six independent prismatic lengths fixes that pose locally. Actual platforms may use different joint architectures; their counts must follow their own joints rather than the word hexapod.

In particular, two platforms connected by six fixed-length rods with spherical joints at both ends give $N = 8$, $J = 12$, and a count of six. Those six freedoms are the axial spins of the rods, not six free motions of the platform. The six fixed distance conditions generically fix the platform pose. This example shows why total mechanism mobility, internal passive motion, and output mobility must be distinguished. The [joint-counting derivation](#) provides the general starting point; the physical interpretation still requires examining the particular linkage.

9.3 Grip Closure With Separated Hand Frames

A rigid two-handed grip fixes each hand frame relative to a different location and orientation on the club. It does not require the hands to coincide. Let $T_L(q)$ and $T_R(q)$ be hand poses in a common world frame, T_C the club pose, and H_L, H_R the constant poses of the hand frames in club coordinates. Then

$$T_L = T_C H_L, \quad T_R = T_C H_R, \quad T_L^{-1} T_R = H_L^{-1} H_R. \quad (9.4)$$

The last relation supplies six local independent restrictions at a regular spatial grasp. Pose coordinates require a nonsingular local representation; three position equations alone do not enforce relative orientation. Grip compliance, rolling, or slip replaces some of these ideal equalities with an appropriate contact model.

For two seven-coordinate arms attached to a fixed torso, there are 14 arm coordinates before the grip closes. Seven is a modeling choice; it commonly resolves shoulder rotation, elbow flexion, forearm rotation, and wrist motion. It is not a claim that the wrist alone supplies all three distal rotations. With six independent relative-pose restrictions, the assembly has $14 - 6 = 8$ local freedoms. Equivalently, adding a free club contributes six coordinates and fixing each hand to it adds twelve independent restrictions:

$$d = 14 + 6 - 12 = 8. \quad (9.5)$$

The fixed torso was already assumed. Subtracting additional pelvis or spine restrictions from this upper-body count is double counting. A whole-body model must instead add its pelvis, trunk, legs, and contact coordinates explicitly. Eight kinematic freedoms also do not mean eight independently available neural commands or eight controllable directions over a finite time interval.

9.3.1 A Planar Example You Can Differentiate

For a deliberately simpler coincident-point grip, fix shoulder bases at $b_L = (-1, 0)^T$ and $b_R = (1, 0)^T$, in meters. Give each arm two unit-length links and relative elbow coordinates. Its hand position is

$$p_i = b_i + \begin{bmatrix} \cos q_{i1} + \cos(q_{i1} + q_{i2}) \\ \sin q_{i1} + \sin(q_{i1} + q_{i2}) \end{bmatrix}, \quad i = L, R. \quad (9.6)$$

The point closure is $\Phi = p_L - p_R = 0$, with $J_c = [J_L \ -J_R]$ and $J_i = D_{q_i} p_i$. The difference is essential: fixing the sum of hand positions would allow the hands to move apart while their midpoint stayed fixed.

At $q_L = (0, \pi/2)$ and $q_R = (\pi, -\pi/2)$, both hands are at $(0, 1)$ and

$$J_L = \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}, \quad J_R = \begin{bmatrix} -1 & -1 \\ -1 & 0 \end{bmatrix}, \quad J_c = \begin{bmatrix} -1 & -1 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}. \quad (9.7)$$

Here J_c has rank two, leaving two freedoms. The velocity $\dot{q} = (1, -1, -1, 1)^T$ satisfies $J_c \dot{q} = 0$, yet both hands move upward at 1 meter per second. Preserving closure does not mean holding the club still. A separated rigid club requires its pose and attachment offsets as above; the point example is a lower-fidelity model, not an anatomical measurement.

9.4 Three Different Null Spaces

The constraint null space $\ker J_c$ contains compatible velocities. For a stationary holonomic closure,

$$J_c \dot{q} = 0, \quad J_c \ddot{q} + \dot{J}_c \dot{q} = 0. \quad (9.8)$$

The second equation includes the change in tangent directions along the motion. Dropping $\dot{J}_c \dot{q}$ can remove required normal acceleration even when the velocity satisfies closure perfectly.

A task has a different Jacobian. Let $y = \psi(q)$ describe the chosen output and $J_y = D_q \psi$. Motions that preserve both closure and this output lie in $\ker J_c \cap \ker J_y$. If the columns of Z form a basis of $\ker J_c$, then $\dot{q} = Z\nu$ and $\dot{y} = J_y Z\nu$. The task-preserving dimension is

$$(n - r) - \text{rank}(J_y Z). \quad (9.9)$$

For the regular spatial arm example, an independent six-dimensional club-pose task leaves two posture freedoms out of eight. Holding only club-center position leaves orientation unspecified. A face-angle change can preserve that position task while changing the impact task. It is not invisible to every meaningful output, and kinematic freedom does not establish negligible actuation cost.

For a feasible desired task velocity, a weighted minimum-velocity construction uses $H = J_y Z$ and a positive-definite coordinate weight W :

$$\nu_* = W^{-1} H^T (H W^{-1} H^T)^{-1} \dot{y}_{\text{des}}. \quad (9.10)$$

This formula requires full row rank of H . It minimizes $\frac{1}{2} \nu^T W \nu$ subject to $H \nu = \dot{y}_{\text{des}}$. It does not minimize joint torque, muscle activation, or metabolic cost. Rank-deficient or

bounded problems need a suitable pseudoinverse or constrained optimization, with explicit handling of infeasible requests.

Finally, a grasp map G maps contact-load coordinates to a resultant club wrench. Its null space $\ker G$ contains self-equilibrated load changes. Those are forces, not joint velocities. Even the multiplier null space $\ker J_c^T$ is a different object: it describes redundant multiplier representations that produce the same generalized reaction. When J_c has independent rows, $\ker J_c^T = \{0\}$, although a bilateral grip can still have unresolved internal loads when applied inputs are unknown.

9.5 Constraint Reactions and the Forward Dynamics

Use minimal open-tree coordinates with $M(q)$ positive definite, input map $B(q)$, and input vector u . Put velocity-dependent inertial terms and gravity in $h(q, \dot{q})$. Keep passive tissue loads and other applied loads explicitly in $p(q, \dot{q}, z)$, where z denotes any additional constitutive states. For a fixed, ideal contact mode,

$$M\ddot{q} + h = Bu + p + J_c^T \lambda. \quad (9.11)$$

Do not put h on the left and then subtract it again inside an additional right-hand-side drift torque. Different bookkeeping conventions are possible, but each physical term must occur exactly once.

Set $a_0 = M^{-1}(Bu + p - h)$ and $A = J_c M^{-1} J_c^T$. If J_c has independent rows, A is positive definite. Substitution into the acceleration constraint yields

$$\lambda = -A^{-1}(J_c a_0 + \dot{J}_c \dot{q}), \quad \ddot{q} = a_0 - M^{-1} J_c^T A^{-1} (J_c a_0 + \dot{J}_c \dot{q}). \quad (9.12)$$

The acceleration correction depends on the mass matrix as well as the geometry. The matrix $I - M^{-1} J_c^T A^{-1} J_c$ acts on velocities or acceleration increments; its transpose acts on generalized-force covectors. They cannot be interchanged merely because both are called projections.

This solution is unique for the specified model, state, applied inputs, and regular active equality constraints. Redundant constraints can leave multiplier coordinates nonunique. Unilateral contacts can be infeasible for the computed reaction; frictional contact can require a different mode or contact law. Calling a problem forward dynamics does not remove those conditions.

9.5.1 Reactions Depend on Applied Muscle Loads

For a two-coordinate illustration take $M = \text{diag}(2, 1)$, $J_c = [1 \ -1]$, $h = p = 0$, and $B = I$. The two coordinates must accelerate together. With $u = (1, 0)^T$, the solution is

$$\ddot{q} = (1/3, 1/3)^T, \quad \lambda = -1/3. \quad (9.13)$$

With $u = (2, -1)^T$, the acceleration is unchanged and $\lambda = -4/3$. An additional applied load in $\text{range}(J_c^T)$ is balanced by a changed reaction. Geometry determines the admissible reaction directions, not their magnitudes independently of actuation. A reaction may be zero or large depending on the state and inputs; the existence or number of loops alone gives no force scale.

In inverse dynamics, the observed motion supplies the balance

$$[B \ J_c^T] \begin{bmatrix} u \\ \lambda \end{bmatrix} = M\ddot{q} + h - p. \quad (9.14)$$

Uniqueness depends on the rank of this identification map and on additional measurements and restrictions. Joint-level net torques can be ambiguous as well as individual muscle forces and hand loads. Known segment inertias and motion fix inertial demands; they do not generally separate all unknown sources.

9.6 Club Wrenches and Bilateral Load Sharing

For a rigid club, express all vectors in a common inertial frame and take moments about its center of mass C . Let r_i point from C to hand contact i , f_i be its force on the club, and m_i its contact couple. Then

$$\begin{aligned} m_c a_C &= f_L + f_R + m_c g + f_{\text{other}}, \\ I_C \dot{\omega} + \omega \times (I_C \omega) &= r_L \times f_L + r_R \times f_R + m_L + m_R + m_{\text{other}, C}. \end{aligned} \quad (9.15)$$

The current rotational inertia tensor I_C is expressed in that same frame. Other loads include aerodynamic forces or ball contact when applicable. The hand points of a rotating club generally have different linear accelerations: $a_i = a_C + \dot{\omega} \times r_i + \omega \times (\omega \times r_i)$. Sharing a rigid body does not make those accelerations equal.

Stack each hand load as $w_i = (m_i, f_i)$ and transport its moment to C before summing. The resulting 6×12 grasp map G has rank six if both contacts are modeled as unrestricted full wrenches. For a compatible required wrench,

$$w = G^+ w_{\text{req}} + N_G \eta, \quad \text{range}(N_G) = \ker G, \quad (9.16)$$

so there are six unconstrained load-sharing parameters before friction, unilateral pressure, moment capacity, and actuator limits are imposed. Those physical restrictions can reduce the feasible set or make it empty; an arbitrary algebraic split need not be realizable by a hand.

If each contact transmits only a point force, the map is 6×6 . For two distinct points it generically has rank five, because equal opposite forces along the joining line produce zero resultant wrench. The null dimension is then one, and arbitrary required spatial wrenches cannot be supported. Grip models that allow contact couples and distributed pressure have different capabilities.

A simple scalar illustration is $\tau_L + \tau_R = 10$ N m, with both torques expressed about the same axis and reference point after all other terms are accounted for. The pairs (4, 6) and (7, 3) satisfy the same balance. A minimum-squared-torque choice gives (5, 5), but that is an optimization assumption, not an observation of how a golfer shared the load. Relative EMG amplitudes do not by themselves supply the missing torque ratio. Calibrated instrumented grips, muscle models, or additional experimental perturbations can constrain the inference.

9.6.1 Force Transmission, Power, and Internal Loading

A twist is a six-component angular/linear velocity representation, not generally a unit vector. With the matching wrench ordering, $V = (\omega, v)$ and $w = (m, f)$ pair to give power $w^T V =$

$m \cdot \omega + f \cdot v$. Both must refer to compatible points and frames. A change of point changes the moment and linear velocity together, preserving power.

For an ideal rigid internal attachment, the two bodies have the same contact twist and experience opposite wrenches. Their contact powers cancel when both bodies are included in the system. Each body separately can receive or lose power through that attachment. Thus the grip can transfer energy to the club while ideal closure reactions do zero total work on the golfer-plus-club system. In generalized coordinates this cancellation is

$$\dot{q}^T J_c^T \lambda = \lambda^T J_c \dot{q} = 0. \quad (9.17)$$

A constraint can redirect acceleration while doing no total work. An internal load need not produce mechanical work merely because it stresses tissue. Muscle maintenance can have a metabolic cost even when modeled joint power is zero; local tissue deformation and injury outcomes require their own measurements and constitutive models.

9.7 Ground Contact: Wrench, Momentum, and Energy

Ground reaction is an external contact wrench on the golfer-plus-club system. Its value depends on the motion, geometry, muscle loading, and contact conditions. It is not independent of muscle action, nor is it an additional muscle force. For a simple unilateral contact with normal gap ϕ_n , the ideal conditions include $\phi_n \geq 0$, $\lambda_n \geq 0$, and $\phi_n \lambda_n = 0$. A sticking Coulomb contact also requires $\|f_t\| \leq \mu \lambda_n$. Distributed foot contact has moment and center-of-pressure restrictions in addition to this force cone. Slip, lift-off, shoe deformation, and moving support change the equations.

For the center of mass C of the selected whole system and angular momentum H_C , the external balances are

$$\begin{aligned} m_{\text{tot}} a_C &= m_{\text{tot}} g + \sum_i f_i + f_{\text{other}}, \\ \dot{H}_C &= \sum_i ((r_i - C) \times f_i + m_i) + m_{\text{other}, C}. \end{aligned} \quad (9.18)$$

Here r_i is the contact position in world coordinates, whereas the preceding club balance used offsets from the club center. Uniform gravity contributes no net moment about the whole-system center of mass. These equations constrain whole-system momentum, not the pelvis acceleration alone. Hip, spine, and joint reactions exchange angular momentum among segments. Even with zero total external moment, one segment can rotate while others counter-rotate. A free moment measured at a foot is therefore not a complete explanation of pelvis motion or proof of a universally passive shoulder.

9.7.1 A Force Plate's Reference Point Matters

Declare a right-handed frame with z upward, and let (F, M_O) be the wrench of the ground on the golfer measured about an origin O in the contact plane. For $F_z \neq 0$, a center of pressure $r = (x, y, 0)$ is defined so that the remaining moment there has only a vertical component. Since $M_O = r \times F + T_z e_z$,

$$x = -\frac{M_y}{F_z}, \quad y = \frac{M_x}{F_z}, \quad T_z = M_z - (xF_y - yF_x). \quad (9.19)$$

Thus the plate-origin yaw moment M_z is not generally the free moment T_z . If the contact plane is at height h relative to O , use $x = (hF_x - M_y)/F_z$ and $y = (M_x + hF_y)/F_z$. Vendor outputs may use the opposite force-on-plate sign or different axes; transform the full wrench before applying these formulas. Near zero normal load the division is ill-conditioned, so state a justified contact threshold and report unloaded intervals separately.

For a numerical check, a force $F = (20, 30, 800)$ N applied at $r = (0.2, -0.1, 0)$ m with free moment $T_z = 5$ N m gives $M_O = (-80, -160, 13)$ N m. The formulas recover $x = 0.2$ m, $y = -0.1$ m, and $T_z = 13 - (6 + 2) = 5$ N m. Calling the measured 13 N m the free moment would conflate the horizontal-force lever arm with the residual couple.

COP is pressure-weighted position, not the center of mass or a direct measure of transferred body mass. A changing foot-load share can move COP while the center of mass barely moves. With two coplanar plates, transform the wrenches to one frame and origin before combining them. Separate plates supply each foot's boundary wrench; one combined plate generally leaves bilateral sharing unresolved. Six-axis measurements are wrench components, not six motion DOFs. Calibration, cross-talk, sampling, synchronization, and filtering affect accuracy; a universal instrument-accuracy percentage cannot substitute for calibration.

With each foot wrench known, segment poses and inertial parameters allow a Newton–Euler recursion from foot to shank to thigh. Transfer both force and moment, including their lever arms, at each step. This gives model-conditioned net joint loads when other external loads and joint assumptions are specified. It does not uniquely identify individual muscle forces or tissue stress.

9.7.2 Support Can Enable Work Without Supplying Energy

A fully stationary rigid foot has zero velocity at its ground-contact points and zero angular velocity. In this idealization, the ground wrench has zero mechanical power, although its force and moment can change the whole-system momentum. Muscle work can increase body and club energy while ground support prevents an incompatible foot motion. The resultant external force dotted with center-of-mass velocity describes center-of-mass translational-energy change; it is not the contact power of that force on a deformable or articulated system.

For time-independent elastic potentials and ideal stationary contacts, an energy balance has the form

$$\frac{d}{dt}(T + V_g + V_e) = \dot{q}^T B u + P_{\text{other}} - P_{\text{diss}}, \quad P_{\text{diss}} \geq 0. \quad (9.20)$$

Moving support, slipping friction, changing elastic rest states, and contact impacts require their corresponding boundary or constitutive power terms. Neither a large ground force nor an early force peak establishes the source, amount, or direction of energy transferred to the club. A segment power audit must state the system boundaries, force points, velocities, and sign conventions.

There is no single force-plate curve in this model that defines a good golfer. Lead-foot and combined forces, normalization by body weight, club choice, swing phase, and target all change a comparison. A claim that the best peak must occur exactly at impact needs an identified study and outcome definition, not an illustrative sketch. The chapter's equations specify what must balance; they do not provide a coaching threshold or an injury-risk cutoff.

9.8 X-Factor, Compliance, and Muscle State

In a planar illustration, define θ_p and θ_t as pelvis and thorax angles about the same axis. Then $\Delta = \theta_t - \theta_p$ is a relative rotation, not a new loop-closure equation. For $\theta_p = 60$ degrees and $\theta_t = 20$ degrees, $\Delta = -40$ degrees under this convention. In three dimensions, use the relative orientation $R_p^T R_t$ and declare its reported angle convention. Subtracting two Euler angles or projected shoulder-line angles need not measure the same quantity.

Brown, Selbie, and Wallace’s X-factor methods study examines this measurement issue. A reported intersegment angle is not a direct measurement of lumbar rotation, disc compression, tissue strain, or stored elastic energy. Distribution among spinal levels, rib-cage/shoulder motion, posture, and muscle activation matter. An angle associated with performance is also not by itself evidence that increasing that angle improves an individual swing or is safe for that individual.

For a clearly declared toy torsional spring,

$$V_e = \frac{1}{2}k(\Delta - \Delta_0)^2, \quad Q_t = -k(\Delta - \Delta_0), \quad Q_p = +k(\Delta - \Delta_0). \quad (9.21)$$

The two torques are opposite, and their combined power is $-k(\Delta - \Delta_0)\dot{\Delta} = -\dot{V}_e$ when k and Δ_0 are fixed. Doubling displacement quadruples this model’s energy only if the stiffness and rest angle stay the same. It does not establish a universal human tissue law or prove that the stored energy reaches the club rather than another segment or dissipation. If stiffness is actively changed, the energy derivative also contains $\frac{1}{2}\dot{k}(\Delta - \Delta_0)^2$; changing a rest angle adds another term. An impedance schedule is not energetically free by definition.

Joint range of motion, tangent stiffness, damping, geometric constraint rank, and neural feedback are distinct. Finite stiffness resists displacement without reducing the exact kinematic dimension. It can make some motions small over a specified loading range and time scale. A rigid-constraint approximation must be justified against those scales, contact margins, and output errors.

Muscle excitation, activation, fiber length, shortening velocity, and tendon stretch must likewise be distinguished. Eccentric contraction means active lengthening; active shortening is concentric. A tendon can recoil while its muscle fibers behave differently. Treating all release-phase tissue forces as passive because club speed is high is not justified. For a muscle model, $\tau = R(q)F_m(q, \dot{q}, z)$ maps muscle forces through moment arms; multiplying moment arms by dimensionless activation alone does not produce torque.

9.9 Constrained Drift and Forward Counterfactuals

At a fixed admissible state and contact mode, reaction elimination gives

$$\begin{aligned} \ddot{q} &= f_c(q, \dot{q}, z) + G_c(q)u, \\ G_c &= (I - M^{-1}J_c^T A^{-1}J_c)M^{-1}B. \end{aligned} \quad (9.22)$$

The input-independent term includes gravity, velocity-dependent dynamics, retained passive forces, and the geometric acceleration correction. It depends on the declared inputs and state variables. This is the useful local affine structure: closure reshapes both drift and the effect of an input. Removing u does not remove the reactions needed to satisfy the remaining contacts.

Computing these terms requires a model, inertial parameters, calibrated kinematics, and any relevant tissue or contact state. Motion capture alone does not measure masses, inertias, tendon preload, or the chosen decomposition. Although gravitational force can be evaluated before solving for reactions, the resulting constrained gravitational acceleration differs from free fall. Observable motion and model-computable force terms are different evidential categories.

For a particle observed in a rotating frame, the centrifugal and Coriolis fictitious forces are

$$F_{\text{cf}} = -m\omega \times (\omega \times r), \quad F_{\text{cor}} = -2m\omega \times v_{\text{rel}}. \quad (9.23)$$

An accelerating origin and changing frame angular velocity introduce translational and Euler terms as well. These expressions cannot be added to inertial-frame loads without the matching frame transformation; that would double count the kinematics. Generalized velocity-dependent terms in an articulated model are not automatically identical to one particle's rotating-frame force.

For a hypothetical 3 kg point mass rotating at 15 rad/s on a 0.6 m radius, $m\omega^2 r = 405$ N. About that same center, however, the radial force has zero moment because $r \times F_{\text{cf}} = 0$. It has zero power on purely tangential relative motion. About another joint it can have a moment, but the perpendicular lever arm and vector directions must be computed. Multiplying 405 N by an arbitrary segment distance does not establish a shoulder torque.

Similarly, $F_{\text{cor}} \cdot v_{\text{rel}} = 0$: the Coriolis term is perpendicular to the relative velocity and is not dissipative resistance to it. For mutually perpendicular ω and v_{rel} , with magnitudes 10 rad/s and 8 m/s and mass 3 kg, its magnitude is 480 N. That conditional number is not an observed golf force. A 3 kg segment weighs 29.43 N in 9.81 m/s² gravity. Neither example establishes a universal ratio of inertial to muscle torque, and ratios of load magnitudes are not fractions of work or clubhead-speed causation.

A forward zero-torque counterfactual (forward ZTCF) branches from a specified initial state and changes a declared torque input or policy. It then evolves the state and recomputes compatible reactions, contact events, and retained passive forces. It must not reset the measured state at every time sample and call the resulting instantaneous accelerations a new trajectory. If the inferred baseline torques or contact model are uncertain, that uncertainty carries into the forward prediction. Forward simulation avoids solving for unspecified inputs during integration; it does not establish that its initial inputs, parameters, or biological interpretation are correct.

9.10 Kinematics, Singularities, and Feasible Control

When all compatible open-tree joint coordinates are known, their link poses can be computed recursively and the closure residual checked. If only some actuated coordinates are specified, passive coordinates and platform pose may require solving nonlinear closure equations. Multiple isolated assembly modes are not the same as continuous redundancy or a Jacobian singularity. Inverse kinematics asks for coordinates achieving a specified pose; its difficulty depends on the architecture. For a Stewart-type platform, computing leg lengths from a pose can be much simpler than computing pose from lengths. Constraint forces are not unknowns in a purely kinematic solve.

For local input coordinates s and output coordinates x , write differentiated closure as $A_x \dot{x} + B_s \dot{s} = 0$. In a square nonredundant mechanism, the [Gosselin–Angeles classification](#) distinguishes rank loss of the input-side matrix B_s (first kind), output-side matrix A_x

(second kind), and simultaneous loss (third kind). When A_x is singular, a nonzero output velocity may satisfy the linearized equations with locked inputs. Architecture singularity concerns geometric design that imposes persistent rank deficiency; it is not a name for any transient unusual posture.

Standing still at address does not demonstrate a singularity, because these ranks depend on geometry rather than whether velocity happens to be zero. A straight two-link planar arm has a singular endpoint-position Jacobian, but its elbow coordinate still exists. A mechanical stop, contact inequality, or a separately imposed lock changes the admissible motion in another way. Neither near extension nor a small singular value proves maximum impact efficiency or safe tissue loading. A first-order rank loss may require nonlinear analysis to determine actual finite self-motion.

Report which Jacobian is tested, its coordinate and unit scaling, the posture, and the numerical threshold. Conditioning of $J_y Z$ concerns task velocity; conditioning of $J_c M^{-1} J_c^T$ concerns reaction elimination; rank of $[B \ J_c^T]$ concerns inverse load identification. Their numerical values answer different questions. Local input rank also does not alone determine finite-time reachability, because drift, input bounds, time horizon, and contact transitions matter.

9.11 An Investigation That Connects the Pieces

The defensible research hypothesis is that preparation and changing geometry alter how feasible muscle inputs and stored energy produce the impact state. The two-handed grip and ground contacts are essential parts of that mapping. They can transmit loads, restrict relative motion, and alter the effectiveness of an input without supplying unlimited energy or determining a unique neural strategy. To investigate the hypothesis:

1. Define the outcome: clubhead velocity and face orientation at contact, strike location, ball launch, or dispersion. A club-center path alone is not a complete impact task.
2. Declare the graph, joint coordinates, contact modes, hand-frame offsets, inertias, muscle/input representation, and elastic states. Check closure, velocity, and acceleration residuals with measurement uncertainty.
3. Synchronize segment kinematics, separate foot wrenches, and any hand-load measurements. Identify which aggregate loads are determined and which bilateral or muscle distributions remain compatible with the data.
4. Audit linear and angular momentum separately from segment power, total energy, and tissue loading. Report sign conventions and reference points. An acceleration contribution, work contribution, and injury hypothesis must not share a percentage without an explicit derivation.
5. Compare a baseline trajectory with declared forward interventions on torque, timing, grip compliance, or contact. Preserve the same starting state when testing a late intervention; change it deliberately when testing preparation. Include uncertainty in unmeasured load sharing and parameters.
6. Test predictions on held-out trials and controlled perturbations where feasible. Similar observed kinematics do not prove identical forces, and a model that reproduces a swing does not identify the golfer's intention.

This approach can reveal when a change in grip loading mainly affects internal stress, when it changes club orientation, or when a different contact condition changes the admissible acceleration. It also allows a passive mechanism to be supported where the evidence supports it, while retaining the muscle work, feedback, and prior preparation needed to explain the entire swing. Chapter 11 develops the next question: how energy moves between the chosen segments and storage elements.

9.12 Exercises and Checks

1. Draw the upper and lower loops and the connecting spine path. Remove one grip edge. Which cycle disappears? Why does defining a pelvis–thorax relative angle not restore it?
2. Reproduce the UPS platform count and explain the six axial rod spins in the fixed-length SPS example. State which freedoms move the platform.
3. Differentiate the planar two-arm positions. Verify both closure and common upward hand motion for the stated configuration and velocity. Then compute the stacked closure/task rank when the hand point is fixed.
4. Solve the two-coordinate reaction example for both applied loads. Verify identical accelerations, different reactions, and zero total ideal constraint power for a common admissible velocity.
5. Construct the two-point-force grasp map. Show that opposite forces along the joining line lie in its null space. Explain why opposite forces in an arbitrary direction generally leave a couple.
6. Recover COP and free moment from the numerical wrench. Repeat after moving the reference origin, transporting the moment consistently.
7. Derive the complete energy derivative of the torsional spring when both stiffness and rest angle vary. Identify which terms require a source or sink beyond the fixed-spring mechanical exchange.
8. Specify an experiment that distinguishes a grip-load change preserving the resultant club wrench from one that changes club orientation. List the measurements needed; explain why motion alone and raw EMG ratios do not resolve every candidate load distribution.

Chapter 10

Passive Stabilization in Parallel Loops

10.1 What Needs to Be Stabilized?

A golfer must tolerate variability while delivering an accurate clubhead state at impact. Geometry, muscle activation, tissue elasticity, and feedback all affect that task. The useful question is how these mechanisms share the work of rejecting a specified disturbance, at what energetic and loading cost. Neither maximum stiffness nor maximum compliance is a universal answer.

Bosch's attractor-fluctuation framework offers a way to discuss robust features of movement [Bosch \[2020\]](#). Its application to golf is a hypothesis to operationalize. A repeatable impact configuration is not automatically an attractor of an autonomous mechanical system, and an energy minimum is not automatically evidence of reduced muscle activity.

10.2 Constraints, Impedance, and Activation

For ideal stationary loop constraints $\phi(q) = 0$, let $A = \partial\phi/\partial q$. The equations are

$$M(q)\ddot{q} + h(q, \dot{q}) = Bu + A^T\lambda, \quad A\dot{q} = 0, \quad A\ddot{q} + \dot{A}\dot{q} = 0. \quad (10.1)$$

The generalized reaction $A^T\lambda$ is determined jointly by geometry, motion, applied forces, and the closure conditions. It depends on actuation even though it is not an independently commanded input. For these ideal stationary constraints, $\dot{q}^T A^T\lambda = 0$. Reactions can transfer energy between connected bodies while doing zero net work on the complete system. Real ground contacts also require friction, unilateral contact, and possible slip or separation. Pelvis–thorax separation is a measured kinematic relation, not by itself an extra rigid loop constraint.

For small perturbations around a fixed operating condition, a local model is

$$M\delta\ddot{q} + D\delta\dot{q} + K\delta q = B\delta u + J^T\delta w, \quad Z_q(s) = Ms^2 + Ds + K. \quad (10.2)$$

Here the local equation uses independent perturbation coordinates; its matrices are the corresponding reduced matrices when rigid loops are present. If q instead retains redundant loop coordinates, retain the linearized constraint and reaction equations. Constraint reduction is detailed below. Here Z_q maps displacement to generalized force. Mechanical impedance is also often defined relative to velocity, in which case it is $Z_q(s)/s$. State the convention. Along a swing these coefficients vary; delayed reflex feedback generally gives a frequency-dependent contribution rather than a constant viscous damper. Neural delay can reduce stability margins.

Intrinsic muscle response begins without waiting for a new neural command. That does not make activated muscle metabolically passive. Existing activation, short-range stiffness, force-length and force-velocity behavior, tendon compliance, and contraction history affect the response. They cannot all be represented by an activation-independent spring **Rack and Westbury** [1974]. The descending limb of the active force-length curve does not guarantee restoring stiffness, and a shortened muscle does not abruptly lose all force as soon as it is below optimal length.

10.3 How Muscle Stiffness Becomes Joint Stiffness

Let muscle length be $l_i(q)$, tension F_i , and $r_i = -\partial l_i / \partial q$ be its signed moment-arm vector. The generalized muscle force is $\tau = \sum_i r_i F_i$. For this quasistatic constitutive approximation, take $F_i = F_i(l_i)$ with activation and internal states held fixed, and no independent shortening-velocity variation. The restoring tangent stiffness is

$$K_q = -\frac{\partial \tau}{\partial q} = \sum_i \left[k_i r_i r_i^T - F_i \frac{\partial r_i}{\partial q} \right], \quad k_i = \frac{\partial F_i}{\partial l_i}. \quad (10.3)$$

The first term includes squared moment arms; without them a muscle stiffness in force per length cannot become joint stiffness in torque per angle. The second is a geometric or prestress contribution and need not be positive semidefinite. This local formula needs modification for tendon states, muscle coupling, history-dependent response, and neural feedback. During motion, changing geometry can also change shortening velocity at fixed joint velocity. A force-velocity law then contributes additional derivatives; it cannot be included merely by calling k_i the force-length slope.

Balanced antagonist forces can yield small net torque while both muscles carry large tension. Co-contraction commonly raises intrinsic joint stiffness and loading; it does not define a low-impedance mode. Finite stiffness changes compliance, not the rank of the kinematic Jacobian or the number of exact degrees of freedom. Treating a stiff direction as locked is an approximation that needs a time scale and a tolerated deflection.

At an unloaded static operating point with $K_q \succ 0$ and task displacement $\delta y = J \delta q$, compliance is

$$C_y = J K_q^{-1} J^T, \quad K_y = C_y^{-1} \quad (10.4)$$

when the task compliance is invertible. Only for square nonsingular J does this become $K_y = J^{-T} K_q J^{-1}$. The familiar $J^T K_y J$ maps task stiffness back to joint coordinates; it is not the general forward map. External preload adds geometric terms, and rigid constraints require a reduced-coordinate or constrained solve.

For $J = [1 \ 1]$ and $K_q = \text{diag}(2, 3)$ in consistent units, $C_y = 1/2 + 1/3 = 5/6$. Multiplying both joint stiffnesses by ten divides compliance by ten and leaves the rank of J unchanged. Geometry and stiffness interact without being the same mathematical object.

10.3.1 Loaded Loops: Geometry Enters the Restoring Stiffness

To compute the preload contribution, consider a smooth conservative static equilibrium q_* with fixed activation and fixed spring parameters. Include all conservative applied loads in $U(q)$ and use regular stationary constraints $\phi(q) = 0$, with A of full row rank. With

the reaction sign used above, $\nabla U(q_*) = A(q_*)^T \lambda_*$. Let the columns of N form a basis of $\ker A(q_*)$, so that $\delta q = N\delta\eta$ to first order. Then

$$\begin{aligned} K_r &= N^T H_L N, \\ H_L &= U_{qq} - \sum_i \lambda_{*,i} \phi_{i,qq}. \end{aligned} \quad (10.5)$$

All derivatives are evaluated at q_* . This is the Hessian of $U - \lambda_*^T \phi$ restricted to admissible directions, with the multiplier held fixed when taking the Hessian. It is the equality-constrained second-variation test Freund [2004]. The minus sign follows from this chapter's reaction convention; changing the multiplier convention changes both signs.

The constraint-curvature term is essential. For any feasible curve $q(s)$, differentiating $\phi(q(s)) = 0$ twice gives $Aq'' + \{q'^T \phi_{i,qq} q'\}_i = 0$. Substitution of $\nabla U = A^T \lambda_*$ into the second derivative of $U(q(s))$ produces the displayed K_r . Merely projecting U_{qq} omits the work of the applied preload along the curved feasible path. Ideal reactions still do zero net work; this geometric effect creates no energy.

If $K_r \succ 0$, a small additional task load has static compliance $C_y = J_r K_r^{-1} J_r^T$, where $J_r = JN$. Its inverse defines task stiffness only when this compliance is nonsingular. This result concerns local static response, not finite-frequency impedance. A change of tangent basis must transform both J_r and K_r ; their combined task compliance is unchanged. Nonconservative follower loads, delayed feedback, moving constraints, and a moving swing require their own linearized dynamics, not this potential test.

10.3.2 A Preload Example With Unchanged Degrees of Freedom

Consider a manufactured point constrained to a circle, with Cartesian coordinates $q = (x, y)$ in metres:

$$\begin{aligned} \phi(q) &= \frac{1}{2}(x^2 + y^2 - R^2) = 0, \\ U(q) &= \frac{1}{2}k_x(x - a)^2 + \frac{1}{2}k_y y^2. \end{aligned} \quad (10.6)$$

At $q_* = (R, 0)$ choose $N = (0, 1)^T$, so η measures tangent displacement in metres. Then $\lambda_* = k_x(R - a)/R$ and $K_r = k_y - \lambda_*$. For $R = 1$ m, $k_x = 100$ N/m, and $k_y = 40$ N/m, choosing $a = 0.8$ m gives $K_r = 20$ N/m and tangent compliance 0.05 m/N. Choosing $a = 0.5$ m instead gives $K_r = -10$ N/m: the equilibrium is unstable for a positive-mass conservative particle even though the ambient Hessian $\text{diag}(100, 40)$ is positive definite. Its inverse is not a stable static compliance measurement. The constraint rank and tangent dimension have not changed.

The exact path $q(\theta) = (R \cos \theta, R \sin \theta)$ checks the result:

$$\left. \frac{d^2 U(q(\theta))}{d\theta^2} \right|_{\theta=0} = R^2(k_y - \lambda_*). \quad (10.7)$$

Dividing the angular energy curvature by R^2 gives tangent-displacement stiffness. These numbers illustrate a mechanism, not human tissue parameters or a golf posture recommendation. They show why activation-induced loading and loop geometry must be assessed together rather than ranking postures by muscle stiffness alone.

10.4 Energy Minima, Stability, and Attractors

For a conservative mechanical model, a strict potential minimum with positive reduced Hessian supplies restoring forces and local Lyapunov stability under the usual regularity and positive-inertia assumptions. Without dissipation, nearby trajectories can oscillate forever; they need not converge to that minimum. An attracting equilibrium requires an additional dissipation argument. An attractor can also be a periodic orbit or another invariant set, rather than an equilibrium. Address, transition, and impact must not all be called potential wells merely because they are recognizable landmarks.

For $M\ddot{e} + D\dot{e} + Ke = 0$ with scalar $M > 0$, $K > 0$, and $D > 0$,

$$\omega_n = \sqrt{K/M}, \quad \zeta = \frac{D}{2\sqrt{KM}}, \quad \lambda_{1,2} = \frac{-D \pm \sqrt{D^2 - 4MK}}{2M}. \quad (10.8)$$

The equilibrium is asymptotically stable. At $D = 0$ it is stable but not attracting. Increasing K with fixed M, D raises natural frequency and lowers damping ratio. Consequently, “more stiffness reduces bandwidth” is not a general law. Disturbance rejection depends on frequency and on which transfer function is being measured.

For a constrained equilibrium, stability uses admissible perturbations and the reduced Hessian of the constrained potential, including preload terms. Along a moving swing, assess the time-varying variational dynamics. Stable eigenvalues of every frozen-time matrix alone do not establish stability of the full time-varying system. A change in stiffness is not evidence of a bifurcation; one must identify the solution family and demonstrate a change of stability.

10.5 What Pre-Tension Changes in an Affine Model

A mechanical state includes position and velocity, not configuration alone: $x = (q, v, a, z)$ may also include activation a and internal tissue states z . For a specified input and contact mode, a model may take the form

$$\dot{x} = f(x) + G(x)u. \quad (10.9)$$

If u denotes residual torque around a maintained baseline activation, that baseline remains in the zero-input dynamics. If u is neural excitation, setting it to zero does not instantaneously erase activation or tendon force. If total actuator torque is set to zero, the intervention removes a different quantity. A zero-torque counterfactual must specify which model is used.

Pre-tension changes state, elastic forces, and local tangent dynamics. In a fixed-geometry rigid model with direct torque input, changing a spring does not add a stiffness matrix to M or directly change $M^{-1}B$. Through later changes of posture, constraints, activation, or actuator model, it can change the input-to-task response. This is why same-state acceleration authority and finite-duration control performance must be assessed separately.

A zero-input branch near a target suggests a potentially useful initial state; it does not establish low total effort. Earlier actions created that state, maintained activation may remain, and small errors in difficult directions may require substantial correction. Compare the complete task and preparation cost under matched input limits and measurement uncertainty.

10.5.1 Why a Drift Ratio Is Not an Efficiency Measurement

A ratio of drift magnitude to a specified bounded-input magnitude summarizes model terms in a chosen norm. It is not a work fraction, an energy source fraction, or a measurement of metabolic efficiency. Norms of sums contain cross terms; the norm of one arbitrarily signed sum of actuator columns need not equal maximum available control. State the admissible input set and use its actual optimization when making an authority claim.

Magnitude also omits direction. Fast motion away from the target can have a large drift ratio and poor performance. Use task error, sensitivity over the remaining horizon, and explicit work or energetic costs to evaluate whether the drift is helpful.

10.6 Stiffness Modulation Has an Energy Budget

For a spring with $e = q - q_0(t)$ and storage $V_s = \frac{1}{2}k(t)e^2$,

$$\dot{V}_s = ke\dot{q} - k\dot{e}q_0 + \frac{1}{2}\dot{k}e^2. \quad (10.10)$$

Changing stiffness or rest position can exchange energy with the mechanism. A scheduled stiffness change is not automatically a passive operation. Report where this energy goes in the actuator or tissue model.

Isometric activated muscle can consume metabolic energy at zero shortening velocity. Thus $(F_{\text{ago}} + F_{\text{ant}})v$ is not a general metabolic-power model. Joint power is $\tau^T \dot{q}$; individual muscle work, tendon storage, activation and maintenance heat, and joint work differ. Net torque can conceal large opposing forces and energetic expenditure.

Fascia and other connective tissues can carry load and store energy, but a tensegrity analogy does not quantify their contribution to club speed or prove a universal low-cost posture. A model needs tissue properties, load paths, boundary conditions, and validation. Likewise, reduced modeled loading is not by itself clinical evidence of reduced injury risk.

10.7 Testing Coordination With the Uncontrolled Manifold

For task output $y = h(x)$, the nonlinear task-equivalent set is $\{x : h(x) = y^*\}$. At a regular point its tangent is $\ker J$, where $J = \partial h / \partial x$. This is not the stationary set of a scalar cost. If the task is clubhead speed, the state must include velocities: joint angles alone do not determine speed. The UCM method originated in task-specific coordination analysis [Scholz and Schöner \[1999\]](#); that study examined sit-to-stand, not golf. Its framework does not by itself validate clubhead-speed regulation.

Declare characteristic scales before combining position, velocity and other state components. For constant nonsingular S and $\xi = S^{-1}(x - x_*)$, use $J_\xi = J_x S$ and $\Sigma_\xi = S^{-1} \Sigma_x S^{-T}$. In the following formulas, J and Σ mean these consistently scaled quantities. With the Euclidean metric in ξ , define $P_\parallel = I - J^+ J$, $P_\perp = J^+ J$, and sample covariance Σ . With $r = \text{rank } J$ and $0 < r < n$, variance per dimension is

$$V_\parallel = \frac{\text{tr}(P_\parallel \Sigma)}{n - r}, \quad V_\perp = \frac{\text{tr}(P_\perp \Sigma)}{r}. \quad (10.11)$$

Parallel means within the task-equivalent tangent; perpendicular means changing the task to first order. Test $V_{\parallel} > V_{\perp}$ with an appropriate subject-level statistical design. Coordinate scaling, measurement noise, phase alignment, and nonlinear curvature affect the result. Such covariance structure supports a task-stabilization interpretation but does not distinguish passive mechanics from feedback or prove a neural algorithm.

10.8 An Experimental Program for the Golf Swing

The earlier chapter asserted a universal elite-golfer stiffness pulse, minimal late-downswing activation, and lower metabolic cost without identifying studies that established those claims. Treat these as proposed comparisons, not results. No numerical co-contraction schedule is prescribed here.

1. Define the task: impact location, clubhead velocity, face orientation, or their joint distribution. Specify the tested club, intent, and skill groups.
2. Synchronize kinematics, external forces, and relevant EMG. EMG is an electrical measurement; it is not a direct force or stiffness measurement.
3. Identify impedance using controlled perturbations and a model separating inertia, damping, stiffness, delays, and measurement error. A dynamic torque/displacement ratio alone conflates these contributions.
4. Compare preparation, swing, and recovery costs. A baseline posture measurement does not determine stiffness throughout a transient swing.
5. Distinguish ordinary force-plate loading from ball-contact transmission. Do not assume a universal ground-force spike exactly at impact without synchronized measurements and adequate bandwidth.
6. Test alternative explanations: geometry, starting speed, net torque, passive tissue loading, activation, and predictive or delayed feedback. Report unfavorable outcomes as well as favorable ones.

10.9 Optimizing a Stiffness Strategy

Let α parameterize a specified activation or stiffness mechanism and let $e_k = h_{\text{task}}(x_k) - y_k^*$ be the declared task error. For step size $h > 0$, use a piecewise-linear parameter schedule with $\nu_k = (\alpha_{k+1} - \alpha_k)/h$, or an explicit rate input with its own dynamics. A finite-horizon model comparison can minimize

$$J = \ell_T(x_N) + \sum_{k=0}^{N-1} h \{ e_k^T Q e_k + u_k^T R u_k + c_{\alpha}(x_k, \alpha_k) + c_{\text{rate}}(\nu_k) \}, \quad (10.12)$$

subject to state dynamics, contact and loop constraints, input bounds, and admissible activation/stiffness rates. Every running-cost term carries the time step. The quantities c_{α} and c_{rate} are stated penalties unless calibrated as physical energetic costs.

Use a state containing velocities and activation dynamics where required. LQR applies to a linearization with a quadratic objective; a finite horizon uses a differential or discrete

Riccati recursion, not automatically an algebraic steady-state solution. Joint optimization of state, inputs, and stiffness is generally nonlinear and constrained. Check feasibility, derivative accuracy, integration convergence, stationarity, and sensitivity to starting guesses. Penalty terms alone do not exactly enforce loop closure.

The optimal profile is an output of this specified comparison. It need not peak at transition or decline with increasing load. Those are hypotheses to test, not conditions to build into the answer while calling it a prediction.

10.10 How the Mechanisms Fit Together

Geometry directs forces and couples joints. Activation changes force capacity and local response. Elastic tissues store and return part of earlier work. Feedback and prediction manage deviations that mechanics alone does not reject. Impact precision selects which deviations matter. The technical argument for coordination is strongest when these contributions are computed in one consistent model and tested with measurements, including the cost of preparing the state. Apparent effortlessness alone cannot identify any of them.

Exercises

1. For $M = 0.02$, $D = 0.5$, and $K = 100$ in rotational SI units, compute natural frequency, damping ratio, and static deflection under a 5 Nm torque. Double K and explain which quantities change.
2. Show that the undamped spring conserves energy and therefore cannot attract all nearby nonzero-energy states to its equilibrium.
3. Verify the task-compliance example and preserve the Jacobian rank while scaling stiffness. Derive the circle example from the exact constrained energy and explain why projecting the ambient Hessian gives the wrong answer.
4. Derive the variable-spring energy identity, including moving rest length. Identify the terms an actuator must supply or absorb.
5. For $J = [1 \ 1]$ and a specified covariance, compute both UCM variances per dimension. Repeat after rescaling one coordinate and discuss the metric.
6. Design a perturbation study that can distinguish intrinsic stiffness from delayed feedback without interpreting EMG amplitude as force.

Part IV

Forces in Detail

Chapter 11

Energy Transfer: How Power Flows Through the Kinetic Chain

A useful explanation of the golf swing must answer two different questions: what changes the motion, and where does the mechanical energy come from? Constraint forces, muscular moments and gravity all affect acceleration, but their energy accounts depend on the system boundary. A force can accelerate a body without doing work on the complete system. A joint motor can add energy while an adjacent segment loses energy. An angular-speed peak does not identify either process by itself.

This chapter develops a consistent account from whole-system work to individual segment power, then applies it to sequencing, lag and impact. The aim is to make the kinetic-chain idea testable: identify the bodies, forces, moments, storage and measurements that would support an explanation. Two explicit teaching mechanisms and six worked exercises separate mechanical identities from hypotheses about a golfer. The idealized examples establish possibilities and counterexamples; their parameter values are not measured human physiology.

11.1 Energy Fundamentals: Define the System

For a chosen collection of rigid bodies and elastic elements, write

$$E_{\text{mech}} = \sum_i K_i + V_g + U_e. \quad (11.1)$$

Here V_g is gravitational potential, U_e is recoverable elastic energy, and chemical energy is outside this mechanical account. Muscles can therefore add mechanical energy during the downswing even if the total energy of a larger thermodynamic system is conserved. Neglecting air resistance does not make an actively driven swing mechanically conservative. Active shortening, lengthening, elastic deformation and dissipation may coexist throughout the motion.

In one inertial laboratory frame, the kinetic energy of rigid segment i is

$$K_i = \frac{1}{2}m_i\|\mathbf{v}_{C_i}\|^2 + \frac{1}{2}\boldsymbol{\omega}_i^T I_{C_i} \boldsymbol{\omega}_i. \quad (11.2)$$

The translational velocity is that of its center of mass; the angular velocity and central inertia tensor must use the same axes. A relative wrist rate is not the club's absolute angular velocity. The scalar expression $I\omega^2/2$ alone applies to rotation about a specified fixed axis with the appropriate inertia about that axis. A translating club generally needs both terms above. For a deforming segment this rigid-body approximation omits internal motion; whether that omission matters requires validation for the intended output.

Near the ground, with upward height y_i and constant gravitational acceleration of magnitude g ,

$$V_g = \sum_i m_i g y_i + \text{constant}. \quad (11.3)$$

Changing the height origin changes no force or energy difference. It does change a ratio such as gravitational energy divided by total mechanical energy. Such a percentage is therefore not an intrinsic efficiency measure. A segment's share of nonzero total kinetic energy is well-defined in the declared frame; at a completely stationary instant that share is 0/0, not zero percent.

11.1.1 What Is Stored, and What Is Merely a Configuration?

Kinetic energy describes motion. Elevation changes gravitational storage. A specified elastic constitutive law assigns energy to deformation. A visible angle, a constrained degree of freedom or a desired movement pattern does not by itself define an additional store. The club, shaft, tendons and body can all participate in an energy account, but including the same elastic element both inside a segment model and as a separate spring would count it twice.

In generalized coordinates q , a rigid multibody model gives

$$K = \frac{1}{2} \dot{q}^T M(q) \dot{q}. \quad (11.4)$$

The mass matrix contains cross terms because a body's motion depends on more than one joint rate. Those terms are required by the coordinate representation; they are not independent packets of energy traveling between anatomical parts. Physical segment energies remain the center-of-mass expressions above. Changing coordinates changes the appearance of the cross terms without changing those energies for the same physical state and observer.

11.2 Power and the Whole-System Work Balance

Power is the rate of energy change through a specified mechanism:

$$P = \frac{dE}{dt}, \quad W = \int_{t_0}^{t_1} P dt. \quad (11.5)$$

A positive input power increases the chosen mechanical account; negative input power extracts energy from it. A dissipative loss is reported here as a nonnegative quantity. For conservative storage $V = V_g + U_e$, use

$$M\ddot{q} + C(q, \dot{q})\dot{q} + \nabla V = B(q)u + Q_{nc} + J^T \lambda. \quad (11.6)$$

The motor map B converts actuator efforts u into generalized forces. Q_{nc} contains the declared remaining nonconservative forces, and $J^T \lambda$ represents an ideal constraint reaction. Do not place the same muscle, spring or contact force in two of these terms. A standard Christoffel choice of C satisfies the skew-symmetry identity for $\dot{M} - 2C$; an arbitrary matrix factorization of the velocity bias need not satisfy that matrix identity.

Differentiating kinetic and potential energy gives

$$\dot{K} = \dot{q}^T M \ddot{q} + \frac{1}{2} \dot{q}^T \dot{M} \dot{q}, \quad \frac{1}{2} \dot{q}^T \dot{M} \dot{q} = \dot{q}^T C \dot{q}, \quad (11.7)$$

$$\dot{V} = \dot{q}^T \nabla V. \quad (11.8)$$

Consequently,

$$\dot{E}_{\text{mech}} = \dot{q}^T B u + \dot{q}^T Q_{\text{nc}} + \lambda^T J \dot{q}. \quad (11.9)$$

The cancellation is between the metric-derivative term and the velocity-bias term. It does not require $\dot{q}^T C \dot{q} = 0$, which is generally false. The general manipulator formulation and a simpler point-mass double pendulum are also developed in Tedrake's multibody notes [Tedrake, n.d.b].

For a stationary ideal constraint and a decomposition into supplied power P_{in} and dissipation $D \geq 0$, the integrated account is $E(t) - E(t_0) = W_{\text{in}} - \int D dt$. Gravity is already represented by V_g . Alternatively one can track kinetic plus elastic energy and include gravitational power on the right. Adding gravitational work while also retaining its potential-energy change counts the same conversion twice. Time-dependent stiffness, an imposed spring rest angle, or a moving prescribed base requires its corresponding parameter or support work; the autonomous formula alone is then incomplete.

11.2.1 Signed Work, Positive Work and Metabolic Cost

For separately modeled actuator powers P_a , distinguish $W_{\text{net}} = \sum_a \int P_a dt$ from $W_+ = \sum_a \int \max(P_a, 0) dt$. The latter is positive work; it does not cancel energy supplied at one time against energy absorbed at another. Taking the positive part after summing powers gives a different quantity when some actuators supply energy while others absorb it. Declare which convention is used before comparing swings.

Net joint moments inferred from motion do not resolve antagonistic muscle forces, tendon storage or activation costs. A joint with zero net power can still involve active muscle and metabolic expenditure. Neither a ratio to net joint work nor a ratio to positive joint work is automatically a metabolic efficiency. Estimating that efficiency requires a separately justified chemical energy or metabolic model and a consistent measurement interval.

11.3 Constraint Power: Zero for Which Boundary?

An ideal reaction to a position constraint $\Phi(q, t) = 0$ has generalized force $Q_c = J^T \lambda$, where $J = \partial \Phi / \partial q$. Its actual power is

$$P_c = Q_c^T \dot{q} = \lambda^T J \dot{q}. \quad (11.10)$$

Differentiating the constraint gives

$$J \dot{q} + \Phi_t = 0, \quad (11.11)$$

and therefore

$$P_c = -\lambda^T \Phi_t. \quad (11.12)$$

It vanishes for a stationary ideal holonomic constraint. A moving support can exchange energy with the modeled bodies. Conversely, some ideal nonholonomic constraints, such as homogeneous no-slip rolling constraints, also have zero reaction power. Holonomic versus nonholonomic is not the test by itself: ideality, the admissible velocities and explicit support motion matter. Friction, tissue damping and impact losses require their own models.

11.3.1 Why Energy Can Cross an Ideal Joint

Consider an arm and a club connected at hinge H , with coincident material-point velocities $\mathbf{v}_H$ in the ideal attachment. Let $\mathbf{F}$ act on the club at H , with $-\mathbf{F}$ on the arm. Then

$$P_{F,\text{club}} = \mathbf{F} \cdot \mathbf{v}_H, \quad P_{F,\text{arm}} = -\mathbf{F} \cdot \mathbf{v}_H. \quad (11.13)$$

The pair sums to zero while either term may be nonzero. This is energy crossing an internal boundary, with equal and opposite entries in the two ledgers. Equality of attachment-point velocities enables the cancellation; it does not prevent transfer. Substituting the two centers' different velocities for the contact-point velocity would calculate the wrong contact power.

In a planar model let a hinge motor apply moment n to the club and $-n$ to the arm. If gravity is included in each segment's potential, the club balance is

$$\dot{E}_{\text{club}} = \mathbf{F} \cdot \mathbf{v}_H + n\omega_{\text{club}}, \quad P_{\text{hinge motor}} = n(\omega_{\text{club}} - \omega_{\text{arm}}). \quad (11.14)$$

The moment's power on the club uses its absolute angular velocity, whereas the motor's net mechanical power uses relative angular velocity. Their difference is the equal-and-opposite moment's power on the arm. An ideal unactuated hinge transmits attachment force while allowing relative rotation; setting its motor moment to zero does not disconnect the club.

An Internal Force Transfers Energy; a Hinge Motor Can Also Supply It

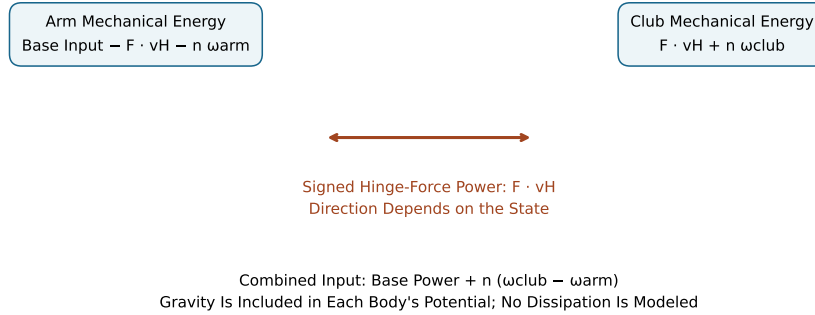


Figure 11.1: Both Sides of the Internal Boundary Must Balance. Hinge-Force Power Cancels Between Bodies, While Motor Power Depends on Their Relative Angular Velocity.

These balances do not select a universal direction of transfer or maximize club energy. Geometry, motion and applied efforts determine the signed power. The opposite signs of the interface entries establish accounting consistency; they do not identify what would happen if the golfer deliberately changed an effort or movement. That requires a new dynamically consistent trajectory.

11.3.2 Reporting Point, Observer and Ground Contact

A wrench's power is $\mathbf{F} \cdot \mathbf{v}_O + \mathbf{n}_O \cdot \boldsymbol{\omega}$ when both the moment and velocity refer to the same rigid body's reporting point O . Moving that point changes the moment and linear velocity

together; the power is unchanged. Changing the inertial observer by a constant velocity changes individual kinetic energies and force powers consistently. A segment energy fraction is therefore an explicitly frame-dependent diagnostic, not an observer-independent property of the swing.

For an ideal fixed no-slip foot contact, local contact velocity is zero and the contact force does no work there, although it can change the golfer's momentum. Multiplying ground-reaction force by whole-body center-of-mass velocity produces a center-of-mass power quantity, not necessarily actual foot-contact power. Deforming footwear, slipping contacts and moving ground alter the boundary. Gravity, air and the ball can also supply external forces; ground reaction is not the only external interaction in a complete swing and collision.

11.4 Energy Partition During a Declared Passive Motion

Before interpreting a golfer's downswing, test the transfer claim on a mechanism whose assumptions and trajectory are known. Take two uniform rods, each of mass 1 kg and length 1 m, a fixed base, ideal hinges, zero gravity and no motors. Absolute angles are measured from vertically downward. Initially the inner rod has angle zero, the outer rod angle $-\pi/2$, and both absolute angular rates are 2 rad/s. Their kinetic energies are $2/3$ J and $8/3$ J, respectively. Total energy is $10/3$ J.

The initial hinge-force power into the outer rod is +1 W. Reflecting the outer angle to $+\pi/2$, without changing either angular rate, gives -1 W. Both states have the same segment kinetic energies. The transfer direction therefore cannot be inferred from the energies or speed ordering alone. The force and attachment-point velocity matter.

Numerical integration of the first state gives the following rounded outer-rod energies at 0, 0.5, 1, 1.5 and 2 s: 2.6667, 3.3327, 2.9780, 2.5007 and 2.3421 J. At all five times the total remains $10/3$ J within integration error. The outer fraction rises from 80 percent to about 99.98 percent at 0.5 s and subsequently falls. The signed power changes direction again later. There is no active braking and no dissipative sink in this model: a decrease of outer-rod energy is an increase of inner-rod energy.

These times are integration times for this declared mechanism, not five measured phases of a golf swing. A realistic model needs measured or justified segment properties, gravity, multiple joints, applied efforts and boundary conditions. A sequence of arbitrarily selected joint rates is not a simulated trajectory; it cannot establish a missing-work budget or a universal club-energy percentage.

11.5 Sequential Speed Peaks and Their Interpretation

The phrase summation of speed can obscure three different statements. Velocities combine vectorially through kinematics. Powers add algebraically in a declared energy balance. Peak times are features of a time series. None of these implies that scalar peak angular speeds should simply be added, or that the observed ordering proves a one-way energy cascade.

For an isolated rotor about a fixed axis, equal assigned energy gives

$$\omega = \sqrt{2K/I}. \quad (11.15)$$

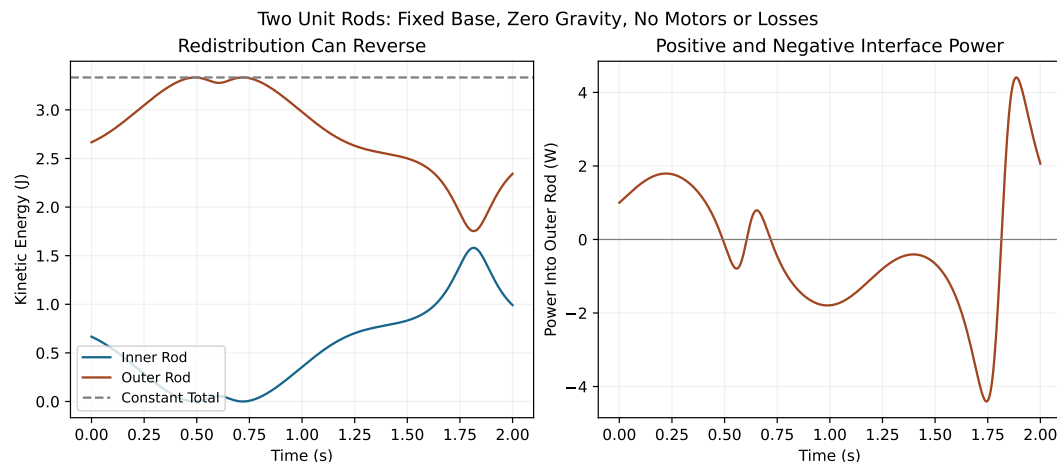


Figure 11.2: An Explicit Passive Mechanism Redistributes Energy in Both Directions. The Outer Rod Gains Energy, Loses Energy and Later Gains It Again; Total Energy Remains Constant.

For example, separate rotors with inertias 2, 0.5 and 0.1 kg m², each assigned 100 J, have angular speeds 10, 20 and approximately 44.72 rad/s. This algebra does not describe three connected segments at successive swing phases. Their energies need not be equal; their centers translate; their physical inertia tensors and the full joint mass matrix have different roles. A smaller scalar inertia alone does not guarantee a later or higher speed peak in the chain.

An especially useful counterexample uses the same two unit rods, now aligned vertically downward with both rates 2 rad/s, zero gravity, a base moment of -1 N m and zero hinge moment. The inner and outer absolute angular accelerations are $-12/7$ and $18/7$ rad/s². Yet their kinetic-energy rates are $-8/7$ and $-6/7$ W. The outer rod accelerates rotationally while losing energy because its translational energy falls more rapidly. Total power is -2 W, exactly the base moment times its angular velocity.

A proximal segment slowing while the club speeds up can accompany transfer, active work, gravity conversion, elastic release or combinations of them. It does not show that the proximal segment has exhausted its energy. Testing a sequencing explanation requires synchronized physical segment energies, interface powers and actuator powers, together with a model that reproduces the measured motion. Prescribing the observed speed peaks in a model and then claiming that the model explains those peaks would be circular.

11.6 Storage, Active Work and a Balanced Swing Ledger

The following five-state account is deliberately hypothetical. It illustrates bookkeeping, not a typical golfer. Start from rest at the chosen reference height with no elastic deformation.

The successive stored energies, in joules, are given by the three rows

$$\begin{aligned} K &: (0, 0, 80, 115, 115), \\ V_g &: (0, 15, 5, 0, 0), \\ U_e &: (0, 10, 2, 0, 0). \end{aligned} \tag{11.16}$$

Their sums are 0, 25, 87, 115 and 115 J. Choose cumulative active mechanical work of 0, 25, 90, 120 and 120 J and cumulative dissipative loss of 0, 0, 3, 5 and 5 J. Every state satisfies stored energy equals initial energy plus supplied work minus losses. Between the second and third states, stored energy increases 62 J while active work increases 65 J and dissipation increases 3 J. Falling gravitational and elastic storage explains part of the kinetic-energy rise; it does not replace the active-work contribution.

The model could instead start at the top with its already stored 25 J. Then that amount is initial energy, while only subsequent work belongs in the downswing input. Adding both the initial store and all work previously used to create it would double count the backswing. A claim that impact energy exceeds downswing muscular work can be consistent with energy conservation, but it requires an explicit initial store and other inputs rather than an efficiency greater than one inferred from incomplete accounting.

11.6.1 Impact Is a New Boundary and Time Interval

For an isolated one-dimensional head-ball collision, assume head mass $m_h = 0.2$ kg, ball mass $m_b = 0.04593$ kg, initially stationary ball, head speed $u_h = 45$ m/s and restitution $e = 0.8$. These are teaching assumptions, including the effective isolated head mass; a shaft-coupled, spinning or oblique impact requires additional dynamics. Momentum conservation and relative restitution give

$$v_h = \frac{m_h - em_b}{m_h + m_b} u_h, \quad v_b = \frac{(1 + e)m_h}{m_h + m_b} u_h. \tag{11.17}$$

The final speeds are approximately 29.8724 and 65.8724 m/s. Initial head energy is 202.5 J; residual head energy is 89.2360 J; ball energy is 99.6491 J; the mechanical loss in this collision is 13.6148 J. Ball energy, residual head energy and collision loss sum to the initial energy. Residual head motion is not collision dissipation, and swing-phase air or tissue losses are not added again to this impact account.

With reduced mass $\mu = m_h m_b / (m_h + m_b)$, the loss is $\mu(1 - e^2)u_h^2/2$. The ratio of ball energy to pre-impact head energy is $m_h m_b (1 + e)^2 / (m_h + m_b)^2$, approximately 49.21 percent here. This is an impact energy-transfer ratio for the declared model, not a measured swing efficiency. A halving of a body's speed quarters its kinetic energy at fixed mass; it does not halve it. These distinctions are essential when interpreting residual club speed, ball launch and claims about efficient energy delivery.

11.7 Lag, Elasticity and Release Timing

A lag angle is a relative configuration. An ideal freely rotating hinge has no elastic potential associated with that angle. If a torsional spring is explicitly included, with stiffness k and rest angle q_{20} , then

$$U_e = \frac{1}{2}k(q_2 - q_{20})^2, \quad \tau_e = -k(q_2 - q_{20}). \tag{11.18}$$

At a 0.5 rad displacement, stiffness zero stores zero joules; stiffness 40 N m/rad stores 5 J. The angle alone cannot distinguish these cases. A deforming shaft or tendon needs its own strain coordinate and constitutive law; its energy is not automatically a function of the visible wrist angle.

If the rest angle or stiffness changes, the explicit parameter contribution is $\partial U_e/\partial t = \dot{k}(q_2 - q_{20})^2/2 - k(q_2 - q_{20})\dot{q}_{20}$. That term represents work associated with changing the modeled spring, rather than free passive release. A viscous hinge damper with coefficient $d \geq 0$ dissipates $d\dot{q}_2^2$. Active moment, elastic moment and damping moment must be separated if the question concerns their individual contributions.

Holding a relative angle fixed is a different idealization from storing energy in a spring. A stationary ideal lock can transmit forces and moments with zero net constraint power. Removing such a lock need not change velocity or kinetic energy instantaneously, although subsequent dynamics change. Engaging a lock when the relative rate is nonzero generally requires an impulsive transition and a specified loss or actuator-work model. Releasing a rotational lock leaves the translational attachment intact; it does not remove the hinge force.

11.7.1 A Fair Release-Timing Comparison

Compare strategies from the same full initial state and with declared actuator limits, allowed controls and delivery event. The event should include relevant club position, face orientation, path and contact conditions, rather than only the largest angular speed anywhere in the trajectory. Report terminal kinetic and elastic energy, signed and positive work, losses and residual motion. Equal commanded torque histories do not generally imply equal work, because the trajectories and rates may differ.

If one strategy begins with 90 J and another with 140 J, their final speeds cannot isolate the effect of timing. If terminal energies are 60 and 150 J for the same isolated fixed-axis rotor, the speed ratio is $\sqrt{150/60}$, approximately 1.581, not two. For a translating multibody club, even that ratio requires checking the relationship between the reported speed and total kinetic energy. There is no universal release delay or speed multiplier established by these examples.

Setting one motor effort to zero at a frozen state is a useful instantaneous acceleration comparison. It is not the complete trajectory produced by a different release strategy. Passive tissue forces, gravity, constraints and other motors can remain active. This is where energy analysis connects to counterfactual dynamics and control: the intervention, retained terms, starting state and event must all be specified before assigning a causal benefit.

11.8 Two-Link Dynamics and Physical Energy Transfer

To derive the coupling explicitly, take a fixed planar base, inner-link length L_1 , masses m_1, m_2 , center distances c_1, c_2 from each proximal hinge and central planar inertias I_1, I_2 . Define relative coordinates $q_1 = \theta_1$ and $q_2 = \theta_2 - \theta_1$, where the absolute angles θ_i are measured from downward. Let $u = \dot{q}_1$, $w = \dot{q}_2$ and $a = I_1 + m_1 c_1^2 + m_2 L_1^2$, $b = I_2 + m_2 c_2^2$, $d = m_2 L_1 c_2$. Directly differentiating the center positions gives

$$K = \frac{1}{2}au^2 + \frac{1}{2}b(u+w)^2 + d\cos q_2 u(u+w). \quad (11.19)$$

The club rotates at $u+w$, not w . The inner contribution is $(I_1 + m_1 c_1^2)u^2/2$; the rest belongs to the translating and rotating outer body. Hence the physical outer-body energy includes

the cross term and an $m_2 L_1^2 u^2 / 2$ contribution. Assigning only $bw^2 / 2$ to the club would omit its motion induced by the inner link.

Euler–Lagrange equations with base and relative-hinge motor efforts $\tau = (\tau_1, \tau_2)^T$ give $M\ddot{q} + h + G = \tau$, where

$$M = \begin{bmatrix} a + b + 2d \cos q_2 & b + d \cos q_2 \\ b + d \cos q_2 & b \end{bmatrix}, \quad h = \begin{bmatrix} -d \sin q_2 (2uw + w^2) \\ d \sin q_2 u^2 \end{bmatrix}. \quad (11.20)$$

Both velocity-bias components matter. For upward gravitational potential,

$$G = \begin{bmatrix} (m_1 c_1 + m_2 L_1) g \sin q_1 + m_2 c_2 g \sin(q_1 + q_2) \\ m_2 c_2 g \sin(q_1 + q_2) \end{bmatrix}. \quad (11.21)$$

Multiplying by the rates and retaining the metric derivative recovers

$$\dot{E} = \tau_1 u + \tau_2 w. \quad (11.22)$$

In particular,

$$\dot{q}^T h = -d \sin q_2 u w (u + w) = \frac{1}{2} \dot{q}^T \dot{M} \dot{q}. \quad (11.23)$$

This scalar can be nonzero. It is canceled in the total energy derivative, not identified as a separate physical input or equated to the club’s energy rate. A term called Coriolis power in one coordinate decomposition must not be promoted into an invariant segment-transfer measurement.

To compute the actual hinge-force transfer, let $\mathbf{a}_{C_2}$ be the outer center acceleration and $\mathbf{g} = (0, -g)^T$. With no other outer-body force, Newton’s equation gives

$$\mathbf{F} = m_2(\mathbf{a}_{C_2} - \mathbf{g}), \quad P_{F,2} = \mathbf{F} \cdot \mathbf{v}_H. \quad (11.24)$$

The physical body rates follow from center translation, rotation and gravity:

$$\dot{E}_2 = P_{F,2} + \tau_2(u + w), \quad \dot{E}_1 = \tau_1 u - P_{F,2} - \tau_2 u. \quad (11.25)$$

Their sum agrees with the generalized balance. This independent Newton-force calculation is the check that a coordinate-based interpretation needs. In three dimensions replace scalar moments and rates by compatible moment/angular velocity pairings; in a two-arm closed chain include both attachment wrenches and avoid duplicating the shared club energy.

11.8.1 Reproducible Calculation

The shared `rod_energy_ledger` routine computes center energies and their derivatives from `RodPair` dynamics. Its state uses absolute angles and rates; motor inputs remain base and relative-hinge moments. For uniform rods the central inertias are $m_i L_i^2 / 12$ and $c_i = L_i / 2$. Run this example from the repository root:

```
1 import numpy as np
2
3 from src.tools.energy_ledger_examples import rod_energy_ledger
4 from src.tools.lagrangian_examples import RodPair
5
6 model = RodPair(gravity=0)
7 state = np.array([0, -np.pi / 2, 2, 2])
8 ledger = rod_energy_ledger(model, state, np.zeros(2))
9 np.testing.assert_allclose(ledger.kinetic, [2 / 3, 8 / 3])
```

```

10 np.testing.assert_allclose(ledger.mechanical_rates, [-1, 1])
11 assert np.isclose(ledger.force_power, 1)
12 assert np.isclose(ledger.actuator_power, 0)
13
14 braking = rod_energy_ledger(
15     model, np.array([0, 0, 2, 2]), np.array([-1, 0])
16 )
17 np.testing.assert_allclose(braking.mechanical_rates, [-8 / 7, -6 / 7])

```

The calculation has been independently checked against directional derivatives of each physical body’s energy, total generalized motor power and the force power at the hinge. Conservation alone would be insufficient verification: two incorrectly assigned segment rates could cancel in the total. Conversely, a small integration drift is a numerical diagnostic, not evidence of unmodeled muscular work or physical dissipation.

11.9 Evidence, Measurement and an Investigable Golf-Swing Argument

Nesbit and Monika Serrano’s work-and-power study used four selected amateur golfers and a selected swing from each, with model-derived mechanical quantities [Nesbit and Serrano, 2005]. It supports examining coordinated body and club work within that study’s model. It does not establish a population-wide energy percentage, directly measure metabolic efficiency or identify the causal effect of an instruction to delay release. Those questions need additional evidence.

To investigate an energy-transfer explanation, first define the outcome: clubhead delivery speed and orientation, ball launch, repeatability or effort. Then choose the body and time boundaries. Estimate segment properties and synchronized motion; state coordinate, frame and joint-moment conventions. Use the inverse-dynamics and constraint-force chapters (Chapters 20 and 7) to construct physically compatible forces. Close the whole-system ledger and both sides of each important interface before interpreting a sequence of peaks. The complete-swing and synthesis chapters (Chapters 34 and 33) connect these local mechanisms to finite-time delivery and competing task objectives.

Power estimates amplify synchronization and differentiation errors. To first order, the error in a wrench power includes

$$\delta P \simeq \delta \mathbf{F} \cdot \mathbf{v} + \mathbf{F} \cdot \delta \mathbf{v} + \delta \mathbf{n} \cdot \boldsymbol{\omega} + \mathbf{n} \cdot \delta \boldsymbol{\omega}. \quad (11.26)$$

These errors can be correlated; uncertainty in integrated work also depends on timing, filtering, interval endpoints and their covariance. Compare a reported transfer difference with that uncertainty, test plausible segment properties, and inspect energy residuals without treating residual magnitude alone as a diagnosis of the missing mechanism. Force-plate, motion-capture and club measurements must share time and frame conventions.

The resulting argument is interconnected but specific: configuration and inertia determine how efforts and constraints affect acceleration; actual velocities determine their power; constitutive behavior determines storage and loss; control changes the trajectory and delivery event; observations constrain which of those explanations can be distinguished. A convincing coaching claim requires a controlled intervention or a validated counterfactual model with uncertainty, not just a mechanically plausible story. The mechanics makes better questions possible without prescribing a universal hip-braking or lag strategy.

11.10 Chapter Exercises: Energy Transfer

1. Derive the mechanical-energy balance from the manipulator equation. Explain why setting the scalar Coriolis contraction to zero is unnecessary. Then derive the power of an ideal constraint with explicit time dependence.
2. At an arm-club hinge, force power into the club is 12 W. The base motor moment is 7 N m, hinge motor moment 3 N m, and arm and club absolute angular rates are 2 and 5 rad/s. Find both body mechanical-energy rates, hinge-motor power and total motor power. Assume all other interactions are conservative.
3. Check the five-state storage ledger between the second and third states. If the second state is the beginning of the downswing, explain which initial energy and work interval belong in its balance. Can a video lag angle determine the initial elastic store?
4. For the stated isolated impact, derive the speeds, energy-transfer ratio and loss from momentum and restitution. Explain why residual head energy and swing losses cannot be relabeled as collision dissipation.
5. Derive M , h and G for the two-link relative chart. For the aligned unit-rod braking case, compute both absolute angular accelerations and both body-energy rates. Explain how the outer rod can gain angular speed while losing energy.
6. Design a comparison of two release strategies that could distinguish active work, elastic release and interface transfer. State the common initial conditions, actuation constraints, delivery event, measured quantities and uncertainties. Explain why equal torques or a sequence of speed peaks is insufficient evidence of equal effort or one-way transfer.

11.11 Worked Answers

1. Differentiate $K = \dot{q}^T M \dot{q}/2$, substitute the equation of motion, and add $\dot{V} = \dot{q}^T \nabla V$. The terms $\dot{q}^T \dot{M} \dot{q}/2$ and $\dot{q}^T C \dot{q}$ cancel by the compatible Christoffel identity, leaving actuator, nonconservative and constraint powers. From $J\dot{q} + \Phi_t = 0$, constraint power is $-\lambda^T \Phi_t$; only the stationary ideal case guarantees zero through this argument. A moving support must remain in the work account.
2. Club moment power is $3(5) = 15$ W, so its energy rate is $12 + 15 = 27$ W. The arm receives base power $7(2) = 14$ W and loses 12 W through the force and $3(2) = 6$ W through the hinge moment: its rate is -4 W. Hinge-motor power is $3(5 - 2) = 9$ W; total motor power is $14 + 9 = 23$ W. The body rates sum to the same 23 W. Equating club moment power to motor power would miss the arm's opposite moment entry.
3. Stored energy rises from 25 to 87 J, a 62 J increase. Active work over that interval is $90 - 25 = 65$ J and loss is 3 J. Kinetic energy rises 80 J, supplied by 65 J active work plus 10 J gravitational release plus 8 J elastic release, minus 3 J loss. Starting at the second state means 25 J initial storage and work counted only afterward. Lag alone supplies neither a spring stiffness nor its rest deformation; an ideal hinge can have that angle with zero elastic energy.

4. Solve $m_h u_h = m_h v_h + m_b v_b$ and $v_b - v_h = e u_h$. The stated speeds follow. Divide $m_b v_b^2/2$ by $m_h u_h^2/2$ to obtain the ratio, about 0.492094. Subtract both final kinetic energies from the initial energy to obtain $\mu(1 - e^2)u_h^2/2$, about 13.6148 J. Residual head energy is still mechanical energy, and earlier swing dissipation occurred outside the collision interval.
5. Differentiate the kinetic-energy quadratic twice with respect to rates to obtain M ; Euler–Lagrange differentiation gives both components of h . Differentiate gravitational potential to obtain both entries of G . In the aligned unit-rod case the absolute-coordinate mass matrix is $\begin{bmatrix} 4/3 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}$, with zero instantaneous velocity bias and gravity. Solving against $(-1, 0)^T$ gives $(-12/7, 18/7)^T$. Inner energy rate is $(1/3)(2)(-12/7) = -8/7$ W; total motor power is -2 W, so outer energy rate is $-6/7$ W. Independent center and hinge-power calculations give the same result. Increasing rotational energy is outweighed by decreasing translational energy; angular speed alone does not measure total outer-body energy.
6. Use identical positions, velocities and internal elastic states, with the same stated motor capacities and support conditions. Define release as a specific control or constraint change, including any switching impulse or parameter work. Evaluate at the same physical delivery event and report its orientation and position as well as speed. Integrate separate active, elastic, dissipative and interface powers; compare signed work, positive work, residual energy and uncertainty across strategies. Equal torque commands can yield different rates and therefore different work. Peak order alone leaves those accounts unidentified. Repeat the comparison under plausible measurement and model variations before attributing a robust benefit to timing.

Chapter 12

Ground-Reaction Forces and Constrained Dynamics

Ground-reaction force (GRF) is the force exerted by the ground on a body in contact with it. In golf, the practically useful observable is usually a **ground-reaction wrench**: three force components and three moments expressed about a declared force-plate origin. That wrench changes whole-body linear and angular momentum. It is not a direct neural command, a direct measure of muscular effort, or by itself a measure of energy delivered to the club.

12.1 Scope of This Chapter

This chapter distinguishes three operations that are often conflated:

1. **Measurement:** a force plate measures an external contact wrench.
2. **Inverse dynamics:** a model reconciles measured kinematics, inertia, and external wrenches to estimate generalized forces [Nesbit, 2005, Gatt et al., 1998].
3. **Counterfactual attribution:** the same declared model is reevaluated at a measured state after selected velocity or control terms are set to zero.

The third operation is model dependent. It does not turn a force-plate trace into a unique estimate of muscle activation.

The central question is how internally generated motion and external support jointly shape club delivery. A foot constraint changes the accelerations that joint inputs can produce; those accelerations determine the required contact reaction. Contact and muscle action therefore cannot be assigned independent roles merely by naming one a reaction and the other a control. Momentum, work, and input attribution describe different aspects of the same motion.

12.2 Declare the System and the Frame

Use an inertial laboratory frame with z upward. Declare the horizontal axes explicitly, for example x along the target line and y completing a right-handed frame. Anatomical anterior-posterior and medial-lateral directions need a stated transformation; their names do not automatically identify the laboratory axes for a golfer whose body rotates.

Unless stated otherwise, the system below includes the **golfer and club**, ends before ball contact, and neglects aerodynamic loads. Hand-club interaction forces and couples are then internal and cancel in whole-system balances. If the club is excluded, its wrench on the hands is external and must be included. During ball impact, include the ball's contact

impulse or enlarge the system again. Use the mass and center of mass of the chosen system consistently.

The third-law partner of the ground's force on the shoe is the shoe's force on the ground. Gravity and the normal reaction both act on the golfer, so they are **not** a third-law pair. Their equality in magnitude in a simple static stance follows from zero acceleration. Dynamic contact need not equal weight.

12.3 Force, Moment, Center of Pressure, and Free Moment

Let a force plate report force

$$\mathbf{F} = (F_x, F_y, F_z)$$

and moment about its origin

$$\mathbf{M}_O = (M_x, M_y, M_z).$$

The axes, signs, origin, and units must accompany the data. Suppose the flat contact surface is $z = z_s$ relative to the reporting origin O . Transport the moment to $C = (x_C, y_C, z_s)$ using

$$\mathbf{M}_C = \mathbf{M}_O - \mathbf{r}_{OC} \times \mathbf{F}.$$

The **center of pressure** (COP) is the surface location where the two tangential moment components vanish. For sufficiently large F_z ,

$$\begin{aligned} x_C &= \frac{z_s F_x - M_y}{F_z}, \\ y_C &= \frac{M_x + z_s F_y}{F_z}, \\ M_{C,z} &= M_z - (x_C F_y - y_C F_x). \end{aligned}$$

At a surface origin, $z_s = 0$, these reduce to $x_C = -M_y/F_z$ and $y_C = M_x/F_z$. The remaining $M_{C,z}$ is the vertical **free moment**. Thus the complete wrench generally requires a force **and** a couple at the COP: not all moments vanish there. The free moment represents a distributed contact couple and cannot be reconstructed from a planar point-contact force alone. COP is also a ratio, so it becomes unstable as F_z approaches zero. Low-load samples need an explicit mask rather than a silently reported COP.

For example, a surface point $C = (0.1, -0.05, 0.04)$ m, force $\mathbf{F} = (100, -50, 800)$ N and free moment $M_{C,z} = 8$ N m produce $\mathbf{M}_O = (-38, -76, 8)$ N m. Omitting the 0.04 m surface offset gives the wrong COP even though the force and moment data are otherwise correct.

Pressure insoles measure plantar pressure, primarily the normal component. Shear forces and free moment require additional sensing or an empirically validated estimator. A study that predicts six-axis force-plate quantities from pressure data must label them as predictions, not direct measurements [Joo et al., 2016].

12.4 Whole-Body Momentum Balance

For a golfer-plus-club system of total mass m , before ball impact and after neglecting small aerodynamic loads, the linear momentum balance is

$$\mathbf{F}_{\text{GRF}} + m\mathbf{g} = m\ddot{\mathbf{r}}_{\text{COM}}, \quad (12.1)$$

where $\mathbf{g}$ points downward. With a vertical axis positive upward and $g = 9.81 \text{ m s}^{-2}$,

$$F_z - mg = m\ddot{z}_{\text{COM}}, \quad \ddot{z}_{\text{COM}} = \frac{F_z}{m} - g.$$

For example, $F_z = 1200 \text{ N}$ for a 90 kg system gives $\ddot{z}_{\text{COM}} = 1200/90 - 9.81 = 3.52 \text{ m s}^{-2}$ upward. A formula yielding 23.1 m s^{-2} would have added gravity twice. If $F_x = 200 \text{ N}$ at the same instant, $\ddot{x} = 2.22 \text{ m s}^{-2}$. These are instantaneous accelerations, not velocities or distances. Downward motion can coexist with upward acceleration: braking a descent requires $F_z > mg$ under this system boundary. Downward acceleration requires $F_z < mg$, irrespective of whether the system is currently moving up or down.

Integrating Equation 12.1 over $[t_1, t_2]$ gives

$$\int_{t_1}^{t_2} \mathbf{F}_{\text{GRF}} dt = m\Delta\dot{\mathbf{r}}_{\text{COM}} - m\mathbf{g}(t_2 - t_1).$$

This is a strong data-quality check: the measured resultant impulse should agree with the change in whole-system momentum under the same filtering, events, mass, and coordinate convention. Impulse has units N s ; mechanical work has units N m . An impulse is not an energy quantity and does not, by itself, specify an ordering of segment-speed peaks. The legs, trunk, arms and club form a coupled system with simultaneous interactions. A proximal-to-distal sequence is an outcome to explain with kinematics, joint powers, inertia and inputs, not a theorem of the whole-body impulse balance. A smaller distal mass alone does not prove greater speed.

The angular-momentum balance about the system COM is

$$\dot{\mathbf{H}}_{\text{COM}} = \sum_i (\mathbf{r}_i - \mathbf{r}_{\text{COM}}) \times \mathbf{F}_i + \sum_i \mathbf{M}_{i,\text{free}}.$$

Consequently, the same resultant force can have different rotational effects when its line of action, bilateral allocation, or free moment differs. Uniform gravity has zero resultant moment about this COM. If another external wrench is retained, include its moment too. The complete resultant force and moment determine the whole-system momentum rates; they generally do not determine every individual foot wrench or internal joint load.

12.5 Contact Work Is a Conditional Statement

For a rigid contacting body, contact power can be represented at a point C using its wrench and the body's instantaneous rigid-body velocity evaluated at that same location:

$$P_c = \mathbf{F}_c \cdot \mathbf{v}_c + \mathbf{M}_c \cdot \boldsymbol{\omega}_c.$$

For an **ideal rigid, stationary, no-slip contact**, the constrained contact velocity is zero, and the ideal constraint wrench does no work. This useful idealization does not imply

zero muscular work. It also does not apply without qualification when the shoe or surface deforms, the support moves, or slip occurs. Ideal rolling without slip on a stationary rigid surface can still have zero contact power; rolling alone is not proof of nonzero work. A COP is an equivalent-wrench location, not necessarily a material point whose velocity can be inserted into a power calculation.

In particular, differentiating the COP coordinates gives the velocity of a moving **location**, not the contacting body's material velocity there. On a stationary planted rigid foot, the COP can travel while all contact material points remain stationary. For a deformable interface, the general power is the surface integral $\int_S \mathbf{t} \cdot \mathbf{v} dA$ of traction times local material velocity; one resultant wrench is insufficient without assumptions about the velocity field.

The wrench formula is independent of the reference point when transported consistently. With $\mathbf{r} = \mathbf{r}_{OC}$,

$$\begin{aligned}\mathbf{v}_C &= \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}, \\ \mathbf{M}_C &= \mathbf{M}_O - \mathbf{r} \times \mathbf{F}, \\ \mathbf{F} \cdot \mathbf{v}_C + \mathbf{M}_C \cdot \boldsymbol{\omega} &= \mathbf{F} \cdot \mathbf{v}_O + \mathbf{M}_O \cdot \boldsymbol{\omega}.\end{aligned}$$

The cancellation follows from the scalar triple product. Mixing a moment about one origin with the velocity at another violates this identity.

Muscles and tendons may perform positive, negative, and elastic work while an ideal external contact wrench has zero instantaneous power at its constrained point. GRF can still redirect momentum and alter how internally supplied energy is distributed among segments.

Multiplying the COM balance by $\dot{\mathbf{r}}_{\text{COM}}$ yields

$$\frac{d}{dt} \left(\frac{1}{2} m \|\dot{\mathbf{r}}_{\text{COM}}\|^2 \right) = (\mathbf{F}_{\text{GRF}} + m\mathbf{g}) \cdot \dot{\mathbf{r}}_{\text{COM}}.$$

This is a translational COM kinetic-energy identity. The term $\mathbf{F}_{\text{GRF}} \cdot \dot{\mathbf{r}}_{\text{COM}}$ is not generally the actual contact power. Total mechanical energy also includes segmental motion relative to the COM, gravitational potential, and retained elastic storage. Muscle work and release of previously stored energy can increase kinetic energy while a stationary ideal contact does no work. Deformation may store or dissipate energy, and a moving support may exchange work with the system.

12.6 Constrained-Reaction Mechanics

At one state, write the multibody equations as

$$\begin{aligned}M(q)\ddot{q} + h_0(q) + h_v(q, \dot{q}) \\ = B(q)u + Q_{\text{ext}} + J(q)^\top \lambda.\end{aligned}\tag{12.2}$$

with acceleration constraint

$$J(q)\ddot{q} + \gamma(q, \dot{q}) = 0.$$

Use independent local generalized coordinates q so that M is positive definite. If a model uses redundant quaternion coordinates, formulate these equations in independent tangent velocities with the corresponding kinematic map rather than blindly inverting a redundant coordinate mass matrix. Here h_0 is velocity independent, h_v is velocity dependent, u is the

declared controllable generalized input, Q_{ext} contains other known applied loads, J is the active-contact Jacobian, and λ is the reaction in contact coordinates. Define

$$A = JM^{-1}J^T.$$

For the formulas immediately below, contact constraints are stationary and holonomic, $\Phi(q) = 0$, with $J = \partial\Phi/\partial q$ and $\gamma = \dot{J}\dot{q}$. Define $h_0 = h(q, 0)$ and $h_v = h(q, \dot{q}) - h(q, 0)$, so $h_v(q, 0) = 0$. Hold the declared Q_{ext} fixed in the pointwise comparisons. The matrix B and retained external loads must be independent of u for the stated affine split.

Solving the motion equation for $\ddot{q}$, substituting into the acceleration constraint and collecting the $JM^{-1}J^T\lambda$ term gives the following result. If J has full row rank, A is positive definite; adequate conditioning is also needed for a reliable numerical solution:

$$\begin{aligned} d &= h_0 + h_v - Bu - Q_{\text{ext}}, \\ \lambda &= A^{-1} [JM^{-1}d - \gamma]. \end{aligned} \tag{12.3}$$

The reaction is affine in the declared generalized inputs, even though it is not itself an independently commanded input. A reaction is therefore not automatically a drift-only effect. Eliminating the constraint reactions changes both the drift and input terms of the constrained state equation. It can be separated as

$$\lambda = \lambda_0 + \lambda_v + \lambda_u + \lambda_e,$$

with

$$\begin{aligned} \lambda_0 &= A^{-1}JM^{-1}h_0, \\ \lambda_v &= A^{-1}(JM^{-1}h_v - \gamma), \\ \lambda_u &= -A^{-1}JM^{-1}Bu, \\ \lambda_e &= -A^{-1}JM^{-1}Q_{\text{ext}}. \end{aligned}$$

These components are exact only for the declared model and state. Errors in kinematics, inertial parameters, contact assignment, external loads, and filtering enter the residual. Dynamic inconsistency is therefore a primary falsification outcome, not a nuisance to hide [Sturdy et al., 2022, Werling et al., 2023].

If constraint rows are redundant, do not use the ordinary inverse of A . Determine the independent constraints and the admissible reaction set. A minimum-norm pseudoinverse can select one solution but cannot establish a unique physical distribution without additional measurements or a justified contact/compliance model.

12.6.1 A Coupled Example With Zero Contact Work

Consider a deliberately simple two-coordinate translational model, with both coordinates measured in meters, M in kilograms and generalized forces in newtons:

$$M = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad J = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Let $h_0 = h_v = Q_{\text{ext}} = \gamma = 0$, and constrain $q_1 = 0$. The direct equations are

$$2\ddot{q}_1 + \ddot{q}_2 = \lambda, \quad \ddot{q}_1 + 2\ddot{q}_2 = u.$$

Consequently $\ddot{q}_1 = 0$, $\ddot{q}_2 = u/2$ and $\lambda = u/2$. The Schur formula gives the same result because $A = 2/3$ and $JM^{-1}B = -1/3$. The contact reaction depends on input, yet its power $\lambda\dot{q}_1$ is zero. This is an algebraic counterexample, not a calibration of shoulder torque into foot force. A human estimate requires the whole body's inertial coupling, joint geometry, state and active contacts. Planted feet do not in themselves prohibit torso rotation.

12.6.2 Pointwise Ground-Reaction ZTCF

The pointwise ground-reaction zero-torque counterfactual (ZTCF) sets the declared controllable input to zero at the achieved $(q, \dot{q})$ while retaining declared non-control external loads:

$$\lambda_{\text{ZTCF}} = \lambda_0 + \lambda_v + \lambda_e.$$

It answers what reaction the model requires at this state with zero declared control. It is not a no-muscle experiment and does not describe what the body would do after controls were removed. The retained state may contain activation or elastic energy from prior action; zero generalized torque and zero muscle excitation are different interventions.

12.6.3 Pointwise Ground-Reaction ZVCF

The pointwise ground-reaction zero-velocity counterfactual (ZVCF) evaluates the same configuration and declared inputs after velocity is zeroed:

$$\lambda_{\text{ZVCF}} = \lambda_0 + \lambda_u + \lambda_e.$$

ZTCF and ZVCF are not forward simulations. They overlap in $\lambda_0 + \lambda_e$ and must not be added to reconstruct the total. The additive attribution is configuration plus velocity plus control plus other external load.

12.6.4 Feasibility and Moving Constraints

An equality-constrained calculation is not sufficient to establish a physical contact. A unilateral contact requires a nonnegative compressive normal reaction; sticking additionally requires a tangential force inside the declared friction cone, for example $\|\mathbf{F}_t\| \leq \mu F_n$. A finite support patch also bounds admissible moments and COP. Test these conditions for the full model and for each recomputation. A negative normal reaction means that the retained mode would require adhesion, not that the ground actually pulls the shoe downward. Lift-off or slip requires a different mode.

For the toy model above, interpreting $q_1 \geq 0$ as a normal gap makes a negative u incompatible with the assumed closed, non-adhesive contact because $\lambda = u/2 < 0$. Analogous failures can occur when zeroing an input in a loaded model. For example, retain $Q_{\text{ext}} = (1, 0)^T$ N in the toy model. Then $\lambda = u/2 - 1$ N: $u = 4$ N gives an admissible 1 N reaction, while zero input would require -1 N. Releasing the contact instead gives $\ddot{q}_1 = 2/3 \text{ ms}^{-2}$, consistent with separation. Report an infeasible fixed-mode diagnostic as such; do not silently relabel it a realizable swing.

For a moving support $\Phi(q, t) = 0$, the velocity constraint is $J\dot{q} + \Phi_t = 0$, and the acceleration bias contains prescribed-motion terms that may survive at zero velocity. Zeroing all velocities may violate the velocity constraint itself. Reevaluate the complete equations,

specify which support motion and external loads are retained, and label any velocity projection as a different intervention. The compact ZVCF expression above is valid only under its stationary-constraint and held-load assumptions. Even for a stationary support, a drag load reevaluated at zero velocity changes λ_e unless the protocol explicitly freezes that load.

12.7 What a GRF Residual Cannot Establish

Measured GRF minus modeled ZTCF is a residual containing input-induced reaction, model error, measurement error, and omitted external loads. It cannot uniquely identify muscle torques. Even with a perfect resultant wrench, the model cannot uniquely determine the bilateral allocation when the contact system is redundant. A pseudoinverse supplies a numerical allocation, not new physical information.

With known kinematics, inertia, and individual external wrenches, inverse dynamics estimates the **net generalized joint load** required by the model. That load can combine muscle, tendon, passive tissue and other modeled effects. It is not a measurement of each muscle force, activation, effort, or injury risk. Differentiation and filtering errors matter, as do unmeasured club loads and contact allocation. Multiple muscle-force combinations can produce the same net joint moment.

12.8 Interpreting Common Golf Measurements

12.8.1 Vertical Force

A vertical peak may combine configuration support, velocity-dependent whole-body acceleration, and input-induced reaction. Peak magnitude alone does not identify which term dominates. ZTCF and ZVCF provide model-based tests of which features persist after declared terms are removed.

12.8.2 Horizontal Shear

Anterior-posterior and medial-lateral components are particularly sensitive to axis definitions, stance orientation, foot contact, and filtering. Report the laboratory-to-golfer coordinate transform and avoid changing signs implicitly for left-handed participants.

12.8.3 Center of Pressure

COP locates an equivalent force together with the residual free couple. It is not the center of mass, not “weight transfer,” and not energy transfer. Ball and Best observed more than one recurring COP strategy across skill levels [Ball and Best, 2007]. The literature therefore does not justify one universal COP or GRF waveform as the correct pattern.

12.8.4 Bilateral Forces and Free Moment

Two force plates provide more information than a combined resultant, but bilateral decomposition remains sensitive to each foot’s frame, crossover, partial contact, and plate assignment. The vertical free moment represents a distributed torsional contact effect. It

should not be equated with joint torque or with the two-hand force couple in a club model merely because the signs look similar.

12.8.5 Association With Clubhead Speed

Golf studies have associated selected lead- and trail-foot force/moment features with clubhead speed [Han et al., 2019], and shoe-ground torque differs across some skill and footwear conditions [Worsfold et al., 2008]. These associations do not establish that maximizing a peak will increase speed for an individual. Rachnavy and colleagues reported improved clubhead-speed association when foot-ground variables were combined with trunk and transfer variables [Rachnavy et al., 2026], but their cross-sectional mediation analysis is not a controlled causal intervention. The systematic review documents substantial methodological heterogeneity [Watson et al., 2026].

For the Rachnavy study, the initial regression block includes skill group as well as COP factors; its explained variance must not be credited to foot variables alone. Its constructed transfer factor is not automatically a dimensionless mechanical efficiency. Establish numerator, denominator, system boundary and power/impulse definitions before interpreting such a quantity as energy conversion. Neither that association nor a force peak establishes the benefit of a particular coaching intervention.

Coaching descriptions such as sway, hanging back and early extension require operational kinematic definitions and synchronized observation. A GRF or COP trace alone does not uniquely identify one of these patterns or prove that it caused a poor strike. Changes should be evaluated against the golfer's task outcome, repeatability and relevant constraints, rather than inferred from agreement with a single reference waveform.

12.9 Falsifying a Drift-Based GRF Model

A defensible human study needs synchronized bilateral six-axis force plates, whole-body and club kinematics, participant-specific or uncertainty-bounded segment inertial parameters, and declared filtering and event rules. Model development and threshold selection should be separated from evaluation on held-out participants.

At minimum, report for every force and moment component:

- bias, RMSE, and an explicitly normalized RMSE;
- waveform R^2 with amplitude error reported alongside it;
- vector impulse error over preregistered phases;
- peak magnitude and timing error;
- COP error only above a declared vertical-force threshold;
- free-moment error for three-dimensional contact models;
- residual pelvis forces and moments as dynamic consistency diagnostics; and
- sensitivity to inertial parameters, filtering, coordinate transforms, and contact allocation.

First specify the tested hypothesis: for example, a declared zero-input recomputation predicts a particular phase’s vertical impulse within a preregistered tolerance. Reject or narrow it when the full model fails its validation, the zero-input prediction fails held-out tests, or the attributed share changes qualitatively under plausible assumptions. Do not convert one failed implementation into a disproof of every possible drift model.

Include a center-of-mass acceleration baseline. For the total resultant force, the COM equation is already an exact balance under its assumptions, not an approximation that added model complexity must beat. A richer model should justify additional claims about individual foot wrenches, internal loads or interventions on appropriately observed outcomes. Compare shared outputs with the same available inputs, filtering, events and error metrics.

Declare whether force measurements were used to fit kinematics, inertial parameters or residual corrections. In-sample agreement after such fitting is a consistency check, not an independent prediction of the fitted force. AddBiomechanics uses measured ground forces in its dynamics optimization [Werling et al., 2023]; RRA parameter optimization reduces manual tuning of dynamic consistency [Sturdy et al., 2022]. These methods are useful, but neither method’s success alone validates a golf-specific drift attribution. Separate calibration from evaluation, and keep participant-specific calibration data distinct from held-out evaluation trials. In a predictive comparison, infer inputs without using the evaluation wrench that the model is supposed to predict.

12.10 Key Takeaways

- GRF is an external contact wrench and a constraint reaction, not a direct motor command.
- Whole-body resultant GRF is strongly constrained by center-of-mass acceleration, but bilateral allocation, COP, and free moment require additional spatial contact information.
- Ideal stationary no-slip contact does zero work at the constrained point; this conditional statement does not eliminate muscular, tendon, shoe, or surface work.
- Pointwise reaction ZTCF and ZVCF are overlapping diagnostics, not additive causes and not forward simulations.
- A measured-minus-ZTCF residual combines control-induced reaction with model and measurement error; it does not reveal unique muscle torques.
- Golfers exhibit multiple viable force and COP patterns. Evaluate individual task outcomes and uncertainty, and test claims against evidence.

12.11 Chapter Exercises

1. A 95 kg golfer-plus-club system stands still, with only gravity and ground forces external. Compute vertical GRF with upward positive.
2. For the same system, $F_z = 1350$ N during the downswing. Compute vertical COM acceleration. Can it be moving downward at that instant? Also compute the velocity change if this force remains constant for 0.10 s.

3. Explain when an ideal contact wrench does zero work and list three reasons a real shoe-ground interface may violate the idealization.
4. Derive Equation 12.3 from Equation 12.2 and the acceleration constraint.
5. Show algebraically why ZTCF plus ZVCF double-counts configuration and retained external-load reaction. State the assumptions needed for the compact ZVCF.
6. Give two different bilateral foot-force allocations with the same combined resultant and explain what additional measurements distinguish them.
7. Design a held-out test that compares a COM-only GRF baseline, ZTCF, and the full constrained model without splitting samples from the same participant across training and test sets.
8. Recover COP and free moment from the offset-surface numerical example. Explain why differentiating that COP path does not generally give contact power, and why inverse joint moments do not uniquely determine muscle force.

12.12 Worked Solutions

1. Static vertical balance gives $F_z = mg = 95(9.81) = 931.95$ N. If the 95 kg excluded a held club, its hand load would have to be included instead of silently applying the golfer-plus-club equation to golfer mass alone.
2. With upward positive,

$$\begin{aligned}
 \ddot{z} &= 1350/95 - 9.81 \\
 &= 4.4005 \text{ m s}^{-2}, \\
 \Delta \dot{z} &= (1350 - 931.95)(0.10)/95 \\
 &= 0.4401 \text{ m s}^{-1}.
 \end{aligned}$$

A negative initial vertical velocity is compatible with that positive acceleration: the descent is being braked. Acceleration alone does not reveal the current direction of travel or displacement.

3. For an ideal rigid, stationary, no-slip contact, the reaction pairs with zero constrained velocity, so its power is zero. A moving support, material deformation with nonzero local traction power, or tangential slip can invalidate the idealization. Pure rolling without slip need not do so. Muscle or elastic work elsewhere can remain nonzero. A particle constrained to a stationary frictionless curved track provides a simple example: the normal force redirects velocity while remaining perpendicular to it.
4. Write $h = h_0 + h_v$. Substitution gives

$$\begin{aligned}
 0 &= JM^{-1}(Bu + Q_{\text{ext}} + J^T \lambda - h) + \gamma, \\
 A\lambda &= JM^{-1}(h - Bu - Q_{\text{ext}}) - \gamma.
 \end{aligned}$$

Invert A only for independent constraint rows. In the two-coordinate example, direct solution and this formula both give $\lambda = u/2$. If the input channel has more independent

components than the measured reaction, the reaction-to-input inverse cannot generally be unique.

5. Under the stated stationary-contact, zero-velocity-bias and held-load assumptions,

$$\begin{aligned}\lambda_{\text{ZTCF}} + \lambda_{\text{ZVCF}} &= 2\lambda_0 + \lambda_v + \lambda_u + 2\lambda_e \\ &= \lambda + \lambda_0 + \lambda_e.\end{aligned}$$

They double-count the retained baseline. Moving supports or reevaluated velocity-dependent loads require recomputing the relevant terms, and any retained unilateral/sticking mode must pass its feasibility conditions.

6. Place point contacts at $(-a, 0, 0)$ and $(a, 0, 0)$, with total vertical force $F > 0$. Allocations $(F/2, F/2)$ and $(3F/4, F/4)$ have the same resultant $(0, 0, F)$ but different moments about the midpoint: $M_y = 0$ and $aF/2$. A measured total moment or separate plates distinguishes these cases. Even a complete total wrench need not fix all individual loads: add equal and opposite horizontal forces directed along the line joining the contacts. Their total force and moment are zero. Separate shear measurements can distinguish this self-equilibrated allocation, provided friction permits it.
7. Split by participant before choosing model structure, thresholds or regression parameters. Define any allowed personal calibration separately. Obtain synchronized kinematics and plate data, but reserve the evaluation plate wrench as an outcome. Use kinematics and declared inertial parameters for the COM resultant baseline; use independently specified or estimated inputs for the full constrained prediction and zero the declared channel for ZTCF. Report errors, uncertainty and contact feasibility on matching outcomes. Individual-foot moments provide additional tests beyond the COM resultant. If inverse dynamics used the same force trace to infer the inputs, label the reconstructed full-model agreement as an identity or consistency check and obtain separate evidence for predictive claims.
8. With $z_s = 0.04$ m,

$$\begin{aligned}x_C &= [0.04(100) - (-76)]/800 \\ &= 0.10 \text{ m}, \\ y_C &= [-38 + 0.04(-50)]/800 \\ &= -0.05 \text{ m}, \\ x_C F_y - y_C F_x &= 0.10(-50) - (-0.05)(100) \\ &= 0, \\ M_{C,z} &= 8 - 0 = 8 \text{ N m}.\end{aligned}$$

For power use the rigid body's velocity field at C , not the rate at which the equivalent COP location changes. For deformable contact, use local traction power or a justified reduction. A net joint moment similarly summarizes combined actions: antagonistic muscle forces can change together while leaving their net moment unchanged, so inverse dynamics alone cannot identify individual muscle forces.

Chapter 13

From Muscle Forces to Joint Torques

In Plain Language 13.1. The Bridge Between Anatomy and Physics

A muscle pulls on a changing anatomical path. That pull can turn several joints, resist their motion, or transmit power between them. The same net joint torque can arise from different muscle forces. This chapter connects those observations through geometry, virtual work and force limits, then asks what they let us infer about a golf swing. The numerical examples are constructed teaching cases, not measurements of a golfer.

13.1 The Problem: Muscles Pull, Joints Rotate

A skeletal muscle contains fibers with different orientations and internal structures. A tractable musculoskeletal model often represents a muscle-tendon unit by one or more tensile paths. A compartment's transmitted tendon tension $F_j \geq 0$ acts along its modeled path; it is not automatically the force of every fiber in the anatomical muscle. Fiber force, pennation, tendon compliance and activation dynamics determine how that tension develops. The muscle-force-generation chapter treats those physiological steps.

Here the immediate question is geometric: given a configuration and a set of path tensions, what generalized forces do they exert on the skeleton? We use independent coordinates $\mathbf{q}$, increasing path length ℓ_j to mean lengthening, and positive tension to mean pulling. These conventions fix the signs of torque and power. A torque is not itself a rotation: acceleration also depends on inertia, other forces and constraints.

For the initial derivation, paths are massless and frictionless, carry uniform tension, and depend only on configuration. Their routing constraints do no additional work. Moving guides, friction, path inertia and independently prescribed attachment motion require additional force or power terms. Declaring this boundary is part of the model, not an anatomical assertion about every tissue.

13.2 Moment Arms and the Geometry of Force

13.2.1 The Basic Definition

Let $\boldsymbol{\rho}$ be a position vector from a point on a fixed rotation axis to a force application point, let $\mathbf{e}$ be the axis unit vector, and let $\mathbf{f} = F\mathbf{d}$ with $\|\mathbf{d}\| = 1$. The moment vector and its

signed axial component are

$$\mathbf{m} = \boldsymbol{\rho} \times \mathbf{f}, \quad \tau_e = \mathbf{e} \cdot \mathbf{m} = Fr_e, \quad r_e = \mathbf{e} \cdot (\boldsymbol{\rho} \times \mathbf{d}).$$

The scalar r_e has units of length and can be positive, negative or zero. Changing the reference point along the same axis leaves τ_e unchanged. Reversing the positive axis reverses both r_e and τ_e .

Definition 13.1. Moment Arm

A signed moment arm is the torque about a specified positive rotational coordinate per unit positive tensile force. For a single rigid rotation and point force it is $r_e = \mathbf{e} \cdot (\boldsymbol{\rho} \times \mathbf{d})$. In a planar problem its magnitude is the perpendicular distance from the pivot to the force line. In three dimensions, distance from a force line to an axis alone does not determine the axial torque; force direction and axis projection matter.

The magnitude $\|\mathbf{m}\| = \|\boldsymbol{\rho}\|F \sin \alpha$, where α is the angle between the position and force vectors, describes the full moment about the reference point. It need not equal $|\tau_e|$. For a wrench, a force perpendicular to the handle and tangential to the intended rotation is effective. Rotating the handle in its plane does not reduce leverage if the applied force stays tangential. A force directed along the handle produces no moment about the pivot.

In Plain Language 13.2. A Force Needs the Right Direction

A long handle helps only to the extent that the force has a component that turns the specified axis. Pulling directly along the handle adds load without the intended turning effect. Always draw the force direction as well as the attachment location.

13.2.2 Example: The Biceps at the Elbow

Suppose a modeled elbow flexor has a positive moment arm of 0.04 m and transmits 500 N. Its flexion torque contribution is 20 Nm. These are illustrative inputs. The result is neither a measured biceps maximum nor the net elbow torque; antagonists, passive tissues and other muscles may also contribute.

Murray, Delp and Buchanan measured elbow-muscle excursion in two cadaver specimens and compared configuration-dependent moment arms with a model. Their results support dependence on elbow and forearm position, not a universal maximum at full extension or one optimum angle for every person. The small specimen sample and modeled paths also limit population inference. [Murray et al., 1995] A subject-specific estimate needs the relevant geometry, coordinate convention and uncertainty.

13.2.3 Moment Arms Are Configuration-Dependent

At fixed tension, changing r_e changes torque. At fixed desired torque, a smaller nonzero $|r_e|$ requires larger tension in a one-actuator example. Neither statement establishes the largest torque a person can produce: attainable tension also changes with fiber length, velocity, tendon state and activation. A posture with greater geometric leverage can have lower force capacity. Geometry and physiology must be evaluated together.

13.2.4 Moment Arm as a Function of Joint Angle

Write the length of path j as $\ell_j(\mathbf{q})$. In a smooth coordinate region, a consistent moment-arm field comes from one length function. Independently fitting each moment arm can violate this requirement. For example, the proposed two-coordinate field

$$r_1 = 0.02 + 0.01q_2, \quad r_2 = 0.01$$

cannot equal $-\nabla\ell$ throughout a simply connected region: its cross derivatives disagree. Around the counterclockwise unit square in (q_1, q_2) , $\oint \mathbf{r} \cdot d\mathbf{q} = -0.01$ m. A fixed 100 N tension would then appear to do -1 J while a single-valued path returns to the same length. That is a geometric inconsistency for the declared workless path, not evidence of muscle dissipation. A real muscle can dissipate energy when its tension changes during a cycle.

This check is local to smooth, configuration-only routing. Contact transitions and wrapping changes require appropriate piecewise treatment. Consistent length, moment-arm and velocity representations are preferable to unrelated fits. The Rice NMSM surrogate-model documentation makes this relationship explicit. [[Rice University NMSM Pipeline](#), n.d.]

13.3 The Muscle Jacobian: From Muscle Space to Joint Space

13.3.1 The Muscle Jacobian Matrix

For m tensile paths and n independent coordinates, define the length Jacobian and torque map by

$$L_{ji}(\mathbf{q}) = \frac{\partial \ell_j}{\partial q_i}, \quad \mathbf{L} \in \mathbb{R}^{m \times n}, \quad \mathbf{r} = -\mathbf{L}^T \in \mathbb{R}^{n \times m}.$$

Rows of $\mathbf{L}$ identify muscles; rows of $\mathbf{r}$ identify coordinates. The length rate is $\dot{\ell} = \mathbf{L}\dot{\mathbf{q}}$. For the assumed workless paths, positive tension pulls toward shorter length, so skeletal virtual work is

$$\delta W_{\text{skel}} = -\mathbf{F}^T \delta \ell = -\mathbf{F}^T \mathbf{L} \delta \mathbf{q} = \boldsymbol{\tau}_M^T \delta \mathbf{q}.$$

Because the independent virtual displacements are arbitrary,

$$\boxed{\boldsymbol{\tau}_M = -\mathbf{L}^T \mathbf{F} = \mathbf{r} \mathbf{F}}, \quad \boxed{P_{\text{skel}} = \boldsymbol{\tau}_M^T \dot{\mathbf{q}} = -\mathbf{F}^T \dot{\ell}}.$$

The minus sign follows from positive tension and positive lengthening. A spatial Jacobian transpose maps a force acting in the positive endpoint direction; its familiar plus sign does not justify applying $+\mathbf{L}^T$ to a positive tensile magnitude. The force variable and its work-conjugate displacement must match.

Key Principle 13.1. The Muscle Jacobian

Declare lengthening, tensile-force sign, matrix dimensions and coordinate order together. With the conventions used here, the signed moment-arm matrix is $\mathbf{r} = -\mathbf{L}^T$ and torque is $\mathbf{r} \mathbf{F}$. Check the power identity as well as matrix multiplication. The same convention is used in the following muscle-force-generation chapter and in the cited OpenSim path documentation. [[OpenSim Contributors](#), 2024]

Sherman, Seth and Delp explain why coupled coordinates and workless-path assumptions matter when defining a muscle moment arm. [Sherman et al., 2013] Their excursion equations use a convention that should not be copied without reconciling signs. The explicit virtual-work derivation above determines the convention throughout this chapter.

13.3.2 Numerical Example: A Two-Joint Arm

Example 13.1. Moment Arms and Joint Torques in a Two-Joint Arm

Consider two rotational coordinates and two abstract muscle paths. Muscle 1 acts only at coordinate 1; muscle 2 acts at both. At the configuration of interest,

$$\mathbf{L} = \begin{pmatrix} -0.05 & 0 \\ -0.02 & -0.04 \end{pmatrix}, \quad \mathbf{r} = -\mathbf{L}^T = \begin{pmatrix} 0.05 & 0.02 \\ 0 & 0.04 \end{pmatrix}, \quad \mathbf{F} = \begin{pmatrix} 500 \\ 300 \end{pmatrix} \text{ N.}$$

The rotational length derivatives have units m/rad, conventionally written as moment-arm lengths in metres. Multiplication gives

$$\boldsymbol{\tau}_M = \begin{pmatrix} 25 + 6 \\ 0 + 12 \end{pmatrix} = \begin{pmatrix} 31 \\ 12 \end{pmatrix} \text{ Nm.}$$

At $\dot{\mathbf{q}} = (2, -3)^T$ rad/s, the path rates are $(-0.10, 0.08)^T$ m/s. Thus $\boldsymbol{\tau}_M^T \dot{\mathbf{q}} = 62 - 36 = 26$ W, also equal to $-500(-0.10) - 300(0.08)$. The second muscle is lengthening despite its positive torque at both coordinates, because the second coordinate rotates negatively fast enough. These nonsymmetric matrices expose a mistaken transpose that a symmetric example would hide.

13.3.3 Configuration-Dependent Jacobian

The map is linear in tension at a fixed configuration. It generally varies nonlinearly with configuration. That statement does not make muscle tension freely adjustable or constant during a swing; the activation and contraction equations supply its time evolution.

If generalized speeds satisfy $\dot{\mathbf{q}} = \mathbf{N}(\mathbf{q})\mathbf{v}$, power-conjugate muscle forces in speed coordinates are $\mathbf{Q}_v = -\mathbf{N}^T \mathbf{L}^T \mathbf{F}$. Do not replace $\dot{\mathbf{q}}$ by $\mathbf{v}$ unless $\mathbf{N}$ is the identity. For a holonomic coupling $\mathbf{q} = \boldsymbol{\phi}(\mathbf{s})$ with $\mathbf{T} = \partial\boldsymbol{\phi}/\partial\mathbf{s}$, the reduced torque is $\boldsymbol{\tau}_s = \mathbf{T}^T \boldsymbol{\tau}_M$. In the preceding example, $\mathbf{q} = (s, 2s)^T$ gives effective arms $(0.05, 0.10)$ m and $\tau_s = 31 + 2(12) = 55$ Nm. A partial derivative holding q_2 fixed would answer a different mechanical question.

For a prescribed time-dependent path, $\dot{\boldsymbol{\ell}} = \mathbf{L}\dot{\mathbf{q}} + \partial\boldsymbol{\ell}/\partial t$. Skeletal coordinate power is still $-\mathbf{F}^T \mathbf{L}\dot{\mathbf{q}}$, but $-\mathbf{F}^T \dot{\boldsymbol{\ell}}$ also contains the work associated with prescribed guide or attachment motion. A powered guide cannot silently be treated as workless.

13.4 Redundancy: More Muscles Than Joints

The relevant count is the number of modeled force channels compared with independent generalized coordinates, not a universal inventory of anatomical muscles and joints. If $m > n$, the linear map has a nontrivial null space, but anatomical counts alone do not

determine feasible recruitment. Model subdivisions, coupling constraints, posture and force bounds all matter.

13.4.1 The Null Space of the Muscle Jacobian

Given one algebraic solution $\mathbf{r}\mathbf{F}_0 = \boldsymbol{\tau}^*$, every algebraic solution is $\mathbf{F} = \mathbf{F}_0 + \mathbf{z}$ with $\mathbf{r}\mathbf{z} = \mathbf{0}$. A null vector is a change in forces; its negative components need not represent compressive muscle action if they reduce a previously positive tension. Nevertheless, the final forces must satisfy physiological constraints. Even with a nontrivial null space, the bounded feasible set can be empty or contain just one point.

Example 13.2. Algebraic Freedom and Feasible Tension

At one configuration, let

$$\mathbf{r} = \begin{pmatrix} 0.04 & 0.02 & -0.03 \\ 0 & 0.03 & 0.02 \end{pmatrix}, \quad \mathbf{F}_0 = \begin{pmatrix} 100 \\ 200 \\ 50 \end{pmatrix} \text{ N}, \quad \bar{\mathbf{F}} = \begin{pmatrix} 400 \\ 300 \\ 250 \end{pmatrix} \text{ N}.$$

Then $\mathbf{r}\mathbf{F}_0 = (6.5, 7)^T$ Nm and $\mathbf{n} = (13, -8, 12)^T$ is a null direction. Parameterize $\mathbf{F}(t) = \mathbf{F}_0 + t\mathbf{n}$, with t in newtons. Intersecting the three conditions $0 \leq F_j(t) \leq \bar{F}_j$ gives

$$-\frac{25}{6} \text{ N} \leq t \leq \frac{50}{3} \text{ N}.$$

At $t = -5$ N the third force is negative; at $t = 20$ N it exceeds its declared capacity. Both still satisfy the torque equation. This is why solving an unconstrained linear system is insufficient.

For a fixed box of force bounds, the torque set

$$\mathcal{T}(\mathbf{q}) = \{\mathbf{r}(\mathbf{q})\mathbf{F} : \mathbf{0} \leq \mathbf{F} \leq \bar{\mathbf{F}}\}$$

is the linear image of that box, a convex polytope. Here the separate coordinate maxima are 22 and 14 Nm. The pair (22, 14) is unattainable: achieving 14 Nm at coordinate 2 requires $F_2 = 300$ N and $F_3 = 250$ N, leaving at most 14.5 Nm at coordinate 1. Separate scalar torque limits therefore overestimate simultaneous capability. This box model is illustrative; activation, tendon and velocity constraints can further restrict attainable forces.

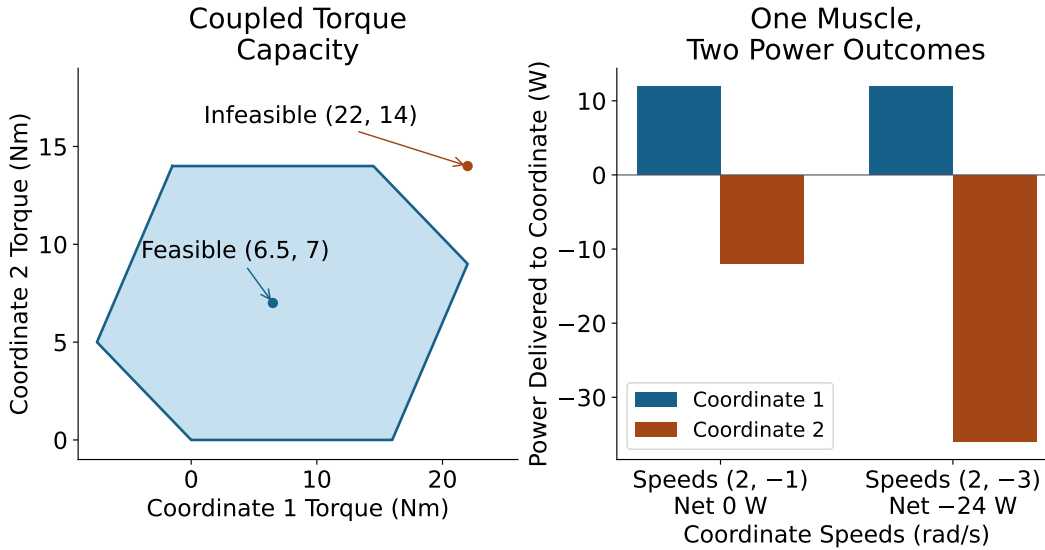


Figure 13.1: The Feasible Torque Set and Signed Joint Powers

Definition 13.2. Co-Contraction

Co-contraction is simultaneous activity of muscles with opposing torque effects about a specified coordinate. A null-space perturbation is not necessarily co-contraction: it can redistribute force between synergists. Conversely, changing antagonist activity need not preserve net torque. State the muscles, coordinate, baseline forces and compensation when interpreting a recruitment change.

13.4.2 Why Redundancy Matters for Golf

The same net torque can be associated with different internal forces, contact loads, activation costs and responses to perturbations. This makes muscle sharing important when studying tissue loading or robustness. It does not show that more co-contraction is always useful, or that a golfer should maximize stiffness at impact. Those are task-specific claims requiring a model of the relevant disturbance, contact and time interval.

To see why stiffness requires a separate calculation, consider fixed internal states and conservative path tension $F_j(\ell_j)$. Let $U = \sum_j U_j(\ell_j)$ with $dU_j/d\ell_j = F_j$. Since $\tau_M = -\nabla_q U$, define restoring tangent stiffness as $K_q = -\partial\tau_M/\partial q$. Then

$$K_q = L^T K_\ell L + \sum_j F_j \nabla_q^2 \ell_j, \quad K_\ell = \text{diag}(dF_j/d\ell_j).$$

The first term reflects material stiffness; the second reflects geometry under tension. A positive first term does not ensure a positive sum. With nonconservative or feedback-controlled muscles, use the appropriate state-dependent linearization instead of assuming an energy Hessian.

For a concrete counterexample, attach a spring from a fixed anchor $a = 0.1$ m from a pivot to a point rotating at radius $b = 0.2$ m. Its length is $\ell(q) = \sqrt{a^2 + b^2 - 2ab \cos q}$.

With stiffness $k = 1000$ N/m and slack length 0.2 m, the spring is taut at $q = \pi$: $\ell = 0.3$ m and $F = 100$ N. There $\ell' = 0$ and $\ell'' = -ab/(a+b)$, so

$$K_q = k(\ell')^2 + F\ell'' = -20/3 \text{ Nm/rad}.$$

The equilibrium is an energy maximum despite positive spring stiffness. Finite differences of spring energy verify the negative curvature. This is a mechanics counterexample, not a quantitative wrist model. Positive restoring stiffness is itself distinct from asymptotic stability, which also depends on inertia, damping, other states, constraints and feedback. Undamped oscillations do not decay merely because stiffness is positive.

13.4.3 In the Control-Affine Framework

A joint-torque input replaces detailed recruitment by its aggregate generalized-force effect. It is useful for a declared rigid-body question, but hides the feasible force set and internal states unless they are explicitly restored. In general, the controls in $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{G}(\mathbf{x})\mathbf{u}$ need not be joint torques. They may be motor currents, muscle excitations or other inputs if the chosen state equations have the required dependence on them. An activation model that is affine in excitation can still have nonlinear force generation downstream.

In Plain Language 13.3. What the Torque Description Leaves Open

Two force patterns can provide the same turning effect while loading the body differently. A torque-based result can identify a useful mechanical demand. The muscle model must still show how that demand can be produced and what the internal consequences are.

13.5 Joint Torques as a Modeling Choice

A constrained whole-body model can be written in generalized-speed coordinates as

$$\dot{\mathbf{q}} = \mathbf{N}(\mathbf{q})\mathbf{v}, \quad \mathbf{M}(\mathbf{q})\dot{\mathbf{v}} + \mathbf{h}(\mathbf{q}, \mathbf{v}) = \mathbf{B}\mathbf{u} + \mathbf{Q}_{\text{passive}} + \mathbf{Q}_{\text{known}} + \mathbf{J}_v^T \boldsymbol{\lambda}.$$

Here $\mathbf{h}$ includes modeled inertial velocity terms and gravity; $\mathbf{B}$ maps actuator inputs to forces conjugate to $\mathbf{v}$; and $\mathbf{J}_v$ maps $\mathbf{v}$ to the contact velocity components conjugate to $\boldsymbol{\lambda}$. Contact reactions must satisfy the declared contact equations and inequalities. If a coordinate-rate Jacobian $\mathbf{J}_c$ instead maps $\dot{\mathbf{q}}$ to those same velocity components, then $\mathbf{J}_v = \mathbf{J}_c \mathbf{N}$. Thus $\mathbf{J}_v^T \boldsymbol{\lambda} = \mathbf{N}^T \mathbf{J}_c^T \boldsymbol{\lambda}$ preserves contact power. In a muscle-driven model, replace the actuator term by $-\mathbf{N}^T \mathbf{L}^T \mathbf{F}$ together with the muscle state equations. A floating base has unactuated directions; it cannot be assigned arbitrary independent joint-torque controls.

13.5.1 Sufficiency

For fixed inertial and geometric parameters, complete initial conditions, specified external forces and a well-posed contact mode, the same total generalized forces give the same instantaneous rigid-body acceleration. The same complete force histories then give the same motion when the initial-value problem has a unique solution. This is a conditional equivalence, not proof that torques alone predict every aspect of the human body. A change in contact mode, compliance or internal actuator state can invalidate the reduced description or alter its future inputs.

13.5.2 Redundancy Avoidance

Torque-driven optimization can study mechanical coordination without first selecting individual muscle forces. The cost is an omitted recruitment problem. An unconstrained torque trajectory may be impossible to realize with tensile actuators, especially with biarticular coupling, delays or state-dependent capacity. A useful workflow first identifies a mechanical opportunity, then asks whether a muscle-driven model can realize it under the same task and boundary conditions. Agreement must be checked rather than assumed.

13.5.3 Experimental Accessibility

Motion capture supplies observations from which kinematics are estimated; force plates measure contact forces and moments over their instrumented surfaces. Inverse dynamics combines such data with inertial parameters, coordinate definitions and a contact model to estimate net generalized forces. It does not directly measure every joint torque or uniquely identify individual muscle tensions. Unmeasured hand-club forces, filtering, differentiation, segment parameters and multiple contacts can affect the result.

A dynamometer measures an external force or moment under specified test conditions. Surface electromyography measures electrical activity associated with muscle excitation; it is neither an invasive force transducer nor a direct measure of tendon tension. EMG-informed force estimates require additional assumptions about activation, geometry and contraction dynamics. Even accurate net-moment agreement cannot identify all muscle forces when the map has a null space.

The upper-extremity model of Holzbaur and colleagues illustrates both the value and limits of such models. Its geometry and muscle parameters support specific comparisons, and some parameters were adjusted against joint-moment data. Those fitted comparisons should be distinguished from other validation tasks. Prescribed scapular motion, generic geometry and omitted intrinsic hand muscles limit transfer to golf-grip questions. [Holzbaur et al., 2005]

13.5.4 Connection to Control

Solving a fixed-contact rigid-body system for acceleration can produce affine dependence on applied torque when the dynamics and constraint equations are linear in those inputs. This does not ensure a globally affine model through contact switching, friction transitions or muscle-state changes. The model must specify the chosen inputs and operating domain. A coordinate's generalized force is power-conjugate to its speed; it need not be an independently selectable control. Translational coordinates have conjugate forces in newtons, rather than rotational torques in newton-metres.

Key Principle 13.2. What a Torque Model Establishes

A declared torque-driven model can explain how aggregate inputs influence motion and impact conditions. Feasibility, recruitment, muscle energy, internal load and perturbation response require additional information. A mechanically successful solution becomes a physiological hypothesis only after those missing constraints and observations are addressed.

13.5.5 What Is Lost?

A net torque history does not specify opposing muscle forces, tendon energy, metabolic expenditure, activation reserve or tissue stress. Instantaneous cancellation of opposing torques does not imply relaxed muscles or zero metabolic cost. Conversely, a nonzero torque need not represent active muscle work: passive tissue can contribute. Keep each quantity tied to its own measurement or constitutive model. The chapter on muscle force generation develops the tendon, fiber and activation states needed to make this connection more explicit.

13.6 Biarticular Muscles: Coupling Between Joints

A muscle path can have nonzero moment arms at several coordinates. Biceps brachii spans the shoulder and elbow region and also has a role in forearm rotation. Only the long head of triceps crosses the shoulder; the other heads should not be assigned that same geometry. Wrist flexors and extrinsic finger flexors are different groups: a muscle named for wrist flexion does not thereby cross the finger joints. A model must represent specific heads, attachments and coordinates rather than assign every member of a group identical leverage.

13.6.1 The Moment Arm Matrix for Biarticular Muscles

For muscle j , its contribution at coordinate i is $\tau_{ij} = r_{ij}F_j$. Two nonzero entries in the same muscle column of $\mathbf{r}$ imply coupled torque contributions. At a fixed configuration their ratio is fixed by geometry when the denominator is nonzero. Coupled torque does not imply equal angular velocities or a prescribed phase order; the entire multibody system determines the motion.

Definition 13.3. Biarticular Coupling

One tensile path can exert torques at several joints. Their signs and magnitudes follow its moment arms; their instantaneous powers also require the respective coordinate velocities. Anatomical coupling alone establishes neither the direction of energy transfer nor a universal proximal-to-distal sequence.

13.6.2 Energy Transfer via Biarticular Muscles

For the configuration-only workless path,

$$P_{ij} = r_{ij}F_j\dot{q}_i, \quad \sum_i P_{ij} = -F_j\dot{\ell}_j.$$

Positive P_{ij} denotes mechanical power delivered to that coordinate by the muscle path; negative power denotes absorption from it. For the second muscle of the two-joint example, $\mathbf{r}_j = (0.02, 0.04)^T$ m and $F_j = 300$ N. At speeds $(2, -1)^T$ rad/s, the powers are $(12, -12)^T$ W. The path absorbs power at coordinate 2 and delivers it at coordinate 1, with zero net path power at that instant. The direction reverses if both velocities reverse. Calling this automatically proximal-to-distal would be incorrect.

At speeds $(2, -3)^T$ rad/s, the same torque contributions instead produce $(12, -36)^T$ W and net absorption of 24 W. Positive powers at both joints would indicate net generation.

Thus a path can transmit, generate or absorb mechanical power depending on its state and motion. Joint-coordinate power partitions are representation-dependent; their sum must preserve total work under a consistent coordinate transformation.

Zero total path power does not imply zero fiber power or zero metabolic expenditure. A compliant tendon can exchange stored energy with shortening or lengthening fibers while total muscle-tendon length stays instantaneously fixed. Include tendon energy and the fiber constitutive model before attributing the power to active contraction, elastic return or dissipation. Segment-level energy transfer also requires a consistent segment force-and-moment power balance; a sequence of joint-speed peaks is not that balance.

In Plain Language 13.4. Reading the Direction of Power

A connection between two joints is like a transmission whose direction of power flow depends on how it is loaded and moving. Knowing that the connection exists is only the beginning. Measure or model force and motion together before deciding which side supplies power and which receives it.

13.6.3 Example: The Wrist Flexors in Golf

Wrist flexion-extension, radial-ulnar deviation and forearm pronation-supination are different coordinates. Their axes also differ from the club shaft axis. A muscle's contribution to club rotation therefore depends on anatomical posture, the hand-club contact geometry and the rest of the body. It cannot be deduced by relabeling wrist flexion as shaft roll. The upper-extremity modeling literature provides specific muscle paths and coordinate definitions, but a generic arm model is not a validated model of a two-handed golf grip. [Holzbaur et al., 2005]

13.7 The Inverse Problem: From Desired Torques to Muscle Forces

13.7.1 The Underdetermined Inverse

Suppose inverse dynamics supplies a net generalized-force estimate and modeled passive and other contributions have been separated consistently. The remaining muscle demand is τ_M^* . Solving $r\mathbf{F} = \tau_M^*$ first requires a feasible target. Being in the column space of r is necessary for an unconstrained real solution, but insufficient for nonnegative capacity-bounded tension. A pseudoinverse can return negative muscle forces. A feasible target can also have multiple force realizations, each with different internal consequences.

The illustrative set $\mathbf{0} \leq \mathbf{F} \leq \overline{\mathbf{F}}$ should not be read as instantaneous voluntary access to every force in that box. With tendon compliance, the current tendon state constrains tension; excitation changes subsequent activation and contraction. Passive tension can also set a nonzero lower bound. The reachable force set depends on state and time horizon. A sequence of feasible static snapshots can fail to form one dynamically realizable muscle trajectory.

13.7.2 Optimization Approaches

One explicit static criterion is normalized squared force:

$$\min_{\mathbf{F}} \sum_j (F_j / \bar{F}_j)^2 \quad \text{subject to} \quad \mathbf{r}\mathbf{F} = \boldsymbol{\tau}_M^*, \quad \mathbf{0} \leq \mathbf{F} \leq \bar{\mathbf{F}}.$$

With positive capacities and a nonempty convex feasible set, this strictly convex objective has a unique minimizing force vector. Uniqueness follows from the stated mathematical criterion, not from evidence that the nervous system uses it. An unnormalized sum of squared forces is a different criterion and weights muscle capacities differently. Neither is automatically metabolic energy.

An energy or fatigue objective requires a specified constitutive model with the appropriate states and units. A trajectory optimization may include activation dynamics, force-length-velocity behavior, tendon energy and history-dependent fatigue. Its result is conditional on those equations, parameter choices, boundary conditions and objective. Comparing alternative plausible objectives and checking sensitivity is more informative than presenting a single optimizer output as identified human recruitment.

Definition 13.4. Static Optimization

Static optimization solves a force-sharing problem at each sampled instant using declared constraints and a criterion. Some formulations include state-dependent force capacities, but separate instantaneous problems do not by themselves enforce a consistent activation or tendon trajectory. They can estimate recruitment under assumptions; they do not establish the brain's algorithm.

13.7.3 Why This Matters for Golf (And Why It Is Hard)

An observed swing and its net torques can constrain hypotheses without identifying one muscle strategy. EMG, imaging, perturbation responses and contact measurements add different information, each with its own limitations. A stronger study calibrates model parameters on designated data, checks predictions on separate conditions, and reports sensitivity to measurement and model uncertainty. Good net-moment fit alone cannot remove a null space in the muscle-force map.

In Plain Language 13.5. From a Swing Outcome to a Testable Hypothesis

A player reaches a given clubhead speed. One proposed change uses a different muscle pattern to obtain the same speed with lower cost. To assess it, specify the complete task, the force and contact constraints, the cost model and the observations that could distinguish the patterns. The common clubhead speed establishes the shared outcome; it does not establish the proposed recruitment or its benefit.

13.8 Practical Example: The Golf Grip

13.8.1 Anatomy of the Grip

Finger flexors, finger extensors, wrist muscles and intrinsic hand muscles contribute different actions. Their anatomical paths determine joint torques, while the contact distribution between hand and handle determines the wrench applied to the club. A count of a few named muscles cannot replace a complete model of the fingers, thumb, palm and forearm. For the present discussion, the contact equations clarify what must be connected without pretending to supply a personalized recruitment model.

13.8.2 Grip Force vs. Grip Torque

Represent hand contact by patches k at positions $\boldsymbol{\rho}_k$ relative to a declared club reference point. With patch forces $\mathbf{f}_k$ and any modeled patch couples $\mathbf{m}_k$, the applied wrench is

$$\mathbf{f}_{\text{hand}} = \sum_k \mathbf{f}_k, \quad \mathbf{m}_{\text{hand}} = \sum_k (\boldsymbol{\rho}_k \times \mathbf{f}_k + \mathbf{m}_k).$$

A normal compressive load and a net wrench are different summaries of that distribution. Opposing normal forces can largely cancel in the net force while increasing pressure and friction capacity. Under a simple Coulomb patch model, tangential force is limited by $\|\mathbf{f}_{t,k}\| \leq \mu_k f_{n,k}$ with $f_{n,k} \geq 0$. This is a declared contact approximation, not an established constant friction coefficient for every hand, glove and grip. Contact deformation and torsional resistance require further modeling.

If $\mathbf{v}_c = \mathbf{J}_c \dot{\mathbf{q}}$ maps skeletal motion to a contact-point velocity, $\mathbf{J}_c^T \mathbf{f}_c$ maps the force acting on the skeleton to generalized force. When $\dot{\mathbf{q}} = \mathbf{N}\mathbf{v}$, the speed-coordinate force is $\mathbf{N}^T \mathbf{J}_c^T \mathbf{f}_c$. Include the angular-velocity Jacobian when retaining contact couples. Force on the club has the opposite sign at the same interface. The same work-conjugacy discipline used for muscle tension prevents a hidden sign error here. A measured pressure map or net wrench does not uniquely identify all muscle forces.

13.8.3 Moment Arm Changes With Hand Configuration

A strong or weak grip describes a hand-club arrangement, not a single specified wrist angle or muscle moment arm. A grip change can alter anatomical posture, contact locations and club orientation in different combinations. Predicting the change requires those configurations and a muscle/contact model. There is no basis in this chapter for universally assigning improved leverage to flexor carpi ulnaris in a strong grip or flexor carpi radialis in a weak grip.

The useful modeling question is conditional: at the measured posture, how do feasible wrist and finger forces project through the contact geometry into the desired club wrench? Then ask whether a changed posture alters contact retention, available torque, comfort or repeatability under the same task. Geometry may suggest a hypothesis; it does not establish a coaching prescription.

13.8.4 Grip Force Modulation During the Swing

Grip loading can vary with motion, club inertia and interaction forces. To quantify its time history, identify the sensor, measured quantity, calibration, temporal resolution and

normalization. Instrumented handles or pressure sensors supply information that a ground force plate cannot directly provide. Surface EMG requires an additional force-estimation model. A percentage of maximum is uninterpretable without the reference test, posture and measured quantity.

Accordingly, this chapter does not assign universal grip-force percentages to backswing, transition, downswing or impact. An experimentally supported account would report the actual study population, task and uncertainty. The absence of such evidence does not mean grip force is constant; it means a particular numerical pattern has not been established here.

13.8.5 Grip Stiffness and Impact

Clubhead-ball contact force is not identical to force transmitted to the hands. The shaft's distributed inertia and compliance, contact duration, grip conditions and pre-impact state govern that transmission. A large clubhead force alone cannot establish an optimal wrist stiffness. Nor does a brief collision imply that an arbitrary new neural command can generate a useful muscle-force change within the remaining interval.

Distinguish pre-existing muscle activation and mechanical impedance from later feed-back responses. A defensible impact hypothesis specifies the coupled hand-club-ball model, perturbation, time horizon and performance outcome. It then compares feasible preparations or interventions from matched initial states. Increased co-contraction may change load and impedance, but its effect on face orientation, speed or variability must be evaluated. The goal is to explain which mechanisms matter under which conditions, rather than turn an unspecified impact-force estimate into a universal instruction to grip harder.

Key Principle 13.3. Key Takeaways

Muscle geometry, force generation, contact mechanics and control are linked through work and state-dependent constraints. With positive tension and positive lengthening, $\tau_M = -\mathbf{L}^T \mathbf{F}$. Torque feasibility requires bounds and dynamics, not only a matrix inverse. Biarticular powers require velocities as well as moment arms. Net torque does not uniquely determine recruitment, stiffness, energy or tissue load. A golf hypothesis becomes stronger when its proposed causal chain is explicit and each link has an appropriate calculation or observation.

Exercises 13.1. Chapter Exercises

1. Define a signed axial moment arm. Explain when a perpendicular distance is sufficient, and derive the sign of $\tau_M = -\mathbf{L}^T \mathbf{F}$ from tensile virtual work.
2. A hypothetical actuator transmits 600 N at moment arms 0.05 m and 0.03 m. Compute both torque contributions. Explain why this does not locate a person's maximum-strength posture.
3. For the nonsymmetric two-joint example in this chapter, compute both muscle contributions, the total torque, path rates at speeds $(2, -3)^T$ rad/s and total skeletal power. Explain why transposing the displayed torque map again changes the answer.

4. For the three-muscle feasible-family example, verify the null direction and derive the allowable interval of t . Are the independent coordinate maxima simultaneously feasible?
5. Define co-contraction and distinguish it from a null-space force redistribution. Explain how the taut-spring example disproves the claim that positive material stiffness guarantees joint stability.
6. Two synergists have positive moment arms 0.04 m and 0.02 m, each with capacity 1000 N. Characterize all nonnegative force pairs producing 50 Nm. Explain whether this freedom is antagonist co-contraction.
7. State a force-sharing optimization and the assumptions under which its optimum is unique. Explain why that uniqueness does not identify the nervous system's objective or guarantee a dynamically feasible trajectory.
8. Compute the second muscle's joint powers at speeds $(2, -1)^T$ and $(2, -3)^T$ rad/s. Identify the direction of instantaneous transfer in the zero-sum case and explain what remains unknown about tendon and metabolic energy.
9. Design the measurements needed to support a phase-dependent grip-force claim. Distinguish handle pressure, net wrench, ground reaction and surface EMG.
10. Explain why changing from a strong to a weak grip does not specify a unique change in the wrist moment-arm matrix. Identify the anatomical and contact information required for a prediction.
11. For the constraint $\mathbf{q} = (s, 2s)^T$, compute the reduced torque in the two-joint example. Explain why a torque-input model also needs actuator feasibility and appropriate floating-base/contact equations.

13.9 Worked Answers and Modeling Checks

1. $r_e = \mathbf{e} \cdot (\boldsymbol{\rho} \times \mathbf{d})$. Its planar magnitude reduces to the pivot-to-force-line distance. With $\delta W = -\mathbf{F}^T \mathbf{L} \delta \mathbf{q}$, matching the coefficient of independent $\delta \mathbf{q}$ gives $\boldsymbol{\tau}_M = -\mathbf{L}^T \mathbf{F}$. Axis orientation and tension convention must be stated.
2. The contributions are 30 Nm and 18 Nm. The assumed tension is held equal; a person's attainable tension need not be equal at the two postures. No angle, force-length state or measured capacity has been supplied.
3. The contributions are $(25, 0)^T$ and $(6, 12)^T$ Nm, totaling $(31, 12)^T$ Nm. Path rates are $(-0.10, 0.08)^T$ m/s and total power is 26 W. Multiplying $\mathbf{r}^T \mathbf{F}$ would instead give $(25, 22)^T$ for these equal-sized matrices and mismatched row meanings; it is not the declared force map.
4. $\mathbf{r}(13, -8, 12)^T = \mathbf{0}$. The force inequalities give $t \in [-25/6, 50/3]$ N. Coordinate maxima 22 and 14 Nm cannot coexist: the force choice required for 14 Nm at coordinate 2 leaves at most 14.5 Nm at coordinate 1.

5. Co-contraction concerns opposing actions about a named coordinate; a null direction can instead exchange synergist forces. At the taut-spring equilibrium, the material contribution vanishes because $\ell' = 0$ and the geometric contribution is $F\ell'' = -20/3$ Nm/rad. The equilibrium is an energy maximum. Even a positive restoring stiffness would not alone establish asymptotic stability without the remaining dynamics.
6. $F_2 = 2500 \text{ N} - 2F_1$, with $750 \text{ N} \leq F_1 \leq 1000 \text{ N}$ and correspondingly $500 \text{ N} \leq F_2 \leq 1000 \text{ N}$. This is a continuum of feasible synergist sharing, not an antagonist pair. A demand above 60 Nm would be infeasible under these bounds; at 60 Nm both forces must be at capacity.
7. Positive-capacity normalized squared force is strictly convex, so a nonempty convex bounded feasible set has one minimizer. Changing the criterion changes the inference. Independent samples do not enforce activation or tendon dynamics; a neural interpretation requires additional evidence.
8. Powers are $(12, -12)^T \text{ W}$ and $(12, -36)^T \text{ W}$. In the first case, power is absorbed at coordinate 2 and delivered at coordinate 1; in the second, net path power is -24 W . Fiber work, tendon storage and metabolic cost remain separate quantities.
9. Use calibrated instrumented-handle or pressure measurements synchronized with kinematics and a defined phase convention. Specify whether the result is contact pressure, integrated normal load or a full wrench, and define the normalization test. Ground forces and EMG are complementary observations, not substitutes for direct handle-force measurement.
10. Measure posture and hand-club placement, define joint and shaft axes, specify muscle paths and a contact model, and propagate their uncertainty. The verbal grip category alone does not supply these quantities or identify a universally advantaged muscle.
11. The reduced torque is $31 + 2(12) = 55 \text{ Nm}$, with effective moment arms $(0.05, 0.10)$ m. This preserves virtual work along the allowed coordinate motion. It does not make the underlying tensions freely controllable or eliminate unactuated base directions, contact feasibility or actuator state dynamics.

The nine independent checks in the accompanying verification module reproduce the geometry, transpose, power, feasible-force, coupled-coordinate, integrability and stiffness examples. They test these constructed derivations. Empirical claims about a particular golfer still require suitable observations and model validation.

Chapter 14

Muscle Force Generation: The Biological Engine

In Plain Language 14.1. Why Muscle Biology Matters for Your Swing

The nervous system prepares muscle activity; muscles pull on tendons; tendon forces act through changing moment arms; the coupled body and club respond. Each link matters. A fast clubhead does not tell us how fast a particular muscle fiber is shortening. A small net joint torque does not tell us that the surrounding muscles are relaxed. Understanding these distinctions lets us ask a useful golf question: which changes in preparation or ongoing activation can still change the impact conditions, by how much, and with what consequences elsewhere in the body?

This chapter develops a reproducible teaching model before discussing measurements and training. Its numerical examples are manufactured calculations, not estimates of a particular golfer. The purpose is to connect biological limitations to mechanics without turning a plausible mechanism into an unsupported claim that every player must use one sequence. The model omits fatigue, injury, detailed neural circuitry and many tissue effects; those omissions define the questions it can answer.

14.1 Muscle Architecture: What a Muscle Actually Is

Definition 14.1. The Muscle Hierarchy

A skeletal muscle contains fascicles, which contain muscle fibers. Each fiber is a cell containing myofibrils with sarcomeres arranged along their length. Actin and myosin interact within these repeating units. Sarcomeres in series contribute length and shortening excursion; contractile structures in parallel contribute force. A muscle can produce active tension while shortening, holding approximately constant length or being lengthened by its load. The word contraction does not require visible shortening.

An electrical signal initiates processes that raise calcium availability within the fiber. Cross-bridge cycling couples chemical energy to force and motion. ATP binding allows myosin to detach from actin, hydrolysis prepares another cycle, and calcium handling also consumes energy. This mechanism is different from stretching a passive rubber band. The band can return energy previously stored in it; an activated muscle can convert chemical energy into mechanical work. A loaded muscle also contains passive structures, so measured

total tension need not disappear when neural excitation stops. [OpenStax, 2022a]

Key Principle 14.1. Muscle Fiber Types and Their Roles

Fiber classifications summarize differences in contractile speed and metabolic properties. Slow oxidative fibers tend to resist fatigue; fast oxidative and fast glycolytic categories describe different combinations of speed and energy supply. These categories do not give a universal percentage composition for every muscle or golfer. Nor does a fast clubhead establish the player's fiber composition. Fiber size, recruitment, architecture, training history and task coordination are separate quantities. [OpenStax, 2022b]

Definition 14.2. Muscle Architectural Parameters

Let ℓ^M be fiber length in meters, ℓ_{opt}^M its reference optimal length, and α the angle between fibers and the tendon line. For approximately parallel, equal-length fibers at the reference configuration, a volume-based physiological cross-sectional area is

$$A_{\text{PCSA}} = \frac{V_M}{\ell_{\text{opt}}^M} = \frac{m_M}{\rho_M \ell_{\text{opt}}^M}, \quad F_0 = \sigma_0 A_{\text{PCSA}}.$$

Here m_M is muscle mass, ρ_M density, and σ_0 specific tension in compatible units. We define F_0 along the fiber direction. Its tendon-direction projection at the reference pennation is $F_0 \cos \alpha_0$. Some sources incorporate this projection in their area or maximum-force convention; applying it again would double-count pennation. These reference parameters are not recalculated by substituting the instantaneous fiber length into the area formula.

For an explicitly hypothetical area of 3 cm^2 and specific tension of 30 N/cm^2 , $F_0 = 90 \text{ N}$. At $\alpha_0 = 20^\circ$ its projection is about 84.6 N . This arithmetic is transparent; interpreting either input as a measured value for a named golf muscle would require anatomical evidence. Increasing fiber length at fixed muscle volume trades parallel area for excursion capacity in this simplified architecture. Increasing pennation changes both packing and projection, so the cosine alone does not establish whether one architecture is stronger.

The source of anatomical parameters matters. Holzbaur and colleagues' upper-extremity model includes shoulder-to-finger geometry and comparisons with moment arms and isometric moment capacity. Its tendon slack lengths were selected using additional constraints, and the authors identify sensitivity and shoulder-geometry limitations. It is a starting model, not validation of muscle force during a golf swing. [Holzbaur et al., 2005] Rajagopal and colleagues' gait model instead uses muscle-tendon actuators in the lower limbs and ideal torque actuators for the upper body. It cannot supply evidence for detailed forearm muscle parameters merely because its title says full-body. [Rajagopal et al., 2016]

14.2 The Force-Length Relationship

In Plain Language 14.2. Strength Changes With Joint Angle

Changing joint angle changes muscle paths, fiber lengths and moment arms. Active force depends in part on filament interaction, while passive tension increases as passive structures are loaded. Joint strength therefore need not peak at the angle where one muscle's active fiber force peaks. Antagonists, other muscles crossing the joint and the test apparatus also contribute to the measured moment.

Normalize fiber length as $\lambda = \ell^M / \ell_{\text{opt}}^M$. A convenient illustrative active curve is

$$f_L(\lambda) = \exp \left[- \left(\frac{\lambda - 1}{\gamma} \right)^2 \right], \quad \gamma = 0.5. \quad (14.1)$$

This Gaussian has a maximum of one at $\lambda = 1$. It has neither an exactly flat plateau nor a finite length at which it becomes exactly zero. At $\lambda = 0.8$ and 1.2 , it gives $e^{-0.16} \approx 0.8521$; at 0.5 and 1.5 , it still gives $e^{-1} \approx 0.3679$. Thus it should not be extrapolated as a literal prediction of filament overlap at extreme lengths. The parameter γ is a declared teaching choice, not a universal tissue measurement.

Key Principle 14.2. Regions of the Force-Length Curve

Distinguish the active-force curve from total force. Activation can scale active force while passive structures contribute separately. A low active-force multiplier at short length is not the same mechanism as low neural recruitment. At longer lengths, decreasing active force can coexist with increasing passive tension. A model should state both contributions and the lengths over which its fitted curves are supported.

For a nonnegative passive teaching curve, choose $k_P = 4$ and $\epsilon_P = 0.6$:

$$f_P(\lambda) = \begin{cases} 0, & \lambda \leq 1, \\ \frac{e^{k_P(\lambda-1)} - 1}{e^{k_P\epsilon_P} - 1}, & \lambda > 1. \end{cases} \quad (14.2)$$

The threshold, shape and normalization are explicit: $f_P(1) = 0$ and $f_P(1.6) = 1$. At 1.2 it is about 0.12227 . The expression never generates passive compression below its slack threshold. Its right derivative at the threshold differs from the left derivative; a differentiable simulation model would require an appropriate transition if a solver or sensitivity calculation needs one.

With dimensionless activation $a \in [0, 1]$ and the velocity multiplier defined next, total fiber-direction force is

$$F_M = F_0 \{ a f_L(\lambda) f_V(w) + f_P(\lambda) \}. \quad (14.3)$$

This expression deliberately omits viscous damping and history effects. It distinguishes the active component from passive tension but does not resolve every structure inside a real muscle.

Example 14.1. Latissimus Dorsi Through the Swing

A statement that the latissimus is near its optimal length at a particular swing phase requires more than an image of the pose. The shoulder girdle configuration, muscle path, tendon extension, pennation and reference fiber length all affect the inference. Suppose a model gives $\lambda = 1.2$ and $a = 0.7$ at zero fiber velocity. Our chosen curves give

$$F_M/F_0 \approx 0.7(0.8521) + 0.12227 = 0.7188.$$

That is a calculation conditional on model inputs; it does not establish either input for a golfer. Testing the claim requires checking the modeled geometry and measured motion, and reporting sensitivity to uncertain parameters.

14.3 The Force-Velocity Relationship: Hill's Equation**In Plain Language 14.3. The Central Constraint: Force Depends on Fiber Speed**

A muscle fiber shortening rapidly generally has less active force capacity than the same fiber shortening slowly under comparable conditions. This statement concerns local fiber velocity. A muscle can move quickly through the room while its fiber length barely changes, and a moving tendon can stretch while the fibers shorten. Clubhead speed alone identifies neither motion.

Choose the shortening-positive convention

$$w = -\frac{\dot{\ell}^M}{V_{\max}}, \quad V_{\max} = \nu \ell_{\text{opt}}^M,$$

where $\dot{\ell}^M$ is positive for lengthening, $V_{\max}$ is in m/s, and ν is in fiber lengths per second. For example, 2–4 lengths/second for a 0.10 m reference fiber means 0.2–0.4 m/s. Confusing these units produces order-of-magnitude errors.

A normalized concentric Hill-type relation is

$$f_V(w) = \frac{1-w}{1+w/k}, \quad 0 \leq w \leq 1, \quad k = 0.4. \quad (14.4)$$

It is hyperbolic, not linear. It equals one at zero shortening velocity and zero at the declared unloaded shortening limit. Extrapolating it above $w = 1$ would produce negative active force; that extrapolation is outside this teaching law's domain. A simulation encountering that condition needs a stated treatment and a check that the assumed force equilibrium is physically and numerically admissible.

Key Principle 14.3. Concentric vs. Eccentric: The Asymmetry

For lengthening, use the following explicitly constructed illustration:

$$f_V(w) = 1 + (F_{\infty} - 1) \frac{-w}{c - w}, \quad w \leq 0,$$

with $F_\infty = 1.4$. Choose $c = 4/35$ so that $c(1 + 1/k) = F_\infty - 1$. It approaches 1.4 as $w \rightarrow -\infty$, equals one at zero and has the same derivative there as the concentric branch. These choices make the joined velocity curve continuously differentiable at zero. They are not the published Thelen equation and do not establish a universal eccentric-force ceiling. Particular muscle models use different fitted curves, constraints and numerical regularizations. [Millard et al., 2013]

Active shortening power is $P_{CE} = -F_{CE}\dot{\ell}^M$. At full activation and optimal length, normalized power is

$$P_{CE}/(F_0 V_{\max}) = w f_V(w).$$

Differentiating gives the interior maximum

$$w_* = \sqrt{k^2 + k} - k \approx 0.34833, \quad \frac{P_*}{F_0 V_{\max}} \approx 0.12133.$$

Maximum force and maximum shortening power occur at different speeds. Isometric force can be substantial while fiber shortening power is zero. During active lengthening, this sign convention gives negative mechanical power: the contractile component absorbs mechanical energy while still requiring metabolic processes.

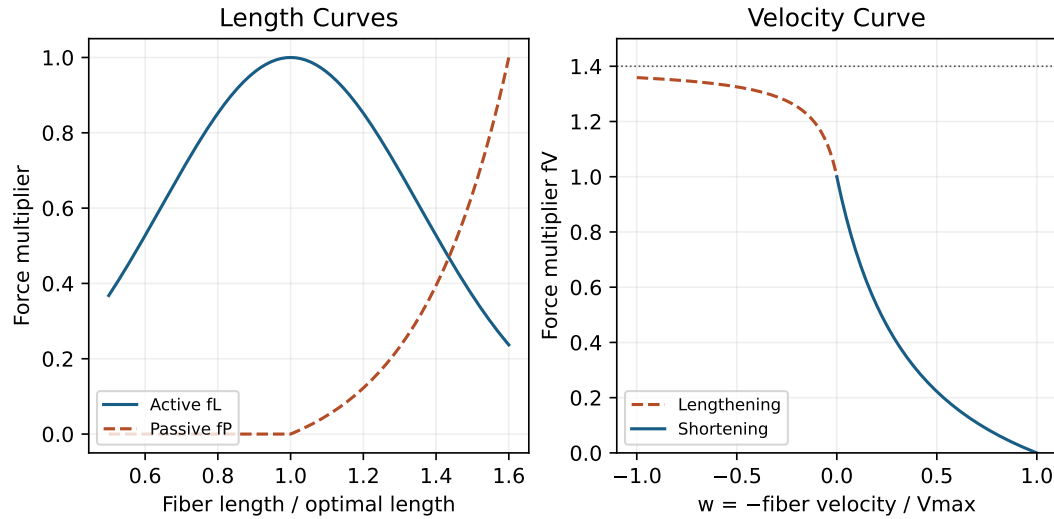


Figure 14.1: Declared Teaching Curves: Active and Passive Length Multipliers, and the Joined Velocity Law. The Parameters Are Illustrative, Not Golfer Measurements.

Key Principle 14.4. What High Club Speed Does and Does Not Establish

High club speed changes the mechanical problem, but it does not insert that speed directly into f_V . First obtain each muscle-tendon path velocity from joint motion, then separate tendon and fiber motion. Likewise, mv^2 has units of energy, not force; a centripetal force estimate needs a radius, mv^2/R , and a specified trajectory. Gravitational force near Earth's surface does not grow with swing speed. Configuration-

dependent inertial and velocity-coupling effects require the equations of motion, not a comparison of unlike speed scales.

Example 14.2. Numerical Estimate: Local Wrist Muscle Force

Consider a hypothetical unpennated muscle with a rigid tendon, constant moment arm $r = 0.02$ m, reference fiber length 0.10 m and $\nu = 10 \text{ s}^{-1}$. If the relevant joint alone rotates at 8.75 rad/s in the shortening direction, $-\dot{\ell}^M = r\dot{q}$, giving 0.175 m/s and $V_{\max} = 1 \text{ m/s}$. Therefore $w = 0.175$ and

$$f_V = 0.825/1.4375 \approx 0.5739.$$

With $F_0 = 100 \text{ N}$, $a = 1$, $\lambda = 1$ and no passive force, the tendon force is 57.39 N and moment 1.148 Nm . These inputs do not describe measured extensor carpi radialis brevis (ECRB) force at impact. Dividing clubhead speed by a guessed lever length cannot replace the required joint and path kinematics.

14.4 Muscle Activation Dynamics

In Plain Language 14.4. Why Muscles Are Slow to Respond

Neural excitation u is an input to the muscle model. Activation a describes its evolving active state; it is not identical to the command or to measured electromyography (EMG) amplitude. Tendon force follows from activation together with length, velocity and elastic loading. Consequently, a command change, an EMG onset and an external force onset are different events.

For an elementary model, choose a fixed time constant $\tau_a > 0$:

$$\dot{a} = \frac{u - a}{\tau_a}, \quad 0 \leq u, a \leq 1. \quad (14.5)$$

For a constant command starting at $t = 0$, direct integration gives

$$a(t) = u + (a_0 - u)e^{-t/\tau_a}. \quad (14.6)$$

The same equation describes decay when $u < a_0$. This fixed-constant model is intentionally simpler than activation-dependent rise and fall laws. Published implementations differ; the chosen law and any minimum activation must be specified before comparing simulations. [OpenSim, 2020]

For $a_0 = 0$, $u = 1$ and $\tau_a = 25 \text{ ms}$, activation reaches 63.2% after one time constant, 50% after 17.33 ms and 95% after 74.89 ms. At 50 ms it is about 0.8647. Exact equality to one occurs only asymptotically. A time constant therefore is not a period of complete inactivity followed by full force. Nor does 95% activation imply 95% of isometric joint torque during a moving, compliant task.

A numerical solver adds another issue. Forward Euler with a 5 ms step gives $a_n = 1 - 0.8^n$, reaching 95% at step 14, or 70 ms. The exact continuous solution crosses that threshold

at 74.89 ms. Refining the time step or using a suitable integrator reduces this numerical discrepancy. It is not evidence for a faster physiological response.

Definition 14.3. Electromechanical Delay

An experimentally reported delay depends on the selected start and end events, detection thresholds, contraction protocol and loading. Nordez and colleagues used ultrasound during electrically evoked gastrocnemius contractions to distinguish aspects of fascicle and tendon onset. That experiment does not supply a universal visual-reaction delay or golf-wrist latency. [Nordez et al., 2009] Adding an EMG-to-force delay to an activation time constant can double-count processes if both quantities already describe overlapping parts of the response. A pure delay may be added to a model, but its location and measurement basis must be explicit.

Example 14.3. Reaction Time in the Golf Downswing

Suppose a command changes 70 ms before impact and a separately specified pure transmission delay is 20 ms. Our activation model then has 50 ms to respond. Its activation increment from zero is 0.8647 at impact. If the same command had already established activation earlier, the starting state would be different. Conversely, a sensory correction requiring additional detection and processing could leave less time. This timing ledger is a conditional example, not a measured sequence of events in every downswing. Preparation and a new response must be compared from their actual initial states.

14.5 The Hill-Type Muscle Model

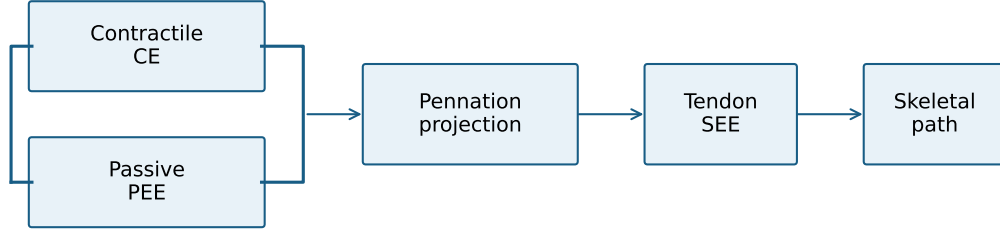
Definition 14.4. Hill-Type Muscle Model Components

The contractile element (CE) and passive parallel element (PEE) act along the fiber direction. Their combined force is projected through pennation onto the series tendon, also called the series elastic element (SEE). With a massless connection and no extra damper in this teaching model,

$$F_T = (F_{CE} + F_{PE}) \cos \alpha = F_M \cos \alpha. \quad (14.7)$$

Equality of CE and tendon force alone would neglect loaded passive fibers and pennation. A generic Hill-type model is a family of approximations, not a unique set of equations. Equilibrium, damping and rigid-tendon variants can behave differently numerically. [Millard et al., 2013]

Parallel Fiber Forces, Then a Series Tendon



$$F_T = (F_{CE} + F_{PE}) \cos \alpha$$

Fiber output = skeletal output + tendon storage rate

Figure 14.2: Force and Energy Connections in the Teaching Model. Active and Passive Fiber Forces Add Before Pennation Projection; the Tendon Is in Series.

The active and passive forces used here are

$$F_{CE} = aF_0 f_L(\lambda) f_V(w), \quad (14.8)$$

$$F_{PE} = F_0 f_P(\lambda). \quad (14.9)$$

Let tendon slack length be L_s and strain be $\epsilon = (\ell^T - L_s)/L_s$. Define a tendon force law independently of the passive fiber curve:

$$F_T(\epsilon) = \begin{cases} 0, & \epsilon \leq 0, \\ F_{\text{ref}} \frac{e^{k_T \epsilon / \epsilon_{\text{ref}}} - 1}{e^{k_T} - 1}, & \epsilon > 0. \end{cases} \quad (14.10)$$

Here F_{ref} is the tendon force at positive reference strain ϵ_{ref} , and k_T is dimensionless. Tendon tangent stiffness is $dF_T/d\ell^T$, measured in N/m; it is not k_T . On the loaded branch,

$$\frac{dF_T}{d\ell^T} = \frac{F_{\text{ref}} k_T e^{k_T \epsilon / \epsilon_{\text{ref}}}}{L_s \epsilon_{\text{ref}} (e^{k_T} - 1)}.$$

The idealization stores energy without hysteretic loss. Actual loading and unloading may require a richer constitutive model. As with the passive fiber curve, the slack transition is continuous but not differentiable here.

14.5.1 Geometry Connects Fiber, Tendon and Joint Motion

Write the total path length as $\ell^{MT} = \ell^T + \ell^M \cos \alpha$. To differentiate it, specify how pennation changes. In a fixed-height model, $h = \ell^M \sin \alpha$ is constant, so

$$\ell^M \cos \alpha = \sqrt{(\ell^M)^2 - h^2}.$$

Differentiating gives

$$\dot{\ell}^M = \cos \alpha (\dot{\ell}^{MT} - \dot{\ell}^T).$$

The relation applies for $\ell^M > h$; approaching 90° pennation is a degenerate geometry, not an ordinary operating state. Assuming constant angle instead would give a different velocity relation. Combining fixed-height kinematics with the force projection gives the power identity

$$F_T \dot{\ell}^{MT} = F_M \dot{\ell}^M + F_T \dot{\ell}^T.$$

For example, $\ell^M = 0.10$ m and $h = 0.06$ m give $\cos \alpha = 0.8$. If the path shortens at 0.10 m/s while the tendon lengthens at 0.02 m/s, the fiber shortens at 0.096 m/s. With $F_M = 100$ N and $F_T = 80$ N, the identity is $-8 = -9.6 + 1.6$ W. Of the fiber's mechanical output, 1.6 W is going into tendon storage at that instant. Tendon compliance therefore changes the fiber's velocity and its force capacity as well as storing energy.

For independent joint coordinates q with $v = \dot{q}$, virtual work connects path length to moment arms:

$$r_{ji}(q) = -\frac{\partial \ell_i^{MT}}{\partial q_j}, \quad \dot{\ell}^{MT} = -r^T v, \quad \tau_M = r F_T.$$

For angular coordinates, entries of r have units of meters per radian, conventionally written meters. Generalized coordinates that include translations require the corresponding force units; a model using $\dot{q} = N(q)v$ must include that mapping. The convention above ensures

$$\tau_M^T v = -F_T^T \dot{\ell}^{MT}.$$

Shortening under positive tension can deliver positive power to the skeleton. Tendon lengthening can instead absorb part of that work. These equations also reveal multi-joint coupling: a muscle crossing two joints can generate moments at both, and motions at the two joints can reinforce or cancel its path-length change.

Take the local path in meters, with angles in radians:

$$\ell^{MT} = 0.30 - 0.02q_1 + 0.01q_2.$$

Its moment arms are $(0.02, -0.01)$ m. At $v = (10, 20)$ rad/s, the path velocity is zero despite motion at both joints. At $v = (10, 0)$, it is -0.20 m/s. Rigid translation of the entire arm and muscle together contributes no intrinsic path-length change. These examples explain why a clubhead or hand speed is insufficient to calculate muscle force.

Key Principle 14.5. Energy Storage and Release in the Tendon

Stored energy is the integral of force over extension, not force over dimensionless strain alone:

$$U_T(\epsilon) = L_s \int_0^\epsilon F_T(s) ds.$$

For positive strain in our law,

$$U_T = \frac{F_{\text{ref}} L_s}{e^{k_T} - 1} \left[\frac{\epsilon_{\text{ref}}}{k_T} \left(e^{k_T \epsilon / \epsilon_{\text{ref}}} - 1 \right) - \epsilon \right].$$

Set $F_{\text{ref}} = 100$ N, $L_s = 0.20$ m, $\epsilon_{\text{ref}} = 0.04$ and $k_T = 3$. At the reference strain, extension is 0.008 m and stored energy is 0.22475 J. Differentiating energy with

respect to extension recovers force. This provides a direct check on the units and expression.

For the lossless model, $\dot{U}_T = F_T \dot{\ell}^T$. Substituting into the power identity shows that tendon recoil can add to instantaneous power delivered to the skeleton while its stored energy falls. It does not create energy. A shaft also stores and returns elastic energy, but its distributed bending and torsion, boundary conditions and contact with the golfer are different from a tendon model. Neither elastic mechanism by itself proves a universal benefit from a particular wrist action or shaft flexibility. The timing and direction of energy delivery must serve the impact task.

Force equilibrium must hold throughout this calculation. Arbitrarily prescribing activation, fiber length, fiber velocity and tendon extension independently can violate it. Some equilibrium formulations solve an inverse force-velocity relation for fiber velocity; zero activation or extreme geometry can make that inversion singular or impossible. A simulator must document how it handles such states. A numerical minimum activation is a modeling choice, not evidence that a person cannot cease neural excitation. [Millard et al., 2013]

14.6 How Muscle Biology Maps to Control Theory

Mechanical position and velocity alone do not generally specify the future of an activated, compliant muscle system. Two otherwise identical poses can have different activation and tendon loading because of their histories. A useful state for this teaching model is

$$x = (q, v, \mathbf{a}, \ell^M), \quad \dot{x} = f(x) + G(x)\mathbf{u}. \quad (14.11)$$

For n independent mechanical coordinates and m modeled muscle compartments,

$$\mathbf{u} \in [0, 1]^m, \quad (14.12)$$

$$\mathbf{a} \in [0, 1]^m, \quad (14.13)$$

$$\mathbf{F}_T \in \mathbb{R}_{\geq 0}^m, \quad (14.14)$$

$$\boldsymbol{\tau}_M = r(q)\mathbf{F}_T \in \mathbb{R}^n. \quad (14.15)$$

There is no fixed universal number of actuators per joint. One anatomical muscle may be split into compartments, and one compartment may span multiple joints. Excitations are bounded; tendon forces are tensile; admissible forces also depend on equilibrium and the current state. They are not arbitrary unconstrained control vectors.

Away from an impact event, a specified unconstrained mechanical model may take the form

$$\dot{v} = A(x) = M(q)^{-1} \{r(q)\mathbf{F}_T(x) + \mathbf{b}_{\text{ext}}(x) - \mathbf{c}(q, v) - \mathbf{g}(q)\}.$$

M is the mass matrix, $\mathbf{c}$ the velocity-dependent inertial terms, $\mathbf{g}$ the gravitational terms and $\mathbf{b}_{\text{ext}}$ specified external generalized forces. Contact constraints, unknown reaction forces or a flexible shaft require the corresponding additional equations. A collision may require an impulse or compliant-contact treatment; the smooth equation should not silently be applied across it.

Suppose a well-defined muscle equilibrium solution gives $\dot{\ell}^M = \Phi(x)$, and activation constants form the diagonal matrix T_a . Then the actual augmented control-affine expression is

$$f(x) = \begin{bmatrix} v \\ A(x) \\ -T_a^{-1}a \\ \Phi(x) \end{bmatrix}, \quad G(x) = \begin{bmatrix} 0_{n \times m} \\ 0_{n \times m} \\ T_a^{-1} \\ 0_{m \times m} \end{bmatrix}. \quad (14.16)$$

Excitation enters the activation derivative, not the mechanical acceleration directly. Its mechanical effect develops as the state evolves. The component units differ appropriately: position rate, acceleration, activation rate and fiber-length rate. Equating $G\mathbf{u}$, a state-rate vector, with a scalar joint torque would be dimensionally incorrect. This expression also depends on a fixed T_a and an admissible Φ independent of instantaneous excitation. If the activation time constant switches with u and a , global affinity in u no longer follows from this display.

In Plain Language 14.5. Why Effective Control Depends on State and Time

At fixed length and velocity, the active force's sensitivity to activation is proportional to $F_0 f_L f_V$. During shortening, that sensitivity can decline as local fiber speed rises. The resulting joint effect also depends on moment arms, the mass matrix, tendon dynamics and other joints. A new excitation change has no instantaneous jump effect on activation in this model. A previously prepared activation can still produce force, and a loaded tendon can still recoil, even if the current excitation is zero.

Example 14.4. Numerical Estimate: Finite-Horizon Joint Response

Consider a deliberately simple joint with constant inertia $I = 0.10 \text{ kg m}^2$. An added actuator torque is $\tau_{\max} a(t)$, with $\tau_{\max} = 2 \text{ N m}$, $a_0 = 0$, $u = 1$ and $\tau_a = 0.025 \text{ s}$. Holding all other torques equal between two trajectories, the changes after a remaining response duration T are

$$\Delta v = \frac{\tau_{\max}}{I} \left[T - \tau_a (1 - e^{-T/\tau_a}) \right],$$

$$\Delta q = \frac{\tau_{\max}}{I} \left[\frac{T^2}{2} - \tau_a T + \tau_a^2 (1 - e^{-T/\tau_a}) \right].$$

At $T = 0.050 \text{ s}$, $\Delta q = 0.0108083 \text{ rad}$, or about 0.6193° . This is a finite-horizon response from explicit initial conditions, not a predicted golf correction. A specified pure delay is removed from the available interval before using T ; if no response time remains, this command produces no pre-impact effect in this model. Existing activation, elastic loading and configuration-dependent dynamics require a different calculation.

14.6.1 Drift, Cancellation and Reachability

In the mechanical equation, a conventionally defined non-muscle generalized term is

$$\mathbf{b}_{\text{mech}} = \mathbf{b}_{\text{ext}} - \mathbf{c} - \mathbf{g}. \quad (14.17)$$

Its definition must say which external forces and passive effects it includes. In the augmented state equation, the zero-input drift also includes ongoing activation and stored elastic state. Therefore zero excitation is not equivalent to zero muscle force or purely gravitational motion. And $I\ddot{q}$ inferred from observed acceleration is the total required inertial balance in a scalar model, not an independently measured free-drift torque.

Suppose a scalar mechanical term is -80 N m and an added control moment can range from -4 to 4 N m . With positive constant inertia, all available accelerations have the same sign. Nevertheless, their magnitudes differ and their velocity and position trajectories separate over time. Inability to cancel the current acceleration is not inability to influence the eventual state. Conversely, a favorable instantaneous torque ratio does not guarantee that the required impact state is reachable within the remaining time and actuator bounds.

For the golf question, compare admissible policies over a stated horizon with the same starting mechanical, activation and elastic states. Measure their effects on clubface orientation, impact location and velocity, while checking other joint loads and contact conditions. A linearized sensitivity of these outputs to excitation can be informative, but it must follow the trajectory and its state transitions. A ratio evaluated at one joint at one instant cannot establish nonlinear controllability, global optimality or injury risk.

14.7 Muscle Force in the Golf Swing: What Numbers Establish

Key Principle 14.6. Peak Torques at Major Golf Joints

Three kinds of number answer different questions. A maximum isometric test estimates capacity in a specified posture and protocol. Inverse dynamics estimates the net generalized moment consistent with measured motion and external forces under a chosen model. A force plate measures an external ground wrench, which must be transformed and combined with segment dynamics before assigning internal joint moments. Comparing them requires matched coordinate conventions, units, populations and loading conditions. None is automatically an individual muscle-force measurement.

For two antagonists with moment arms $(0.02, -0.02)\text{ m}$, tendon forces $(100, 0)\text{ N}$ and $(200, 100)\text{ N}$ both produce 2 N m net torque. Their summed tensions are 100 and 300 N. Thus a low net torque can coexist with substantial opposing forces. Joint compression, stiffness and metabolic cost require their own geometry and constitutive models; they cannot be recovered from that torque equality. An optimization algorithm may select one force allocation, but its objective and constraints provide information that inverse dynamics alone does not contain.

EMG is useful for investigating electrical activity, but it is not a calibrated force meter. Roberts and Gabaldón's direct force, fascicle and EMG measurements in turkey hindlimb muscles illustrate that their relationships can vary across movement conditions. That finding supports distinguishing the signals; it does not calibrate human forearm force during golf. [Roberts and Gabaldón, 2008] A golf study needs its own measurement model, normalization choices and uncertainty analysis.

Key Principle 14.7. Muscle Capacity and Mechanical Demand

Earlier actions can change later geometry, velocity, activation and tendon loading. Later actions can change the trajectory within their remaining response time and bounds. A sequence that reduces one muscle's shortening speed may improve its active-force capacity while increasing demand elsewhere. A motion that increases elastic storage may also change joint loading or mistime recoil. These are coupled tradeoffs, so no universal early-effort/late-coast rule follows from a single force curve.

To evaluate a proposed sequence, first state the desired impact outputs and acceptable variability. Next estimate the mechanical and muscle states consistently, identifying parameters inferred rather than measured. Finally compare alternatives with the same task and initial conditions, including sensitivity to moment arms, slack lengths, activation constants and measurement error. Agreement with a net torque curve is useful but insufficient validation of every internal force. A stronger claim should survive independent observations relevant to the proposed mechanism.

14.8 Implications for Swing Mechanics and Training

Key Principle 14.8. Interpreting the Cue to Accelerate Through Impact

A coaching cue is not a direct report of muscle force or clubhead acceleration. Continuing effort can coexist with a declining shortening-force multiplier, and clubhead speed can increase through coupled motion while a particular muscle does little positive work. Evaluate the cue by its repeatable effect on the intended impact outcomes and by an appropriate mechanical explanation. The equations here neither require every muscle to accelerate the club until contact nor prove that all late effort is harmful.

Training can alter capacity, coordination and how a player prepares the state entering the downswing. These mechanisms need different tests. Greater isometric strength does not by itself establish greater available force at the relevant fiber lengths and velocities; a successful practice swing does not establish improved retention or transfer. A useful investigation links the proposed change to a measured outcome and checks whether the inferred mechanism is consistent with the full movement. This keeps practical instruction informative while leaving uncertainty visible.

Key Principle 14.9. Key Takeaways: Muscle Force Generation

Trace the chain from excitation to activation, fiber force, tendon force, joint moments and task outputs. Keep local fiber velocity distinct from club speed. Preserve force equilibrium and virtual work when joining geometry to mechanics. Account for the energy and activation already present at the start of a comparison. Interpret net moments, EMG and capacity measurements according to what each measures. Judge control over the remaining trajectory, and test proposed golf sequences against alternatives rather than declaring one optimal from an instantaneous ratio.

14.8.1 Looking Ahead

The muscle-to-joint and movement-control chapters use these distinctions at larger scales. Multi-joint geometry determines how muscle forces influence motion; activation and tendon memory determine what the same current command can do; learning determines how a player prepares and adjusts those states. Together they make the golf swing a question about coordinated trajectories and impact outcomes, not an isolated contest between one muscle and one inertial number.

Exercises 14.1. Chapter Exercises

All numerical parameters below belong to the teaching model, not a golfer dataset.

1. For the Gaussian with $\gamma = 0.5$, calculate f_L at $\lambda = 0.8, 1.2$ and 1.5 . Does it have any finite zero? At $\lambda = 1.2$, calculate total force divided by F_0 for $a = 0.7$, $w = 0$ and the stated passive curve.
2. Use $r = 0.02$ m, $\dot{q} = 8.75$ rad/s, $\ell_{\text{opt}}^M = 0.10$ m and $\nu = 10$ s⁻¹. Assume a rigid tendon, zero pennation and shortening in the positive joint direction. Calculate w , f_V and torque for $F_0 = 100$ N at optimal length and full activation.
3. For $\tau_a = 25$ ms, $a_0 = 0$ and $u = 1$, find the 50% and 95% activation times. Compare the latter with forward Euler using 5 ms steps. Explain why neither is a pure neural delay.
4. A path shortens by 0.02 m per radian of q_1 and lengthens by 0.01 m per radian of q_2 . Find both moment arms and path velocity at $v = (10, 20)$ rad/s. With $F_T = 100$ N, calculate the two joint powers and their sum. Assume no other path dependence.
5. Derive the finite-horizon formulas by integrating $\Delta\ddot{q} = Ka(t)$, where $K = \tau_{\text{max}}/I$ and $a(t) = 1 - e^{-t/\tau_a}$. Use zero initial position and velocity differences. Evaluate Δq after 50 ms using the chapter's parameters. Explain why a small instantaneous torque ratio does not set this result to zero.
6. Use $F_{\text{ref}} = 100$ N, $L_s = 0.20$ m, $\epsilon_{\text{ref}} = 0.04$ and $k_T = 3$. Find force, extension and energy at the reference strain. Verify $dU_T/d\ell^T = F_T$, and identify the dimensional error in integrating force over strain without L_s .
7. Design a comparison of two proposed swing sequences. State the common initial conditions, task outcomes, muscle-state estimates and uncertain parameters. Identify one observation that would distinguish a change in muscle timing from a change in geometry or tendon loading.

14.8.2 Worked Answers

1. The multipliers are 0.8521, 0.8521 and 0.3679; none is zero. Total normalized force is approximately 0.7188, including 0.12227 passive force. The Gaussian remains positive at every finite argument.
2. $V_{\text{max}} = 1$ m/s, $w = 0.175$, $f_V \approx 0.5739$, force 57.39 N and moment 1.148 N m. A linear

calculation would give a different answer because it is a different law.

3. $t_{50} = 17.33$ ms and $t_{95} = 74.89$ ms. Euler first exceeds 95% at step 14, 70 ms. The continuous response begins immediately in this model; the time constant controls its rate of approach.
4. The moment arms are $(0.02, -0.01)$ m and path velocity is zero. Torques $(2, -1)$ N m produce powers $(20, -20)$ W, summing to zero. Opposite joint powers can coexist with no net muscle-tendon shortening power.
5. Integrate once for velocity and again for position, retaining both integration constants fixed by the initial conditions. The position difference is 0.0108083 rad, about 0.6193° . A torque too small to cancel another term can still accumulate a trajectory difference.
6. Force is 100 N, extension 0.008 m and energy 0.22475 J. Since $d\epsilon/d\ell^T = 1/L_s$, differentiating the energy cancels the slack-length factor and returns force. Without that factor, the strain integral has units of newtons rather than joules.
7. One defensible design matches the task and starting body, club, activation and elastic states, then compares feasible excitation histories. Record impact orientation, location and velocity as well as relevant loads. Use synchronized motion and electrical signals, with a validated force/geometry model and, where feasible, independent fascicle or tendon measurements. Test parameter sensitivity and predictions not used for fitting. A timing claim should explain observations that an alternative geometry or elastic-state change does not explain equally well; otherwise the mechanism remains underdetermined.

Chapter 15

Aerodynamic Loads in Swing and Flight

Air loads belong in a golf model when their effect matters for the question and accuracy being investigated. The useful question is not whether drag is always negligible or always dominant. It is how a plausible aerodynamic load changes a measured-load estimate, a forward prediction, or the delivered ball flight, relative to the uncertainty and effect size that matter to the study.

That question connects several parts of the swing. Geometry determines local air-relative velocity and orientation. Fluid mechanics determines distributed tractions and moments. Joint and contact geometry maps those loads into the multibody equations. The chosen input policy and initial state determine the forward response. Impact then supplies the ball's initial flight state; ball aerodynamics determines how that state develops. A force estimate at one point does not settle all of these questions at once.

15.1 Define the Relative Airflow

At a point on the moving body, let v be its velocity in an inertial reporting frame and w the undisturbed ambient-air velocity evaluated at that point. Define $v_r = v - w$. In a quasi-steady drag model,

$$F_D = -\frac{1}{2}\rho C_D A \|v_r\| v_r. \quad (15.1)$$

The force opposes motion relative to the air. It need not oppose velocity relative to the ground. Density ρ , reference area A , and coefficient C_D require declared units and conventions. This expression represents the drag component; lift, side force, and an aerodynamic couple may be needed to represent the full load on an oriented or rotating clubhead.

The coefficient summarizes the effects of shape, orientation, Reynolds number, surface condition, and other features of the flow. It is not a universal constant belonging to a shape name. The [NASA drag-equation explanation](#) emphasizes that a measured coefficient and its reference area must be used together. Doubling the reporting area halves the coefficient for the same force. A fixed nominal area with an orientation-dependent coefficient is a valid convention; repeatedly changing area as well as using a coefficient already calibrated for that change can double-count the orientation effect.

For a ball, $A = \pi r_b^2$ is a common convention. For a cylinder in normal crossflow, projected area is diameter times exposed length. For a clubhead, state the area used in its calibration rather than assuming it always equals the clubface area. Its airflow incidence changes during the swing.

15.1.1 Quadratic Scaling and Its Limits

The force magnitude is

$$D = \frac{1}{2}\rho C_D A v_r^2. \quad (15.2)$$

It increases fourfold when speed doubles only if ρ, C_D, A remain fixed. A momentum-flux argument motivates a scale $\rho A v_r^2$ in inertia-dominated flow. It does not prove that the coefficient is speed-independent. At low Reynolds number, a sphere's Stokes drag is proportional to speed; expressing it using the quadratic formula gives $C_D = 24/\text{Re}$. Thus the same coefficient notation can encode different speed dependence.

Use

$$\text{Re} = \frac{\rho \|v_r\| L}{\mu} \quad (15.3)$$

to state the flow regime, with the relevant characteristic length L and dynamic viscosity μ . For illustrative values $\rho = 1.225 \text{ kg/m}^3$, $\mu = 1.81 \times 10^{-5} \text{ Pa s}$, and speed 50 m/s , a 0.1 m head scale gives $\text{Re} \approx 3.4 \times 10^5$; a 0.01 m shaft diameter gives about 3.4×10^4 . They are different flow problems. Reynolds number alone does not tell whether every boundary layer or wake is turbulent. Surface roughness, orientation, unsteadiness, and separation affect the result.

A quasi-steady coefficient approximation also needs the airflow to adjust sufficiently quickly relative to the changing motion. Rapid orientation changes, shaft-head interference, and wake history can require unsteady measurements or a more detailed fluid model. CFD is a way to calculate a candidate load model; mesh, time-step, turbulence-model, and experimental validation remain necessary.

15.2 A Force Estimate With a Clear Scope

Choose $\rho = 1.225 \text{ kg/m}^3$, $C_D = 0.4$, $A = 0.005 \text{ m}^2$, and $\|v_r\| = 50 \text{ m/s}$ as an illustrative coefficient-area-speed combination. Then

$$D = \frac{1}{2}(1.225)(0.4)(0.005)(50)^2 = 3.0625 \text{ N}. \quad (15.4)$$

This is a conditional calculation, not a measured typical driver force. The same force results from $C_D = 0.2$ with area 0.01 m^2 . Comparing coefficients across products without matching area, incidence, and flow conditions can be misleading.

In still air, the instantaneous translational drag power for this example is $-D\|v\| = -153.125 \text{ W}$. If the same speed and load were maintained for 0.05 s , 7.65625 J would be transferred out through drag. Maintaining speed requires other loads to supply that work. This calculation is an energy scale, not a claim that a freely evolving club loses exactly that fraction of its speed. A 0.2 kg point-mass head at 50 m/s has translational kinetic energy 250 J ; the complete club also has its distributed translational and rotational energy. A percentage loss for an actual swing requires integration along that swing.

Similarly, impulse is $\int F_D dt$, while work is $\int F_D^T v dt$ with the relevant point velocity and any aerodynamic-couple power included. A peak force multiplied by total downswing duration does not reproduce either integral unless the corresponding simplifying assumptions hold. Comparing newtons to newton metres without a defined moment arm has no physical meaning.

15.3 Distributed Shaft Loads and the Actual Pivot

For a slender shaft with centerline $p(s, q)$, local velocity $v(s)$, and tangent unit vector $t(s)$, a simple normal-crossflow model uses

$$v_{\perp} = (I - tt^T)(v - w), \quad dF_D = -\frac{1}{2}\rho C_n d(s) \|v_{\perp}\| v_{\perp} ds. \quad (15.5)$$

The coefficient C_n is calibrated for the chosen crossflow model. This approximation omits axial skin friction and some three-dimensional or unsteady effects. Flexible motion enters through the actual centerline and local velocity; a coating or aerodynamic shape is not itself shaft compliance.

As an exactly specified toy geometry, take a straight uniform shaft of length L , rotating in a plane at signed angular speed ω about a fixed pivot on its extended axis. The near end is distance $a \geq 0$ from that pivot, and $s \in [0, L]$ measures distance along the shaft. Its speed is $|\omega|(a + s)$. With still air and constant diameter d and C_n , integration gives

$$D_{\text{shaft}} = \frac{1}{2}\rho C_n d \omega^2 \frac{(a + L)^3 - a^3}{3}, \quad (15.6)$$

$$\tau_{D, \text{pivot}} = -\frac{1}{2}\rho C_n d \omega |\omega| \frac{(a + L)^4 - a^4}{4}. \quad (15.7)$$

The moment sign opposes the signed rotation. For $a = 0$, the pivot is the shaft's near end; it is not automatically the shoulder. A real golfer's grip translates and rotates, so a single fixed-pivot shaft example does not represent all shoulder, wrist, and club angular speeds.

For $L = 1.1$ m, $d = 0.01$ m, $C_n = 0.5$, $\rho = 1.225$ kg/m³, $\omega = 25$ rad/s, and $a = 0$, the resisting moment magnitude is 0.700595 N m. The near-end-pivot head speed in this same toy geometry would be 27.5 m/s. It would be inconsistent to combine that shaft calculation with a 50 m/s head while claiming one rigid fixed-pivot motion. A different motion model can produce different local speeds, but must specify how.

The generalized load contribution of a segment depends on its velocity Jacobian and the chosen output, not merely its speed or mass. Arm, torso, shaft, and head contributions should therefore be integrated with compatible kinematics. There is no universal 40/40/20 percentage split. The component that dominates force need not dominate a selected joint moment, work integral, or change in endpoint club speed.

15.4 From Air Loads to Generalized Loads

If point i has velocity $v_i = J_{v_i}(q)\dot{q}$ and segment angular velocity $\omega_i = J_{\omega_i}(q)\dot{q}$, its aerodynamic force F_i and couple m_i about that same point contribute

$$Q_{\text{aero}} = \sum_i (J_{v_i}^T F_i + J_{\omega_i}^T m_i), \quad (15.8)$$

with an integral replacing the sum for distributed loads. The formula follows from virtual power:

$$\dot{q}^T Q_{\text{aero}} = \sum_i (v_i^T F_i + \omega_i^T m_i). \quad (15.9)$$

All components must use compatible frames and reference points. For one revolute coordinate, the generalized moment is the projection of the force-induced moment and couple onto that joint's axis. A sum of three-vectors $r_i \times F_i$ is not a general n -coordinate load vector.

Use one consistent sign convention in the equations:

$$M\ddot{q} + h = Bu + p + Q_{\text{aero}} + J_c^T \lambda. \quad (15.10)$$

The drag force already contains its resisting sign. It enters the right-hand side as written; adding an extra minus sign there would reverse its action. Here h contains the declared inertial bias and gravity, p other constitutive loads, and $J_c^T \lambda$ the ideal closure reactions. Other external loads must also be included where appropriate.

For a regular fixed contact mode, eliminating reactions gives an acceleration map $a = f_a + KBu$, where

$$K = M^{-1} - M^{-1} J_c^T (J_c M^{-1} J_c^T)^{-1} J_c M^{-1}. \quad (15.11)$$

An input-independent aerodynamic load contributes KQ_{aero} to the zero-input acceleration at a fixed state. It does not change KB at that state merely because it depends on velocity. It can change the drift linearization, the trajectory, required effort, and reachable outcomes under bounded inputs. Those are different from changing the instantaneous input coefficient. Actively controlled aerodynamic surfaces or actuator-dependent flow would require a model that represents that additional input dependence.

The influence of drag on a norm ratio is also not fixed in sign. It is a vector addition that can reinforce or cancel other drift components. Sums of scalar load magnitudes do not establish a drift/control ratio, and a large norm ratio by itself is not a reachability or neural-strategy certificate.

15.5 Power, Frames, and System Boundaries

For the quasi-steady translational drag law with $C_D \geq 0$,

$$F_D^T v_r = -\frac{1}{2} \rho C_D A \|v_r\|^3 \leq 0. \quad (15.12)$$

It is zero at zero relative speed. In an inertial ground frame,

$$F_D^T v = F_D^T v_r + F_D^T w. \quad (15.13)$$

A tailwind faster than a body can push it forward and add body kinetic energy, while drag still opposes relative motion. For a one-dimensional illustrative law $F_D = -|v - w|(v - w)$, take body speed $v = 1$ and wind $w = 3$ in consistent units: $F_D = 4$, ground-frame body power is 4, and relative-motion power is -8 . The difference comes from energy exchange with the moving air.

In still air, for a declared whole mechanical system with conservative gravity and elastic storage included in $E = T + V_g + V_e$, a suitable balance is

$$\dot{E} = \dot{q}^T Bu + P_{\text{aero}} + P_{\text{other}} - P_{\text{diss,other}}. \quad (15.14)$$

This assumes the remaining passive terms have been classified consistently and ideal stationary constraints do zero total power. Moving constraints, activation-dependent constitutive parameters, and additional stored-energy states require their own terms. Gravity is already

in V_g ; it is not counted again as an energy source in this total-energy derivative. Inertial coupling redistributes the kinetic-energy rate and is not an extra supply of energy.

Aerodynamic work is not metabolic expenditure. Isometric force, co-contraction, activation, and muscle efficiency can consume metabolic energy even when the net external mechanical work is small. A drag estimate therefore cannot explain range-session fatigue or diagnose effort by itself. Likewise, lower head drag may affect attainable speed under a specified swing policy; it does not by itself establish forgiveness on off-center impacts.

15.6 The Sign of an Inverse-Dynamics Correction

For $B = I$ and known reactions and other loads, rearrangement gives

$$u = M\ddot{q} + h - p - Q_{\text{aero}} - J_c^T \lambda. \quad (15.15)$$

There is no leading M^{-1} . The result has generalized-load units, not acceleration units. With the same known boundaries, omitting Q_{aero} gives $u_{\text{omit}} = M\ddot{q} + h - p - J_c^T \lambda$, hence

$$u - u_{\text{omit}} = -Q_{\text{aero}}. \quad (15.16)$$

For an isolated positive rotation with inertia 2 kg m^2 , acceleration 3 rad/s^2 , and resisting aerodynamic moment -4 N m , the true applied moment is 10 N m . Omitting drag gives 6 N m . The omission underestimates the positive applied moment in this example. With a different axis, motion, measured quantity, or signed load, the numerical bias can differ; a universal claim about every inferred joint torque is unjustified.

If hand or ground reactions are also unknown, changing the aerodynamic model can change the fitted allocation between inputs and reactions. Then the simple difference above cannot be used while silently assuming those unknown loads remain fixed. Analyze the joint balance and measurement map as in Chapter 20. Net hand force, net hand couple, net joint load, and individual muscle force are different inferred quantities.

The relative error must name its denominator. Near a zero crossing, a small absolute moment discrepancy can produce an arbitrarily large percentage. Sensitivity to aerodynamic uncertainty should be reported in physical units and compared with measurement uncertainty and the scientific effect being tested. Unsupported universal wrist, ankle, or shoulder error percentages do not provide an accuracy budget.

15.7 A Reproducible Forward Counterfactual

A forward zero-torque counterfactual (ZTCF) sets a declared effective input to zero and integrates from a specified complete state. It retains the stated aerodynamic and passive-load laws and recomputes them along the evolving branch. It is not obtained by recycling measured drag and reaction histories after the trajectory has changed. Zero joint input also does not mean zero muscle activation or no previously stored elastic energy.

Consider an illustrative pendulum, with angle θ measured from its stable bottom equilibrium, angular velocity ω , inertia $I = 2 \text{ kg m}^2$, and gravity moment $-5 \sin \theta \text{ N m}$. Choose quadratic drag coefficient $c \geq 0$ in moment per squared angular-speed units. The zero-input model is

$$\dot{\theta} = \omega, \quad 2\dot{\omega} = -5 \sin \theta - c\omega|\omega|. \quad (15.17)$$

Angular acceleration is $\dot{\omega} = \ddot{\theta}$; $\ddot{\omega}$ would be angular jerk. The signed drag law opposes either direction of rotation.

Starting at $\theta = 0, \omega = 0$ gives an exact equilibrium. No spontaneous swing appears. To investigate a falling motion, start instead at $\theta_0 = \pi/2, \omega_0 = 0$. Its initial potential energy relative to the bottom is 5 J. With

$$E = \omega^2 + 5(1 - \cos \theta), \quad \dot{E} = -c|\omega|^3, \quad (15.18)$$

the first bottom crossing without drag has speed $|\omega| = \sqrt{5}$, about 2.23607 rad/s. A speed of 85 rad/s would require 7225 J of kinetic energy for this inertia, which the stated zero-input initial state cannot supply.

For a second independent check, set $y(\theta) = \omega^2$ along the falling branch, where $\omega < 0$. Write the gravity coefficient as $k = 5$ N m and define $\kappa = 2c/I$. With angles in radians, the resulting angle equation is

$$\begin{aligned} \frac{dy}{d\theta} - \kappa y &= -\frac{2k}{I} \sin \theta, & y(\pi/2) &= 0, \\ y(0) &= \frac{2k}{I} \int_0^{\pi/2} e^{-\kappa s} \sin s \, ds. \end{aligned} \quad (15.19)$$

For the specified inertia, $c = 0.5$ gives $\kappa = 0.5$ and $2k/I = 5$ in the corresponding angular-speed units. It yields

$$y(0) = 4 - 2e^{-\pi/4}, \quad |\omega_{\text{bottom}}| \approx 1.75731 \text{ rad/s}. \quad (15.20)$$

Numerically integrating angle, velocity, and accumulated dissipated work reproduces both bottom-crossing results:

Drag Coefficient	Bottom Time (s)	Bottom Speed (rad/s)	Dissipated Work (J)
$c = 0$	1.17262	2.23607	0
$c = 0.5$	1.26570	1.75731	1.91188

The checks use adaptive integration with relative tolerance 10^{-10} , absolute tolerance 10^{-12} , and maximum step 0.01 s; they also test the energy identity and the independent angle integral. These parameters are reproducible numerical settings, not evidence that the pendulum represents a measured golf swing. It omits the multisegment geometry, real initial motion, muscle and tissue states, and club mechanics that a golf investigation needs.

In this one-coordinate still-air example, energy loss lowers speed at the same bottom angle. With drag, the speed maximum occurs before the bottom: along the falling branch, angular acceleration becomes zero when $5 \sin \theta = c\omega^2$, and the resisting drag remains nonzero at the bottom. Thus a comparison of bottom-crossing speeds is not a comparison of both branches' peak speeds. It does not prove a universal ordering of every segment's speed at a fixed time in a coupled golfer. Changed timing and configuration can redistribute energy and alter endpoint outcomes. Report whether branches are compared at the same time, a matched event, or a matched configuration. Differences between already diverged trajectories also do not reconstruct instantaneous applied torque; the inverse-dynamics chapter gives an explicit counterexample to that inference.

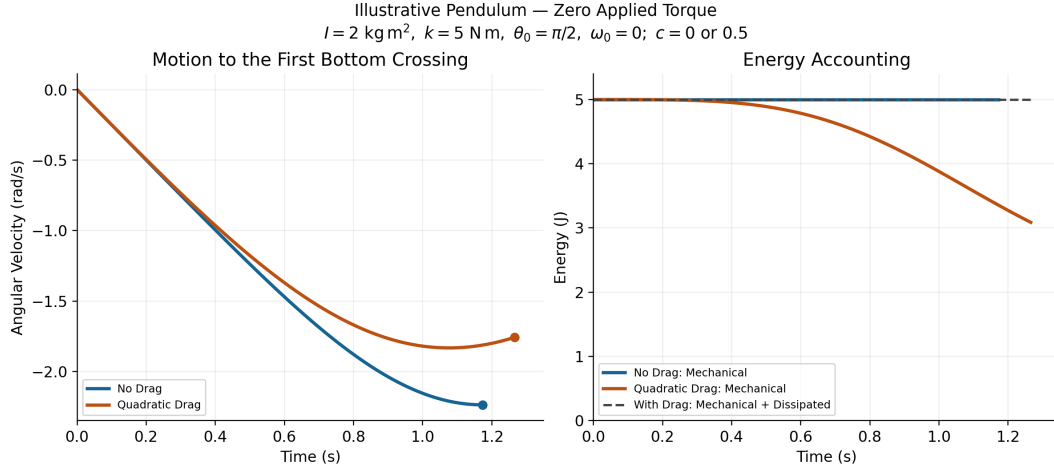


Figure 15.1: Illustrative Zero-Input Pendulum Motion and Energy With and Without Quadratic Drag. Each Curve Ends at Its First Bottom Crossing.

15.8 Ball Flight Connects the Swing to the Outcome

At impact, club motion, impact location, contact mechanics, and ball properties jointly establish launch position, velocity, and spin. Aerodynamics then changes the flight. A swing study that predicts clubhead speed has not automatically predicted carry distance or dispersion. Conversely, a launch-monitor endpoint does not uniquely reconstruct the golfer's joint loading.

For a simplified spherical-ball flight model, let $v_r = v - w$ and use

$$m_b \dot{v} = m_b g + F_D + F_L, \quad \dot{r} = v. \quad (15.21)$$

The drag law uses ball-specific C_D and area conventions. A quasi-steady lift model uses the scalar component $\frac{1}{2} \rho C_L A \|v_r\|^2$, with a signed coefficient and a stated lift direction. For ordinary transverse-spin Magnus lift, a candidate direction is $(\omega_b \times v_r) / \|\omega_b \times v_r\|$. Handle zero cross product explicitly. Coefficients calibrated for spin perpendicular to the flow should not be applied to arbitrary spin-axis angles without an appropriate extension. Spin decay requires an aerodynamic moment law; constant spin is a separate approximation, not a consequence of the translational equation.

A usual transverse-spin parameter is

$$S = \frac{r_b \|\omega_b\|}{\|v_r\|}. \quad (15.22)$$

For illustrative radius 0.02135 m, speed 50 m/s, and transverse spin 200 rad/s, $S = 0.0854$. A coefficient cannot be inferred from that number alone without its Reynolds number, ball geometry, surface condition, and calibrated relation. Report the chosen ball's dimensions and mass rather than mixing approximate radii and masses from different examples.

Dimples can promote boundary-layer transition and delay separation in relevant flow regimes, reducing pressure drag despite additional skin friction. Their benefit depends on Reynolds number, dimple geometry, spin, and surface condition; there is no universal

constant coefficient or fixed percentage range gain. The primary wind-tunnel study [Bearman and Harvey, Golf Ball Aerodynamics \(1976\)](#) provides evidence about tested configurations, not a coefficient table for every ball or driver. A smooth sphere's drag coefficient also changes across flow regimes; labeling a single value as its universal laminar coefficient is incorrect.

Backspin changes lift and may also change drag. Depending on launch speed, angle, spin, coefficient curves, and environment, adding spin can increase or decrease carry. Lift perpendicular to relative velocity does not directly do work in that relative frame, but changes the flight path, duration, and drag history. Height, carry, stopping behavior, and dispersion are different outputs. Neither lower drag nor higher lift alone establishes greater accuracy or stability.

15.9 Environment and Model Uncertainty

For dry ideal-gas air, $\rho = p/(R_{\text{air}}T)$ with absolute temperature T . The [NASA equation-of-state treatment](#) explains the pressure, density, and temperature relation. At the same pressure, the density ratio between 35 and 15 degrees Celsius is

$$\frac{\rho_{35}}{\rho_{15}} = \frac{288.15}{308.15} \approx 0.93510. \quad (15.23)$$

That is a decrease of about 6.49 percent. Pressure, humidity, and viscosity changes need their own treatment; an altitude alone does not specify a day's air density. A reference atmosphere is a model, not a local weather measurement. Lower density reduces both drag and lift for fixed coefficients and relative speed; the change in ball range requires solving the flight problem.

A 5 m/s headwind exactly opposite a 50 m/s point velocity changes relative speed to 55 m/s and, with all other factors fixed, raises drag magnitude by $(55/50)^2 - 1 = 21$ percent. During a swing the velocity direction changes; wind must be subtracted as a vector at each point, rather than adding a scalar headwind to every segment for the entire motion. Crosswind can change both incidence and the direction of aerodynamic force.

In a scalar coefficient model, away from zero speed and for small changes in the separately represented factors, the differential sensitivity is

$$\frac{\delta D}{D} \simeq \frac{\delta \rho}{\rho} + \frac{\delta C_D}{C_D} + \frac{\delta A}{A} + 2\frac{\delta v_r}{v_r}. \quad (15.24)$$

This is a first-order identity, not a rule for adding uncertainty variances. Covariances matter; C_D and A may be linked by calibration, and coefficient variation with speed adds its derivative when speed changes. Near transitions or rapidly changing orientation, local linear sensitivity can be inadequate. Propagate the joint parameter/measurement uncertainty through the quantity actually reported.

15.10 Choosing the Aerodynamic Model

A useful inclusion decision starts with a scientific output and a tolerance. For example, an aerodynamic-moment uncertainty of a fraction of a newton metre might be immaterial to one question and larger than the claimed effect in another. A universal five-percent threshold, a joint's name, or calm weather cannot make that decision on their own.

1. Specify the measured or predicted output, system boundary, coordinate conventions, and tolerable discrepancy in physical units.
2. Begin with a transparent force and moment model. Record coefficient-area pairs, incidence and Reynolds-number ranges, air properties, wind, and the evidence supporting each parameter.
3. Use compatible segment velocities and map loads by virtual power. Check force and moment signs, zero-relative-speed behavior, units, and energy balance before interpreting the output.
4. Compare omitted, nominal, and plausible alternative aerodynamic models through the same inverse or forward analysis. Preserve the same input policy and initial state when comparing forward branches, or explicitly define a different intervention.
5. Check numerical convergence and compare predictions with independent measurements. Add aerodynamic couples, flexibility, wake interaction, or CFD-derived loading when the simpler model's discrepancy matters for the question and the added detail is supportable.
6. Report the resulting sensitivity and remaining uncertainty. Treat a measured performance difference, a model prediction, an illustrative calculation, and a proposed design change according to their evidence.

This places aerodynamic loading within the broader investigation of the golf swing: it is a physical load that interacts with geometry, constrained dynamics, control policy, and impact conditions. Its importance is an output-dependent result to establish, rather than a fixed percentage or a proof that either muscles or passive dynamics must dominate.

15.11 Chapter Exercises

1. Reproduce the 3.0625 N example. Change the reporting area while preserving force; calculate the new coefficient.
2. Derive shaft force and moment for pivot offsets $a = 0$ and $a = 0.6$ m. Explain why a near-end pivot is not a shoulder model.
3. Derive generalized aerodynamic load from virtual power, including the single revolute-axis moment as a special case.
4. Verify the tailwind example in both velocity frames and identify the energy supplied by the moving air.
5. Reproduce the 10 versus 6 N m inverse example. Explain what changes when contact reactions are also unknown.
6. Verify the pendulum equilibrium, bottom-crossing speeds, and energy loss. Explain why peak speed can occur before the bottom.
7. Compute the temperature and headwind examples. Distinguish the factors held fixed from those that can change outdoors.
8. Compare ball flights with equal launch states and different coefficient curves. Explain why range does not identify prior muscle loading.

Chapter 16

Damping, Friction, and Energy Dissipation in the Kinematic Chain

16.1 The Messy Reality of Motion

The golfer, club and ball exchange mechanical energy while tissues, interfaces and materials can dissipate some of it. Energy remaining in body motion, elastic storage or gravity is not automatically heat. Muscle metabolic expenditure is also distinct from the mechanical work delivered at the modeled joints. A useful damping model starts with a stated system boundary and an energy ledger.

This chapter separates three questions: where mechanical energy is lost, how a local loss changes coupled motion, and what can be inferred about stability, impact or perceived sweetness. Passive loss does not guarantee better control, worse ball speed or a preferred sound. Those outcomes require a specified trajectory, boundary condition and comparison.

16.2 The Hidden Friction in Your Body

16.2.1 Viscous Damping

For a translational coordinate, $F_d = -cv$ has c in N s/m. For an angular coordinate, $\tau_d = -c_\theta \dot{q}$ has c_θ in N m s/rad. The corresponding power is $-cv^2$ or $-c_\theta \dot{q}^2$. The interpretation of a fitted joint coefficient depends on the mechanisms represented by that model; it is not a direct measurement of one tissue's viscosity.

16.2.2 Synovial Fluid Viscosity

Lubrication and deformation at a joint can contribute resistance, but the phenomenological approximation $\tau_{\text{fluid}} = -c_{\text{fluid}} \dot{q}$ requires identification over a declared posture, loading, temperature and frequency range. It does not establish a constant coefficient for a whole golf swing or a universal monotone dependence on joint size.

16.2.3 Muscle Viscosity and the Hill Model

An active muscle can deliver positive mechanical work. Its force-velocity relation must therefore not be relabeled as a passive damper. Hill's force-velocity experiment concerns active contraction [Hill, 1938]; a local slope around a particular activation and operating state is an incremental impedance, not a proof of global passivity. A Hill-type implementation must declare its contraction sign, active and passive elements, moment arms and

activation dynamics. A dashpot is a modeling choice, not a mandatory element implied by the historical equation.

For a passive translational dashpot with moment arm r and local shortening speed $v = r\dot{q}$, power consistency gives $c_\theta = cr^2$. One cannot convert N s/m to N m s/rad without geometry. With multiple muscles, the corresponding Jacobian congruence retains cross-joint coupling.

16.2.4 Tendon Viscoelasticity and Hysteresis

For a closed measured force-extension cycle, the loop integral $\oint F d\ell$ is the mechanical energy not returned over that cycle. A synthetic storage of 10 J with an explicitly assumed 8% loss would return 9.2 J; this arithmetic supplies no measured tendon percentage or golfer energy budget.

As an illustrative nonlinear loss model, a memoryless quadratic resistance $-b|\dot{q}|\dot{q}$ is passive for $b \geq 0$, but it does not by itself represent viscoelastic memory or identify a measured hysteresis loop across frequencies. Internal-state or hereditary models need their own stored energy, loss law and calibration. Elastic recoil, dissipation and active muscle work remain separate.

16.2.5 Ligament Damping and Cartilage Deformation

Deformation can store and return energy as well as dissipate it. A model must distinguish these contributions before assigning any residual work to friction. Tissue-specific parameters and injury or clinical conclusions require appropriate identification; this chapter supplies neither a universal joint-loss hierarchy nor an injury-risk model.

16.2.6 Soft Tissue Wobble Damping

Relative soft-tissue motion adds inertial and constitutive states, as discussed in Chapter 23. A reduced damping coefficient can approximate a response only in the range over which that reduction is checked. Omitted tissue inertia or resonance must not be concealed by increasing a numerical damping parameter.

16.2.7 Damping by Joint

There is no qualified table of golfer-wide joint damping coefficients in this chapter. The identification requirements are instead:

Modeled Element	Required Coordinates and Evidence
Shoulder and scapular motion	Declared relative rotations, posture, activation and coupled torque response
Elbow and wrist	Joint-axis definitions, tendon routing and independent perturbation data
Trunk	Distributed or reduced coordinate model and the retained mechanical boundary
Hand-club interface	Colocated force/torque and relative motion, grip pressure, phase and frequency band
Shaft and head	Specimen, boundary conditions, mode shapes, damping and instrument correction

Passive lower-limb studies are not substitutes for measured shoulder, wrist or grip coefficients. Synthetic sensitivity ranges must be labeled as assumptions and cannot be promoted to physiological uncertainty bounds.

16.3 Adding Damping to the Equations of Motion

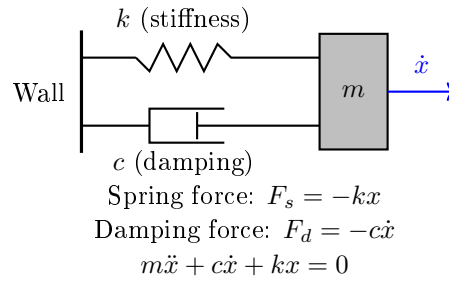


Figure 16.1: Parallel Spring and Damper Acting on a Translating Mass. Both Elements Share the Same Displacement and Velocity; Their Forces Add. An Angular Analogue Uses Rotational Inertia and Torques.

Recall from Chapter 9 that a kinematic chain with actuator and constraint forces is governed by:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \mathbf{B}(\mathbf{q})\mathbf{u} + \mathbf{J}_c^T(\mathbf{q})\boldsymbol{\lambda}$$

This is a second-order force balance. Its first-order drift field is obtained after defining the state and setting the control input to zero; actuator forces belong to the input term. A phenomenological viscous model adds a separate dissipative force:

Key Principle 16.1. Damped Kinematic Chain Equations

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{D}(\mathbf{q})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \mathbf{B}(\mathbf{q})\mathbf{u} + \mathbf{J}_c^T(\mathbf{q})\boldsymbol{\lambda}$$

where the symmetric viscous model used here assumes $\mathbf{D}(\mathbf{q}) = \mathbf{D}(\mathbf{q})^T \succeq 0$. Non-negative diagonal entries alone do not establish passivity for a coupled matrix. The dissipated power is $P_d = \dot{\mathbf{q}}^T \mathbf{D} \dot{\mathbf{q}} \geq 0$.

The matrix $\mathbf{D}$ maps the declared generalized velocities to generalized dissipative forces. Geometry, activation and internal material states can change an identified response. Neither configuration dependence nor increasing damping with flexion follows merely from choosing the notation $\mathbf{D}(\mathbf{q})$.

16.3.1 Types of Damping Models

Linear viscous damping is useful for a local, frequency-limited approximation. A coupled passive form can be parameterized as $\mathbf{D} = \mathbf{R}^T \mathbf{R}$; this preserves nonnegative loss without assuming independent joints. Positive diagonal entries alone do not suffice.

Dry sliding friction, rate-dependent drag and viscoelastic memory are different constitutive models. For an angular coordinate, a quadratic torque law uses a coefficient in N m s²/rad²; an aerodynamic translational force coefficient cannot be inserted unchanged. The friction section below states the sliding and sticking conventions explicitly. No universal dominance of linear damping at swing speeds is assumed.

16.3.2 Rewriting in Control-Affine Form

Use local independent joint coordinates with $\mathbf{v} = \dot{\mathbf{q}}$ and state $\mathbf{x} = (\mathbf{q}, \mathbf{v})$. Evaluate the coefficient matrices and gravity force at the current state. For a smooth mode without unresolved constraint forces,

$$\begin{aligned}\dot{\mathbf{x}} &= f_0(\mathbf{x}) + f_d(\mathbf{x}) + G(\mathbf{x})\mathbf{u}, \\ f_0 &= \begin{bmatrix} \mathbf{v} \\ -\mathbf{M}^{-1}(\mathbf{C}\mathbf{v} + \mathbf{g}) \end{bmatrix}, \\ f_d &= \begin{bmatrix} \mathbf{0} \\ -\mathbf{M}^{-1}\mathbf{D}\mathbf{v} \end{bmatrix}, \\ G &= \begin{bmatrix} \mathbf{0} \\ \mathbf{M}^{-1}\mathbf{B} \end{bmatrix}.\end{aligned}$$

Every vector field has $2n$ components; joint damping forces enter the acceleration block through the inverse mass matrix. A quaternion or other redundant orientation representation instead needs its declared kinematic map $\dot{\mathbf{q}} = \mathbf{N}(\mathbf{q})\mathbf{v}$. In a closed chain, retain $\mathbf{J}_c^T \boldsymbol{\lambda}$ in the force balance and solve the compatible constraint equations, or eliminate the reactions consistently. The unconstrained G above cannot simply be reused after that elimination.

16.3.3 Conservative Coupling and Dissipated Power

The [manipulator-equation derivation in Tedrake's MIT notes](#) identifies the mass and Coriolis terms together as the contribution of kinetic energy. For a time-independent Lagrangian model with a consistent Coriolis representation,

$$\begin{aligned}T &= \frac{1}{2}\mathbf{v}^T \mathbf{M}\mathbf{v}, \\ \mathbf{v}^T \mathbf{C}\mathbf{v} &= \frac{1}{2}\mathbf{v}^T \dot{\mathbf{M}}\mathbf{v}.\end{aligned}$$

One standard choice makes $\dot{\mathbf{M}} - 2\mathbf{C}$ skew-symmetric; the matrix $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ itself is not unique. With conservative potential V and $\mathbf{g}(\mathbf{q}) = \nabla V$, differentiating $E = T + V$ and substituting the force balance gives

$$\dot{E} = \mathbf{v}^T \mathbf{B}\mathbf{u} + \mathbf{v}^T \mathbf{J}_c^T \boldsymbol{\lambda} - \mathbf{v}^T \mathbf{D}\mathbf{v}.$$

The Coriolis contribution cancels the changing kinetic metric; it is not heat loss. Ideal stationary constraints do no work because $\mathbf{J}_c \mathbf{v} = 0$. Moving constraints can supply or remove energy, so their work must remain in the ledger. Other physical loss laws require their own power terms.

Linearization about a driven trajectory can contain velocity-dependent terms that affect perturbation growth or decay. Those terms are not, by that fact alone, passive viscous damping. For [impact and acoustics](#), distinguish this inertial redistribution and moving-grip work from identified material/hand losses and acoustic radiation. A quieter or faster-decaying response does not by itself identify which mechanism caused it.

16.4 How Damping Propagates Through a Chain

A key observation is that damping at one joint does not affect only that joint. It propagates through the entire chain via the mass matrix coupling, Coriolis coupling, and constraint coupling.

16.4.1 Direct Effect: Local Damping

The most obvious effect is local. Damping at joint i directly opposes angular velocity at that joint:

$$-D_{ii}\dot{q}_i$$

This is a force contribution. The resulting acceleration requires the mass and constraint solve; other simultaneous forces can still accelerate the joint.

16.4.2 Coupling Through the Mass Matrix

Here is where it gets interesting. Rearranging the damped equation of motion:

$$\ddot{\mathbf{q}} = \mathbf{M}(\mathbf{q})^{-1} [\mathbf{B}\mathbf{u} - \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{D}\dot{\mathbf{q}} - \mathbf{g}(\mathbf{q})]$$

Because $\mathbf{M}(\mathbf{q})$ is *not diagonal*, the terms inside the brackets get "mixed" when inverted. Let's illustrate with a simple two-joint chain: shoulder (joint 1) and elbow (joint 2).

Example 16.1. Shoulder-Elbow Damping Coupling

For a planar two-link illustration with massless links, a point mass m_1 at the first link's end and m_2 at the second link's end, use absolute first-link angle and relative elbow angle q_2 . Its exact mass matrix is:

$$\mathbf{M}(\mathbf{q}) = \begin{pmatrix} m_1 L_1^2 + m_2(L_1^2 + L_2^2 + 2L_1 L_2 \cos q_2) & m_2(L_2^2 + L_1 L_2 \cos q_2) \\ m_2(L_2^2 + L_1 L_2 \cos q_2) & m_2 L_2^2 \end{pmatrix}$$

The off-diagonal terms $M_{12} = M_{21}$ are nonzero and depend on the elbow angle q_2 . When we invert this matrix to solve for $\ddot{\mathbf{q}}$, we get:

$$\ddot{\mathbf{q}} = \mathbf{M}(\mathbf{q})^{-1} \cdot (\dots - \mathbf{D}\dot{\mathbf{q}} - \dots)$$

Suppose we have damping only at the elbow: $\mathbf{D} = \text{diag}(0, c_2)$. The damping term

entering the inverse is:

$$-D\dot{\mathbf{q}} = \begin{pmatrix} 0 \\ -c_2\dot{q}_2 \end{pmatrix}$$

After multiplication by $\mathbf{M}(\mathbf{q})^{-1}$, the shoulder acceleration $\ddot{q}_1$ receives a contribution from this elbow damping force, even though the shoulder has zero damping directly. Quantitatively:

$$(\ddot{q}_1)_d = -(\mathbf{M}(\mathbf{q})^{-1})_{12}c_2\dot{q}_2 = \frac{M_{12}}{\det(\mathbf{M}(\mathbf{q}))}c_2\dot{q}_2$$

This is only the damping contribution to shoulder acceleration. The cross-term $(\mathbf{M}(\mathbf{q})^{-1})_{12}$ couples the two joints; its sign depends on configuration. The determinant already occurs in the inverse and must not be applied a second time. Total passive loss remains $c_2\dot{q}_2^2$, even when the shoulder accelerates in the direction of its own motion.

This is crucial: in a multi-joint chain, you cannot treat each joint's dynamics in isolation. The mass matrix creates a network of inertial coupling. Damping at a distal joint propagates proximally.

16.4.3 Coupling Through Coriolis Effects

The Coriolis force depends quadratically on the joint velocities at a fixed configuration:

$$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}$$

Uniformly scaling all velocities by a factor scales this force by the factor squared. Changing one velocity component or the configuration can instead increase, decrease or reverse individual force components. Damping therefore changes subsequent inertial coupling through the trajectory, but it creates no additional Coriolis dissipation beyond the physical loss terms in Section 16.3.3.

16.4.4 Energy Cascade in Serial Chains

At a prescribed state with diagonal passive joint damping, the local loss is $P_i = c_i\dot{q}_i^2$. The mass matrix affects the resulting accelerations and trajectory, but does not insert an extra downstream inertia multiplier into this instantaneous loss. Neither the proximal nor the distal joint must dissipate more. Relative joint rates must not be replaced by absolute segment rates.

For a synthetic pair with equal $c_i = 0.2$ N m s/rad and rates 10 and 30 rad/s, the losses are 20 and 180 W. Reversing the rates reverses the ranking. Integrating along matched trajectories gives a loss comparison; recomputing a freely evolving swing can also change actuator work and delivered geometry. The previous shoulder/wrist speed estimate and its power ratio are withdrawn because they did not establish those coordinates or measurement conditions.

16.4.5 Constraint Coupling in Parallel Mechanisms

For two hands on different points of a rigid club, hand-point velocities are generally different: a rotating club contributes $\boldsymbol{\omega} \times \mathbf{r}$ at each offset. Express all twists and wrenches in declared

compatible frames. Let $\mathbf{q} = (\mathbf{q}_L, \mathbf{q}_R, \mathbf{q}_c)$ include both arms and the club, and let the no-slip closure be $\mathbf{J}_c \dot{\mathbf{q}} = 0$. Ideal reactions are $\mathbf{J}_c^T \boldsymbol{\lambda}$, with

$$P_{\text{constraint}} = \dot{\mathbf{q}}^T \mathbf{J}_c^T \boldsymbol{\lambda} = 0.$$

Separate generalized torques of arms with different coordinate spaces cannot simply be added and set to zero. Club force and moment balance retains its inertia, external loads and both contact wrenches. Ideal internal constraints redistribute work between components without creating extra total loss.

If $\dot{\mathbf{q}} = \mathbf{N}\dot{\mathbf{s}}$ with $\mathbf{J}_c \mathbf{N} = 0$, then $\mathbf{D}_r = \mathbf{N}^T \mathbf{D} \mathbf{N}$ and $\dot{\mathbf{s}}^T \mathbf{D}_r \dot{\mathbf{s}} = \dot{\mathbf{q}}^T \mathbf{D} \dot{\mathbf{q}}$. This proves preserved nonnegative loss while allowing off-diagonal reduced coupling. A changing basis also contributes kinematic terms to the dynamics. Moving constraints have prescribed nonzero velocity and can perform work; compliant or slipping contacts require explicit port laws rather than ideal no-slip closure.

16.5 The Damping Paradox

16.5.1 Damping and Stability

For an autonomous linear perturbation model,

$$\mathbf{M}\ddot{\mathbf{x}} + (\mathbf{G} + \mathbf{C})\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = 0,$$

assume constant symmetric positive definite $\mathbf{M}, \mathbf{K}$, skew $\mathbf{G}$, and symmetric positive semidefinite $\mathbf{C}$. Then $E = \frac{1}{2}\dot{\mathbf{x}}^T \mathbf{M} \dot{\mathbf{x}} + \frac{1}{2}\mathbf{x}^T \mathbf{K} \mathbf{x}$ is positive definite and $\dot{E} = -\dot{\mathbf{x}}^T \mathbf{C} \dot{\mathbf{x}} \leq 0$. This proves Lyapunov stability. Asymptotic stability additionally requires that the largest invariant set with zero dissipation contain only the equilibrium. Positive definite $\mathbf{C}$ suffices under these assumptions; semidefinite damping can leave undamped modes.

With negative stiffness, even the scalar dimensionless equation $\ddot{x} + \dot{x} - x = 0$ retains the growing rate $(\sqrt{5}-1)/2$. A stronger counterexample has $\mathbf{M} = \mathbf{I}$, $\mathbf{K} = -\mathbf{I}$ and $\mathbf{G} = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix}$.

With $\mathbf{C} = 0$ its four distinct rates are $\pm i(3 \pm \sqrt{5})/2$. Adding the passive $\mathbf{C} = 0.1\mathbf{I}$ produces a pair with positive real parts. Energy loss and instability coexist because the equilibrium energy is indefinite. Dissipation-induced instability is a documented phenomenon [Bloch et al., 1994], not a claim that this particular instability occurs in a golf club.

A nonsymmetric loaded tangent, accelerating frame or time-varying swing requires its own stability analysis. Frozen eigenvalues and a nonsingular frequency response do not establish stable time evolution. Physical damping must remain separate from numerical stabilization.

16.5.2 Underdamped vs. Overdamped Joints

For one constant-coefficient scalar oscillator with positive inertia I , stiffness k and damping c , define $\omega_n = \sqrt{k/I}$ and $\zeta = c/(2\sqrt{kI})$. Values $0 < \zeta < 1$, $\zeta = 1$ and $\zeta > 1$ distinguish oscillatory, repeated-real and distinct-real decay rates. At $c = 0$ oscillations do not decay. Critical damping gives the familiar nonoscillatory step response for zero initial conditions; arbitrary initial velocity can still cause a crossing of equilibrium, and no universal optimal energy-dissipation objective follows.

A joint embedded in a coupled active chain need not have a single independently meaningful damping ratio. Identify modes or a matrix impedance over the relevant operating range. The chapter does not infer that all human joints are underdamped or that a wrist swing exploits a particular resonance.

16.5.3 Active Damping and Co-Contraction

Impedance control concerns the dynamic relationship between motion and applied wrench [Hogan, 1985]. For a local restoring force on the modeled limb, one convention is $\mathbf{f} = -\mathbf{K}(\mathbf{x} - \mathbf{x}_d) - \mathbf{C}(\dot{\mathbf{x}} - \dot{\mathbf{x}}_d)$. The sign reverses for the reaction exerted on its environment. A moving target can perform work even when the relative loss is passive.

Co-contraction can alter an identified impedance, but does not supply a universal independent adjustment of stiffness and damping. The identified change depends on activation, reflex delay, posture and task. Passive tissue loss and work by active control must be retained separately; an assumed change during transition is a hypothesis requiring measurement.

16.5.4 The Loose Grip Debate

The relevant question is how the identified hand-club impedance depends on pressure, posture, frequency and amplitude. A grip can change storage, inertia, damping, contact area and slip together. It is not enough to assign one scalar damper or assume that tighter grip always improves control and reduces ball speed.

Chimentin and colleagues studied one participant and one hybrid club. Their modal results generally show increased bending damping with firmer grip, but their conclusion states the opposite trend. Their numerical impact response also contains algorithmic damping that is not independently separated from material loss. Preserve these limitations; this study does not establish a universal optimum grip pressure or broadband impact sound prediction [Chimentin et al., 2019].

The guitar analogy depends on mode-dependent contact: touching where a mode moves can change its decay, while a node can reduce coupling to that mode. A flexible head and shaft have different modes and radiation efficiencies. A shorter ringdown at the grip is not automatically quieter sound at the listener or a sweeter hit. Compare calibrated sound and hand vibration separately, then use blinded perception experiments.

16.6 Friction Models for Joints

16.6.1 Coulomb Friction

For a scalar angular sliding coordinate, use $\tau_f = -\tau_k \text{sign}(\dot{q})$ when $\dot{q} \neq 0$, with $\tau_k \geq 0$ in N m. At a simple interface, $\tau_k = \mu_k N r$ requires the effective moment arm r . Distributed contact requires integration of traction and lever arm. The product $\mu_k N$ alone is a force, not a joint torque.

16.6.2 Stiction and Reversal Points

At zero slip velocity, static friction is set-valued: $\tau_f \in [-\tau_s, \tau_s]$. Solve force balance and the sticking constraint together; slip begins when the required reaction exceeds the admissible bound. Setting $\text{sign}(0) = 0$ cannot represent nonzero sticking force. Different joints need

not reverse simultaneously, and a swing transition is not evidence that an anatomical joint behaves like a dry bearing.

16.6.3 The Stribeck Effect

An explicitly phenomenological sliding law is

$$\begin{aligned} h(v) &= \tau_k + (\tau_s - \tau_k)e^{-(|v|/v_s)^2}, \\ \tau_f(v) &= -h(v)\operatorname{sign}(v) - cv, \quad v \neq 0. \end{aligned}$$

Here $\tau_s \geq \tau_k > 0$, $v_s > 0$, and $c \geq 0$. The dry-friction magnitude approaches τ_s at vanishing nonzero speed and τ_k at high speed; the viscous term can still grow. Power is $-h(v)|v| - cv^2 \leq 0$. The separate zero-speed inclusion handles sticking. This fixes the reversed limits in the previous expression and does not identify a golfer's joint law.

16.6.4 LuGre Friction Model

A bristle-state model can represent presliding memory [Canudas de Wit et al., 1995]. Under a rotational convention with z in rad and $h(v)$ in N m, one possible notation is

$$\dot{z} = v - \sigma_0|v|z/h(v), \quad \tau_{\text{resistance}} = \sigma_0 z + \sigma_1 \dot{z} + \sigma_2 v.$$

The torque applied against motion is $-\tau_{\text{resistance}}$. State storage means instantaneous returned power need not be zero: passivity must be checked against a declared storage function and admissible parameters. Define initialization, units, steady sliding and numerical handling before use. This is a model option, not a validated anatomical law; its necessity depends on the measured phenomena and validation target.

With $\sigma_0 > 0$ in N m/rad and $\sigma_1, \sigma_2 \geq 0$ in N m s/rad, the candidate storage $S = \frac{1}{2}\sigma_0 z^2$ gives

$$\tau_{\text{resistance}}v - \dot{S} = \frac{\sigma_0^2|v|z^2}{h(v)} + \sigma_1 \dot{z}v + \sigma_2 v^2.$$

For $\sigma_1 = 0$ this is nonnegative. With bristle damping, the cross term need not be nonnegative, so coefficient positivity alone does not make this storage a certificate for the full output. For a synthetic instantaneous state with $\sigma_0 = \sigma_1 = 1$, $\sigma_2 = 0$, $z = 0.2$, $v = 1$ and $h(v) = 0.1$ in the stated units, $\dot{z} = -1$ and the right-hand side is -0.6 W. Check the permitted initial states and constitutive operating range before accepting or rejecting a complete passivity claim.

16.6.5 Friction Criticality at the Transition

At sticking, friction can support a force with zero mechanical power. During sliding, loss depends on slip displacement and traction, not on counting reversals. A smooth-looking or jerky transition alone cannot quantify dissipation. No universal 20 ms transition, simultaneous peak tissue stretch or optimal friction level is inferred here. Measure motion, forces and internal states over a defined interval before assigning a transition loss.

16.7 Energy Dissipation Budget

16.7.1 Sources of Dissipation

For a closed mechanical model with explicitly accounted boundary inputs,

$$E(t_1) - E(t_0) = W_{\text{actuators}} + W_{\text{boundary}} - E_{\text{losses}}.$$

Include translational and rotational kinetic energy of every retained component, gravity, strain energy and internal constitutive storage in E . Define the system boundary before allocating aerodynamic, interface and radiated-acoustic energy: a flux leaving one subsystem can be stored in another. Avoid counting the same fitted residual as both tissue damping and grip loss.

16.7.2 Interpretation

For a synthetic ledger, $E(t_0) = 20$ J, actuator work 100 J, moving-boundary work 5 J and losses 15 J give $E(t_1) = 110$ J. If clubhead kinetic energy is 40 J, the other 70 J may remain in the modeled body, shaft and potential energy. It is not an additional 70 J dissipation. These numbers are an accounting example, not an amateur-golfer energy budget or a metabolic efficiency estimate.

16.8 Implications for Swing Modeling

16.8.1 When Damping Is Negligible

Neglect a loss only after a sensitivity or error-budget check for the quantity of interest. A small change in ball speed can coexist with a large change in a narrow resonance or ringdown. A zero instantaneous velocity removes a viscous power term at that instant; it does not establish negligible damping over the surrounding interval or eliminate static forces.

16.8.2 When Damping Is Critical

The importance of a loss depends on the excited modes, their frequencies and motion at dissipative ports, the duration of contact and the observation window. Separate pre-impact trajectory changes, forces during contact and vibration after separation. Post-separation damping cannot change the launch state already established, although it can change sound and feel.

16.8.3 Sensitivity Analysis

For a declared scalar damping parameter d and reproducible comparison protocol,

$$\Delta v_{\text{club}} \approx \frac{\partial v_{\text{club}}}{\partial d} \Delta d.$$

There is no extra minus sign: the derivative already carries its sign. A synthetic derivative of -0.02 m/s per N m s/rad with $\Delta d = 5$ N m s/rad gives -0.1 m/s, approximately -0.224 mph using $1 \text{ mph} = 0.44704 \text{ m/s}$. This is neither a measured derivative nor a 2 mph

effect. State whether controls, delivery state, trajectory or external impulse are held fixed, and check finite-difference step convergence where used. A chosen 50% parameter sweep is scenario analysis, not a confidence interval.

16.8.4 Recommendations for Modelers

Model choice depends on the requested observable and on reproducing the specified measurements within their uncertainty. Record coordinates, force signs, units, boundary work, source data and the qualified frequency/amplitude range. Distinguish calibration from held-out validation and report null effects.

Separate physical damping from integration damping; refine time step, mesh and retained modes without retuning coefficients to hide numerical error. Test stationary, moving-support, free and clamped limits. Inspect stability and passivity independently of solver convergence. Unknown losses must remain unknown rather than becoming default golfer coefficients.

16.9 Damping in the Shaft

16.9.1 Material Damping

A recorded decay can combine shaft material loss, head and adhesive loss, grip/fixture dissipation, radiation and instrumentation. It is not automatically intrinsic material damping. Identify the specimen, layup, mode and boundary condition. A material name or flex rating alone cannot rank full-club decay or ball speed.

16.9.2 Quality Factor (Q-Factor)

For a lightly damped isolated mode, $Q \approx 1/(2\zeta)$ relates a conventional quality factor to modal damping ratio. With $x(t) = Ae^{-\zeta\omega_n t} \cos(\omega_d t + \phi)$ and $\omega_d = \omega_n \sqrt{1 - \zeta^2}$, the logarithmic decrement is $\delta = 2\pi\zeta/\sqrt{1 - \zeta^2}$. The energy lost between corresponding peaks is a fraction $1 - e^{-2\delta}$. Thus defining Q directly as 2π times starting energy divided by that finite-cycle loss agrees with $1/(2\zeta)$ only in the weak-loss limit.

For a synthetic 100 Hz mode with $\zeta = 0.01$, the conventional $Q = 50$, the amplitude time constant is about 0.159 s, and the corresponding peak-to-peak energy loss is about 11.8%. Higher Q means weaker fractional damping under these conventions; it does not mean higher stiffness. The previously unsourced steel/graphite ranges and contradictory material ranking are withdrawn.

16.9.3 Lead-Lag Dynamics and Damping

Before impact, mass, stiffness, pre-existing shaft deformation and velocity, applied wrenches and losses determine the delivered state together. After impact, observed motion depends on which bending, torsional and head modes were excited. A claim that shaft damping is negligible during the downswing needs a qualified calculation or measurement for that model and observable.

16.9.4 Practical Implications

For a frozen small perturbation with point-motion map $\mathbf{L}$ and the $e^{i\omega t}$ convention, the impact-point mobility is

$$\mathbf{Y}(\omega) = i\omega\mathbf{L} [\mathbf{K} - \omega^2\mathbf{M} + i\omega(\mathbf{G} + \mathbf{C})]^{-1} \mathbf{L}^T.$$

The transpose maps point wrench back to generalized force and preserves power. Translational and rotational rows have different units. Replacement of a constant-coefficient boundary by an identified frequency-dependent grip impedance depends on its qualification, but its storage and loss must not be counted twice. At a finite preload, differentiating the configuration-dependent port map also contributes geometric tangent terms. A singular or unresolved pencil requires refusal or a qualified alternative formulation, not an arbitrary pseudoinverse.

16.9.5 Material Damping and Shaft Behavior

Rotation-induced stress can change the tangent stiffness, while Coriolis terms change dynamic coupling. Tension, centrifugal terms, bending and the grip boundary must be derived from one consistent loaded configuration; they do not universally increase every frequency or create extra dissipative loss. A held pose in a rotating observer does not imply a stationary support in the laboratory.

The implementation program in Tools #5072 retains these distinctions in its loaded chain and finite grip ports. Its synthetic balance and operator tests establish numerical verification only. Contact duration and offset, flexible-head response, acoustic radiation, microphone transfer and physical validation remain separate steps. A claim that one player's strike is sweeter requires matched-delivery analysis and blinded listening evidence, not a ranking of shaft material names.

16.10 The Control-Affine Perspective on Damping

In the control-affine framework, damping modifies the drift field. Recall:

$$\dot{\mathbf{x}} = f_0(\mathbf{x}) + G(\mathbf{x})\mathbf{u}$$

For the independent-coordinate viscous model in Section 16.3.2, the drift field becomes:

$$f_{\text{damped}}(\mathbf{x}) = f_0(\mathbf{x}) + \begin{bmatrix} \mathbf{0} \\ -\mathbf{M}(\mathbf{q})^{-1}\mathbf{D}\mathbf{v} \end{bmatrix}.$$

Configuration-dependent passive coefficients may represent identified losses within their calibration range. Coulomb friction, viscoelastic memory and active muscle dynamics need their own constitutive states or force laws; they cannot generally be folded into a constant viscous matrix. Coriolis terms remain in the conservative drift and are excluded from the dissipated-power budget.

Drift vs. Control 16.1. Damping as Drift Modification

For an unforced system with stationary ideal constraints, passive damping makes total mechanical energy nonincreasing. Gravity and joint coupling can still increase an individual joint's speed, and energy decay alone does not prove stability of every equilibrium or driven trajectory. When a particular trajectory is prescribed, inverse dynamics determines the additional actuator work associated with damping. This bookkeeping does not establish a universal muscular strategy for reversing the swing.

16.11 Summary: The Damping Principle

Passive loss, energy redistribution, active work and numerical damping are distinct. Coupling changes which motions and ports are excited, while geometry and operating state change the loaded response. Neither energy decay nor a successful solve establishes stability, a launch advantage or perceived sweetness. Report those quantities with their own assumptions and evidence.

16.12 Chapter Exercises

All numerical inputs below are synthetic verification cases, not golfer parameter estimates.

- **Damping Coefficient Estimation.** For this synthetic case, use $c_w = 0.1$ N m s/rad and $\dot{q}_w = 50$ rad/s. Find the resisting torque and instantaneous loss power. Compare the torque magnitude with an assumed available 20 N m; explain why this fraction is not a metabolic efficiency.
- **Energy Dissipation in a Two-Joint Chain.** For this synthetic case, use rates 10 and 30 rad/s and coefficients 0.5 and 0.15 N m s/rad. Compute total loss power and its ratio to an assumed 400 W mechanical actuator input. Explain why joint location alone does not determine the ratio.
- **Damping Ratio and Step Response.** For this synthetic case, a scalar angular oscillator has inertia $I = 0.05$ kg m², stiffness $k = 5$ N m/rad and damping $c = 0.2$ N m s/rad. Compute ω_n , ζ and the approximate 2% step-response settling estimate $4/(\zeta\omega_n)$. State its assumptions and distinguish rotational inertia from mass.
- **Coulomb vs. Viscous Friction.** For this synthetic case, an idealized contact has $\mu_k = 0.1$, normal force 50 N and moment arm 0.02 m. Find its sliding torque and the angular speed at which $c = 0.5$ N m s/rad gives equal viscous torque. Explain why zero-speed sticking needs a force-balance inclusion.
- **Configuration-Dependent Damping.** For this synthetic case, on the explicitly restricted interval $0 \leq q \leq 1$ rad, let $c(q) = 0.1 + 0.02q$ N m s/rad and $\dot{q} = 50$ rad/s. Compute torque at $q = 0, 0.5, 1$. Check nonnegative loss and distinguish this assumed law from measured posture dependence.
- **Stability Counterexample.** Calculate the roots of $\ddot{x} + \dot{x} - x = 0$ and identify why the positive-stiffness energy argument fails. Repeat for the two-coordinate gyroscopic example with and without 0.1I damping.

- **Grip and Sound.** Design a comparison that separates changed delivery, changed impact-point mobility, radiated sound and listener preference. State which data would falsify a proposed grip-damping mechanism and which quantities a shaft accelerometer alone cannot identify.

Chapter 17

Impact: The Collision That Matters

17.1 The State Delivered to the Ball

Impact converts part of the club's incident mechanical energy into ball translation, spin, deformation, and dissipation. The golfer and club retain energy afterward; the entire swing energy is not transferred in the collision. Launch also depends on the local face normal, contact-point velocity, strike location, ball and face properties, and the club's response during contact. Ball speed, launch direction, launch elevation, and the full spin vector are needed to initialize a flight model. Three scalars omitting speed and spin axis cannot determine the trajectory.

For the rigid clubhead, let O be its centre of mass and P the contact point:

$$v_P = v_O + \omega_c \times r, \quad r = p_P - p_O. \quad (17.1)$$

Both velocities and all offsets must use the same frame. A path reported at the face centre need not equal the centre-of-mass path. Face attitude is a different quantity from velocity; static loft plus attack angle does not determine dynamic loft. Declare whether delivery is measured immediately before contact, at maximum compression, or at another instant.

17.2 Normal Impact: Momentum, Restitution, and Energy

For an isolated one-dimensional collision with stationary ball, club mass M , ball mass m , and incident club speed v , momentum and restitution give

$$Mv = Mv_c^+ + mv_b^+, \quad (17.2)$$

$$e = \frac{v_b^+ - v_c^+}{v}. \quad (17.3)$$

This is relative separation speed divided by relative approach speed. A coefficient from a fixed-surface rebound experiment requires its own setup definition before being compared with a moving-club result. Solving gives

$$v_b^+ = \frac{(1+e)M}{M+m}v, \quad v_c^+ = \frac{M-em}{M+m}v. \quad (17.4)$$

For $M = 0.200$ kg, $m = 0.04593$ kg, and illustrative $e = 0.83$, the speed ratio is about 1.488. This explains the scale of driver smash factor within the model. It is not a universal equipment ceiling or a direct energy ratio.

The ball's fraction of initial club translational energy is

$$\frac{E_b^+}{E_c^-} = \frac{m}{M} \left[\frac{(1+e)M}{M+m} \right]^2, \quad (17.5)$$

about 0.508 for those values. The total kinetic-energy loss is

$$E^- - E^+ = \frac{1}{2} \frac{Mm}{M+m} (1 - e^2) v^2. \quad (17.6)$$

Thus e^2 describes retained relative-motion energy in the centre-of-mass frame for this model; it is not the fraction of club energy delivered to the ball. Oblique impact also partitions energy into rotation and deformation. These distinctions underlie standard golf impact treatments [Jorgensen, 1994, Penner, 2003].

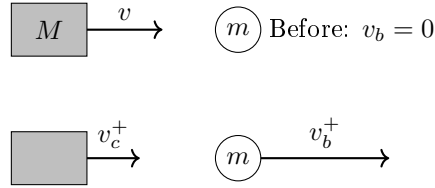


Figure 17.1: Isolated One-Dimensional Impact Before and After Contact. Momentum and Relative-Speed Restitution Determine Both Outgoing Velocities.

17.3 Effective Mass and Off-Centre Impact

For a free rigid clubhead, an impulse $J\hat{n}$ produces translational and rotational changes. With inertia tensor I_c about its centre of mass,

$$\frac{1}{M_{\text{eff}}} = \frac{1}{M} + (r \times \hat{n})^T I_c^{-1} (r \times \hat{n}). \quad (17.7)$$

The second term is nonnegative, so $M_{\text{eff}} \leq M$. It vanishes when the impulse line passes through the centre of mass, even if the contact point is some distance in front of it. Effective mass is directional inertial response, not the amount of material physically reached by a wavefront.

For a multibody model with mass matrix M_q and contact Jacobian J_P , the corresponding directional inverse mass is $\hat{n}^T J_P M_q^{-1} J_P^T \hat{n}$, modified by active constraints. Flexible modes, boundary conditions, and contact duration require a compatible dynamic model. There is no general rule that shaft and arm coupling always add 10–20 percent to clubhead effective mass.

Off-centre impact changes head rotation, contact velocity, and tangential impulse. This produces gear effect. Local face bulge and roll also change the normal, while face compliance can vary with strike position. Ball-speed loss cannot universally be assigned to only one of these mechanisms.

17.4 Oblique Contact and Spin

Let $\hat{n}$ point from face into ball and split impulse into normal and tangential components. Neglecting other impulses over the short contact,

$$m\Delta v_b = J_n \hat{n} + J_t, \quad (17.8)$$

$$I_b \Delta \omega_b = r_b \times (J_n \hat{n} + J_t). \quad (17.9)$$

For an ideal sphere of radius R , $r_b = -R\hat{n}$. A centred normal impulse then produces no spin, while the tangential impulse produces angular momentum. A finite contact patch can also transmit a torsional moment that the point contact model omits.

Coulomb friction supplies a bound $\|F_t\| \leq \mu F_n$, with equality in ideal gross sliding. Sticking does not imply zero friction. Tangential deformation, partial slip, and restitution can continue after gross slip ceases, so a rigid rolling condition does not universally fix outgoing spin. Contact properties must be identified at relevant speed, loading, and surface conditions. A low-speed ball-on-granite experiment is not a direct driver calibration.

More obliqueness does not guarantee monotonically increasing spin. Sliding, normal impulse, tangential compliance, moisture, and strike location can change the trend. Grooves help manage contamination, but a universal spin increment from groove count or loft alone is not supplied by this model.

17.5 Exact Spin-Loft Geometry

Use a forward/up/right frame, face azimuth ϕ , path azimuth ψ , dynamic loft λ , and attack angle α . The unit vectors are

$$\hat{n} = (\cos \lambda \cos \phi, \sin \lambda, \cos \lambda \sin \phi), \quad (17.10)$$

$$\hat{v} = (\cos \alpha \cos \psi, \sin \alpha, \cos \alpha \sin \psi). \quad (17.11)$$

Spin loft is their angle:

$$\cos \Phi = \sin \lambda \sin \alpha + \cos \lambda \cos \alpha \cos(\phi - \psi). \quad (17.12)$$

For zero face-to-path, $\Phi = |\lambda - \alpha|$ in the relevant range. Otherwise, multiplying $\cos(\phi - \psi)$ by $\cos(\lambda - \alpha)$ is not the exact identity. Nor can a horizontal path angle be subtracted from loft to obtain spin loft.

With measured dynamic loft 12° , attack 2.5° , and square face-to-path, spin loft is 9.5° . If an illustrative vertical launch weighting is 0.84 toward the normal and 0.16 toward the velocity, launch elevation is $0.84(12) + 0.16(2.5) = 10.48^\circ$. The weighting is empirical, not a universal law. Spin rate and carry require additional contact and flight inputs. With face azimuth fixed, changing path from -3° to $+3^\circ$ preserves spin loft by symmetry but reverses the ideal horizontal spin tendency; it does not subtract three degrees of loft.

For a centred symmetric impact on a nonspinning ball, the ideal spin direction is parallel to $\hat{v} \times \hat{n}$. The cross product degenerates at zero obliquity. Use its projections in a declared launch frame to calculate axis tilt; a tan/tan rule is not a general three-dimensional identity. Gear effect and asymmetric or rotating contact require additional terms.

In a normal-only model with $C = (1 + e)/(1 + m/M_{\text{eff}})$, $v_b = C(v_P \cdot \hat{n})\hat{n}$. Total ball speed scales with $\cos \Phi$; its projection onto incident velocity scales with $\cos^2 \Phi$. Those are different outputs. A tangential impulse modifies both direction and speed.

17.6 How Isolated Is the Collision?

The relevant comparison is impulse. An external force $F(t)$ contributes $\int F dt$, which must be compared with the ball-contact impulse in the same direction and model boundary. For an illustrative 300 N acting unchanged over 0.5 ms, the impulse is 0.15 N s. A 0.04593 kg ball leaving at 67 m/s gains approximately 3.08 N s. This comparison is a scale estimate, not a measured grip-to-ball sensitivity: grip forces may not reach the head with that magnitude or direction.

Axial wave speed $c = \sqrt{E/\rho}$ in a uniform slender rod differs from frequency-dependent bending-wave propagation. Wave arrival time establishes neither force amplitude nor the resulting change in launch speed. Existing shaft deformation, head inertia, grip impedance, reflection, and contact dynamics must be modeled together. An assumed spring deflection multiplied by stiffness cannot by itself prove a percentage change in effective mass.

The short collision leaves no time for a new consciously selected response to that collision to influence it. Existing activation, preprogrammed commands, intrinsic tissue forces, and mechanical boundary conditions nevertheless remain. This distinction is especially important in the preceding tens of milliseconds: brief ball contact does not establish an unactuated final 50–100 ms of downswing.

17.7 A Precise Counterfactual Question

A smooth pre-impact model may be written $\dot{x} = f(x) + G(x)u$ for declared state, inputs, and constraints. The collision can be represented by a reset $x^+ = \Delta(x^-)$ or by resolved compliant contact. It is not automatically the same smooth drift field continued through an impulsive event.

To isolate earlier actuation, branch before impact, specify which torques or excitations are removed, and evolve each branch with its own contact event. To isolate grip boundary mechanics during impact, hold the pre-impact state fixed and vary only the declared boundary model. These are different experiments. A peak collision-force/grip-force ratio is not a bound on reachable states, a work fraction, or evidence that muscle control has ceased.

17.8 Measurement and Model Checks

1. Declare the contact point, normal, reference point for club velocity, and measurement instant. Include their uncertainty and time synchronization.
2. Compare impulse integrals with linear and angular momentum changes. Do not infer moment solely from a force magnitude without its line of action.
3. Check both bodies' post-impact energy. A fitted restitution coefficient must not create unexplained energy in an otherwise passive model.
4. Validate ball speed, launch angles, spin magnitude and axis together. Fitting one outcome does not validate the others.
5. Vary integration step and contact resolution for compliant models; report convergence of impulses, energy residual, and launch conditions.

6. Separate equipment conformance tests from ball-contact experiments. Characteristic time from a steel pendulum is not golf-ball contact duration.

17.9 Connecting Swing Mechanics to Ball Flight

Earlier joint work and energy transfer help create incident club motion. Geometry and wrist control help determine local contact velocity and face attitude. Shaft deformation contributes to the delivered state and boundary response. The collision maps those inputs into launch; aerodynamics, wind, and the landing surface determine later flight and roll. This chain explains why a swing theory should predict a distribution of impact states and measured launch outcomes, rather than declaring one isolated motion variable optimal.

17.10 Exercises

1. Use $M = 0.200$ kg, $m = 0.04593$ kg, $v = 50$ m/s, $e = 0.83$ to compute both outgoing velocities, the ball's energy fraction, and total energy loss. Verify momentum conservation independently.
2. At fixed inertia, compare directional effective mass for an impulse through the centre of mass and one with a perpendicular offset. Explain why distance from the face to the centre of mass alone is insufficient.
3. Compute exact spin loft for $\lambda = 40^\circ$, $\alpha = 20^\circ$, and face-to-path 30° , then quantify the error in the product-of-cosines shortcut.
4. Distinguish total ball speed from its forward projection at 30° obliquity in the normal-only model.
5. Design a matched-state grip-boundary experiment. Specify which measured changes would falsify the claim that grip mechanics are negligible at the chosen accuracy.

Chapter 18

Swing Plane, Clubface Control and Launch Optimization

In Plain Language 18.1. From a Swing to a Shot

Clubhead speed is one part of delivery. Two swings with the same speed can produce different ball speeds, launch directions, spin vectors and landing positions because their face orientations, strike locations and contact conditions differ. A useful explanation follows the whole chain: preparation and continuing actuation produce club motion; collision produces ball motion; airflow and gravity produce flight; the chosen target determines whether the result is useful. Geometry connects these stages, but cannot replace their mechanics or their evidence.

This chapter develops that chain with explicit coordinates and independently checkable examples. The geometric identities and impulse balances are derived results under stated assumptions. The launch-ratio observations are measurements from a particular study. The quadratic distance surface introduced later is a deliberately manufactured example, not a fitted golf-ball model or a suggested launch window. These distinctions let the reader reuse the mathematics without mistaking a teaching example for a validated prescription.

18.1 Defining the Swing Plane

Use a right-handed frame with x horizontal toward the target, y horizontal to the target's left and z upward. Azimuths in this chapter are positive to the left. Convert a device's positive-right angles by changing the sign after checking its handedness and target convention. Angles inside trigonometric functions are in radians unless explicitly marked in degrees.

Let $r_C(t)$ and $v_C(t)$ denote the position and velocity of a specified point on the clubhead. A geometric center, center of mass and face contact point are generally different points. For a rigid head,

$$v_C = v_G + \omega_H \times (r_C - r_G).$$

Consequently, a change of reporting point can change speed, path and attack angle even when the underlying motion is identical. A flexible shaft does not invalidate rigid-head transport, but its deformation changes the head's pose and velocity. Specify the reporting epoch too: first touch, maximum compression and separation are not interchangeable events.

Definition 18.1. A Plane Requires More Than One Direction

A plane through r_0 with unit normal n_P satisfies $n_P^T(r - r_0) = 0$. An instantaneous velocity supplies one tangent direction, not a unique plane. Adding the vertical direction defines a vertical plane, whose inclination to horizontal is 90° ; it does not define an arbitrary tilted swing plane. The inclination of a non-oriented plane can be reported as $\beta = \arccos(|n_P^T e_z|)$ in $[0, 90^\circ]$.

An address construction might use a shaft line and a separate noncollinear point, or the shaft and a horizontal target direction. Name that construction. A ball position alone does not resolve the plane if it lies on the chosen line. A shoulder-and-ball plane can be constructed from three noncollinear points, but it changes when those points move. None of these planes must coincide with the plane fitted to a segment of clubhead motion.

For a measured arc, subtract the sample mean from its three-dimensional positions and form a covariance matrix. Its least-variance eigenvector is a least-squares plane normal. Report the fitting interval, point definition, weighting, residual distances and sensitivity to measurement error. A nearly straight short arc does not identify a stable normal: two small eigenvalues leave many planes almost equally plausible. An instantaneous osculating plane instead uses $v_C \times a_C$, when that cross product is nonzero. It need not equal the finite-arc fit, and differentiated acceleration amplifies noise.

Trackman's published swing-plane metric uses a finite segment of clubhead motion, whereas attack angle describes a velocity direction at an impact reference event. These are different measurements. The vendor also cautions against using its swing-plane value as a direct lie-fitting measurement [Trackman, 2024, 2020].

18.1.1 Why the Swing Plane Tilts

Key Principle 18.1. Geometry Constrains the Available Motion

Club length, stance, reach, posture and hand path affect the motions a golfer can make. They do not supply a universal formula mapping height or shoulder tilt to a unique delivery plane. A plane fitted after observing the motion is a description of that motion, not proof of which anatomical variable caused it.

The conventional club lie angle is measured between the shaft and the horizontal ground or waterline in the specified reference configuration. It is not measured from vertical. Thus a 57° lie corresponds to a 33° complement to vertical. Forward shaft lean and sideways lie are distinct aspects of three-dimensional orientation. Dynamic lie, static lie, face orientation and arc inclination should therefore be reported separately. Shaft bending can make the local shaft direction differ along its length.

Shorter clubs often accompany a more upright delivery, but player adaptation and the metric definition matter. A static specification cannot establish the attack angle, the energy transferred to the ball or an optimal technique. Comparisons should match the task, define the arc and report actual data instead of assigning all golfers a club-dependent plane table.

18.1.2 Attack Angle and the Vertical Component of Club Path

Definition 18.2. Attack Angle and Path From the Same Velocity

For nonzero horizontal speed, define

$$\alpha = \text{atan2}(v_{Cz}, \sqrt{v_{Cx}^2 + v_{Cy}^2}), \quad \psi = \text{atan2}(v_{Cy}, v_{Cx}).$$

Attack angle α is positive upward and negative downward. Path azimuth ψ is positive left in this frame. Attack angle is the elevation of a velocity vector, not an acceleration, a shaft angle or a displacement from the top of the backswing. At zero speed the direction is undefined; at purely vertical speed the azimuth is undefined.

A finite drop divided by a finite horizontal displacement gives the slope of a chord. It does not determine the instantaneous tangent at impact. To connect path with an ideal fixed swing plane, let its horizontal base direction have azimuth d and define orthonormal in-plane vectors

$$b = (\cos d, \sin d, 0)^T, \quad u = (-\sin d \cos \beta, \cos d \cos \beta, \sin \beta)^T.$$

Choose the unit tangent $\hat{v}_C = \cos \chi b + \sin \chi u$, where χ describes its direction within that plane. Direct projection gives

$$\sin \alpha = \sin \beta \sin \chi, \quad \psi - d = \text{atan2}(\cos \beta \sin \chi, \cos \chi).$$

Equivalently, on the forward-moving branch for $0 < \beta < 90^\circ$,

$$\tan \alpha = \tan \beta \sin(\psi - d).$$

This is a conditional geometric identity. It does not require constant speed or prove that a real head follows one plane. For small χ , $\alpha \simeq \sin \beta \chi$; replacing $\sin \beta$ by β additionally requires small plane inclination, which is inappropriate for a 60° plane.

At $\beta = 60^\circ$ and $\alpha = -5^\circ$, the forward solution is $\psi - d = -2.8953^\circ$: the path is right of the plane's base direction in the stated coordinate convention. At the bottom of an ideal circular arc, $\chi = 0$, so attack is zero and path equals the base direction. Moving the impact location along that fixed arc changes both quantities together. Moving the arc itself is a different intervention.

18.1.3 Variation by Club: Driver vs. Short Irons

An upward driver strike and a downward iron strike are possible delivery choices, not definitions of the clubs. The sign and useful magnitude depend on the lie of the ball, intended flight, strike quality and equipment. Population averages are not fitting targets. Even a fixed-plane identity cannot determine performance without face orientation and collision mechanics.

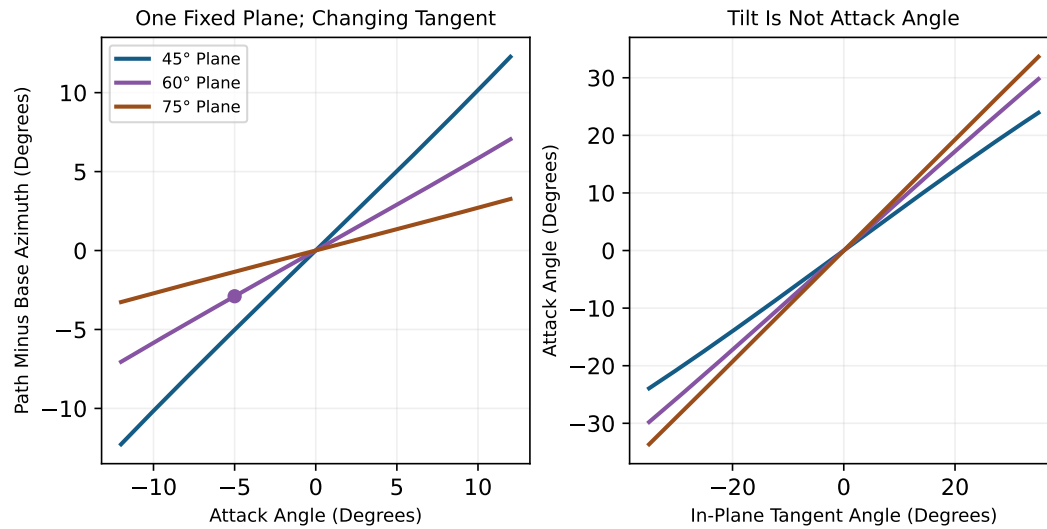


Figure 18.1: Fixed-Plane Geometry Links Path and Attack Angle. Curves Use the Declared Positive-Left Frame and Forward Tangent Branch; They Are Geometric Calculations, Not Measured Swings.

Quantity	Required Comparison
Static Lie	Club specification, reference sole position and shaft axis.
Dynamic Lie	Local shaft direction at a declared event and reference plane.
Swing Plane	Point trajectory, fitting interval, inclination and residual.
Attack Angle	Elevation of the declared point's instantaneous velocity.
Club Path	Horizontal azimuth of that same velocity in the target frame.

Table 18.1: Separate Definitions Before Comparing Clubs or Golfers. None of These Quantities Substitutes for Another.

Reference 18.1. Source Contract for Launch and Direction Numbers

An instrument's reported value combines observations, calibration, processing and sometimes a fitted model. Identify the device, mode, point, event and uncertainty rather than describing every output as directly measured. The Wood study below supplies bounded launch-ratio evidence. The distance examples later in this chapter are explicitly manufactured response surfaces. They are not vendor regressions, population norms or empirical fitting advice.

18.2 Clubface Control in Three Dimensions

The local face normal controls the direction of the contact normal force. It is essential to the collision, but cannot by itself determine the shot. The local surface velocity, head mass and inertia, ball properties, impact offset, friction and deformation also matter. On a curved driver face, the normal at the strike location need not equal the normal at the face center.

Key Principle 18.2. Face Orientation and Joint Contributions

Let $R_H \in SO(3)$ map head coordinates to the reporting frame, and let n_0 be a unit face normal in head coordinates. For a fixed local surface point, $n = R_H n_0$. Define face azimuth and normal elevation by

$$\phi = \text{atan2}(n_y, n_x), \quad \ell = \text{atan2}(n_z, \sqrt{n_x^2 + n_y^2}).$$

The latter is the local dynamic loft convention used here. These two angles determine a normal direction, not the entire three-dimensional head orientation. A third independent direction, such as a groove direction, completes the frame. Toe-up/heel-up rotation and shaft-axis rotation are not definitions of face azimuth. Near a vertical normal, azimuth becomes ill-conditioned.

A planar approximation may express delivered loft using static loft and shaft lean, with carefully declared signs. In three dimensions, ordered rotations replace unrestricted scalar addition. Shaft twist, bending, grip geometry and both arms enter through their actual kinematic relations, not through fixed percentages of a final face angle.

18.2.1 Face Angle Determination: The Joint Contributions

For an illustrative serial chain, write

$$R_H(q) = R_1(q_1)R_2(q_2) \cdots R_k(q_k)R_{\text{grip}}R_{\text{head}}.$$

The axes and frame of each rotation must be specified. A real two-handed grip creates a closed-chain constraint; joint coordinates cannot generally be varied independently while preserving both contacts. Compliance adds state variables and changes the relation between hand and head frames.

Finite rotations do not commute. Acting on $(1, 0, 0)^T$, a 90° rotation about y followed by 90° about x gives $(0, 1, 0)^T$. Reversing the order gives $(0, 0, -1)^T$. A scalar sum of the

two angles cannot distinguish these different normals. Neither forearm pronation nor wrist deviation has a universal signed contribution independent of arm, grip and club orientation.

18.2.2 Sensitivity Analysis: Joint Angle Errors

For a small spatial rotation $\delta\theta$ about unit axis k , $\delta n = (k \times n)\delta\theta$. Differentiating face azimuth yields

$$\frac{\partial\phi}{\partial\theta} = \frac{n_x(k \times n)_y - n_y(k \times n)_x}{n_x^2 + n_y^2} = k_z - \tan\ell(k_x \cos\phi + k_y \sin\phi).$$

The derivative is dimensionless when both angles use the same units. It is local: a large finite error requires recomputing the orientation. The axis k is the joint axis transported to the reporting frame at the current configuration, not merely its anatomical name.

Example 18.1. A Joint's Axis Changes Its Face-Angle Gain

At $\phi = 0$ and $\ell = 15^\circ$, rotations about spatial z , x and y have azimuth gains 1, $-\tan 15^\circ \simeq -0.26795$ and 0 respectively. These are geometric examples, not estimates of three anatomical joints. A zero local azimuth derivative does not mean zero effect on loft, impact position, future motion or higher-order face-angle change. A joint's useful authority also depends on feasible torque, inertia, constraints and time.

For several feasible small coordinate changes, assemble $\delta\phi = J_\phi\delta q$. The corresponding variance approximation is $\text{Var}(\phi) \simeq J_\phi \Sigma_q J_\phi^T$, retaining covariance. If measured angles share errors, or if constraints correlate their motion, their separate contributions cannot be added as independent percentages. A large observed range of one joint does not establish its causal dominance.

18.2.3 The Gate Concept: The Narrow Window at Impact

Definition 18.3. A Gate Is an Inverse Image of an Objective

Let $y = F(z, \eta)$ predict landing outcomes from delivery variables z and environment/model parameters η . For an acceptable outcome set $\mathcal{A}$, the delivery gate is $\{z : F(z, \eta) \in \mathcal{A}\}$. A face-angle interval is a slice of that set with the other variables held fixed. With uncertainty, one might instead require $\Pr(F(z, \eta) \in \mathcal{A}) \geq p_0$.

There is no universal three-to-five-degree gate implied by club speed alone. The target width, carry, spin, strike location, wind and acceptable probability must be supplied. A joint change may alter several delivery variables together, so a face-only slice need not describe a physically achievable intervention.

18.2.4 Drift vs. Control: Remaining Time Before Impact

Key Principle 18.3. Feedback Timing and Continuing Actuation

Muscle activation and force may continue through impact even when a newly perceived error cannot produce a useful late correction. A predetermined command, a response to earlier information and a new sensory correction are different inputs. The computational-brain chapter derives a delayed-activation example in which remaining time continuously changes the same command's effect. It does not establish a universal instant at which all control switches off.

For a declared input-affine model, $\dot{x} = f(x) + G(x)u$, the drift term f depends on the chosen state and input. Zero neural excitation does not mean zero retained muscle force. A local fixed-time delivery sensitivity has the form

$$\delta z = C\Phi(T, 0)\delta x_0 + \int_0^T C\Phi(T, s)B(s)\delta u(s) ds.$$

Delay, activation dynamics and feasible commands must be included in this calculation. The state includes velocities and relevant activation or elastic history; club pose at the top alone cannot determine angular momentum. Impact time is also an outcome. For a transverse contact event $g(x(T)) = 0$,

$$\delta T = -\frac{g_x \delta x(T)}{g_x \dot{x}(T)}, \quad \delta x_{\text{event}} = \delta x(T) + \dot{x}(T)\delta T.$$

This first-order expression assumes a differentiable event with nonzero denominator and no intervening mode change. Near grazing contact it becomes ill-conditioned. It explains why a sensitivity computed at fixed clock time can miss an important delivery effect, without proving a neural strategy.

18.3 The D-Plane and Modern Ball Flight Laws

18.3.1 Launch Direction: Measurements and Their Limits

For small horizontal angles, a local empirical approximation is

$$\lambda \simeq w\phi + (1 - w)\psi,$$

where λ is launch azimuth. The weight summarizes a particular data range and measurement protocol. It is neither a force fraction nor a causal percentage of an entire shot.

Wood and colleagues reported horizontal face weights of $76\% \pm 8\%$ for drivers, $69\% \pm 3\%$ for 7-irons and $61\% \pm 7\%$ for wedges, with the table identifying these intervals as 95% confidence intervals. The respective filtered sample sizes were 89, 172 and 28 shots, not the full 1,575 recorded shots. The 157-player protocol included centered-strike and minimum face-to-path filters; a separate 15-shot robot test gave $63\% \pm 4\%$. The study was funded by PING and used PING clubs. Its vertical ratios differ from the horizontal ratios, so a value from one projection cannot validate a claim about the other [Wood et al., 2018].

The practical conclusion is qualified face-dominant initial direction in those tests, not a universal rejection or confirmation of every vendor rule. The robot also changed delivered

loft as face angle changed. Measurement alignment, strike location and small denominators in launch-ratio estimates limit interpretation. A reported confidence interval is not the spread of individual golfers' future shots or a universal uncertainty band.

18.3.2 Mathematical Formulation of Launch Direction

Definition 18.4. The D-Plane as a Geometric Construction

For nonparallel local face normal n and incoming relative surface-velocity direction $\hat{v}$, their span defines a plane with normal parallel to $\hat{v} \times n$. This construction remains possible at an off-center strike. What can fail there is the simplified prediction that the complete collision remains planar. If the two vectors are parallel, no unique plane normal or nonzero spin axis follows from them.

Under an ideal planar contact model, with an initially stationary ball, resolve the impulse on the ball as $J = J_n n + J_t t$, where $t = (\hat{v} - (n^T \hat{v})n) / \|\hat{v} - (n^T \hat{v})n\|$. Then

$$v_B^+ = \frac{J_n n + J_t t}{m_B}, \quad \hat{v}_B^+ = \frac{J_n n + J_t t}{\|J_n n + J_t t\|}. \quad (18.1)$$

This is impulse balance, not a constant-weight direction law. A convex combination of unit vectors generally has norm below one and must be normalized if used as a direction. Moreover, a horizontal angle weight does not automatically specify a three-dimensional vector weight.

Spin loft is the three-dimensional obliqueness $\Phi = \arccos(n^T \hat{v})$. With the earlier angle definitions,

$$\cos \Phi = \cos \ell \cos \alpha \cos(\phi - \psi) + \sin \ell \sin \alpha.$$

Only when the azimuths coincide does this reduce to $|\ell - \alpha|$ on the relevant branch. Obliqueness is important, but does not eliminate dependence on contact material, friction, velocity, deformation and head motion.

Henrikson and colleagues used a contact model with tangential compliance and partial slip to explain a declining launch ratio at increasing obliqueness even with fixed friction. Their parameter study also changed the ratio by changing friction; in a low-incidence regime, lower friction could move launch closer to path. This is a finite-regime result, not a statement about all friction values. Their comparison reused the earlier player data, with a separate ball-on-plate experiment supporting the contact model. The fixed surface approximation and its treatment of normal inelasticity limit transfer to a freely rotating head [Henrikson et al., 2020].

Thus neither a single coefficient per club nor an obliqueness-only formula is a universal law. A strictly frictionless normal impulse on an initially nonspinning sphere produces velocity along the local normal and no spin; extrapolating a finite-friction trend to that limit can give the wrong answer.

18.3.3 Spin Axis and the Vector Perpendicular to the D-Plane

The contact point lies on the ball's surface, not at its center. In a spherical illustration with radius R_B , take $r_{B/C} = -R_B n$ from the ball center to contact. With isotropic rotational

inertia I_B and zero initial spin,

$$\omega_B^+ = \frac{r_{B/C} \times J}{I_B} = -\frac{R_B J_t}{I_B} n \times t. \quad (18.2)$$

The impulse model supplies J_t and therefore both spin magnitude and sign. The unit axis is $\omega_B^+ / \|\omega_B^+\|$ when spin is nonzero. A bare cross product of directions cannot specify RPM. An actual ball's inertia must be measured or modeled; $I_B = 2m_B R_B^2 / 5$ describes a homogeneous solid sphere, not every layered golf ball.

For a level incoming path along $+x$ and a face normal tilted upward, positive tangential impulse along the projected path gives spin toward $-y$. Then $\omega_B \times v_B$ points upward, consistent with backspin lift in the usual positive-lift convention. With no tangential impulse the spin is zero and an axis should not be reported. Pre-existing spin, a varying normal, nonplanar friction and head rotation require the fuller contact calculation.

Gear effect concerns coupled rotation and tangential motion associated with an eccentric clubhead impact. It is not caused simply by the fact that a ball is touched on its surface. Off-center impact can rotate the head, change the local normal and contact velocity, and alter the ball's spin. Bulge and roll introduce an additional local-normal dependence.

18.3.4 The Face-to-Path Difference and Shot Shape

The horizontal difference is $\phi - \psi$ in the chosen frame. Under an ordinary centered backspin-dominated impact it is useful for interpreting draw or fade tendencies. It does not determine curvature independently of loft, attack angle, spin magnitude, strike location and wind. Curvature during flight depends on the full aerodynamic force history.

Using $w = 0.76$ as an illustrative local weight, a path 2° right and square face give $\lambda = -0.48^\circ$ in this chapter's positive-left frame: the ball starts 0.48° right. This calculation predicts initial direction only. A lower speed does not universally produce more time aloft or more curvature. Even in vacuum at fixed elevation, flight time is proportional to speed; real aerodynamic flight requires its own calculation.

18.4 Launch Condition Optimization

Key Principle 18.4. Launch State and the Flight Model

Ball speed, launch elevation and spin rate are useful summaries. A full flight state includes position, the three-dimensional velocity vector and the spin vector; ball orientation may also matter in a seam- or surface-sensitive model. Predicting landing additionally requires gravity, air density, wind, aerodynamic coefficients and a landing-surface definition. Bounce and roll require ground-contact models. The objective may be carry, total distance, a landing region, stopping behavior or expected score.

For a representative quasi-steady model, let $v_r = v_B - w(r_B, t)$ be air-relative velocity, A_B a reference area and ρ air density. Then

$$m_B \dot{v}_B = m_B g - \frac{1}{2} \rho A_B C_D \|v_r\| v_r + \frac{1}{2} \rho A_B C_L \|v_r\|^2 \hat{l}, \quad \dot{r}_B = v_B,$$

where $\hat{l}$ is a declared lift direction, commonly along $\omega_B \times v_r$ when that cross product is nonzero. A model must handle its zero limit consistently. Coefficients depend on the chosen ball, flow regime and spin parameters; rotational drag may evolve ω_B . The aerodynamics chapter develops these boundaries. A coefficient table from one ball and operating range should not silently become a universal flight law.

18.4.1 Ball Speed and the Smash Factor

Define smash factor S using specified ball and club speed conventions:

$$S = \frac{v_{\text{ball}}}{v_{\text{club}}}, \quad v_{\text{ball}} = S v_{\text{club}}. \quad (18.3)$$

This is a definition, not a mechanism or an energy efficiency. At club speed 100 mph and $S = 1.48$, ball speed is 148 mph. Ball energy scales with ball mass and speed squared; a speed ratio larger than one does not violate energy conservation. Collision restitution, effective mass, obliqueness and contact losses determine which ratios are achievable. Club speed measured at a different head point changes the reported ratio.

Improving strike location can change ball speed, launch and spin together. Treating smash factor as independently adjustable while holding everything else fixed is an intervention assumption to be tested, not a property of its definition. Likewise, an increase in club speed is not numerically identical to an increase in ball speed.

18.4.2 Optimal Launch Angle for Distance

Specify a prediction $D(z; \eta)$ and feasible launch set $\mathcal{Z}$ before asking for $\arg \max_{z \in \mathcal{Z}} D$. Distinguish a ball-flight optimum at fixed speed and spin from an equipment or swing intervention that changes all three together. Carry and total-distance optima need not coincide. An interior stationary point, a constrained boundary solution and a robust choice under uncertainty obey different conditions.

For a checkable limiting example, neglect air resistance, spin effects and initial height, and land at the launch elevation. With speed v ,

$$D = \frac{v^2}{g} \sin(2\theta), \quad T_f = \frac{2v \sin \theta}{g}.$$

Here 45° maximizes range at every positive speed. This is not a driver fitting recommendation: it is a counterexample to a universal speed-only rule for optimal angle. Lift, drag, launch height and feasible launch conditions change the optimization problem.

For the remaining worked examples, define a manufactured local response surface, expressed in yards, around 140 mph ball speed, 14° launch and 2,400 RPM spin:

$$\delta v = v_{\text{ball}} - 140 \text{ mph}, \quad \delta \theta = \theta - 14^\circ, \quad \delta \Omega = \Omega - 2400 \text{ RPM}.$$

Use numerical deviations in the listed units, with dimensioned coefficients,

$$D = 230 + 2\delta v - \frac{1}{2}a\delta\theta^2 - b\delta\theta\delta\Omega - \frac{1}{2}c\delta\Omega^2, \quad (18.4)$$

where $a = 0.8$ yards/degree², $b = 0.002$ yards/(degree RPM), and $c = 0.00002$ yards/RPM². The speed coefficient is 2 yards/mph. These values define the example completely; they

Ball Speed	Launch	Spin	Model Carry
140 mph	16°	2,200 RPM	228.8 yd
140 mph	13°	2,200 RPM	228.8 yd
140 mph	14.5°	2,200 RPM	229.7 yd
140 mph	14°	2,400 RPM	230.0 yd

Table 18.2: Manufactured Local Surface: Coupled Launch and Spin Change Which Angle Is Best. These Are Calculated Examples, Not Golf Fitting Data.

were not estimated from launch-monitor data. The model is used locally, and should not be extrapolated to negative spin, arbitrary speeds or extreme launch angles.

At fixed 2,200 RPM the optimum angle is 14.5°, because $\delta\theta^* = -b\delta\Omega/a = 0.5^\circ$. Calling a reduction from 16° to 13° an improvement would be wrong in this explicit example: both points have the same predicted carry. An observed carry value at one launch condition cannot determine the response to an unmeasured change.

18.4.3 Spin Rate Trade-Off: Lift and Air Resistance

Key Principle 18.5. A Claimed Optimum Must Follow From the Model

Backspin can support lift, while its associated aerodynamic changes can also increase drag. The resulting carry optimum must be calculated using the declared flight model and constraints. A polynomial $Av^2 - B\Omega - C\Omega^2$ with positive B, C decreases for all nonnegative spin; it cannot demonstrate a positive-spin interior optimum.

In the manufactured surface, the launch-spin curvature matrix is

$$H = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad ac - b^2 = 0.000012 > 0$$

in the corresponding compound units. Since $a > 0$ and the determinant is positive, the quadratic loss is positive definite. The chosen center is therefore a strict joint maximum at fixed speed. Fixing launch away from its central value shifts the spin optimum to $\delta\Omega^* = -b\delta\theta/c$. This coupling is exactly what separate one-variable target windows can conceal.

18.4.4 The Launch Window

A performance window should describe a level set or probability requirement, not an unexplained collection of independent tolerances. For fixed speed in the manufactured example, accepting at most one yard of modeled loss gives

$$\frac{1}{2}(\delta\theta, \delta\Omega)H(\delta\theta, \delta\Omega)^T \leq 1 \text{ yard.}$$

It is an ellipse with coupled axes, not a rectangular launch-spin box. At zero spin deviation the boundary is $|\delta\theta| = \sqrt{2/a} \simeq 1.581^\circ$; at zero launch deviation it is $|\delta\Omega| = \sqrt{2/c} \simeq 316.23$ RPM. Taking both individual limits simultaneously generally violates the joint one-yard requirement.

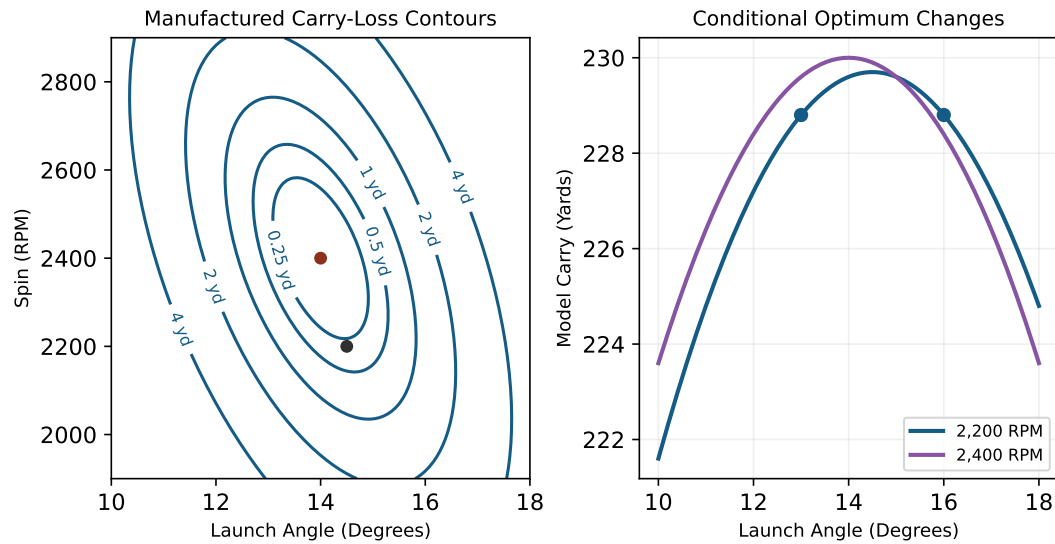


Figure 18.2: A Manufactured Launch Surface Shows Coupling and Curvature. The Elliptical One-Yard Loss Region and Conditional Optima Follow From the Stated Quadratic, Not a Golf Fitting Dataset.

Window Choice	Meaning
Deterministic Level Set	Outcomes under one declared model and environment.
Feasible Delivery Set	Launch combinations reachable by the golfer and equipment.
Chance Constraint	Required success probability under declared uncertainty.
Fitting Target	A tested intervention balancing outcome, repeatability and cost.

Table 18.3: Define the Purpose of a Launch Window Before Using It as a Target.

The useful target lies in the intersection of acceptable outcomes and achievable delivery. It may favor a slightly different mean condition when that condition reduces harmful variability or avoids a severe miss. Such a choice requires a stated loss function, not a universal preference for either maximum mean distance or minimum variability.

18.4.5 Equipment Design and Launch Optimization

Moving a clubhead's center of mass changes moment arms and inertia; it does not by itself prescribe a change in launch angle or spin. Distinguish fixed delivery from the golfer's adaptation to a changed club, and distinguish a fixed location on the face from a fixed location relative to the center of mass. Local loft, vertical and horizontal gear effect, shaft dynamics and impact offset can change together.

A frictionless normal-impact illustration makes one effect explicit. For ball mass m_B , head mass M_H , head inertia tensor I_H about its center, and contact offset r_H , the normal effective inverse mass for a spherical ball is

$$\kappa_n = \frac{1}{m_B} + \frac{1}{M_H} + (r_H \times n)^T I_H^{-1} (r_H \times n).$$

The spherical ball's normal impulse passes through its center, so it adds no ball rotational term in this normal direction. For relative closing speed $v_n > 0$ and restitution e_n , $J_n = (1 + e_n)v_n/\kappa_n$ under the stated free-body impulse approximation. External hand/shaft impulses, tangential coupling or deformable-body effects require a more complete model.

For $M_H = 0.20$ kg, $m_B = 0.04593$ kg, $I_H = 0.0004$ kg m² about the relevant rotation axis and a 0.020 m perpendicular offset, the added inverse mass is 1 kg⁻¹. The normal impulse is about 0.9640 times the centered value at the same closing speed and restitution. This is a mechanics example, not a measured mishit penalty or a prediction of spin.

Face flexibility affects the transient collision and cannot be summarized as the energy efficiency of the face alone. Restitution, characteristic-time equipment tests and smash factor are different quantities. Apply the current equipment rules and their test procedures when assessing conformity; a launch fitting table cannot establish it. The impact chapter treats the collision and equipment boundaries in more detail.

18.4.6 Wind Effects on Optimal Launch Conditions

Wind enters through $v_r = v_B - w$, not through an assumed ground-effect bonus. A headwind generally increases air-relative speed for an outgoing shot and a tailwind generally decreases it initially. Drag, lift and spin evolution then alter the entire flight. Because those changes depend on the ball and launch state, a fixed wind speed does not imply a universal degree adjustment to the optimal launch angle.

For a real fitting calculation, specify the wind field, density, ball coefficients, landing elevation and objective. Recompute feasible launch choices and their uncertainty. If optimizing total distance, include the landing angle, landing speed and ground behavior. A claim that more time aloft is always beneficial is incompatible with these coupled costs.

18.5 Sensitivity Analysis: What Matters Most?

18.5.1 Partial Derivatives of Distance

Sensitivity is local and conditional on which quantities are held fixed. For the manufactured surface,

$$\frac{\partial D}{\partial v_{\text{ball}}} = 2 \text{ yards/mph}, \quad (18.5)$$

$$\frac{\partial D}{\partial \theta} = -a\delta\theta - b\delta\Omega, \quad (18.6)$$

$$\frac{\partial D}{\partial \Omega} = -b\delta\theta - c\delta\Omega. \quad (18.7)$$

At a smooth interior optimum with the other variables held fixed, the relevant first derivative is zero. A maximum is not generally the point of maximal first-order sensitivity. Near it, curvature determines the leading loss. At a boundary optimum, a nonzero gradient can be balanced by constraints.

At $(16^\circ, 2200 \text{ RPM})$, the angle derivative is $-1.2 \text{ yards/degree}$. A linear prediction for a -3° change gives a $+3.6 \text{ yard}$ gain. The quadratic correction is $-\frac{1}{2}a(3)^2 = -3.6 \text{ yards}$, exactly canceling it. The endpoints therefore have equal carry in this model. A local derivative does not justify a large finite extrapolation without a remainder estimate.

Comparing the numbers 2 yards/mph and 0.002 yards/RPM without scaling is meaningless as a ranking. Compare effects over plausible changes, achievable interventions or uncertainty ranges. The parameter with the largest raw derivative may not be the one that contributes most to actual error.

18.5.2 Directional Sensitivity

Ignore aerodynamic curvature for a moment and let R be a fixed downrange distance. The geometric lateral offset is $Y = R \tan \lambda$. Under the local angle-weight model with constant w ,

$$\frac{\partial Y}{\partial \phi} = R \sec^2 \lambda w, \quad (18.8)$$

$$\frac{\partial Y}{\partial \psi} = R \sec^2 \lambda (1 - w). \quad (18.9)$$

These derivatives use radians. At $R = 250 \text{ yards}$, $\lambda = 0$ and $w = 0.76$, they are approximately $3.316 \text{ yards/degree}$ of face change and $1.047 \text{ yards/degree}$ of path change. The familiar $250 \tan 1^\circ \simeq 4.364 \text{ yards}$ is for a one-degree *launch* change, not a one-degree face change under this weighting.

Neither result includes curvature, changes in carry or a changing weight. For actual flight, write $Y = F_Y(z, \eta)$ and propagate all delivery changes, including spin-axis and speed changes. A path change can strongly affect spin while contributing less to initial launch azimuth. It therefore does not follow that path has a universally small effect on final lateral error.

18.5.3 The Consistency Argument

For small perturbations and a scalar outcome, the first-order variance is

$$\text{Var}(D) \simeq g_D \Sigma_z g_D^T, \quad g_D = \nabla_z D.$$

Correlations can increase or decrease this variance. Measurement noise, calibration bias, model discrepancy and genuine shot variability should be estimated separately where the data permit. Outcome repeatability alone does not identify which joint, muscle or controller produced it.

At fixed club speed 100 mph, a smash-factor *standard deviation* of 0.05 gives ball-speed standard deviation 5 mph. A smash-factor variance of 0.05 would instead give $100\sqrt{0.05} \simeq 22.36$ mph. If both club speed V and smash factor S vary independently, the exact product variance is

$$\text{Var}(VS) = \mu_V^2 \sigma_S^2 + \mu_S^2 \sigma_V^2 + \sigma_V^2 \sigma_S^2.$$

For correlated variables, additional mixed moments enter the exact result; the first-order approximation includes $2\mu_V\mu_S \text{Cov}(V, S)$. One cannot infer the total spread by holding a varying factor constant.

At the manufactured surface's launch-spin maximum, first-order propagation alone gives zero sensitivity to these two variables, but nonzero fluctuations still reduce mean carry. For mean-zero deviations with covariance Σ ,

$$E[D] = 230 - \frac{1}{2} \text{tr}(H\Sigma)$$

at fixed 140 mph. This identity is exact for the quadratic model. With launch standard deviation 1° , spin standard deviation 200 RPM and correlation $+0.5$, mean loss is 1.0 yard. Changing only the correlation to -0.5 gives 0.6 yard. Equal marginal spreads do not imply equal performance.

Mean distance and spread are distinct objectives. If a manufactured linear distance model has slope 2 yards/mph, a 3 mph increase in mean ball speed gives 6 yards of mean gain. Increasing speed standard deviation from 4 to 5 mph changes distance standard deviation from 8 to 10 yards; reducing it to 2.5 mph gives 5 yards. Which intervention is preferable depends on the target and the cost of misses. The second is more consistent in this example, but that does not prove it has higher expected scoring value.

18.6 The Zero-Torque Counterfactual Perspective on Launch

Key Principle 18.6. Define the Counterfactual Intervention

A zero-torque counterfactual sets specified inputs to zero in a declared model, at a specified time and state, while stating what happens to the remaining inputs, contacts and controller. Its predicted launch is a comparison within that model. It does not establish that a golfer actually stops applying force, that all neural feedback ends at a common time, or that the same result would follow from changing grip or equipment.

For a useful golf investigation, propagate the complete pre-impact state to the contact event, resolve the collision and then simulate flight. Compare the specified intervention with a baseline under the same reported conventions. Repeat with plausible parameter and measurement uncertainty. If a contact mode changes or the club misses the ball, report that result rather than silently comparing the state at an unrelated clock time.

18.6.1 What a Late Correction Can Change

Example 18.2. New Corrections and Continuing Force

Suppose a newly detected error would require 60 ms before its corrective command reaches an actuator, while impact is 30 ms away. That new command cannot influence this impact through the stated delayed pathway. Existing activation, prior commands and faster pathways are separate questions. This timing example neither proves a universal golf delay nor says that the club is mechanically free. It also does not determine the benefit of a prepared response to information available earlier.

The proper calculation combines pathway timing with output gain, remaining time and feasible force. Reaching a desired face orientation may also require preserving strike position, speed and contact constraints. A useful controller must solve this coupled task rather than minimize face-angle error alone.

18.6.2 Determinants of Delivery: State, Inputs and Constraints

Grip geometry establishes kinematic relations and possible contact forces. Changing grip force can affect friction capacity and effective boundary conditions; they do not change the intrinsic mass distribution of the club. Posture changes moment arms, feasible directions and motion. Activation and elastic deformation retain history. Ground reactions participate in the multibody balance and may depend on the applied muscle forces and constraints. Continuing actuator work and energy redistribution can coexist.

The model's ability to reproduce one club trajectory does not identify all these internal mechanisms. Seek measurements that distinguish competing explanations, or report the family of compatible models. A claim about a particular grip's performance requires a defined task and intervention, not only a geometric example in which one torque component is reduced.

18.7 Integration of Impact Conditions With Equipment and Swing Dynamics

The explanatory chain is now explicit. A state and input history generate a constrained head trajectory. The selected point, frame and event determine reported delivery quantities. Local surface orientation and velocity, mass, inertia and contact constitutive behavior determine impulse and ball spin. Initial ball state and the environment determine flight. The target and loss function determine whether the outcome is favorable.

Every link admits both uncertainty and a useful test. Check geometric identities against independent coordinates; compare local gains with finite perturbations; validate contact

predictions against suitable impact data; test flight models within their coefficient domains; evaluate interventions with held-out outcomes and matched definitions. These steps strengthen the argument for golf as a coupled control problem without asserting one universal plane, release pattern, late-control cutoff or launch window.

18.8 Problems and Thought Experiments

1. **Swing Plane Geometry.** Use a fixed 60° plane and attack angle -5° . Find the forward-branch path offset from the base direction. Explain why a two-inch total drop over fifteen horizontal inches does not determine this attack angle, and convert a 57° conventional lie to its complement to vertical.
2. **Face Angle Sensitivity.** At $\phi = 0$, $\ell = 15^\circ$, derive the azimuth gains for small spatial rotations about x, y, z . With fixed other delivery variables and the geometric $R = 250$ yard, $w = 0.76$ approximation, estimate lateral standard deviations for face standard deviations 2° and 4° . State what has been omitted.
3. **Launch Optimization.** In the manufactured quadratic model, hold ball speed at 140 mph and spin at 2,200 RPM. Calculate carry at 16° , 13° and the conditionally optimal launch. Explain why a first-order estimate for the three-degree change fails.
4. **D-Plane and Impulse.** For a path 2° right and square face, apply the declared horizontal weight $w = 0.76$. Separately derive spin from $r_{B/C} \times J$ for a level path and an upward-tilted face. What happens when the tangential impulse is zero? Does an off-center strike make the geometric span of two nonparallel vectors cease to exist?
5. **Control and Event Timing.** A new correction is delayed 60 ms and impact is 30 ms away. Identify what this rules out and what it does not. Explain why a fixed-time face-angle sensitivity can differ from an impact-event sensitivity.
6. **Equipment and Effective Mass.** Use the masses, inertia and 20 mm offset in the equipment example. Compute the ratio of off-center to centered normal impulse at fixed restitution and closing speed. Explain why this does not predict the spin change from moving a driver's center of mass.
7. **Wind and Feasible Launch.** Specify the additional inputs needed to compare optimal shots in a 15 mph headwind and tailwind. Explain why the wind speed alone cannot supply a numerical angle correction or a complete total-distance prediction.
8. **Consistency and Objective.** Reproduce the quadratic mean losses for correlations $+0.5$ and -0.5 with the specified launch and spin spreads. Then compare the 3 mph mean-speed gain and the standard-deviation reduction in the linear example. Distinguish variance, standard deviation, mean outcome and expected scoring value.

18.8.1 Worked Checks and Interpretation

1. The path offset is $\arcsin(\tan(-5^\circ)/\tan 60^\circ) = -2.8953^\circ$, or right of the base direction. A finite chord does not specify an instantaneous tangent; the lie complement is 33° and does not supply the missing motion data.

2. The gains are -0.26795 , 0 and 1 respectively. The face-to-offset gain is $250(0.76)\pi/180 \simeq 3.316$ yards/degree, giving approximately 6.63 and 13.26 yards of lateral standard deviation under the local independent face-error model. Curvature, correlated path, strike and speed changes, wind and model error are omitted; these are not predicted golfer dispersions.
3. The two endpoint carries are both 228.8 yards. The conditional optimum is 14.5° , giving 229.7 yards. The initial derivative predicts a 3.6 -yard gain for the finite reduction, canceled by the exact 3.6 -yard quadratic loss. This is why a local derivative cannot be used without scale.
4. Launch azimuth is -0.48° , or 0.48° right. With the defined tangent, $\omega_B = -R_B J_t(n \times t)/I_B$; positive tangential impulse gives backspin toward $-y$. At $J_t = 0$ the assumed initially nonspinning ball remains nonspinning. Nonparallel directions still define a plane off center, but head rotation and contact changes can invalidate the simplified planar outcome prediction.
5. The stated new delayed command arrives after contact. Earlier commands and activation remain possible; other pathways need their own latency and gain. A perturbation may move the contact time, adding $\dot{x}\delta T$ to the fixed-time state perturbation before collision.
6. The extra inverse mass is $(0.020)^2/0.0004 = 1 \text{ kg}^{-1}$. The impulse ratio is

$$\frac{5 + 1/0.04593}{6 + 1/0.04593} \simeq 0.963993.$$

The calculation is normal and frictionless. Predicting spin needs tangential contact mechanics, local velocity and orientation, and the complete change in head motion and strike geometry.

7. Supply initial state, ball aerodynamic and spin-decay model, wind field, density, landing geometry, feasible delivery and objective. Add a ground-contact model for bounce and roll. Evaluate both environments rather than assigning a universal positive or negative angle adjustment.
8. The covariance terms give $\frac{1}{2} \text{tr}(H\Sigma) = 0.8 + 0.4\rho$ yards: 1.0 or 0.6 yard. In the linear example the mean gain is 6 yards, while distance standard deviations are 8 yards before intervention, 10 after the faster but more variable intervention, and 5 after the consistency intervention. These are different trade-offs. A target-specific loss or scoring model is needed to rank them as overall performance changes.

Chapter 19

The Physics of Putting

In Plain Language 19.1. Connect the Stroke, Impact, and Green

A putting model has three connected stages: the golfer moves the putter, impact establishes the ball's velocity and spin, and contact with the green changes both. A reproducible stroke does not guarantee a reproducible putt if the impact or surface changes. Conversely, the same rollout can result from different strokes and launch conditions. We derive simple models for each stage, state their limits, and keep calculated examples separate from measurements of golfers.

19.1 Rolling Versus Sliding Dynamics

Initial spin is an outcome of impact, not universally zero. Loft, impact location and contact conditions can also produce a short airborne or bouncing phase. Penner explicitly distinguishes those possibilities before adopting pure rolling as an approximation for his path calculations [Penner \[2002\]](#). The following contact model begins after any flight: a rigid spherical ball remains on a stationary, level plane, with constant kinetic friction and no rolling-resistance moment during the sliding interval.

Let R be ball radius, $I = \kappa m R^2$ its central inertia, and choose the positive angular rate so forward rolling satisfies $v = R\omega$. A uniform solid sphere has $\kappa = 2/5$; a layered golf ball need not have exactly that value. The signed slip speed is $s = v - R\omega$. For $s \neq 0$,

$$\dot{v} = -\mu_k g \operatorname{sgn}(s), \quad \dot{\omega} = \frac{\mu_k g}{\kappa R} \operatorname{sgn}(s). \quad (19.1)$$

Friction opposes *slip*, not necessarily translation: an overspinning ball can accelerate forward while losing total kinetic energy. The slip sign stays fixed until it reaches zero. Integrating gives

$$t_r = \frac{\kappa |v_0 - R\omega_0|}{(1 + \kappa)\mu_k g}, \quad v_r = \frac{v_0 + \kappa R\omega_0}{1 + \kappa}, \quad (19.2)$$

$$x_r = v_0 t_r - \frac{1}{2} \mu_k g \operatorname{sgn}(s_0) t_r^2, \quad E_0 - E_r = \frac{m\kappa}{2(1 + \kappa)} s_0^2. \quad (19.3)$$

Here x_r is signed displacement, not path length if translation reverses. Pure rolling initially gives $t_r = 0$. These formulas require $\mu_k > 0$ and do not cover a bouncing, deforming or intermittently contacting ball. The combined rotational and translational energy is $E = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$; during slip, $\dot{E} = -\mu_k mg|s|$.

For the illustrative values $v_0 = 2.0$ m/s, $\omega_0 = 0$, $\mu_k = 0.25$ and $g = 9.81$ m/s², the uniform-sphere model gives

$$t_r = 0.2330 \text{ s}, \quad x_r = 0.3994 \text{ m}, \quad v_r = 1.4286 \text{ m/s}.$$

It dissipates 2/7 of the initial kinetic energy. This example already contradicts treating every skid as a few centimeters or a fixed 0.1 s. Its fraction of the *whole putt* also depends on the subsequent rolling resistance and target distance; no universal 20% follows from the equations.

19.2 Putt Paths on Slopes

After rolling begins, model the green locally as a fixed plane with slope angle α , downslope unit vector $\hat{\mathbf{s}}$ and sufficient static traction to maintain rolling. Represent resistance by a moment of magnitude $c_r NR$ opposing the rolling angular velocity, where $N = mg \cos \alpha$ and $c_r \geq 0$ is dimensionless. This is a declared effective contact law, not a kinetic-sliding coefficient. Neglect aerodynamic forces and spin about the plane normal.

Eliminating the tangential contact force between Newton's and Euler's equations, using the no-slip relation, gives

$$\ddot{\mathbf{r}} = \frac{g \sin \alpha}{1 + \kappa} \hat{\mathbf{s}} - a_0 \cos \alpha \frac{\dot{\mathbf{r}}}{\|\dot{\mathbf{r}}\|}, \quad a_0 = \frac{c_r g}{1 + \kappa}, \quad \|\dot{\mathbf{r}}\| > 0. \quad (19.4)$$

Here a_0 is the level-green deceleration in this model. The factor $1/(1 + \kappa)$ accounts for rotational inertia. With $\kappa = 2/5$ and no resistance, downslope acceleration is $(5/7)g \sin \alpha$, not $g \sin \alpha$. Static friction still supplies the torque needed to spin up the ball; *zero rolling resistance does not mean zero tangential contact force*. The Newton–Euler derivation of no-slip rolling is reviewed in [Moebis et al. \[2016b\]](#).

At rest the velocity direction is undefined. Equation (19.4) must end at a stopping event or be completed by a static contact/resistance law; assigning an arbitrary direction at zero speed does not determine whether the ball stays at rest on a slope. The rolling assumption must also be rejected when the required tangential force exceeds available traction. Real turf introduces anisotropy, speed dependence, local slope changes and stochastic deviations.

For initial motion across the fall line, set $\mathbf{r}(0) = 0$ and $\dot{\mathbf{r}}(0) = (V, 0)$, with y downslope. Locally,

$$x(t) = Vt - \frac{1}{2}a_0 \cos \alpha t^2 + O(t^3), \quad (19.5)$$

$$y(t) = \frac{g \sin \alpha}{2(1 + \kappa)} t^2 + O(t^3). \quad (19.6)$$

Thus the leading small-time relation is approximately parabolic, $y \simeq g \sin \alpha x^2 / [2(1 + \kappa)V^2]$. It is not an exact finite-putt parabola when resistance changes the velocity direction and speed. Do not extrapolate this expansion to the stopping time to predict the aim point of a long putt. Hole capture additionally depends on arrival speed and offset; crossing the hole's location is not a capture model.

19.3 The Stimpmeter and Effective Deceleration

A Stimpmeter reading is a rollout *distance*, conventionally reported in feet, rather than a velocity. The USGA procedure uses repeated rolls in opposite directions on a sufficiently level area and averages the distances; its instructions also distinguish the standard and

alternate notches [United States Golf Association \[n.d.\]](#). It is a surface-comparison procedure, not a direct measurement of a microscopic friction coefficient.

If the established-rolling entry speed v_s is known and the subsequent level deceleration is constant, elementary kinematics gives

$$a_0 = \frac{v_s^2}{2L_s}, \quad \mu_{\text{eff}} = \frac{a_0}{g}, \quad c_r = (1 + \kappa)\mu_{\text{eff}}. \quad (19.7)$$

The last equality belongs to the resistance-moment model above. μ_{eff} is a deceleration ratio; it is neither μ_k nor a universal static-friction coefficient. Energy is dissipated during rollout, so this is not an assertion of mechanical-energy conservation.

Penner uses a historical entry-speed estimate of 1.83 m/s [Penner \[2002\]](#). Using that value as an *illustrative calibration*, an 8-foot rollout gives $a_0 = 0.687 \text{ m/s}^2$, whereas 13 feet gives 0.423 m/s^2 . Their ratio is $13/8 = 1.625$, not two. These selected distances do not define universal course categories. Device/notch choice, the launch-to-roll transition and surface conditions must be included in a measured calibration rather than assigning 1.83 m/s to every modern setup.

19.4 A Compound Pendulum Is a Conditional Stroke Model

For a rigid assembly rotating about a fixed horizontal pivot, let I_p be inertia about that pivot and d the pivot-to-COM distance. With a point torque $\tau(t)$ and no other losses,

$$I_p \ddot{\theta} + mgd \sin \theta = \tau(t), \quad L_{\text{eff}} = \frac{I_p}{md}. \quad (19.8)$$

The equivalent pendulum length L_{eff} is not generally the actual pivot-to-putterhead radius ℓ_h . Linearizing near the downward equilibrium and setting $\tau = 0$ gives

$$\omega_n = \sqrt{g/L_{\text{eff}}}, \quad T = 2\pi\sqrt{L_{\text{eff}}/g}.$$

For $L_{\text{eff}} = 0.8 \text{ m}$, $T = 1.794 \text{ s}$, $T/2 = 0.897 \text{ s}$ and $T/4 = 0.449 \text{ s}$. Comparing the full period with a measured half-cycle is a factor-of-two error. Larger amplitude changes the passive period; $T/T_0 = 1 + \theta_{\text{max}}^2/16 + O(\theta_{\text{max}}^4)$ for angles in radians. Compound-pendulum assumptions and equivalent length are described in [Moebs et al. \[2016a\]](#).

A passive bottom-to-extremum-to-bottom excursion has equal outgoing and return durations. It does not derive a universal 2:1 backswing-to-downswing ratio. Starting from rest at the bottom requires an input to initiate motion at all. A real golfer can move the pivot, change joint configuration and stiffness, and apply time-dependent torque; event definitions also affect reported ratios. Grober proposes a *driven* oscillator and fixed timing/velocity-shape factors as a distance-control model [Grober \[2011\]](#). Such a conditional account should not be replaced by a claim that an unforced pendulum proves optimal human tempo or minimal neuromuscular noise.

19.5 Connect Putter Speed to Ball Speed and Accuracy

For passive release from angle θ_0 , the ideal assembly's head speed at its lowest point is

$$U_h = \ell_h \sqrt{\frac{2g}{L_{\text{eff}}}(1 - \cos \theta_0)} \simeq \ell_h \omega_n |\theta_0|. \quad (19.9)$$

This is *putterhead speed*, not ball launch speed. In an ideal one-dimensional central collision of a moving free putter mass M with a stationary ball of mass m , coefficient of restitution e gives $v_b^+ = (1 + e)MU_h/(M + m)$. Real impact geometry, inertia, shaft/hand coupling, spin and the subsequent contact transition require a richer model or measured calibration. Do not silently identify M with total putter mass or treat the central-collision formula as a complete golf impact law.

For a declared impact-to-established-roll speed gain k_{imp} , write $v_r = k_{\text{imp}}U_h$. Combining this calibration with level constant deceleration yields

$$D = \frac{k_{\text{imp}}^2 U_h^2}{2a_0}, \quad (19.10)$$

where D starts at established rolling, not necessarily at the impact point. Holding the impact transfer, surface and timing fixed explains why stroke amplitude can control distance in a small-angle model. It does not prove that golfers must keep timing fixed.

Direction and speed errors must also be separated from aiming and green reading. Karlsen, Smith and Nilsson studied 1,301 approximately 4-m putts from 71 skilled players and distinguished stroke direction from the aim line [Karlsen et al. \[2008\]](#). Their stroke-specific variability estimates should not be promoted to universal hole-out probabilities. Standard deviation has units of degrees; angular variance has units of degrees squared. Even a deterministic 0.5° direction error gives approximately $6.096 \tan(0.5^\circ) = 0.0532$ m lateral displacement at 20 feet before slope or speed effects. That calculation alone cannot establish capture, perceptual thresholds, or a coaching prescription.

Exercises 19.1. Problems and Thought Experiments

1. **Spin and Slip:** Reproduce the zero-spin example above. Then set $R\omega_0 = 3.0$ m/s while keeping $v_0 = 2.0$ m/s. Show that translation speeds up during contact slip while total kinetic energy decreases. Identify which assumptions fail if the ball bounces.
2. **A Local Slope Prediction:** Use $\kappa = 2/5$, $V = 2.0$ m/s, $\alpha = 2^\circ$, $a_0 = 0.15$ m/s² and $t = 0.10$ s. Compute the second-order x, y approximation and compare it with a numerical integration of equation (19.4). Explain why this is not an aim recommendation for a 20-foot putt with zero arrival speed.
3. **Separate the Three Stages:** For illustrative Stimp distances 9 and 12 feet, use $v_s = 1.83$ m/s to calculate a_0 . For $D = 15$ feet of established rolling, obtain v_r , then U_h using a supplied $k_{\text{imp}} = 1.5$. Use $L_{\text{eff}} = 0.8$ m and $\ell_h = 0.9$ m to compute small-angle release amplitudes. Compare their ratio and the exact passive-energy correction. Which measurements would be needed before transferring this calculation to a golfer?

Part V

The Inverse Problem and Its Limits

Chapter 20

Inverse Dynamics and the Parallel Loop Problem

A swing can reveal how the club moved without revealing how each hand loaded it. A measured hand wrench can reveal an arm's net joint loading without identifying every muscle force. A forward simulation can predict a specified intervention without establishing that a golfer actually used that strategy. These are three different questions, and each deserves a complete answer at its own level.

The purpose of inverse dynamics is to determine loads consistent with measured motion, declared mechanics, and known boundary conditions. Its value does not depend on recovering every muscle's contribution. The research challenge is to identify which combinations of loads the available measurements determine, which remain ambiguous, and which additional measurements would distinguish competing explanations of the golf swing.

20.1 Start With the Quantity Being Investigated

An analysis should name its target before choosing an algorithm. Clubhead speed, clubface orientation, net hand wrench, shoulder moment, muscle force, mechanical work, metabolic demand, and tissue stress are not interchangeable outputs. They require different boundaries, models, and evidence. A joint moment is not an electrical activation signal; a ground moment is not a direct measurement of pelvis acceleration; and neither one alone measures energy supplied to the club.

For example, suppose two golfers deliver the same club pose and velocity at impact. They need not have followed the same earlier trajectory. Even identical full club trajectories need not imply identical hand loading. Identical skeletal trajectories and boundary loads can still admit multiple muscle-force patterns. Conversely, additional sensors or sufficiently restrictive mechanics can make a particular load identifiable. The conclusion must follow from the specified inverse problem rather than the slogan that closed loops are always ambiguous.

Motion capture supplies image or marker observations. Segment poses, joint coordinates, and their derivatives are estimates obtained through calibration, kinematic fitting, smoothing, and differentiation. External forces require instruments or a declared force model; they are not automatically another output of differentiating joint angles. A complete report carries that chain of inference into its final uncertainty estimates.

20.2 Equations, Inputs, and Boundary Conditions

Use regular local coordinates $q \in \mathbb{R}^n$ for an articulated-tree model, with any additional ideal, time-independent closures $\Phi(q) = 0$. Let $J_c = D_q \Phi \in \mathbb{R}^{k \times n}$ contain independent rows

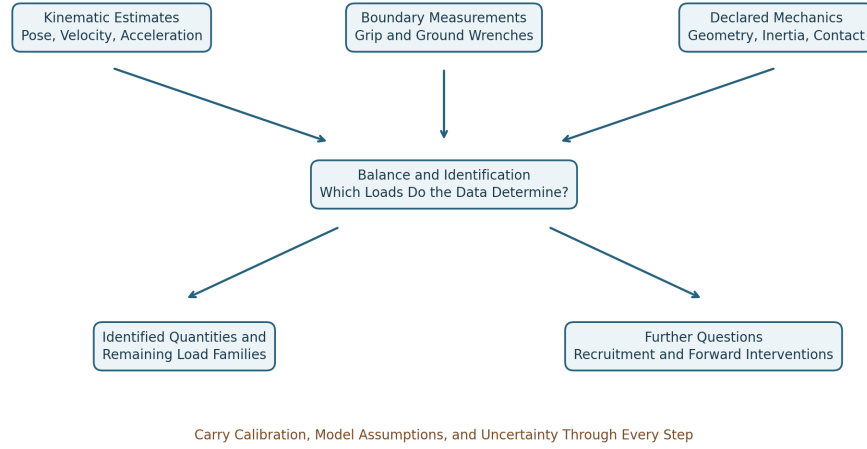


Figure 20.1: From Measurements and Declared Mechanics to Identified Loads and Further Research Questions.

in the contact mode under consideration. Write

$$M(q)\ddot{q} + h(q, \dot{q}) = B(q)u + p(q, \dot{q}, z) + Q_e + J_c(q)^T \lambda. \quad (20.1)$$

Here h contains the declared inertial bias and gravitational generalized loads; $u \in \mathbb{R}^p$ contains the selected applied inputs; B maps them into generalized loads; p contains separately modeled passive or other constitutive loads, with internal states z when needed; and Q_e contains known external loads. A load belongs in exactly one term. Unknown hand or ground loads cannot be placed in Q_e and simultaneously treated as already measured. Their parameters must instead join the unknown vector.

For example, a segment wrench maps to generalized load by the transpose of its compatible velocity Jacobian. Generalized components may have units of force or moment according to their coordinates; they are not all Cartesian forces. The matrix often written $C(q, \dot{q})$ is a representation of an inertial bias vector $C\dot{q}$, not a list of independent physical forces at the hands.

For a regular fully actuated open tree with $B = I$ and known p, Q_e , inverse dynamics evaluates

$$u = M\ddot{q} + h - p - Q_e. \quad (20.2)$$

There is no mass-matrix inversion in this expression. A positive-definite mass matrix follows when every nonzero independent coordinate velocity produces positive kinetic energy in the modeled bodies. Redundant coordinates, massless idealizations, or a singular coordinate chart require separate treatment. A straight elbow can make a hand-position Jacobian singular while the physical joint-space mass matrix remains positive definite.

Even the net load u needs interpretation. A revolute joint's generalized moment is the load component about its allowed axis; its ideal reaction components are separate. A full transmitted joint wrench, a generalized joint moment, and the muscle contribution to that moment are different quantities. Passive tissue loads already included in the net balance must be subtracted consistently before calling the remainder active muscle moment.

20.3 A Complete Newton–Euler Recursion

The recursive Newton–Euler algorithm computes open-tree inverse dynamics without explicitly assembling M . Its linear computational cost for a tree with fixed-size joints does not remove measurement uncertainty or repair an incorrect contact boundary. The essential ingredients are spatial coordinate transport, inertial bias, external wrenches, and propagation of child loads.

For one-freedom revolute or prismatic joints, use angular-first spatial motion $v_i = [\omega_i; v_{O_i}]$ and its power-dual wrench $w_i = [m_{O_i}; F_i]$, both expressed in body frame i . Let $\pi(i)$ be the parent, X_i the motion transform from parent frame to child frame, S_i the joint motion subspace, and $\mathcal{I}_i$ the body’s spatial inertia about its frame origin. The fixed-base recursion is

$$\begin{aligned} v_i &= X_i v_{\pi(i)} + S_i \dot{q}_i, \\ a_i &= X_i a_{\pi(i)} + S_i \ddot{q}_i + \text{crm}(v_i) S_i \dot{q}_i, \\ w_i &= \mathcal{I}_i a_i + \text{crf}(v_i) \mathcal{I}_i v_i - w_i^{\text{ext}}. \end{aligned} \quad (20.3)$$

Here $\text{crm}(v)$ is the spatial motion cross-product operator and $\text{crf}(v) = -\text{crm}(v)^T$ its force counterpart. For $v = [\omega; v_O]$,

$$\text{crm}(v) = \begin{bmatrix} [\omega]_{\times} & 0 \\ [v_O]_{\times} & [\omega]_{\times} \end{bmatrix}, \quad [a]_{\times} b = a \times b. \quad (20.4)$$

Set $v_0 = 0$ and $a_0 = [0; -g]$ to incorporate uniform gravity through this recursion; do not then subtract gravity again as an external wrench. With this convention the recursively stored acceleration includes the gravity offset. For more general joints, include the appropriate joint bias acceleration. A moving or floating base requires its own specified motion or balance equations.

After forming each body’s inertial and external contribution, accumulate from tips to base:

$$w_i \leftarrow w_i + \sum_{j: \pi(j)=i} X_j^T w_j, \quad \tau_i = S_i^T w_i. \quad (20.5)$$

The transpose transport follows from power invariance: $(X_j v_i)^T w_j = v_i^T X_j^T w_j$. It carries force-induced moments as well as rotating components. Adding moments from different points without this transport loses the lever-arm contribution. These conventions follow the [Featherstone spatial-vector implementation](#); its explicit inverse-dynamics recursion also shows the velocity-product terms.

For a closed chain, one may cut a contact to obtain a tree, introduce its unknown cut wrench, and solve the resulting coupled balance and measurement equations. A tree algorithm with an arbitrarily assigned cut wrench gives an answer for that assignment. It does not establish that the assignment was measured or uniquely implied by motion.

20.4 What the Two-Handed Grip Constrains

The two hand frames occupy different locations on the club. If T_C is the club pose and H_L, H_R are fixed hand-to-club attachment transforms under an ideal rigid grip, closure is

$$T_L = T_C H_L, \quad T_R = T_C H_R, \quad T_L^{-1} T_R = H_L^{-1} H_R. \quad (20.6)$$

The final relation supplies six independent local restrictions only where the chosen representation is regular. Equality of the two hand poses would describe coincident attachment frames, not the ordinary separated grip. Compliance, sliding, or changing finger contacts require a different model.

Rigid-body angular velocity is common to its points when expressed in one frame, but their linear velocities differ:

$$v_R = v_L + \omega_C \times (r_R - r_L). \quad (20.7)$$

The corresponding acceleration relation includes angular acceleration and centripetal terms. A Jacobian closure of the form $[J_L, -J_R]\dot{q} = 0$ is valid only after both Jacobians describe the same transported task frame and point.

The general compatibility equations are

$$J_c \dot{q} = 0, \quad J_c \ddot{q} + \dot{J}_c \dot{q} = 0. \quad (20.8)$$

They test whether a candidate trajectory respects the declared constraint; they contain no independent measurement of λ . At constant rank r , feasible local configuration dimension is $n - r$. At a singular point, extra linearized directions need not extend to finite feasible motions. Low speed at address or transition is not evidence of rank loss; finite tissue stiffness is not a kinematic rank calculation.

With two seven-coordinate arm models and a regular six-restriction rigid-grip closure, eight configuration freedoms remain. They include motion of the club. If the full club-pose task has rank six on the feasible tangent space, only two posture freedoms remain while holding that task fixed. These counts assume the specified shoulder bases and arm models; they are not universal counts of the human body.

Thus $\ker J_c$ contains compatible velocities; task-preserving velocities also satisfy $J_y \dot{q} = 0$. Neither space says that movement is energetically free. Grip pressure is a contact-load variable, not a joint velocity. With independent rows, $\ker J_c^T$ is zero, even though unknown actuator and reaction loads can still be exchanged without changing the observed motion.

20.5 Identification Is a Rank Question

At a specified state and acceleration, define the known model residual and unknown vector by

$$r = M\ddot{q} + h - p - Q_e, \quad z_\ell = \begin{bmatrix} u \\ \lambda \end{bmatrix}, \quad A = \begin{bmatrix} B & J_c^T \end{bmatrix}, \quad Az_\ell = r. \quad (20.9)$$

Use z_ℓ for load unknowns to distinguish them from constitutive states. For a consistent linear problem, the solution set is $z_{\ell,0} + \ker A$. It is a singleton exactly when A has full column rank. Having more unknowns than equations rules out a unique unconstrained solution; a square matrix alone does not guarantee uniqueness. Bounds, contact inequalities, and nonlinear measurements can further restrict the feasible set and must be analyzed as part of that different problem.

Suppose additional calibrated linear measurements are $Sz_\ell = y_m$. Stack $A_m = [A; S]$. More sensor channels help identification only when they add independent information about the unresolved directions. A particular linear quantity Cz_ℓ is identifiable even without the whole vector precisely when

$$\ker A_m \subseteq \ker C. \quad (20.10)$$

Indeed, all indistinguishable solutions differ by a vector in $\ker A_m$; the reported quantity is unique only if C annihilates every such difference. For finite-dimensional linear maps this is equivalent to the rows of C lying in the row space of A_m . Noise replaces an exact affine set by an uncertainty region; poor singular values then matter as well as rank.

20.5.1 An Example Where the Sensor Choice Matters

Take $M = \text{diag}(2, 1)$, $J_c = [1, -1]$, $B = I$, and zero other loads. For the compatible acceleration $\ddot{q} = (1/3, 1/3)^T$,

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \lambda \end{bmatrix} = \begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix}. \quad (20.11)$$

Its null direction is $(-1, 1, 1)^T$. Loads $(1, 0, -1/3)$ and $(2, -1, -4/3)$ therefore produce the same acceleration. Measuring $u_1 + u_2 = 1$ adds no rank: it repeats the sum of the existing equations. Measuring $\lambda = -1/3$ selects $(u_1, u_2) = (1, 0)$ in this ideal example. Measurement precision still limits the precision of that inference.

If the same mechanism has only one actuator, $B = (1, 0)^T$, the inverse matrix for (u_1, λ) is square with determinant -1 . The corresponding loads are uniquely determined for this measured acceleration. A loop therefore does not by itself make inverse dynamics nonunique. Conversely, multiple muscles can make recruitment nonunique in a completely open limb.

20.6 Hand Wrenches and Club Balance

A hand's force and moment must be referenced to a point and expressed in a coordinate frame. Use the same angular-first ordering as the spatial recursion. For an inertial-frame force F_i and moment m_i about attachment point i , transport to club center of mass C with $r_i = p_i - p_C$:

$$w_i^C = X_i^* w_i, \quad X_i^* = \begin{bmatrix} I & [r_i]_\times \\ 0 & I \end{bmatrix}, \quad w_i = \begin{bmatrix} m_i \\ F_i \end{bmatrix}. \quad (20.12)$$

Rotate components first if the sensor axes differ from the reporting axes. The club's net hand wrench is

$$\begin{bmatrix} I_C \alpha + \omega \times (I_C \omega) - m_{\text{other}}^C \\ m_c(a_C - g) - F_{\text{other}} \end{bmatrix} = G \begin{bmatrix} w_L \\ w_R \end{bmatrix}, \quad G = [X_L^* \quad X_R^*]. \quad (20.13)$$

Here I_C is rotational inertia about the club COM expressed in the same inertial axes as ω, α ; it changes orientation with the club. The gyroscopic term remains necessary. Other forces include modeled aerodynamic and impact loads. A moving-grip-point expression $I_{\text{grip}} \alpha$ alone is not a general three-dimensional club balance. During impact, finite sensor bandwidth and flexible shaft motion may invalidate a simple rigid-body instantaneous-acceleration estimate; an impulse balance requires its own interval, boundary, and measurements.

Each X_i^* is invertible, so the full-wrench grasp map has rank six and a six-dimensional load null space before contact restrictions. Motion can identify the resultant under the declared model while leaving six combinations of hand wrenches unresolved. If the model permits only point forces and no hand moments, the corresponding six-by-six map has rank five at two distinct points. Its one internal direction is an equal-and-opposite force pair

along the line joining them. A transverse opposing pair produces a couple and is visible in angular balance. Counting only resultant force misses that distinction.

Arm joint-torque vectors live in their own generalized-coordinate spaces. For a six-coordinate two-arm model, its six required joint loads cannot be written as the sum of two three-component hand forces. Hand wrenches enter the arm equations through their Jacobian transposes and the appropriate opposite-action signs. Equal and opposite numerical joint-torque entries do not automatically cancel the club wrench.

20.6.1 A Transport and Power Check

Take $r = (0.2, 0, 0)$ m, $F = (0, 100, 0)$ N, and a local hand moment $m = (0, 0, 3)$ N m. About C , the moment is $(0, 0, 23)$ N m. If $v_C = (1, 0, 0)$ m/s and $\omega = (0, 0, 2)$ rad/s, the hand-point velocity is $(1, 0.4, 0)$ m/s. Power computed either way is

$$m^T \omega + F^T v_{\text{hand}} = (m + r \times F)^T \omega + F^T v_C = 46 \text{ W}. \quad (20.14)$$

Using the untransported 3 N m moment together with COM velocity would report 6 W and lose 40 W. The discrepancy is a bookkeeping error, not a new energy source. At an ideal maintained attachment, the equal-and-opposite wrench acting on the hand has opposite power when both sides use the same contact twist. It can transfer energy between golfer and club while canceling in the combined system's internal power balance.

20.6.2 What an Instrumented Grip Actually Measures

A single six-axis sensor below both hands measures a section wrench. After accounting for the intervening club/grip inertia and other loads, it can constrain their aggregate action. It generally cannot resolve twelve independent hand-wrench components merely because it reports six channels. A second measurement of the same resultant improves precision but adds no independent load-sharing rank.

Separate mechanically isolated hand interfaces can supply distinct boundary wrenches. Multiple shaft sections may also distinguish loads if their placement, section balances, and calibration make the combined observation map informative. A pressure array usually measures a subset of contact traction; integrating normal pressure alone does not recover all tangential forces or free moments. A contact's net wrench also does not uniquely reconstruct the pressure and shear distribution across the skin.

The experiment must state exactly which bodies lie on each side of each sensor, where its wrench is referenced, how its axes are registered to motion capture, and how sensor mass changes the instrumented club's inertia. Timing offsets between motion and force signals can create apparent balance violations or false power peaks. Calibration, cross-talk, saturation, bandwidth, hysteresis, and repeated-load validation belong in the measurement uncertainty.

20.7 Sensor Physics With Defined Units

A bonded resistance strain gauge measures deformation through $\Delta R/R = GF \varepsilon$ in its calibrated small-strain range. It measures local strain first; a structural model and calibration map that strain to a load. Multiple gauges can separate bending, axial loading, and torsion only if their geometry and calibration support those components.

For a declared Wheatstone bridge, put R_1 above R_2 in the left divider and R_3 above R_4 in the right divider, with excitation V_{ex} from the common top node to the bottom node. Measure left midpoint minus right:

$$\frac{V_{\text{out}}}{V_{\text{ex}}} = \frac{R_2}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} = \frac{R_2 R_3 - R_1 R_4}{(R_1 + R_2)(R_3 + R_4)}. \quad (20.15)$$

Reversing the measurement leads reverses the sign. If only $R_1 = R(1 + \delta)$ changes and the others equal R , then

$$\frac{V_{\text{out}}}{V_{\text{ex}}} = -\frac{\delta}{4 + 2\delta} \simeq -\frac{\text{GF} \varepsilon}{4}. \quad (20.16)$$

With $\text{GF} = 2$, strain 500 microstrain, and 5 V excitation, the exact output is approximately -1.249 mV. Gauge orientation, active bridge arms, temperature compensation, lead resistance, and self-heating affect the actual installation. The bridge conventions and calibration choices should be reported, as emphasized in the [NI strain-measurement documentation](#).

For a piezoelectric material, a constitutive relation in stress-charge form is

$$D = d \sigma + \epsilon^T E. \quad (20.17)$$

D is electric displacement (charge per area), σ is stress, E is electric field, and d is the piezoelectric coupling tensor under the stated material convention. This is not an equation for voltage. In a calibrated one-dimensional approximation, generated charge is $Q_{\text{ch}} = d_F F$. An ideal open-circuit voltage magnitude is Q_{ch}/C for the relevant capacitance; an ideal inverting charge amplifier gives $V_{\text{out}} = -Q_{\text{ch}}/C_f$ over its operating band. For $d_F = 4$ pC/N and $F = 100$ N, the charge is 400 pC: this corresponds to 2 V across 200 pF or -0.4 V with a 1 nF feedback capacitor. These are illustrative circuit values, not a specification for a particular instrument.

Leakage, amplifier time constants, and insulation limit sustained measurements; preload and mounting affect the force path and dynamics. Piezoelectric systems can measure dynamic and suitably conditioned quasi-static loading. They are not universally faster or more sensitive than every strain-gauge system. See the [Kistler force-sensor documentation](#) for the distinction between a sensor's calibrated force response and its measurement conditions. Six independent load components require an appropriate mechanical and electrical sensing map, not necessarily six crystals or a hexapod.

20.8 Optimization Selects a Hypothesis

When measurements leave a load family, optimization can select a member. It adds a criterion; it does not turn the selected member into an observation of neural intent. State the decision variables, units, constraints, physiological bounds, and whether the objective is a hypothesis or a computational regularizer.

For example, minimize

$$\frac{1}{2}(z_\ell - z_0)^T W (z_\ell - z_0) \quad \text{subject to} \quad A_m z_\ell = b_m, \quad W \succ 0. \quad (20.18)$$

Here z_0 is a declared reference load vector, not a measured fact unless it actually is measured. For full-row-rank A_m , stationarity of the Lagrangian and substitution into the constraints give

$$z_\ell^* = z_0 + W^{-1} A_m^T (A_m W^{-1} A_m^T)^{-1} (b_m - A_m z_0). \quad (20.19)$$

Redundant consistent rows can be removed or handled with an appropriate pseudoinverse. Inconsistent noisy measurements require a residual or statistical model rather than exact constraints that cannot all hold. Contact bounds and nonlinear muscle laws alter the optimization problem and may destroy feasibility or convexity; a numerical optimizer returning a vector is not proof of a valid physiological solution.

In a deliberately scalar example, $u_L + u_R = 10$ N m. Equal weights choose (5, 5) N m. Minimizing $u_L^2 + 4u_R^2$ chooses (8, 2) N m, with objective 80 in the declared squared-moment units. The balance did not change; the preference did. A proper norm needs scaling when the vector mixes force, moment, or different capacity scales. Squared joint moment is not metabolic power. The sum of measured EMG amplitudes is constant with respect to unknown loads and cannot select a solution unless an explicit decision-dependent measurement or physiological model connects them.

20.9 Muscles, Activation, and EMG

Given a net generalized moment, muscle recruitment is another inverse problem. If F_m collects musculotendon forces and $R(q)$ their moment-arm map, a simplified moment balance is

$$\tau_{\text{net}} = R(q)F_m + \tau_{\text{passive, other}}. \quad (20.20)$$

The sign convention for R follows the chosen coordinate and muscle-length conventions. A muscle spanning several joints contributes to several entries; net joint moments do not identify all tensile muscle forces when the map has unresolved directions. Known external hand wrenches improve the joint-load estimate without automatically eliminating muscle redundancy.

Surface EMG records a superposition of electrical potentials detected through electrodes and intervening tissue. Its amplitude depends on motor-unit activity and recording conditions. It is not a direct recording of motor-cortex commands, activation as used in a muscle model, or contractile force. The [Farina, Merletti, and Enoka analysis](#) explains why neural interpretation requires care. Electrode placement, cancellation, cross-talk, normalization, fatigue, and tissue motion all matter. A larger normalized signal in one arm does not compel a larger resultant hand wrench from that arm.

A model can link processed EMG to excitation with calibration and uncertainty, then evolve activation and musculotendon states. For example,

$$\dot{a} = \frac{e - a}{\tau(e, a)}, \quad F_{\text{CE}} = F_{\text{max}} a f_\ell(\ell_f) f_v(\dot{\ell}_f). \quad (20.21)$$

This illustrative contractile-element relation must be completed with passive fiber behavior, pennation, tendon mechanics, and force equilibrium or the chosen dynamic muscle model. It is not by itself a complete musculotendon force law. The [OpenSim activation-dynamics documentation](#) explicitly distinguishes excitation from activation. Different force-length and force-velocity states can produce different forces at the same activation.

EMG-informed modeling can be quantitatively useful when calibrated and validated; calling EMG merely qualitative would also be too strong. Report which muscles are observed, which deep or unmeasured muscles remain free, the activation and tendon assumptions, and prediction errors on independent conditions. Neither an EMG ratio nor a minimum-activation objective uniquely establishes how two hands share a club load without that intervening mechanics.

A model of internal tissue stress requires geometry and constitutive behavior in addition to boundary loads. An internal-load ambiguity identifies a limit on inference, not an injury diagnosis or a universal coaching prescription.

20.10 Ground Loads and Segment-by-Segment Balance

Separate calibrated force plates can provide a wrench for each foot when each foot's contact belongs to its assigned plate and all relevant contacts are captured. One plate under both feet usually gives their combined wrench; it does not generally determine the bilateral split. Vertical force alone is less information than a full force-and-moment measurement.

For a segment with COM C , known distal wrench (F_d, m_d) at point d , and unknown proximal wrench (F_p, m_p) at point p , all acting on the segment in inertial axes, the balances are

$$\begin{aligned} F_p &= m(a_C - g) - F_d - F_o, \\ m_p &= I_C \alpha + \omega \times (I_C \omega) - m_d - m_o^C \\ &\quad - (p_d - p_C) \times F_d - (p_p - p_C) \times F_p. \end{aligned} \quad (20.22)$$

F_o, m_o^C collect any other nongravitational external loads with moments already referenced to C . The segment's moment about its COM includes the moment arms of both distal and proximal forces. To proceed from foot to shank to thigh, reverse the interaction wrench on the neighboring segment and transport it to the appropriate point. Passing only an ankle torque upward while omitting its joint force is incomplete.

These equations estimate a transmitted net wrench under the segment model. Projecting onto a declared joint axis gives its generalized moment. Anatomical axis definitions, soft-tissue artifact, joint-center locations, inertial parameters, and synchronization affect the result. The estimate does not assign all of that wrench to one muscle or one tissue.

20.10.1 Yaw Moment Is Not a Vertical Force Offset

For a force F at COP r_{COP} and a free moment T_{free} , the moment about a reporting point P is

$$M_P = (r_{\text{COP}} - r_P) \times F + T_{\text{free}}. \quad (20.23)$$

In right-handed axes with z vertical, its yaw component is

$$(M_P)_z = (x_{\text{COP}} - x_P)F_y - (y_{\text{COP}} - y_P)F_x + (T_{\text{free}})_z. \quad (20.24)$$

A purely vertical force at a horizontal offset creates roll/pitch moments, not yaw. For $r = (0.2, -0.1, 0)$ m and $F = (0, 0, 1400)$ N, $r \times F = (-140, -280, 0)$ N m. A reported plate moment about its origin must be distinguished from the free moment after moving the resultant to COP. Vendor signs and whether the wrench is force-on-plate or force-on-golfer must be converted consistently.

For a 100 kg system, 1400 N is about 1.43 body weights using $g = 9.81$ m/s². If it is the only nongravitational vertical force on that system, its COM vertical acceleration is $1400/100 - 9.81 = 4.19$ m/s². Neither this force nor an 80 N m free moment alone determines pelvis acceleration. Dividing 80 N m by a declared 2 kg m² fixed-axis inertia gives 40 rad/s² for that isolated toy rotation, not a torque and not a complete multisegment pelvis model.

Whole-body angular momentum about the whole-body COM changes with the net external moment. Pelvis angular momentum also exchanges with legs and trunk through

internal interactions. Different segments can counter-rotate while preserving whole-body angular momentum. A ground-moment contribution assigned to a pelvis coordinate therefore depends on the coupled dynamics and the chosen decomposition; it is not read directly from a force plate.

20.10.2 Momentum Exchange and Mechanical Power

At an ideal stationary no-slip contact on a rigid stationary ground, the contact velocity is zero, so contact force supplies zero mechanical power there. It can nevertheless change the golfer's momentum and permit muscle work to be expressed as segment and club motion. Real shoe deformation, sliding, moving supports, and contact compliance require their actual velocities and energy terms. A force multiplied by COM velocity gives a contribution to COM translational kinetic-energy rate, not automatically the work rate at the physical contact.

Assigning active ground power to every downswing and passive ground braking to every follow-through combines unsupported physiological interpretation with an incorrect contact-power shortcut. Measure the moment direction, state the system boundary, and compute power with its conjugate velocity. The energy source, route of transfer, and momentum constraint are related but distinct parts of the swing explanation.

20.11 Conditioning and Measurement Uncertainty

Inverse dynamics often uses acceleration estimates, so differentiation quality matters. Independent position-sample errors of variance σ_q^2 passed through the centered second difference have acceleration-error variance $6\sigma_q^2/\Delta t^4$. Real smoothing and correlated errors change this expression. It illustrates why quoting frame rate alone does not establish acceleration precision. Filtering choices should be evaluated against relevant motion and force bandwidth, not chosen solely to make torque traces smooth.

For an open-tree net-load evaluation at fixed model parameters, a first-order perturbation includes

$$\delta\tau = M \delta\ddot{q} + (\delta M)\ddot{q} + \delta h - \delta p - \delta Q_e. \quad (20.25)$$

Cross-correlations matter when these quantities come from the same fitted trajectory. Covariance propagation or repeated sampling through the complete pipeline is preferable to treating all errors as independent by default. Model discrepancy and missing external loads are not necessarily random measurement noise.

20.11.1 A Counterexample to the Small-Inertia Argument

Let $M = \text{diag}(1, \epsilon)$ in consistent coordinate and inertia units, $0 < \epsilon \leq 1$. Its two-norm condition number is $1/\epsilon$. But for a fixed acceleration error $(0.1, 0.1)^T$,

$$\delta\tau = M \delta\ddot{q} = (0.1, 0.1\epsilon)^T. \quad (20.26)$$

The second torque error is 0.01 at $\epsilon = 0.1$ and 0.001 at $\epsilon = 0.01$. It decreases. By contrast, a fixed torque error $(0.1, 0.1)^T$ in a forward solve produces acceleration error $(0.1, 0.1/\epsilon)^T$, whose second component increases. The condition number describes relative worst-case

sensitivity with the specified norm and scaling; it is not the absolute gain of every inverse-dynamics perturbation. A matrix $1000I$ has condition number one yet multiplies an absolute acceleration error by 1000 in a load evaluation.

For a planar two-link arm with uniform links, each of unit length and unit mass, at a straight configuration the joint inertia and endpoint-position Jacobian can be

$$M = \begin{bmatrix} 8/3 & 5/6 \\ 5/6 & 1/3 \end{bmatrix}, \quad J_y = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}. \quad (20.27)$$

The task Jacobian has rank one; the mass matrix has positive diagonal entries and determinant $7/36 > 0$. The endpoint singularity has not made the joint inertia singular. In constrained and sensor-augmented inverse problems, inspect the correctly scaled identification matrix and its singular values as well as the underlying inertial model. State angle/translation scaling before comparing condition numbers across models.

20.12 Forward Dynamics and a Declared Counterfactual

For specified inputs, constitutive states, and external loads, the acceleration and ideal reaction in a fixed contact mode satisfy

$$\begin{bmatrix} M & -J_c^T \\ J_c & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} Bu + p + Q_e - h \\ -\dot{J}_c \dot{q} \end{bmatrix}. \quad (20.28)$$

If $M \succ 0$ and J_c has full row rank, the matrix is nonsingular. To see this, a homogeneous solution obeys $M\delta a - J_c^T \delta \lambda = 0$ and $J_c \delta a = 0$. Multiplying the first equation by δa^T gives $\delta a^T M \delta a = 0$, so $\delta a = 0$ and then $\delta \lambda = 0$.

This proves unique instantaneous acceleration for those inputs. Local trajectory existence and uniqueness also require compatible initial conditions and sufficient regularity of the reduced differential equations and input policy. A smooth fixed-mode vector field is locally Lipschitz; contact switching, friction, impacts, and rank changes require event laws and a fresh analysis. Matrix invertibility at one instant is not a global guarantee for an entire swing.

20.12.1 Same-State Acceleration and a Forward Branch

A zero-torque counterfactual (ZTCF) must specify which input is set to zero. Zero commanded joint torque, zero muscle excitation, zero activation, and removal of muscle/tissue forces are different interventions. In an effective joint-input model, setting $u = 0$ retains the stated p, Q_e, h laws and their states. If preload or activation-dependent impedance is not measured, the baseline inherits that uncertainty. Gravity can be computed from geometry and mass assumptions; that does not make every elastic or physiological drift term observable from skeletal motion.

At the same complete state and fixed mode, with other loads held under the same law, write the acceleration map as

$$a(q, \dot{q}, z, u) = a(q, \dot{q}, z, 0) + K(q)B(q)u, \quad (20.29)$$

$$K = M^{-1} - M^{-1}J_c^T(J_c M^{-1}J_c^T)^{-1}J_c M^{-1}.$$

This identifies the input's acceleration effect under the model. Even if it is known exactly, KBu need not identify u , because projection and actuator redundancy can remove load

directions. Reactions must be recomputed when the input changes. Retaining measured reactions while zeroing u generally breaks constraint compatibility.

A forward ZTCF instead starts at the chosen initial state and integrates the zero-input policy along its own trajectory. It recomputes geometry, inertia, velocity bias, passive states, reactions, and events along that branch. Once it has diverged from the actual trajectory, subtracting the two accelerations no longer compares the same state. Multiplication by one trajectory's M cannot undo that difference or recover muscle torques.

For an unconstrained dimensionless toy oscillator,

$$\ddot{q} = -q + u, \quad q(0) = \dot{q}(0) = 0, \quad q_{u=1}(t) = 1 - \cos t, \quad q_{u=0}(t) = 0. \quad (20.30)$$

At $t = \pi/2$, the two trajectories' accelerations are both zero, although the actual input remains one. The acceleration difference across branches is zero; the same-state difference is $0 - (-1) = 1$. This elementary example shows why a trajectory residual is a model intervention outcome, not an instantaneous input reconstruction.

For finite-horizon clubface or speed outcomes, compare the outcomes of the two branches with their uncertainty and initial-state matching explicitly stated. When using a local variational approximation instead, propagate perturbations along its evolving baseline and identify its validity range. A forward counterfactual asks a useful different question from inverse muscle inference; it is not universally more accurate simply because it runs forward.

20.13 A Golf Study That Can Distinguish Explanations

Suppose the question is whether two movement patterns achieve similar club outcomes through different hand loading, and whether an effective zero-input branch predicts different continuation of the motion. A defensible study can connect these questions without treating any one measurement as sufficient.

1. Define the primary output: for example, clubhead velocity and face orientation just before impact, a net hand-wrench time history, or work transferred to the club over a specified interval. Define tolerances for calling two outcomes similar and distinguish impact conditions from ball flight, which also depends on collision and aerodynamic variables.
2. Specify the body graph, joint coordinates, separated grip frames, club flexibility assumptions, contact modes, inertial parameters, and input channels. Validate measured trajectory compatibility; report closure residuals instead of hiding them inside large artificial reactions.
3. Calibrate and synchronize kinematics, bilateral ground wrenches, and the selected grip measurements. Derive the augmented identification map and show which quantities are determined. Reserve measured channels or trials for validation rather than using all data both to fit and to confirm.
4. Solve the declared net-load problem and test segment and whole-system force, moment, and power balances. Propagate plausible measurement and parameter variation. If competing load distributions remain possible, report that family or interval rather than only one optimizer output.

5. Add muscle modeling and EMG only for the recruitment questions they can inform. Compare alternative calibrated recruitment assumptions and inspect prediction error on independent data. Do not translate inferred muscle forces into tissue stress or injury likelihood without the additional validated model required for that output.
6. Branch the forward counterfactual from a declared event or state, specify zeroed inputs and retained states, and integrate each branch independently. Report the event rules, numerical convergence, and uncertainty in its endpoint. Compare models and repeat the analysis across trials before drawing a general conclusion about a golfer or a population.

This links geometry to admissible motion, dynamics to required net loading, measurement to identification, muscle models to recruitment hypotheses, and power balance to energy transfer. The central scientific gain is an argument that makes clear what would change its conclusion. A successful fit to one swing is useful evidence about that fitted model; independent predictions and controlled comparisons are needed for stronger claims about strategy or benefit.

20.14 Chapter Exercises

1. Verify the two-coordinate inverse matrix's null vector. Compare ranks after measuring the actuator sum, measuring the multiplier, and removing the second actuator.
2. Derive wrench transport from invariant power. Reproduce the 46 W example and identify the lost lever-arm term when reference points are mixed.
3. Derive grasp-map ranks for two full hand wrenches and for two separated point forces. Explain why a transverse opposing force pair creates a couple.
4. Derive the weighted minimum from its Lagrangian and verify (8, 2). Explain why the selected objective does not establish neural intent.
5. Derive the bridge voltage from its divider voltages and reverse the output leads. Convert the piezoelectric charge using each stated capacitance.
6. Reproduce the inverse and forward error calculations for $M = \text{diag}(1, \epsilon)$. Verify the straight arm's singular task Jacobian and positive-definite inertia.
7. Use the oscillator solution to compare cross-branch and same-state acceleration differences at $t = \pi/2$. State what each comparison holds fixed.
8. A paper reports separate left and right spinal muscle moments. Identify required boundary loads, moment arms, physiological laws, and independent measurements. Explain why muscle redundancy needs no spinal kinematic loop.
9. Design a sensor addition for a remaining golf load-sharing null direction. State its rank benefit, calibration needs, and a quantity it cannot identify.

Part VI

The Physical System

Chapter 21

The Flexible Shaft: Elastic Energy and the Catapult Effect

In Plain Language 21.1. Why the Shaft Matters

A shaft connects a moving golfer to a moving clubhead through a deformable structure. Its bending changes the position and orientation of the head, stores some of the work exchanged through the grip, and returns or dissipates that energy as loading changes. The reaction also acts on the golfer. Understanding this two-way connection is essential: motion of the head relative to the grip is not automatically an increase in the head's speed relative to the ground.

This chapter connects beam mechanics, coupled dynamics, energy accounting and control. The word “catapult” describes temporary elastic storage and subsequent motion; it does not establish a universal speed bonus. The useful question is which shaft, golfer, input history and impact outcome are being compared, and which measurements can distinguish the proposed mechanisms.

21.1 When a Rigid Model Is Adequate

A rigid club model is a deliberate approximation. It can explain gross segment motion or provide a reference for a more detailed calculation. Its adequacy depends on the output: an error acceptable for overall swing timing may be unacceptable for face orientation, strike location or strain. Assess the difference against an explicit tolerance and measurement uncertainty. Adding flexibility without identifying its parameters does not by itself improve predictive accuracy.

Separate bending in two transverse directions, torsion about the shaft, and axial extension. Composite construction can couple these deformations. A planar bending model is useful for deriving principles, but it cannot independently predict three-dimensional face delivery. Even the direction called “lead” needs a reference frame; a camera projection alone can mix shaft motion, perspective and grip rotation.

Example 21.1. Shaft Deflection Magnitudes

For a straight, uniform, nonrotating cantilever under a static transverse tip force F , let L be its free length and EI its bending rigidity. Integrating the moment-curvature

relation gives

$$w(z) = \frac{Fz^2(3L - z)}{6EI}, \quad \delta = w(L) = \frac{FL^3}{3EI}, \quad w_z(L) = \frac{FL^2}{2EI}.$$

These relations require small slopes, a fixed clamp, no tip moment and negligible shear deformation. They establish the strong length dependence and connect displacement to tip rotation; they are not a measured range for all golf shafts. For the illustrative values $L = 1$ m, $EI = 40$ N m² and $F = 2$ N, the tip deflects 16.67 mm and rotates 0.025 rad. Changing the support or loading changes the answer. Elementary cantilever theory is developed in [Parks, 2004].

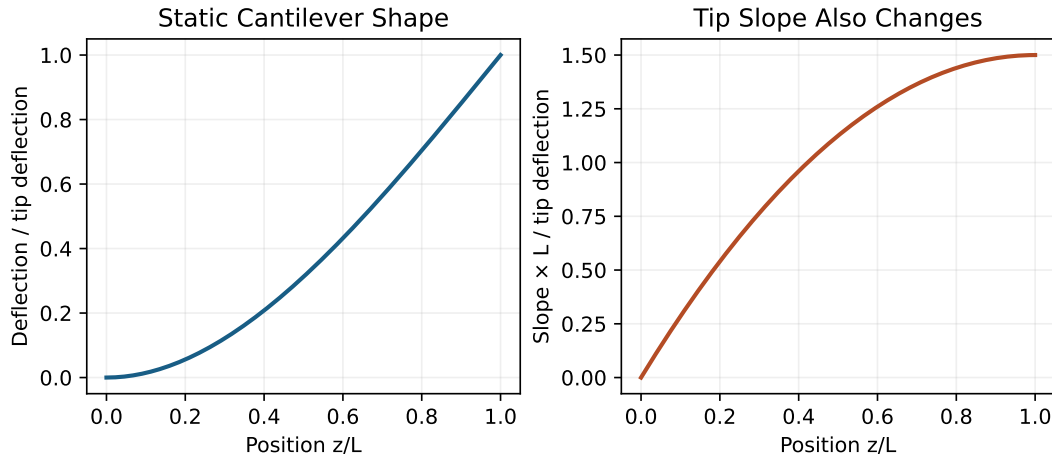


Figure 21.1: A Tip-Normalized Static Cantilever Shape and Its Slope. This Is an Admissible Approximation, Not an Exact Vibration Eigenmode.

An acceleration magnitude does not specify a bending load. A force along the centerline of an ideal straight beam produces axial loading. A transverse force or an offset force can produce a bending moment. The clubhead center of mass is generally offset from the shaft axis, so a grip force described as radial can participate in bending. MacKenzie and Sprigings varied this offset and replayed selected grip-force components in a three-dimensional model, demonstrating the importance of geometry and the declared loading intervention [MacKenzie and Sprigings, 2010]. A large centripetal acceleration alone cannot establish the sign or size of shaft bend.

21.2 Beam Theory Basics: How Shafts Deform

Let $z \in [0, L]$ denote distance along a straight reference shaft, $w(z, t)$ a transverse displacement, and $\mu(z)$ mass per length. For small bending about a stationary configuration, an Euler–Bernoulli teaching model with prescribed tensile prestress $N(z)$ is

$$\mu w_{tt} + \partial_z^2(EI w_{zz}) - \partial_z(N w_z) = q(z, t).$$

Here q is transverse load per length in N/m. Spatial variation of EI stays inside the derivatives. For constant EI , zero prestress and a nonrotating frame this reduces to $\mu w_{tt} + EI w_{zzzz} = q$. This scalar model neglects shear deformation and cross-section rotary inertia within the beam; it is not a complete equation for a rotating, anisotropic golf club. Moving-frame inertial terms, boundary motion, torsion and head inertia must be added consistently when those effects matter.

At a fixed clamp, $w(0) = w_z(0) = 0$. For an unloaded bare tip with $N = 0$, the bending moment and shear vanish: $EI w_{zz}(L) = 0$ and $\partial_z(EI w_{zz})(L) = 0$. A head with mass and rotational inertia supplies dynamic end forces and moments instead. A compliant grip supplies different boundary relations again. A frequency measured with a bench clamp therefore belongs to that assembled test, including clamp length, tip attachment and measurement procedure.

In Plain Language 21.2. Why EI Matters

Bending rigidity EI has units N m^2 ; a translational spring stiffness k has units N/m ; a rotational spring coefficient has units N m/rad . They cannot be substituted for one another. In the static cantilever above, $k = 3EI/L^3$. A tapered composite shaft has a stiffness profile, and two shafts with similar tip compliance can have different local curvature, tip slope, torsion or vibration response.

The elastic bending energy is

$$U_b = \frac{1}{2} \int_0^L EI(z) w_{zz}^2 dz.$$

For a fixed tensile prestress, the quadratic transverse contribution is $U_N = \frac{1}{2} \int_0^L N(z) w_z^2 dz$. Varying these functionals yields the stiffness terms in the PDE, including the boundary terms. Positive tension penalizes transverse slope in this idealization. Compression contributes with the opposite sign and can invalidate a stable small-deflection approximation near buckling. A rotating shaft requires the prestress and associated work to be derived from the same mechanical system.

Definition 21.1. Modal Coordinates

Approximate $w(z, t) = \sum_{j=1}^n \phi_j(z) \eta_j(t)$ using declared admissible shapes. For a mode with nonzero tip displacement, choose dimensionless ϕ_j with $\phi_j(L) = 1$; its amplitude η_j is then in metres. With several such modes, the tip displacement is their sum. The amplitudes are signed coordinates, and zero displacement in one coordinate does not imply that every deformation is absent.

For a stationary planar beam with a centered point head mass m_h , projection gives

$$M_{ij} = \int_0^L \mu \phi_i \phi_j dz + m_h \phi_i(L) \phi_j(L),$$

$$K_{ij} = \int_0^L EI \phi_i'' \phi_j'' dz + \int_0^L N \phi_i' \phi_j' dz.$$

A tip rotational inertia J_h , when included about the appropriate bending axis with coincident mass center and attachment, adds $J_h \phi'_i(L) \phi'_j(L)$ to M_{ij} . An offset mass needs the full tip kinematics and its translational-rotational coupling. These expressions show why the loaded modes need not equal those of a bare clamped-free shaft.

The static shape in the first example, normalized by its tip displacement, is $\phi = (3s^2 - s^3)/2$ with $s = z/L$. It is an admissible single-coordinate approximation, not an exact vibration eigenmode. For uniform μ and zero prestress it gives $M = 33/140 \mu L + m_h$ and $K = 3EI/L^3$. This is a useful consistency check linking the static compliance, distributed kinetic energy and modal stiffness.

Mode normalization is bookkeeping. Replacing ϕ by $a\phi$ and η by η/a multiplies the corresponding single-mode mass and stiffness by a^2 , while preserving displacement, energy and frequency. Reporting an effective mass without the coordinate normalization is incomplete. Add modes until the outputs of interest converge: tip slope and rapid transients may need more resolution than gross displacement. Validate against measurements after numerical convergence; one does not substitute for the other.

21.3 The Extended State Vector

Let $\mathbf{q}$ contain skeletal coordinates and $\boldsymbol{\eta}$ the retained elastic coordinates. One ordering is

$$\mathbf{x} = (\mathbf{q}, \dot{\mathbf{q}}, \boldsymbol{\eta}, \dot{\boldsymbol{\eta}}).$$

Example 21.2. State Dimension Growth

Two joint coordinates plus one bending coordinate give six mechanical states. In the order $(q_1, q_2, \dot{q}_1, \dot{q}_2, \eta, \dot{\eta})$, the fifth derivative is $\dot{x}_5 = x_6$. Its value is a velocity, not x_5 itself. Additional bending directions, torsion, muscle activation or tendon dynamics require additional states. Count coordinates separately from their velocities and actuator states.

With a stationary, uncoupled single mode,

$$m\ddot{\eta} + c\dot{\eta} + k\eta = Q_\eta, \quad \omega_n = \sqrt{k/m}, \quad \zeta = \frac{c}{2\sqrt{km}}.$$

For a displacement coordinate, Q_η is a force. Dividing by m gives an acceleration on the right of $\ddot{\eta} + 2\zeta\omega_n\dot{\eta} + \omega_n^2\eta = Q_\eta/m$. In a coupled model, skeletal accelerations also enter the elastic equations; these accelerations depend on applied input. They cannot generally be hidden in a function of joint position and velocity alone.

A rating of 240–280 cycles per minute converts to 4.00–4.67 Hz, or approximately 25.13–29.32 rad/s. This conversion says nothing about whether that bench frequency equals an in-swing mode. Nor does multiplying a free-vibration frequency by downswing duration establish how many loading/recovery cycles a driven shaft executes. Forcing, initial state, damping and changing configuration all matter.

21.4 How Shaft Flexibility Enters the Drift Field

Collect all mechanical coordinates into $\mathbf{r} = (\mathbf{q}, \boldsymbol{\eta})$. For an ideal torque-input model, write

$$\mathbf{M}(\mathbf{r})\ddot{\mathbf{r}} + \mathbf{h}(\mathbf{r}, \dot{\mathbf{r}}) + \nabla U(\mathbf{r}) + \mathbf{D}\dot{\mathbf{r}} = \mathbf{B}(\mathbf{r})\mathbf{u}.$$

Here $\mathbf{h}$ contains the remaining velocity and load terms, and U includes the retained conservative potentials. The acceleration input map is $\mathbf{M}^{-1}\mathbf{B}$. An elastic coordinate can have a zero row in $\mathbf{B}$ but a nonzero row in this map. Virtual work identifies where forces are applied; solving the coupled inertia problem identifies how every coordinate accelerates.

An exact translation example makes the distinction concrete. Let a base of mass m_b have position x , and a head of mass m_h have position $x + \eta$. Connect them with a spring and damper, and apply an external force u to the base. There is no gravity along this axis. Then

$$T = \frac{1}{2}m_b\dot{x}^2 + \frac{1}{2}m_h(\dot{x} + \dot{\eta})^2, \quad U = \frac{1}{2}k\eta^2, \quad \delta W = u\delta x.$$

The mass matrix and generalized input are

$$\mathbf{M} = \begin{bmatrix} m_b + m_h & m_h \\ m_h & m_h \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{M}^{-1}\mathbf{B} = \begin{bmatrix} 1/m_b \\ -1/m_b \end{bmatrix}.$$

Solving gives

$$\ddot{x} = \frac{u + k\eta + c\dot{\eta}}{m_b}, \quad \ddot{\eta} = -\frac{u}{m_b} - \left(\frac{1}{m_b} + \frac{1}{m_h} \right) (k\eta + c\dot{\eta}).$$

At $m_b = 1$ kg, $m_h = 0.2$ kg, $u = 2$ N and initially zero bend and relative velocity, the base acceleration is 2 m/s^2 and relative acceleration is -2 m/s^2 . The head's absolute acceleration is initially zero. There is no actuator in the shaft, yet the input immediately changes the bending-coordinate acceleration.

In Plain Language 21.3. The Catapult Mechanism

Recoil pushes on both sides of the connection. A head can move forward relative to the grip while the grip slows relative to another trial. Comparing relative recoil alone omits that reaction. In the two-mass example, zero external force conserves total momentum, while damping removes mechanical energy. Internal motion can redistribute energy and momentum among the bodies without creating either.

This distinction also clarifies the affine form $\dot{\mathbf{x}} = f(\mathbf{x}) + G(\mathbf{x})\mathbf{u}$. Elastic restoring forces contribute to the drift under the declared input convention. Flexibility changes the state space, and inertia coupling can give elastic accelerations direct dependence on the same input. If muscle excitation is the input, activation and force dynamics must be included before making statements about immediacy or control bandwidth. An ideal torque channel and a physiological excitation channel are different models.

21.5 Elastic and Damping Terms

For a fixed linear modal stiffness $k > 0$ and viscous damping $c \geq 0$,

$$Q_{\text{elastic}} = -k\eta, \quad Q_{\text{damp}} = -c\dot{\eta}, \quad U = \frac{1}{2}k\eta^2.$$

Restoring force opposes signed displacement; damping force opposes relative velocity. Their powers are $-\dot{U}$ and $-c\dot{\eta}^2$, respectively. A shaft can therefore absorb mechanical work while loading, return some while unloading, and dissipate some throughout motion. A

single viscous coefficient is a chosen approximation to measured dissipation, not a universal material constant independent of frequency or assembly.

Key Principle 21.1. Elastic and Damping Forces

For the complete two-mass example,

$$\frac{d}{dt}(T + U) = u\dot{x} - c\dot{\eta}^2.$$

For multiple elastic coordinates, passive viscous dissipation requires the symmetric part of the damping matrix to be positive semidefinite. Check the full power balance before interpreting a plot of individual energy components. Nonzero elastic energy is storage at that instant; release over an interval is a decrease in storage, and head kinetic-energy gain also depends on the other bodies and external work.

At a fixed deflection, increasing k increases U . At a fixed force in static equilibrium, $\eta = F/k$ and $U = F^2/(2k)$, so increasing k decreases U . These are different experiments. With stiffnesses 80 and 160 N/m, a fixed deflection of 0.02 m stores 0.016 and 0.032 J. A fixed force of 1.6 N instead gives deflections 0.02 and 0.01 m, storing 0.016 and 0.008 J. In a swing, force and deformation evolve together; neither static comparison alone predicts the result of changing shafts.

21.6 Shaft Loading During the Downswing: Centrifugal Stiffening

In an inertial frame, centripetal acceleration is produced by real applied forces. A rotating-frame description introduces inertial terms often described using centrifugal loading. Those descriptions must agree when transformed consistently. Gravity remains approximately constant near the ground; changes in its projection onto a shaft-fixed direction arise from rotation, not an increase in gravitational acceleration during the downswing.

Tensile prestress can increase tangent bending stiffness through U_N . It does not remove transverse forcing or the moment of an offset head mass. In the deflection model discussed earlier, radial loading and head geometry affected lead and toe-down deflection near impact; the model did not reduce to a spring freely snapping straight [MacKenzie and Sprigings, 2010]. The magnitude of prestress effects must be compared with bending stiffness and the other inertial terms over a declared trajectory.

Example 21.3. Centrifugal Stiffening in a Driver Swing

If a reduced model prescribes $k(t)$ externally, its storage is $U = \frac{1}{2}k(t)\eta^2$. Therefore

$$\dot{U} = k\eta\dot{\eta} + \frac{1}{2}\dot{k}\eta^2.$$

The second term is parameter power. Increasing stiffness while holding a nonzero bend requires energy from the mechanism changing that stiffness. For $\eta = 0.02$ m and $\dot{k} = 10$ N/(ms), this contribution is 0.002 W. A full rotating model accounts

for the corresponding exchange through its coupled coordinates. Inserting a speed-dependent stiffness into a constant-parameter energy balance omits that exchange.

21.7 The Catapult Effect: Energy Release Near Impact

For an isolated, undamped oscillator, speed is maximal at its equilibrium position. A driven, rotating golf club with moving supports and an offset head is a different system. Peak head speed need not occur when the shaft is straight, and the sign of bend at impact need not be lag. An equilibrium position under current loading can itself be deflected.

To connect deformation to observable motion, let $\mathbf{r}_g$ and R_g be the grip position and rotation, and let $\mathbf{a}(\boldsymbol{\eta})$ locate a selected head point in the grip frame. Its inertial position and velocity are

$$\mathbf{r}_h = \mathbf{r}_g + R_g \mathbf{a}, \quad \mathbf{v}_h = \mathbf{v}_g + R_g (\boldsymbol{\omega}_g \times \mathbf{a} + J_\eta \dot{\boldsymbol{\eta}}),$$

where $\boldsymbol{\omega}_g$ is expressed in the grip frame and $J_\eta = \partial \mathbf{a} / \partial \boldsymbol{\eta}$. Include tip slope, twist and the offset from shaft attachment to the chosen head point when constructing $\mathbf{a}$. A clubface center, head center of mass and impact point are different points. The moving-frame rotation term remains even when relative deflection stops changing.

The norm $\|\mathbf{v}_h\|$ is not generally the sum of the norms of these terms. A projected deformation contribution is a kinematic decomposition within one trajectory. A speed gain from changing equipment is a difference between two complete trajectories and may include different grip motion, timing and impact states.

In Plain Language 21.4. Why Shaft Flex Profiles Matter

In MacKenzie and Sprigings' optimized model, a sizable relative kick velocity coexisted with a much smaller difference between flexible- and rigid-club speed; the grip motion also differed. The tested flexible shafts produced very similar maximum speeds despite different deflections and presentation [MacKenzie and Sprigings, 2009b]. This supports measuring the coupled response. It does not establish a universal percentage benefit, a universal best stiffness, or that flexibility never matters.

21.8 Shaft Flex Profiles and Forward ZTCF Trajectories

A forward zero-torque counterfactual (ZTCF) begins at a declared full state and sets a specified active torque channel to zero from that instant. The shaft retains its initial deformation and deformation velocity, passive forces remain, and the grip evolves under the coupled equations. A measured or prescribed grip trajectory supplies continuing external influence and is not that full-system counterfactual.

In the two-mass example with $u = 0$, the center of mass travels at constant velocity and the internal coordinate obeys

$$m_{\text{red}} \ddot{\eta} + c \dot{\eta} + k \eta = 0, \quad m_{\text{red}} = \frac{m_b m_h}{m_b + m_h}.$$

This reduced mass differs from the head mass used when the base is externally fixed. With $m_b = 1 \text{ kg}$, $m_h = 0.2 \text{ kg}$ and $k = 80 \text{ N/m}$, the undamped internal frequency is 21.91 rad/s ; a fixed-base model gives 20 rad/s . Boundary freedom changes dynamics even though the spring and head are unchanged.

Example 21.4. Comparing Shaft Profiles

Specify whether a comparison holds joint torques, muscle excitations, measured grip motion, initial mechanical energy or task objective fixed. Holding the same bend across different stiffnesses changes initial elastic energy. Holding the same force changes initial bend. Reoptimizing each swing changes the controller. Each comparison answers a legitimate but different question. Report its initial state, input channel, constraints, event time and output before attributing an improvement to the shaft.

Changing stiffness alone can leave G unchanged at a matched state in a particular fixed-mass model, while altering its later trajectory and therefore task sensitivity. Changing shaft mass or geometry can also change the mass matrix and input map directly. Neither possibility licenses an inference about control authority from the size of the zero-input motion alone.

21.9 Shaft Fitting Through the Lens of Drift

Shaft fitting is a coupled equipment-and-movement comparison. Relevant outcomes include clubhead velocity vector, face presentation, strike location, ball launch, dispersion and the golfer's ability to reproduce the task. A deterministic model can predict a trajectory or local sensitivity. Shot dispersion requires a distribution of inputs, states or parameters, together with an observation and impact model; oscillation in a single trajectory is not dispersion.

MacKenzie and Boucher studied 33 golfers using two shaft-stiffness conditions with matched club inertial properties. Greater relative kick in the more flexible condition accompanied lower grip angular velocity, while the group clubhead-speed comparison was not statistically significant. Individual results varied. This was not an equivalence test, and individual significance in repeated trials is not proof of a stable fitting response across future sessions. The study also distinguished bend-induced loft from total delivered loft [MacKenzie and Boucher, 2017].

Key Principle 21.2. Shaft Fitting as Coupled System Design

Define the desired outcome and an improvement threshold before comparing shafts. Record the assembly and measured properties, counterbalance trial order, allow familiarization, retain repeated shots and quantify uncertainty. To distinguish passive reaction from altered motor behavior, grip/head kinematics alone may be insufficient; additional force or physiological measurements and a validated model are needed. A simulator's reoptimized solution is a hypothesis about achievable behavior, not evidence that a golfer has learned it.

This connects shaft dynamics to the muscle and motor-learning chapters. Muscle force depends on activation, length, shortening velocity and tendon state. Shaft reaction changes

skeletal loading and can change those states even with the same neural command. The golfer may also adapt commands through practice and sensation. Separating these pathways requires an explicit intervention; a changed shot outcome alone cannot identify which pathway caused it.

21.10 The Shaft as an Energy Buffer

For a declared golfer-club mechanical system, energy accounting over an interval has the form

$$\Delta(T + U_g + U_{\text{shaft}} + U_{\text{tissue}}) = W_{\text{active}} + W_{\text{external}} - E_{\text{diss}}.$$

Choose the system boundary before classifying grip work as internal or external. If the golfer and club are both inside, their grip interaction is internal and must not be counted again as an external energy source. If only the club is inside, grip work crosses the boundary. Avoid double-counting active work and the release of elastic energy previously stored by that work.

An isometric fiber contraction is not itself a reservoir of potential energy: zero fiber shortening implies zero fiber mechanical power in the idealized description, although metabolic energy is consumed and series tissues may change length. Muscle, tendon and shaft need separate storage and power terms. During recoil, head kinetic energy can rise while another segment slows, stored energy falls, external work continues and dissipation remains positive.

In Plain Language 21.5. Why Swing Speed Alone Cannot Select a Shaft

A slower swing does not by itself establish greater elastic benefit. Available work, loading direction, head offset, grip response, shaft properties, timing and the chosen performance outcome are intertwined. Static energy comparisons reverse when the controlled quantity changes. Treat a swing-speed fitting rule as an empirical rule to test under stated conditions; it does not follow from the spring-energy equation or from centrifugal stiffening alone.

21.11 Worked Example: A Prescribed Base and a Flexible Head

21.11.1 Rigid Shaft Case

Use a one-dimensional horizontal translation model to isolate base excitation. The axis is a declared transverse teaching direction, not simultaneously a vertical coordinate and a rotating swing plane. A massless spring-damper connects a prescribed base $y_b(t)$ to a point head of mass m . The rigid comparison imposes $y_h = y_b$ and requires whatever actuator force maintains that constraint. It is not a free golfer-club simulation.

Choose the smooth observed interval

$$y_b(t) = A \sin(\Omega t), \quad A = 0.02 \text{ m}, \quad \Omega = \frac{\pi}{0.2 \text{ s}}, \quad 0 \leq t \leq 0.4 \text{ s}.$$

The base starts and ends with nonzero velocity $A\Omega$. We do not abruptly stop it at either endpoint. At $t = 0.1 \text{ s}$ its velocity is zero and acceleration is maximally negative; negative acceleration already occurs throughout $0 < t < 0.2 \text{ s}$. Calling 0.1 s the onset of deceleration would misdescribe the function.

21.11.2 Flexible Shaft Case

Define the relative displacement $\eta = y_h - y_b$. Newton's law on the head is

$$m(\ddot{y}_b + \ddot{\eta}) = -k\eta - c\dot{\eta}, \quad \ddot{\eta} + \frac{c}{m}\dot{\eta} + \frac{k}{m}\eta = -\ddot{y}_b.$$

Set $\eta(0) = \dot{\eta}(0) = 0$. The head initially shares the base velocity, so the initial absolute kinetic energy is nonzero. Use $m = 0.2 \text{ kg}$, $k = 80 \text{ N/m}$ and $c = 0.64 \text{ N s/m}$, giving $\omega_n = 20 \text{ rad/s}$ and $\zeta = 0.08$. These are manufactured teaching parameters, not fitted golfer measurements. By comparison, combining $k = 7000 \text{ N/m}$ with the same mass would give $\omega_n = 187.08 \text{ rad/s}$, not 15 rad/s .

21.11.3 Numerical Example

The forcing is $A\Omega^2 \sin(\Omega t)$. Introduce $s_\Omega = \sin(\Omega t)$ and $c_\Omega = \cos(\Omega t)$. A reproducible independent solution uses the state $\mathbf{z} = (\eta, \dot{\eta}, s_\Omega, c_\Omega)$:

$$\dot{\mathbf{z}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -400 & -3.2 & A\Omega^2 & 0 \\ 0 & 0 & 0 & \Omega \\ 0 & 0 & -\Omega & 0 \end{bmatrix} \mathbf{z}, \quad \mathbf{z}(0) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Thus $\mathbf{z}(t) = \exp(Ht)\mathbf{z}(0)$ for the displayed matrix H . Its rows have the units appropriate to the mixed state components; the matrix is not dimensionless. The figure uses a direct ODE integration, checked against this matrix exponential. Displacement, velocity and energy must agree under tighter tolerances, rather than merely producing a plausible-looking curve.

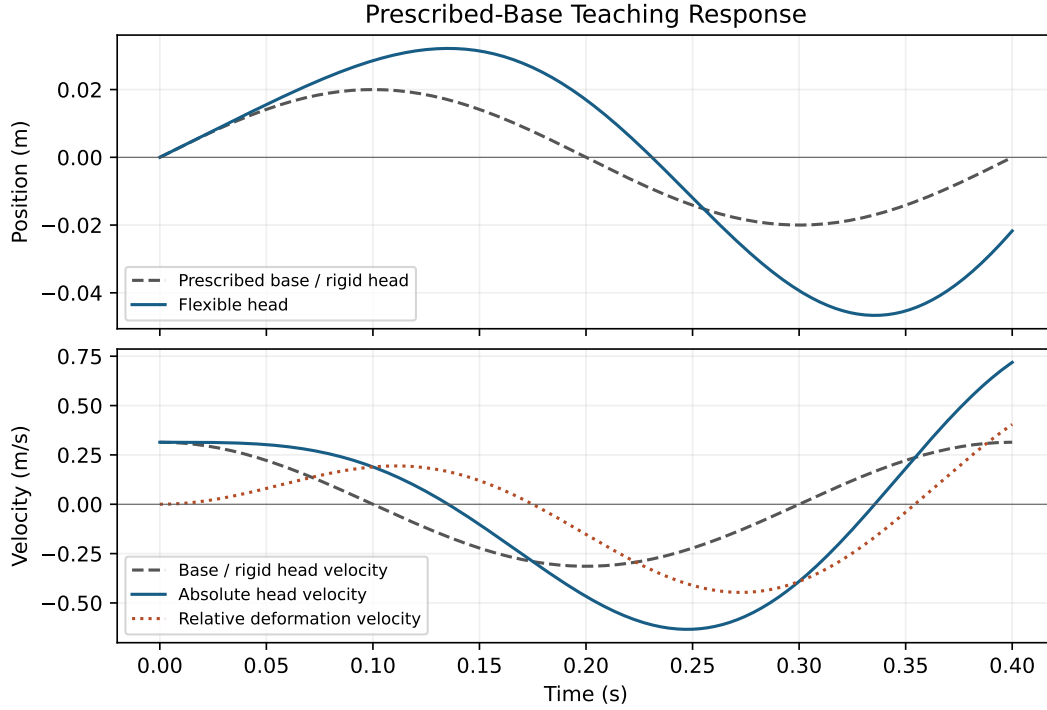


Figure 21.2: Position and Velocity Under the Declared Prescribed Base. The Flexible Response Includes Continuing Boundary Work; No Ball-Impact Event Is Modeled.

Two energy descriptions answer different questions. Relative oscillator energy satisfies

$$E_{\text{rel}} = \frac{1}{2}m\dot{\eta}^2 + \frac{1}{2}k\eta^2, \quad \dot{E}_{\text{rel}} = -m\ddot{y}_b\dot{\eta} - c\dot{\eta}^2.$$

The effective forcing term is work in the base-relative description. For absolute head kinetic energy plus spring storage,

$$E_{\text{abs}} = \frac{1}{2}m(\dot{y}_b + \dot{\eta})^2 + \frac{1}{2}k\eta^2,$$

$$\dot{E}_{\text{abs}} = P_b - c\dot{\eta}^2, \quad P_b = -(k\eta + c\dot{\eta})\dot{y}_b.$$

Here P_b is power supplied through the prescribed base to the spring-damper/head system. It can have either sign. Integrating this balance, including initial head kinetic energy, checks the forcing sign and reveals the source of motion. The example shows a response to continuing boundary input. It provides neither a ZTCF of the full golfer nor a percentage speed benefit at ball impact, which is absent from this model.

21.12 Passive Flexibility and Ongoing Control

Drift vs. Control 21.1. Shaft Flexibility: Passive Forces and Coupled Inputs

The golfer does not command a bending coordinate independently. The golfer can influence its evolution through forces and motion at the grip, subject to muscle dynamics, constraints and finite time. Restoring and damping forces are passive, while their coordinate accelerations can still respond to active inputs through $\mathbf{M}^{-1}\mathbf{B}$. These facts are compatible.

Evaluate control over a specified remaining time, task output and admissible input set. A large passive contribution is not proof that an active perturbation has no effect, and a nonzero input-map row is not proof that an arbitrarily large correction is feasible. With excitation inputs, propagation through activation, muscle force, skeletal motion and shaft deformation can limit the attainable change before impact. This is where the beam, muscle and control descriptions must meet within one model.

Key Principle 21.3. Key Takeaways

The shaft stores and exchanges energy; it creates no mechanical energy. Its deformation changes position and orientation, and its reaction changes the grip response. Specify boundary conditions, normalized coordinates, inertia and input channels before discussing drift. Distinguish relative recoil from equipment-induced speed differences, a prescribed boundary from zero input, and deterministic sensitivity from measured dispersion. Model checks establish internal consistency; measured comparisons establish how well the model describes golfers.

Exercises 21.1. Chapter Exercises

1. An unforced underdamped mode has $\omega_n = 12 \text{ rad/s}$, $\zeta = 0.05$, $\eta(0) = 0.03 \text{ m}$ and $\dot{\eta}(0) = 0$. Find the time for the exponential envelope to shrink by a factor e^{-1} , and distinguish it from the first zero of displacement.
2. Compare spring energy for $k = 80$ and 160 N/m under a fixed deflection of 0.02 m and under a fixed force of 1.6 N . Explain why a dynamic fitting conclusion requires more information.
3. If total shaft mass changes but its spatial distribution, stiffness profile and grip/head boundaries are unknown, can the new modal frequency be computed? State the conditional result when only a known modal mass increases.
4. A repeated-shaft trial reports a change in distance and dispersion. Identify measurements and controls needed to distinguish relative shaft recoil, altered grip motion and impact-location effects. Does a nonsignificant group speed test demonstrate equivalence?
5. In the prescribed-base example, translate the whole forcing and its matched

initial state earlier by Δt . Contrast this with changing only a local part of the acceleration history while keeping the initial state fixed. Must the latter shift peak deformation by Δt ?

6. For externally prescribed $k(t)$, derive the additional energy term. Explain why tensile stiffening cannot establish that the shaft becomes straight at impact.

21.13 Worked Answers

1. With $\omega_d = \omega_n \sqrt{1 - \zeta^2}$, the displacement is $\eta_0 e^{-\zeta \omega_n t} [\cos(\omega_d t) + (\zeta \omega_n / \omega_d) \sin(\omega_d t)]$. The envelope reduction time is $1/(\zeta \omega_n) = 1.6667$ s. Writing $\alpha = \arctan(\zeta \omega_n / \omega_d)$, the first zero is $(\pi/2 + \alpha)/\omega_d \approx 0.13524$ s. The sine term enforces release from rest; the exponential alone is not the displacement.
2. Fixed deflection gives 0.016 and 0.032 J; fixed force gives 0.016 and 0.008 J. During a swing, changes in loading, timing, grip response and initial state must also be solved or measured.
3. No. For a fixed single-coordinate stiffness and all other model assumptions unchanged, $\omega_n = \sqrt{k/m}$ falls as modal mass increases. Total shaft mass does not determine that modal mass without its spatial weighting and normalization.
4. Record grip and head kinematics, delivered face and velocity, strike location and ball launch, with repeated, counterbalanced trials and uncertainty estimates. Force measurements and model interventions can test competing mechanisms. Nonsignificance alone does not establish equivalence; that requires a prespecified practical margin and an appropriate uncertainty analysis.
5. Translating the complete input and matched state history of a time-invariant system translates its response. A local change adds a new forced response and can change both amplitude and phase. Its peak must be recomputed; it need not move by the same amount as the edited feature.
6. Differentiation gives $\dot{U} = k\eta\dot{\eta} + \frac{1}{2}\dot{k}\eta^2$. The second term accounts for parameter power. Transverse loads, head offsets, initial state and ongoing boundary motion can sustain bend even when tangent stiffness increases.

Chapter 22

Fascia and Connective Tissue: Material Mechanics and Evidence

In Plain Language 22.1. What Connective Tissue Contributes

Connective tissue helps determine how a golfer's muscles load the skeleton, how neighboring structures move against one another, and how deformation is sensed. Those roles connect anatomy to both mechanics and control. They do not establish a universal percentage of swing power supplied by fascia, nor do they make the tissue mechanically unimportant.

A stretched elastic structure can pull and can accelerate another structure while it recoils. The energy came from earlier loading. A tissue can also dissipate energy, change its properties over time, and contain cells and nerve endings. The useful questions concern the amount, direction and timing of these effects in a specified task.

This chapter replaces an earlier overly categorical account. In particular, stiffness is not the opposite of elasticity, force transmission is not necessarily force loss, and a single hypothetical tissue volume cannot establish whole-body golf energy storage. We will derive those distinctions and identify the measurements needed to test their importance.

22.1 What Fascia Actually Is

The word fascia needs an anatomical scope. A named fascial sheet, such as fascia lata around the thigh, is a specific structure. The broader term fascial system can include several interconnected connective-tissue compartments. A model that counts tendon, aponeurosis, intramuscular connective tissue and deep fascia separately must define non-overlapping volumes before adding their energy. Otherwise, a broad label can count the same material twice.

Collagen fibers contribute tensile strength and recoverable deformation. Elastin, hydrated matrix, cells and interstitial fluid also contribute, in proportions that depend on the tissue and site. Epimysium surrounds a muscle, perimysium organizes fascicles, and endomysium surrounds individual muscle fibers. Deep fascial sheets and these intramuscular coverings are related structures, but they are not interchangeable names for a single interface. Nor are tendon and bone successive layers underneath every patch of skin.

Definition 22.1. Anatomy Before Parameters

A reproducible tissue model specifies its anatomical region, constituent boundaries, fiber directions, thickness, reference configuration and attachments. Its parameters also require a loading protocol and material law. The name fascia alone does not supply these quantities.

For example, Bonaldi and colleagues tested composite fascia-lata specimens from four fresh-frozen human donors. Their measured response depended strongly on loading direction relative to collagen layers. Their preparation and relatively slow test protocols must accompany any quoted modulus; these are not universal constants for a golfer's shoulder or torso. [Bonaldi et al., 2023]

Elastic deformation stores recoverable mechanical energy within the model's applicable range. A large elastic modulus means that more stress is required for a specified small strain. A material can therefore be both stiff and elastic. Viscosity introduces rate-dependent dissipation; plasticity and damage introduce other forms of irreversible change. Biological remodeling changes tissue over still other time scales. These mechanisms should not all be described as a tissue becoming looser or releasing.

22.2 Four Claims That Need Separate Tests

22.2.1 Can Fascial Structures Supply Swing Power?

Myth vs. Reality 22.1. Force, Power and Metabolic Generation

A passive stretched spring exerts force. If its tensile force is F and its length is ℓ , the power supplied to the spring by its attachments is $F\dot{\ell}$. During unloading, this quantity can be negative: the spring is returning power to the surrounding system. Calling the spring passive does not make its instantaneous force or output power zero.

For a passive element with recoverable energy S , let P_{in} be the total mechanical power entering its boundary and $\mathcal{D} \geq 0$ its dissipation rate. Then

$$\dot{S} = P_{\text{in}} - \mathcal{D}.$$

Over a cycle returning every internal state to its initial value, net mechanical input is nonnegative. During part of that cycle, however, output power can be substantial. Neither a measured force nor a peak-power percentage establishes how much new metabolic energy the tissue supplied.

The categorical biological statement that fascial tissue cannot contract is also incorrect. Schleip and colleagues examined human tissue histology and pharmacologically stimulated rat fascial strips containing myofibroblasts. Their study discusses responses on minutes-to-hours scales and explicitly does not support a substantial direct contractile contribution on the seconds scale of locomotion. These findings justify distinguishing cellular contractility from rapid skeletal-muscle actuation; they do not measure golf-swing power. [Schleip et al., 2019]

Thus a claim such as half the swing power comes from fascia needs a defined tissue

boundary, power signal, time interval and denominator. It cannot be established or refuted by saying that fascia only transmits force. The defensible energy question is how earlier loading, later recoil and dissipation alter club delivery.

22.2.2 Can Fascia Store and Return Elastic Energy?

Myth vs. Reality 22.2. Energy Requires a Loading History

Collagenous structures can store elastic energy. Their contribution cannot be inferred from the elastin fraction, nor ranked from modulus alone. For a linear material with modulus E , strain ϵ and volume V , the stored energy is $S = VE\epsilon^2/2$. This formula assumes homogeneous uniaxial strain, a fixed reference volume and linear elasticity. At equal strain and volume, increasing E increases stored energy. At equal stress σ , however, $\epsilon = \sigma/E$ and $S/V = \sigma^2/(2E)$: increasing E decreases stored energy. These are different experiments. Neither comparison determines which tissue has more energy during an actual swing, because the anatomical volumes, loading and deformations also differ.

For a nonlinear elastic law, energy density is the area under the elastic stress-strain curve. For a viscoelastic cycle, some input work is dissipated and the returned work depends on the path and rate. A whole-body fascial energy budget would need the relevant fields across all counted tissue regions and a consistent muscle-tendon accounting. No such calibrated golf budget is established here.

22.2.3 Do Anatomical Continuities Establish Force Chains?

Myth vs. Reality 22.3. Continuity, Direction and Coupling

Connective-tissue continuity is a legitimate anatomical observation. It does not by itself establish an exclusive path carrying a specified fraction of force or energy. Conversely, a network need not spread loading equally in every direction: aligned fibers and attachments can produce strong anisotropy. Dismissing every proposed continuity as anatomical fiction replaces one unsupported generalization with another.

In a study of 37 participants, Carvalhais and colleagues altered latissimus-dorsi tension while measuring passive hip torque and resting position, with muscle activity monitored. Passive and active tensioning produced different changes in the measured hip variables. This supports a bounded interpretation of mechanical interaction across connected regions. It does not establish an exclusive transmission route, a universal map of meridians, or a golf-swing energy fraction. [Carvalhais et al., 2013]

A proposed load path becomes useful when it predicts an observable change under a declared perturbation and performs better than alternatives. An anatomical drawing, a remote range-of-motion change and a validated dynamic power budget are different levels of evidence. Preserve that distinction when using a coaching image of a sling or line.

22.2.4 What Would Establish a Training Effect?

Myth vs. Reality 22.4. Outcomes and Mechanisms

A change in range of motion, soreness or performance does not identify the tissue or mechanism that changed. Nor does failure to isolate one mechanism prove that connective tissue cannot adapt. The intervention, comparison group, measurement and time scale determine the inference.

Konrad and Tilp's six-week stretching study reported increased range without detected changes in the measured muscle-tendon structural parameters. A later acute study by Konrad and colleagues reported changes in muscle and muscle-tendon stiffness after two stretching protocols. Taken together, these examples caution against explaining every improvement either as permanent tissue deformation or as a purely neurological effect. Neither study isolated fascia or tested a golf outcome. [Konrad and Tilp, 2014, Konrad et al., 2017]

To claim that a fascia-targeted intervention improves a swing through a particular tissue change, measure that change and the golf outcome, compare a credible alternative intervention, and address other adaptations. A short-term response to rolling or stretching cannot by itself establish scar removal, faster healing, or improved club-face control. These are hypotheses requiring their own evidence, not consequences of the word myofascial.

22.3 Mechanical Models and Their Measurable Predictions

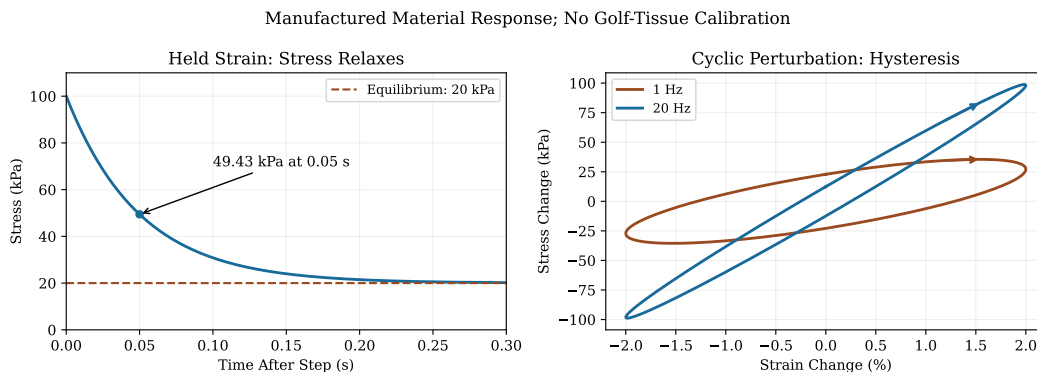


Figure 22.1: Manufactured Standard-Linear-Solid Response With Equilibrium Modulus 1 MPa, Branch Modulus 4 MPa and Relaxation Time 0.05 s. Left: An Ideal 2 Percent Strain Step. Right: Steady Cycles of 2 Percent Strain Amplitude About a Tensile Bias; Arrows Show Time Direction. These Curves Are Model Examples, Not Golf-Tissue Measurements.

22.3.1 Force Transmission and Virtual Power

Consider tissue paths with lengths $\ell(q)$, length Jacobian $J_\ell = \partial\ell/\partial q$, and tensile force vector F . Positive F_i resists increasing path length. The generalized force exerted by those paths on the modeled body is

$$\tau_f = -J_\ell^T F, \quad \dot{\ell} = J_\ell \dot{q}, \quad \tau_f^T \dot{q} = -F^T \dot{\ell}.$$

These equations follow from virtual work. The same force can appear at several joint coordinates through its moment arms. That redistribution does not imply energy has disappeared. Dissipation requires a material mechanism and a nonnegative loss term in a consistent power balance.

Example 22.1. A Coupled Pair of Motions

For a manufactured small-motion spring with extension proportional to $q_1 - q_2$, let $S = k(q_1 - q_2)^2/2$. Its stiffness matrix is

$$K = k \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

It resists relative motion, with stiffness eigenvalue $2k$, while common motion $(1, 1)^T$ has eigenvalue zero. The spring changes directional response; it does not reduce every possible motion by the same percentage. This example is not an anatomical rotator-cuff model. It explains why the sign and direction of a coupling matter before making such a model.

Mechanical tension in a neighboring tissue also does not establish that its muscle has received an additional neural command. Activation, passive tension and any reflex response must be represented separately. The muscle-force (Chapter 14) and passive-control (Chapter 32) chapters develop those distinctions.

22.3.2 Elastic and Viscoelastic Elements

Whether a tissue is represented in series, in parallel, or through a distributed network depends on its attachments and on the model boundary. An aponeurosis along a muscle's force path can contribute series compliance. Intramuscular and neighboring connective tissues can provide additional parallel or lateral paths. Treating all surrounding fascia as a compulsory series spring is an anatomical assumption, not a general consequence of connective-tissue mechanics.

For a uniform, linear, uniaxial strip with reference length L , area A , extension δ and strain $\epsilon = \delta/L$,

$$F = EA\epsilon = k\delta, \quad k = \frac{EA}{L}, \quad S = \frac{1}{2}k\delta^2 = \frac{1}{2}E\epsilon^2 AL.$$

Here E has units Pa, k has units N/m and S has units J. Doubling area or halving length changes structural stiffness even if the material modulus is unchanged. A joint stiffness additionally includes the attachment geometry and moment arms.

For a small-strain nonlinear elastic law $\sigma_e(\epsilon)$, define

$$\psi(\epsilon) = \int_0^\epsilon \sigma_e(s) ds, \quad E_{\text{tan}} = \frac{d\sigma_e}{d\epsilon}.$$

The reference stress and energy have been set to zero here. If one writes $\sigma_e = E_{\text{sec}}(\epsilon)\epsilon$, then

$$E_{\text{tan}} = E_{\text{sec}} + \epsilon \frac{dE_{\text{sec}}}{d\epsilon}.$$

The coefficient multiplying strain is a secant modulus; it is not generally the tangent modulus. Likewise, substituting a strain-dependent modulus into $E\epsilon^2/2$ does not generally give the correct stored energy. For $\sigma_e = E_0(\epsilon + b\epsilon^3)$, the integral is $\psi = E_0(\epsilon^2/2 + b\epsilon^4/4)$.

The familiar compliant toe region of a collagenous tensile response can reflect progressive fiber recruitment before a stiffer range. Failure and damage require further modeling. For large deformation, a three-dimensional law must use a consistent stress-strain pair and reference volume, with fiber directions, near-incompressibility and shear as appropriate. The scalar formulas here are controlled illustrations, not a universal constitutive law for all fascia.

A Kelvin-Voigt approximation adds a dashpot in parallel:

$$\sigma = E\epsilon + \eta\dot{\epsilon}, \quad \sigma\dot{\epsilon} = \dot{\psi} + \eta\dot{\epsilon}^2, \quad \psi = \frac{1}{2}E\epsilon^2.$$

The viscosity η has units Pa.s. With $E > 0$ and $\eta \geq 0$, the material stores elastic energy and dissipates energy. During a prescribed constant-rate ramp from zero to ϵ_0 in time T , its final stored energy density is $E\epsilon_0^2/2$, independent of ramp time, while the dissipated energy density is $\eta\epsilon_0^2/T$. Faster loading at the same final strain increases this model's loss; it does not eliminate its stored energy. Under prescribed force rather than prescribed strain, the deformation history is different.

In Plain Language 22.2. Why Tissue Memory Matters

Hold a stretched material at the same length and its force may change with time. Its current length then does not tell you everything needed to predict its current force. The loading history matters.

This does not mean tissue becomes a liquid whenever stretching is slow. It means the chosen model needs enough internal state to represent its measured response. A stiffness measured over minutes cannot simply be inserted into a millisecond loading calculation without checking the rate range. Likewise, a rapid test cannot establish the effects of weeks of training.

Kelvin-Voigt has no gradual stress relaxation once strain is held constant. A standard linear solid can represent one relaxation time. Let z be a viscous strain-like internal state, $E_\infty > 0$ an equilibrium modulus, $E_1 > 0$ a branch modulus and $\tau_r > 0$ a relaxation time:

$$\begin{aligned} \sigma &= E_\infty\epsilon + E_1(\epsilon - z), \\ \dot{z} &= (\epsilon - z)/\tau_r, \\ \psi &= \frac{1}{2}E_\infty\epsilon^2 + \frac{1}{2}E_1(\epsilon - z)^2. \end{aligned}$$

The branch viscosity is $\eta_1 = E_1\tau_r$. Direct differentiation gives the energy balance

$$\sigma\dot{\epsilon} - \dot{\psi} = \frac{E_1}{\tau_r}(\epsilon - z)^2 \geq 0.$$

This is a check on the model's passivity with fixed parameters, not proof of whole-body stability or of a particular tissue's parameter values. For an ideal imposed strain step ϵ_0 from an initially relaxed state,

$$\sigma(t) = \left(E_\infty + E_1 e^{-t/\tau_r}\right) \epsilon_0, \quad t \geq 0.$$

An actual testing machine applies a finite ramp, which must be included when fitting data. The equilibrium spring leaves a nonzero stress at long times. The memory variable can also leave two specimens at the same strain with different stresses after different histories.

As a manufactured example, take $E_\infty = 1$ MPa, $E_1 = 4$ MPa, $\tau_r = 0.05$ s and $\epsilon_0 = 0.02$. Stress begins at 0.100 MPa, is about 0.04943 MPa after one relaxation time, and approaches 0.020 MPa. These numbers illustrate the equations; they are not fitted golf-tissue measurements. Multiple relaxation modes, nonlinear response or fluid transport may be necessary for actual data.

For a steady sinusoidal strain perturbation $\delta\epsilon = \epsilon_a \sin \omega t$ about a tensile bias, the same linear model gives

$$\begin{aligned} \delta\sigma &= \epsilon_a (E' \sin \omega t + E'' \cos \omega t), \\ E' &= E_\infty + \frac{E_1 (\omega \tau_r)^2}{1 + (\omega \tau_r)^2}, \\ E'' &= \frac{E_1 \omega \tau_r}{1 + (\omega \tau_r)^2}. \end{aligned}$$

Here ω is angular frequency in rad/s. The storage modulus E' and loss modulus E'' describe this harmonic experiment. The dissipated energy per cycle per reference volume is $\pi E'' \epsilon_a^2$. With this one-mode model, that loss is greatest at $\omega \tau_r = 1$ and decreases at still higher frequencies; a blanket assertion that higher rate always means greater cyclic loss is incorrect. The figure uses perturbations about a sufficiently tensile baseline, not a claim that slack fascia supports arbitrary compression.

22.3.3 What Material Numbers Establish

The earlier table combined unsupported site-independent values for fascia, passive muscle and other tissues. It also contradicted its own cortical-bone value. A defensible comparison keeps the experiment attached to each number. Bonaldi and colleagues' Table 1 reports, for one donor's composite fascia lata, linear-region moduli of 191.9 MPa in the L1 direction and 6.9 MPa in the T1 direction, with standard deviations 54.5 and 2.0 MPa respectively. L1 and T1 designate specimen orientation relative to the first layer's fibers; the specimens retain the composite layers. The study included four donors aged 54–89 and used a 0.5%/s failure ramp after preconditioning. [Bonaldi et al., 2023]

Table 22.1: One Donor's Composite Fascia Lata: Directional Linear-Region Moduli

Orientation	Mean (MPa)	Standard Deviation (MPa)
L1	191.9	54.5
T1	6.9	2.0

Those values refute a universal claim that deep fascia must have a modulus below 2 MPa. They do not supply a constitutive model for a living golfer, nor a safe strain limit.

A modulus measured in one part of a nonlinear failure curve cannot automatically be used across an entire loading cycle. Donor variability and test uncertainty also matter; a quoted mean is not a guaranteed value for an individual.

Example 22.2. An Energy Estimate With Correct Units

Retain the earlier example's hypothetical area $A = 10 \text{ cm}^2$, path length $L = 10 \text{ cm}$, strain 5% and linear modulus 1 MPa. The conversion is

$$A = 10(10^{-2} \text{ m})^2 = 10^{-3} \text{ m}^2, \quad V = AL = 10^{-4} \text{ m}^3.$$

Consequently,

$$S = \frac{1}{2}(10^6)(0.05)^2(10^{-4}) = 0.125 \text{ J}.$$

A 0.2 kg clubhead traveling at 50 m/s has translational kinetic energy $K = mv^2/2 = 250 \text{ J}$. The ratio is 0.05%, for this hypothetical region and these assumptions. The earlier 1.25 J result used ten times the stated area. Neither result is a measured golf contribution.

Geometry can also be expressed as sheet width, length and thickness. A hypothetical 10 cm by 10 cm by 1 mm sheet has volume 10^{-5} m^3 . With a selected modulus of 50 MPa and strain 2%, a linear calculation gives 0.1 J. These values are a second manufactured example, not a re-interpretation of the first geometry or an extrapolation of a donor measurement.

To estimate returned work at impact, replace uniform strain by measured fields and include unloading history, material loss and the boundary power transferred to other structures. Adding several guessed regions does not create a credible whole-body total. The original claim of a few-percent fascial contribution to golf energy is therefore withdrawn.

22.3.4 Fascia in the Drift and Control Model

A mechanical model can write

$$M(q)\ddot{q} + c(q, \dot{q}) + g(q) = \tau_m(q, \dot{q}, a) + \tau_f(q, \dot{q}, z) + J_c^T \lambda.$$

Here a denotes muscle activation, z tissue memory variables, and $J_c^T \lambda$ declared contact or ideal-constraint forces. Elastic fascial loading belongs in τ_f as well as dissipative loading; it is not represented completely by a scalar drag coefficient. If $S_f(q)$ is an elastic potential, its contribution is $-\nabla_q S_f$. For a path-dependent model, the internal variables need their own evolution equations.

Tissue compliance can affect ranges of motion, contact and load paths. An ideal constraint fixing a tissue path length may be useful in a stated limit, but actual stiffness does not automatically create or remove a holonomic degree of freedom. The decision to resolve a compliant state, eliminate it quasi-statically, or replace it by a constraint changes what the model can predict.

Key Principle 22.1. Declare the Input Before Assigning Drift

Suppose the input is neural excitation u and activation follows $\dot{a} = (u - a)/\tau_a$ with constant $\tau_a > 0$. Then muscle force depends on an activation state, and a change in u does not instantly equal a change in joint torque. A possible augmented state is $x = (q, \dot{q}, a, z)$.

Under this particular activation law and constitutive choices, the augmented equations may be control-affine, $\dot{x} = f(x) + G(x)u$. Muscle activation already present in x contributes to the zero-new-input baseline. That baseline is not necessarily mechanically passive. If activation time constants or force laws depend nonlinearly on excitation, a control-affine form requires additional justification.

If the declared input instead is generalized torque, its input map is different. Tissue coupling may change force transmission, passive response and task sensitivity without being representable solely as a reduction of G . Setting new excitation to zero, setting activation to zero, and deleting elastic tissue forces are distinct counterfactuals. The muscle-force, passive-control and DCR chapters explain why comparing them requires a common state and boundary definition.

Even at a frozen state, coupling has no universal effect on useful capacity. In a manufactured two-input acceleration map $\ddot{q} = Bu$, choose

$$B = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}, \quad |u_i| \leq 1.$$

The maximum magnitude of acceleration in the sum coordinate $q_1 + q_2$ is 3; in the difference coordinate $q_1 - q_2$ it is 1. With $B = I$, both capacities are 2. The coupled map remains full rank, since its determinant is 0.75. This example establishes only that coupling can help one task direction while hindering another. To infer a golfer's capacity, identify the real input bounds, force-length geometry, activation and time available.

22.4 Connecting Tissue Mechanics to the Golf Task

22.4.1 Load Distribution and Local Stress

Distributing a resultant force over a larger effective area can reduce average stress, F/A . It does not guarantee a smaller peak stress everywhere. Attachment geometry, fiber orientation, prestress and load-sharing constraints can move or create concentrations. A model must resolve the quantity it claims to predict: a resultant joint force is not a tissue stress field, and a clubhead acceleration is not the load in a shoulder tendon.

The swing and ball-contact phases also require distinct time resolutions and boundary forces. One cannot transmit an estimated peak clubhead force unchanged through every tissue to the ground. Segment inertia, joint reactions, grip compliance and wave propagation affect that relation. Those effects connect this chapter to the flexible-shaft (Chapter 21) and force-balance (Chapter 4) chapters; a tissue model should be coupled to the same declared whole-body dynamics.

Example 22.3. What a Shoulder Model Would Need

Replacing the earlier claim that healthy fascia automatically prevents rotator-cuff injury requires more than a softer adjective. A mechanical shoulder model would need anatomical attachments, loading histories, activation, contact and material properties. A local stress estimate then requires spatial resolution or a validated reduction. A risk claim requires an additional relationship between those exposures and observed injury, including relevant patient and loading factors.

Reducing average stress in one idealized region does not establish that the maximum stress, accumulated damage or clinical risk decreases. Similarly, a strength improvement or a change in motion does not identify a fascial mechanism. These are separate predictions, with different measurements needed for validation.

22.4.2 Sensing and Position Estimation

Stecco and colleagues found free nerve endings and encapsulated receptors in sampled deep fascia from 20 upper limbs, with regional differences. Their anatomical study proposed a proprioceptive role; it did not quantify how much golf-swing state estimation comes from fascia. [Stecco et al., 2007]

Sensory and mechanical functions are intertwined. Tissue deformation is part of the physical stimulus reaching a receptor. A change in local loading can therefore alter a measurement as well as a force. The resulting signal is processed with other sensory information and predictions from motor commands. Describing it as purely sensory, with no mechanical relationship, misses this connection. Describing it as a second nervous system adds an unsupported interpretation.

In Plain Language 22.3. What the Golfer Can Know

The nervous system does not need a camera view of every joint to estimate its state. Signals from muscles, tendons, skin and other tissues can be combined with prior information and motor predictions. Their usefulness depends on noise, delay and the movement being performed.

A tissue's nerve endings establish an anatomical route for information; they do not establish that a particular drill improves that route or that the signal can correct every error before impact. To connect sensing to performance, specify which error is detected, when it becomes detectable and what change remains physically achievable.

For example, a modeling choice might be a measurement $y = h(q, \dot{q}, a, z) + \nu$ with noise ν , followed by a delayed controller. The same memory variable z that affects mechanical force may affect h . Adding an estimator and controller expands the state again. The closed-loop response then depends on tissue mechanics, sensing, activation and policy together. A scalar tissue stiffness or DCR cannot identify that whole response.

22.4.3 Injury and Recovery: Limits of a Mechanical Account

A tissue model can predict deformation or stress under specified loading. That output alone does not diagnose the source of pain, establish a clinical lesion, or prescribe recovery time. Different anatomical regions, disease processes and treatment contexts cannot be reduced

to a generic poorly vascularized fascia category. The original universal treatment sequence and recovery claims exceeded the evidence presented in this chapter and are removed.

Changes in a measured range can result from several interacting factors: passive mechanical resistance, active muscle behavior, perception and the test protocol. Reduced resistance in a model is not synonymous with healing. A clinical intervention's effect also cannot be assigned to fascia merely because pressure or stretch was applied near it.

Key Principle 22.2. Keep Clinical and Mechanical Endpoints Distinct

Use separate endpoints for tissue structure, mechanical response, symptoms, function and golf performance. A change in one may motivate a hypothesis about another, but does not establish it. Intervention claims require appropriate clinical evidence in the relevant population and time frame. This chapter supplies mechanical reasoning and evidence boundaries; it does not derive a universal injury-prevention or treatment program from a material law.

22.5 An Integrated, Testable Account

The central question is how an interacting golfer-club system produces a useful impact state. Connective tissue contributes to that question through load paths, stored energy, dissipation and sensing. Its role can be modest or consequential for a particular outcome without being an independent metabolic engine. A small energy contribution may still change a sensitive output; a large stored-energy quantity may be delivered in the wrong direction or at the wrong time.

A useful investigation starts with a specified change in club delivery, such as a change in impact position, orientation or speed. It then declares the model boundary and inputs. Tissue lengths, strains, loading history and attachment geometry connect local material laws to joint forces. Muscle activation and sensing determine how the golfer responds. The impact event defines when the output is sampled, which can change the sensitivity relative to a fixed-time calculation.

Before attributing an effect to fascia, compare the full model with a clearly defined alternative. Removing a tissue, changing its modulus, suppressing its viscous term and setting its stored strain to zero are different interventions. Some alter equilibrium or contact, making identical initial configurations mechanically inconsistent unless the balancing forces are also specified. Report that difference rather than interpreting every simulation contrast as a biological intervention.

Parameter fitting and validation are also different steps. Matching a force-relaxation curve checks one loading protocol; it does not establish high-rate unloading, multi-axial response or golfer-level prediction. Use separate loading histories and outcomes to test transfer. Report which parameters can actually be identified from the measurements, and compare competing models when several mechanisms fit equally well. Uncertain mechanisms can still support useful bounded predictions if those predictions survive the uncertainty.

Key Principle 22.3. Key Takeaways

Fascia is not one universal material with one modulus. Specify anatomy, direction, geometry and protocol. Stiffness and elasticity are compatible properties. Passive tissue can exert force and return energy; net energy production and cellular contractility are separate questions.

Energy estimates require correct units, measured deformation and a material law. The numerical examples here are manufactured and do not establish a whole-body golf percentage. Viscous loss and recoverable storage must be computed from the same loading history. Memory needs internal state when current deformation alone does not determine force.

Mechanical coupling can redistribute task capacity without destroying force or causing neural activation. Its usefulness depends on the desired correction, time remaining, contact and sensory feedback. Anatomical, experimental and clinical findings should be carried only as far as their measurements support. The strength of the framework is its ability to make those assumptions explicit and test their consequences.

Exercises 22.1. Chapter Exercises and Worked Answers

Exercise 1. Assess a claim that fascia supplies 50% of swing power. What measurements and definitions are missing?

Answer 1. Specify the counted tissue, the mechanical boundary, time interval, power signal and denominator. A peak output-power fraction differs from integrated work and from net metabolic production. A passive element can return previously stored energy. Test its loading and unloading balance, including dissipation; the percentage is not determined by tissue identity alone.

Exercise 2. For two hypothetical linear materials with moduli 200 MPa and 1 MPa, compare strains at the same applied stress. Then compare energy density under fixed stress and under fixed strain.

Answer 2. At equal stress, $\epsilon = \sigma/E$, so the 1 MPa material has 200 times the strain within the assumed linear range. Its energy density $\sigma^2/(2E)$ is also 200 times greater. At equal strain, the 200 MPa material has 200 times the energy density $E\epsilon^2/2$. These are prescribed material examples, not a validated tendon/fascia ranking or a prediction of actual tissue strains.

Exercise 3. A tissue transmits a nonzero tensile force while its endpoints remain fixed. Is it producing mechanical power? What changes during recoil?

Answer 3. With both endpoints fixed, their mechanical power contributions vanish even if force is large. During shortening, $F\dot{\ell} < 0$ in the sign convention used here, so the tissue can supply power to the surroundings. That power must be accounted for by decreasing stored energy or a declared active source, minus any losses. Force alone does not specify power.

Exercise 4. In a hypothetical shoulder model with increased capsular resistance, why might intact muscles fail to achieve a desired angle? Does that explanation diagnose a frozen shoulder?

Answer 4. Bounded muscle torque may be unable to overcome passive resistance along a feasible path, or may be unable to do so within the available time. The model

must specify its resistance law, activation and initial state. This is a mechanical possibility, not a diagnosis: observed motion restriction does not uniquely identify its cause, and the model does not determine treatment.

Exercise 5. A rolling intervention increases a measured range of motion. What can be concluded about fascia or golf performance?

Answer 5. The reported endpoint changed under that test. Without further evidence, one cannot determine whether tissue mechanics, muscle activity, perception or another factor mediated the change. Measure candidate mechanisms and compare a suitable control. Golf performance requires its own outcome test; a range change alone does not establish improved club delivery.

Exercise 6. Use the original exercise geometry, 10 cm by 10 cm by 5 cm, with linear modulus 1 MPa and homogeneous strain 5%. Compare stored energy with a 0.2 kg clubhead moving at 50 m/s.

Answer 6. The volume is $0.1 \times 0.1 \times 0.05 = 5 \times 10^{-4} \text{ m}^3$. Therefore $S = 0.5(10^6)(0.05)^2(5 \times 10^{-4}) = 0.625 \text{ J}$. Clubhead translational kinetic energy is 250 J, so the ratio is $0.625/250 = 0.0025$, or 0.25%. This geometry differs from the corrected 10 cm² cross-sectional example. Neither is an anatomical measurement or a returned-work estimate.

Exercise 7. Explain why rapid loading does not imply zero elastic energy storage. What can the standard linear solid predict that Kelvin-Voigt cannot?

Answer 7. At the same final prescribed strain, Kelvin-Voigt stores $E\epsilon_0^2/2$ regardless of ramp duration, although faster loading increases dashpot dissipation in the constant-rate example. The standard linear solid includes an internal state and predicts gradual stress relaxation at fixed strain, toward $E_\infty\epsilon_0$. A finite-ramp experiment and its time scale are needed before fitting either model to tissue.

Exercise 8. Does fascia belong in the drift field, control map, or measurement model?

Answer 8. The answer depends on the state, input and reduction. Elastic and viscous forces contribute to mechanical dynamics; memory variables require evolution equations. Tissue geometry can alter how muscle forces map to joint forces. Local deformation can influence sensory measurements. If excitation is the input, activation is generally a separate state. A declared control-affine representation may include all these effects, but its zero-input drift is not automatically passive and its scalar DCR is not a measure of golf success.

Chapter 23

Soft Tissue and Pliable Systems: Beyond the Rigid Body

In Plain Language 23.1. Why Soft Tissue Matters in Golf

A body segment can move approximately as a rigid object while its skin, muscle and other tissues move relative to its bones. That relative motion matters in two different ways: it changes forces and stored energy, and it changes what a surface-mounted sensor reports. A marker can therefore give a repeatable measurement without being a faithful measurement of the underlying bone.

Abdominal pressure adds another connection. Pressure acts on the surrounding structures, while muscles and connective tissues help contain that pressure. The resulting support depends on the whole force balance. A large pressure reading does not, by itself, reveal the load on a spinal joint, the energy reaching the club, or the likelihood of an injury.

This chapter builds small models that make those distinctions calculable. Their purpose is to decide which effects a golf or humanoid model must resolve and what evidence would validate it. The numerical examples are declared teaching systems, not measured anatomical parameters or instructions to change breathing or muscle activation.

23.1 The Rigid Body Assumption: Where It Works and Where It Fails

Rigid-segment modeling assumes that distances between material points assigned to a segment remain fixed. It does not require all joints to be hinges, all muscles to be inactive, or the club shaft to be rigid. A model can combine rigid bones, multiaxis joints, deformable tendons and a flexible club. The choice should follow the question and the available measurements.

Stiffness and strength are different properties. A bone's failure load does not establish how accurately a rigid approximation predicts its deformation under a particular load. Likewise, a rigid segment's center of mass and inertia must include the tissue assigned to it. Adding a wobbling mass to an anthropometric segment without subtracting that same mass from the rigid part counts tissue twice.

A useful rigid approximation reproduces specified outputs over a specified operating range. It may be adequate for a gross pose while inadequate for tissue stress or an impact peak. A small pose error does not guarantee a small force error, especially when forces depend on acceleration or contact deformation. There is no universal 15-degree-of-freedom

model, or universal percentage accuracy, for coaching, inverse dynamics or injury assessment.

Pain and Challis modeled two heel-first drop landings by one man using subject-specific segment estimates and a partly calibrated wobbling-mass model [Pain and Challis, 2006]. Their model and a rigid alternative produced similar gross kinematics but substantially different computed loading. Separate controlled impacts supplied tissue-motion comparisons; the trunk response lacked equivalent independent data. This is evidence that model choice can matter under those impact conditions. It does not establish a correction percentage or tissue-motion amplitude for golf.

The practical decision is therefore output-specific. Predicting where the ball flies requires a sufficiently accurate launch state and flight model; it does not automatically require every internal tissue mode. Predicting how the golfer creates that state, or estimating a particular tissue load, introduces different requirements. Validation must follow that distinction.

23.2 Wobbling Mass: A Coupled Mechanical Model

Begin with two positive masses moving along a horizontal line. Let x_r be the rigid part's position and x_w the effective tissue mass's position. A spring of stiffness $k > 0$ and a damper with coefficient $c \geq 0$ connect them. Coordinates are measured from a configuration in which the spring is unstrained. The applied force u acts on the rigid part. Gravity has no component along this particular line; that is a boundary assumption, not a property of soft tissue.

Define relative displacement $z = x_w - x_r$. The spring and damper exert equal and opposite forces, so the closed equations are

$$\begin{aligned} m_r \ddot{x}_r &= u + kz + c\dot{z}, \\ m_w \ddot{x}_w &= -kz - c\dot{z}. \end{aligned}$$

Adding them gives $(m_r + m_w)\ddot{x}_{\text{COM}} = u$. Internal coupling can redistribute motion but cannot accelerate the complete system's center of mass without an external force. An effective tissue mass represents a selected deformation mode; it is not automatically the mass of every anatomically soft structure in the segment.

The mechanical energy and its rate are

$$E = \frac{1}{2}m_r \dot{x}_r^2 + \frac{1}{2}m_w \dot{x}_w^2 + \frac{1}{2}kz^2, \quad \dot{E} = u\dot{x}_r - c\dot{z}^2.$$

The spring stores energy; the damper removes mechanical energy at a nonnegative rate. Energy entering tissue motion is not necessarily already dissipated. It can return through the coupling. Over a simulation, integrated actuator work must equal the change in mechanical energy plus accumulated dissipation. This balance provides a stronger implementation check than a plausible-looking trajectory.

23.2.1 Prescribed Motion and Free Motion Have Different Modes

If the rigid part follows an externally prescribed trajectory, the tissue equation becomes

$$m_w \ddot{z} + c\dot{z} + kz = -m_w \ddot{x}_r.$$

The base acceleration drives relative motion. Maintaining that prescribed base requires whatever reaction force the model predicts; it is not the same experiment as applying a fixed force to a freely moving rigid part.

For a fixed base, the undamped natural angular frequency and damping ratio are

$$\omega_n = \sqrt{k/m_w}, \quad \zeta = \frac{c}{2\sqrt{km_w}}.$$

Take $m_w = 1$ kg, $k = 5000$ N/m and $c = 200$ N s/m. Then $\omega_n = 70.7107$ rad/s, equivalent to 11.2540 Hz, and $\zeta = 1.4142$. The homogeneous roots are -29.2893 s $^{-1}$ and -170.7107 s $^{-1}$. This system is overdamped. A positive displacement released from rest decays without repeated oscillation. Describing these same parameters as an underdamped system with several cycles would contradict the equation.

If both masses move freely, subtract their acceleration equations. With reduced mass $\mu = m_r m_w / (m_r + m_w)$,

$$\mu \ddot{z} + c \dot{z} + kz = -\frac{m_w}{m_r + m_w} u.$$

The unforced relative mode uses $\sqrt{k/\mu}$, not $\sqrt{k/m_w}$. For $m_r = 1.5$ kg and the preceding tissue parameters, $\mu = 0.6$ kg and the undamped relative frequency is 91.2871 rad/s. Boundary conditions change the mode even when the component properties stay the same.

These examples do not identify a golfer's tissue parameters. A large acceleration also does not specify high-frequency excitation: a sustained acceleration and a short pulse can have the same peak and very different spectra. Oscillation amplitude additionally depends on initial conditions, forcing history, damping and the validity of the linear approximation.

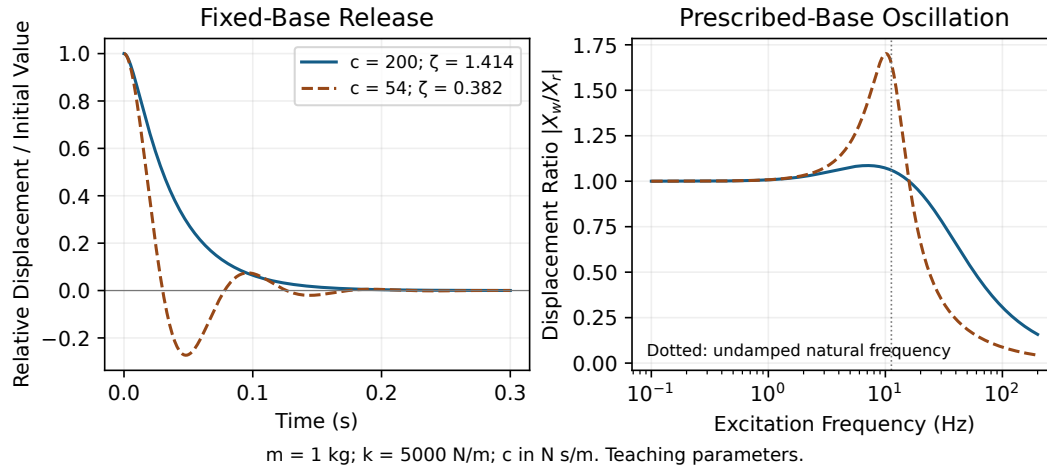


Figure 23.1: Boundary Conditions and Damping Determine Tissue Response. Both Plots Use a Fixed or Prescribed Base, a 1 kg Tissue Mass and 5000 N/m Stiffness; Parameters Illustrate a Model, Not Measured Golf Anatomy.

23.2.2 Effects of Wobbling Mass on Inverse Dynamics

Soft-tissue motion is physical, while its contribution to a bone-motion estimate is a measurement artifact. Those statements are compatible. A surface marker follows its local attachment, not necessarily the bone or the center of mass of an effective tissue mode.

In a local one-dimensional observation model,

$$y(t) = x_r(t) + z_{\text{marker}}(t) + \epsilon(t).$$

The sensor observes their sum. Without additional information, changing the proposed bone trajectory by $a(t)$ and changing marker motion by $-a(t)$ leaves the same observation. A fitted decomposition therefore needs calibration, independent observations or justified constraints. Adding a tissue model does not automatically make the components identifiable.

Differentiation magnifies rapid motion and noise. For a sinusoidal marker displacement of amplitude A and frequency f , acceleration amplitude is $A(2\pi f)^2$. At $A = 0.01$ m and $f = 10$ Hz, it is 39.4784 m/s². The same displacement at another frequency gives a different acceleration; an amplitude alone is insufficient.

Filtering can reduce noise or an identifiable transient, but tissue artifact need not occupy a separate frequency band. Fuller and colleagues compared skin-mounted and bone-pin marker arrays and found overlapping spectral content [Fuller et al., 1997]. Their results oppose a universal rule that low-pass filtering recovers skeletal motion. The task, attachment, smoothing method and reference measurement all matter.

An accelerometer further measures specific force in its sensor frame. Converting it to a point's coordinate acceleration requires orientation, gravity and reference-point conventions. Rotational motion, sensor offset, attachment motion, bias and timing errors can all enter. Neither a marker nor an accelerometer alone directly measures an individual muscle force.

For inverse dynamics, distinguish errors in motion, external forces, inertial parameters and the mechanical model. A net generalized-force estimate remains a combined result of muscles, passive tissues and constraints. Estimating individual tissue loads requires additional constitutive and force-allocation assumptions. The inverse-dynamics chapters (Chapter 20) develop that distinction.

23.2.3 Effects of Wobbling Mass on the Drift Field

A useful frequency-domain calculation exposes what an effective inertia means. Prescribe a small harmonic base displacement $x_r(t) = \text{Re}(X_r e^{i\omega t})$ in the preceding linear model. Substitution into the tissue equation gives

$$H(\omega) = \frac{X_w}{X_r} = \frac{k + i c \omega}{k - m_w \omega^2 + i c \omega}, \quad m_{\text{dyn}}(\omega) = m_r + m_w H(\omega).$$

The required external force amplitude is $U = -\omega^2 m_{\text{dyn}} X_r$. Both the spring and damper transmit base motion, so both belong in the numerator. Omitting its damping term changes the phase, reaction and energy balance.

At low frequency, H approaches one and the dynamic mass approaches $m_r + m_w$. At sufficiently high frequency, H approaches zero and the response approaches that of m_r , within this model's valid range. Around a mode, amplitude and phase depend on damping. A complex dynamic mass is a compact input-output description under harmonic forcing, not a new physical mass; its real part need not behave monotonically.

The mean input power provides an independent check:

$$\bar{P} = \frac{1}{2} \text{Re}[U(i\omega X_r)^*] = \frac{1}{2} c \omega^2 |X_w - X_r|^2.$$

The star denotes complex conjugation. In steady harmonic motion the average stored-energy change is zero, so input power equals damper loss. This identity and direct time integration test the response without relying on the same algebraic rearrangement.

Here ω is the frequency of a small imposed oscillation. It is not simply the golfer's angular speed. A rapid rotation need not excite the same mode as a rapidly varying perturbation, and a measured swing is not a steady sinusoid. The calculation therefore supplies a testable coupling mechanism, not an explanation that elite athletes exploit tissue decoupling.

In a state-space model, adding z and $\dot{z}$ changes the state and the equations. It can also change the input map and observation map. Eliminating those states can introduce memory; a complex harmonic response should not be inserted into a time-domain equation as an ordinary constant inertia. Drift still means the vector field at the declared zero input, with the retained contact, gravity, elastic and activation mechanisms specified.

23.3 Internal Motion in the Torso

The torso contains deformable organs, muscles, connective tissues and fluids, with different attachments and material properties. It is not adequately characterized by assigning a single fraction of its mass to an unrestrained liquid. Total-body blood mass is not all located in the torso cavity, and organ masses cannot be added to an existing whole-torso estimate without checking what that estimate already includes.

For a declared rotating-frame illustration, a point at radius $r = 0.15$ m in a frame rotating steadily at $\omega = 10.5$ rad/s has a centripetal acceleration magnitude $\omega^2 r = 16.5375$ m/s² if it remains fixed in that frame. In rotating-frame force balance the corresponding centrifugal term points outward. This calculation gives an acceleration scale; it does not give an organ displacement or stress.

The complete acceleration transformation is

$$\mathbf{a} = \mathbf{a}_O + R[\ddot{\boldsymbol{\rho}} + 2\boldsymbol{\omega} \times \dot{\boldsymbol{\rho}} + \dot{\boldsymbol{\omega}} \times \boldsymbol{\rho} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{\rho})].$$

Here R maps components from the moving frame to the inertial frame, $\boldsymbol{\rho}$ is position relative to its origin, and the dotted relative quantities and $\boldsymbol{\omega}$ use moving-frame components. Frame translation, angular acceleration and relative motion add terms that a centrifugal-only argument omits. Attachment forces, gravity, contact and material response must balance the resulting dynamics.

Predicting deformation requires a constitutive law and boundary conditions. In a continuum description, local momentum balance is $\rho \mathbf{a} = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{b}$, where ρ is density, $\boldsymbol{\sigma}$ stress and $\mathbf{b}$ body force per unit mass. An approximately incompressible fluid can transmit pressure while having no static shear modulus; surrounding tissues can resist shear and tension. A lumped oscillator, a deformable solid and a fluid model make different assumptions.

23.3.1 Time-Varying Inertia Tensor

A rigid body's inertia about its center of mass is constant in a fixed body frame, while its components rotate in an inertial frame. Deformation can change even the body-frame inertia. Those are separate effects, and neither alone determines the complete angular momentum of a deforming body.

Choose an origin at the complete modeled system's center of mass and a rotating frame. With all quantities below expressed in that frame,

$$\mathbf{H} = I\boldsymbol{\omega} + \mathbf{h}_{\text{rel}}, \quad \boldsymbol{\tau}_{\text{ext}} = I\dot{\boldsymbol{\omega}} + \dot{I}\boldsymbol{\omega} + \dot{\mathbf{h}}_{\text{rel}} + \boldsymbol{\omega} \times (I\boldsymbol{\omega} + \mathbf{h}_{\text{rel}}).$$

The relative angular momentum $\mathbf{h}_{\text{rel}}$ comes from material motion relative to the chosen frame. It generally cannot be inferred from the instantaneous inertia alone. Changing the system boundary or using an accelerating origin away from its center of mass requires the appropriate additional accounting.

A special scalar model with no relative angular momentum gives $\tau = d(I\omega)/dt$. At zero external torque, angular momentum is conserved. If I changes from 3.00 to 3.18 kg m² with initial angular speed 10 rad/s, the final speed is 9.43396 rad/s. The associated rotational energy falls from 150 to 141.5094 J. Reversing the shape change reverses that energy change in an ideal reversible process. It is not automatically heat.

If instead an actuator holds angular speed at 10 rad/s while that inertia rises uniformly over 0.2 s, it must supply a 9 N m torque in this scalar model. Work associated with the shape change must also be accounted for. The term $\dot{I}\omega$ is not a viscous damper merely because it appears next to a velocity.

23.3.2 Practical Implications

Internal motion can affect momentum, energy, sensor observations and local loading together. Its importance must be estimated from measurements or sensitivity analysis for the intended task. The chapter does not assign a universal percentage change to torso inertia or a measured displacement to organs during a golf transition.

Transition deserves investigation because velocities, activation and contact forces evolve there; a claim that visceral motion makes it the most dangerous or most powerful phase needs separate evidence. The spine chapter (Chapter 24) develops segment-level loading models. The damping chapter (Chapter 16) distinguishes elastic storage, local dissipation and effects on the complete coupled system.

23.4 Intra-Abdominal Pressure: Define the Pressure Boundary

Intra-abdominal pressure is pressure within the abdominal compartment. Its measured value depends on the measurement method, reference pressure, location, posture and task. Gastric or bladder measurements can provide useful information under defined protocols, but a pressure reading is not a direct measurement of every boundary traction.

23.4.1 Anatomy of the Pressure-Generating System

The diaphragm separates thoracic and abdominal compartments; the abdominal wall and pelvic floor complete much of the surrounding boundary. Vertebral, fascial and muscular structures provide additional attachments and constraints. These structures deform and exert forces together. Treating the abdomen as a pressure vessel can clarify force balance, but a literal hydraulic cylinder with rigid end plates is only a simplified model.

Contraction and respiratory motion can change pressure, geometry and muscular tension at the same time. Their mechanical effects must be distinguished rather than assigned to pressure alone. The pressure difference across the diaphragm involves thoracic pressure as well as abdominal pressure. Raising an abdominal pressure relative to atmosphere does not specify that difference if thoracic pressure changes too.

23.4.2 IAP as a Structural Support System

For a surface patch S with unit normal $\mathbf{n}$ pointing from the cavity into its wall, the net pressure force and moment on the wall are

$$\mathbf{F}_p = \int_S \Delta p \mathbf{n} dA, \quad \boldsymbol{\tau}_p = \int_S (\mathbf{r} - \mathbf{r}_O) \times (\Delta p \mathbf{n}) dA.$$

The pressure difference Δp must compare the two sides of that wall. Wall tension, muscle forces and attachments remain separate terms. The force direction varies across a curved surface, so multiplying pressure by its total curved area does not usually give the vertical resultant.

For uniform Δp , the vertical force uses signed projected area:

$$F_{p,z} = \Delta p \int_S n_z dA = \Delta p A_{\text{proj}}.$$

A hemispherical cap of radius R has curved area $2\pi R^2$ but vertical projected area πR^2 . Its axial resultant is therefore $\Delta p \pi R^2$, not twice that value. Numerical surface integration verifies the distinction.

Using 1 mmHg $\approx$ 133.3224 Pa, a declared pressure difference of 100 mmHg over projected area 0.04 m^2 gives approximately 533.29 N. Rounding pressure to 13,300 Pa gives 532 N. Neither number is automatically a reduction in spinal compression; it is one patch's resultant under the stated pressure and geometry.

For a complete closed boundary under uniform pressure,

$$\oint_S p \mathbf{n} dA = \mathbf{0}, \quad \oint_S (\mathbf{r} - \mathbf{r}_O) \times (p \mathbf{n}) dA = \mathbf{0}.$$

Opposing regions balance globally while individual regions remain loaded. A nonuniform pressure field can have a resultant associated with fluid weight or acceleration; the corresponding fluid balance must then be included. For the complete body, internal action-reaction pairs do not create a net external supporting force. They can nevertheless redistribute loads between structures.

This is why a diaphragm force cannot simply be subtracted from a joint load. The muscles and connective tissues containing the pressure contribute their own forces and moments. Their effects depend on geometry and activation. A comparison with one supposed typical back-extensor force omits that balance.

23.4.3 IAP and Torso Stiffness

Pressure can contribute to the response of a deformable structure through geometry and wall prestress. The work calculation shows the limit of a pressure-only explanation. For uniform internal pressure relative to a constant external reference, the pressure power delivered to a moving boundary is

$$P_p = \int_S p \mathbf{n} \cdot \mathbf{v} dA = p \dot{V}, \quad Q_{p,j} = p \frac{\partial V}{\partial q_j}.$$

Here V is cavity volume and q_j a boundary-deformation coordinate. If a proposed ideal twist preserves volume, its direct hydrostatic pressure torque is zero. A fluid does not acquire a static shear modulus merely because its pressure is high.

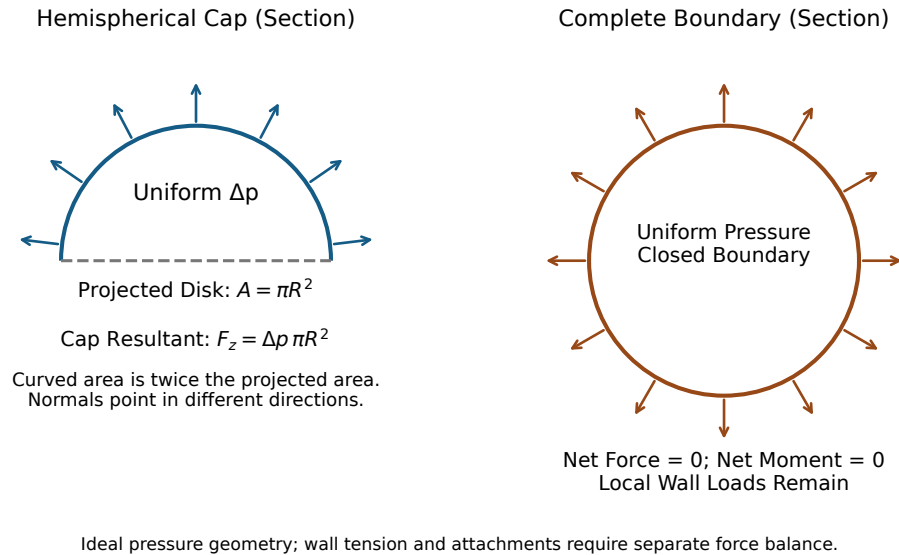


Figure 23.2: Pressure Acts Normal to a Surface. A Hemispherical Cap Has an Axial Resultant Based on Its Projected Disk; a Complete Uniform-Pressure Boundary Has Zero Net Force and Moment Despite Local Loading.

The surrounding wall can still resist that twist through its material stiffness, tension and constrained geometry. Pressure can change those quantities and the resulting tangent response. A model must include them to predict the change. It cannot replace that derivation with a universal stiffness proportional to pressure or assume that damping increases by the same rule.

Hodges and colleagues used phrenic-nerve stimulation to raise pressure in three prone subjects and measured an increase in posteroanterior lumbar indentation stiffness [Hodges et al., 2005]. This supports a mechanical contribution under their controlled conditions. It does not supply a golf-specific torsional stiffness, prove stability of a moving swing, or establish an optimum pressure profile.

23.4.4 Breathing Coordination and the Exhale Cue

Inspiration, expiration, glottal state, abdominal activation and posture interact. Inhalation does not universally reduce IAP, and exhalation does not uniquely determine it. The diaphragm's action and pressure difference depend on what the surrounding structures do.

Shirley and colleagues measured a composite posteroanterior spinal response during respiratory tasks in eight subjects [Shirley et al., 2003]. Both inspiratory and expiratory efforts could increase stiffness. Their analysis distinguished pressure, muscle activity and the measurement's composite nature. This contradicts a simple account in which inhalation necessarily removes support while exhalation guarantees it.

A breathing cue is therefore a behavioral intervention with several possible pathways. It may change timing, attention, muscle activity, pressure or comfort together. To explain its effect, measure the intended intermediate variables and the outcome. A plausible physiological account does not establish increased club speed or protection from a cardiovascular

event.

23.5 Pressure, Support and Golf Performance

The central question is how pressure and muscle action jointly change a specified outcome. Local stiffness, stability, tissue load, club speed and perceived effort are different outcomes. Evidence about one can guide the next experiment but does not settle it.

23.5.1 Spine Stabilization and Load Distribution

Stokes and colleagues' static model included curved abdominal muscle paths, specified pressures and a force-allocation objective [Stokes et al., 2010]. It predicted lower spinal compression when pressure increased under its tested loads. The authors also explained that optimized activation may underestimate real coactivation and compression. The result is conditional on the complete model, not on pressure alone.

An experiment by Nachemson and colleagues measured pressures and muscle activity in four subjects performing five isometric tasks [Nachemson et al., 1986]. Four tasks showed increased rather than decreased compression during Valsalva maneuvers. The two studies ask related questions under different activation and task conditions. Together they motivate explicit force balance and experimental validation rather than a universal unloading percentage.

Stability additionally concerns how perturbations evolve. A larger measured indentation stiffness does not by itself establish stability for a multiaxis moving system with delays, muscle limits and changing contacts. Nor does lower estimated compression prove lower injury risk: the relevant tissue, loading direction, duration, exposure and capacity still matter.

23.5.2 Torso Stiffness and Energy Transmission

An elastic deformation can store recoverable energy. A dissipative process converts mechanical energy into other forms. A compliant connection can therefore transmit and return energy without being a source of loss, while a stiff connection can still dissipate energy or demand substantial muscle work.

For a linear torsional spring with fixed reference,

$$\tau_{\text{spring}} = -K\theta, \quad V = \frac{1}{2}K\theta^2.$$

At fixed deformation, increasing K increases stored energy. At fixed applied torque magnitude T in static equilibrium, however, $\theta = T/K$ and $V = T^2/(2K)$, so increasing K decreases the stored energy. The comparison changes when its boundary condition changes.

Neither result establishes that a stiffer torso produces greater shoulder speed for the same hip torque. The delivered motion also depends on timing, inertia, geometry, muscle activation, damping and constraints. A finite-horizon simulation or experiment should compare the desired output under matched conditions and include the work needed to produce the changed mechanical response.

23.5.3 The X-Factor Stretch Mechanism

A measured thorax-pelvis separation depends on the segment definitions and angle convention. It is not automatically the strain of a single spring or the sum of lumbar joint rotations. A three-dimensional motion can change that reported angle through several coupled rotations.

To calculate elastic storage, identify a tissue or an effective constitutive element, its reference configuration and its force-deformation relation. For a nonlinear one-dimensional elastic element, stored energy is the integral of its restoring-force magnitude over deformation along the defined loading path. Hysteresis or active force production requires additional accounting.

If stiffness changes while the element is deformed,

$$\frac{d}{dt} \left(\frac{1}{2} K(t) \theta^2 \right) = K \theta \dot{\theta} + \frac{1}{2} \dot{K} \theta^2.$$

The last term is not free energy. Raising K from 100 to 200 N m/rad at a clamped 0.1 rad deformation adds 0.5 J to this ideal storage element. The source of that energy and any losses must be modeled. Pressure or muscle activation can participate in such a change, but naming the mechanism does not remove its work cost.

23.5.4 Breathing Coordination as a Performance Cue

A useful test starts with a reproducible outcome and a tolerable intervention that the participant can perform consistently. Record delivery, ball outcome and the relevant timing or pressure measurements over comparable trials. Separate an immediate performance change from retention, adaptation and changes in reported effort.

Within-person comparisons should address order, fatigue, familiarization and strike variability. Generalization requires additional golfers and sessions. If a cue changes pressure and activation together, the result identifies the combined intervention unless the design can distinguish their contributions. A single better shot cannot establish the proposed internal mechanism.

23.6 Pressure and Health Outcomes

Mechanical models are useful for formulating loading hypotheses. They do not establish a safe pressure threshold, a disease mechanism or a rehabilitation prescription without relevant clinical evidence. The chapter's teaching examples consequently do not prescribe breath holding, maximal bracing or a core-training protocol.

23.6.1 IAP and Cardiovascular Pressures

Abdominal, thoracic, arterial and intracranial pressures are different quantities, with different reference boundaries and time courses. Blood flow depends on pressure gradients and vascular properties; a pressure reading in one compartment cannot simply be assigned to another. Changes during a strain and its release need not have the same cardiovascular effect.

Haykowsky and colleagues reported three cases of subarachnoid hemorrhage associated with weight training and discussed a possible relation to straining [Haykowsky et al., 1996].

A case report can identify a concern but cannot estimate a golfer's event probability or show that a particular exhalation cue prevents it. The physiological and clinical question requires its own measurements and study design.

23.6.2 Pelvic-Floor and Hernia Claims

Pressure loads pelvic-floor and abdominal-wall structures, but loading is not identical to injury. Tissue properties, geometry, activation, exposure and individual history affect response. A mechanical force calculation does not show that repeated golf swings cause pelvic-floor weakness, hernia or incontinence.

Measurement also matters. Bø and Sherburn's review concerns methods for assessing pelvic-floor function and strength [Bø and Sherburn, 2005]; those are distinct outcomes, measured with different instruments. It does not establish a long-term golfer injury rate or an optimum swing-pressure schedule. A study claiming prevention must measure the relevant health outcome, not substitute a pressure peak or a model's preferred activation pattern.

23.6.3 Testing Pressure Strategies

Specify what the proposed strategy should improve and how that effect will be measured. Pressure control may compete with mobility, comfort, fatigue or task performance, and a policy can shift load between structures. A numerical optimum is only an optimum for the chosen objective, constraints and model.

Clinical testing requires an appropriate population, exposure accounting, comparison and follow-up. Mechanical verification should precede that step, but it cannot replace it. The synthesis chapter (Chapter 33) distinguishes a plausible mechanism, an identified intervention effect and evidence of improved health outcomes.

23.7 Modifying the Equations of Motion for Pliable Systems

Start from a coordinate and energy definition. For independent generalized coordinates $\mathbf{q}$ and a declared input $\mathbf{u}$, a mechanical model can take the form

$$M(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{c}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}(\mathbf{q}) + \mathbf{f}_{\text{passive}} = B(\mathbf{q})\mathbf{u} + J(\mathbf{q})^T \boldsymbol{\lambda} + \mathbf{Q}_{\text{ext}}.$$

The generalized mass matrix is not merely a list of segment inertia tensors. The vector $\mathbf{c}$ contains the velocity-dependent terms consistent with those coordinates. If $\mathbf{u}$ contains muscle forces, B can contain moment-arm mappings; if it contains generalized torques, the corresponding map has a different meaning. Contact reactions, external loads and passive forces must be assigned once to the intended system boundary.

23.7.1 Wobbling Mass Extension

For the earlier two-mass example, choose coordinates (x_r, z) , with $x_w = x_r + z$. Substitution into the kinetic energy gives

$$M = \begin{bmatrix} m_r + m_w & m_w \\ m_w & m_w \end{bmatrix}.$$

Its determinant is $m_r m_w > 0$, and its off-diagonal terms express inertial coupling. The equations are

$$M \begin{bmatrix} \ddot{x}_r \\ \ddot{z} \end{bmatrix} + \begin{bmatrix} 0 \\ kz + c\dot{z} \end{bmatrix} = \begin{bmatrix} u \\ 0 \end{bmatrix}.$$

The zero second input does not mean that the relative mode is uncontrollable; force on the rigid part acts through the coupled dynamics. It also does not mean that living tissue is passive in general. An active contractile element would require an explicit force law and activation state.

If these same coordinates instead point vertically upward, gravitational potential is $g[(m_r + m_w)x_r + m_w z]$. The generalized gravity vector is therefore $(g(m_r + m_w), gm_w)^T$. Its soft-coordinate component is not zero. Coordinate transformations determine which equations receive gravity and other forces.

23.7.2 Time-Varying Inertia

For a Lagrangian $L(\mathbf{q}, \dot{\mathbf{q}}, t)$, derive

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = Q_i.$$

If $L = \frac{1}{2} \dot{\mathbf{q}}^T M(\mathbf{q}, t) \dot{\mathbf{q}} - V(\mathbf{q}, t)$, this gives

$$\sum_j M_{ij} \ddot{q}_j + \sum_{j,k} \Gamma_{ijk} \dot{q}_j \dot{q}_k + \sum_j \frac{\partial M_{ij}}{\partial t} \dot{q}_j + \frac{\partial V}{\partial q_i} = Q_i,$$

where $\Gamma_{ijk} = \frac{1}{2}(\partial_k M_{ij} + \partial_j M_{ik} - \partial_i M_{jk})$. The explicit partial time derivative is separate from the coordinate-dependent terms. Adding the full total $\dot{M}\dot{\mathbf{q}}$ to an already consistent Coriolis vector generally counts part of the inertia variation twice.

For a nondimensional example, $L = \frac{1}{2}(1 + q^2)\dot{q}^2$ gives $(1 + q^2)\ddot{q} + q\dot{q}^2 = u$. At $q = 1$, $\dot{q} = 2$ and $u = 0$, acceleration is -2 . Appending an extra $\dot{M}\dot{q}$ incorrectly gives -6 and violates conservation of this system's kinetic energy.

When shape coordinates evolve dynamically, including them in the augmented state is often clearer than prescribing an unexplained $M(t)$. If their motion is prescribed, the actuation and work maintaining that prescription remain part of the model boundary. The general angular-momentum relation above is another way to expose omitted relative-motion terms.

23.7.3 IAP Effects

Pressure can enter generalized forces through boundary tractions and can influence wall geometry, prestress and activation-dependent material response. It is not necessarily an independent control input. A pressure state requires a closure relation involving cavity volume, wall motion and the chosen physiological approximation.

A fitted local torsional impedance may summarize small perturbations around one posture and activation condition. Its coefficients require identification, uncertainty estimates and a stated frequency range. Labeling coefficients $k_{\text{IAP}}(P)$ and $d_{\text{IAP}}(P)$ does not establish their signs, linearity, causal interpretation or portability to a golf swing.

23.7.4 Model Error Requires Validation

More detail can improve representation while increasing uncertainty or making parameters harder to identify. A model with many soft-tissue states can fit surface motion without correctly estimating internal forces. A simpler model can also be adequate for one output and inadequate for another.

Use a sequence of checks: conservation and units, numerical convergence, parameter sensitivity, independent measurements, and performance on held-out tasks or participants. Validation should specify the output, reference method, error measure, sampling and uncertainty. A universal error percentage attached to a model class conceals those choices.

23.8 When to Use Which Model

For gross kinematics, begin with segment and joint definitions that match the observations and assess attachment error. For inverse dynamics, add synchronized external forces, consistent inertial properties and sensitivity to filtering and tissue motion. For local loading, specify the tissues, constitutive laws, geometry and force-allocation assumptions. For impact vibration, resolve the modes and contacts relevant to the measured bandwidth.

For a pressure hypothesis, compare measured pressure and activation with the complete modeled boundary forces; test whether the predicted intermediate response occurs. For a clinical claim, add the corresponding exposure and health outcomes. For a ball-flight question, first determine whether uncertainty in the launch state already dominates the proposed internal-model refinement.

These choices can be connected without forcing every investigation into one maximum-detail model. The strongest connection is a shared record of coordinates, system boundaries, measurements, uncertainty and the outputs each model has actually reproduced.

23.8.1 Looking Ahead

Soft tissues link dynamics, estimation and load distribution. The spine chapter develops a more detailed anatomical model, while the interdisciplinary synthesis explains how to connect its predictions to control, measurement and outcomes. Neither a rigid approximation nor a soft-tissue extension earns general authority from complexity alone.

23.9 Chapter Exercises

1. For a tissue mass of 1 kg attached to a fixed base by $k = 5000 \text{ N/m}$ and $c = 200 \text{ N s/m}$, calculate ω_n , frequency in hertz, damping ratio and characteristic roots. Does release from a positive displacement at zero velocity ring? How does the unforced relative natural frequency change if a 1.5 kg base is free to move?
2. Derive the closed two-mass momentum and energy balances. Then use (x_r, z) coordinates with a vertically upward axis to derive the mass matrix and gravity vector. Explain why zero direct input to z does not mean zero response to force applied at x_r .
3. Derive $H(\omega)$ for prescribed harmonic base motion and verify that its mean input power equals damper loss. For a skin displacement of 0.01 m at 10 Hz, calculate acceleration amplitude. Explain why filtering alone need not identify bone motion.

4. A hemispherical cap has radius 0.1 m and a uniform pressure difference of 100 mmHg. Calculate its axial pressure resultant using 133.3224 Pa/mmHg. Explain why total curved area gives the wrong answer and why a closed surface under the same uniform pressure has no net force or torque.
5. In the declared zero-external-torque scalar model, inertia increases from 3.00 to 3.18 kg m² from an initial 10 rad/s. Calculate final speed and rotational-energy change. Contrast this with constant-speed motion over a 0.2 s inertia change, and explain why neither calculation proves viscous dissipation.
6. Compare a torsional spring with stiffness 100 and 200 N m/rad under a fixed 0.1 rad deformation and under a fixed 10 N m torque. Calculate the energy in each case. What additional energy term appears when stiffness changes while deformed?
7. Design an evidence chain for the claim that a breathing intervention reduces a specified spinal load and improves a golfer's outcome. Distinguish pressure, activation, load estimation, performance and clinical outcomes; identify a measurement ambiguity and an alternative explanation that the experiment must address.

23.10 Worked Answers

1. The fixed-base values are 70.7107 rad/s, 11.2540 Hz and $\zeta = 1.4142$. Roots are -29.2893 and -170.7107 s^{-1} ; the stated release decays without ringing. A free base gives reduced mass 0.6 kg and relative undamped frequency 91.2871 rad/s. This comparison changes the boundary condition, not the material.
2. Adding the two force equations gives external force equal to total momentum rate. Multiplying each by its velocity and adding gives actuator power equal to mechanical-energy rate plus $c\dot{z}^2$. The transformed mass matrix is the one derived above, and the gravity vector is $(g(m_r + m_w), gm_w)^T$. Off-diagonal inertia couples the base force to the relative motion.
3. Substitution of $e^{i\omega t}$ motion gives the stated numerator $k + i c \omega$ and denominator $k - m_w \omega^2 + i c \omega$. Averaging force times velocity over a cycle gives the nonnegative loss $\frac{1}{2} c \omega^2 |X_w - X_r|^2$. The marker acceleration amplitude is 39.4784 m/s². Bone and marker-relative motions can share a frequency and sum to the same observation, so removing high frequencies cannot uniquely separate them.
4. Projected area is $\pi(0.1)^2 \text{ m}^2$, giving approximately 418.84 N. Curved area is twice as large, but its normals do not all point axially. On a complete uniform-pressure boundary, opposite contributions cancel in the vector force and moment integrals. Local loading remains even though the resultant is zero.
5. Angular momentum is 30 kg m²/s, final speed is 9.43396 rad/s and rotational energy changes by -8.49057 J . Holding speed constant during the specified change instead requires 9 N m in the scalar angular equation. Internal shape work and actuator work complete the energy budget. Reversing an ideal shape change can restore the rotational energy; a viscous-loss explanation would require an actual dissipative mechanism.

6. At fixed angle the energies are 0.5 and 1.0 J. At fixed torque they are 0.5 and 0.25 J. The stiffness-change contribution to storage rate is $\frac{1}{2}\dot{K}\theta^2$; clamping the angle while increasing stiffness in this example adds 0.5 J. Identify its source before attributing later motion to passive recoil.
7. Calibrate the pressure reference and synchronize pressure, activation, motion and external load measurements. Validate the load estimate against independent information and vary uncertain force allocations. Compare matched intervention and control trials, including order, fatigue and strike variability. A change in pressure can accompany altered muscle coactivation, so reduced compression cannot be inferred from pressure alone. Better ball performance is another outcome, and a prevention claim additionally requires appropriate prospective health outcomes and exposure accounting.

Part VII

The Musculoskeletal Model

Chapter 24

Modeling the Spine: Motion, Load, and Evidence

In Plain Language 24.1. One Swing, Several Different Questions

A golfer can reproduce a similar shoulder turn with different distributions of motion, muscle activity and spinal loading. A video may resolve the large-scale motion well while leaving those internal differences uncertain. The purpose of a spine model is to make the connection explicit: which parts move, which forces produce that motion, how those forces are shared, and what evidence tests the calculation.

The spine participates in a coupled system that includes the pelvis, rib cage, abdominal wall, head, limbs and club. It transmits forces and moments, deforms, stores and dissipates mechanical energy, and receives active muscle forces. None of those roles alone establishes an efficient swing or predicts pain. This chapter develops the mechanics needed to investigate them together, from measured segment orientations to multibody dynamics and tissue-level questions.

24.1 Why the Spine Matters for Golf

24.1.1 The Mechanical Link

The pelvis and thorax interact through several anatomical load paths: vertebral structures, muscles spanning multiple levels, fascia, and the pressurized abdominal wall. A model that represents their interaction by one joint wrench aggregates those paths. It does not show that every joule arriving at the club passed through one disc, or that the legs generated all the energy later present in the upper body.

For a chosen body or subsystem, linear momentum changes according to its external forces and angular momentum according to its external moments about a fixed inertial point or the subsystem center of mass. An arbitrary accelerating reference point requires the corresponding transport terms. Internal muscle and joint forces redistribute momentum within the whole golfer–club system. They do not generate its total angular momentum without a compensating exchange. Ground contact can supply an external moment even when a stationary contact point does no mechanical work. Muscle work can occur throughout the body. Thus a sequence of segment-speed peaks is a useful kinematic observation, but it is not by itself a measurement of energy transferred along an anatomical chain.

At an interface point O , the instantaneous power delivered to a rigid segment is

$$P_O = \mathbf{F} \cdot \mathbf{v}_O + \mathbf{M}_O \cdot \boldsymbol{\omega}. \quad (24.1)$$

Force, moment, velocity and angular velocity must refer to the same frame and compatible points. Across a compliant connection, the two endpoint velocities can differ; the signed sum of powers delivered into the connection contributes to stored energy, dissipation and any active work inside it. Deformation is therefore not synonymous with lost power.

24.1.2 What the X-Factor Measures

An X-factor is a declared measure of pelvis–upper-trunk separation. A difference between projected shoulder and hip lines is not generally the same as a three-dimensional relative axial angle. A shoulder-line marker set can also move relative to the rib cage through scapular motion and soft-tissue artifact. Report the segment definitions, projections, rotation sequence and sign convention before interpreting its magnitude.

If R_{WP} and R_{WT} map pelvis and thorax coordinates into a common world frame, the relative orientation is

$$R_{PT} = R_{WP}^T R_{WT}. \quad (24.2)$$

This describes the thorax relative to the pelvis, without yet assigning motion to individual vertebrae. Cervical motion changes the head relative to the thorax; it cannot be added as a source of this pelvis–thorax relative rotation. Similarly, a global change in thorax orientation can include pelvic rotation. Subtracting loosely defined rotation magnitudes does not separate these contributions.

Increasing separation may change muscle lengths, tissue deformation, moment arms and feasible future motion. Whether it improves club delivery, changes spinal load or aggravates symptoms requires additional information. There is no universal target angle implied by the definition.

24.1.3 Pain and Injury Statistics Need Denominators

Low-back problems are a substantial concern in golf, but the proportion of injuries affecting the back is different from the proportion of golfers with pain. A retrospective survey of 1,634 Australian amateur golfers reported that 17.6% had sustained at least one golf-related injury in the preceding year; the lower back accounted for 25% of reported injuries [McHardy et al., 2007b]. A separate one-year prospective survey of 588 amateurs reported 15.8 injuries per 100 golfers, with the lower back comprising 18.3% of injury sites [McHardy et al., 2007a]. These results do not establish that half of all golfers have chronic spinal injury.

A history of pain, an imaging finding, a modeled load and a diagnosed injury are different observations. A person may change a swing because of pain, so an association between movement and pain does not by itself establish which came first. Mechanical models help formulate testable explanations; prospective observation and appropriate clinical outcomes are needed to evaluate injury-related predictions.

24.2 Anatomy and the Model Boundary

24.2.1 Vertebrae, Articulations, and Discs

The usual description has seven cervical, twelve thoracic and five lumbar vertebrae above the sacrum. The sacrum ordinarily comprises five fused elements; the coccygeal count and degree of fusion vary. The familiar developmental count of about 33 is not a count of 33

independently moving adult joints. The atlas has no vertebral body, and upper-cervical articulations differ from typical disc-bearing levels [Betts et al., 2022].

In typical anatomy there are 23 intervertebral discs, from C2–C3 through L5–S1. There is no disc between the skull and atlas or between atlas and axis. A model must list its actual connections: L1 through L5 supplies four internal lumbar intervertebral levels; including L5–S1 gives five, and adding T12–L1 gives six. Counts depend on the endpoints. An articulated chain with a fixed base and n modeled three-rotation joints has $3n$ configuration coordinates; using compliant six-coordinate connections gives a different model. A floating base adds its own coordinates and constraints.

24.2.2 Structures Carry Different Loads

A typical vertebra has a body, pedicles, laminae and processes. Intervertebral discs join neighboring bodies; paired facet articulations and their capsules provide posterior contact and restraint. Endplates, discs, facets and ligaments share loads according to geometry, deformation and contact state. Their contributions cannot be assigned from the net joint moment alone. The rib cage also alters thoracic mechanics; a model without it requires a correspondingly limited interpretation.

24.2.3 The Intervertebral Disc

The nucleus pulposus, annulus fibrosus and endplates form a heterogeneous, hydrated structure. The annulus has alternating collagen-fiber families rather than a universal single fiber angle. Microscopy documents spatial gradients in lamellar organization [Cassidy et al., 1989]. The angle must be stated relative to its reference plane or axis; an angle measured from a transverse plane is not the same angle measured from the spine's long axis.

A water-rich nucleus is not mechanically identical to a volume of pure water. Experiments show time-dependent shear behavior, including different responses in stress relaxation and dynamic loading [Iatridis et al., 1997]. Whole-disc response additionally depends on the surrounding matrix, annular restraint, fluid transport, swelling and endplate conditions. An elastic spring can approximate a specified operating range; it does not reproduce all those mechanisms. A viscoelastic or poroelastic model needs internal states or a history law if loading duration and recovery matter.

24.2.4 Ligaments and Contact

Ligaments provide passive tensile restraint, while facet contact is unilateral: a surface can push when engaged but cannot pull another surface through a gap. Different ligaments have different attachments, slack lengths and prestrain. It is inaccurate to assume that every ligament is slack in every neutral posture. The anterior longitudinal ligament resists extension; posterior structures contribute to restraint in flexion, with load sharing depending on posture and level [Betts et al., 2022].

The activation of a muscle is a different mechanism. A balanced pair of active muscles can exert little net joint moment while still producing substantial compression and stiffness. A small net moment does not mean the connection is passive or unloaded.

24.2.5 Curvature, Posture, and Hip Flexion

Lordosis and kyphosis describe anatomical sagittal curvatures. They should not be confused with the differential-geometric curvature of a configuration-space metric. Changes in posture affect lever arms, muscle lengths and contact, but a visible spinal curve does not directly determine tissue stress.

Forward inclination at address can combine hip flexion, pelvic tilt and regional spinal motion. Hip flexion is relative motion between pelvis and femur; lumbar flexion is relative motion within a specified spinal region. They are not numerically interchangeable. Report these separately rather than converting a global trunk inclination into an assumed lumbar deformation. Neither a drawing of a neutral posture nor a universal flexion cutoff supplies a clinical safety boundary.

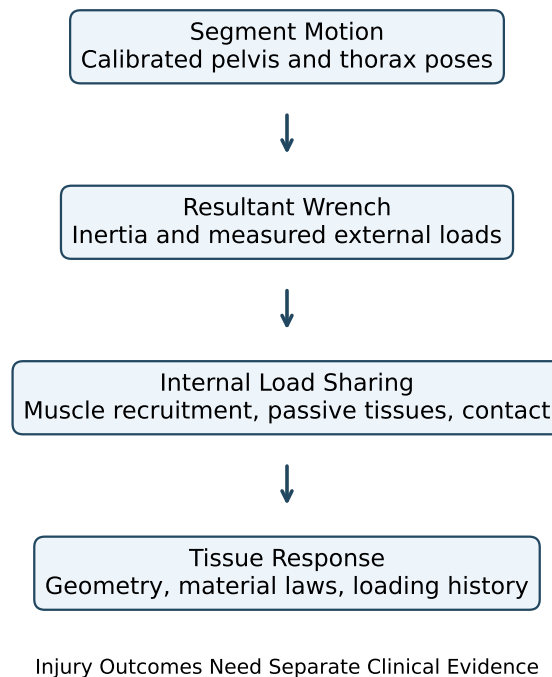


Figure 24.1: From Motion to Tissue Response: Each Stage Requires Its Own Evidence. A Single Measured Shoulder Turn Does Not Identify Every Internal Load.

24.3 Motion Segments and Measured Mobility

24.3.1 Three Rotations and Three Translations

A typical functional spinal unit contains two adjacent vertebrae and their connecting disc, ligaments and facet articulations. Their relative rigid-body pose has three rotational and

three translational components. Those six components are not six unrestrained, independently actuated motions: stiffness, contact, loading and neighboring anatomy govern their response.

For this chapter use right-handed axes x anterior, y left and z superior in a declared local reference posture. Flexion–extension is associated with rotation about a mediolateral axis, lateral bending with an anteroposterior axis, and axial rotation with a superior–inferior axis. A motion occurring in a sagittal plane rotates about an axis normal to that plane; naming the plane as the rotation axis creates a convention error. Once the segment tilts, its local superior axis need not coincide with world vertical.

Anterior–posterior translation is along the local anterior direction, not along the spinal long axis. Compression and distraction concern the local axial component. Small translations can matter for contact, disc strain and load sharing even when they have little effect on a coarse reconstruction of shoulder turn. Omitting them is a task-dependent approximation, not a consequence of their being unimportant mechanically.

24.3.2 Range of Motion Is a Test Result

A range-of-motion value requires a level or region, reference posture, direction, applied load, preload and measurement method. One-sided excursion from a reference differs from total excursion between opposite endpoints. Passive cadaveric loading, voluntary in-vivo movement and a particular golf swing are different experiments.

For example, Wilke and colleagues tested 68 isolated thoracic functional units using pure moments of $\pm 7.5 \text{ Nm}$, with short rib remnants and no axial preload. Reported mean one-sided axial rotations included 5.9° at T6–T7 and 3.3° at T11–T12 [Wilke et al., 2017]. These observations expose the error in assigning $20\text{--}30^\circ$ to every thoracic level. They are not individual golfer limits, and the preparation does not represent an intact loaded rib cage.

For coaxial rotations about the same fixed axis, signed angles add. For general spatial rotations, regional orientation is a product,

$$R_{0n} = R_{01}R_{12} \cdots R_{n-1,n}. \quad (24.3)$$

Adding independently measured maximum excursions may combine incompatible postures and loading conditions. Even if a scalar sum is a useful approximation for a particular test, its assumptions must be stated. A model result outside an expected range is a prompt to examine calibration, conventions and evidence, not an automatic diagnosis or a proof of anatomical impossibility.

24.4 Coordinate Coupling and Physical Coupling

24.4.1 A Tilted Axis Changes Components

Let R map local vector components into the world frame. Angular velocity components satisfy $\omega_b = R^T \omega_W$. For a segment tilted by ϕ about y , take $R = R_y(\phi)$. A world-vertical angular velocity $\omega_W = \Omega e_z$ then has local components

$$\omega_b = \Omega(-\sin \phi, 0, \cos \phi)^T. \quad (24.4)$$

At $\phi = 30^\circ$ and $\Omega = 4 \text{ rad/s}$ these are $(-2, 0, 2\sqrt{3}) \text{ rad/s}$. The local anteroposterior component appears because the local frame is tilted. This identity does not require a ligament, an intervertebral coupling law or an additional physical rotation imposed by anatomy. It is one vector described in two frames.

Actual anatomical coupling is a different question: for a declared applied wrench and constraints, which relative motions occur? A local linear compliance relation might be $\delta \boldsymbol{\eta} = \mathbf{C} \delta \mathbf{m}$. Off-diagonal entries describe coupled response in those coordinates. A single projected angle cannot identify that matrix. Coupling can also arise through the whole-body inertia, other joint loads and contact changes.

24.4.2 Euler Products Must Match the Stated Sequence

Use active right-handed matrices and column vectors. As a coordinate chart, define

$$\mathbf{R} = \mathbf{R}_z(\alpha) \mathbf{R}_y(\beta) \mathbf{R}_x(\gamma). \quad (24.5)$$

This is the product for successive rotations about fixed x , fixed y and fixed z axes, applied in that order. Equivalently it can describe moving-axis rotations in the reversed axis order. It is not the product for successively rotating about moving x , moving y , then moving z axes. Here β coincides with positive forward flexion only in the corresponding reference configuration; the chart labels are not three universal anatomical hinges.

Writing $c_a = \cos \alpha$, $s_a = \sin \alpha$, and similarly for β, γ , multiplication gives

$$\mathbf{R} = \begin{bmatrix} c_a c_b & c_a s_b s_g - s_a c_g & c_a s_b c_g + s_a s_g \\ s_a c_b & s_a s_b s_g + c_a c_g & s_a s_b c_g - c_a s_g \\ -s_b & c_b s_g & c_b c_g \end{bmatrix}. \quad (24.6)$$

All three angles are represented, although not every entry depends on every angle. Differentiating the product and using $\dot{\boldsymbol{\omega}}_b = \mathbf{R}^T \dot{\mathbf{R}}$ gives

$$\boldsymbol{\omega}_b = \begin{bmatrix} 1 & 0 & -\sin \beta \\ 0 & \cos \gamma & \sin \gamma \cos \beta \\ 0 & -\sin \gamma & \cos \gamma \cos \beta \end{bmatrix} \begin{bmatrix} \dot{\gamma} \\ \dot{\beta} \\ \dot{\alpha} \end{bmatrix} = \mathbf{E} \dot{\mathbf{q}}_e. \quad (24.7)$$

Thus Euler-angle rates are not generally angular-velocity components. Since $\det \mathbf{E} = \cos \beta$, this chart becomes singular at $\beta = \pm \pi/2$. The orientation and physical angular velocity can remain regular there. A different chart, rotation matrix or unit quaternion can represent the motion; constraints and double covering must still be handled correctly.

24.4.3 Moments and Generalized Torques Transform Together

Power invariance supplies the conjugate generalized load:

$$\mathbf{m}_b^T \boldsymbol{\omega}_b = (\mathbf{E}^T \mathbf{m}_b)^T \dot{\mathbf{q}}_e, \quad \mathbf{Q}_e = \mathbf{E}^T \mathbf{m}_b. \quad (24.8)$$

Using a physical moment component as an Euler generalized torque without this mapping changes the work. A consistent coordinate transformation does not make inverse dynamics physically ambiguous; it changes the components used to describe the same result. Muscle forces remain underdetermined for a separate reason: multiple muscles and passive structures can produce the same resultant.

If a local stiffness law is expressed in Euler coordinates, differentiating $E^T \mathbf{m}_b$ introduces derivatives of E as well as the tissue law. A configuration-dependent coordinate matrix can therefore create apparent coupling in the reported tangent stiffness. Comparing stiffness numbers requires the operating load, pose, coordinate definitions and whether the derivative is taken at fixed activation or with feedback operating.

24.4.4 What Motion Capture Can Identify

An isolated marker provides position, not a complete segment orientation. A noncollinear marker cluster plus calibration can estimate a rigid segment frame, with uncertainty from skin motion and model fit. Pelvis and thorax frames yield R_{PT} ; recovering every intermediate vertebral pose requires additional measurements or assumptions. Many products of intervertebral rotations give the same endpoint orientation.

There is no single Euler sequence that reveals the historical order in which the anatomy moved. A declared convention makes measurements comparable within its domain. It does not remove missing observations. For a golf investigation, preserve the measured poses, document the decomposition and test conclusions against reasonable alternative calibrations. Distinguish a change of representation from a change in the physical model.

24.5 Choose a Model That Can Answer the Question

24.5.1 From Segment Motion to Tissue Deformation

A rigid thorax connected to a separate pelvis can represent an aggregate trunk rotation. A single rigid body containing both cannot: their relative pose is fixed by construction. Neither model identifies the motion of individual lumbar levels. An aggregate joint is useful for whole-body momentum and club-delivery questions when its geometry and inertial properties adequately represent the task. Its joint torque is a resultant, not the force in a disc.

A model with separate lumbar and thoracic regions allows their contributions to vary and can include bending, twist and translation. Each additional freedom needs an inertia, a constraint or force law, and information to identify its state. More segments do not automatically supply those data. If only pelvis and thorax motion are observed, adding intermediate joints may create multiple equally plausible distributions of motion and load.

A vertebral chain exposes level-specific joint reactions and muscle moment arms. Its joint count follows the included interfaces: L1 through L5 supplies four internal interfaces, adding the sacrum supplies L5–S1, and adding T12 supplies T12–L1. The chosen fixed base, floating base, rotational joints and translational constraints determine the number of generalized coordinates. An anatomical count alone does not establish the model's degrees of freedom.

A finite element model resolves deformation within discs, vertebrae, ligaments or cartilage. It can estimate spatial stress and strain given boundary conditions, contact laws and constitutive properties. It cannot repair an incorrect applied load or an unidentified material law. Mesh convergence addresses discretization error; agreement with independent measurements addresses a different question. A detailed model with uncertain loading can give precise-looking stress maps with large physical uncertainty.

The useful hierarchy is therefore a sequence of questions: Does the model reproduce measured motion? Does it satisfy the measured external wrench balance? Are muscle and

contact-force estimates supported? Are local deformation and material laws validated for the loading regime? Does any proposed injury relationship have separate clinical evidence? Success at one step does not establish all later steps.

24.5.2 A Coupled Equation for the Golfer and Club

For independent joint coordinates $\mathbf{q}$ in a nonsingular chart, consider a fixed, ideal contact mode:

$$M(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{c}(\mathbf{q}, \dot{\mathbf{q}}) + \nabla_{\mathbf{q}}(V_g(\mathbf{q}) + U(\mathbf{q})) = B(\mathbf{q})\boldsymbol{\tau} - D(\mathbf{q})\dot{\mathbf{q}} + J_c^T \boldsymbol{\lambda}. \quad (24.9)$$

Here M is the coupled mass matrix; $\mathbf{c}$ contains velocity-dependent inertial terms; V_g and U are gravitational and passive elastic potentials; $B\boldsymbol{\tau}$ is the generalized active load; and D is a positive semidefinite damping matrix. The contact multiplier $\boldsymbol{\lambda}$ and acceleration must satisfy the constraint equations together. Applied club, ground or apparatus loads belong in the appropriate generalized-force terms if they are not already represented by included bodies and contacts. Do not count them twice.

The equation makes an important connection visible. Changing a hip or arm torque can change spinal acceleration through M^{-1} , even if no spine torque changes. Changing posture also changes inertia, moment arms, gravity, passive forces and contact geometry. A collection of uncoupled scalar springs cannot represent all those effects. Conversely, including a coupled mass matrix does not identify the individual muscles that generated a measured resultant.

For quaternion or other redundant orientation coordinates, write $\dot{\mathbf{q}} = N(\mathbf{q})\mathbf{v}$ and formulate dynamics in generalized velocity $\mathbf{v}$. One must then use the corresponding force and energy gradients rather than silently identifying $\mathbf{v}$ with $\dot{\mathbf{q}}$. For contact impacts, lift-off or slip, the active equations and possibly the state reset change. A smooth equation applies within its stated mode.

24.5.3 Inverse Dynamics and Internal Load Sharing

Choose a cut through the model and declare which tissues cross it. Measured motion, inertia and external loads can constrain the resultant wrench required across that cut. A simplified axial balance illustrates the remaining ambiguity:

$$N_{\text{contact}} + N_{\text{muscle}} + N_{\text{ligament}} = N_{\text{required}}, \quad (24.10)$$

with each quantity signed in the same free-body convention. Muscle forces also contribute moments through their attachment geometry. A joint reaction from one software model may include or exclude explicit muscle, disc or constraint forces differently from another. Compare definitions before comparing numbers.

For an illustrative antagonist pair with equal moment-arm magnitude r , the net moment is $\tau = r(F_+ - F_-)$. Increasing both tensions by ΔF leaves that moment unchanged. If both have compressive components at the joint, the compression increases. Thus identical motion and net moment can coexist with different internal loads. EMG, muscle geometry, recruitment criteria and physiological constraints can narrow the possibilities; none should be hidden inside an unexplained “joint force.”

24.6 Spinal Loads and Injury Evidence

24.6.1 Measured Inputs and Modeled Contact Forces

Lim, Chow and Chae studied five male collegiate golfers using an EMG-assisted optimization model. Their estimated L4–L5 mean peak compression exceeded six body weights during the downswing; reported shear peaks occurred during follow-through. These are model-based internal contact estimates informed by measurements, not forces measured by an instrument inside the disc. The participant group, recruitment assumptions, modeled level and definition of peak all matter. [Lim et al., 2012]

The peak of an ensemble-average waveform need not equal the average of individual peaks: individual maxima can occur at different times. An anterior shear peak, a compression peak and a torsional-moment peak also need not coincide. Reporting only the largest absolute value discards sign, timing and concurrent loads. For transition, early downswing and follow-through, compare complete time histories with uncertainty and a common phase definition. No universal “most dangerous phase” follows from one model’s largest force.

Body-weight normalization is useful for reporting scale, but it is not a tissue-strength criterion. For an illustrative 75 kg person with $g = 9.81 \text{ m/s}^2$, eight body weights correspond to 5886 N. Dividing by an assumed 1800 mm^2 area gives a mean normal traction magnitude:

$$\frac{N}{A} = \frac{5886 \text{ N}}{1800 \text{ mm}^2} = 3.27 \text{ MPa.} \quad (24.11)$$

The area and load here are teaching inputs, not an anatomical measurement or a safe-load recommendation. This quotient cannot locate the peak annular stress, partition disc and facet load, or determine whether a golfer will develop pain.

24.6.2 Normal and Shear Stress Are Different Tensor Components

Consider a homogeneous circular beam only to clarify the mechanics. Let z run along the beam, x, y be principal centroidal cross-section axes, and $N > 0$ denote compression. With bending moments defined by the traction moment on the positive- z face,

$$\sigma_{zz}(x, y) = -\frac{N}{A} + \frac{M_x y}{I_x} - \frac{M_y x}{I_y}. \quad (24.12)$$

For Saint-Venant torsion of this circular beam,

$$\sigma_{xz} = -\frac{Ty}{J}, \quad \sigma_{yz} = \frac{Tx}{J}, \quad J = I_x + I_y. \quad (24.13)$$

The normal and shear components act in different directions. Their signs depend on the declared face and axis conventions. They must enter a stress tensor and the traction relation $\mathbf{t} = \boldsymbol{\sigma}\mathbf{n}$; adding their scalar magnitudes does not produce a general “total disc stress.” Transverse shear from a net transverse force would require additional terms, and these circular-beam torsion equations are not formulas for an arbitrary cross-section.

A two-dimensional example makes the distinction concrete. If the nonzero components are a normal stress σ and shear stress s , the matrix

$$\begin{bmatrix} \sigma & s \\ s & 0 \end{bmatrix} \quad (24.14)$$

has principal stresses

$$\sigma_{1,2} = \frac{\sigma}{2} \pm \sqrt{\left(\frac{\sigma}{2}\right)^2 + s^2}. \quad (24.15)$$

These follow from its eigenvalues, not from $\sigma + s$. Even a correctly computed isotropic equivalent stress would not establish failure of an anisotropic, hydrated, heterogeneous disc. Fiber direction, matrix behavior, fluid transport, endplates, degeneration and loading history require their own constitutive description and evidence.

24.6.3 Loading, Damage and Pain Require Separate Evidence

Cadaver experiments by Adams and Hutton produced different lesions under compression and flexion depending on the loading conditions and disc degeneration. They help investigate mechanisms under controlled conditions; they do not supply a universal human golf-swing failure threshold. [Adams and Hutton, 1982]

An inferred high contact force is not a diagnosis of herniation, and pain is not a direct readout of one stress component. An injury argument needs a specified outcome, exposure duration and repetition, participant characteristics, competing explanations, and a study design that can test the proposed relationship. Cross-sectional differences between people with and without pain can reflect causes, consequences or adaptations. A model can generate a testable hypothesis about tissue loading without claiming that its preferred swing prevents injury.

This distinction strengthens the golf investigation. Instead of declaring a particular X-factor harmful, ask whether a change in the measured movement alters a validated load estimate under comparable club delivery, whether that result survives uncertainty in muscle recruitment and tissue properties, and whether the relevant exposure predicts the stated outcome in independent observations. Each link can succeed or fail separately.

24.7 Elasticity, Active Stiffness and the Drift Field

24.7.1 A Passive Chain Does Not Create Extra Torque

Take n coaxial, linear torsional springs in series, with positive stiffnesses k_i , no intermediate applied moments and negligible intermediate acceleration. The same moment τ passes through each spring, so

$$\theta_i = \frac{\tau}{k_i}, \quad \Theta = \sum_i \theta_i, \quad k_{\text{eq}} = \left(\sum_i \frac{1}{k_i} \right)^{-1}. \quad (24.16)$$

Thus $\tau = k_{\text{eq}}\Theta$. Adding an identical spring in series decreases the equivalent stiffness. At a fixed total angle it decreases the moment and stored energy. At a fixed applied moment it increases the total deflection and energy. Specifying what is held fixed resolves the apparent contradiction.

Angles in an energy calculation must be in radians. Consider five springs, each with stiffness 12 N m/degree. Their combined stiffness is $k_{\text{eq}} = 2.4$ N m/degree. At a total angle of 20°, the moment is 48 N m and

$$U_{\text{total}} = \frac{1}{2}\tau\Theta = \frac{1}{2}(48 \text{ N m})\frac{20\pi}{180} \approx 8.38 \text{ J}. \quad (24.17)$$

These are synthetic teaching values. Real spinal joints have nonlinear, coupled responses; muscles can bridge several levels, facets can engage, and intersegmental inertia makes transmitted moments differ. The series result is a limiting example, not a method for multiplying a regional range of motion into an energy reserve.

24.7.2 A Ligament Law Must Define Length and Units

One illustrative tension-only law uses current length L , slack length L_0 and k with units N/m^2 :

$$F(L) = \begin{cases} 0, & L \leq L_0, \\ k(L - L_0)^2, & L > L_0. \end{cases} \quad (24.18)$$

The continuous potential is

$$U_L(L) = \frac{k}{3} [\max(L - L_0, 0)]^3, \quad \mathbf{Q}_L = -F(L) \nabla_{\mathbf{q}} L. \quad (24.19)$$

The second relation includes the geometry that converts tension to generalized load. If strain is used instead, its dimensionless definition and the coefficient's units must change consistently. A neutral joint pose does not imply every ligament is at its slack length; prestrain can produce tension there. The quadratic law illustrates a conservative nonlinear spring and omits hysteresis, damage and rate dependence. Those effects need additional state or constitutive terms.

24.7.3 Changing Stiffness Can Require Mechanical Work

For a one-coordinate example, let

$$U(\theta, t) = \frac{1}{2} k(t) (\theta - \theta_0(t))^2, \quad \delta = \theta - \theta_0(t). \quad (24.20)$$

With constant inertia I , applied torque τ_u , and damping coefficient $d \geq 0$, suppose

$$I\ddot{\theta} = \tau_u - d\dot{\theta} - k\delta. \quad (24.21)$$

For mechanical energy $E = \frac{1}{2} I \dot{\theta}^2 + U$, differentiation gives

$$\dot{E} = \tau_u \dot{\theta} - d\dot{\theta}^2 + \frac{1}{2} \dot{k} \delta^2 - k\delta \dot{\theta}_0. \quad (24.22)$$

Changing stiffness or equilibrium angle can therefore add or remove mechanical energy. For example, holding a nonzero deformation fixed while increasing k increases stored energy; the mechanism changing stiffness supplies that energy. This is not free concentration of an unchanged energy store. Nor does simultaneous acceleration of adjacent segments necessarily dissipate energy: dissipation is associated with the relevant constitutive losses, such as $d\dot{\theta}^2$, not with acceleration itself.

Active muscle tension, short-range mechanical response, activation dynamics and feedback can all affect an estimated joint stiffness. Co-contraction can change stability, stiffness and compression together. Calling it entirely passive hides the maintained activation and its physiological cost. A mechanical energy equation also does not measure metabolic expenditure. These distinctions matter when comparing a golfer who stiffens the trunk with one who changes timing: a similar endpoint motion may arise from different force, energy and control histories.

24.7.4 Gravity and Elastic Return Have Signed Contributions

An elastic restoring moment is $-\partial U/\partial\theta$, not an unsigned amount that always helps the downswing. Gravity contributes $-\partial V_g/\partial\theta$. Depending on pose, the two can reinforce or oppose one another, and their power depends on the sign of velocity. In a planar illustration, let θ be a mass's angle from the downward vertical, with center-of-mass distance ℓ . Then

$$V_g = -mg\ell \cos\theta, \quad \tau_g = -mg\ell \sin\theta. \quad (24.23)$$

For a spring centered at zero, $\tau_e = -k\theta$. At a positive angle both moments point toward zero; if the angle is increasing, both have negative mechanical power. Their magnitudes alone do not identify energy transfer into the club. A full model must also include coupled motions and moving attachment points.

24.7.5 Define the Zero Input Before Computing Drift

For a free, independent-coordinate model with fixed passive properties, let $\mathbf{x} = (\mathbf{q}, \mathbf{v})$ and $\mathbf{v} = \dot{\mathbf{q}}$. Within a smooth mode,

$$\dot{\mathbf{x}} = \underbrace{\begin{bmatrix} \mathbf{v} \\ M^{-1}(-\mathbf{c} - \nabla V - D\mathbf{v}) \end{bmatrix}}_{\mathbf{f}(\mathbf{x})} + \underbrace{\begin{bmatrix} 0 \\ M^{-1}B \end{bmatrix}}_{\mathbf{G}(\mathbf{x})} \boldsymbol{\tau}, \quad V = V_g + U. \quad (24.24)$$

The drift is a state-velocity field. A torque expression alone has the wrong units and omits both the kinematic component and the inverse inertia. In a constrained mode, eliminate or solve the contact reaction consistently for both drift and controlled dynamics; simply deleting $J_c^T \boldsymbol{\lambda}$ changes the physical problem.

If muscle excitation is the input, include activation and other force-generating states. Zero excitation need not mean zero activation or immediate disappearance of muscle force. If stiffness or reference posture is commanded, their evolution and the associated work belong in the model. The same physical behavior can be split differently into drift and input after shifting the reference control. Consequently, a “passive percentage” is not established merely by assigning co-contraction to $\mathbf{f}$.

For a golf counterfactual, state which inputs become zero, which contacts remain, and which initial muscle, shaft and tissue states are retained. Compare forward trajectories over a declared interval. Removing an instantaneous torque term at one recorded posture does not by itself predict the subsequent delivery or the subsequent spinal load.

24.7.6 Abdominal Pressure Requires a Free-Body Diagram

Pressure acts through distributed surface traction. On a chosen surface S with consistent normal convention, the pressure-force and moment contributions involve

$$\mathbf{F}_p = \int_S p \mathbf{n} dA, \quad \mathbf{M}_{p,O} = \int_S (\mathbf{r} - \mathbf{r}_O) \times p \mathbf{n} dA. \quad (24.25)$$

Uniform pressure on a closed surface has zero net force and moment, although a portion of that surface can experience a nonzero resultant. The choice of subsystem is therefore essential. Multiplying pressure by one projected area computes one contribution under stated geometry; it does not automatically give the change in spinal compression. Abdominal wall tension and the forces needed to generate pressure must satisfy the same equilibrium.

Hodges and colleagues used artificial diaphragm stimulation to measure a trunk extension moment associated with pressure under controlled laboratory conditions. Stokes and colleagues modeled pressure together with curved abdominal wall muscles and recruitment optimization, predicting unloading under their specified static conditions. The latter authors also discuss limitations of optimized muscle recruitment. Neither result establishes that a particular breath during a dynamic golf swing reduces an individual's disc load. [Hodges et al., 2001, Stokes et al., 2010]

The mechanical question is useful without a breathing prescription: when pressure changes, how do muscle forces, trunk moment, contact reactions and stiffness change together under the actual task constraints? A model that includes only the apparent extension moment and omits its muscular support cannot answer that question.

24.8 A Reproducible Spine Investigation in Golf

Start with a stated comparison. For example: does redistributing relative motion between modeled lumbar and thoracic joints change estimated lumbar contact load while preserving a specified club-delivery task? Define the delivery tolerances, participant population, measured coordinates, phase events and load outputs before optimizing a movement. Otherwise a lower load may simply accompany a slower or different swing.

Measure and retain the quantities needed for the model: calibrated segment poses, external ground and club interactions, timing and, where relevant, EMG with its processing and normalization. Document sampling, filtering and differentiation because acceleration estimates and inverse dynamics depend on them. Check whole-body force and moment balance before interpreting a level-specific internal force. A successful fit to thorax markers is not validation of every vertebral motion.

Use the simplest model that resolves the proposed mechanism, then test sensitivity to added degrees of freedom and uncertain parameters. Compare plausible muscle recruitment, joint stiffness, attachment geometry, anthropometry and contact assumptions. If the sign of a proposed improvement changes across reasonable choices, report that dependence. Increasing model detail is valuable when new data constrain it or it changes a validated prediction, not because a larger model must be more accurate.

Separate calibration from evaluation. Fit parameters on one set of observations and assess predictions on independent trials or participants appropriate to the intended claim. Report errors in the actual measured outputs; do not invent a blanket accuracy percentage for a model class. Internal contact and tissue stress remain conditional estimates unless the corresponding quantities have independent validation. Any proposed clinical benefit requires outcome evidence beyond a mechanical simulation.

The resulting account can connect the swing's components without reducing them to one ideal angle. Preparation establishes posture, activation and elastic state. Ground interaction, muscle action and coupled inertia shape the subsequent motion. That motion changes force directions and moment arms; muscle recruitment and passive structures partition the required wrench. Tissue properties and history govern local deformation. The club-delivery objective constrains which alternatives are comparable. A rigorous explanation follows these links, names the assumptions at each one, and shows which conclusions survive uncertainty.

The muscle-force model in Chapter 14 and the closed-chain mechanics in Chapter 9 supply complementary details about force generation and constraints. Chapter 34 connects these mechanisms to preparation, delivery and outcome across the complete swing.

Exercises 24.1. Chapter Exercises

1. **Joint Count.** A chain contains rigid T12, L1–L5 and sacrum bodies, with the sacrum fixed. Count adjacent interfaces. Determine the configuration dimension if each interface is an ideal three-rotation joint, and if instead each allows six independent relative motions. Explain why neither is an anatomical claim that all those motions are equally free.
2. **Tilted Spin.** For $R = R_y(\phi)$ and $\omega_W = \Omega \mathbf{e}_z$, derive ω_b . Evaluate it for $\phi = 30^\circ$ and $\Omega = 4 \text{ rad/s}$. Show that its norm is unchanged, and explain why the nonzero local x component does not identify a tissue compliance coefficient.
3. **Euler Kinematics and Power.** Derive the displayed matrix E from $R_z(\alpha)R_y(\beta)R_x(\gamma)$. Verify numerically at a nonsingular nonzero pose that $\mathbf{m}_b^T \omega_b = (E^T \mathbf{m}_b)^T \dot{\mathbf{q}}_e$. What fails at $\beta = \pi/2$, and what physical quantity remains well defined?
4. **Finite Rotation Composition.** Compare $R_x(30^\circ)R_y(20^\circ)$ with the reversed product. Compute their action on $\mathbf{e}_z$. Explain why summing maximal regional angles cannot generally predict a thorax orientation.
5. **Load and Mean Traction.** Reproduce the 75 kg, eight-body-weight, 1800 mm^2 calculation. If the assumed area decreases by 20% at fixed force, what happens to the mean traction magnitude? List the additional information needed to interpret local disc stress and clinical risk.
6. **Stress Components.** For $\sigma = -3 \text{ MPa}$ and $s = 1 \text{ MPa}$ in the displayed two-dimensional stress tensor, calculate its principal stresses. Compare them with $\sigma + s$. Explain why neither this tensor nor an isotropic equivalent stress is a validated annular failure law.
7. **Series Compliance.** Derive the equivalent stiffness and stored energy for five identical 12 N m/degree springs at a total 20° rotation. Repeat with ten identical springs at the same total rotation, then at the same applied 48 N m moment. State the mechanical assumptions behind the comparison.
8. **Ligament Geometry.** For the tension-only law in this chapter, let $k = 2 \times 10^7 \text{ N/m}^2$, $L_0 = 0.020 \text{ m}$, and $L = 0.022 \text{ m}$. Calculate tension, tangent length stiffness and stored energy. If $dL/d\theta = 0.010 \text{ m/rad}$, determine the signed generalized elastic moment. These are illustrative parameters.
9. **Controlled Stiffness Work.** Hold $\delta = 0.10 \text{ rad}$ fixed while increasing k from 100 to 200 N m/rad . Calculate the change in stored energy. Reconcile the result with zero joint velocity and identify the energy term omitted by treating stiffness changes as free.
10. **Signed Gravity and Spring Moments.** Use the planar convention above with $m = 10 \text{ kg}$, $\ell = 0.30 \text{ m}$, $g = 9.81 \text{ m/s}^2$, $k = 20 \text{ N m/rad}$ and $\theta = 30^\circ$. Calculate gravity and elastic moments, then their total power for $\dot{\theta} = +1$ and -1 rad/s . Explain the energy interpretation.

11. **Muscle Redundancy.** Two ideal antagonists have $r = 0.050$ m, tensions $F_+ = 300$ N and $F_- = 100$ N, and equal compressive projection factors 0.8. Compute their net moment and combined compressive contribution. Repeat after adding 150 N to each tension. What could motion and net-moment measurements alone distinguish?
12. **Pressure and Subsystems.** A uniform 5 kPa pressure acts on a planar projected area 0.10 m^2 . Calculate that surface's force contribution. Explain why the net force on the complete closed surface is zero and why neither result alone gives a 500 N reduction in lumbar compression.
13. **Coupled Acceleration.** At an instant in a two-rotation model, take $M = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \text{ kg m}^2$, zero other generalized loads, and applied torque $\tau = (1, 0)^T \text{ Nm}$. Find both accelerations. Explain why a zero second torque does not imply zero second acceleration, and state how fixed contact constraints would alter the calculation.
14. **Peak Definitions and Injury Claims.** Two normalized compression histories at three common times are $(2, 6, 2)$ and $(2, 2, 6)$ body weights. Compare the peak of their mean history with the mean of their individual peaks. Explain what additional evidence would be required to turn either number into a claim about an injurious phase.
15. **A Testable Swing Comparison.** Design a study or simulation comparing two distributions of trunk motion under matched club delivery. Specify measurements, the input/activation convention, contact assumptions, uncertain parameters, independent validation outputs and the limit of the intended conclusion. Explain how you would report a predicted load reduction that disappears under another plausible muscle-recruitment model.

Chapter 25

Anatomy and Joint Modeling: Choosing the Right Idealization

In Plain Language 25.1. What a Joint Model Promises

A knee model can answer several different questions: where the foot moves, what net moment acts between the leg segments, how cartilage carries contact load, or which tissue might be injured. A model adequate for the first question may be inadequate for the others. The useful simplification is therefore defined by the prediction and the evidence available to test it.

Joint primitives give us a precise language for allowed relative motion. They are one part of a biomechanical model, alongside muscles, passive tissues, segment inertia, grip and ground contact. This chapter connects those choices: how a change in anatomy or constraints propagates into motion, force, work and the conclusions we can defend about the golf swing.

25.1 The Modeling Problem: Choose the Question First

Consider the knee. A rigid-body model may replace the tibiofemoral articulation with a hinge. A more detailed model can include coupled rotations and translations, patellar motion, ligament forces and contact between deformable surfaces. These models represent different aspects of the same structure. Neither the number of coordinates nor the realism of a rendered skeleton establishes predictive accuracy.

A hinge has one configuration coordinate, its relative angle q . That angle is not automatically an actuator input. In forward dynamics, a net moment or a set of muscle excitations can drive the motion; the angle evolves from the initial state and the equations of motion. In a prescribed-motion calculation the angle trajectory is supplied, and inverse dynamics can estimate the net forces needed to produce it. Identifying individual muscle forces or tissue stress requires additional assumptions and observations.

For a fixed hinge axis u expressed in the parent joint frame, one possible orientation convention is

$$R(q) = \exp(q[u]_{\times})R_0,$$

where u is a unit vector, $[u]_{\times}v = u \times v$, and R_0 is the relative orientation at $q = 0$. This formula describes rotation about the specified axis. The joint-origin constraint and the placement of each segment's center of mass are also needed to determine the segment motion.

A detailed finite-element model can resolve deformation and contact effects that this hinge removes, provided its geometry, materials, boundary conditions and numerical approximation are supported. It does not become accurate merely by adding detail. Conversely, a reduced model is not automatically identifiable or fast enough for a particular optimizer. Calibration, sensitivity analysis, convergence checks and validation against independent observations establish which predictions are trustworthy.

For golf, begin by declaring the output and system boundary. Club delivery requires the coupled body–hand–club chain and its support conditions. A knee contact-force question also requires the forces that realize the measured motion. A tissue-injury question adds constitutive behavior, exposure and clinical evidence. The following sections develop useful reductions and show where one question requires information that another can omit.

25.2 Joint Types in Robotics — A Taxonomy

A joint primitive specifies admissible relative motion. It does not specify the actuator, muscle recruitment, friction, contact pressure or tissue tolerance. For two otherwise free rigid bodies, relative configuration has six dimensions. A regular joint with f local freedoms imposes $6 - f$ independent constraints. Count the rank of the constraints, not the number of entries in a rotation matrix [Lynch and Park, 2017].

In this section, $R \in SO(3)$ maps vector components from body B's joint frame into body A's joint frame. The vector p is the displacement from A's joint-frame origin to B's, expressed in A. Attach those origins at the modeled joint center, not at the segment centers of mass. A statement that $p = 0$ therefore permits a segment center of mass to move on an arc. Joint axes are unit vectors and all angular velocities below are relative to A and expressed in A.

25.2.1 Revolute (1 DOF)

A revolute joint permits rotation about one common axis. Let u_A and u_B be its axis directions fixed in the respective bodies. Its constraints are

$$p = 0, \quad Ru_B = u_A.$$

The first equation gives three translational constraints. The equality of two unit axis directions gives two independent rotational constraints: its three written components have differential rank two at a compatible orientation. The total is five, leaving one freedom. With coincident, aligned reference frames and the axis denoted by u , one convenient parameterization is $R(\theta) = \exp(\theta[u]_{\times})$. A general reference orientation introduces a constant rotation factor.

For an arbitrary segment-fixed point with offset a from the joint, its relative position is Ra . Thus a hinge does not prevent every point of one body from translating relative to the other. It prevents separation of the chosen joint origins. A knee hinge is a possible reduction of biological motion; its adequacy must be assessed for the target output rather than inferred from the primitive's name.

25.2.2 Prismatic (1 DOF)

A prismatic joint permits sliding along a unit axis u . Choose orthonormal e_1, e_2 perpendicular to u . In aligned reference frames,

$$\begin{aligned} R &= I_3, & p &= du, \\ e_1^T p &= 0, & e_2^T p &= 0. \end{aligned}$$

Fixed orientation imposes three independent conditions on $SO(3)$, not nine. The two scalar position conditions remove both transverse translations, leaving the distance d . One unspecified perpendicular vector would remove only one transverse direction. A sliding rail is a literal engineering example; a translation coordinate can also be useful in a biological model without implying that an anatomical joint is a perfect rail.

25.2.3 Spherical (3 DOF)

A spherical joint imposes $p = 0$ and permits any $R \in SO(3)$ before anatomical limits or other loads are added. It has three rotational freedoms. A unit quaternion uses four components with one normalization constraint, and the two signs represent the same orientation; it does not add a fourth rotational freedom.

One local representation is the ZYX sequence

$$R = R_z(\phi)R_y(\theta)R_x(\psi).$$

The rotations act about successive axes, whose spatial directions depend on preceding rotations. They are not three fixed, mutually orthogonal axes throughout the motion. At $\theta = \pm\pi/2$, this Euler coordinate chart loses rank. The ideal spherical joint still has three physical rotational freedoms. A physical three-axis gimbal, in contrast, has a mechanism Jacobian whose rank can fall when its outer and inner axes coincide. A coordinate singularity, a mechanism singularity, a joint limit and loss of task controllability require different analyses.

The hip is often reduced to a spherical joint with an admissible configuration envelope. The primitive's unrestricted orientation set is not a claim that a biological hip can attain every orientation.

25.2.4 Universal (2 DOF)

An ideal universal joint consists of two intersecting revolute axes held perpendicular by the intermediate member. Let u_A be the first axis fixed in A and u_B the second axis fixed in B. Its finite constraints are

$$p = 0, \quad u_A^T R u_B = 0.$$

The scalar orthogonality condition supplies one rotational constraint; together with the three position constraints it leaves two freedoms. It does not require a fixed third axis to remain unchanged by R .

For reference axes $u_A = e_x$, $u_B = e_y$, use

$$\begin{aligned} R &= R_x(\alpha)R_y(\beta), \\ \omega &= W(\alpha)\dot{q}, \quad \dot{q} = (\dot{\alpha}, \dot{\beta})^T, \end{aligned}$$

$$W(\alpha) = \begin{bmatrix} 1 & 0 \\ 0 & \cos \alpha \\ 0 & \sin \alpha \end{bmatrix},$$

$$W(\alpha)^T W(\alpha) = I_2.$$

Both columns remain orthogonal for every α . The ideal two-axis joint therefore does not lock because its own axes align. A driveshaft assembled using a Cardan joint has additional shaft-axis and transmission geometry; its operating limits are not a proof that this two-coordinate joint loses rank. Likewise, a wrist-club task map can be singular even when the local joint rate map has rank two.

The forbidden instantaneous angular-velocity direction is the moving normal $n = e_x \times R_x(\alpha)e_y$. For example, at $\alpha = \pi/2$, the allowed second-axis velocity points along A's z direction. A rule that permanently forbids rotation around A's z direction would reject legitimate universal-joint motion. At $\alpha = \pi/2, \beta = 0$, $Re_z = -e_y$, directly disproving the fixed-normal rule.

This rate map also determines which moment components drive the coordinates. If m is an applied relative moment expressed in A, virtual power gives

$$Q = W^T m, \quad Q^T \dot{q} = m^T \omega.$$

The generalized moments are projections onto the current axes; they are not arbitrary fixed Cartesian components. An ideal reaction moment along n does no relative work because $n^T \omega = 0$, although it can transmit load. This distinction matters when interpreting wrist torques, grip reactions and clubhead acceleration.

25.2.5 Cylindrical (2 DOF)

A cylindrical joint permits translation and rotation along the same axis. With aligned reference frames, $p = du$ and $Ru = u$ impose two translational and two rotational constraints. The coordinates are d and θ . This is a kinematic idealization, not an assertion that one biological joint realizes an exact cylinder.

25.2.6 Planar (3 DOF)

A planar joint permits two translations in a plane and rotation about its normal. For a fixed unit normal n , aligned reference normals give $n^T p = 0$ and $Rn = n$: one translational and two independent rotational constraints. A foot kept flat on a plane can be modeled this way only within that assumed mode. The primitive alone imposes neither friction nor nonnegative normal force. Sticking, slipping, edge contact and lift-off require additional contact laws and feasibility checks.

25.2.7 6-DOF (Free, Floating)

A free body has three translational and three rotational freedoms relative to ground. A floating-base model uses those freedoms to represent overall pose; it does not assert absence of gravity, muscle forces or contact reactions. Quaternion storage may require seven position entries for this six-dimensional pose. Position-array length, velocity dimension and actuator count need not agree.

25.2.8 Constraint Counting

For a spatial linkage with N bodies including ground and joints whose constraints are independent, the mobility estimate is

$$d = 6(N - 1) - \sum_j (6 - f_j).$$

Count three moving links attached in series to a fixed base. If the base is merged with ground, $N = 4$ and there are three revolute joints:

$$d = 6(4 - 1) - 3(6 - 1) = 3.$$

If the fixed base is kept as a separate body, $N = 5$ and its weld to ground must be counted as a fourth joint with $f = 0$. Then $d = 24 - (6 + 15) = 3$. Omitting that weld gives nine freedoms, which describes the floating chain, not the intended fixed-base arm. The error is missing topology, not the use of a spatial formula on a planar example. The formula can be applied to trees as well as loops when its assumptions hold.

For a fixed-base tree, independent joint coordinates give $d = \sum_j f_j$; a floating base adds six. In a loop, let $q \in \mathbb{R}^{n_q}$ be local tree coordinates and $c(q) = 0$ the closure equations. At a regular configuration,

$$d = n_q - \text{rank } J_c, \quad J_c = \frac{\partial c}{\partial q}.$$

Redundant constraints cannot each remove another freedom. At a singular configuration, the linearized tangent space may be larger than the set of realizable finite motions, so a pointwise rank loss must be checked against the nonlinear closure. For a two-handed golf grip, specify the relative hand–club transforms and evaluate that closure rank; do not simply subtract a number called “the loop.” Mobility is also distinct from the number of independently controlled inputs and the dimension of a chosen hand or clubhead task.

25.3 The Knee — A Revolute (Mostly)

25.3.1 Anatomy Summary

The tibiofemoral joint carries load between the femoral condyles and tibial plateau through cartilage and the menisci. The menisci distribute contact load and contribute to stability; they are not simply interchangeable shock absorbers. The patella articulates with the femur and changes the geometry of the knee extensor mechanism. A model that merges it into an effective quadriceps moment arm has made a further reduction beyond replacing the tibiofemoral articulation with a hinge.

The cruciate and collateral ligaments resist combinations of displacement and rotation. In a simple anatomical description, the ACL is a major restraint to anterior tibial translation, the PCL to posterior translation, the MCL to valgus loading and the LCL to varus loading. These are load-dependent restraints, not exact kinematic welds. Valgus and varus refer to opposite frontal-plane angulations; they should not be confused with axial twisting. Muscle forces, articular geometry and the capsule also contribute to stability [Neumann, 2017, Nordin and Frankel, 2012].

25.3.2 Range of Motion and the Hinge Approximation

Flexion–extension is an important knee motion, but dominance in angular excursion is not sufficient to establish that other motions have negligible effects on a particular output. A hinge may describe a sagittal-plane teaching model well while being unsuitable for estimating axial knee moments, contact loads or how a planted leg accommodates pelvis rotation. Lead and trail knees need not follow the same trajectory. A video projection, an anatomical joint angle and an external knee moment are different measurements.

A commonly quoted flexion range near 140 degrees is an anatomical reference, not the knee excursion used by every golfer [Neumann, 2017]. Assessment posture, active versus passive testing, subject anatomy and the angle convention matter. An accuracy claim for a golf model needs a declared dataset and error metric, such as hand-position residuals or differences in estimated joint moments.

25.3.3 Where Revolute Breaks Down

Real tibiofemoral motion includes coupled axial rotation, translation and frontal-plane motion. The screw-home description concerns coupling near terminal extension; tibial rotation relative to the femur and femoral rotation relative to a planted tibia must be distinguished. It does not imply an identical rotation in every loading condition or a mechanically irreversible lock.

Comparing an omitted axial rotation with the total flexion range is not an error calculation: these rotations act about different axes. For a point at perpendicular distance ρ from an omitted rotation axis, an angle error δ produces the displacement

$$\|\Delta p\| = 2\rho \left| \sin \frac{\delta}{2} \right|.$$

A point on the axis has zero displacement from this rotation even though its attached segment changes orientation. A point 5 cm off the axis moves about 8.72 mm for a 10-degree rotation. Tibial length alone is insufficient to calculate the ankle-position error. For club delivery, the relevant effect also depends on the remaining kinematic chain, support constraints and compensating motion.

25.3.4 Choosing the Knee Model

Start with the outputs to be explained. For a simplified planar swing, a hinge is a useful explicit assumption. For spatial kinematics, compare it with a model that permits measured coupled rotations or translations, and evaluate residuals over the whole swing. A one-coordinate knee can even have translation and axis orientation prescribed as functions of flexion; one degree of freedom does not require a geometrically fixed hinge.

For tissue loading, additional contact, ligament and muscle models may be necessary, but adding six free coordinates does not itself create a validated injury model. More coordinates require more constraints, constitutive information or measurements to identify their behavior. Computation time must be benchmarked for the actual solver and task; a hinge is not universally required for optimization.

25.4 The Hip — A Ball-and-Socket (Almost)

25.4.1 Anatomy and Spherical Reduction

The femoral head articulates with the acetabulum. Cartilage, the labrum, capsule, ligaments and surrounding muscles contribute to load transmission and stability. A spherical reduction uses a fixed relative center and three rotational freedoms. Its purpose is to approximate the motion required for a particular analysis, not to assert exact sphericity, zero biological translation or unrestricted motion.

Avoid assigning a universal center shift from an anatomical sketch. A measured shift may reflect real articular motion, a fitted-center definition, marker movement or model mismatch. Its consequence is also task dependent: a center-location error affects segment motion, moment arms and the conversion of a force into a moment. Those effects should be propagated into the quantities being reported.

25.4.2 Range of Motion and Golf Measurements

Pelvis turn in the laboratory frame is not anatomical hip internal rotation. Hip motion describes the femur relative to the pelvis. Its components depend on anatomical frames, rotation sequence, reference posture and sign convention. Cheetham and colleagues' pelvis–upper-torso separation near transition does not establish a universal lead-hip internal-rotation requirement [Cheetham et al., 2001].

Standard reference values such as approximately 120 degrees of hip flexion describe particular assessments [Neumann, 2017]. They are not independent limits that can be combined into a rectangular feasible region without checking combined posture. Knee position, loading and active versus passive testing can change the measured range. Measured swing motion must be expressed in the model's frames and coordinates before comparing it with any such limits.

For example, with laboratory orientations R_P and R_F for pelvis and femur, respectively, the relative orientation is

$$R_{PF} = R_P^T R_F.$$

Changing both laboratory orientations by the same rigid rotation leaves this relative orientation unchanged. Increasing a pelvis-turn angle therefore does not uniquely prescribe a change in hip rotation: the femur and foot may also move. Coordinate components obtained from different Euler sequences are not directly interchangeable.

25.4.3 Limits, Impingement and Model Selection

Bony geometry, capsuloligamentous resistance and muscle state can restrict combined hip configurations. A model may describe them through geometric contact, passive forces or an empirically measured admissible envelope. A hard angle bound and a passive resistance law make different predictions near the boundary and should not be used interchangeably without explanation.

Femoroacetabular impingement syndrome is a clinical condition requiring the combination of symptoms, clinical signs and imaging findings; a morphology or a rotation limit alone is not the diagnosis [Griffin et al., 2016]. It is therefore unjustified to assign every affected golfer a 20–30-degree limit or predict a compulsory spinal compensation from that number.

Subject-specific observations may motivate a model constraint, whose effects can then be studied without claiming that it diagnoses the cause of pain.

A three-rotation hip with calibrated geometry and configuration limits is a defensible starting point for many whole-body models. Evaluate whether additional translation or contact detail changes the target prediction beyond the uncertainty already present. Motion-capture angles cannot simply be copied into a model with different frames or conventions.

25.5 The Shoulder Complex — A Moving Girdle

25.5.1 The Four Articulations

The glenohumeral (GH) joint connects humerus to scapula. The acromioclavicular (AC) joint connects scapula to clavicle, and the sternoclavicular (SC) joint connects clavicle to the axial skeleton. The scapulothoracic (ST) articulation describes the scapula's motion against the thorax rather than an additional ordinary synovial joint in series with all three. Its soft-tissue interface and the clavicular linkage constrain how the scapula can move.

This topology matters for counting. One cannot add separately assigned GH, AC, SC and ST freedoms and then announce an approximate independent total. In a model with thorax–clavicle–scapula–humerus segments, ST pose is related to the SC and AC transformations. Additional scapulothoracic contact or coupling equations further restrict the motion. An alternative reduced model may parameterize scapular motion directly, but its independent coordinates and omitted structures must be stated.

25.5.2 Moving Centers and Measured Angles

Let $r_{GH}(q_s)$ locate the GH center relative to the thorax, where q_s describes shoulder-girdle configuration, and let $R_H(q_s, q_g)$ orient the humerus. All components are expressed in the thorax frame; the angular velocity below is relative to that frame. For a humerus-fixed point with offset a ,

$$r = r_{GH}(q_s) + R_H(q_s, q_g)a,$$

$$\dot{r} = \frac{\partial r_{GH}}{\partial q_s} \dot{q}_s + \omega_H \times (R_H a).$$

The moving-center term remains even if a reduced model fits the humeral orientation exactly. With an attached forearm and hand, their additional relative-motion terms must also be included. This is one way scapular motion changes hand paths and grip geometry: it changes both the location from which the arm moves and its orientation.

Humerothoracic orientation describes humerus relative to thorax; GH orientation describes humerus relative to scapula. Widening a fixed-center model's angle limits may help it fit some observed poses, but does not reconstruct the missing center translation, anatomical moment arms or scapular muscle work. A scapula rotating at a prescribed rate is not a fixed scapula.

25.5.3 Scapulohumeral Rhythm

Scapulohumeral rhythm describes coordinated scapular and humeral motion under a specified task. It is not a universal constant. Direct bone-pin measurements by McClure and

colleagues in eight healthy volunteers found substantial scapular rotations about three axes during arm elevation; their reported mean GH-to-ST ratio was 1.7:1 [McClure et al., 2001]. That study does not supply a phase-by-phase golf-swing law.

The familiar 2:1 ratio can still serve as a declared planar illustration: if total arm elevation is 90 degrees and GH elevation is twice the scapular contribution, the components are 60 and 30 degrees. These angles add only in that simplified common-axis construction. General three-dimensional relative orientations compose as rotation matrices; componentwise angle addition is not valid.

In a golf investigation, measure or estimate the relevant shoulder-girdle motion and propagate its uncertainty. Similar hand positions can be reached through different combinations of trunk, scapular, humeral and elbow motion. That kinematic redundancy does not establish which muscles supplied work or how much load reached the GH joint.

25.5.4 Choosing the Shoulder Model

A fixed-center spherical surrogate can be adequate for a deliberately coarse whole-arm task, provided its limitations are tested against observed hand positions and orientations. Label its angles as effective humerothoracic coordinates when that is what they represent, rather than reporting them as anatomical GH rotations.

For scapular or muscle-work questions, use an explicitly described shoulder-girdle model. Seth and colleagues provide a dedicated scapulothoracic formulation [Seth et al., 2016]. Their later musculoskeletal study illustrates why prescribing scapular motion as a function of humeral motion can alter work attribution: the coupling can allow muscles crossing the GH joint to do work that the model assigns to scapular motion [Seth et al., 2019]. This is a modeling mechanism to examine, not evidence that a particular golf phase is driven by one muscle group.

Their experimental comparison involved one healthy subject performing shrugging and arm-elevation tasks. Muscle work was estimated from the model, with kinematics and EMG used for comparison; muscle forces and velocities were not measured directly. These limits matter when applying its lessons to a golf swing.

Prescribed kinematic coupling also deserves an energy check. If $q_s = h(q_g)$ is imposed and $H = \partial h / \partial q_g$, then $\dot{q}_s = H\dot{q}_g$. Generalized forces Q_g, Q_s reduce to

$$\begin{aligned} Q_{\text{reduced}} &= Q_g + H^T Q_s, \\ Q_g^T \dot{q}_g + Q_s^T \dot{q}_s &= Q_{\text{reduced}}^T \dot{q}_g. \end{aligned}$$

This identity preserves power for the imposed coupling; it does not establish that the coupling is physiological. Compare models on the same observed task, loads and initial state. A diagnosis or observed weakness does not by itself specify a reduced scapular range, a passive stiffness value or a unique compensatory movement.

25.6 The Wrist — A Universal Joint (Approximately)

25.6.1 Anatomy and Primary Motions

The radiocarpal and midcarpal articulations involve multiple carpal bones rather than one rigid intermediate member. The familiar wrist coordinates describe net hand motion relative to the forearm: flexion–extension and radial–ulnar deviation. Their axes and centers are

approximations that depend on the measurement and model. A two-coordinate universal reduction can be useful without implying that individual carpal bones execute ideal hinges.

Radial deviation means movement toward the thumb side; ulnar deviation means movement toward the little-finger side. Flexion and extension should be defined anatomically, not as “downward” and “upward” in the laboratory, because forearm orientation changes during the swing. Specify the hand and forearm frames and the order of coordinate rotations before comparing wrist angles across software or studies.

Pronation–supination is primarily forearm rotation involving both the proximal and distal radioulnar articulations. It should not be assigned exclusively to a single elbow-region hinge or counted again as an independent wrist twist. A reduced model may implement it as one forearm rotation coordinate, with an axis appropriate to that model.

25.6.2 The Dart-Thrower’s Motion

The dart-thrower’s path runs from radial extension toward ulnar flexion. Crisco and colleagues studied its *in vivo* carpal kinematics and found positions along this path where scaphoid and lunate motion approached zero [Crisco et al., 2005]. That finding concerns the measured carpal motions; it does not prove that this is always the strongest or fastest wrist action, or that every golfer should follow that path.

Two-coordinate models can represent a directional stiffness preference. Let $\Delta q = q - q_0$ be displacement from a chosen reference posture, with coordinates positive in flexion and ulnar deviation. Consider the illustrative elastic law

$$U = \frac{1}{2} \Delta q^T K \Delta q, \quad \tau_{\text{passive}} = -K \Delta q,$$

$$K = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \text{ N m/rad.}$$

The eigenvalues are 1 and 3 in the stated angular units. Equal positive changes in both coordinates follow the lower-stiffness direction; opposite changes follow the higher-stiffness direction. This is a constructed example of coupling, not a measured human wrist parameter set. Nonlinear stiffness, damping, activation and changing axes can be added when supported by data. The choice of two kinematic coordinates does not forbid coupling between their forces.

25.6.3 Wrist Motion, Grip Motion and Club Orientation

Club roll in space does not imply slip inside the hands. With a rigid hand–club transform,

$$R_C = R_H R_{HC}, \quad R_{HC} = \text{constant.}$$

R_H maps hand-frame components to the laboratory and R_{HC} maps club-frame components to the hand frame. As the forearm and hand rotate, the club rotates in the laboratory even though it has no relative grip freedom. Add a hand–club roll coordinate only if the model actually permits grip slip, compliance or a bearing-like connection. A shaft-torsion coordinate is another distinct possibility: it describes deformation between locations along the shaft rather than a freely slipping handle.

For an ideal rigid grip, the club’s angular velocity equals the hand’s angular velocity. Its component along the current shaft direction can vary because of several upstream joint

motions. That shaft-axis component, a clubface angle and a chosen Euler roll rate need not be the same observable. With two hands attached to one club, changes at one wrist must also satisfy the other hand's closure constraints; independent angle edits generally break the loop.

25.6.4 Choosing the Wrist Model

Use the two-coordinate reduction for a declared gross-motion question, then test whether its axis geometry, coupled stiffness and grip assumptions explain the measured trajectory and loads. Descriptions such as “cocking,” “bowing” and “release” require a mapping to those coordinates; a projected arm–shaft angle is not a single anatomical wrist angle.

More detailed carpal models can represent motion hidden by a net wrist coordinate, but assigning two independent rotations to both radiocarpal and midcarpal joints is itself a new model assumption, not an anatomical identity. Calibrate the additional coupling and contact information before interpreting individual-bone or tissue loads.

25.7 The Elbow — A Revolute (Very Good Approximation)

25.7.1 Anatomy and Two-Coordinate Reduction

The elbow complex includes humeroulnar, humeroradial and proximal radioulnar articulations. Humeroulnar motion is often reduced to a flexion–extension hinge. Forearm pronation–supination involves the radius and ulna along the forearm, including their distal articulation. A gross upper-limb model can use one flexion coordinate and one forearm rotation coordinate, but two independently specifiable coordinates do not imply dynamically independent motions.

For example, off-diagonal inertia terms allow a moment at one coordinate to change acceleration at another. Muscles can span more than one coordinate, and their moment arms change with geometry. Separating the coordinates in the model is useful bookkeeping; it does not isolate muscle action or energy transfer.

25.7.2 The Carrying Angle

The carrying angle describes the relative alignment of the upper arm and forearm in a specified extended posture, conventionally with the forearm supinated. It is not a complete measurement of the flexion axis. Segment landmark error, forearm rotation and soft-tissue motion affect an estimate. A static angle alone cannot identify a unique three-dimensional axis or muscle moment–arm function.

Estimate segment frames and, where needed, a functional axis from several poses. State which parameters came from anatomical landmarks, fitted motion or imaging. Reference ranges in anatomy texts provide context, but a subject's active range, passive range and task trajectory are different data. Neither a 150-degree flexion reference nor an 80-degree forearm-rotation reference should be treated as a universal swing requirement [Neumann, 2017].

25.7.3 Choosing the Elbow Model

A hinge plus a forearm rotation coordinate is a practical starting reduction. Test axis alignment and motion residuals before using it for detailed joint moments. Muscle-force or tendon-load questions require a musculotendon model and additional assumptions; the net elbow moment does not uniquely identify the flexors, extensors or co-contraction.

25.8 The Ankle/Foot — The Ground Connection

25.8.1 Anatomy and Simplified Motion

The talocrural articulation principally contributes dorsiflexion–plantarflexion; subtalar and other foot articulations contribute coupled three-dimensional motion. Tarsometatarsal joints connect tarsals to metatarsals; metatarsophalangeal joints connect metatarsals to the proximal phalanges. They should not all be described as toe joints. A rigid-foot model removes deformation of this articulated structure.

Two effective ankle coordinates are a useful reduction, but treating their axes as exactly perpendicular sagittal and frontal hinges is an assumption. The foot can stay nearly stationary while ankle moments and contact forces change substantially. Conversely, heel lift, toe pivot and local slip can change support geometry even when a coarse video description calls the foot planted.

25.8.2 Contact, Work and Deformation

Ground contact needs a separate specification: contact locations or surfaces, normal compliance or an ideal nonpenetration constraint, friction, and rules for mode changes. A permanently welded foot, a flat sliding foot and unilateral frictional contact predict different reactions and admissible motion. Count only constraints active in the selected mode, and check whether their resulting reactions are physically feasible.

For the entire golfer–club system, ground reaction is an external wrench that changes momentum. A point force does zero power at an ideal stationary contact point. For a contact wrench, total power also includes its moment paired with angular velocity; an ideal fixed contact has zero translational and angular velocity. This does not make the ground irrelevant to power generation: support changes the motions and loads through which muscles do work. It does mean that force-plate force multiplied by center-of-mass velocity is not automatically the local work done at the foot–ground interface.

A deformable arch may store and return elastic energy. In an illustrative passive rotational spring–damper,

$$U = \frac{1}{2}k\eta^2, \quad \tau = -k\eta - c\dot{\eta},$$

$$\tau\dot{\eta} = -\dot{U} - c\dot{\eta}^2,$$

for constant $k > 0$, $c \geq 0$. The spring returns previously stored energy; it does not create energy. If muscle activation changes effective stiffness, a variable parameter can exchange energy and must be included in the ledger. Identifying arch behavior requires deformation and load data, not simply a change in force-plate center of pressure.

25.8.3 Choosing the Ankle/Foot Model

Choose a rigid foot for a task whose predictions remain adequate under that reduction. Add articulated toes, arch deformation or distributed contact when they materially change the measured output. “Weight shift” should be translated into observables such as individual foot forces, center of pressure, center-of-mass motion or changing contact mode; those quantities are not synonyms.

25.9 Summary Table and Modeling Guidelines

Region	Candidate Reduction	What Still Needs Checking
Hip	Spherical, three rotations	Frames, center, combined configuration limits
Knee	Hinge or coupled one-coordinate joint	Axial motion, translations, task sensitivity
Ankle/foot	Two effective rotations, rigid foot	Axis geometry, deformation and contact modes
Shoulder	Effective spherical arm root or explicit girdle	Moving center, GH versus thorax angles, muscle work
Elbow/forearm	Flexion and forearm rotation	Axis identification and dynamic coupling
Wrist	Two rotations with coupled force laws	Carpal reduction, stiffness and grip closure

25.9.1 Decision Flowchart

1. Define the prediction: pose, club delivery, net moment, tissue load or clinical outcome. These require different evidence.
2. Specify bodies, frames, independent coordinates, inputs, constitutive laws and contact modes. A joint name alone is not a model.
3. Fit the relevant measurements and inspect residuals across the full task. Include measurement uncertainty and parameter identifiability.
4. Compare plausible model refinements using the same outputs and observations. Additional complexity is useful when it changes an important prediction and its parameters can be supported.
5. Validate on observations not used for fitting, and report which conclusions remain unchanged across model choices. Good kinematic fit does not independently validate muscle forces or injury predictions.

25.9.2 A Complete Counting Example

Consider an explicitly chosen tree: a floating pelvis (six freedoms), a three-rotation pelvis–thorax joint, two legs with hip/knee/ankle freedoms $3 + 1 + 2$, and two arms whose shoulder, elbow and forearm, and wrist freedoms are $3 + 2 + 2$. Attach the club rigidly to one hand. Before adding ground or second-hand closure, the local velocity dimension is

$$d_{\text{tree}} = 6 + 3 + 2(3 + 1 + 2) \\ + 2(3 + 2 + 2) = 35.$$

The rigid club attachment adds a body and a weld, with no new relative freedom. Let J_c stack the active ground and second-hand closure constraints on these tree coordinates. At a regular feasible configuration,

$$d = 35 - \text{rank } J_c.$$

A six-component hand-pose closure removes six freedoms only if those constraints are independent on this model. Ground conditions and their rank must also be stated. The count is neither “13 per arm” nor a universal 30-DOF golfer. Independent muscle inputs, activation states and shaft deformation may change other dimensions without changing this particular rigid kinematic count.

25.9.3 How the Choices Interact

Changing a joint’s center or admissible motion changes hand and club kinematics, muscle moment arms and the feasible contact reactions. Changing passive stiffness changes forces and stored energy, without necessarily changing the number of coordinates. Changing the grip model changes loop closure and wrench transmission. Changing the chosen input from net torque to neural excitation adds activation and force-generation dynamics. These decisions interact, but none substitutes for the others.

For a task output $y(q, \pi)$, with configuration q and geometry parameters π , the first-order sensitivity is $\delta y \simeq J_q \delta q + J_\pi \delta \pi$. The second term accounts for geometry changes even when the coordinate values stay fixed. Dynamic predictions require the associated changes to inertia, loads and constraints as well. At impact, event timing and the full club delivery state matter. A small anatomical angle can have a material effect on a sensitive task, while a large angle difference can be irrelevant to another output. State the output and calculate the sensitivity before calling a simplification negligible.

25.10 Injury Biomechanics of the Golf Swing

25.10.1 From Motion to Tissue Load

Anatomical modeling can help investigate loading mechanisms, but a kinematic limit is not a tissue-failure criterion. An ideal constraint is a mathematical restriction on motion; biological injury depends on the affected tissue, force or stress history, rate, exposure and individual capacity. Net joint moments, internal contact forces, muscle forces and local stresses are different quantities. Their uncertainties accumulate across the inference chain.

25.10.2 Low Back Injury

The lumbar region can experience compression, shear and torsional loading during a swing. Muscle forces and co-contraction can contribute substantially to internal loading; it is not determined just by ground force, visible torso rotation or the X-factor. The pelvis–spine–thorax chain is an articulated path, not by itself a closed kinematic loop.

Lim, Chow and Chae estimated L4–L5 loads using measured motion, force plates and an EMG-assisted optimization model in five male collegiate golfers. Their abstract reports mean peak compression above six body weights during the downswing, with peak anterior and medial shear in follow-through [Lim et al., 2012]. These are model estimates under that study’s assumptions, not direct instrumented measurements inside professional golfers’ spines or universal phase-specific limits.

An address angle or per-segment rotation limit cannot be used as a prospective golf injury threshold without evidence linking that defined measurement to subsequent outcomes. A model comparison should specify segment geometry, muscle recruitment, passive tissue laws, loading history and the actual outcome of interest. A calculated load change is a mechanical result; showing that a swing modification reduces pain or future injury requires separate clinical evidence.

25.10.3 Wrist and Hand Injuries

Club–ball contact force is not the force transmitted unchanged to a wrist. For a club considered as a separate system, with mass and center-of-mass velocity denoted below by the subscript C, the external-force balance in an inertial frame is

$$F_{\text{hands}} + F_{\text{ball}} + F_{\text{other}} = m_C \dot{v}_C.$$

The angular-momentum balance is also needed, and a flexible shaft and distributed grip introduce further dynamics. Even when expressed about the same point in the same frame, two hand wrenches generally cannot be uniquely inferred from their sum. A contact impulse or peak ball force therefore does not establish the load on a particular wrist tendon, carpal structure or grip location.

Club angular velocity, shaft-axis spin and anatomical wrist joint rates are distinct measurements. A club rate cannot be relabeled as the rate of one anatomical wrist coordinate. To study a specific hand injury mechanism, measure the relevant contact history, estimate the appropriate tissue loads with uncertainty, and distinguish acute loading from repeated exposure. A list of possible injured structures does not validate a single causal mechanism for all of them.

25.10.4 Medial Epicondylitis and Muscle Inference

The common flexor–pronator origin near the medial epicondyle provides an anatomical reason to investigate repeated loads in that region. It does not establish a universal trail-arm cause or prove eccentric contraction from observed forearm rotation. Fiber shortening or lengthening depends on musculotendon geometry, activation and tendon behavior; surface EMG indicates electrical activity rather than force or contraction mode by itself.

Clinical labels and injury surveys should be used with their population, case definition and exposure denominator. A model of a symptomatic player can test mechanical alternatives, but it cannot diagnose tendon pathology or assign causation from one motion pattern.

25.10.5 Control Authority Is a Separate Question

A high drift-to-control norm ratio is not an injury criterion and does not demonstrate that control is ineffective in the direction that matters. Finite-horizon task authority depends on input bounds, dynamics, initial state, contact constraints, available time and the selected

output. An instantaneous zero-torque comparison changes a declared actuator input; it does not remove all activation, stored elastic energy or prior muscle work.

One can formulate a hypothesis that certain states and load histories leave fewer feasible low-load delivery options. Testing it requires a model with defined tissue-load quantities, an admissible control set, uncertainty and independent outcome evidence. Describing a swing as “on autopilot” supplies none of these. The mechanical investigation and the clinical conclusion should remain connected through explicit evidence rather than a metaphor.

25.11 Chapter Exercises

1. **Knee Analysis.** Model an omitted 10-degree rotation about the tibial long axis. Does a tibial length of 40 cm determine ankle-point displacement? Calculate it for points on the axis and 5 cm from it. **Worked answer:** Length along the axis is irrelevant to this pure rotation. The two displacements are zero and $2(0.05)\sin(5^\circ) \simeq 0.00872$ m. Ankle orientation still changes in both cases. Translation, a different axis and chain closure require additional information.
2. **Hip Limits.** An individual’s measured hip internal-rotation envelope is narrower than the model’s default. Does that uniquely determine a compensatory spinal motion? **Worked answer:** No. Express the measured restriction in compatible femur–pelvis frames and posture, then impose it on the coupled model. Foot orientation, pelvis motion, knee motion, task targets and other bounds determine whether feasible alternatives exist. A hip limit alone neither diagnoses FAI syndrome nor identifies a unique compensation.
3. **Shoulder Rhythm.** Under a deliberately planar 2:1 GH-to-scapular ratio, divide 70 degrees of total humerothoracic elevation into its components. Explain the limitation. **Worked answer:** GH contributes $140/3 \simeq 46.67$ degrees and scapula $70/3 \simeq 23.33$ degrees. This is arithmetic under the assumed same-axis ratio, not a prediction for a measured golf backswing. In three dimensions use relative rotation composition and measured or modeled girdle motion.
4. **Wrist Coupling.** State the dart-thrower’s direction and calculate the energy in the illustrative stiffness matrix for $\Delta q = (0.1, 0.1)^T$ and $(0.1, -0.1)^T$ radians. **Worked answer:** The path goes from radial extension toward ulnar flexion. The energies are 0.01 J and 0.03 J. Direction-dependent resistance can be represented with two coordinates; this example supplies no empirical force or speed ranking.
5. **Elbow Carrying Angle.** Does a static 10-degree valgus alignment identify the full flexion axis? **Worked answer:** No. It constrains an alignment measurement in the chosen posture, not all axis parameters. Define segment frames, obtain several flexion poses, fit an axis with residuals and uncertainty, and control for forearm rotation and marker motion. A muscle moment-arm estimate needs further attachment/path information.
6. **Grübler for a Golf Body.** Fix the pelvis to ground. Add distinct lumbar and thoracic bodies through two spherical joints, then two arms each with a spherical shoulder, hinge elbow and universal wrist. Weld a rigid club to one hand. Count before and after a generic full-rank second-hand pose closure. **Worked answer:** Merging pelvis

with ground gives ten bodies including ground and nine tree joints, with freedoms $3 + 3 + 2(3 + 1 + 2) = 18$. The spatial count is $6(10 - 1) - (54 - 18) = 18$. A regular independent rank-six second-hand closure leaves twelve. This illustrative arm/spine model omits legs and forearm pronation; it is not the 35-coordinate model above. At singular or infeasible configurations the generic conclusion must be reconsidered.

7. **Model Refinement.** A model matches the backswing but releases a projected wrist angle too early. Does this identify insufficient wrist stiffness? **Worked answer:** No. Alternative explanations include coordinate mismatch, initial velocity or activation error, grip or shoulder geometry, shaft dynamics, loading and timing. Compare predicted anatomical angles, hand paths and measured forces, then vary independently supported parameters and test held-out swings. A stiffness change that improves one fitted trace is not unique mechanistic identification.

Chapter 26

Degrees of Freedom and Robot Models of the Human Body

How many independent numbers describe a golf swing? The answer depends on what the model retains. Joint angles describe configuration; velocities describe how that configuration is changing; muscle activation and elastic deformation may require additional states. Two hands gripping one club and two feet contacting the ground connect these descriptions through constraints and forces. Counting the entries in a motion-capture file does not settle any of these questions.

This chapter develops a consistent route from anatomy to computation. We first count admissible motions, then ask which of them change a declared club task, then derive how applied torques affect that task. Finally, we encode a physical mechanism and check the imported dynamics against independent mechanics. Each step answers a different question. Their connection is the foundation for interpreting a swing, designing an experiment and testing a proposed control explanation.

A rigid-body model uses the same mechanical laws for a robot and a golfer, but this does not make its joints, actuators or objectives biologically complete. Our upper-body teaching model has 20 internal freedoms before the two-hand loop is closed. That is a declared approximation, not a universal anatomical count. The executable example is a complete two-link mechanism whose limitations are explicit; extending it to a golfer requires the additional modeling decisions developed below.

26.1 Counting Degrees of Freedom — The Robotics Way

26.1.1 The Formula and Its Rank Assumption

For a spatial assembly containing N bodies including one fixed reference body, begin with $6(N - 1)$ unconstrained freedoms. A joint permitting f_i relative freedoms nominally imposes $6 - f_i$ constraints. If those constraints are independent at a regular configuration, the mobility is

$$d = 6(N - 1) - \sum_{i=1}^J (6 - f_i). \quad (26.1)$$

This is the spatial Grübler–Kutzbach count. The independence assumption is essential. A negative result signals an inappropriate or dependent-constraint count; it is not a physical negative mobility. Special geometric arrangements and closed loops require a rank calculation. The [Modern Robotics mobility lesson](#) illustrates precisely this limitation.

With generalized velocities $v \in \mathbb{R}^{n_v}$ before additional closures, let $A(q)v = 0$ describe independent, stationary constraints. At a regular configuration,

$$d = n_v - \text{rank } A(q). \quad (26.2)$$

For holonomic constraints $\phi(q) = 0$ in local coordinates with $\dot{q} = v$, $A = D\phi$. With quaternion coordinates, $\dot{q} = E(q)v$ and $A = D\phi E$. Do not subtract the quaternion normalization constraint a second time from n_v : it has already been accounted for in the velocity representation.

26.1.2 Bodies, Coordinates and Reference Frames

N freely floating bodies have $6N$ absolute freedoms relative to an external world frame. Choosing one body as a reference describes only their relative configurations, with $6(N - 1)$ freedoms. Physically fixing that body removes its world motion. These are different experimental assumptions even though the relative count is the same.

A free rigid body can be stored as three translation entries and a unit quaternion: seven configuration entries, six independent velocity components. Its dynamical state then uses seven plus six stored entries, subject to quaternion normalization. Likewise, three Euler angles locally parameterize orientation but their rates are not generally the Cartesian angular-velocity components. The number of coordinates, the dimension of configuration and the number of independent controls need not agree.

26.1.3 Worked Example 1: A Serial Robot Arm

A fixed base plus three moving links gives $N = 4$. Three revolute joints nominally impose 15 spatial constraints, leaving $18 - 15 = 3$ freedoms. More directly, for a tree without additional coupling constraints, sum the joint freedoms. Three revolute joints give three.

Whether the endpoint can realize three independently prescribed planar task components is a separate question. A planar position-and-orientation task has three components, but its Jacobian may lose rank at particular configurations. A task dimension cannot prove the mechanism's mobility or guarantee reachability.

26.1.4 Worked Example 2: A Closed Loop

For a planar four-bar mechanism, there are four links including ground and four revolute joints. Each unconstrained planar body has three freedoms and each planar revolute removes two. The nominal mobility is $3(4 - 1) - 4(2) = 1$. Away from singularities, and within a chosen feasible assembly branch, one input parameter determines the others locally.

The spatial expression $6(4 - 1) - 4(5) = -2$ fails because the spatial constraints are dependent in this special planar arrangement. The planar count avoids that overcount, but it too needs regularity. At a toggle configuration, first-order constraint rank may change. Extra infinitesimal motions there need not extend to independent finite motions: second-order closure can restrict them. Inspect the actual constraint equations instead of applying the regular-manifold theorem at a singular point.

26.2 Worked Example — The Golf Swing Model

26.2.1 Model Definition

Use a pelvis fixed in position and orientation. Attach a lumbar body through an ideal three-DOF spherical joint, then a thorax body through another. Attach two identical seven-DOF arm reductions to the thorax. Each arm has shoulder orientation (3), elbow flexion (1), forearm pronation/supination (1), and two wrist angles (2). Fingers, scapular motion independent of the shoulder model, distributed spinal deformation and club flexibility are omitted. Pronation is a relative rotation between forearm bones in anatomy; assigning it to a compound elbow/forearm joint is a reduction, not a claim that elbow flexion and pronation share one anatomical axis.

The physical segment tree is shown schematically below. Joint names belong on the connections; they are not additional massive anatomical segments.

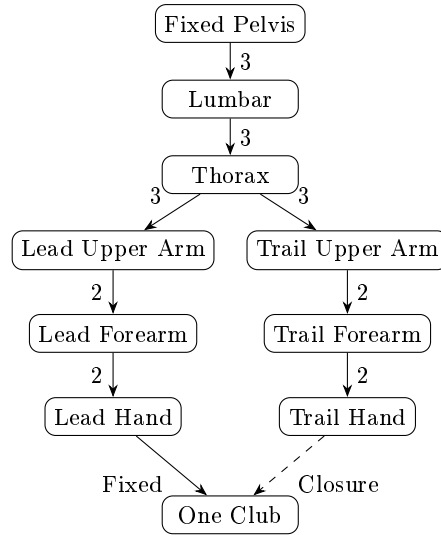


Figure 26.1: Upper-Body Tree and Two-Hand Closure. Numbers Indicate Compound-Joint Freedoms; the Dashed Connection Closes the Second Grip.

26.2.2 A Reconciled Upper-Body Count

The nine bodies are pelvis, lumbar body, thorax, and three segments in each arm. Treat the fixed pelvis as the reference body: $N = 9$. There are four three-DOF joints and four two-DOF joints. Their sum is 20 and their total nominal constraint count is $48 - 20 = 28$. Thus $6(9 - 1) - 28 = 20$. Alternatively, introduce a separate ground body and one fixed pelvis-to-ground joint: both the extra body's six freedoms and the extra joint's six constraints cancel.

This count is for both arms, not for one arm plus both legs. A serial representation of the compound joints introduces virtual intermediate links. It changes the XML link and joint counts together without creating extra independent anatomical motion. Adding an

independent pelvis yaw would add one freedom; changing to a fully floating pelvis would add six. Neither change is consistent with still calling the pelvis fixed.

26.2.3 The Grip Constraint and the Club Body

First rigidly attach the club to the lead hand. A rigid attachment adds a body and a fixed joint but no relative freedom. Requiring the trail-hand frame to match its prescribed grip frame on that club then adds a loop-closure condition. A full rigid relative pose supplies up to six independent scalar constraints. For a feasible configuration with closure rank six, the mobility becomes $20 - 6 = 14$.

The same result follows by initially making the club independent. A rigid club adds six freedoms once, giving $20 + 6 = 26$. Two rigid hand-to-club attachments nominally add twelve constraints. If their combined Jacobian has rank twelve, the result is $26 - 12 = 14$. These are alternative formulations of the same model; subtracting both counts in one formulation would count closure twice.

Finite-stiffness grips are different. Two compliant attachments exert forces and moments but do not each create a new independent club body. If both permit finite relative displacement, the rigid-segment configuration still has $20 + 6 = 26$ freedoms before any other exact constraints. Their force laws determine how those freedoms evolve. In a very stiff limit they may approximate rigid closure, but numerical stiffness, stored energy and contact reactions require separate analysis. Internal grip states or deformable shaft modes add further states only when explicitly modeled.

A point-coincidence grip supplies up to three constraints, a full rigid grip up to six, and a slipping or rolling contact has its own permitted directions. The rank of the combined Jacobian decides the count. Neither the word “grip” nor the number of hands specifies a unique constraint model.

26.2.4 Adding Legs and Ground Contact

For each leg choose hip (3), knee (1) and ankle (2): six freedoms. Adding both legs to our fixed-pelvis upper-body tree gives $20 + 12 = 32$ internal freedoms. Releasing the pelvis to float gives 38. An earlier, different reduction using pelvis orientation (3), one spine joint (3), two legs (12) and two arms (14) also sums to 32; equal arithmetic totals do not make these the same model.

For illustration, give the floating 38-DOF body one independent rigid club, then impose two rigid feet and two rigid grips. If the combined 24 closure rows are independent and geometrically compatible, mobility is $38 + 6 - 24 = 20$. This is a local count for an ideal contact mode. It says nothing about whether the required ground wrenches lie inside friction and support limits, whether a foot should lift, or whether the golfer can supply the required joint torque.

Stack foot and hand constraints before evaluating their rank. When a heel rises or a foot slips, the contact mode and constraint matrix change. Position continuity alone does not settle velocity jumps at impact. A fixed-pelvis upper-body simulation, a floating-body simulation with ground contact, and a prescribed-pelvis simulation answer different questions about how ground interaction influences the club.

26.3 The Human Arm — A Seven-DOF Redundant Manipulator

26.3.1 DOF Count and the Selected Task

The seven-DOF arm is useful because it separates joint motion from a hand task. A hand position lies in $\mathbb{R}^3$. A full rigid hand pose lies in $SE(3)$, a six-dimensional manifold, rather than globally in a single $\mathbb{R}^6$ coordinate chart. Locally one can differentiate task coordinates or use an appropriate spatial/body twist, keeping the frame and angular/linear ordering consistent.

For a seven-DOF arm with a feasible, regular full-pose task of rank six, the instantaneous task-null space has dimension one. For a regular position-only task of rank three it has dimension four. These dimensions assume the shoulder base is fixed and no additional loop or anatomical constraints remove motion. A singular or mechanically restricted arm can have a lower task rank.

More detailed anatomy changes the chosen configuration. For example, [Holzbaur et al. \[2005\]](#) include 15 freedoms spanning shoulder, elbow, forearm, wrist, thumb and index finger, together with 50 muscle compartments. That model is evidence of an explicit modeling choice, not justification for assigning a universal number of freedoms to every golfer or every coaching description.

26.3.2 The Null-Space Motion: Elbow Swivel

Hold the shoulder center S and wrist center W fixed. Idealize upper arm and forearm as rigid segments with lengths a and b . Let $D = \|W - S\|$, with $|a - b| < D < a + b$. The elbow center P satisfies $\|P - S\| = a$ and $\|P - W\| = b$. The two spheres intersect in a circle. With $e = (W - S)/D$ and orthonormal e_1, e_2 perpendicular to e , define

$$c = \frac{a^2 - b^2 + D^2}{2D}, \quad r = \sqrt{a^2 - c^2}, \quad (26.3)$$

$$P(\psi) = S + ce + r(e_1 \cos \psi + e_2 \sin \psi).$$

This describes a workspace circle of possible elbow centers, not necessarily a globally closed curve in joint coordinates. If elbow flexion $\beta = 0$ denotes a straight arm, the cosine law gives $\cos \beta = (D^2 - a^2 - b^2)/(2ab)$. Fixed shoulder-to-wrist distance fixes the flexion angle in this ideal geometry. Swivel therefore requires coordinated rotations; it is not isolated elbow flexion.

Holding hand orientation as well as wrist position requires the shoulder, forearm and wrist rotations to compensate compatibly. Anatomical limits and collisions can truncate the circle to arcs or disconnected feasible pieces. At a straight or fully folded limit the circle degenerates. If $D = 0$ and $a = b$, the above formula is undefined and the sphere-intersection geometry is different. These exceptions are part of the model, not small numerical inconveniences.

26.3.3 Why Redundancy Matters for Golf

At a particular club pose, different feasible internal configurations can give different mass matrices, moment arms, joint loads, collision clearances and future reachable motions. This

is a mechanical reason to study internal configuration rather than judging a swing only by the club's appearance. It does not establish that every configuration is equally useful or that every golfer can realize it.

Two arms gripping the same club cannot generally swivel independently while holding the torso and both grip frames fixed. The combined closure and task Jacobians decide which coordinated variations remain. If a model predicts alternatives, their dynamic feasibility must still be tested with torque, contact, activation and speed constraints over the relevant time interval.

26.3.4 The Pseudoinverse and Null-Space Projection

In local joint coordinates let $\dot{y} = J\dot{q}$. For a wide, full-row-rank J , its Euclidean Moore–Penrose inverse is $J^+ = J^T(JJ^T)^{-1}$. The expression $(J^TJ)^{-1}J^T$ instead requires full column rank and is invalid for this redundant case. In general use an SVD, with an explicit tolerance and physically meaningful coordinate scaling.

If $\dot{y}_d$ lies in the range of J , every exact velocity solution has the form

$$\dot{q} = J^+\dot{y}_d + (I - J^+J)w. \quad (26.4)$$

Here w has joint-velocity units. The first term has minimum Euclidean norm in the chosen coordinates, not automatically minimum energy, muscle activation or joint load. If the target is outside the range, $JJ^+\dot{y}_d$ is a projection and the remaining task residual is nonzero. A damped inverse bounds amplification near rank loss by accepting task error; its nominal “null” projector generally leaks into the task.

26.4 Kinematic Redundancy and the Manifold of Solutions

26.4.1 The Redundancy Problem

Let $\mathcal{Q}_c$ be the admissible configuration manifold after exact closures and let F map it to the declared task:

$$F : \mathcal{Q}_c \longrightarrow \mathcal{Y}, \quad y = F(q). \quad (26.5)$$

For club pose, $\mathcal{Y} = SE(3)$; for a selected point's position, $\mathcal{Y} = \mathbb{R}^3$. A speed objective, impact event or target ball flight adds further structure. In particular, “square face with maximum speed” is not fully specified by the six club-pose coordinates. Speed belongs to velocity state and an optimization objective; impact also depends on time and collision conditions.

With $\dim \mathcal{Q}_c = d$ and task differential rank r at a regular point, local redundancy is

$$d_{\text{null}} = d - r. \quad (26.6)$$

Thus the fixed-pelvis, rigid-two-grip upper-body model has $d = 14$ under our rank assumptions. A further regular six-dimensional club-pose task would leave eight local configuration directions. Using the original open-tree count $20 - 6 = 14$ for that same closed model would omit the grip closure.

26.4.2 The Task Jacobian on Admissible Velocities

Choose a full-column-rank basis $Z(q)$ for $\ker A(q)$, so every admissible velocity can be written $v = Z\nu$. The task velocity is

$$\dot{y} = J(q)v = J(q)Z(q)\nu. \quad (26.7)$$

The relevant rank is that of JZ , not the unconstrained J . The physical task-null velocities satisfy both conditions:

$$\mathcal{N}_q = \{v : A(q)v = 0, J(q)v = 0\}. \quad (26.8)$$

Equivalently their dimension is $n_v - \text{rank}[A^T J^T]^T$. The choice of basis Z changes the numerical representation but not this physical subspace. Choosing a metric is additional: it determines what counts as a small velocity, a costly torque or a statistically large deviation.

26.4.3 The Solution Manifold

For a prescribed attainable target y_0 , define the level set

$$\mathcal{S}(y_0) = \{q \in \mathcal{Q}_c : F(q) = y_0\}. \quad (26.9)$$

If F has constant rank r in a neighborhood of the point, the constant-rank theorem gives a local level-set manifold of dimension $d - r$. This is a local statement. It neither proves the target is reachable nor makes disconnected inverse-kinematics branches one smooth global surface. Joint limits add boundaries and corners; singular points need separate treatment.

An instantaneous null vector preserves the task to first order. For a finite step $q + \epsilon v$, the second-order term in the Taylor expansion can change F . To maintain the task during finite motion, update the null space as the configuration changes and solve or stabilize closure throughout the integration. The curvature of the level set explains why a straight step tangent to it can leave it.

26.4.4 Different Golfers, Different Paths

Equal impact pose does not imply equal impact velocity, equal timing or equal trajectories. Even equal pose and velocity of a chosen club frame do not by themselves establish equal ball flight without matching strike location, ball state, surface interaction and the collision model. Conversely, matching a complete club trajectory can still leave internal motion unobserved.

The existence of this redundancy is a mathematical property of the chosen map. Showing that golfers organize variability along it requires data. Myers et al. [2008] measured torso/pelvis kinematics and ball velocity in 100 recreational golfers; those associations do not establish equality of club trajectories or identify null-space torques. Glazier [2011] discuss competing interpretations and methods for movement variability. Neither source supplies a universal count of freedoms “used” by novices and experts.

26.4.5 Bernstein’s Degrees of Freedom Problem

The coordination question is how an organism produces reliable task behavior with many available internal variables. A mathematical redundancy creates room for coordination; it does not identify the nervous system’s algorithm. Reduced variation in some joint angles can reflect a chosen strategy, physical limits, an imposed task, measurement noise, or correlations among coordinates.

Likewise, the number of principal components explaining a dataset's variance is not the number of anatomical freedoms. It depends on normalization, sampled movements, noise and the retained variance threshold. A learned low-dimensional controller can command a mechanism whose configuration remains high-dimensional. Any claim that skill “releases” a particular number of freedoms must define and measure that number rather than substitute a suggestive metaphor.

26.4.6 Self-Organization, Compensation and Evidence

Constraints and feedback can organize coordinated motion without a controller explicitly solving every equation in this chapter. Optimization is one candidate description, alongside other control models. Its success in a simulation is a prediction under assumed dynamics, information and costs, not direct evidence that the brain computes that optimization.

Restricted motion in one joint can change the feasible set and the effort needed elsewhere. Whether an alternative movement preserves a particular golf outcome is an empirical question involving the individual, task and available control. Kinematic redundancy alone does not establish that compensation succeeds, reduces tissue load or prevents injury. A useful technical account reports the predicted alternatives and the measurements that could distinguish them.

26.5 The Seven-DOF Arm — Redundancy in Miniature

26.5.1 A Concrete Elbow-Circle Calculation

Take $a = 0.30$ m, $b = 0.25$ m and $D = 0.40$ m. The circle center lies $c = 0.234375$ m from the shoulder along the shoulder-to-wrist axis and its radius is approximately 0.18727 m. Sampling the circle preserves both segment lengths; the elbow flexion is constant because $\cos \beta = 0.05$. The wrist orientation still requires compatible compensating rotations. These are dimensions of a teaching geometry, not a participant's measurements.

Adding a prescribed elbow height can remove one additional local degree of redundancy if its differential is independent on the remaining level set. For a full-pose, rank-six arm, that can leave isolated solutions locally. It need not select a unique global configuration: a horizontal plane can intersect the elbow circle at two points. At a tangent intersection or when the entire circle has that height, the rank argument changes. With a position-only task, the four-dimensional task-null space is not exhausted by describing one elbow circle.

26.5.2 Application to the Golf Swing

Suppose two feasible address configurations place both grips and the club identically. Their future behavior under the same torque history need not match: inertia and drift depend on configuration, and contact reactions may differ. If one configuration permits a better future speed/accuracy tradeoff, the reason must be demonstrated through the dynamics and the stated objective.

This also clarifies a common verbal mistake. Once a particular club velocity is prescribed, a task-null velocity cannot increase that same velocity at that instant. It can change posture while preserving the prescribed velocity, and that posture may affect subsequent acceleration or feasible control. “Preserve the present task while improving a future option” is a precise claim that can be investigated.

26.6 Redundancy and Null-Space Control

26.6.1 From Joint Velocities to Applied Torques

For the moment use unconstrained local coordinates with $v = \dot{q}$, symmetric positive-definite mass matrix $M(q)$ and generalized dynamics

$$M(q)\dot{v} + h(q, v) = B(q)u. \quad (26.10)$$

h collects the specified inertial, gravitational and passive terms on the left. B maps the chosen inputs into generalized forces. With independent joint torques, $B = I$. With muscle forces, B contains moment-arm information; with neural commands, activation dynamics may prevent the command from acting directly in this acceleration equation.

For $x = (q, v)$ the torque-driven state equation is

$$\dot{x} = f(x) + G(x)u, \quad G = \begin{bmatrix} 0 \\ M^{-1}B \end{bmatrix}. \quad (26.11)$$

If $B = I$, G is injective: no nonzero joint torque disappears from the full-state derivative. Task redundancy therefore does not imply a nontrivial kernel of the full-state actuation map. A zero-torque counterfactual also retains the terms assigned to h and any retained activation or elastic states according to its declared intervention. Removing the specified torque commands is not synonymous with removing every biological or stored-energy contribution.

26.6.2 The Acceleration-Level Task Map

Differentiating $\dot{y} = Jv$ gives

$$\ddot{y} = JM^{-1}Bu - JM^{-1}h + \dot{J}v. \quad (26.12)$$

At the same configuration and velocity, changing the input by δu changes task acceleration by $JM^{-1}B\delta u$. The input-null condition is therefore $JM^{-1}B\delta u = 0$. It differs from the velocity-null condition $J\delta v = 0$. Dropping $\dot{J}v$ or gravity would also produce an incorrect absolute acceleration even if the input difference were computed correctly.

For an elementary counterexample, take $M = \text{diag}(2, 1)$ and $J = [1 \ 1]$ in consistent chosen units. The velocity $(1, -1)^T$ lies in $\ker J$. Applying the same components as a generalized torque gives $JM^{-1}(1, -1)^T = -1/2$, so the task accelerates. This calculation isolates the mass-matrix distinction without appealing to a complicated human simulation.

26.6.3 A Dynamically Consistent Projection

When J has full row rank, define

$$\begin{aligned} \Lambda &= (JM^{-1}J^T)^{-1}, \\ \bar{J} &= M^{-1}J^T\Lambda, \quad N = I - \bar{J}J. \end{aligned} \quad (26.13)$$

Then $J\bar{J} = I$ and $JN = 0$. The velocity $\bar{J}\dot{y}$ minimizes instantaneous kinetic energy among velocities realizing $\dot{y}$, under this unconstrained model. The corresponding torque decomposition uses the transpose projector:

$$\tau = J^TF + N^T\tau_0, \quad JM^{-1}N^T = 0. \quad (26.14)$$

Indeed $JM^{-1}N^T = JM^{-1} - JM^{-1}J^T\bar{J}^T = 0$ because $\bar{J}^T = \Lambda JM^{-1}$. This is instantaneous dynamic consistency, as developed in [Khatib's operational-space analysis](#). F still must account for drift and the desired task acceleration. A projection does not automatically construct the complete tracking controller.

For the numerical example above, $\bar{J} = (1/3, 2/3)^T$. N maps velocities into $\ker J$, while N^T maps torques into $\ker(JM^{-1})$. Interchanging them would erase the distinction the derivation was meant to capture. Neither null torque nor null velocity implies zero internal energy change. Power remains $\tau^T v$, and a null input can alter subsequent state, limits and task sensitivity.

26.6.4 Contact, Underactuation and Internal Grip Forces

For stationary ideal constraints use

$$M\dot{v} + h = Bu + A^T\lambda, \quad A\dot{v} + \dot{A}v = 0. \quad (26.15)$$

Assume A has independent rows. Eliminating the constraint reaction gives

$$\begin{aligned} K &= AM^{-1}A^T, \\ W &= M^{-1} - M^{-1}A^TK^{-1}AM^{-1}, \\ \dot{v} &= W(Bu - h) - M^{-1}A^TK^{-1}\dot{A}v. \end{aligned} \quad (26.16)$$

Here $AW = 0$: W maps generalized force changes into admissible acceleration changes. The task input map becomes JWB . Redundant constraints require removing dependent rows or a justified generalized inverse; unilateral/frictional contact also requires checking whether the assumed mode remains valid. A floating body generally lacks direct actuation of its base coordinates. Consequently an arbitrary generalized null torque need not be implementable by its joint actuators.

Inputs that leave all admissible accelerations unchanged may nevertheless change constraint reactions. With two hands, opposing forces can cancel in the net club wrench while changing the loads at each grip. This force redundancy is not the same as an elbow-swivel velocity. The [multicontact analysis of Sentis, Park and Khatib](#) connects support constraints, task control and internal forces under explicit actuation/contact assumptions. It does not provide evidence for a particular golfer's neural strategy.

Ideal stationary constraints do no net work on admissible motion because $(A^T\lambda)^T v = \lambda^T A v = 0$. Individual bodies can exchange power through their common constraint forces. Compliant grips can store or dissipate energy, and prescribed moving constraints can supply power. Distinguishing system boundary and constraint law is necessary before interpreting an internal force as an energy source.

26.6.5 Null-Space Optimization and Its Limits

For an unconstrained kinematic task with an Euclidean projector $P = I - J^+J$, one can choose $w = -k\nabla H(q)$ to favor a posture cost H , with $k > 0$. The null contribution to $\dot{H}$ is $-k\nabla H^T P \nabla H \leq 0$. The task-producing term $J^+\dot{q}_d$ can still increase H , so the full cost need not decrease. Joint limits, non-Euclidean metrics and closed loops require the corresponding constrained formulation.

A torque-level objective requires the acceleration dynamics, not substitution of the same w into a torque port. Force limits, muscle activation dynamics, grip feasibility and future

tasks can defeat a locally attractive direction. For golf, a credible optimization specifies the horizon, impact event, observations, controls and tradeoff between speed and accuracy; then reports feasibility and sensitivity, not only the optimizer’s preferred posture.

26.7 URDF — The Unified Robot Description Format

26.7.1 Basic Structure and Frames

URDF describes a tree of links connected by joints. Every non-root link has one parent joint; the root has none. Each joint specifies a parent, child, origin transform and joint type. A conventional revolute joint axis is expressed in its joint frame. The zero-position transform and rotation sequence are part of the definition: changing them while retaining the same angle values changes the physical pose. Consult the [URDF model API](#) for the underlying tree and joint representation, and the selected importer’s documentation for supported features.

A root link called “world”, “ground” or “pelvis” does not, by its name, specify every simulator’s base behavior. A parser may fix it or add a floating joint according to its conventions and setup. Verify the imported joint types, n_q , n_v and world attachment. Likewise, a displayed mesh is not evidence that the inertia or collision model matches it.

26.7.2 A Complete Two-Link Example

The following complete tree has a base and two uniform solid cylinders of lengths (0.30, 0.25) m, masses (2, 1) kg and radius 0.02 m. Their long axes are local x ; the joints rotate about local z . Their COMs lie halfway along each cylinder. URDF cylinders are aligned with their geometry-frame z axis, so the visual and collision frames are rotated by $\pi/2$ about y . The inertial frames remain aligned with the link frames, with the small longitudinal moment in the xx entry.

Continuous joints are used for this unrestricted teaching mechanism. These are not physiological range-of-motion or torque-limit claims. Gravity, actuation and the time integrator are configured by the simulator separately.

```

1 <robot name="teaching_arm">
2   <link name="base" />
3   <link name="link1">
4     <inertial>
5       <origin xyz="0.15 0 0" rpy="0 0 0" />
6       <mass value="2" />
7       <inertia ixx="0.0004" iyy="0.0152" izz="0.0152" ixy="0" ixz="0" iyz="0" />
8     </inertial>
9     <visual>
10      <origin xyz="0.15 0 0" rpy="0 1.57079632679 0" />
11      <geometry>
12        <cylinder radius="0.02" length="0.3" />
13      </geometry>
14    </visual>
15    <collision>
16      <origin xyz="0.15 0 0" rpy="0 1.57079632679 0" />
17      <geometry>
18        <cylinder radius="0.02" length="0.3" />
19      </geometry>
20    </collision>
21  </link>

```

```

22 <joint name="joint1" type="continuous">
23   <parent link="base" />
24   <child link="link1" />
25   <origin xyz="0 0 0" rpy="0 0 0" />
26   <axis xyz="0 0 1" />
27 </joint>
28 <link name="link2">
29   <inertial>
30     <origin xyz="0.125 0 0" rpy="0 0 0" />
31     <mass value="1" />
32     <inertia ixx="0.0002" iyy="0.0053083333333333" izz="0.0053083333333333" ixy="0" ixz="0"
    iyz="0" />
33   </inertial>
34   <visual>
35     <origin xyz="0.125 0 0" rpy="0 1.57079632679 0" />
36     <geometry>
37       <cylinder radius="0.02" length="0.25" />
38     </geometry>
39   </visual>
40   <collision>
41     <origin xyz="0.125 0 0" rpy="0 1.57079632679 0" />
42     <geometry>
43       <cylinder radius="0.02" length="0.25" />
44     </geometry>
45   </collision>
46 </link>
47 <joint name="joint2" type="continuous">
48   <parent link="link1" />
49   <child link="link2" />
50   <origin xyz="0.3 0 0" rpy="0 0 0" />
51   <axis xyz="0 0 1" />
52 </joint>
53 </robot>

```

The downloadable `teaching_arm.urdf` and the generator `TwoLinkArm.urdf()` use the same model. A content test checks that they agree. This example is a verifiable building block; the open two-link tree does not implement a golfer's two-hand closure, foot contacts or muscle physiology.

26.7.3 Link Inertia and Joint Semantics

An inertial origin specifies the COM position and the orientation of the inertia frame relative to the link. The six independent inertia entries describe the symmetric rotational inertia about that COM in that frame, in kg m^2 . They must not be copied from a tensor about a joint origin without a parallel-axis correction. For a uniform cylinder along x ,

$$I_{xx} = \frac{1}{2}mr^2, \quad I_{yy} = I_{zz} = \frac{m}{12}(3r^2 + L^2). \quad (26.17)$$

The first cylinder has moments (0.0004, 0.0152, 0.0152) kg m^2 . For the second cylinder they are approximately (0.0002, 0.00530833, 0.00530833) kg m^2 . Positive mass and symmetric positive-definite rotational inertia are necessary for nondegenerate solid bodies. Principal moments must also obey the triangle inequalities, such as $I_1 + I_2 \geq I_3$; positive eigenvalues alone do not establish physical realizability.

If R expresses the inertia frame in another frame, rotate the tensor as $RI_C R^T$. To move its reference point by displacement d from COM to the new origin, add $m(\|d\|^2 I - dd^T)$.

Rotation and translation perform different operations. Summing segment tensors requires expressing them about the same point and in the same frame.

26.7.4 Compound Joints and Closed Loops

Standard revolute, continuous, prismatic and fixed joints represent their specified relative motions. Floating and planar joint support varies by importer. A three-axis shoulder represented by serial ZYX revolute needs two intermediate virtual links between the physical parent and child. The axes are successive local axes, and the Euler-angle rate map becomes singular at its chart's gimbal-lock orientations. An ideal spherical joint has three physical rotational freedoms without that particular chart singularity.

Virtual links represent coordinate frames, not new pieces of anatomical tissue. A dynamics engine may reject massless moving links or treat them specially. Adding small masses to make a parser accept them changes inertia; it requires an explicit regularization study. Where supported, native multi-DOF joints or multiple joint primitives on one physical body can avoid this artificial mass assignment.

The URDF tree cannot give a child link two parents. Choose a spanning tree and add the omitted rigid closure through the engine's constraint mechanism, or define a compliant attachment with stated force and energy laws. Do not duplicate the club as two independent bodies, silently weld away intended motion, or expect a second parent tag to create a valid two-hand model.

26.8 Building a Full-Body Golf Model

26.8.1 A Construction Sequence That Preserves the Physics

Begin with the nine-body, eight-compound-joint upper-body blueprint and document every physical segment and reference frame. Decide whether the pelvis is fixed, prescribed or floating before adding ground contact. Add legs only when the research question needs their motion and reaction forces; add a head or detailed hands when their inertia or task role matters. Assign the club once, with measured or explicitly idealized mass, COM and inertia.

Then choose the joint representation, including pronation and shoulder-girdle approximation, and verify forward kinematics at recognizable poses. Add one grip to the spanning tree and the second as an explicit closure, or use two compliant attachments. At a feasible reference pose compute the constraint rank and residual. Finally add actuators and their states, contact parameters and the controller. Each addition should have a mechanical check before interpreting a full swing.

Calling this a musculoskeletal model requires more than a file of rigid links. Muscle paths, moment arms, force-length and force-velocity behavior, activation dynamics and any tendon compliance must be defined when those mechanisms support the argument. A tendon path in a simulator is not by itself a muscle activation model. Parameter provenance and validation must match the scientific claim being made.

26.8.2 Anthropometry: Segment Definitions Come First

De Leva [1996] adjusted segment parameters derived from Zatsiorsky-Seluyanov's gamma-scanning measurements of living subjects, using additional anthropometric information to

change reference landmarks. This was not a new cadaver-measurement study. Population estimates provide a starting model, not an individual’s measured inertia. Using their values requires matching the segment boundaries, sex-specific table, COM endpoints and inertia-axis definitions.

For orientation, selected male segment mass fractions from Table 4 are: upper arm 2.71%, forearm 1.62%, hand 0.61%, thigh 14.16%, shank 4.33% and foot 1.37% of body mass, each for one segment. The whole-trunk value is 43.46%. A table containing whole trunk plus its subparts would double-count mass. Nor can a generic “thorax” fraction and an undefined “lumbar” body be assigned independently and presumed to preserve the original partition.

The same table gives the male forearm longitudinal radius of gyration as 12.1% of the specified segment length. That yields $I_{\text{long}} = m(0.121L)^2 = 0.014641mL^2$, not $0.121mL^2$. Alternative endpoint definitions change the normalized coefficient. The published URDF example uses derived cylinder inertias instead, so its geometry and mass properties can be checked without presenting population means as subject anatomy.

26.8.3 Scaling Mass, Length and Inertia

For an unchanged segment definition, scaling body mass from 70 to 80 kg while retaining mass fractions multiplies segment masses by 80/70. If every segment’s geometry is unchanged and only density changes uniformly, its inertia scales by the same factor. If lengths also scale by s_L with geometrical similarity, inertia scales by $s_m s_L^2$ and COM distances by s_L . With unchanged density, $s_m = s_L^3$, giving inertia scaling s_L^5 .

These are conditional scaling laws, not proof that two people have similar segment shapes or densities. If a lumbar segment is repartitioned, its mass, COM and inertia must be recomputed together. A subject-specific optimization must maintain physical consistency while fitting measurements; independently perturbing all tensor entries can generate nonphysical bodies even when each entered number looks plausible.

26.9 From URDF to Simulation — Drake, MuJoCo and Pinocchio

26.9.1 What the Software Supplies

Drake provides multibody modeling within a systems and optimization framework. Its [Parser](#) adds model instances to a plant; the returned model-instance identifiers are not the plant itself. A usable setup creates the plant, parses into it, specifies world attachment and actuation as needed, finalizes the plant, creates a context and supplies state/input before simulation. This distinction matters when translating a short API sketch into an executable experiment.

MuJoCo compiles URDF or its native MJCF into a model and advances a separate data/state object. MJCF exposes richer engine-specific constraints, tendons and actuators. Its [modeling documentation](#) allows multiple primitive joints on one physical body, avoiding dummy bodies for such compound representations. An RL algorithm remains a separate controller/training procedure; a physics engine does not itself establish policy learning or biological validity.

Pinocchio supplies rigid-body algorithms including inverse dynamics, generalized inertia, kinematics and Jacobians, with URDF parsing, as described in its [official feature](#)

[documentation](#). A dynamics library, numerical integrator, contact solver, derivative evaluator and trajectory optimizer are distinct components even when a larger package integrates them. Select and test the required combination rather than ranking software by an unsupported steps-per-second number.

26.9.2 An Executed Import and Dynamics Check

Run the following from the repository root with Python 3.12, NumPy and MuJoCo 3.4.0. The version is stated because signatures can change; the installed version's `mj_fullM` accepts the model, destination dense matrix and sparse inertia array in that order. The XML path is relative to the repository, and the checked root interpretation is fixed. No GUI or training run is required.

```

1 import mujoco
2 import numpy as np
3
4 model = mujoco.MjModel.from_xml_path(
5     "articles/The_Physics_of_Golf/models/teaching_arm.urdf"
6 )
7 model.opt.gravity[:] = [0, -9.81, 0]
8 model.opt.timestep = 0.0005
9 model.opt.integrator = mujoco.mjtIntegrator.mjINT_RK4
10 data = mujoco.MjData(model)
11 data.qpos[:] = [0.3, 0.7]
12 mujoco.mj_forward(model, data)
13 mass = np.empty((model.nv, model.nv))
14 mujoco.mj_fullM(model, mass, data.qM)
15 print(model.nq, model.nv, model.nu)
16 print(mass, data.qfrc_bias.copy())
17 data.qfrc_applied[:] = [1.0, 0.0]
18 mujoco.mj_forward(model, data)
19 print(data.qacc.copy())
20 mujoco.mj_step(model, data, nstep=200)
21 print(data.time, data.qpos.copy(), data.qvel.copy())

```

The imported dimensions are $n_q = n_v = 2$ and $n_u = 0$: the URDF defines no actuator. Here `qfrc_applied` supplies external generalized torques directly, which is useful for a mechanics check but does not model bounded biological actuation. At $q = (0.3, 0.7)$ rad and zero velocity, the independent and imported results agree:

$$M \simeq \begin{bmatrix} 0.22849650 & 0.04961492 \\ 0.04961492 & 0.02093333 \end{bmatrix}, \quad h \simeq \begin{bmatrix} 6.28565628 \\ 0.66254570 \end{bmatrix}. \quad (26.18)$$

The units are kg m^2 for M and N m for h . Applying $(1, 0)$ N m gives initial accelerations approximately $(-33.50095, 47.75165)$ rad/s^2 . The positive distal acceleration follows from dynamic coupling and gravity; it is not evidence of a distal actuator command. After 200 RK4 steps of 0.0005 s, this model reaches $q \simeq (0.135246, 0.915810)$ rad. A numerical trajectory is a result for the specified example, not a verified golf prediction.

26.9.3 Independent Mechanics Behind the Check

Let $c_i = L_i/2$ and let I_i be each cylinder's centroidal zz moment. Expanding the Cartesian translational and rotational kinetic energies gives

$$\begin{aligned} M_{11} &= I_1 + m_1 c_1^2 + I_2 + m_2 (L_1^2 + c_2^2 + 2L_1 c_2 \cos q_2), \\ M_{12} &= M_{21} = I_2 + m_2 (c_2^2 + L_1 c_2 \cos q_2), \\ M_{22} &= I_2 + m_2 c_2^2. \end{aligned} \tag{26.19}$$

With downward global y gravity, potential is $V = g[(m_1 c_1 + m_2 L_1) \sin q_1 + m_2 c_2 \sin(q_1 + q_2)]$ and the zero-velocity bias is ∇V . At nonzero velocity the bias also contains velocity-dependent terms. The tests compare $v^T M v/2$ against independently assembled Cartesian kinetic energy and compare ∇V with finite differences before checking the importer. Agreement between two calls into the same engine would be a weaker verification.

26.10 Sensitivity Analysis — What Happens When Parameters Are Wrong?

26.10.1 Sources of Uncertainty

Relevant uncertainties include segment boundaries and inertias, marker-to-bone motion, joint axes, club properties, external wrenches, initial velocities, actuator parameters and the contact model. State-estimation filtering can correlate errors across time and across coordinates. Finite differences used to estimate acceleration can magnify noise. Sampling frequency alone does not specify motion-capture accuracy or identify the correct model structure.

Uncertainty statements need units and an evidential basis. An orientation error is measured in radians or degrees; an angular-velocity error in rad/s or degrees/s. A perturbation range is an assumed scenario unless it is calibrated to measurements. There is no model-independent percentage by which a forearm mass error changes club speed, and no universal ordering that makes kinematic measurement accuracy more important than inertia accuracy for every output.

26.10.2 Fixed-Control and Feedback Sensitivities

For a smooth fixed-mode system $\dot{x} = f(x, u, p)$, differentiate with respect to a parameter vector p along a specified input history. With $S = \partial x / \partial p$,

$$\dot{S} = f_x S + f_p, \quad S(0) = \frac{\partial x_0}{\partial p}. \tag{26.20}$$

If the controller is $u = \pi(x, p)$, the corresponding terms become $(f_x + f_u \pi_x)S + f_p + f_u \pi_p$. Holding a torque history fixed, replaying recorded commands and reoptimizing the controller for each parameter are different experiments. Report which is performed. An optimizer's improved value after reoptimization does not measure the same sensitivity as a golfer receiving an unanticipated perturbation.

For a fixed terminal time and output $z = \psi(x(T), p)$, $dz/dp = \psi_x S(T) + \psi_p$. A differentiable impact-time definition $H(x(t_*), p, t_*) = 0$ adds event-time sensitivity:

$$\frac{dt_*}{dp} = -\frac{H_x S(t_*) + H_p}{H_x f(t_*) + H_t}. \quad (26.21)$$

The denominator must be nonzero. Terminal output derivatives then include the time-shift term $\psi_x f dt_*/dp$ and any explicit time dependence of ψ . Grazing events, changing contact modes and discontinuous controllers may invalidate this smooth calculation and require event/reset analysis.

26.10.3 A Bounded Numerical Interpretation

If a stated model gives $\partial v_{\text{club}}/\partial m_f = -0.05$ (m/s)/kg and the only perturbation is $\delta m_f = \pm 0.2$ kg, the first-order speed change is ∓ 0.01 m/s. At 45 m/s this is about 0.0222% of the nominal speed. These are hypothetical supplied numbers, not a measured golf sensitivity. Other parameter and state errors can dominate or cancel this contribution.

For a covariance Σ of a parameter/state error vector and local output sensitivity row s , the linearized variance is $s\Sigma s^T$. Correlations matter. For interval assumptions, propagate the stated admissible set rather than labeling the result a confidence interval. Check non-linear effects by finite perturbations and numerical convergence; do not infer a probability distribution from a convenient percentage sweep.

26.10.4 Sensitivity to Initial Conditions and Robustness

Initial-state perturbations obey the variational equation $\dot{\Phi} = f_x \Phi$ for the specified open-loop system, or the corresponding closed-loop Jacobian. Large finite-time amplification is not by itself evidence of chaos. Proving chaotic dynamics needs a more specific dynamical claim than observing that a small start error changes an impact result.

For a useful robustness assessment, define the output or conclusion, the feasible uncertainty set, the controller and the contact/event rules. Compare multiple time steps and derivative perturbation sizes. Include parameter correlations and physically realizable inertias. State which conclusions persist and which change. A successful arbitrary $\pm 20\%$ sweep establishes only the tested scenarios; it cannot certify robustness outside them or determine the best measurement investment without a sensitivity comparison.

26.11 Building Intuition: From URDF to Golf Physics

26.11.1 The Redundancy Principle

The key connection is now explicit. Constraints determine the motion the body can make; the task map determines which of those motions appear at the club; inertia and actuation determine how forces change those motions. An internal configuration can preserve the present club pose while changing future acceleration authority. Grip-force distribution can preserve a resultant club wrench while changing loads on the arms. Those are complementary mechanical possibilities, each with its own observable consequences.

Redundancy is therefore an opportunity to investigate alternative feasible solutions, not a guarantee of greater efficiency, safety or consistency. A model that omits a material wrist,

grip or elastic state can hide a relevant pathway even if its joint-angle fit is excellent. A model that adds detail without identifiable parameters can create unwarranted confidence. The appropriate resolution is the one needed to answer and test a stated question.

26.11.2 The Identification Problem

Club trajectory alone generally does not recover full-body joint motion. Full-body kinematics alone generally does not recover unknown external contact wrenches. If inertial parameters, full state/acceleration and all external loads are known, inverse dynamics determines the required net generalized force in the specified model. Splitting that net force among muscles remains another inverse problem, especially with multiple muscles and co-contraction.

Likewise, a fitted inverse-optimal-control cost is not uniquely the nervous system's objective. Multiplying a cost by a positive constant preserves its minimizers, and different costs can agree along one observed trajectory while disagreeing elsewhere. Model error, unmeasured constraints and assumed feedback information can all be absorbed into fitted weights. Cross-condition predictions and controlled perturbations are more informative than fitting one successful swing.

A strong study therefore states the forward model and which quantities are measured, reports parameter/state uncertainty and residuals, and tests predictions on conditions withheld from fitting. Depending on the question, discriminating data might include force-plate wrenches, grip loads, additional segment kinematics or activation-related measurements with their own observation model. No single sensor automatically closes every level of this inverse problem.

26.11.3 What We Can Now Claim

We can count regular mobility for a declared model, compute task-null directions on its admissible velocity space, distinguish them from torque-null inputs, and verify a rigid-body file against mechanics. These tools make competing explanations concrete. They do not yet establish a unique control strategy, a complete musculoskeletal representation or a universal best swing.

The next step is to ask how control works with limited information, delay and noisy measurements. Those constraints belong alongside anatomy, mechanics and task geometry in an explanation of the golf swing. Preserving their distinctions makes their interaction more understandable, not less.

26.12 Primary Sources and Reading Guidance

1. [Lynch and Park, Mobility](#), and [Singularities](#): independent constraints, task Jacobians and rank; these are mechanisms lessons, not human motor-control measurements.
2. [Khatib, Dynamic Consistency](#): operational-space inertia and torque projection. Follow the dimensions of every matrix when reading the scanned equations.
3. [Sentis, Park and Khatib \(2010\)](#): multicontact and internal-force control, under the paper's contact and actuation assumptions.

4. [de Leva \(1996\)](#): adjusted segment definitions, Table 4 and reference endpoints. Do not combine whole-trunk and trunk-subsegment fractions.
5. [Holzbaur, Murray and Delp \(2005\)](#): an explicit arm and hand model; compare retained joints and muscles before transferring its parameters.
6. [Myers and Colleagues \(2008\)](#) and [Glazier \(2011\)](#): torso/pelvis associations and the interpretation of movement variability, respectively; neither identifies the torque-null controller of a golfer.
7. [MuJoCo 3.4 Python Bindings](#), the linked modeling manual, URDF API, Drake Parser and Pinocchio feature documentation: verify importer semantics and runtime version independently of scientific validation.

26.13 Chapter Exercises

1. **DOF Counting:** Reconstruct the nine-body upper-body model's 20 freedoms using both joint summation and the spatial constraint count. Add the club first as a rigid lead-hand attachment and then as an independent body. Under full-rank rigid-grip closure, obtain 14 in both formulations. Repeat with two compliant grips, then explain why nominal foot constraints must be stacked with grip constraints before making a full-body count.
2. **Null-Space Motion:** For the regular seven-DOF arm distinguish a rank-three position task from a rank-six pose task. Derive the elbow circle for fixed shoulder/wrist centers and prove flexion is constant. Add a prescribed elbow height, stating the independence assumption and the possible global multiplicity. Explain why one circle does not parameterize all four freedoms of the position-only null space.
3. **URDF Construction:** Starting from the complete two-link file, verify every COM, cylinder orientation and principal inertia. Design a kinematic spine/arm tree with two three-axis compound joints. Count the intermediate virtual links needed for six serial revolute and identify the reference body. Describe the selected engine's treatment of massless frames or native multi-DOF joints before claiming a valid dynamic model; do not assign undocumented artificial tissue mass.
4. **Anthropometric Scaling:** Scale a 70 kg model to 80 kg with unchanged segment geometry, then repeat with an independent uniform length factor s_L . Derive the inertia factors in both cases and under fixed-density geometrical similarity. Explain why changing trunk partitions requires recomputing COM and inertia, not only multiplying masses.
5. **Sensitivity Calculation:** Use the hypothetical derivative $-0.05 \text{ (m/s)}/\text{kg}$ with mass perturbation $\pm 0.2 \text{ kg}$ to calculate the local speed change and its fraction of 45 m/s. State what additional evidence is needed to call this a confidence interval. Derive how a differentiable impact-time condition modifies the output derivative and identify the grazing-event failure condition.

6. **Loop Constraints and Forces:** Contrast a point grip, rigid grip and compliant grip. For $M = \text{diag}(2, 1)$ and $J = [1 \ 1]$, show that $(1, -1)^T$ is velocity-null but not torque-null. Construct $\bar{J}$, N and N^T and verify their respective identities. Explain how two opposing grip forces can alter hand loads without altering the net club wrench.
7. **Redundancy in Golf:** Formulate two dynamically feasible candidate swings that agree in a declared club task but differ internally. Specify whether the agreement is at one impact pose, impact state or the entire trajectory. Identify an observable that could distinguish the candidates and a perturbation that tests their predicted future behavior. Explain why a fitted trajectory alone does not identify the brain's unique objective or prove reduced injury risk.

Part VIII

The Kinetic Chain and Performance

Chapter 27

The Kinetic Chain: Motion, Work, and Control in the Golf Swing

The central question of the kinetic chain is how a golfer's coordinated motion produces a useful club state at impact. The answer involves muscle work, contact forces, changing geometry, inertia, elastic storage, and timing. None of these can replace the others. A pelvis that slows while the club accelerates is an observation worth explaining; it is not, by itself, a measurement of energy moving from pelvis to club. This chapter develops the balances needed to make that explanation testable.

The useful intuition behind the whip analogy is that a connected system can develop high distal speed without every part moving equally fast. Its limitation is equally important: a golfer has active joints, two hands on a club, two changing foot contacts, and a three-dimensional body. A prescribed sequence of peaks is not a mechanical law for that system. We will use simple rotors and a moving-grip club to isolate mechanisms, then reconnect those mechanisms to measurements and control. The examples are deliberately specified teaching models, not fitted swings or prescriptions for an athlete.

27.1 Define the Observation Before Explaining It

27.1.1 Segments, Joints, and Comparable Rates

A segment is a body with a defined mass, center of mass, and inertia. A joint is a connection with relative motion and transmitted force or moment. A wrist is a joint region, not an additional rigid body whose energy can simply be added to forearm and hand energy. Similarly, a measured shoulder-line orientation may describe the thorax; counting both as separate energy reservoirs can count the same mass twice. A whole-body energy calculation requires disjoint segment definitions.

For a segment, the angular velocity vector describes rotation relative to a declared frame. A joint rate describes relative motion between adjacent segments. Neither is clubhead linear speed. The relation between clubhead velocity and generalized velocity is a Jacobian relation, $\mathbf{v}_C = J_C(q)\dot{q}$, including base translation when the base moves. Changing posture changes this map. Comparing a pelvic rate in degrees per second with clubhead speed in miles per hour does not establish amplification without geometry and a unit conversion.

Three-dimensional orientation also requires care. Euler-angle derivatives generally are not the Cartesian components of angular velocity. If $\boldsymbol{\omega} = E(\theta)\dot{\theta}$, a physical moment $\boldsymbol{\tau}$ has generalized force $E^T\boldsymbol{\tau}$, so that power is preserved. Angular-speed magnitude, a selected angular-velocity component, and the derivative of a projected turn angle can have different peak times.

27.1.2 A Sequence Is a Measurement Result

Define downswing onset, the instant immediately before ball contact, the measured quantity, its sign convention, smoothing, and the method for finding peaks. If t_0 and t_I are those events, normalized time is $s = (t - t_0)/(t_I - t_0)$. A peak at a percentage of downswing time is meaningful only under that event definition. Finite sampling and filtering create timing uncertainty; a broad plateau should not be represented as a precisely ordered instant.

Peak segment speed and peak acceleration are distinct events. For a smooth signed velocity, an interior velocity extremum has zero acceleration; an acceleration extremum concerns its derivative. Nothing makes the peak speed of one segment equivalent to the peak acceleration of its neighbor. A speed magnitude introduces another distinction: $d\|\boldsymbol{\omega}\|/dt = \boldsymbol{\omega} \cdot \dot{\boldsymbol{\omega}}/\|\boldsymbol{\omega}\|$ away from zero speed.

Vena and colleagues' five-subject study, using instantaneous screw-axis angular velocities, reported support for a proximal-to-distal kinematic sequence in two subjects, rather than a universal order [Vena et al., 2011]. That result does not say sequencing is useless. It says a proposed invariant sequence must survive explicit measurement choices and between-player variation. We therefore do not assign universal pelvic, thoracic, arm, or wrist peak percentages here. Ball contact also changes the dynamics: pre-impact club speed should not be inferred from a peak several milliseconds into a collision or follow-through.

27.2 Force and Power Balances for a Connected Body

27.2.1 Start at Each Segment's Center of Mass

For a rigid segment of mass m , central inertia tensor I_G , and center G , Newton–Euler equations are

$$m\mathbf{a}_G = \sum_k \mathbf{F}_k + m\mathbf{g}, \quad \left. \frac{d\mathbf{H}_G}{dt} \right|_{\text{inertial}} = \sum_k [(\mathbf{r}_k - \mathbf{r}_G) \times \mathbf{F}_k + \boldsymbol{\tau}_k]. \quad (27.1)$$

In body-fixed principal coordinates the left angular term is $I_G\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I_G\boldsymbol{\omega})$. All vector components must be represented consistently. A scalar $I\ddot{\theta} = \tau$ is a special fixed-axis equation, not the general balance about a moving anatomical joint. Forces at both proximal and distal boundaries contribute moments about the center of mass.

The corresponding kinetic energy and rate are

$$T = \frac{1}{2}m\|\mathbf{v}_G\|^2 + \frac{1}{2}\boldsymbol{\omega}^T I_G \boldsymbol{\omega}, \quad \dot{T} = \sum_k (\mathbf{F}_k \cdot \mathbf{v}_k + \boldsymbol{\tau}_k \cdot \boldsymbol{\omega}) + m\mathbf{g} \cdot \mathbf{v}_G. \quad (27.2)$$

Here $\mathbf{v}_k$ is the material-point velocity at the application point. A force can supply power to a translating segment even if the joint has no relative translation. Moving the point at which a wrench is represented changes its moment and the associated velocity together; the total wrench power stays the same. This is why a club calculation must retain grip force power.

27.2.2 Transfer, Generation, Storage, and Dissipation

At a coincident ideal joint, the two segments share the same point velocity. Equal and opposite joint forces then make powers that cancel when both bodies are included. They

can nevertheless transfer energy across the segment boundary. For an applied joint couple τ acting on the distal body and $-\tau$ on the proximal body, the combined power is

$$P_j = \tau \cdot (\omega_d - \omega_p). \quad (27.3)$$

An ideal constraint couple does zero power along the allowed relative motion. An actuator couple need not. In a planar hinge $P_j = \tau_j \dot{\phi}_j$, where $\phi_j = \theta_d - \theta_p$. Calling every joint moment a “constraint torque” hides precisely the work we need to account for.

A spring and damper are another distinct case. Let $\delta = \theta_p - \theta_d - \delta_0$, and let $k\delta + c\dot{\delta}$ act positively on the distal rotor. The combined spring/damper power is

$$P_p + P_d = -(k\delta + c\dot{\delta})\dot{\delta} = -\frac{d}{dt} \left(\frac{1}{2}k\delta^2 \right) - c\dot{\delta}^2. \quad (27.4)$$

Thus proximal power loss need not equal distal power gain at that instant. With $k = 100$ Nm/rad, $c = 2$ Nm s/rad, $\delta = 0.1$ rad, and angular speeds 5 and 3 rad/s, the two powers are -70 and 42 W. The remaining 28 W is 20 W of storage and 8 W of dissipation. This single example exposes the ambiguity in a claim that all proximal braking becomes distal acceleration.

27.2.3 Whole-System Equations Do Not Identify Individual Muscles

In local generalized coordinates with $v = \dot{q}$, a constrained mechanical model can be written

$$M(q)\dot{v} + h(q, v) = B(q)u + J(q)^T \lambda, \quad J(q)v = 0. \quad (27.5)$$

The term h includes the chosen velocity-dependent, gravitational, elastic, and dissipative contributions. The multipliers λ are solved together with acceleration and constraint consistency. They are not an arbitrary derivative of a separate “constraint Lagrangian.” Multiplying a generalized-force residual by M^{-1} yields acceleration, not torque.

Joint-coordinate mass matrices are generally coupled. A block-diagonal inertia representation is possible for separate bodies before connectivity is imposed; it does not make the assembled joint dynamics independent. An inverse-dynamics result is a model-dependent net generalized force. Many muscle-force combinations, including different amounts of co-contraction, may produce it. Segment kinematics alone do not select one biological realization.

27.3 What Coupled Oscillators Actually Predict

27.3.1 A Solvable Two-Rotor Counterexample

Consider two fixed-axis rotors with absolute angles θ_1, θ_2 , inertias I_1, I_2 , one torsional spring k , no gravity or damping, and a constant external torque τ on rotor 1. Set both angles and speeds to zero at $t = 0$. These are not anatomical hip and shoulder equations; they isolate elastic coupling:

$$I_1 \ddot{\theta}_1 = \tau - k(\theta_1 - \theta_2), \quad I_2 \ddot{\theta}_2 = k(\theta_1 - \theta_2). \quad (27.6)$$

Define the inertia-weighted mean $\bar{\theta} = (I_1\theta_1 + I_2\theta_2)/(I_1 + I_2)$ and separation $\delta = \theta_1 - \theta_2$. Then

$$\ddot{\bar{\theta}} = \frac{\tau}{I_1 + I_2}, \quad \ddot{\delta} + \Omega^2 \delta = \frac{\tau}{I_1}, \quad \Omega^2 = k \left(\frac{1}{I_1} + \frac{1}{I_2} \right). \quad (27.7)$$

The response combines an accelerating rigid mode and an oscillatory relative mode. It is not two independent oscillators with frequencies $\sqrt{k/I_i}$. For $I_1 = 2$, $I_2 = 0.5 \text{ kg m}^2$, $k = 100 \text{ Nm/rad}$, and $\tau = 50 \text{ Nm}$,

$$\Omega = \sqrt{250} \text{ rad/s}, \quad \ddot{\theta}_1 = 20 + 5 \cos(\Omega t), \quad \ddot{\theta}_2 = 20[1 - \cos(\Omega t)]. \quad (27.8)$$

Rotor 1 accelerates positively throughout: its acceleration never falls below 15 rad/s^2 . It therefore has no interior speed peak, although rotor 2 has its first acceleration maximum at $t = \pi/\Omega \approx 0.19869 \text{ s}$. The first proximal speed maximum on $[0, 1] \text{ s}$ is simply the endpoint. Coupling alone has not produced the proposed universal peak order.

Integrating gives $\bar{\theta} = 10t^2$ and $\delta = 0.1[1 - \cos(\Omega t)]$. The individual angles are $\theta_1 = \bar{\theta} + 0.2\delta$ and $\theta_2 = \bar{\theta} - 0.8\delta$. Their total kinetic-plus-spring energy equals $50\theta_1$ joules, the input work from rest. This is a useful independent check on signs and numerical integration. A different torque history, damping, initial preload, or terminal objective can yield different ordering; that is a result to compute, not an exception to hide.

27.3.2 Normal Modes Belong to the Coupled System

For a linear chain $M\ddot{\theta} + C\dot{\theta} + K\theta = \tau(t)$, undamped modes solve $K\psi = \omega^2 M\psi$. Free overall rotation gives a zero-frequency mode when there is no spring to ground. With $M = \text{diag}(2, 1.2, 0.4) \text{ kg m}^2$ and neighboring springs 120 and 100 Nm/rad,

$$K = \begin{bmatrix} 120 & -120 & 0 \\ -120 & 220 & -100 \\ 0 & -100 & 100 \end{bmatrix}. \quad (27.9)$$

There is one rigid mode and two elastic modes. Each mode involves the chain, not a single segment. Relative-joint damping gives a similarly assembled matrix; terms $-c_i \dot{\theta}_i$ instead describe damping to a stationary reference. Neither model automatically imposes a quarter-period intersegment delay. Even a single oscillator's phase depends on whether one compares force, displacement, velocity, or acceleration and whether the forcing is transient or periodic.

For golf, this means that effective inertia, stiffness, posture, muscle activation, and boundary conditions jointly shape the response. A smaller distal inertia alone does not prove a particular timing sequence. Linear modes are local approximations; large rotations, changing contacts, and activation-dependent stiffness can make those modes vary through a swing.

27.4 X-Factor: Relative Motion Before Biological Interpretation

27.4.1 Use a Declared Angular Convention

One workable scalar convention defines $\alpha_X = \theta_T - \theta_P$, with thorax and pelvis turn angles measured from address and positive toward the backswing. Then $\dot{\alpha}_X = \dot{\theta}_T - \dot{\theta}_P$. If the pelvis turns toward the target while the thorax is still turning away, the separation increases in this convention. At a smooth interior maximum the two turn rates are equal; no particular number of degrees of pelvic motion is required.

These turn angles must come from a stated projection or orientation decomposition. The angle between two rotation axes is a different quantity. For example, thorax and pelvis could rotate about parallel vertical axes while their turn angles differ substantially. A full relative orientation $R_P^T R_T$ contains three-dimensional information that a scalar X-factor does not. Marker placement, tilt, side bending, angle unwrapping, and handedness affect interpretation.

Define the reference event t_0 and an early-downswing window $[t_0, t_1]$ before computing stretch:

$$\Delta\alpha_X = \max_{t \in [t_0, t_1]} \alpha_X(t) - \alpha_X(t_0). \quad (27.10)$$

Including t_0 makes this number nonnegative by definition; zero means no increase in that window. It does not prove that every golfer exhibits a positive stretch. A signed change between two specified events can instead be negative. Maximum pelvic and thoracic backswing positions need not occur simultaneously, and two zero angular velocities do not imply zero relative angle or zero elastic preload.

27.4.2 What the Classic Study Measured

Cheetham and colleagues compared 10 highly skilled and 9 less-skilled right-handed golfers using 5-irons and electromagnetic tracking. Their reported 30 Hz sampling and pelvis/T3 rotation measurements concern the transition region. They found a greater early-downswing X-factor increase in the higher-skill group, but no statistically significant between-group difference at the top of the backswing [Cheetham et al., 2001]. This group comparison does not establish that increasing one player's separation will increase distance.

The observed angle change also does not measure tissue strain energy, spindle response, or muscle-fiber shortening. For an assumed linear torsional spring, $U = \frac{1}{2}k(\alpha_X - \alpha_0)^2$ is a model with a specified rest angle and stiffness. In a living trunk, distributed joints, muscle routing, tendon compliance, activation, and passive tissues must justify that approximation. An increased angle can alter storage and muscle operating conditions while also changing loads and the future club path. The measured association is a reason to investigate that interaction, not a warrant to maximize separation.

27.5 Wrist Motion and Work at the Grip

27.5.1 Lag Is Geometry, Not a Muscle State

In a planar approximation, define an absolute arm angle θ_a and absolute club angle θ_c from the same reference; their signed relative angle is $\phi = \theta_c - \theta_a$. An included angle from directed forearm and shaft vectors is another convention and can differ by a supplement. State the directions and zero position. Three-dimensional wrist motion includes more than a single lag coordinate, and a flexible shaft adds deformation. A projected angle does not uniquely specify the clubface orientation.

Neither hand speed nor clubhead speed determines muscle shortening speed. A muscle-tendon length $\ell(q)$ has rate $\dot{\ell} = (\partial\ell/\partial q)\dot{q}$; fiber speed additionally depends on tendon compliance and pennation. Classical force-velocity behavior concerns the contractile element under declared activation and length conditions. It cannot be applied by converting an absolute club speed directly into a universal wrist torque limit. Active lengthening and isometric force are also different operating conditions from concentric shortening.

Sprigings and Neal’s planar three-segment simulation found that suitably timed active wrist torque could increase clubhead speed by about 9 percent in its modeled conditions [Sprigings and Neal, 2000]. That is a simulation result with particular torque generators and constraints, not an experimentally established instruction to add a late wrist action. It is nevertheless enough to reject the universal claim that wrist torque cannot contribute at high club speed. Conversely, a measured small net wrist moment does not prove that all wrist muscles are inactive.

27.5.2 The Club Receives a Wrench, Not Only a Torque

For a rigid club, with grip point H , external grip force $\mathbf{F}_H$, and grip couple $\boldsymbol{\tau}_H$, its pre-impact power balance is

$$\frac{d}{dt}(T_c + U_{g,c}) = \mathbf{F}_H \cdot \mathbf{v}_H + \boldsymbol{\tau}_H \cdot \boldsymbol{\omega}_c + P_{\text{other}}. \quad (27.11)$$

The term P_{other} accounts for effects such as aerodynamic loading. For a flexible club, include shaft strain energy and dissipation and use the grip section’s angular velocity for its applied couple. The power entering the club is not the same as the combined power of an anatomical wrist actuator acting between two moving bodies.

Even when $\boldsymbol{\tau}_H = 0$, a translating grip can do work through $\mathbf{F}_H \cdot \mathbf{v}_H$. The two hands may also produce a resultant couple through their separated forces. A zero net couple at one chosen port does not remove force transmission or make the club an isolated body. Nesbit and Serrano’s analysis explicitly modeled both force and torque work at the club interface [Nesbit and Serrano, 2005]. The balance here follows rigid-body mechanics independently of any biological interpretation of their reconstructed loads.

27.6 A Release Model That Retains the Moving Grip

27.6.1 Derive the Unactuated Equation

Use an inertial vertical plane with horizontal x and upward y . The rigid club’s center is $\mathbf{r}_G = \mathbf{r}_H + \ell \mathbf{e}$, where $\mathbf{e} = (\cos \theta, \sin \theta)$ points from grip to center and $\mathbf{e}_\perp = (-\sin \theta, \cos \theta)$. Let m be mass, I_G central inertia, and $I_H = I_G + m\ell^2$. Eliminating the grip force between translation and rotation gives

$$I_H \ddot{\theta} = \tau_H + m\ell(\mathbf{g} - \mathbf{a}_H) \cdot \mathbf{e}_\perp. \quad (27.12)$$

To see the origin of the forcing, substitute $\mathbf{a}_G = \mathbf{a}_H + \ell \ddot{\theta} \mathbf{e}_\perp - \ell \dot{\theta}^2 \mathbf{e}$ into $m\mathbf{a}_G = \mathbf{F}_H + m\mathbf{g}$, then take moments about G . The radial $\dot{\theta}^2$ term has no moment about the grip; the acceleration of the grip has a tangential projection that does. Omitting $\mathbf{a}_H$ changes the problem after “release.”

For an illustrative instantaneous state, set $m = 0.2$ kg, $\ell = 1$ m, $I_G = 0.02$ kg m², $\theta = 0$, $\dot{\theta} = 2$ rad/s, $\mathbf{v}_H = (0, -3)$ m/s, $\mathbf{a}_H = (0, -20)$ m/s², and $\tau_H = 0$. Then $I_H = 0.22$ kg m² and $\ddot{\theta} \approx 9.2636$ rad/s². The grip force is approximately $(-0.8, -0.18527)$ N and supplies 0.55582 W. Directly differentiating $T_c + U_{g,c}$ gives the same rate. Angular acceleration is present with no applied grip couple because the grip trajectory is still imposed and its force still acts.

Changing only grip acceleration to free fall, $\mathbf{a}_H = \mathbf{g}$, gives zero angular acceleration. A stationary grip instead gives $\ddot{\theta} \approx -8.9182 \text{ rad/s}^2$ at the same horizontal orientation. These are three different boundary conditions, not three explanations of the same swing. Centrifugal or Coriolis terms in other coordinates reorganize the dynamics; they do not introduce an additional energy source.

27.6.2 Holding, Releasing, and Feasibility

Suppose an idealized holding condition imposes $\theta = \theta_a(t) + \phi_0$. The required grip couple is

$$\tau_{\text{hold}} = I_H \ddot{\theta}_a - m\ell(\mathbf{g} - \mathbf{a}_H) \cdot \mathbf{e}_\perp. \quad (27.13)$$

A bound $|\tau_{\text{hold}}| \leq \tau_{\text{max}}$ checks whether this imposed trajectory is feasible in the toy model. It does not assert that a human wrist locks perfectly until its torque limit is reached. A controller could change its command earlier; anatomical joints need not behave as binary latches. For a smooth release without an impulse, initialize the released motion with the angle and angular velocity immediately before release and continue the specified grip trajectory. If a constraint change produces an impulse, its momentum and energy effects require a separate event model.

A claim that later release improves speed must specify which intervention changes, which upstream controls or trajectories remain fixed, the admissible torque/force and joint-range limits, and the endpoint. Compare states at a common impact surface with correct clubface orientation, not just at an arbitrary common clock time. If grip motion is prescribed, its force and work generally change between alternatives. Giving both alternatives the same torque history does not make their input work equal. There is no universal percentage benefit or optimal delay that follows from this pendulum equation alone.

27.7 Ground Reaction: Momentum Channel and Contact Constraint

27.7.1 Compute the Full Moment Vector

Choose a right-handed frame: x along the target line, y horizontally across it, and z upward. About origin O , a force applied at displacement $\mathbf{r} = (r_x, r_y, r_z)$ and a free moment $\mathbf{m}$ produce

$$\boldsymbol{\tau}_O = \mathbf{r} \times \mathbf{F} + \mathbf{m} = \begin{bmatrix} r_y F_z - r_z F_y \\ r_z F_x - r_x F_z \\ r_x F_y - r_y F_x \end{bmatrix} + \mathbf{m}. \quad (27.14)$$

Vertical force at a horizontal offset produces horizontal-axis moments, not a vertical-axis yaw moment. Horizontal force at a vertical offset contributes horizontal-axis moments too. Summing those components as if all were pelvic yaw torque is invalid.

For a numerical example about a stated origin, let $\mathbf{r} = (0.4, 0.2, -1) \text{ m}$ and $F_z = 1.5(80)(9.81) = 1177.2 \text{ N}$. Vertical force alone gives $(235.44, -470.88, 0) \text{ Nm}$. Add $F_x = 100 \text{ N}$, $F_y = 200 \text{ N}$, and a free yaw moment $m_z = 10 \text{ Nm}$: the complete moment, in Nm, is

$$\boldsymbol{\tau}_O = (435.44, -570.88, 70)^T. \quad (27.15)$$

If that net yaw moment stays constant for 0.05 s, its yaw angular impulse is 3.5 N m s. These are declared illustrative inputs; the calculation is not a prediction of hip angular speed.

27.7.2 Whole-Body Angular Momentum Is Not Pelvic Rotation

For the complete golfer-plus-club system about a fixed inertial origin,

$$\dot{\mathbf{H}}_O = \sum \boldsymbol{\tau}_{O,\text{external}}. \quad (27.16)$$

Joint forces and couples are internal to that system. Ground wrenches, gravity, aerodynamic loads, and ball contact where present are external. About the total center of mass, uniform gravity has zero resultant moment, and the external-moment balance also holds for angular momentum defined relative to that center. Other moving reference points require the appropriate transport terms. The force example above gives a change in whole-system angular momentum only when all other relevant external moments are accounted for.

One cannot divide that moment by an isolated pelvic inertia to obtain pelvic acceleration in a coupled swing. Nor does zero external yaw moment prevent the pelvis from rotating: internal actions can produce counter-rotation elsewhere while conserving total angular momentum. Ground contact permits exchange of momentum with the environment and changes feasible motion; it is not the sole possible cause of every observed segment rotation.

27.7.3 Center of Pressure and Ground Work

A force plate's resultant force, moment, and center of pressure are related but not interchangeable. The center of pressure describes a pressure-resultant location on the contact plane under its adopted convention. It does not by itself determine horizontal shear force or the residual free yaw moment. Near vanishing normal force, center-of-pressure estimates can become ill-conditioned. Combining two plates also requires a common origin and frame.

At a perfectly stationary, non-slipping contact, ideal reaction forces do zero mechanical work because the material contact points have zero velocity. Yet they can provide a nonzero moment and angular impulse. Muscles supply mechanical work internally from chemical energy while contact constrains the motion. Real shoes, ground compliance, slipping, and foot deformation can add storage or work terms. A peak ground force before impact does not mean zero force at impact, and it does not determine when the center-of-mass speed peaks. Timing force, impulse, work, and segment velocity answers four different questions.

27.8 Close the Energy Budget Before Naming Efficiency

27.8.1 Choose the System and the Time Interval

For a pre-impact mechanical model including the body, club, gravity, and chosen elastic elements, write

$$E = T_{\text{body}} + T_{\text{club}} + U_g + U_e, \quad E(t_I) - E(t_0) = W_{\text{act}} + W_{\text{ext}} - D. \quad (27.17)$$

Here W_{act} is signed actuator work at the declared joint ports, W_{ext} is work from the other external interactions not already represented by potential energy, and $D \geq 0$ is the modeled dissipation. Avoid counting an elastic element both in U_e and again as an independent actuator energy input. An ideal frictionless model with ongoing positive actuator work does not conserve mechanical energy; it obeys this balance with increasing E .

Backswing history enters through initial position, velocity, and stored energy. The top of a backswing is not generally a state of complete rest. Gravity needs a scale check: raising an 80 kg whole-body center of mass by 0.1 m changes potential energy by 78.48 J. Individual segment and club height changes must also be handled consistently. Small-looking displacement is not a sufficient argument to delete the term.

Energy of a segment depends on both translation and rotation. A decrease in angular speed about one measured axis can coexist with increasing segment kinetic energy. Even a verified decrease in total segment energy does not identify its destination without boundary powers and storage. Negative net joint power also does not uniquely identify eccentric action in every spanning muscle. Tendon energy changes and simultaneous muscle actions can yield the same net value.

27.8.2 Signed Work, Positive Work, and Metabolism

For an actuator power P_j , distinguish signed work $W_j = \int P_j dt$ from positive work $W_j^+ = \int \max(P_j, 0) dt$ and absorbed-work magnitude $W_j^- = \int \max(-P_j, 0) dt$, so $W_j = W_j^+ - W_j^-$. Summing positive work over joints is not the same as taking the positive part after summing their powers. State which denominator a reported ratio uses.

Nesbit and Serrano analyzed one selected swing from each of four amateur golfers with reconstructed body and club models. Their “swing efficiency” was club work divided by total body joint work; their table reports about 20.2–26.8 percent [Nesbit and Serrano, 2005]. Those values are not a measured elite population range, a direct muscle efficiency, or a fraction of metabolic energy. A larger ratio also need not imply a faster club: numerator and denominator can change together.

For a transparent illustrative budget, suppose initial recoverable mechanical energy is 100 J, positive actuator work is 500 J, absorbed actuator work is 80 J, and other dissipation is 20 J, with zero other external work. Final mechanical energy is 500 J. It could comprise 250 J in the club and 250 J in body motion and retained potentials. Club energy divided by positive actuator work is 0.50, while club energy divided by signed actuator work is about 0.595. Neither ratio alone identifies where the rest became heat or establishes metabolic efficiency. Initial energy can even make a club-energy/work ratio exceed one without violating conservation.

Collision adds another boundary and another interval. Pre-impact club kinetic energy is not all transferred to the ball; club motion, deformation, losses, and ball spin remain relevant. A numerical impact-transfer fraction must follow an explicit collision model or measurement. It cannot be appended as a universal percentage to close an incomplete downswing budget.

27.9 How Drift and Control Connect to the Chain

27.9.1 Declare What Zero Input Removes

The full local state must at least include configuration and velocity, $x = (q, v)$. With a direct torque input and consistently eliminated regular constraints, it may admit

$$\dot{x} = f(x) + G(x)u, \quad f(x) = \begin{bmatrix} v \\ a_0(q, v) \end{bmatrix}. \quad (27.18)$$

The first component is configuration rate, not the complete dynamics. The drift is the zero-input field for the declared input port. Its contents depend on which gravity, elastic, dissipative, actuator, and contact effects the model retains. A change of coordinates transforms both the drift and the input fields; numerical component magnitudes need a declared scaling or metric.

If input means neural excitation, zero input need not mean zero activation or zero muscle force. Activation and tendon states must then be included, and a simple affine torque model may no longer represent the same intervention. If input means a net joint torque, removing it also differs from making all muscles inactive. In a moving-grip club model, setting its grip couple to zero retains the imposed grip path and its force. These distinctions are essential to the zero-torque counterfactual framework in Chapter 6.

27.9.2 Attribution and Later Consequences

At a fixed state in a genuinely affine model, $G(x)u$ is the instantaneous difference between actual and zero-input state rates. It does not follow that the difference between two finite trajectories equals its integral along the original trajectory. Once states diverge, gravity, geometry, constraint reactions, and future control effectiveness change too. A counterfactual that crosses a contact boundary requires an event-consistent model.

The first-order endpoint sensitivity along a nominal smooth trajectory is

$$\delta x(t_I) = \Phi(t_I, t_0)\delta x(t_0) + \int_{t_0}^{t_I} \Phi(t_I, s)G(x(s))\delta u(s) ds, \quad (27.19)$$

where Φ is the state-transition matrix of the full linearized dynamics under the specified control policy. For a terminal output $y = h(x)$ at a fixed time, premultiply by Dh . If impact time is determined by a contact event, include its timing sensitivity and any impact map rather than silently treating t_I as fixed.

This gives the site's central argument a precise form: early action changes the state and geometry from which later action operates, and the existing state continues to evolve between inputs. The relevant question is how much admissible future action can change the required impact outputs, with what uncertainty and cost. A large instantaneous drift-control ratio does not measure neural latency, prove an optimal policy, or establish absence of useful distal control. It needs a declared model, coordinate metric, input port, and output horizon before it can inform that question.

27.10 Test the Explanation

27.10.1 Link Each Claim to an Observable

To test sequencing, collect synchronized segment kinematics with declared coordinate frames and timing uncertainty. To test power flow, add external wrenches, body/club inertial estimates, and an inverse-dynamics model whose residuals and filtering sensitivity are reported. To infer individual muscle or tissue action, additional measurements and physiological assumptions are necessary. Electromyography measures an electrical signal related to activation; it does not directly measure muscle force, joint work, or tendon energy.

Check integrated boundary powers against the change in mechanical energy, allowing for measurement uncertainty. Large unexplained residuals can arise from differentiation

noise, mismatched time bases, inaccurate inertias, omitted foot or shaft deformation, or inconsistent moment references. Reproducing measured positions in a kinematically driven model is not an independent validation of all its internal forces.

27.10.2 Test Interventions Against Competing Explanations

Suppose a player gains club speed after changing transition timing. Candidate explanations include different active work, a changed hand path, different segment geometry, altered elastic storage, a different impact time, or changed clubface/contact conditions. Compare repeated trials and quantify the outputs that matter: speed, path, face orientation, strike location, ball result, and consistency. A faster isolated segment peak is not enough.

Use a model intervention to separate hypotheses only after declaring the preserved quantities. Holding the grip path fixed isolates one question; holding actuator effort fixed isolates another. Sensitivity to initial state and model parameters determines whether the predicted advantage survives measurement uncertainty. A simulation that matches an observed sequence can demonstrate compatibility with a mechanism, but competing mechanisms may match the same sequence. Successful explanation requires discriminating observations or interventions.

27.11 What the Reader Can Use in Practice

The mechanics support an investigation organized around the desired club state and the player's repeatability. They do not justify a universal instruction to maximize pelvic speed, freeze the wrists, or seek the largest possible X-factor. A change should be evaluated by its measured effect on the complete shot, including dispersion and strike, rather than by conformity to an attractive sequence diagram.

Slow practice can help a player explore positions and coordination, but it is not automatically dynamically similar to a full-speed swing. Under a time scaling, inertial terms scale differently from gravity, and activation, elastic, and contact effects need not scale together. The motion should be reassessed at the intended speed. Likewise, smoother commanded torque does not guarantee small club jerk near rapidly changing geometry or contact events.

Force-plate feedback can inform contact forces, moments, impulse, and their timing when measured correctly. It does not directly calibrate a drift-control ratio. Feeling "effortless" does not prove that muscles stopped working, and muscle activation does not identify concentric action; active muscle can shorten, remain approximately isometric, or lengthen. These distinctions allow useful coaching descriptions to coexist with technically honest explanations.

The connections are the important result. Ground contact shapes feasible momentum exchange; muscle action changes energy and state; geometry maps that state to club motion; elastic elements store and return work; and the remaining time constrains what later control can accomplish. An account that quantifies those connections is more informative than either a list of peak timings or a claim that one part of the chain supplies all the speed.

27.12 Reproduce the Mechanical Checks

The following Python functions use NumPy and SciPy to evaluate the stated teaching cases. Times are in seconds and other inputs use SI units. The rotor solution can be checked against

independent numerical integration; the modal vectors are mass-normalized. The moving-grip calculation checks instantaneous force, moment, and total mechanical-energy rates, not a complete golf trajectory. The same program is printed in both editions.

```

1 # Kinetic-chain mechanical checks
2 import numpy as np
3 from scipy.linalg import eigh
4
5
6 def rotor_case(time: np.ndarray) -> dict[str, np.ndarray]:
7     """Two fixed-axis rotors, zero initial state, constant 50 Nm input."""
8     frequency = np.sqrt(250.0)
9     phase = frequency * time
10    separation = 0.1 * (1 - np.cos(phase))
11    relative_speed = 0.1 * frequency * np.sin(phase)
12    mean_angle = 10 * time**2
13    mean_speed = 20 * time
14    angle_one = mean_angle + 0.2 * separation
15    angle_two = mean_angle - 0.8 * separation
16    speed_one = mean_speed + 0.2 * relative_speed
17    speed_two = mean_speed - 0.8 * relative_speed
18    energy = speed_one**2 + 0.25 * speed_two**2
19    energy += 50 * separation**2
20    return {
21        "state": np.array([angle_one, angle_two, speed_one, speed_two]),
22        "acceleration": np.array([20 + 5 * np.cos(phase), 20 * (1 - np.cos(
23            phase))]),
24        "energy": energy,
25    }
26
27 def mode_case() -> dict[str, np.ndarray]:
28     """Mass-normalized modes of the stated three-rotor model."""
29     mass = np.diag([2.0, 1.2, 0.4])
30     stiffness = np.array([[120, -120, 0], [-120, 220, -100], [0, -100, 100]])
31     squared, modes = eigh(stiffness, mass)
32     return {"squared_frequencies": squared, "modes": modes}
33
34
35 def grip_case(acceleration: tuple[float, float]) -> dict:
36     """Horizontal rod, zero grip couple, translating/accelerating grip."""
37     mass, length, central_inertia = 0.2, 1.0, 0.02
38     gravity = np.array([0.0, -9.81])
39     grip_velocity = np.array([0.0, -3.0])
40     angular_speed = 2.0
41     polar_inertia = central_inertia + mass * length**2
42     alpha = mass * length * (gravity[1] - acceleration[1]) / polar_inertia
43     center_acceleration = np.array(acceleration) + length * np.array([-4.0,
44         alpha])
45     force = mass * (center_acceleration - gravity)
46     center_velocity = grip_velocity + np.array([0.0, length * angular_speed])
47     kinetic_rate = mass * center_velocity @ center_acceleration
48     kinetic_rate += central_inertia * angular_speed * alpha
49     return {
50         "alpha": alpha, "force": force,
51         "grip_power": force @ grip_velocity,
52         "energy_rate": kinetic_rate - mass * gravity @ center_velocity,
53     }
54

```

```

55 def ground_case() -> dict:
56     """Moments about the declared origin; all inputs are illustrative SI
    values."""
57     arm = np.array([0.4, 0.2, -1.0])
58     vertical_force = np.array([0.0, 0.0, 1.5 * 80 * 9.81])
59     force = vertical_force + np.array([100.0, 200.0, 0.0])
60     moment = np.cross(arm, force) + np.array([0.0, 0.0, 10.0])
61     return {
62         "vertical_moment": np.cross(arm, vertical_force),
63         "full_moment": moment, "yaw_impulse": moment[2] * 0.05,
64     }
65
66
67 def joint_case() -> dict[str, float]:
68     """A spring/damper stores and dissipates work while transferring power."""
69
70     separation, relative_speed = 0.1, 2.0
71     stiffness, damping = 100.0, 2.0
72     torque = stiffness * separation + damping * relative_speed
73     return {
74         "proximal_power": -torque * 5.0,
75         "distal_power": torque * 3.0,
76         "storage_rate": stiffness * separation * relative_speed,
77         "dissipation": damping * relative_speed**2,
78     }

```

For example, evaluate the rotor function at a time grid from zero to one second and compare its energy with $50\theta_1$. For the modal case, verify $K\Psi = M\Psi \text{diag}(\omega^2)$ and $\Psi^T M \Psi = I$; the smallest squared frequency is zero up to floating-point roundoff. Evaluate the grip case with accelerations $(0, -20)$, $(0, -9.81)$, and $(0, 0)$ to reproduce the three boundary conditions. Ground and joint functions reproduce the moment and power examples directly. Small rounding differences should not be interpreted as physical residuals.

27.13 Exercises

1. **Sequence Counterexample.** Starting from the two-rotor equations, derive the mean and relative solutions and reconstruct both angular velocities. On $[0, 1]$ s, locate rotor 1's maximum velocity and rotor 2's first acceleration maximum. Explain why their order refutes a universal proximal-speed/distal-acceleration rule without refuting every possible sequencing mechanism. Verify the input-work identity independently.
2. **X-Factor and Storage.** In the declared backswing-positive convention, let thorax and pelvis angles at the reference event be 90 and 45 degrees. At a later event let them be 88 and 38 degrees. Compute separation at each event and its change. For an assumed torsional spring with rest separation 30 degrees and stiffness 100 Nm/rad, compute the storage change in joules. State why this result is not a measurement of human trunk energy and why both turn rates becoming zero would not remove preload.
3. **A Moving-Grip Release.** Derive the club equation by eliminating grip force from Newton–Euler balances. Reproduce the three acceleration cases and check energy power. Then prescribe a smooth grip trajectory and an arm angle history of your choosing, state them explicitly, and compute the holding couple. Explain the addi-

tional endpoint, feasibility, and input-work checks needed before comparing two release times. Zero grip couple alone is not a complete boundary condition.

4. **Ground Wrench and Reference Point.** Reproduce all three moment components in the ground example. Change the origin by a known displacement and transform the moment consistently. Integrate the original yaw moment over the stated interval. Identify the additional information needed to infer pelvic acceleration from whole-system angular impulse, and explain why center of pressure alone cannot supply it.
5. **Energy and Identifiability.** Close the illustrative 100 J initial-energy budget using the stated positive work, absorbed work, and dissipation. Compute both club-energy ratios. Construct a second consistent budget with the same final club energy but different body energy and actuator work. Explain why club speed alone cannot distinguish the budgets or determine individual muscle action. Add a separately declared collision model only if estimating ball energy.
6. **Modes and Experimental Discrimination.** Solve the three-rotor generalized eigenproblem, identify its rigid mode, and verify mass normalization. Apply a finite torque pulse to one rotor and calculate the response from rest. Repeat with relative-joint damping and changed initial preload. Compare the resulting peak definitions explicitly. Propose a synchronized measurement that could distinguish a change in active work from a change in energy redistribution in a golfer; report which required quantities remain model-dependent.

Chapter 28

Induced Acceleration Analysis

In Plain Language 28.1. The Question Behind This Chapter

A golfer’s wrist can accelerate even when the net wrist moment is zero. Conversely, a large wrist moment need not produce a large increase in clubhead speed. The explanation lies in the coupled mechanics of the whole system: mass distribution, posture, current motion, external support and the point whose acceleration we ask about.

Induced acceleration analysis (IAA) computes how a specified force term enters a model’s instantaneous acceleration. It connects the force accounting of earlier chapters to motion without mistaking a sequence of segment-speed peaks for a measured sequence of energy transfers. Its useful question is precise: at this state, in this model, how do the declared inputs and other terms combine to produce the chosen output acceleration?

The method has a substantial biomechanics history, including the review by [Zajac and Gordon, 1989], throwing analysis by [Hirashima et al., 2008], and walking simulations by [Neptune et al., 2001]. Golf also has prior acceleration-partition research: [Takagi et al., 2021] analyzed pelvis rotation in skilled golfers. This chapter develops the mechanics and their connection to affine control; it does not claim the first application to golf.

Normative Attribution Record

A publishable attribution must identify the model and revision; engine, solver, and revision; generalized coordinates and reference frame; output point; mass matrix; constraint handling; contact model and mode; complete force partition; residual treatment; and numerical tolerance with closure error. Its identifiability contract must state the measurements or assumptions supporting each term. An algebraic term does not by itself identify anatomical source, neural intent, necessity, sufficiency, or intervention effect. Missing fields make a result **unsupported or unqualified**; an unavailable cross-engine state is reported that way rather than treated as solver parity.

Two counterexamples block unique attribution. Under $q = Tz$, the same dynamics use $M_z = T^T M_q T$ and $Q_z = T^T Q_q$: component values change even though $T\ddot{z} = \ddot{q}$. Likewise, if $Q = Q_A + Q_B$, then $Q = (Q_A + r) + (Q_B - r)$: a residual reassignment changes both reported terms while preserving their sum. An intervention must separately re-solve constraints and contact and declare what is held fixed.

28.1 The Mathematical Foundation

28.1.1 Starting From the Manipulator Equation

First consider independent generalized coordinates q , with generalized velocity $v = \dot{q}$. Our sign convention puts all retained non-inertial terms on the right:

$$M(q)\ddot{q} = Q_u + Q_v + Q_g + Q_p + Q_e + Q_r. \quad (28.1)$$

Here Q_u denotes applied control, Q_v the velocity-dependent inertial terms, Q_g gravity, Q_p passive constitutive forces, Q_e specified external loads, and Q_r residual forces used to reconcile model and data. These categories must not overlap. If a conventional equation uses $M\ddot{q} + C(q, v)v + g(q) = Bu + Q_e$, then our convention has $Q_v = -Cv$, $Q_g = -g$, and $Q_u = Bu$.

For a regular independent-coordinate model, M is symmetric positive definite. Solving the coupled system gives

$$\ddot{q} = M^{-1}Q, \quad Q = \sum_k Q_k. \quad (28.2)$$

Implementations should factor and solve with M , rather than explicitly form its inverse. The inverse notation identifies the mathematical operator. A floating base can have an invertible mass matrix even though some coordinates are unactuated; underactuation and singular inertia are different conditions.

$$\ddot{q} = \sum_k a_k, \quad a_k = M^{-1}Q_k. \quad (28.3)$$

This superposition is exact at the *same* state and parameters. It does not add together trajectories simulated under different inputs. Nor does it authorize dividing a joint torque by that joint's isolated inertia: the full coupled solve is essential.

28.1.2 What a Muscle Label Requires

Net joint moments inferred from inverse dynamics are not individual muscle forces. If a muscle model supplies forces F_i and generalized moment-arm columns $r_i(q)$, its actuator terms are $Q_i = r_i F_i$. Many muscle-force combinations can generate the same net moment, including antagonistic co-contraction. Optimization, activation dynamics, EMG constraints and other measurements can narrow that ambiguity, but their assumptions remain part of the result.

The choice of boundary matters equally. A grip wrench is external to a club-only model and internal to a combined golfer-club model. Counting it as both a specified load and a separately solved internal reaction duplicates the same interaction. The reader must be able to trace every force to one place in the accounting.

28.1.3 Constraints Add a Baseline

Suppose a smooth, fixed contact mode imposes independent acceleration constraints $A\ddot{q} = b$. For a stationary holonomic constraint $\phi(q) = 0$, $A = \partial\phi/\partial q$ and $b = -\dot{A}v$. The state must

already satisfy the position and velocity constraints. With the reaction convention $A^\top \lambda$, solve

$$\begin{aligned} M\ddot{q} &= Q + A^\top \lambda, \\ A\ddot{q} &= b. \end{aligned}$$

Substitution yields the Schur complement $W = AM^{-1}A^\top$ and

$$\begin{aligned} \lambda &= W^{-1}(b - AM^{-1}Q), \\ P &= M^{-1} - M^{-1}A^\top W^{-1}AM^{-1}, \\ a_c &= M^{-1}A^\top W^{-1}b, \\ \ddot{q} &= PQ + a_c. \end{aligned}$$

These formulas assume positive-definite M and full row rank of A . Redundant constraints require a consistent rank treatment, not an unexplained inverse of a singular matrix. Quaternion coordinates also require their proper velocity map; the simplifying convention $v = \dot{q}$ is not universal.

The force-to-acceleration map is **affine**, because a_c need not vanish. The correct partition is $\ddot{q} = \sum_k PQ_k + a_c$: count the baseline once. Each source increment satisfies $A(PQ_k) = 0$, whereas $Aa_c = b$. Solving a full constrained problem for every isolated force and summing those solutions would count the curvature baseline repeatedly. Although P is symmetric positive semidefinite, it is not generally an idempotent Euclidean projection; it has inverse-inertia units.

For example, a 2 kg particle at $q = (1, 0)$ m on a fixed 1 m circular guide has velocity $v = (0, 3)$ m/s. Taking $A = (1, 0)$, the acceleration constraint is $\ddot{q}_x = -9$ m/s². The solution is

$$\ddot{q} = (-9, F_y/2)^\top, \quad \lambda = -18 - F_x.$$

The radial acceleration is present even with zero applied force, while changing F_x changes the reaction. This is a bilateral guide example. A unilateral contact must also pass its reaction-sign and friction-capacity conditions. If the required contact cannot persist, the active mode must change and this frozen operator no longer describes the realizable response.

28.1.4 From Joint Acceleration to Clubhead Acceleration

Let $p(q)$ be a point expressed in a fixed inertial frame and $J = \partial p / \partial q$. Then

$$\ddot{p} = J\ddot{q} + \dot{J}v = \sum_k JPQ_k + Ja_c + \dot{J}v.$$

The last term is the kinematic curvature of the output map. Joint angular acceleration can be zero while a rotating clubhead has substantial centripetal acceleration. Thus a joint-coordinate ledger cannot simply be renamed a clubhead-acceleration ledger.

For two force choices at the *same* state and mode, the baseline cancels: $\Delta \ddot{p} = JP\Delta Q$. This is often the clearest sensitivity to report. Use a feasible reference load when zero force would violate the contact conditions. For clubhead speed, project the total acceleration along its velocity: $\dot{s} = \hat{v}_p^\top \ddot{p}$ when $s = \|\dot{p}\| > 0$. A large acceleration normal to motion bends the path without increasing speed at that instant. At zero speed this unit-tangent formula is undefined.

28.1.5 Connection to the Control-Affine Framework

When $Q = Q_0(q, v) + B(q)u$, the local constrained equations give

$$\frac{d}{dt} \begin{bmatrix} q \\ v \end{bmatrix} = \underbrace{\begin{bmatrix} v \\ PQ_0 + a_c \end{bmatrix}}_{f(x)} + \underbrace{\begin{bmatrix} 0 \\ PB \end{bmatrix}}_{G(x)} u.$$

IAA and affine control share this frozen-state input map when they describe the same plant and input. The drift includes the kinematic block v , all retained autonomous forces and the constraint baseline. It is not just a velocity-dependent torque. Passive tissue forces and muscle activation or tendon states can enlarge the autonomous system. A generalized-moment input is not interchangeable with neural excitation; whether the enlarged plant is affine in that excitation must be derived from its actuator equations.

Contact switches, impacts and input-dependent mode changes require a hybrid or non-smooth description. The equation above is a mode-local statement, not a global proof of control affinity through every phase of a golf swing.

28.2 Dynamic Coupling and Posture

28.2.1 Why One Torque Can Accelerate Several Coordinates

Write $H = M^{-1}$ in the unconstrained case. For a two-coordinate torque increment,

$$\delta \ddot{q} = H \begin{bmatrix} \delta \tau_1 \\ 0 \end{bmatrix} = \begin{bmatrix} H_{11} \delta \tau_1 \\ H_{21} \delta \tau_1 \end{bmatrix}. \quad (28.4)$$

Off-diagonal entries allow remote acceleration. They are often nonzero, but physical connection does not guarantee a nonzero response at every coordinate in every posture. The analogous constrained map is P , and the physical point response uses JP .

Consider two uniform planar rods, each of length 1 m and mass 1 kg, pinned at the proximal end. Their center-of-mass inertias are $1/12 \text{ kg m}^2$. Let q_1 measure the first rod from vertically downward, and q_2 the second rod relative to the first. The mass matrix, in kg m^2 , is

$$M(q_2) = \begin{bmatrix} 5/3 + \cos q_2 & 1/3 + \frac{1}{2} \cos q_2 \\ 1/3 + \frac{1}{2} \cos q_2 & 1/3 \end{bmatrix}.$$

At $q_2 = 0$, a unit proximal torque increment produces $(12/7, -30/7) \text{ rad/s}^2$; a unit distal torque produces $(-30/7, 96/7) \text{ rad/s}^2$. The remote gains are equal because H is symmetric. At $q_2 = \arccos(-2/3)$, however, the matrix is diagonal and the same proximal increment produces $(1, 0) \text{ rad/s}^2$. Connected rods can have zero instantaneous cross-acceleration. These manufactured states establish mechanics, not an optimal golf posture.

28.2.2 Normalized Coupling Is Not Nonreciprocal Mobility

Challis introduced an induced-acceleration index for joint coupling [Challis, 2011]. To avoid an ambiguous inverse-element convention, we derive a specifically defined two-coordinate

ratio here. Under a force increment only at coordinate 1, the unforced second row gives $M_{21}\delta\ddot{q}_1 + M_{22}\delta\ddot{q}_2 = 0$. Therefore

$$R_{1\rightarrow 2} = \left| \frac{\delta\ddot{q}_2}{\delta\ddot{q}_1} \right| = \left| \frac{H_{21}}{H_{11}} \right| = \left| \frac{M_{21}}{M_{22}} \right|. \quad (28.5)$$

The denominator is the *mass-matrix entry* M_{22} , not the inverse-matrix entry H_{22} . This ratio excludes the common acceleration baseline. With multiple unforced coordinates indexed by $\mathcal{P}$, use the block relation

$$\delta\ddot{q}_{\mathcal{P}} = -M_{\mathcal{P}\mathcal{P}}^{-1} M_{\mathcal{P}1} \delta\ddot{q}_1.$$

Independent elementwise divisions generally fail because those passive coordinates couple to one another. Additional constraints require the constrained operator instead of this unrestricted block formula.

For the straight two-rod model, $R_{1\rightarrow 2} = 2.5$ and $R_{2\rightarrow 1} = 0.3125$. Their ratio is eight although the equal-torque cross gains are reciprocal. Different normalization denominators explain the apparent directional asymmetry. Ratios are dimensionless only for commensurate coordinates, and rescaling coordinates changes them. Mixed rotational and translational coordinates require explicit units or a justified metric. The more-than-twelfold quiet-standing index difference reported in the abstract of [Challis, 2011] should therefore remain attached to that study's definition and model; it does not establish a golf or neural-control strategy.

28.2.3 Coordinates and Physical Responses

Under a constant invertible change $q = Tz$, virtual work and kinetic energy give

$$M_z = T^T M_q T, \quad Q_z = T^T Q_q, \quad J_z = J_q T.$$

Consequently $J_z M_z^{-1} Q_z = J_q M_q^{-1} Q_q$. A consistently transformed force increment has the same acceleration at the same physical point. The analogous constrained result follows by transforming $A_z = A_q T$. Representation dependence is a reason to report coordinates and units, not a suggestion that physical predictions are arbitrary.

A nonlinear change needs acceleration transport. For a one-dimensional free particle described by $q = z^2$, with $z > 0$,

$$\ddot{q} = 2z\ddot{z} + 2\dot{z}^2 = 0, \quad \ddot{z} = -\dot{z}^2/z.$$

A velocity-dependent coordinate term appears without a physical force or particle acceleration. Complete physical outputs remain consistent when all transport terms are retained. This is one reason to avoid equating the size of a coordinate velocity term with a universal amount of mechanical assistance.

28.3 Instantaneous and Accumulated Effects

28.3.1 Integrating a Ledger Along One Trajectory

If a complete acceleration partition is evaluated along one nominal motion, integration gives

$$\dot{q}(t) - \dot{q}(t_0) = \sum_k \int_{t_0}^t a_k(s) ds + \int_{t_0}^t a_c(s) ds.$$

This is valid bookkeeping for coordinate-velocity change on smooth intervals, with any impulse jumps accounted for separately. The terms can be signed, oppose one another, or exceed the net change in magnitude. A percentage obtained by dividing by a nearly zero net acceleration is unstable and can be misleading.

The velocity-dependent term is an *instantaneous function of the current state*. That state reflects earlier forces and initial conditions, but it does not uniquely identify their history. For the double integrator $\ddot{q} = u$, starting from rest at $q = 0$, both $u_1(t) = 1$ and $u_2(t) = 2 + 6t^2 - 6t$ reach $(q, \dot{q}) = (1/2, 1)$ at $t = 1$. Both the input integral and its first time moment agree, despite different inputs. A single state cannot distinguish them.

28.3.2 Why Removing an Input Is a Different Calculation

For a transparent nonlinear example, let $\dot{x} = x^2 + u$, $x(0) = 0$, and $u = 1$. Before $t = \pi/2$, $x(t) = \tan t$. The nominal input integral is t and the nominal drift integral is $\tan t - t$; they sum exactly. Removing the input yields $x(t) = 0$, so the intervention difference is $\tan t$, not the input ledger entry t . At $t = 0.5$, these are approximately 0.5463 and 0.5.

The removed input changes subsequent states and hence subsequent drift. In a swing, it may also change muscle states, support reactions and contact transitions. A force-removal simulation must specify those rules and integrate the resulting system. Neither an instantaneous term nor its nominal integral supplies that simulation automatically. Describing drift as the accumulated effect of all past torques obscures this distinction.

28.4 What Throwing Studies Establish

[Hirashima et al., 2008] studied six baseball players with a 13-degree-of-freedom model comprising trunk, upper arm, forearm, and hand with ball. It partitioned reconstructed generalized-force dynamics into joint-torque, velocity-dependent and gravity terms. It reported larger joint-torque terms at proximal coordinates and substantial velocity-dependent terms in elbow extension and wrist flexion. These were estimates for net generalized moments, not separately measured muscle forces.

The study also integrated nominal induced accelerations and checked reconstructed motion. Its elbow-torque reversal and distal acceleration terms illustrate coupling; they do not identify a protective neural intention. The paper explicitly discusses omitted segments and lower-body/ground interactions. Marker acquisition at 200 Hz and spline resampling at 2000 Hz are different quantities: interpolation does not supply ten times the independent measurement bandwidth.

These findings motivate a golf study but do not settle it. A two-handed club, compliant shaft and foot contacts change the mechanical system. Golf comparisons need matched outputs, force conventions and contact treatment; a reported pitching joint-angle term cannot stand in for clubhead tangential acceleration. The chapter supplies no numerical human-golf finding that late-swing velocity terms universally dominate.

28.5 Walking, Muscle Function and Energy

28.5.1 Acceleration and Energy Answer Different Questions

In the 1.5 m/s walking simulation of [Neptune et al., 2001], both soleus and gastrocnemius supported and advanced the trunk during parts of stance, yet their modeled energy delivery differed: soleus delivered energy to the trunk, while gastrocnemius delivered most of its generated energy to the leg during the late-stance/pre-swing interval. Calling gastrocnemius exclusively a swing initiator would hide its simultaneous support and progression roles. The study used both acceleration and segment energetics to make these distinctions.

An acceleration vector alone does not determine energy transfer. Generalized-force power is $Q_k^T v$. At a body interface, use the wrench and that body's twist at the same point and in compatible frames: $F \cdot v_P + \tau_P \cdot \omega$. An ideal fixed support can redirect momentum while doing zero work at its stationary contact point. In a multibody system, individual segments can exchange energy through internal forces even when the net internal work on the whole system cancels. Track the chosen boundary and velocities explicitly.

28.5.2 Clinical Examples Are Conditional Model Results

The sit-to-stand study of [Caruthers et al., 2016] used static optimization to estimate muscle forces before induced-acceleration analysis. Its reported forward center-of-mass terms for gluteus maximus and soleus differed from the opposing quadriceps term. That is a useful example of how anatomy alone does not fix the sign of a whole-body output response. It remains conditional on the muscle estimation and model; the present chapter does not independently reproduce it.

IAA has also been applied to stiff-legged gait [Riley and Kerrigan, 1999]. A decomposition can help formulate patient-specific hypotheses, but a computed term does not by itself identify a treatable impairment or predict the effect of rehabilitation. Such claims require clinical evidence beyond algebraic closure. The transferable lesson for golf is to test the force-estimation and contact assumptions, not to import a walking or clinical result as a swing prescription.

28.6 A Worked Golf-Model Investigation

28.6.1 A Fully Specified Two-Link Calculation

Use the uniform rods defined above, with gravity $g = 9.81 \text{ m/s}^2$, no damping, no other external loads and no additional constraints. This is a toy arm-club analogue, not anthropometry. Angles remain *relative joint coordinates*; do not substitute formulas for two absolute rod angles. The right-hand-side terms are

$$Q_v = \begin{bmatrix} \frac{1}{2} \sin q_2 (2v_1 v_2 + v_2^2) \\ -\frac{1}{2} \sin q_2 v_1^2 \end{bmatrix},$$

$$Q_g = -g \begin{bmatrix} \frac{3}{2} \sin q_1 + \frac{1}{2} \sin(q_1 + q_2) \\ \frac{1}{2} \sin(q_1 + q_2) \end{bmatrix}.$$

At $q = (-45, -90)$ degrees, $v = (500, 200)$ degrees/s and $Q_u = (10, -2)$ Nm, first convert angles and rates to radians. The joint acceleration contributions in rad/s^2 are

$$\begin{aligned} a_u &= (9.000, -15.000)^\top, \\ a_v &= (-55.973, 170.205)^\top, \\ a_g &= (7.804, 2.601)^\top, \\ \ddot{q} &= (-39.170, 157.806)^\top. \end{aligned}$$

The distal endpoint is $p = (\sin q_1 + \sin(q_1 + q_2), -\cos q_1 - \cos(q_1 + q_2))^\top$ m. Differentiating this map gives the following point-acceleration terms in m/s^2 :

$$\begin{aligned} Ja_u &= (10.607, -2.121)^\top, \\ Ja_v &= (-120.353, -41.195)^\top, \\ Ja_g &= (-1.839, -12.876)^\top, \\ \dot{J}v &= (159.394, -51.695)^\top, \\ \ddot{p} &= (47.808, -107.887)^\top. \end{aligned}$$

The large positive horizontal curvature term and negative velocity-force term partly cancel. Omitting either changes the answer qualitatively. The gravitational point term is not simply free-fall acceleration because the proximal pin constrains the linked system. These numbers demonstrate why the output projection and complete accounting matter more than an isolated label such as proximal, distal or passive.

28.6.2 From the Toy Model to Human Evidence

The prior golf study of [Takagi et al., 2021], published online in 2019 and in a 2021 journal issue, used 31 skilled golfers, motion capture and two force plates to partition pelvis angular acceleration. Its existence rules out a broad first-golf-application claim here. Its abstract supports this study description; this chapter does not claim to reproduce its full methods or extend its pelvis findings to the clubhead.

A defensible next investigation would combine measured kinematics and support loads with a specified golfer-club model. Report the output point and frame, net-moment or muscle-force estimation, grip and foot contact assumptions, residuals and solver details. Then calculate complete acceleration terms and vary plausible inertial, kinematic, actuator and contact parameters. Sensitivity to differentiation noise is especially relevant because inverse dynamics and output acceleration use time derivatives of measured motion.

Use closure as a necessary implementation check: the sum must match a separate complete forward solve within a stated absolute and relative tolerance. Closure alone is not validation against humans; even an inaccurate model can close algebraically. Compare predicted motion or held-out observables with independent measurements and propagate uncertainty to signed output terms. Keep uncertain terms visible rather than assigning the residual to a favored muscle or mechanism.

28.7 Superposition and Its Limits

Three distinct results should remain distinguishable. An *instantaneous decomposition* explains one calculated acceleration under a stated partition. A *local sensitivity* reports the

change in an output for a small feasible input change at that state. A *trajectory intervention* predicts a changed motion after integrating a specified counterfactual. The first does not supply the third; the second remains local to its contact mode.

There is also a distinction between acceleration authority and energetic cost. Two postures can have different *JPB* maps for the same generalized input while requiring different muscle forces, activation, stability margins and contact reactions. A favorable gain alone therefore does not prove reduced human effort, safety or superior technique. Those are additional outputs with additional evidence requirements.

28.8 Summary

Induced acceleration analysis links a complete force model to a declared motion output. Its core calculation is a coupled solve at one state. Active constraints supply an affine baseline, and point acceleration adds kinematic curvature. Posture changes the coupling map; consistent coordinate transformations preserve physical output increments. Nominal integrals account for a velocity change but do not reconstruct a unique force history or replace a force-removal simulation. Muscle roles require force-estimation evidence, and energy flow requires power and boundary accounting. Together these distinctions make IAA a useful bridge between biomechanics and control theory.

Exercises 28.1. Chapter Exercises

1. Reproduce the two unit-torque responses at $q_2 = 0$ and the zero cross-response at $q_2 = \arccos(-2/3)$. Compute both normalized ratios and explain why their asymmetry does not contradict the symmetric inverse mass matrix.
2. For the circular-guide example, apply $F = (4, 6)$ N. Verify $\ddot{q} = (-9, 3)$ m/s² and $\lambda = -22$ N. Split the applied force into its two Cartesian components and show that adding two complete constrained solutions would double the radial baseline. Explain why reaction feasibility must be checked before interpreting a unilateral version of this example.
3. Reproduce the worked two-link acceleration ledger and differentiate its endpoint map to recover all five point vectors. Compute the endpoint velocity and project each point term onto its unit tangent. State why the resulting signed speed-rate terms differ from vector magnitudes.
4. Integrate both double-integrator input histories from $t = 0$ to $t = 1$. Verify the common final state using $\dot{q}(1) = \int_0^1 u(t)dt$ and $q(1) = \int_0^1 (1-t)u(t)dt$. Explain which force-history information that state fails to preserve.
5. Design a comparison between a net-joint-moment golf model and a muscle-driven model. Specify a shared physical output, contact mode, residual convention and independent measurement. Identify which conclusions need acceleration, power, muscle-force estimation or an integrated intervention.

Part IX

Motor Control and Learning

Chapter 29

Motor Control I: The Brain as Controller

In Plain Language 29.1. What This Chapter Is About

A golf swing joins a physical system to a nervous system that prepares, senses, predicts and learns. Muscles supply and absorb energy; the body's geometry and connections distribute force and motion. A useful controller must work with both. The scientific question is how preparation and ongoing regulation combine to achieve a strike despite uncertainty and limited time.

Engineering models make this question precise. They do not establish that a particular brain region literally solves our equations. We will distinguish a mechanical identity, a proposed control architecture and an experimental finding. The strongest argument for exploiting mechanical structure comes from showing what it changes in the task, not from declaring the swing passive or the brain too slow to participate.

29.1 The Control Problem: What the Brain Must Solve

For an unconstrained rigid model in local generalized coordinates, let $q \in \mathbb{R}^n$ and $v = \dot{q}$. With generalized torque $u \in \mathbb{R}^m$ applied through $B(q)$,

$$\dot{x} = f(x) + G(x)u, \quad x = \begin{bmatrix} q \\ v \end{bmatrix}, \quad f = \begin{bmatrix} v \\ -M^{-1}(c + g) \end{bmatrix}, \quad G = \begin{bmatrix} 0 \\ M^{-1}B \end{bmatrix}. \quad (29.1)$$

Here M is the positive-definite mass matrix, c is the complete velocity-dependent bias and g is gravitational loading. Drift includes the kinematic block v ; it is not a source of free energy or a substitute for muscle work. Contact constraints change the equations and admissible motions. A moving base, both arms, flexible tissues and a flexible club require their corresponding states and force laws.

The input in this equation is generalized torque, not a cortical spike or a muscle excitation. A more physiological model might use neural drive e , activation a , muscle-tendon state z and muscle forces F_m :

$$\dot{a} = \mathcal{A}(a, e), \quad \dot{z} = \mathcal{Z}(z, q, v, a), \quad \tau_m = R(q)F_m(q, v, a, z).$$

The moment-arm matrix R maps muscle forces to joint torques. This map is generally redundant, state dependent and subject to unilateral force and activation bounds. Two activation patterns can produce the same net torque but different stiffness, metabolic cost and future force capacity. Adding activation and material memory to the state changes the

control problem; a torque-level affine representation does not automatically remain affine in every proposed neural input.

The task is also more specific than reproducing a joint trajectory. Club position, velocity, face orientation and contact conditions jointly influence a strike; ball flight and ground interaction then introduce further states and uncertainty. A state estimate, an admissible input set and a defined task loss are needed before an optimal-control comparison is meaningful.

Definition 29.1. Feedback and Feedforward in One Policy

Feedback makes a command depend on available observations or an estimate derived from them. A local engineering example is

$$u(t) = \text{proj}_{\mathcal{U}} [u_{\text{ff}}(t) + K(t)(x_d(t) - \hat{x}(t))].$$

The projection denotes the nearest admissible torque under a declared metric and a closed convex set $\mathcal{U}$; more complicated actuator constraints require their own allocation rule. K maps state error into torque with compatible units. Neither this proportional law nor its stability is a universal property of neural control.

Definition 29.2. Prepared Commands and Feedback Policies

A feedforward component is prepared without depending on the corresponding newly arriving sensory consequence. It can be learned without explicitly identifying a physical model. A prepared command and a prepared feedback policy can operate together: learning the latter does not make execution open-loop.

The sensory system observes outputs rather than an exact instantaneous state. A schematic channel ℓ gives

$$y_\ell(t) = h_\ell(x(t - d_\ell)) + \nu_\ell(t),$$

where d_ℓ is its defined delay and ν_ℓ is measurement error. A delayed observation must be compared with a prediction for the same time and sensory quantity. An estimator may correct its past estimate and propagate the correction forward using intervening commands. It cannot remove uncertainty merely by predicting, and it cannot infer unobservable states uniquely from insufficient measurements.

Example 29.1. Delays in the Golf Swing

Short-latency upper-limb stretch responses are often discussed in approximate onset bands of 20–45 ms, long-latency responses in 50–100 ms, and voluntary responses at greater than 100 ms. These are exemplar EMG-response bands from specific experiments, not universal golf constants or complete times to change club motion [Kurtzer, 2014, Pruszynski and Scott, 2012]. Parallel pathways do not form a single serial sum.

Neural conduction, sensory processing, activation and force transmission must be defined separately. If a measured latency already ends at force onset, adding a second full electromechanical delay would double-count processes. Late visually guided

corrections may have little time to affect impact, while earlier task-dependent feedback may influence muscle activity, impedance or trajectory. Response authority means the attainable change in a specified mechanical outcome, not simply whether an EMG response has begun.

29.2 Internal Models: Prediction, Inversion and Feasibility

An internal forward model is a functional hypothesis about prediction. Evidence for prediction does not imply a conscious simulation, an exact neural copy of rigid-body mechanics or one unique implementation. In a human hand-localization experiment with movements in darkness and externally imposed forces, Wolpert and colleagues tested state estimation that combines sensory information with motor-command information [Wolpert et al., 1995]. That task supports a predictive-estimation account; it does not identify a golf-swing control law.

Definition 29.3. A Dimensionally Consistent Forward Model

If $\hat{f}$ and $\hat{G}$ predict a continuous-time state derivative, a first-order discrete prediction is

$$\hat{x}_{k+1}^- = \hat{x}_k^+ + \Delta t \left[\hat{f}(\hat{x}_k^+) + \hat{G}(\hat{x}_k^+) u_k \right].$$

The superscripts distinguish before and after measurement correction. A derivative must be multiplied by time and added to a state. Alternatively define a discrete transition $\hat{x}_{k+1}^- = F_{\Delta t}(\hat{x}_k^+, u_k)$; it is a different object from the vector field. For a smooth model, one Euler step has local error of order Δt^2 , while accumulated error over a fixed interval is generally first order.

A sensory prediction is $\hat{y}_k = h(\hat{x}_k^-)$, not necessarily $\hat{x}_k^-$. For a linear measurement, an illustrative estimator update is $\hat{x}_k^+ = \hat{x}_k^- + L_k(y_k - H\hat{x}_k^-)$. The innovation is a sensory discrepancy. Its gain depends on modeled uncertainty, observability and the estimation objective; it is not obtained by treating every sensory channel as equally reliable. In nonlinear, constrained or delayed systems this simple update needs appropriate extensions.

Definition 29.4. Inverse Control With a Feasibility Residual

For a desired derivative, define $b = \dot{x}_d - \hat{f}(\hat{x})$. The matrix $\hat{G}$ is $2n \times m$ in the torque model and has a zero position block. It generally has no ordinary inverse. Exact instantaneous realization requires $b \in \hat{G}\mathcal{U}$. A bounded allocation can instead solve

$$u^* = \arg \min_{u \in \mathcal{U}} \left[\frac{1}{2} \|\hat{G}u - b\|_W^2 + \frac{\lambda}{2} \|u\|_V^2 \right], \quad r = b - \hat{G}u^*.$$

W and V specify units, scaling and priorities; $\lambda \geq 0$ sets regularization. The feasibility residual r must be reported. A pseudoinverse alone supplies neither actuator feasibility nor an exact solution outside the range. When an exact solution exists without bounds, null-space inputs may remain undetermined.

For full joint actuation, inverse dynamics can instead compute $u = M\ddot{q}_d + c + g$ along a kinematically consistent desired trajectory. This still requires torque feasibility and a controller for deviations. It does not identify individual muscles. Differentiating a forward model yields sensitivities; turning those into a useful inverse requires a well-posed constrained problem, not merely writing an inverse symbol.

Example 29.2. A Rectangular Input Cannot Set Every Derivative

Use explicitly normalized state and input coordinates, $f = (v_1, v_2, 0, 0)^T$, and $G = (0, 0, 1, 1)^T$. Suppose $b = (0, 1, 3, 0)^T$. No input can realize the requested extra position derivative, and one shared input cannot simultaneously produce accelerations 3 and 0. With identity weighting the unconstrained least-squares input is 1.5; with $|u| \leq 1$ it is 1, leaving $r = (0, 1, 2, -1)^T$. This residual is a physical limitation of the declared map, not a numerical solver failure.

Instantaneous rank deficiency does not by itself prove finite-time uncontrollability. The double-integrator position block is zero, yet acceleration changes later position. That distinction is essential when connecting drift, instantaneous ratios and reachable strike outcomes.

29.2.1 The Cerebellum and the Limits of a Neural Interpretation

The cerebellum contributes to coordination and adaptation, but “the body’s simulator” is a functional metaphor, not a complete anatomical explanation. A lesion does not selectively delete a line of pseudocode. Symptoms and retained skills vary; a diagnosis cannot establish that every affected person is unable to strike a ball or make a smooth reach.

A widely studied hypothesis links climbing-fiber activity, Purkinje-cell activity and plasticity at parallel-fiber synapses. It is not a universal scalar gradient-descent algorithm. In experiments on three genetically altered mouse lines lacking the tested form of cerebellar long-term depression, Schonewille and colleagues found preserved learning in their tested paradigms; they also explicitly considered alternative and compensatory mechanisms [Schonewille et al., 2011]. This limits a claim that that one plasticity mechanism is necessary for all cerebellar learning. It does not show that the cerebellum or LTD is irrelevant to learning.

Example 29.3. A Learning Rule Requires a Defined Objective

For a declared linear predictor $\hat{y} = w^T \phi$ and target y , set $e = \hat{y} - y$ and $\mathcal{L} = \frac{1}{2}e^2$. Gradient descent gives

$$\Delta w = -\eta e \phi. \quad (29.2)$$

This is a mathematical derivation for that predictor. If the output nonlinearity or loss changes, so does the gradient. An additional postsynaptic-rate factor requires a justified model. A signed error can yield positive or negative updates; a minus sign alone does not establish synaptic depression, stable learning or a biological implementation. Temporal credit assignment additionally requires specifying which past activity is eligible for modification.

29.3 Predictive Processing and Distinct Kinds of Error

Prediction enters several theories of motor control. Predictive processing, active inference and optimal feedback control are related research programs with different assumptions; success on one prediction task does not establish them as interchangeable or settled descriptions of the whole brain. A forward model can appear inside a feedback controller without committing to a free-energy account.

Definition 29.5. Prediction, Task and Reward Errors

A sensory prediction error compares an observation with its predicted sensory consequence. A task error compares an achieved outcome with a desired outcome. A reward prediction error compares received and expected value. These quantities can disagree. A golfer may correctly predict a miss, producing little sensory surprise but a large target error. A successful shot may still be unexpectedly good or entirely expected.

For a normalized belief distribution $q(z)$ over latent states and a generative joint probability $p(y, z)$, variational free energy is

$$\mathcal{F}[q] = \mathbb{E}_q[\log q(z) - \log p(y, z)] = D_{\text{KL}}(q(z) \| p(z | y)) - \log p(y).$$

This identity requires suitable support and finite expectations. Its nonnegative KL term makes it an upper bound on negative log evidence. It is an information quantity, not tissue energy measured in joules. Active-inference models must additionally specify actions, beliefs, preferred outcomes and their optimization; simply imagining a target is not a mathematical definition of active inference [Friston et al., 2010].

The identity does not prove that biological neurons minimize this objective. Nor does low prediction error imply a good golf shot: an accurately predicted error remains an error relative to the task. A useful scientific comparison asks competing models to predict different responses to sensory changes, rewards or mechanical perturbations, then tests those responses.

29.4 Ideomotor Theory and Task-Oriented Instruction

Ideomotor accounts propose that learned links between actions and their sensory effects contribute to action selection [Hommel et al., 2001]. This motivates a hypothesis about recalling an action through a desired effect. It is not a theorem that all movement is represented only as effects, that mechanical instruction confuses a motor system, or that thinking about a target selects a uniquely correct muscle sequence.

Key Principle 29.1. Effects Do Not Determine One Motor Solution

Several body motions may produce similar ball flights, while similar felt motions may produce different strikes. Selecting an effect leaves questions about state estimation, feasible actuation, robustness and credit assignment. An outcome goal and mechanical knowledge can inform the same learning process.

In an engineering formulation, a task objective can penalize strike error and effort without requiring every joint to follow a prescribed path. Stochastic optimal feedback control provides one formal account of allowing variation in directions that do not interfere with the goal [Todorov and Jordan, 2002]. That prediction depends on the cost, dynamics and uncertainty model. It does not imply that any joint can be ignored throughout a swing or that all variation is beneficial.

A coaching comparison therefore needs more than two attractive slogans. Specify the cue, skill level, task and outcome; compare retention and transfer as well as immediate performance. A cue about club motion may be an external effect even though it contains mechanics. A cue about bodily feeling may direct attention internally even though it contains no joint terminology. Ideomotor theory alone cannot rank these instructions or prescribe one for every golfer. The chapter's role is to identify testable alternatives.

29.5 Distributed Neural Contributions and the Mechanical System

The nervous system contains interacting cortical, subcortical, brainstem, cerebellar and spinal circuits. A serial diagram from “goal center” to “muscles” omits recurrent communication and sensory dependence. The figure separates functional roles without assigning each role to a single neural structure or inventing one clock rate for each box.

29.5.1 The Motor Cortex

Motor-cortical activity participates in movement preparation, execution and feedback. A preferred-direction population vector is a useful decoding model under specified conditions, not a complete law that every movement is the vector sum of neuronal votes. Churchland and colleagues reported structured population dynamics during monkey reaching [Churchland et al., 2012]. This motivates alternatives to a fixed movement-parameter code; it does not establish a unique neural representation for golf or imply that losing neurons has negligible consequences.

Definition 29.6. Descending Pathways and Feedback

The corticospinal tract is one important descending pathway among several. Its origins and connections are more diverse than one M1 neuron controlling one contralateral muscle. Conduction velocity, path length, synapses, recruitment and muscle mechanics all matter when interpreting a response latency. A fastest-fiber speed alone cannot establish the time from an intention to useful joint torque.

Pruszynski and colleagues used mechanical perturbations in monkeys and stimulation in humans to investigate rapid multijoint feedback. Their findings support an M1 contribution to integrating shoulder and elbow information in responses around 50 ms in those experiments [Pruszynski et al., 2011]. This is evidence that some rapid responses reflect coupled limb mechanics. It does not measure correction of clubface orientation during a golf downswing.

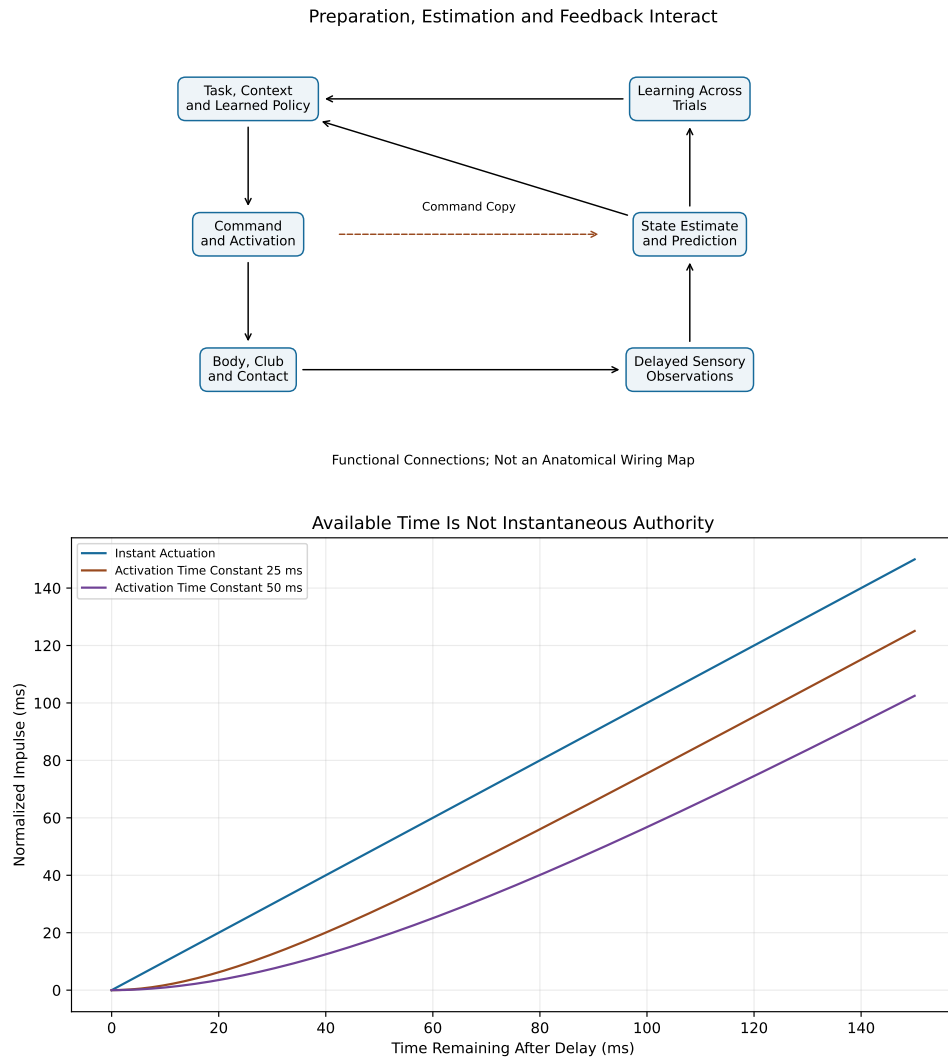


Figure 29.1: Functional Roles in Control and a Manufactured Activation Example. Preparation, Prediction and Feedback Interact; the Diagram Does Not Assign Each Role to One Brain Region. The Curves Show Normalized Impulse After a Declared Delay, Not Measured Golf Responses.

29.5.2 The Spinal Cord: Local Regulation Within a Larger Network

Definition 29.7. Stretch Responses and Prepared Gain

The monosynaptic stretch-reflex pathway includes a muscle-spindle afferent connection to an alpha motor neuron. The observed response also depends on background activation, spinal circuitry and descending modulation. A short response need not await a new conscious decision, but its gain and the state in which it occurs can be prepared beforehand. Calling it wholly independent of the brain loses that distinction.

Definition 29.8. Central Pattern Generators and Scope

A central pattern generator is a circuit capable of generating patterned rhythmic output without a matching rhythmic input. Evidence about rhythmic behaviors does not by itself identify a golf-swing CPG. A single prepared swing, a learned feedback policy and a rhythm-generating circuit are different hypotheses.

Local regulation matters because it acts through the mechanical system. Increasing coactivation may alter impedance before an unexpected perturbation, but also changes force demand, metabolic cost and available torque reserve. Tissue viscosity or stiffness can produce an immediate mechanical response while neural feedback arrives later. Neither guarantees stability: geometry, constraints, damping, delay and active energy supply determine the combined response.

29.5.3 The Basal Ganglia, Reward and Credit Assignment

The basal ganglia participate in interacting circuits involved in action and learning; reducing them to an isolated routine selector leaves out that interaction. Dopaminergic activity has been related to reward prediction errors, not simply whether an outcome is objectively good [Schultz et al., 1997].

Definition 29.9. A Motor Program as a Modeling Abstraction

A learned action can be represented by a command sequence, a parameterized policy or a combination. Familiarity and automaticity do not establish that brain activity ceases after initiation. Different representations make different predictions when the body or task is perturbed.

A simple temporal-difference learning model uses

$$\delta_k = r_k + \gamma V(s_{k+1}) - V(s_k), \quad 0 \leq \gamma < 1.$$

Here r_k is reward, V expected future return and δ_k the reward prediction error. These symbols are not joint angles, muscle energy or sensory discrepancies. Neural reward signals vary by population and task; this model is not a complete dopamine theory.

Example 29.4. Reward Does Not Identify the Effective Part of a Routine

A good shot can follow a poor routine by chance, and a good routine can precede an unlucky outcome. A predicted success need not produce a positive prediction error. Assigning learning to a particular waggle, alignment step or muscle command requires information about what caused the result. Delayed outcomes create a credit-assignment problem; a scalar reward alone does not label the responsible action.

For a terminal outcome with $V(s_{k+1}) = 0$, expected value 0.8 and reward 1 give $\delta_k = 0.2$. Expected value 0.2 with the same reward gives 0.8. The identical result can therefore give different updates in the model. This is not a direct prediction of dopamine firing in a golfer.

Clinical disorders can affect several aspects of movement and learning. They are not controlled deletions of “action selection” or “the internal model,” and cannot establish universal inability to initiate or perform a particular movement. Mechanistic claims require specific tasks, measurements and appropriate comparison groups.

29.5.4 Motor Unit Recruitment and Force Development

A motor unit consists of a motor neuron and the muscle fibers it innervates. Recruitment order and discharge rate both shape the developing force. The size principle describes a tendency for lower-threshold units to be recruited before higher-threshold units under common drive; it is not a serial queue with a compulsory 100–500 ms wait before large units can act. Neural inputs, task and recruitment history matter.

Del Vecchio and colleagues decomposed high-density EMG in tibialis anterior during explosive isometric contractions in 20 men. Recruitment speed and early discharge behavior were associated with rapid force development [Del Vecchio et al., 2019]. Those measurements challenge a universal long recruitment wait. They do not provide recruitment times for every golf muscle, and a motor-unit discharge frequency is not a closed-loop movement bandwidth.

Force development also involves excitation-contraction coupling, contractile behavior, tendon loading and prior activation. Low force does not guarantee precise task control; high force does not prove open-loop control. Signal-dependent variability is a model of how variation changes with drive, not a universal 5–10 percent noise constant. It must specify the measured signal, averaging interval, muscle, task and dependence on force.

29.6 Delay, Bandwidth and Time Remaining Before Impact

Control bandwidth concerns the frequency response of a specified closed loop, with a declared performance criterion. Reaction time is not a neural clock. A flicker-detection threshold is not a motor sampling frequency. Neither the inverse of a response latency nor the frequency of a single shaft mode determines the bandwidth of a coupled golfer-club feedback loop.

Definition 29.10. Delay Adds Phase to a Defined Loop

For a locally linear, time-invariant single-loop approximation,

$$L(s) = C(s)P(s)e^{-sd}, \quad \arg e^{-i\omega d} = -\omega d.$$

The pure delay has unit magnitude and a frequency-dependent phase lag. With $d = 0.05$ s, it contributes -90° at 5 Hz and -180° at 10 Hz. Other controller and plant dynamics also contribute phase and gain. These numbers alone do not establish a crossover frequency, stability margin or a maximum human correction rate.

A shaft frequency also depends on boundary conditions, loading and the mode being measured. The relevant plant includes the body's coupled inertia, contact, actuator dynamics and task output, with properties changing through the swing. For a short nonstationary movement, finite-horizon sensitivity is often more informative than a steady-state bandwidth analogy.

Linearizing an augmented model around a declared nominal motion gives $\delta\dot{x} = A(t)\delta x + B(t)\delta u$. At a fixed terminal time T ,

$$\delta x(T) = \Phi(T, t_0)\delta x(t_0) + \int_{t_0}^T \Phi(T, s)B(s)\delta u(s) ds.$$

The transition matrix Φ transports both initial errors and control effects through the changing dynamics. If new information cannot affect u until $t_0 + d$, its new-response integral starts there; already prepared commands and other pathways still contribute. Torque limits, activation and output sensitivity determine the attainable change. A norm ratio at one instant does not determine this integral.

Example 29.5. An Actuator Can Begin Responding Without Completing a Correction

In a manufactured incremental model, let $\dot{a} = (1 - a)/\tau_a$ with $a(0) = 0$ after a separately specified pure delay. If the remaining time is $h \geq 0$, then

$$a(h) = 1 - e^{-h/\tau_a}, \quad I_a(h) = \int_0^h a(s) ds = h - \tau_a(1 - e^{-h/\tau_a}).$$

I_a has units of time. Multiplication by a constant torque scale gives angular impulse, not directly club-speed change. With $\tau_a = 50$ ms, remaining times of 10, 40 and 100 ms give normalized impulses of approximately 0.937, 12.466 and 56.767 ms. Instant actuation would give 10, 40 and 100 ms. Prior activation, state-dependent torque transmission or a different force law changes this comparison.

The impulse example assumes constant mechanical mapping. In the actual linearized chain the kernel $\Phi(T, s)B(s)$ can rotate, amplify or cancel contributions. An EMG onset before impact therefore establishes opportunity for influence, not its sign, size or usefulness.

Impact may also occur at an event time rather than a prescribed clock time. For a smooth event $\psi(x, t) = 0$ with transverse nominal crossing,

$$\delta T = -\frac{\psi_x \delta x(T)}{\psi_t + \psi_x F(T)}, \quad \delta y_{\text{event}} = y_x \delta x(T) + (y_t + y_x F(T))\delta T.$$

$F(T)$ is the nominal state derivative and $\delta x(T)$ the fixed-time perturbation just before the event. These formulas require a nonzero denominator and differentiability; grazing or changing contact modes need separate treatment. A controller can alter arrival time as well as state. Speed at a fixed time and speed at impact are consequently different comparisons.

29.7 Variability, Energy and Computation

Movement variability does not pass unchanged from a muscle to a club. A local task map with sensitivity J gives $\delta y \simeq J\delta x$, hence

$$\Sigma_y \simeq J\Sigma_x J^T.$$

The covariance includes correlations, not just marginal standard deviations. For two normalized state components with equal standard deviation 0.02 and correlation ρ , take $y = (x_1 + x_2)/\sqrt{2}$. Then $\text{var}(y) = 0.0004(1 + \rho)$. Correlations 0, -0.75 and 0.9 give standard deviations 0.0200, 0.0100 and approximately 0.0276. Equal component variability can therefore produce substantially different task variability.

This is a geometric calculation, not evidence that golfers use those correlations. Large sensitivity in one direction can amplify errors while variation in another direction has little immediate effect. Under the same local approximation, the law of total covariance also separates trial-to-trial state uncertainty from conditional observation noise. Measurement error, changing conditions and systematic bias must be accounted for before attributing consistency to a neural strategy.

A coefficient of variation is standard deviation divided by the magnitude of a nonzero mean. A claimed club-speed percentage needs a defined golfer sample, club, instruction, number of trials and instrument uncertainty. It cannot be derived by comparing an unrelated muscle-force percentage with a proprioceptive detection threshold. Drift can amplify or attenuate perturbations depending on the trajectory; its magnitude alone proves neither robustness nor precision.

Mechanical structure influences the controller through these mappings. The same torque can cause different acceleration in different configurations; free acceleration depends on the full inverse inertia, including a Schur-complement effect when other coordinates respond. A constraint changes admissible response, stiffness changes local dynamics, and an actuator's state changes available force. Preparation can alter all of them, while feedback uses information about the resulting state. These mechanisms are intertwined without being interchangeable explanations.

Key Principle 29.2. Separate Mechanical Work, Metabolic Cost and Computation

Joint mechanical power is torque times joint rate; net negative work, positive work and coactivation have different physiological implications. A controller's metabolic expenditure is a different quantity. An estimate of whole-brain power cannot be assigned entirely to swing computation or converted into a known number of optimization iterations.

Counting neurons, synapses or spikes does not identify the arithmetic operations needed by a digital optimizer. Even a specified state dimension does not determine solve time: horizon, discretization, constraints, sparsity, numerical condition-

ing, solver, hardware and accuracy all matter. A valid efficiency comparison specifies the task, performance, energy boundary and measurement protocol.

Example 29.6. Online Control and Learning Across Trials

A robot with a 1 ms command period needs sensing, estimation, computation and command transmission to meet that deadline, or an explicitly designed asynchronous policy. A 20-state description alone does not show whether this is feasible. A learned policy can be quick to evaluate and still depend on feedback; an optimizer can also be warm-started from previous solutions. Neither architecture follows from the word “robot.”

Likewise, practice can alter prepared commands, prediction, sensory weighting and feedback gains. The relevant test is what changes in behavior and response to perturbations, including retention and transfer, rather than whether the action appears effortless.

Key Principle 29.3. Key Takeaways

The mechanical model defines what force can do; the observation model defines what can be inferred; the actuator model and time remaining define which changes are feasible. A policy connects them to a task, and learning changes the policy or model over experience. No one instantaneous ratio, neural delay or brain region replaces this chain of reasoning.

Golf-specific perturbation evidence is needed to quantify the relative contributions of parallel pathways during a downswing. A useful study measures perturbation time, muscle response, force or torque, body and club motion, event time and strike outcome. Similar motions can be produced by different controllers, so a successful fit is not automatically identification of the neural mechanism.

Exercises 29.1. Chapter Exercises and Worked Answers

1. Correct the Forward Prediction

A model outputs $\dot{x} = (2, -4)^T$ in the appropriate state units per second, with current state $x = (1, 3)^T$ and step $\Delta t = 0.01$ s. Compute one Euler step and explain its limit.

Worked answer. The prediction is $(1.02, 2.96)^T$. It adds a change to the existing state. Equating the next state directly to $(2, -4)^T$ is dimensionally wrong. Smoothness gives local error of order Δt^2 ; this estimate alone does not verify the model or numerical stability over a whole swing.

2. Diagnose an Infeasible Inverse

For the normalized rectangular example above, compute the bounded input and residual. Does failure of instantaneous position control prove that positions can never be controlled?

Worked answer. Minimizing $1 + (u - 3)^2 + u^2$ gives unconstrained $u = 1.5$, or bounded $u = 1$ with $|u| \leq 1$. The desired-minus-achieved residual is $(0, 1, 2, -1)^T$.

It exposes both the kinematic mismatch and conflicting acceleration requests. Over time, acceleration affects position; finite-time reachability must be analyzed through the dynamics, not from the zero position block alone.

3. Compare Delay With Activation

A new response has a separately defined delay of 50 ms. Impact is 90 ms after the perturbation. In the chapter's zero-initial-state actuator with $\tau_a = 50$ ms, what time remains, and what is its normalized impulse?

Worked answer. The remaining time is 40 ms. The normalized impulse is $0.04 - 0.05(1 - e^{-0.8}) \simeq 0.012466$ s. Multiplying by a constant 10 N m scale gives approximately 0.12466 N m s. Computing a club velocity additionally requires the mechanical response map. The example does not erase pre-existing activation or responses in other pathways.

4. Interpret a Frequency Comparison

A pure delay is 50 ms. Calculate its phase at 5 Hz and explain why a shaft mode at that frequency does not determine whether the full loop is stable.

Worked answer. The lag is $-2\pi(5)(0.05) = -\pi/2$ radians, or -90° . Stability depends on the full open-loop gain and phase, including the plant and controller, and on the chosen approximation. A shaft mode under one support condition is neither the whole plant nor a sampling rate. Nonstationary strike tasks also need finite-time analysis.

5. Explain Consistency Without Inventing a Noise Constant

Use the two-component covariance example. What happens when the correlation changes from 0 to -0.75 , and what would be needed to apply the result to golf?

Worked answer. Task variance falls from 0.0004 to 0.0001 and its standard deviation halves. The marginal component deviations remain 0.02. A golf application requires a measured covariance in declared coordinates, a validated task sensitivity and instrument uncertainty. It also needs event-time effects if trials are compared at impact. The example cannot establish a universal club-speed coefficient of variation or an automatic stabilizing role for drift.

6. Distinguish Three Errors

A player correctly predicts that a shot will miss the target. The outcome is nevertheless better than expected in the scoring system. Identify the sensory prediction, task and reward errors.

Worked answer. The observed consequence can have a small sensory prediction error, a substantial task error and a positive reward prediction error. Their dimensions and reference values differ. One scalar reward does not specify which muscle, state estimate or pre-shot action should change.

7. Test the Free-Energy Identity

For joint probabilities $p(y, z) = (0.12, 0.18)$, compute the evidence and the minimizing belief. Does minimizing variational free energy demonstrate a reduction in mechanical energy?

Worked answer. The evidence is 0.30 and the posterior belief is $(0.4, 0.6)$. At that belief the KL term vanishes and $\mathcal{F} = -\log 0.30 \simeq 1.204$ with natural logarithms.

A different belief adds a nonnegative KL term. This is an information-theoretic identity, not a mechanical or metabolic energy budget and not proof of a neural implementation.

8. Evaluate a Recruitment Claim

Does size-ordered recruitment require waiting a fixed interval before higher-threshold units contribute? What measurements distinguish recruitment from force development?

Worked answer. Recruitment order does not prescribe a universal serial waiting time. Recruitment speed, discharge behavior, prior activation and contractile transmission jointly affect force. Measure motor-unit or appropriately interpreted EMG activity and force with synchronized timing. Evidence from an explosive isometric leg task cannot simply be assigned to a wrist during a swing.

9. Design a Controller Comparison

Compare a prepared command sequence with a learned feedback policy. What information is missing from a proposal to solve a 20-state optimization every millisecond?

Worked answer. Perturb the movement in a controlled way and compare subsequent commands and task outcomes, accounting for delays and actuator state. Similar unperturbed trajectories need not reveal which architecture generated them. For computation, specify horizon, objective, constraints, discretization, solver tolerance, hardware and sensing/transmission time. State dimension and movement duration alone cannot yield a minimum computation time.

10. What Would Establish Useful Within-Swing Correction?

Design a golf experiment that separates response onset from useful response authority, while allowing learning between trials.

Worked answer. Apply safe, controlled perturbations at declared phases with randomized timing and appropriate comparison trials. Record synchronized muscle activity, mechanical response, club pose and velocity, impact time and strike outcome. Estimate uncertainty and distinguish local changes from improvements in the task. Compare early and late perturbations, repeated exposure and retention; control equipment, instruction and starting state. An early EMG difference shows a response, while a reliable task improvement requires mechanical and outcome evidence. Attribution to a particular neural pathway requires additional evidence beyond timing alone.

Chapter 30

Motor Control II: Learning the Swing

In Plain Language 30.1. What This Chapter Is About

A well-struck shot leaves several kinds of information: the ball's flight, contact sound, pressure at the hands, effort, balance and the golfer's judgment of the result. Learning involves finding which information predicts useful outcomes, then changing preparation and action so those outcomes become more reliable. The sensation of a good swing can help, but reproducing a sensation does not guarantee reproducing its mechanics or result.

This chapter connects feel to physical state, prediction, task goals and practice design. Its central distinction is between performing well now and acquiring a capability that survives a delay or a change of conditions. Neither a satisfying shot nor a compelling explanation of it is sufficient evidence of lasting learning.

30.1 Stages of Motor Learning

Fitts and Posner's cognitive, associative and autonomous stages describe changing demands on attention and skill organization [Fitts and Posner, 1967]. They are useful descriptions, not a fixed timetable, a neurological diagnosis or a guarantee that improvement proceeds monotonically. A golfer may execute a familiar putt automatically while deliberately learning an unfamiliar bunker shot.

Definition 30.1. Cognitive Stage

The learner is establishing the task, relevant cues and possible actions. Instructions and explicit strategies may occupy substantial attention. What matters is the difficulty of organizing this task for this learner, not the number of weeks since a first lesson.

Definition 30.2. Associative Stage

The learner is refining relationships among preparation, action, feedback and results. Some components become reliable while others remain uncertain. Improvement can involve a better strategy, more accurate prediction, different coordination or better use of feedback.

Definition 30.3. Autonomous Stage

Execution demands less deliberate supervision in a familiar context. Automaticity does not mean perfect repetition, immunity to pressure or absence of neural feedback. A change of equipment, lie or task can require renewed attention.

Example 30.1. The Three Stages in Golf

Consider learning distance control in putting. Initially the golfer may consciously compare stroke size and roll distance. Later, a familiar distance can evoke an effective preparation without that calculation. To assess the change, record both distance error and the attention required. A single lower error does not establish a stage transition; fatigue, warm-up and helpful instructions can temporarily alter performance.

There is no justified one-to-one sequence from prefrontal cortex to premotor cortex to cerebellum. Behavioral stages do not identify the neural structures responsible for them. Likewise, variability need not fall in every joint as skill improves. We need a task-level account before deciding which variation is beneficial or harmful.

30.1.1 Bernstein's Problem: Coordination and Task Redundancy**Definition 30.4. Bernstein's Problem**

A body has many mechanically coupled degrees of freedom and many possible muscle actions. A task usually constrains only some combinations of these variables. The coordination problem is to organize feasible actions that achieve the task while respecting physical and physiological constraints [Bernstein, 1967].

The number of degrees of freedom depends on the model: its joints, constraints, flexible structures and contact assumptions. Position, velocity and acceleration are not three independently selectable commands at every joint. They are related by differentiation and equations of motion. Muscle redundancy introduces another distinction: different activation patterns may produce similar net joint torques while differing in force, stiffness and metabolic demand.

Key Principle 30.1. Three Coordination Strategies

Restricting joint motion, coupling motions and exploiting mechanical interactions are possible strategies. They are not obligatory beginner, intermediate and expert stages. Restricting motion can simplify a task but increase effort; allowing more motion can improve adaptability or make delivery less reliable. No strategy is most energy-efficient without a specified system, task and energy measure.

Example 30.2. Bernstein's Problem in Golf

Suppose two small joint deviations contribute to a scalar delivery error as $y = q_1 + q_2$. All pairs satisfying $q_2 = -q_1$ cancel in this local task model. A golfer can vary both joints while preserving delivery. If a different club or operating point changes the sensitivity to $y = q_1 + 2q_2$, the same cancellation no longer works. A coordination pattern is task- and context-dependent.

More generally, let delivery be $\mathbf{y} = G(\mathbf{x}, c)$, where c specifies club, lie and other conditions. Near a nominal state, $\delta\mathbf{y} \approx J\delta\mathbf{x}$, with $J = \partial G/\partial\mathbf{x}$. For zero-mean deviations with covariance Σ_x ,

$$\Sigma_y \approx J\Sigma_x J^T.$$

For a scalar output with mean error μ_y , mean squared task error is $\mu_y^2 + \text{Var}(y)$. Precision and bias must both be assessed. If $\mathbf{x}$ mixes angles, velocities and forces, its covariance has mixed units; a raw trace is not a coordinate-independent measure of skill.

Even with two identically scaled angular coordinates, smaller joint variability can mean greater task variability. For $J = (1, 1)$, consider manufactured covariance matrices in squared degrees:

$$\Sigma_A = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}, \quad \Sigma_B = \begin{pmatrix} 4 & -3.8 \\ -3.8 & 4 \end{pmatrix}.$$

Both are positive definite. Although $\text{tr}\Sigma_A = 2 < 8 = \text{tr}\Sigma_B$, the task variances are 3.8 and 0.4 square degrees respectively. The second pattern has greater joint variation but better cancellation. This proves a limitation of a variability metric, not that a particular golfer uses a neural synergy. Measurement error, shared mechanics and task constraints can also produce covariance.

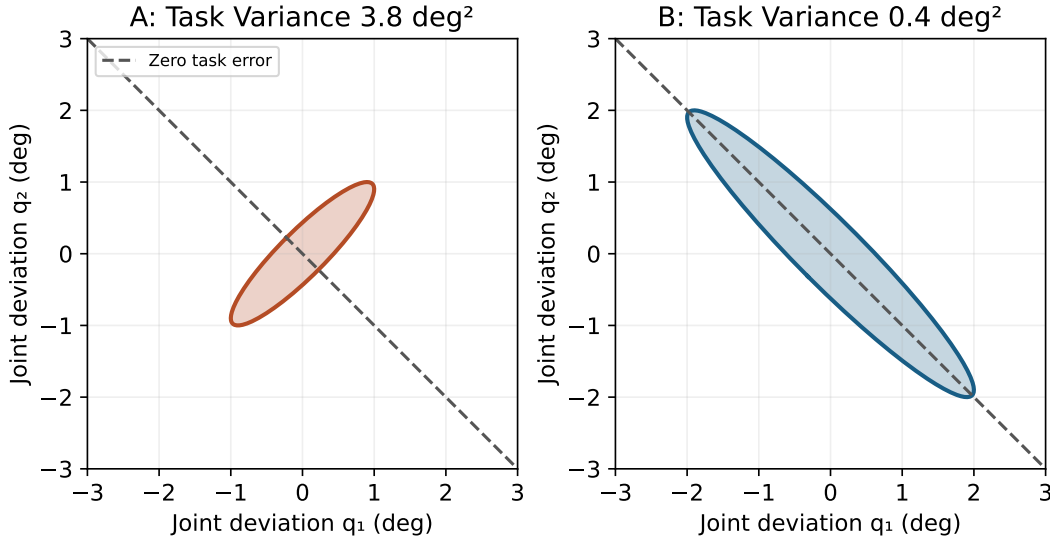


Figure 30.1: Manufactured Covariance Ellipses for the Task $y = q_1 + q_2$. Greater Joint Variation Can Coexist With Smaller Task Variance. Ellipses Have Unit Mahalanobis Radius; They Are Not Human Measurements.

30.2 The Feel of a Good Shot: Proprioception as a Control Cue

Key Principle 30.2. The Setpoint Hypothesis: Feel as a Testable Cue

A remembered sensation may serve as a cue for preparing or evaluating a swing. Calling it a reference or setpoint is a modeling choice. The hypothesis becomes useful when it predicts observable changes under altered sensory feedback or task conditions. It does not establish that the nervous system stores and replays one complete sensory trajectory.

30.2.1 Proprioception: Signals and State Estimates

Key Principle 30.3. Proprioceptive Receptors

Muscle spindle, tendon, skin and joint afferents contribute information related to movement, position and force. Their signals depend on physiological and mechanical conditions. Perception combines populations of signals with central information; there is no single receptor that directly reports a perfectly calibrated joint angle or total muscle force [Proske and Gandevia, 2012].

The golfer's report of feel also includes touch, vision, sound, effort and vestibular information. Vestibular signals concern head motion and orientation, including angular motion and gravito-inertial acceleration; they are not equivalent to eye position. Holding the gaze near the ball does not establish a stationary head or silence vestibular input. A joint's perceptual accuracy cannot be assigned a universal threshold independently of movement, load and measurement protocol.

Let the physical state include configuration $\mathbf{q}$, velocity $\mathbf{v}$, activation $\mathbf{a}$ and relevant elastic or contact states $\mathbf{z}$:

$$\mathbf{x} = (\mathbf{q}, \mathbf{v}, \mathbf{a}, \mathbf{z}), \quad \dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}, c).$$

An illustrative sensory channel is

$$\mathbf{s}_t = h(\mathbf{x}_{t-d}, c) + \boldsymbol{\eta}_t.$$

Here d is that channel's delay and $\boldsymbol{\eta}_t$ represents noise or unmodeled influences. An estimator combines observations, command history and a forward model to form $\hat{\mathbf{x}}_t$. Neither the observer nor the golfer is thereby granted direct access to the complete physical state. Different channels have different delays and reliabilities; treating them as one synchronous measurement can create apparent prediction errors.

An elementary Gaussian example clarifies the role of reliability. Suppose an unbiased scalar prediction is 20° with variance 9 deg^2 , and an independent unbiased observation is 22° with noise variance 1 deg^2 . Combining their precisions gives an estimate of 21.8° and variance 0.9 deg^2 . This is a calculation under stated assumptions, not an asserted neural algorithm. If the prediction and observation share an error, the independence assumption fails and confidence can be overstated.

30.2.2 Reference Trajectories, Predictions and Policies

Constraint Insight 30.1. Pedagogical Simplification

A desired trajectory, a predicted trajectory and an estimated trajectory answer different questions: what should happen, what is expected to happen, and what appears to be happening. A successful controller need not make all three identical. A reference trajectory is one possible representation; a policy can choose actions from estimated state, task and context without replaying one fixed path.

Write a policy as $\mathbf{u}_t = \pi_\theta(\hat{\mathbf{x}}_t, c, g, t)$, with goal g and adjustable parameters θ . Learning may change the estimator, the policy, the valuation of outcomes or the preparation of the initial state. A familiar feeling can cue a policy without uniquely specifying its mechanical trajectory. The same verbal description of feel can accompany different motions; similar motions can feel different when fatigue or sensory conditions change.

Example 30.3. The Feel of a Good Swing

A golfer reports a light grip and an unhurried transition after a centered strike. Record the report alongside impact location, delivery and ball flight. If the cue predicts centered contact across later trials and relevant conditions, it has practical value. If the player reports the same feel while contact moves across the face, the cue needs recalibration. Neither result requires dismissing subjective experience or treating it as an instrument reading.

Impact sensations mostly inform evaluation and later attempts. A response triggered by the collision cannot causally correct an earlier impact condition. Forces and vibrations measured at the clubhead cannot be equated to loads throughout the body without a transmission model. A prepared grip or muscle state may affect impact mechanically; that differs from a new neural correction initiated after contact. The computational-brain and impact chapters treat those timing and coupling questions separately.

30.3 Contemporary Motor Learning Frameworks

Frameworks emphasize different explanatory levels: generalization across tasks, relations between a performer and environment, coordination dynamics, or changes induced by varied practice. Their terminology does not make them mutually exclusive, and a successful drill

rarely identifies one uniquely correct theory.

30.3.1 Schema Theory

Definition 30.5. Motor Schema

In schema theory, experience supports relationships among initial conditions, response specifications, sensory consequences and outcomes, allowing a skill to be parameterized for new requirements [Schmidt, 1975]. This is a theoretical account of generalization, not a claim that distance is converted into force by one universal linear rule.

Example 30.4. Golf Schema

For a projectile launched and landing at equal height in a vacuum, $R = v_0^2 \sin(2\alpha)/g$. At fixed angle α with positive $\sin(2\alpha)$, v_0 scales as $\sqrt{R}$: quadrupling range doubles launch speed. Real golf adds aerodynamic forces, spin, unequal elevations and ground interaction. Even this simplified case disproves a universal proportionality between required speed and distance. Launch speed also does not uniquely determine a muscle-force history.

Practicing several distances can supply information about a response-to-outcome relationship. Whether that information transfers depends on where practice samples the relationship, how noisy the feedback is and whether test conditions preserve it. Interpolating among familiar distances and extrapolating to a different club or lie are different problems. A theory's prediction of useful variation is not proof that every varied schedule beats every constant schedule.

30.3.2 Ecological Dynamics and Constraints-Led Approach

Definition 30.6. Ecological Dynamics

An approach that studies action in relation to the performer, task and environment, emphasizing how information and available opportunities for action shape coordination [Davids et al., 2008].

Definition 30.7. Constraints-Led Approach

Practice design that changes task, environmental or performer-related constraints to encourage exploration of useful solutions. Examples include target size, lie, club and permitted landing area. The resulting solution must be measured; the constraint does not uniquely prescribe it.

A shorter target may change backswing amplitude, speed, launch or club choice. It does not force energy transfer to occur earlier. A narrow corridor penalizes dispersion but does not mechanically require a draw. The design question is whether the drill makes the intended

information and skill relevant while retaining a meaningful relationship to the eventual golf task. A drill can be difficult yet teach an adjustment that does not transfer.

30.3.3 Dynamical Systems Theory and Attractors

Definition 30.8. Movement Attractor

An attractor is an invariant set approached by trajectories from a specified neighborhood in a specified dynamical system. A finite swing that ends at impact is not automatically an attractor. One may instead study finite-time sensitivity, a time-dependent tracking system, or a return map across repeated trials, stating the system and stability definition [Kelso, 1995].

Example 30.5. Attractors in Golf

For a deliberately simplified trial model, let m_n be a learned scalar response and let r be a fixed required response. Set

$$m_{n+1} = am_n + b(r - m_n).$$

The retention factor a and error sensitivity b are dimensionless model parameters, not measured neural constants. For $a = 1$, the task error $e_n = r - m_n$ satisfies $e_{n+1} = (1 - b)e_n$. With $r = 4$ and $m_0 = 0$, $b = 0.25$ gives errors 4, 3, 2.25, 1.6875, ... With $b = 1.5$, errors alternate in sign while shrinking. With $b = 2.1$, their magnitude grows. More aggressive correction can destabilize this model.

The general stability condition is $|a - b| < 1$. When stable and $1 - a + b \neq 0$, the equilibrium is $m_* = br/(1 - a + b)$, leaving error $e_* = (1 - a)r/(1 - a + b)$. For $a = 0.98$, $b = 0.25$ and $r = 4$, this error is $8/27 \approx 0.2963$. These equations separate stability from zero error and immediate change from retention. They are manufactured examples, not fitted rates of golf learning or proof of a particular practice schedule.

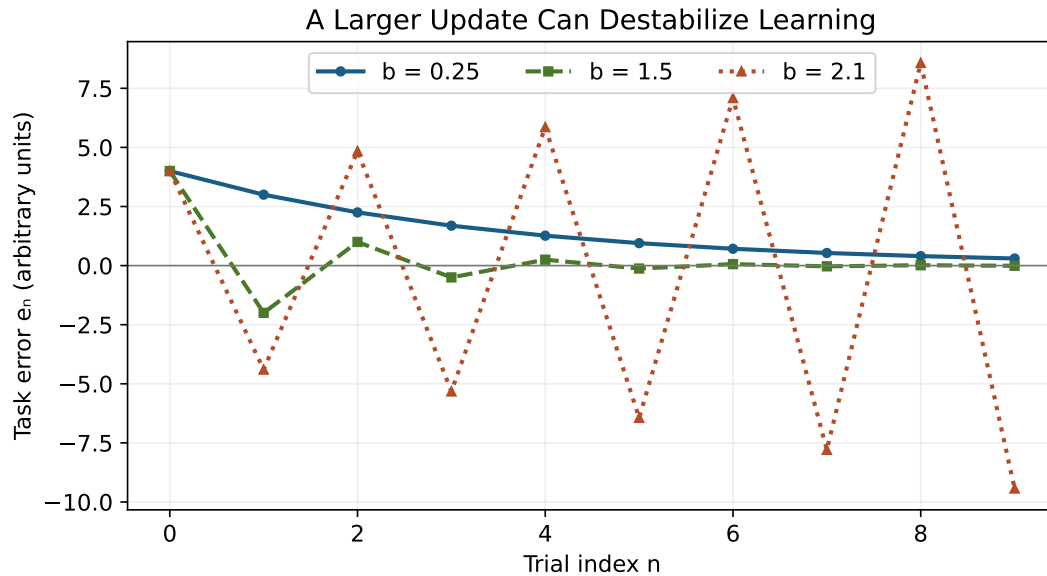


Figure 30.2: Manufactured Trial Updates With $a = 1$, $r = 4$ and $m_0 = 0$. Gains of 0.25 and 1.5 Converge; a Gain of 2.1 Diverges. These Curves Do Not Represent Measured Learning Stages.

30.3.4 Differential Learning and Noise

Definition 30.9. Differential Learning

Wolfgang Immanuel Schöllhorn's differential-learning approach deliberately varies movement conditions and emphasizes fluctuations during practice. It is conceptually distinct from merely randomizing the order of several practiced tasks [Schöllhorn, 2016].

Example 30.6. Differential Learning in Golf

Varying stance or stroke conditions introduces movement variation. Randomizing three target distances changes task order. Repeating one distance in blocks changes neither in the same way. A useful comparison specifies the actual manipulations, total attempts, feedback and later test. Calling a drill differential, variable or random does not establish its effectiveness.

Variation can reveal a useful relationship or obscure it. For example, changing stance, club, target and feedback together makes it difficult to attribute an improvement to one element. This does not forbid combined practice; it limits what that practice can establish scientifically. Added noise is not intrinsically beneficial, and its amplitude and relevance matter.

30.4 Neural Networks in the Brain: What the Evidence Establishes

30.4.1 The Neocortex: Distributed Learning

Key Principle 30.4. The Cortical Hierarchy for Motor Control

Planning, perception and action involve interacting cortical and subcortical systems. A hierarchical diagram can organize computational roles, but arrows in that diagram are not evidence for a literal serial pipeline. Behavioral adaptation, explicit strategy, reinforcement and skill acquisition need to be distinguished before assigning a mechanism [Kitago and Krakauer, 2013].

An observation that two regions are active during learning does not establish which computed the error, which changed a command or where a memory is stored. Those claims require experiments that distinguish the alternatives. Golf adds a further generalization step: a result from a laboratory pointing task may motivate a golf hypothesis without validating it.

30.4.2 The Cerebellum: Plasticity and Prediction

Key Principle 30.5. Cerebellar Learning Mechanism

One influential account involves long-term depression at parallel-fiber–Purkinje-cell synapses, with climbing-fiber activity contributing to plasticity induction. It is not a synapse between parallel and climbing fibers. Nor is that plasticity the only possible learning mechanism. Schonewille and colleagues found preserved performance on several motor-learning tasks in three mouse lines with disrupted expression of this form of LTD. The result limits an obligatory single-mechanism account; it does not establish that LTD is irrelevant in all tasks or species [Schonewille et al., 2011].

A useful mathematical teaching model is gradient learning of a prediction. Let a scalar predicted observation be $\hat{s}_n = \mathbf{w}_n^\top \boldsymbol{\phi}_n$ with loss $L_n = \frac{1}{2}(s_n - \hat{s}_n)^2$. Then

$$\mathbf{w}_{n+1} = \mathbf{w}_n + \eta(s_n - \hat{s}_n)\boldsymbol{\phi}_n. \quad (30.1)$$

This follows by differentiating the specified quadratic loss. The sign depends on defining error as observed minus predicted. It is not an experimentally established synaptic law for golf. Features, timing, learning rate and biological implementation are additional hypotheses.

30.4.3 The Basal Ganglia: Reinforcement Learning

Key Principle 30.6. Dopamine and Reward Prediction Error

Schultz, Dayan and Montague related phasic dopamine responses in conditioning experiments to reward prediction error. A predicted reward need not produce the phasic increase elicited by an unexpected reward; omission can produce a decrease relative to baseline. Their temporal-difference account is a computational interpretation, not a claim that every dopamine neuron reports only one scalar or that a successful golf shot stores its entire trajectory [Schultz et al., 1997].

Using time index t within an episode, a standard temporal-difference expression is

$$\delta_t = R_{t+1} + \gamma V(\chi_{t+1}) - V(\chi_t), \quad 0 \leq \gamma \leq 1.$$

Here χ_t denotes the model's informational state; V predicts discounted future reward and R is a scalar reward in the model. A complete episode or discounting assumption is required for the return to be well-defined. At a terminal state $V(\chi_{t+1}) = 0$. A reward of 1 expected with value 1 gives $\delta = 0$; the same reward expected with value 0.25 gives 0.75. Zero error is not zero reward or absence of baseline neural firing.

Reward design matters. A practice score that rewards carry alone may favor a different behavior from one that penalizes missed fairways or poor contact. A scalar outcome also leaves a credit-assignment problem: which preparation, timing or command helped? It does not directly identify a vector of muscle corrections. Explicit reasoning, structured exploration and sensory information can all help distinguish candidate changes.

30.5 The Setpoint Hypothesis: Errors and Testable Predictions

Key Principle 30.7. The Setpoint Hypothesis: A Mechanistic Model to Test

Suppose a golfer uses a remembered sensory pattern to help prepare a policy and evaluate its execution. This model should be tested against alternatives in which task outcomes, explicit aiming or other sensory cues dominate. Successful prediction in one context does not establish a unique stored pattern, its anatomical location or a universal explanation of swing failure.

Four errors clarify what such a test must measure:

1. Tracking error compares a specified desired state with an estimate: $\mathbf{e}_{\text{track}} = \mathbf{x}_{\text{ref}} - \hat{\mathbf{x}}$. The reference must be defined in compatible coordinates and at a compatible time.
2. Sensory prediction error compares an observation with the predicted observation: $\mathbf{e}_{\text{sens}} = \mathbf{s} - \hat{\mathbf{s}}^-$. Predictions must account for the channel's delay. This is not desired minus actual state.
3. Task error compares the desired and observed outcome: $\mathbf{e}_{\text{task}} = \mathbf{y}_{\text{goal}} - \mathbf{y}_{\text{obs}}$. It depends on the chosen task, such as landing location rather than a joint angle.

4. Reward prediction error compares received and subsequently predicted value with previous expected value. It is not the task-error vector or a synonym for disappointment.

These quantities can disagree. In a manufactured pointing example, a hand at -45° with a $+45^\circ$ displayed rotation places a cursor on a 0° target. Task error is zero. If the predicted cursor is still -45° , sensory prediction error is $+45^\circ$. Mazzoni and Krakauer's visuomotor experiment provides a related empirical dissociation: an explicit compensatory strategy initially achieved the task, yet subsequent implicit adaptation caused drift away from it [Mazzoni and Krakauer, 2006]. That result concerns the tested pointing perturbation, not an inevitability that explicit golf instruction must fail.

Example 30.7. What the Setpoint Hypothesis Can Predict

Before revealing ball-flight feedback, ask the player to rate expected contact and direction. Then compare the prediction with measured contact and delivery. Separately record the felt movement and the actual outcome. Alter one relevant sensory or mechanical condition in a controlled comparison. If feel remains constant while delivery changes, a fixed-feel target is insufficient in that condition. If calibration improves, test whether it survives a delay and transfers. Expectation, attention and feedback changes remain competing explanations unless the design separates them.

A useful protocol therefore records predictions before outcomes, avoids selecting only memorable good shots and includes unsuccessful attempts. Otherwise hindsight can make a vague sensation appear more predictive than it was. Asking about feel may itself change attention, so compare equivalent questioning across conditions.

The model also does not diagnose the yips as a corrupted reference trajectory. Involuntary movements, anxiety-related performance changes and other clinical presentations require appropriate distinctions; the observational yips literature does not support a single universal cause [Adler et al., 2011]. Likewise, a return of an old swing under pressure does not prove that two stored trajectories were activated simultaneously. Describe the observed change first, then test explanations.

30.6 Practice Design: Implications for Learning

Practice performance is measured while training. Retention assesses the learned task after a delay, with the test conditions specified. Transfer assesses performance under a changed task or context. These are related but different outcomes. Temporary guidance, warm-up, fatigue and familiarity can change immediate performance without the same change in retained capability.

For illustration, suppose practice condition A ends with mean error 2 and B with 3 in the same units. On a matched delayed test A has error 6 and B 4. A looked better during acquisition; B is better on that retention test. These invented numbers demonstrate the logic of the comparison. They are not evidence for a particular drill, neural process or schedule. A test with the guidance removed also evaluates dependence on that guidance; report the change explicitly.

30.6.1 Random vs. Blocked Practice: Contextual Interference

Definition 30.10. Blocked Practice

Repeated attempts at one task variant before switching to another, such as a block of putts at each of three distances.

Definition 30.11. Random Practice

Task variants interleaved in an unpredictable order. The task set, number of attempts and feedback can be held constant while order changes.

Constraint Insight 30.2. Contextual Interference Depends on the Task

Contextual interference concerns how mixing practiced tasks affects acquisition, retention and transfer. It is not a universal random-practice advantage. In Porter and Magill's novice putting experiment, a schedule progressing from blocked to serial to random practice produced better retention than the compared blocked and random schedules [Porter and Magill, 2010]. The result supports testing progression; it does not provide a universal dose for every golfer or full swing.

A practical design can start with enough repetition to understand the task and feedback, then introduce decisions representative of play. The transition should follow observed readiness and the purpose of practice, not a rule that difficulty is always desirable. Compare schedules with equal task exposure; otherwise increased variation, extra attempts or different feedback can explain the result.

30.6.2 External vs. Internal Focus of Attention

Definition 30.12. Internal Focus

Attention directed toward the performer's body movement, such as the action of the arms.

Definition 30.13. External Focus

Attention directed toward an effect of movement, such as the club's motion or the intended ball flight. A cue such as "smooth tempo" is not inherently external; its meaning and the player's interpretation matter.

Wulf, Lauterbach and Toole tested 22 golf novices practicing pitch shots. Attention to club motion, compared with arm motion, benefited practice and next-day retention without the instructions [Wulf et al., 1999]. This provides golf-specific evidence for that comparison. It does not show that all technical explanations impair learning, that every external cue is equivalent, or that one neurological mechanism has been uniquely established.

Technical analysis and an execution cue serve different purposes. Between attempts, a player and coach can use mechanics to diagnose delivery. During an attempt, a concise cue can direct attention toward the intended effect. Evaluate both the outcome and whether

the cue is understood. The best-looking explanation is not necessarily the most useful instruction at the moment of execution.

30.6.3 The Quiet Eye: Attention Timing

Definition 30.14. Quiet Eye

A final task-relevant fixation or tracking gaze defined relative to a movement event, using explicit spatial and duration thresholds. It is an operational eye-tracking measure; the definition and relevant event must be reported for the task.

Vine, Moore and Wilson studied 22 skilled male golfers in a putting intervention combining gaze feedback and instructions. The trained group improved some laboratory and competitive measures. Retention differences were not significant for every endpoint, and gaze data from five trained participants were lost. The authors also identified possible motivational confounding and limitations of putts per round as a putting measure [Vine et al., 2011]. These results justify a bounded putting hypothesis, not a prescribed two-to-three-second fixation for every golf shot.

Gaze timing can influence information pickup, preparation and attention, but those are distinguishable candidate mechanisms. A relation between fixation duration and success does not establish how much success was caused by fixation. Extending a putting result to a driver requires a new argument and evidence. Eye stillness alone is not a measure of balance, head stability or vestibular activity.

30.6.4 Deliberate Practice

Key Principle 30.8. Characteristics of Deliberate Practice

For the practice plan here, specify the skill, a measurable criterion, feedback that can inform a change, a manageable challenge and a later test. This is an explicit design proposal, not a claim that every useful learning event requires immediate feedback or continuous conscious attention. Feedback frequency, timing and content are experimental choices.

Example 30.8. Deliberate Practice in Golf

For driver practice, choose a delivery-and-dispersion objective rather than maximum speed alone. Record club, ball, target, wind and measurement method. Obtain a baseline sample, test one defined cue or drill, and record every attempt under a stated inclusion rule. Later, repeat a matched assessment without extra coaching. Add a separate transfer test with another target or relevant condition. Report speed, strike location and dispersion together so improvement in one does not conceal deterioration in another.

Small samples deserve modest conclusions. Under an illustrative independent Bernoulli model, eight successes in ten attempts give an observed rate of 0.8, but a 95% Wilson interval is approximately [0.490, 0.943]. Correlated attempts or changing conditions undermine that

simple model. A single excellent shot provides still less information about reliability. Report uncertainty and the practical size of a change, not just whether a preferred cue produced one memorable result.

There is no arithmetic route from a large practice total to guaranteed expertise. At 52 weeks per year, 10 hours weekly for 10 years is 5,200 hours. Reaching 10,000 hours at that rate takes about 19.23 years; at 20 hours weekly it takes about 9.62 years. These are bookkeeping examples, not thresholds or recommended workloads. Practice histories differ in content, opportunity, feedback and recovery, so hours alone cannot identify the cause of different outcomes.

30.7 Individual Differences: Why Learning Rates Differ

Key Principle 30.9. Sources of Individual Differences

Prior experience, physical capabilities, sensory information, task understanding, practice opportunity, feedback and changing circumstances are candidate contributors to different learning trajectories. Observing a faster learner does not isolate a genetic, motivational or cognitive cause. Specify the comparison and consider baseline ability, task difficulty and measurement uncertainty.

Transfer from another sport may involve perception, timing, physical capacity or useful strategies, but can also involve interference. Similarity of an action's appearance is insufficient: the relevant dynamics, equipment and success criterion may differ. Test what carries over under a defined change rather than assuming that "athleticism" explains every difference.

The affine-control viewpoint contributes a precise question: what must the golfer learn about how state, passive mechanics and commands jointly affect delivery? In a model $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{B}(\mathbf{x})\mathbf{u}$, passive evolution is state-dependent, and the selected definition of zero input matters. A prepared activation or elastic state can remain after subsequent commands are set to zero. Learning may improve preparation or the timing of intervention; that does not make mechanics irrelevant or prove a unique energy-optimal strategy.

To test this connection, compare specified policies or counterfactual interventions from matched initial states and report feasible actuation, contact and uncertainty. Distinguish model prediction from human validation. A reduction in simulated commanded torque is not automatically lower muscular effort, lower joint loading or better transfer to play.

Key Principle 30.10. Key Takeaways

Feel, mechanics and outcome provide related but nonidentical information. State estimates, desired trajectories, sensory predictions and reward predictions must be kept distinct. Coordination is judged against a task, including bias and covariance, rather than the smallest movement variation. Practice designs should make their proposed mechanism testable and assess retention and transfer. Neural and golf-specific claims must stay within the evidence that supports them.

Exercises 30.1. Chapter Exercises

1. Describe a golf skill you are learning. What observations distinguish increasing automaticity from a temporary reduction in task error?
2. After a poor shot, distinguish a useful report of feel from an inferred physical cause. Design one comparison that can test whether the report predicts contact quality.
3. Derive the speed ratio for vacuum ranges R and $4R$ at fixed launch angle and equal launch/landing heights. Explain why it is not a force-scaling rule for golf.
4. Explain how a schedule can look better during practice but worse in retention. Identify two design differences that would confound a comparison of blocked and random practice.
5. Separate a mechanical diagnosis between attempts from an attentional cue during execution. What evidence would justify retaining the cue?
6. Define a quiet-eye measurement and explain why a putting intervention does not establish a universal full-swing fixation duration or a vestibular mechanism.
7. Design a progression from repetition to varied decisions for a beginner. Specify a criterion for advancing and a delayed test that could reveal dependence on guidance.
8. Calculate task variance for $J = (1, 1)$ and the two covariance matrices in this chapter. Which coordination is more task-precise? Can the result rank muscular energy cost?
9. Derive the quadratic-loss prediction update. Explain the distinction between this algorithm and a demonstrated cerebellar synaptic mechanism.
10. Design a driver practice assessment with baseline, feedback, retention and transfer. Explain what eight successes in ten independent trials can and cannot establish.
11. For the scalar trial update, derive the stability condition and steady error. Relate preparation and model learning to the zero-input counterfactual without equating zero commands with zero activation.
12. Give an example of possible positive or negative transfer from another sport. Identify what must remain comparable and what must be measured before attributing the change to transfer.

30.8 Worked Answers

1. Record task error over repeated, comparable assessments and how much deliberate supervision is required. A familiar task becoming easier to execute while preserving performance is compatible with automaticity. One warm-up improvement or helpful

instruction cannot establish lasting change; add a delayed assessment.

2. Record a prediction of contact before seeing measured impact location, using the same questions in each condition. Compare all trials, not only good ones. An association supports the report's predictive use; manipulating a cue or sensory condition with appropriate controls is needed for a stronger causal claim.
3. From $v_0 = \sqrt{Rg/\sin(2\alpha)}$, $v_0(4R)/v_0(R) = 2$. The assumptions exclude golf aerodynamics and ground interaction. Mapping launch speed to force requires mass, impulse, coupling and a time history; distance alone does not provide these.
4. Temporary guidance can improve acquisition while a different condition retains more when guidance is removed. Unequal task exposure and different feedback confound an order comparison. Use matched delayed tests and state whether guidance, task order or context changed.
5. Diagnose measured face delivery, path or strike between shots, then test an understandable cue aimed at the intended effect during execution. Retain it if its practically useful benefit persists under suitable later tests, while tracking other outcomes and uncertainty. A verbal category alone is not proof of effectiveness.
6. Specify gaze location, allowable angular deviation, minimum duration and its timing relative to stroke events. The cited intervention concerns putting and bundles training elements. Neither outcome association nor eye fixation isolates head motion or vestibular processing.
7. Begin with repeated attempts that permit understanding feedback, then mix relevant distances or targets when the golfer can explain the task and respond meaningfully. Use a matched delayed test without extra prompts and a distinct transfer condition. The progression is a proposal to evaluate, not a universal optimum.
8. $J\Sigma_A J^T = 1 + 1 + 2(0.9) = 3.8$ and $J\Sigma_B J^T = 4 + 4 - 2(3.8) = 0.4$. B is more precise for this unbiased scalar task despite greater joint variation. Neither matrix supplies muscle work, activation or joint loading, so no energy ranking follows.
9. $\nabla_{\mathbf{w}} L = -(s - \mathbf{w}^T \phi)\phi$; subtracting $\eta \nabla L$ gives the stated update. Physiological interpretation additionally requires evidence for the features, teaching signal, timing, plasticity and task. A mathematical gradient is not evidence that those biological requirements hold.
10. Fix the target criterion and measurement conditions, record a baseline, compare a specified intervention, and repeat a delayed assessment plus a separate transfer test. The independent-trial Wilson interval for 8/10 is approximately [0.490, 0.943]. It shows substantial uncertainty; it does not establish an 80% stable success probability or validity of independence.
11. Subtract the fixed point to get $m_{n+1} - m_* = (a - b)(m_n - m_*)$, so asymptotic stability requires $|a - b| < 1$. Solving the fixed-point equation gives $e_* = (1 - a)r/(1 - a + b)$. Preparation changes the initial physical state; setting later commands to zero preserves that prepared state and is not necessarily zero torque or zero muscle force.

12. A prior sport might help target selection or timing but encourage an unsuitable equipment-specific motion. Compare relevant baseline skill and practice exposure, then assess the new task under controlled conditions. Shared joint motion alone cannot identify what transferred, and a failure to transfer is informative too.

Chapter 31

Motor Control III: The Computational Brain

In Plain Language 31.1. What This Chapter Is About

A golf swing asks the nervous system to prepare a moving body, estimate what it is doing, and influence a club whose delivery depends on the whole evolving system. The interesting question is how those operations cooperate. A successful swing does not show that the brain solved a particular optimization problem, stored a complete physics engine, or stopped using feedback near impact.

This chapter connects the control and learning models to mechanics. We will count the variables in an explicitly chosen model, calculate how much time a delayed correction has to act, distinguish drift from energy supply, and examine what stiffness and muscle coordination can accomplish. We will then ask which experiments distinguish competing explanations. Every numerical controller below is an illustrative model, not a measured neural controller or a prescription for grip pressure.

The resulting picture is more useful than a contest between conscious control and passive physics. Preparation changes the state from which the swing develops; mechanics changes the consequences of commands; prediction helps interpret delayed observations; feedback can alter the remaining trajectory; learning changes future preparation and responses. Each contribution has limits that can be investigated.

31.1 The Scale of the Problem the Brain Solves

31.1.1 The Dimensionality of the System

Start by distinguishing a body from a model of that body. A coordinate count depends on which segments, joint motions, contacts and flexible modes the model retains. A shoulder represented by a ball joint has three rotational coordinates; a restricted two-axis shoulder is a different approximation. Two hands holding one club form a closed chain, and fixed-foot contact further restricts feasible motion. The club's pose is not an additional set of independent coordinates if those same coordinates have already been determined by the hands and shaft model.

Let $\mathbf{q}$ describe a configuration, $\mathbf{v}$ its independent generalized velocity, $\mathbf{a}$ muscle activation, and $\mathbf{z}$ additional tissue or actuator state. Flexible coordinates and their velocities must also be retained when their history affects delivery. A possible state is $\mathbf{x} = (\mathbf{q}, \mathbf{v}, \mathbf{a}, \mathbf{z})$. Its precise dimension follows from a declared model and constraint rank, not from counting named body parts. Orientation coordinates may be redundant even when the physical velocity dimension is minimal.

A count that mixes bilateral joints with aggregate trunk or lower-limb coordinates is incomplete until each coordinate and constraint is specified. A defensible report supplies the actual joints, axes, constraints and retained states so that another reader can reproduce the count.

Key Principle 31.1. Count Schedules, Dimensions and Feasible Motions Separately

Suppose, purely for a combinatorial illustration, that a controller chooses among seven command levels for each of 200 channels over 30 intervals. The number of discrete command schedules is

$$N = (7^{200})^{30} = 7^{6000}, \quad \log_{10} N = 6000 \log_{10} 7 \approx 5070.58824.$$

There are 6000 scalar command entries, not 10^{5070} dimensions. The continuous command cube would be $[0, 1]^{6000}$; at one instant it is $[0, 1]^{200}$, a set with a boundary and corners. Neither number is an empirical count of neural commands or distinct successful swings. Activation dynamics, force limits, collisions and contacts make many schedules infeasible, and different schedules can produce indistinguishable observations.

Thirty 10ms intervals cover 300ms; a trajectory sampled at both endpoints has 31 sample instants. Selecting a numerical discretization does not establish a 100Hz neural clock. Evaluating every schedule at one nanosecond each would take approximately $10^{5061.58824}$ seconds. This illustrates why that particular enumeration is impractical, not why all other solution methods are impossible.

31.1.2 Constraints Reduce the Space

A smooth parameterization, a learned policy and a physical constraint simplify different parts of a problem. Five coefficients at each of 30 intervals, each with seven levels, give 7^{150} schedules, with logarithm 126.76471. Five unconstrained cubic time functions have 20 coefficients before boundary conditions, amplitude bounds and shared timing are imposed. These are alternative parameterizations, not equivalent estimates of what the brain stores.

Reducing the number of decision variables can make a numerical search easier while excluding a necessary motion. Conversely, a high-dimensional feedback policy may evaluate quickly after training, without searching over whole trajectories during execution. Learning cost, planning cost, online evaluation cost and physical response time are separate quantities. A large schedule count therefore motivates exploiting structure; it does not identify the structure or demonstrate human optimality.

For golf, the meaningful reduction is tied to the task. A family of motions that preserves clubhead speed but changes face orientation is not redundant for a directional shot. A family that preserves the full delivery state may still differ in balance, contact feasibility, tissue load or repeatability. The chosen outputs and constraints determine which differences can be ignored.

31.1.3 The Constraint of Time

Feedback uses information about a state or disturbance to change subsequent action. It need not be conscious, visual, or organized into a sequence of complete serial correction cycles. A continuous delayed loop can have many signals in transit simultaneously. Counting how often the movement duration can be divided by a latency is not a general measure of its control capacity.

Constraint Insight 31.1. Separate Delay, Activation and Remaining Time

For a perturbation at t_p , let L be the total delay before its corrective command reaches the actuator. Let T be the impact time of the nominal example. Then $T - t_p - L$ is the time available for that command's mechanical effect, if positive. Muscle activation and force development can further attenuate the response during that interval.

A first-order activation time constant is not a dead time. In $T_a \dot{a} + a = e$, a step in excitation starts changing activation immediately and reaches about 63.2% of its eventual change after T_a . Measured latency to an EMG change, force change or detectable movement are different observables. Adding published values for all three may count the same physiological processes more than once.

Primary experiments show why the distinction matters. Pruszynski and colleagues found rapid integration of shoulder and elbow information in arm perturbation tasks, with responses on a timescale of about 50ms [Pruszynski et al., 2011]. Saunders and Knill found online corrections to virtual-hand perturbations during reaching. Their mean detected latency was 163ms; their analysis estimated a positive detection bias of about 25ms [Saunders and Knill, 2003]. These are different tasks, signals and measurements. Neither result supplies a universal downswing latency, but both contradict the premise that fast movement categorically excludes feedback.

31.2 How Prediction, Feedback and Mechanics Cooperate

31.2.1 A Worked Remaining-Time Calculation

Consider the change in a single rotational coordinate caused by a corrective torque command. The illustrative actuator and rotor satisfy

$$I\delta\ddot{\theta} = \tau_m, \quad T_a \dot{\tau}_m + \tau_m = U H(t - t_p - L),$$

where H is the unit step, all perturbation states initially vanish, and the command remains on until T . This is a response calculation after a correction has been chosen; it is not a complete feedback policy. Set $h = \max(T - t_p - L, 0)$. Integration gives

$$\delta\theta(T) = \frac{U}{I} \left[\frac{h^2}{2} - T_a h + T_a^2 (1 - e^{-h/T_a}) \right].$$

The bracket has units of time squared. For $h \ll T_a$, expansion of the exponential gives $h^3/(6T_a)$ to leading order: activation filtering makes a very late angular correction smaller than the ideal instantaneous-torque result $Uh^2/(2I)$.

Take $T = 0.300\text{s}$, $L = 0.060\text{s}$, $T_a = 0.040\text{s}$, $I = 0.050\text{kgm}^2$ and $U = 0.50\text{N m}$. Perturbations at 40, 180, 220 and 260ms give available times of 200, 60, 20 and 0ms. Their final angular changes are approximately 0.135892, 0.006430, 0.000296 and 0rad, respectively. The first is 7.786 degrees; the third is only 0.01693 degrees. Those differences arise with the same actuator and command amplitude. They are not measured golf corrections.

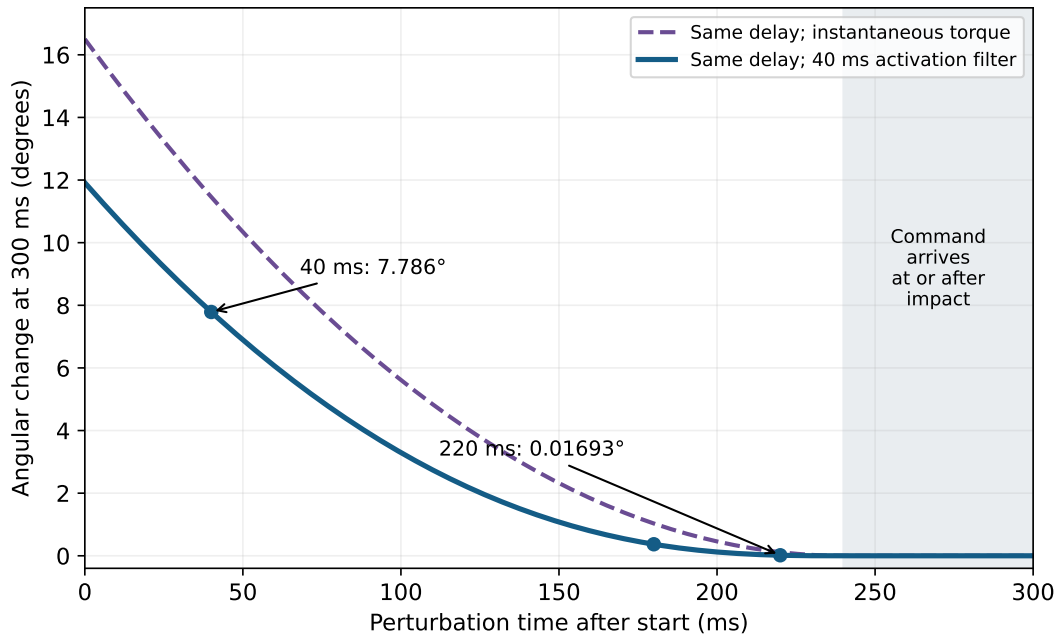


Figure 31.1: Remaining Time Changes the Same Command's Effect. The Declared Rotor Model Compares Instantaneous Torque With Activation Filtering After the Same 60ms Delay. These Are Illustrative Responses, Not Measured Golf Data.

This example explains the value of advance preparation without declaring all feedback useless. Early errors can have an appreciable correction window; late errors may have little or none. Mechanical stiffness already present at the time of a disturbance can respond before a newly generated command arrives. A correction that changes final angular velocity may have little effect on final angle, so response capacity must name the output. The real golfer also has coupled coordinates, changing leverage, force bounds, measurement uncertainty and an impact time that may itself shift.

31.2.2 Predictive Control and Anticipatory Activation

Key Principle 31.2. Prediction Uses Information Already Available

A forward model predicts a future state from a state estimate and a command. An estimator can propagate an old observation using intervening commands to improve feedback despite sensory delay. Prediction cannot reveal an unpredictable disturbance before information about it is available.

For example, a model may propagate $\hat{\mathbf{x}}(t - L)$ through $\dot{\mathbf{x}} = F(\mathbf{x}, \mathbf{u})$ using recorded commands to estimate $\hat{\mathbf{x}}(t)$. A discrepancy between predicted and observed signals can then update the estimate. Errors in initial state, activation, contact and the model accumulate during propagation. Replacing the unknown current state with a prediction is therefore useful only together with an account of prediction error.

Anticipatory activation changes the state before an expected event. It shifts work and information requirements earlier; it does not abolish neural conduction or muscle dynamics. A model of advance stabilization must say which muscles or actuators change, how that changes torque and impedance, and what the preparation costs. Neither address posture nor the duration of a pre-shot routine identifies those hidden quantities by itself.

Dependencies in a Control Model, Not a Map of Brain Regions

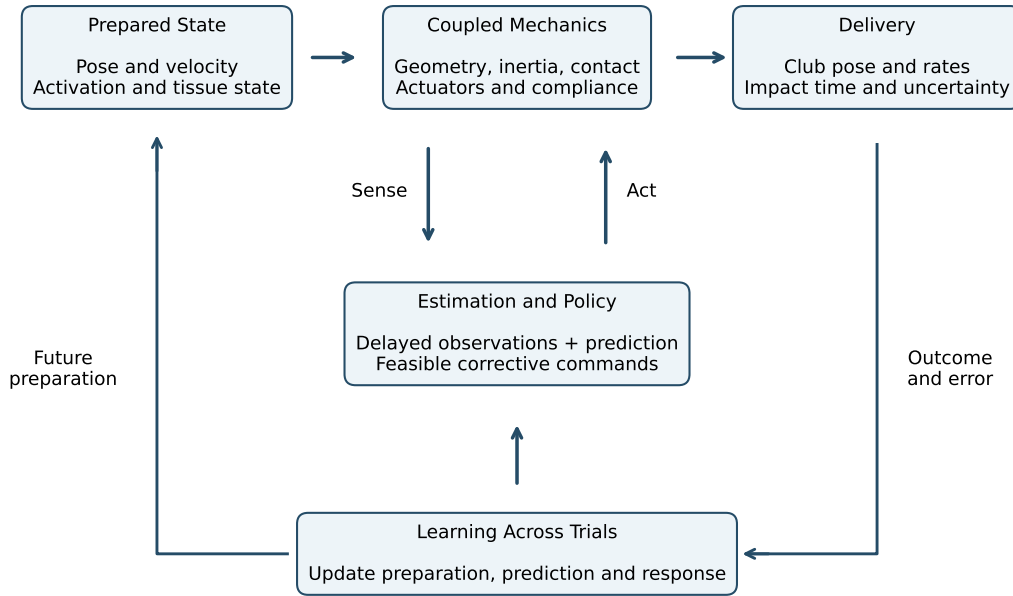


Figure 31.2: Preparation, Mechanics, Estimation, Feedback and Learning Interact. Within-Trial Responses and Changes Across Trials Must Be Distinguished Experimentally.

31.2.3 Drift Is a Model Term, Not a Free Actuator

For an input-affine model,

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{G}(\mathbf{x})\mathbf{u},$$

$\mathbf{f}$ is the evolution when the declared input coordinates are zero. If the input is joint torque, this differs from a model whose input is neural excitation and whose state includes nonzero muscle activation. Zero new excitation does not instantly eliminate active force. Contact reactions and tissue forces must be treated consistently with the chosen coordinates and state.

An instantaneous ratio between drift and applied input can describe a trajectory under a declared scaling. It does not prove weak available control, low metabolic cost, or low sensitivity at impact. Linearizing a smooth, fixed-contact model around a nominal trajectory gives

$$\delta\dot{\mathbf{x}} = \mathbf{A}(t)\delta\mathbf{x} + \mathbf{B}(t)\delta\mathbf{u},$$

$$\delta\mathbf{y}(T) = \mathbf{C}_T\Phi(T, 0)\delta\mathbf{x}_0 + \int_0^T \mathbf{C}_T\Phi(T, s)\mathbf{B}(s)\delta\mathbf{u}(s) ds.$$

Here Φ propagates perturbations, and $\mathbf{C}_T$ maps state changes into the selected delivery outputs. The kernel $\mathbf{C}_T\Phi(T, s)\mathbf{B}(s)$, together with feasible inputs and their delays, determines first-order influence on those outputs. A norm of $\mathbf{f}$ alone omits that propagation and output geometry. Impact-event sensitivity adds a time-shift term when T changes.

An abstract counterexample makes the logical issue explicit. Let $\dot{x} = c + ax + u$, with $x(0) = 0$, $a = 10\text{s}^{-1}$, $c = 1\text{rad/s}$ and $T = 0.3\text{s}$. The sensitivity of $x(T)$ to a constant input u is $(e^{aT} - 1)/a \approx 1.90855\text{s}$. Near the end, the nominal drift is about 20.0855rad/s . An input of 0.01rad/s is small compared with that drift but still changes the endpoint by about 0.01909rad . Large drift and appreciable input sensitivity coexist. This is a mathematical counterexample, not a proposed golf trajectory.

The [zero-torque counterfactual framework](#) can test a specified intervention in a golfer model. It must start from the same complete state, define the channels removed, evolve the resulting trajectory and verify that the assumed contacts remain feasible. Agreement with an observed phase ordering does not establish that the golfer silenced muscles or consciously relinquished control. Those are separate hypotheses requiring separate observations.

31.2.4 Mechanical Impedance During a Moving Task

Mechanical impedance describes the relation between motion and interaction force or torque. In a local linear model it can include inertia, damping and stiffness. Stiffness is a derivative of restoring force with respect to displacement under specified conditions, not simply a synonym for grip pressure. Impedance control addresses the dynamics of interaction with an environment [[Hogan, 1985](#)].

Key Principle 31.3. A Moving Reference and Its Local Response

An illustrative joint-torque law is

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{\text{ff}}(t) - \mathbf{K}(t)(\mathbf{q} - \mathbf{q}_d(t)) - \mathbf{D}(t)(\dot{\mathbf{q}} - \dot{\mathbf{q}}_d(t)).$$

The feedforward term supports the nominal motion; the other terms respond to position and velocity errors. This is a generalized-torque model, not a claim that the nervous system independently commands every matrix entry. If the displayed linear law is valid, larger errors produce larger restoring torques. Saturation or a limited range of validity must be modeled explicitly.

For a scalar spring-damper with fixed reference, positive constant stiffness K , nonnegative damping D , and no feedforward torque, write $e = q - q_d$ and $V = Ke^2/2$. Then

$$\tau\dot{q} + \dot{V} = -D\dot{q}^2 \leq 0.$$

This storage balance shows passivity at that mechanical port. Positive stiffness stores energy; it does not dissipate it. A compliant element may store and return energy, whereas damping dissipates it. Nor does this passive mechanical description imply zero metabolic cost in the muscles maintaining the state.

If K varies, $\dot{V}$ contains $\dot{K}e^2/2$. At a clamped error of 0.1rad, increasing stiffness from 10 to 30N m/rad increases stored energy by 0.1J despite zero joint motion. That energy has to come from the mechanism changing stiffness. If the reference moves, its work also enters. With constant K, D , no feedforward term and $\dot{e} = \dot{q} - \dot{q}_d$,

$$\tau\dot{q} + \dot{V} = -D\dot{e}^2 + \tau\dot{q}_d.$$

The moving-reference term can be positive. A controller cannot be called passive merely because its torque equation contains a spring and a damper.

Example 31.1. What This Means for Grip and Balance

Changing grip can change local hand-club compliance, available friction, muscle activation, posture and transmission of disturbances. These changes need not have the same effect on clubhead speed, face orientation and tissue loading. A firmer grip does not necessarily lock a wrist coordinate; a lighter grip does not necessarily destabilize the club. The comparison requires a specified load, contact model, perturbation direction and outcome.

In a whole-body model, an apparently helpful local correction may alter the contact wrench needed at the feet. A controller that corrects face angle while violating friction or losing balance has not solved the original task. Joint and endpoint impedance also differ through the configuration-dependent Jacobian and, under load, geometric stiffness terms.

Burdet and colleagues investigated arm movement in an unstable robotic environment and found adaptation of impedance geometry [Burdet et al., 2001]. That supports investigating directional stabilization. It does not establish a universal optimal grip stiffness, a fixed impedance throughout a swing, or an injury-prevention rule. Those require golf-specific measurements and appropriately matched comparisons.

31.2.5 Synergies, Feasible Commands and Neural Evidence

One meaning of a muscle synergy is a low-dimensional description of recorded muscle activity. Another is a hypothesis that neural circuitry restricts commands to coordinated groups. A third is a deliberately chosen control basis in a simulation. These uses should not be interchanged.

For a data matrix $\mathbf{E}$ of processed EMG signals, a factorization $\mathbf{E} \approx \mathbf{WH}$ can summarize patterns. Its result depends on the recorded muscles, filtering, normalization, task set, chosen rank and validation procedure. EMG is an electrical observation, not a direct measurement of muscle force, excitation, joint torque or mechanical work. A good reconstruction of the training recordings need not predict another posture or a novel perturbation.

Kutch and Valero-Cuevas used cadaveric measurements and computational models to show that biomechanics and task selection can produce low-dimensional patterns without a controller that groups muscles [Kutch and Valero-Cuevas, 2012]. This is a counterexample

to a particular inference, not a disproof of all neural coordination. It motivates testing the mechanical alternatives alongside the neural hypothesis.

In a model, let $\mathbf{R}(\mathbf{q})$ map muscle forces to generalized torque, with moment-arm sign convention $R_{ji} = -\partial l_i / \partial q_j$ for musculotendon length l_i . Even the simplified isometric relation

$$\boldsymbol{\tau} = \mathbf{R}(\mathbf{q}) \text{diag}(\mathbf{F}_{\max}) \mathbf{a}, \quad 0 \leq a_i \leq 1,$$

shows why equal activation need not mean zero net torque. Antagonists with moment arms 0.04 and -0.03m , force capacities 1000 and 800N, and equal activation 0.2 produce $8 - 4.8 = 3.2\text{N m}$. Force-length-velocity effects, pennation, tendon state and history make the real relation more involved.

Example 31.2. A Compact Basis Can Remove the Needed Action

Consider two normalized opposing actuators with $\tau = a_1 - a_2$. Independent commands in $[0, 1]^2$ can produce any torque in $[-1, 1]$. Restricting activation to the single coactivation vector $\mathbf{a} = c(1, 1)^\top$ gives zero torque for every c . The basis has one dimension, yet it loses both directions of net torque. The same coactivation could change stiffness in a richer muscle model; that property is absent from this simple torque map and cannot be inferred from it.

For $\mathbf{a} = \mathbf{W}\mathbf{c}$, the admissible coefficient set must satisfy $0 \leq \mathbf{W}\mathbf{c} \leq 1$ and any excitation-rate or activation constraints. Rank alone does not describe its feasible torque set. A claim that a small basis simplifies learning should therefore report both held-out prediction and the motions, forces and recovery actions the restriction excludes. No fixed number of golf synergies or universal explained-variance percentage is assumed here.

31.3 The Brain as an Internal Model: What It Learns

Predicting sensory consequences, selecting a command and evaluating an outcome are different computational functions. They can be represented by a forward model, a policy and a value or cost model, respectively. A controller may combine learned dynamics, cached actions, explicit aiming strategies and feedback corrections. Good performance does not uniquely identify which combination produced it.

Writing $\hat{\mathbf{f}}$ and $\hat{\mathbf{G}}$ is useful when building an estimator for an affine mechanical model. It does not establish that the brain separately stores those functions or that a particular brain region implements each one. Many parameter combinations can explain the same observed movement. Changes in posture, activation or contact can also look like a change in the fitted dynamics if the state is incomplete.

31.3.1 What Adaptation Experiments Can Establish

Example 31.3. Separate Task Error From Prediction Error

Mazzoni and Krakauer introduced a visuomotor rotation and supplied an explicit aiming strategy that initially corrected it. Continued implicit adaptation subsequently interfered with that strategy [Mazzoni and Krakauer, 2006]. This demon-

strates that explicit and implicit processes can coexist and conflict in that task. It does not show that all rapid learning is unconscious or that successful golf requires a uniquely identified internal physics model.

A task error compares achieved and desired outcomes. A sensory prediction error compares an observation with its expected consequence. They can differ: a deliberate aiming offset may put a cursor on target while the cursor still appears away from the position predicted from the hand command. Removing a perturbation and observing an aftereffect helps distinguish adaptation from immediate compensation, but aftereffects alone do not reveal every learned variable or neural location.

Switching clubs is a useful proposed experiment, not a completed experiment with a universal trial count. The new club may change inertia, geometry, contact, visual cues and the shot objective. Comparing a driver swing with a putt changes the task as well as the tool. A stronger design perturbs a measured club property within a matched task, randomizes conditions, measures the initial response and learning curve, includes washout and retention tests, and assesses transfer to conditions withheld from training.

For a simple trial-to-trial model $z_{n+1} = az_n + be_n$, retention a and error sensitivity b are parameters to estimate under a declared observation model. Similar performance curves can arise from different hidden states, explicit strategies or changes in exploration. Fit quality is insufficient: compare predictions for error clamps, altered feedback, changed loads and delayed retention. The investigation should expose where the models disagree.

31.4 Comparison With Artificial Intelligence: What Can Machines Learn?

31.4.1 How Brain Learning Differs From AI Learning

Key Principle 31.4. Specify a Comparable Benchmark

A meaningful comparison declares the task, body and actuators, observations, prior experience, allowed feedback, training data, computation budget and success criterion. It separates real interactions from simulated interactions and separates training from execution. A trial, an episode, a sensor sample and a gradient update are not interchangeable units of experience.

An experienced adult learning a shot brings prior motor and perceptual experience. A pretrained robot can also bring demonstrations or learned representations. Counting only the final practice session on one side and all training on the other gives a misleading efficiency comparison. Conversely, a simulator can generate information unavailable to the physical agent; privileged state or perfect contact parameters must be disclosed.

There is no generic rule that reinforcement learning uses joint angles only, lacks internal models, or learns only from a final scalar reward. Dreamer, published in 2020, learned a world model from images and trained behavior using imagined latent trajectories [Hafner et al., 2020]. This is a concrete counterexample to that characterization, not evidence that it reproduces human golf learning or a claim about the best current algorithm.

31.4.2 What Human Performance Contributes to the Comparison

Key Principle 31.5. Measure Adaptation and Transfer

Useful comparisons ask how performance changes with altered club properties, uncertain ground contact, restricted sensing, fatigue or a new target. Report error distributions, recovery time, failures and retained learning. Transfer from another sport is a hypothesis to measure, not an assumed benefit for every learner. A demonstration that communicates a goal is also different from one that supplies an executable full-body policy.

Humans and machines have different embodiments, sensing and learning histories. Those differences are scientifically interesting and need not be hidden to make a comparison. A specialized machine that repeats one swing may answer a repeatability question while saying little about general adaptation. A human who adapts successfully to a changed task does not thereby establish global optimality or robustness to every disturbance.

31.4.3 What Computation Does and Does Not Guarantee

Key Principle 31.6. Deterministic Evaluation Is Not a Deterministic Shot

A deterministic policy returns the same command for an identical represented input. Sensor noise, hidden tissue or actuator state, contact variability, model error and numerical or hardware differences can still change the physical outcome. Likewise, evaluating many candidate actions does not guarantee a global optimum for a nonlinear constrained problem. Such a guarantee requires a suitable problem class, algorithm and certificate.

Frequency comparisons need equal care. A neural firing rate, a sampled command update rate, a feedback bandwidth and a shaft natural frequency describe different processes. For the delay-plus-activation part of our illustrative controller,

$$H(i\omega) = \frac{e^{-i\omega L}}{1 + i\omega T_a}, \quad \arg H = -\omega L - \arctan(\omega T_a).$$

With $L = 0.06\text{s}$ and $T_a = 0.04\text{s}$, the phase contributions sum to approximately -35.7 , -101.8 and -159.5 degrees at 1, 3 and 5Hz. The mechanical plant, sensor processing and controller add their own dynamics. These numbers are not a stability certificate, but they show why comparing an alleged neural clock with a shaft mode does not determine correction capacity. Sampling faster cannot by itself remove propagation delay.

Power comparisons must use a common system boundary. An illustrative processor drawing 20W for 0.3s consumes 6J. A 0.2kg clubhead moving at 50m/s has 250J of translational kinetic energy. Neither calculation measures human brain power or whole-body metabolic expenditure, and the 6J cannot be substituted for the mechanical energy supplied to the club. A fair embodied comparison includes the actuator energy, preparation, losses, support equipment and any energy recovered, with training energy reported separately when relevant.

31.5 What Artificial Intelligence Could Learn From the Brain

31.5.1 Hierarchical Control

Key Principle 31.7. Organize Decisions by Their Role

A practical controller can separate shot selection, desired delivery, feasible motion generation and actuator feedback. Each layer passes constraints and uncertainty as well as a target. The lower layer must be able to reject an infeasible request. This is an engineering decomposition; a one-to-one map from those software layers to prefrontal, premotor and motor regions is not established by drawing the diagram.

For example, an outer planner may request a delivery distribution rather than one exact club pose. A motion planner then considers contact, balance and muscle limits; a faster estimator and feedback law respond to the current observations. If the estimator detects a changed support condition, that information may need to modify the motion goal itself. Hierarchy does not mean that task information disappears from lower-level responses.

31.5.2 Internal Models and Model-Based RL

Example 31.4. Use Prediction Where It Is Reliable

A model-based controller predicts consequences before selecting an action; a model-free policy or value estimate can select actions without an explicit dynamics rollout at decision time. Learned models, policies and value functions can be combined. The practical question is which combination performs well with the available data and computation, not whether every biological or artificial learner belongs to one category.

An inaccurate model can recommend actions that exploit its own errors. Evaluate multi-step prediction at the speeds and contacts relevant to delivery, not only one-step accuracy on a familiar trajectory. Uncertainty estimates should widen outside measured conditions, and optimization should respect actuator and contact feasibility. A cached policy can reduce online computation while still relying on substantial prior optimization or training.

31.5.3 Embodiment and Physics Exploitation

Key Principle 31.8. Mechanics Redistributes and Stores Energy

Geometry, inertia, compliant elements and contact determine how commands affect motion. They can make a task easier to execute or harder to recover from. Inertial coupling transfers influence and energy between coordinates; it is not an independent energy source. Elastic return requires earlier storage. A low-dimensional control description does not remove those energy requirements.

For the illustrative 0.2kg head above, a 1m drop supplies only $mg\Delta h = 1.962\text{J}$ of gravitational work, about 0.785% of its 250J kinetic energy at 50m/s. This calculation concerns the head alone. A complete golfer-club budget must include all segments, their height changes, initial kinetic energy, elastic storage, actuator work and losses. It does not follow that gravity is negligible everywhere; it does rule out crediting the head's own 1m fall with supplying its full final energy.

Similarly, a natural frequency is a property of a specified linearization and boundary condition. A rapid, nonperiodic swing with changing supports and nonlinear muscle dynamics cannot be optimized simply by matching one resonance. A compliant design that filters high-frequency disturbances may also delay the desired action or amplify another frequency range. Mechanical design and control design must be assessed together against delivery and load constraints.

31.5.4 Prediction-Driven Learning

Example 31.5. Learn From Consequences, Then Test Transfer

A robot can record commands and subsequent observations to train a predictive model without manually labeling every desired action. That does not make the task objective unnecessary, guarantee useful exploration, or eliminate model bias. The learned representation still needs testing on the loads, viewpoints and contact transitions relevant to the task.

The analogous scientific question for human learning is which error signals alter which future behaviors. A changed sensory prediction, a missed target and an undesirable effort cost need not drive the same update. Experiments that manipulate them separately provide more information than describing every improvement as the brain learning physics. Predictive models are powerful tools for posing these questions; their success is not automatic evidence that the biological implementation uses the same variables.

31.6 The Golf Swing as a Window Into Intelligence

31.6.1 Why Golf Is a Useful Scientific Tool

Key Principle 31.9. Measure the Chain From Action to Outcome

Golf supplies a visible target and measurable delivery and ball-flight outcomes, but a stationary ball does not make the environment constant. Lie, friction, wind, club properties, contact location and the golfer's state vary. Differences between experts and novices also confound practice, age, strength, selection and other history unless the study design addresses them.

Ball landing position alone is a poor identifier of the preceding control. Different combinations of face, path, strike location, speed and spin can produce similar landing points. Synchronized club, body, force and shaft measurements can narrow the possibilities, but estimating internal muscle state and separating causes still requires assumptions. Intro-

spective reports are observations about experience; they are not direct measurements of neural commands.

Example 31.6. A Discriminating Golf Experiment

To investigate a claimed grip-related reduction in face-angle sensitivity, measure the grip condition and a calibrated perturbation, randomize its timing and direction, and retain the same declared shot objective. Record pre-perturbation state, delivery, contact feasibility and trial-to-trial dispersion. Compare immediate mechanical response, delayed correction and adaptation across later trials. A smaller error can then be tested against alternatives such as changed initial state, increased stiffness, different input gain or a changed policy.

A feasible experiment need not apply a hazardous disturbance during a full-speed swing. Instrumented slow tasks and controlled simulator studies can establish parts of the mechanism first, with their task boundaries stated. A simulator can verify a derivation and generate competing predictions; independent physical measurements are needed to validate those predictions for people.

31.6.2 Open Questions That Golf Helps Address

Key Principle 31.10. Ask Questions That Could Change the Model

Which delivery directions remain correctable at each perturbation time? Does an apparent improvement come from a more favorable prepared state, altered passive response or delayed feedback? Does a compact activation basis predict responses to unfamiliar loads? Which parameters transfer between clubs, and which must be relearned? Do competing estimators predict different effects when visual and proprioceptive information disagree?

For small errors, a useful analysis combines the propagated initial-state covariance, disturbance covariance and the delivery Jacobian. More repeated swings do not cure systematic calibration bias. Correlated state and command errors must be retained when present; treating them as independent can misstate outcome variance. Near a grazing contact or a change of contact mode, a smooth linear covariance approximation may fail. Those limits are part of the model's domain, not inconvenient exceptions to ignore.

The same investigation applies to humanoid modeling. Contact constraints shape feasible accelerations and forces; geometry maps joint motion to an end-effector task; internal actuator state changes future force; estimation determines which corrections can be chosen; delay determines when they act. A golf swing is a demanding instance of this general coupled problem, not proof of one universal neural solution.

31.7 Why Skilled Performance Sometimes Fails

31.7.1 The Yips: Describe the Phenotype Before the Mechanism

The term yips is used for task-related disruptions such as involuntary movements or loss of control during putting and other skilled actions. It does not by itself identify one cause. A missed putt, performance anxiety and a task-specific movement disorder are not interchangeable observations.

Key Principle 31.11. Evidence Does Not Support a Single Corrupted Program

Adler and colleagues studied 50 golfers using movement recordings and surface EMG. Findings depended on whether grouping used self-report or video-defined involuntary movement; the latter groups differed in pronation-supination movement, while the co-contraction comparison was a statistical trend [Adler et al., 2011]. This supports careful phenotype measurement, not a universal diagnosis or a demonstrated corrupted cerebellar trajectory.

A model with an unstable reference, excessive gain or delayed correction can generate jerky motion. That establishes what the model can do, not what caused a particular golfer's symptoms. Several mechanisms can produce similar traces, and psychological and neurological factors need not form mutually exclusive categories. Persistent involuntary movement is a clinical assessment question; this chapter's equations do not identify its cause or treatment.

31.7.2 Choking Under Pressure

Example 31.7. A Miss Is Not Its Own Explanation

Imagine an all-square match on the last hole, with a putt that would win it. The golfer misses. That one outcome does not establish choking: a skill with a high success probability can still fail. Investigating a pressure effect requires a comparison with an appropriate performance distribution under matched conditions, preferably with pressure manipulated or measured independently.

Key Principle 31.12. Compare Attention and Pressure Hypotheses

Explicit-monitoring accounts propose that attention to execution can disrupt some well-practiced skills. Distraction, altered arousal, changed risk preferences and mechanical changes in preparation are other candidate pathways. Beilock and Carr's putting experiments investigated skill knowledge, pressure and training conditions [Beilock and Carr, 2001]. Their task-specific evidence does not establish that technical thought always harms performance or that one explanation covers every pressure-related miss.

Studying mechanics between trials, learning a new movement and monitoring a highly practiced action during execution are different activities. The fact that one attention instruction changes performance cannot justify a blanket instruction to stop analyzing the

swing. Nor can an observed benefit from a routine identify a particular amygdala or cerebellar mechanism without matching neural evidence.

Key Principle 31.13. Test an Intervention Against the Intended Outcome

Compare a specific routine or attention instruction with a matched control condition, randomize order, account for skill and practice effects, and measure both performance and the intended manipulation. Distinguish immediate success from retention and transfer. A technique can be useful for a particular person or task without being a universal prevention strategy or proving the proposed neural explanation.

31.8 What the Combined Account Establishes

Key Principle 31.14. The Computation Is Distributed Across Time and Mechanisms

Preparing the state can reduce what must be corrected later. Existing mechanical impedance can shape a disturbance before a new command arrives. Prediction can improve estimates from delayed observations. Feasible feedback can change the remaining trajectory, and learning can improve future preparation and correction. The usefulness of each contribution depends on the task, state, uncertainty and time remaining.

This account does not require gravity to supply most swing energy, the brain to enumerate all muscle schedules, or feedback to switch off at a universal phase boundary. It also does not require a perfect internal model. It asks for a model detailed enough to make a discriminating prediction and measurements capable of testing it.

Key Principle 31.15. Key Takeaways

The coordinate dimension, command parameter count and number of discretized schedules are different. A finite delay limits when an observed disturbance can affect delivery; it does not forbid every correction. Drift magnitude, energy supply and terminal sensitivity require different calculations. Impedance includes dynamics and energy exchange, and its modulation can require work. A compact EMG factorization is not unique evidence of neural grouping. Adaptation, transfer and performance under pressure need controlled comparisons. Human and artificial controllers should be compared on declared tasks with common information and energy boundaries.

The purpose of understanding these connections is practical as well as scientific: to know which change a model predicts, why it predicts it, and which observation could show that the explanation is wrong. That is the footing on which a useful reference for golf and general humanoid control can be built.

31.9 Chapter Exercises

1. Compute the schedule count for 200 channels, seven levels and 30 intervals. Repeat for five channels. Explain why neither result is a dimension or a count of successful swings.
2. For $\mathbf{a} = \mathbf{W}\mathbf{c}$, derive the coefficient constraints induced by $0 \leq \mathbf{a} \leq 1$. Use $\tau = a_1 - a_2$ to explain why the basis $(1, 1)^\top$ loses torque authority despite reducing the command dimension.
3. Derive the constant-input endpoint sensitivity for $\dot{x} = c + ax + u$. Why does a large nominal drift not prove a small endpoint response? Identify the additional quantities needed for a golf delivery calculation.
4. Integrate the delayed actuator/rotor model. Reproduce the four perturbation-time examples and derive the $h^3/(6T_a)$ short-time limit. State precisely when this model predicts zero response before impact.
5. Derive the spring-damper storage balance for fixed and moving references. Calculate the energy added when stiffness increases from 10 to 30 N m/rad at a clamped 0.1 rad error.
6. Explain how task error can vanish while sensory prediction error remains. Propose measurements that distinguish explicit aiming, implicit adaptation and a change in mechanical impedance after a club perturbation.
7. Compare model-based rollout, a cached policy and a learned value function. Which observations would be needed before attributing one of these computational descriptions to a neural implementation?
8. Propose a hierarchy for a humanoid golf controller. Describe information flowing upward as well as downward, and specify how the controller handles an infeasible delivery request.
9. A delayed feedback model produces jerky putting. Give two competing explanations and design a measurement that could distinguish them. Explain why simulation alone does not diagnose the yips.
10. Contrast explicit-monitoring and distraction accounts of performance under pressure. State a result that would count against each account in a specified task.
11. Design an attention experiment with novice and experienced putters. Include randomization, a pressure manipulation, a manipulation check, a performance outcome, and retention or transfer testing.
12. Derive the magnitude and phase of the delay-plus-activation transfer function at 1, 3 and 5 Hz. Explain why neither neural firing rate nor shaft resonance alone specifies closed-loop bandwidth or stability.
13. Specify a human/robot golf benchmark with at least five matched conditions. Include prior experience, sensing, actuation, delivery uncertainty and energy boundaries. Distinguish processor energy from actuator work.

14. Design one experiment that separates prepared-state sensitivity, immediate mechanical response, delayed feedback and adaptation across trials. State which parts require additional measurements and where the inference could remain ambiguous.

31.9.1 Worked Checks and Answer Guidance

For Exercise 1, the logarithms are 5070.58824 and 126.76471; the continuous schedule dimensions are 6000 and 150 before further constraints. Exercise 2 requires $0 \leq \mathbf{W}\mathbf{c} \leq 1$, plus any separately declared sign, rate or physiological bounds on $\mathbf{c}$; its example gives $\tau = 0$ for every feasible coefficient.

For Exercise 3, differentiate $x(T) = (c + u)(e^{aT} - 1)/a$ with respect to u , taking the limit T when $a = 0$. In a multi-state swing, use the propagated input kernel, the output Jacobian, feasible controls, latency and any event-time correction. For Exercise 4, integrate $\tau_m = U(1 - e^{-s/T_a})$ over the remaining interval twice. Zero response occurs when $t_p + L \geq T$ under the stated zero initial perturbation and causal command assumptions. It does not assert zero motion or zero pre-existing muscle force.

For Exercise 5, differentiate $Ke^2/2$ and retain $\dot{K}e^2/2$ and the reference-work term. The clamped stiffness change adds 0.1J. For Exercise 6, record both aiming behavior and the sensory transformation; include aftereffects and controlled loads rather than interpreting a single improved outcome. Exercises 7 and 8 require distinguishing a useful computational architecture from evidence for its neural implementation, and passing contact/actuator feasibility information back to the planner.

Exercises 9–11 need explicit competing predictions, not a presupposed corrupted motor program. Baseline variability, involuntary movement, attention and pressure must be measured separately. For Exercise 12, the illustrative actuator magnitudes are approximately 0.970, 0.798 and 0.623; the unwrapped phases are -35.7 , -101.8 and -159.5 degrees. These omit the remaining loop dynamics. Exercise 13 must account for the 6J processor example and the separate 250J clubhead energy without treating their ratio as an efficiency. Exercise 14 should use perturbation timing and within-trial versus between-trial responses to distinguish mechanisms while acknowledging hidden-state and measurement ambiguity.

Chapter 32

Passive and Distributed Control: Mechanics, Feedback and Stability

In Plain Language 32.1. What You'll Learn

A golfer's response to a disturbance depends on more than a corrective command. The initial posture, muscle activation, tendon strain, club flexibility and contact conditions already influence what moves next. Sensory feedback then changes the response on its own time scales. The useful question is how these mechanisms cooperate to deliver the club while resisting relevant disturbances.

This chapter distinguishes mechanical passivity, muscle impedance and distributed neural feedback. It derives what each model can establish, explains how to measure the distinctions, and separates those results from hypotheses about expert golf. Apparent effortlessness is an observation to explain, not a measurement of passive contribution.

32.1 Time Scales and the Distribution of Control

Conscious instructions, descending motor commands, spinal processing, muscle activation and mechanical deformation are different processes. Their rates cannot be compared by treating a joint sample as a conscious decision. Multiplying a model's joint count by its numerical sampling frequency gives a sample count; it does not establish the brain's information capacity or the computational cost of biological control. A behavioral information-rate estimate is not a universal ceiling on neural activity.

Prediction, learned feedforward commands, intrinsic mechanics and feedback can operate together. A command prepared before movement can continue to require neural activity and muscle energy during movement. Automatic execution does not mean zero active input. Conversely, an immediate force response to imposed stretch need not await a newly generated neural correction.

Key Principle 32.1. Separate Preparation From Execution

A useful model can place a learned policy above a fast neuromechanical system. This is a modeling architecture, not proof that all gains are selected once, that the cortex is uninvolved during the downswing, or that every joint receives an independent millisecond instruction. Compare architectures using measured inputs, responses and prediction error.

In a primary study of humans and rhesus monkeys, Pruszynski and colleagues found

evidence for multi-joint feedback processing through primary motor cortex at roughly 50 ms after a mechanical perturbation [Pruszynski et al., 2011]. This establishes that fast feedback need not be exclusively spinal. It does not identify the controller of a golf swing or imply that a response triggered by ball contact can alter the preceding impact state.

32.2 Passive Mechanical Systems: Springs and Dampers

Definition 32.1. Mechanical Passivity

Declare an input/output power port and a nonnegative storage function S . A mechanically passive model satisfies

$$S(t) - S(0) \leq \int_0^t \tau_{\text{ext}}(s) \dot{q}(s) ds.$$

It cannot deliver more mechanical energy through that port than it initially stores plus the energy supplied to it. Passivity is a work inequality; it is not a synonym for lack of consciousness, absence of neural feedback or convergence to a desired trajectory.

For a fixed-base translational spring and damper,

$$m\ddot{x} + c\dot{x} + kx = F_{\text{ext}}, \quad S = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2,$$

with $m, k > 0$ and $c \geq 0$, differentiation gives

$$\dot{S} = F_{\text{ext}}\dot{x} - c\dot{x}^2.$$

The spring can return stored energy; the damper dissipates it. With no input and no damping, motion can oscillate indefinitely. A damped pendulum near its downward equilibrium can converge, whereas its upright equilibrium can be unstable despite the same passive components. Storage, dissipation and stability require separate statements.

In Plain Language 32.2. What Mechanical Preparation Can Do

Preparation changes the conditions under which a disturbance acts. At a declared activation and geometry, muscle-tendon mechanics can resist some imposed motions immediately. Maintaining activation still uses physiological resources. The response also depends on stretch direction, amplitude, velocity and history. A local spring approximation should not be extended to an entire swing without checking those dependencies.

32.3 Impedance as Part of Motion Control

Hogan's impedance-control framework treats motion and mechanical interaction together. His 1985 manipulation paper explicitly includes both physical hardware and a real-time controller; it does not establish that the cortex controls impedance exclusively [Hogan,

1985]. Earlier work examined coactivation of antagonist muscles as a way of modulating mechanical impedance [Hogan, 1984].

Definition 32.2. Specify the Mechanical Port

For a small rotational perturbation e about a declared operating condition, a constant-coefficient teaching model is

$$I\ddot{e} + D\dot{e} + Ke = \tau_{\text{ext}}.$$

Here I is rotational inertia, K has units N m/rad, and D has units N m s/rad. With zero initial perturbation, its dynamic stiffness and mechanical impedance are respectively

$$\frac{\tau_{\text{ext}}(s)}{e(s)} = Is^2 + Ds + K, \quad Z(s) = \frac{\tau_{\text{ext}}(s)}{\dot{e}(s)} = Is + D + \frac{K}{s}.$$

The first is torque per angle; the second is torque per angular velocity. They differ by a factor of s . State the convention when comparing reports. A static stiffness number alone does not specify the full frequency-dependent response.

Real joint response can include activation-dependent intrinsic muscle mechanics, passive tissues, tendon compliance, reflex feedback, inertia and device dynamics. The fitted components depend on the observation window. A matrix model must also retain cross-joint effects when they matter. A single label such as “high impedance” is incomplete without direction, frequency and operating state.

For idealized opposing muscles with a constant moment arm r , net torque is $r(F_1 - F_2)$. Increasing both forces equally can leave that torque unchanged while changing the local stiffness. If each muscle has positive local length stiffness k_i , the corresponding restoring joint stiffness in this simplified geometry is $r^2(k_1 + k_2)$. Variable moment arms add geometric terms; tendon compliance changes the series response. This example explains why net torque cannot uniquely identify coactivation. It does not imply that stiffness and damping increase by the same factor or that coactivation has no metabolic cost.

Key Principle 32.2. A Testable Impedance Hypothesis

The nervous system may adjust mechanical response to the task’s instability, uncertainty and costs. Test this with imposed perturbations and an identified response, while controlling for posture, mean force and movement. A plausible mechanism becomes stronger when it predicts a response that an alternative model does not.

32.3.1 The Impedance Profile of the Golf Swing

There is no universal proximal-high, distal-low schedule established by the equations above. Different phases impose different objectives: preparation, transport, club delivery, collision and deceleration. Those objectives motivate measurements, not predetermined stiffness values. A phase-dependent response may reflect changing activation, geometry, feedback gains and shaft state simultaneously.

Burdet and colleagues studied horizontal arm reaching in a robot-created divergent force field. Participants adapted endpoint stiffness geometry in the destabilized direction; the setup constrained the trunk and wrist, and the learning analysis used five naive participants. The stiffness estimate used the interval 120–180 ms after perturbation onset [Burdet et al., 2001]. That measured response should not be described as purely immediate intrinsic tissue stiffness. The experiment supports task-dependent impedance adaptation, not a measured golf-phase prescription.

In Plain Language 32.3. Interpreting Grip Tension

A firm grip can change contact and muscle activation, but visible tension does not identify a joint's dynamic impedance. A compliant wrist can still experience substantial torque, and a stiff joint can move under sufficient load. Compare club delivery and measured response within a golfer before inferring that a particular grip pattern improves speed or control.

32.4 Preparation, Reference Motion and Power

A commonly used torque model is

$$\tau = \tau_{ff}(t) - K(t)(q - q_d(t)) - D(t)(\dot{q} - \dot{q}_d(t)).$$

It specifies a feedforward drive and a response around a moving reference. It is a candidate controller or local approximation, not a physiological identity. Even if its parameters are prepared in advance, its time-varying command remains an input. Updating gains during execution, or through feedback, changes the model again.

For $e = q - q_d$, the spring-like storage is $U_s = K(t)e^2/2$, giving

$$\dot{U}_s = Ke\dot{q} - Ke\dot{q}_d + \frac{1}{2}\dot{K}e^2.$$

The second term accounts for motion of the reference; the third is parameter power. Holding a displaced spring while increasing its stiffness requires an energy source. For $e = 0.1$ rad and $\dot{K} = 20$ N m/(rad s), the parameter contribution is 0.1 W. If $K = 50$ N m/rad and $\dot{q}_d = 0.2$ rad/s, the reference contribution is -1 W at the same error. The signs depend on the state and the prescribed change.

A damper relative to a moving reference also exchanges power with that reference. Its torque on the joint is $-D(\dot{q} - \dot{q}_d)$, so power delivered to the joint is $-D(\dot{q} - \dot{q}_d)\dot{q}$. This need not be negative. Dissipation in the relative damper coordinate is $D(\dot{q} - \dot{q}_d)^2$; the reference supplies or receives the remaining power.

Example 32.1. What Parameter Counts Establish

Twenty diagonal stiffnesses and twenty diagonal dampers give 40 scalar parameters. Two general symmetric 20×20 matrices instead have $2(20 \cdot 21/2) = 420$ independent entries. Neither count includes a moving reference, a drive, muscle state or an estimator. Parameter count is not decisions per second, and dividing these quantities does not establish a neural speedup. Reduced models are useful when their

predictions remain adequate for the chosen task.

32.5 Passive Forces in the Forward Counterfactual

Let $x = (q, v)$ and consider an ideal generalized-torque-input model

$$M(q)\dot{v} + h(q, v) = \tau_{\text{imp}}(q, v) + B(q)u, \quad \dot{q} = v.$$

Then the additive impedance vector field is

$$f_{\text{imp}}(x) = \begin{bmatrix} 0 \\ M^{-1}(q)\tau_{\text{imp}}(q, v) \end{bmatrix}, \quad \dot{x} = f_{\text{bare}}(x) + f_{\text{imp}}(x) + G(x)u.$$

The zero in the top block matters: the bare field already contains $\dot{q} = v$. Adding another copy would double the kinematic velocity. Coupled inertia also means an impedance torque at one coordinate can alter accelerations elsewhere.

Definition 32.3. Declare What Is Held and What Is Zeroed

A forward zero-torque counterfactual (ZTCF) sets a specified active torque channel to zero from a declared initial state and integrates the resulting equations. Stored elastic energy and passive forces remain. If a maintained muscle controller is included in a closed-loop baseline, label that controller explicitly: zero additional command about this baseline is not zero muscle activation. In an excitation-driven model, zero excitation does not instantaneously erase activation or tendon strain.

An augmented closed-loop drift can be a useful comparison, but it is a different counterfactual from removing all modeled active torque. A feedback law depends on measured or estimated state; representing it inside an autonomous augmented model does not make it physiologically passive. Delays require history or an appropriate augmented realization. Counterfactual comparisons must retain the initial activation, deformation, velocity and contact conditions relevant to the chosen intervention.

ZTCF denotes a counterfactual construction, not a fraction of the swing. If comparing trajectories, define an observable, time horizon, units, alignment and error statistic. A smaller trajectory error after adding fitted impedance does not uniquely identify the true mechanism or a percentage of motion caused by passive forces. It may also reflect additional fitting flexibility; evaluate prediction on held-out perturbations.

32.6 Muscle Tone and Distributed Feedback

Resistance to imposed movement can contain non-neural tissue mechanics and ongoing neural activity. A relaxed instruction does not establish zero activation, and measurable resistance does not prove active contraction. Distinguish EMG, muscle force, short-range stiffness, tendon strain and the complete joint response; each measurement constrains a different part of the model.

32.6.1 The Stretch Reflex Loop

Muscle spindles provide information affected by muscle length, velocity, history and fusimotor activity. The homonymous short-latency pathway includes spindle afferents exciting motor neurons to the same muscle. Actual limb responses also involve other muscles, spinal interneurons, descending modulation and longer pathways. It is not a set of independent identical controllers with one latency for every muscle.

An intrinsic force response, an EMG response and a later change in force have different onset times. Neither an EMG latency nor a conscious reaction-time estimate can stand in for all of them. Rapid multi-joint responses involving cortex further weaken a simple contrast between “spinal reflex” and “slow conscious control” [Pruszynski et al., 2011].

32.6.2 Gamma Drive and Task-Dependent Sensitivity

Alpha motor neurons innervate force-producing extrafusal fibers; gamma motor neurons act on intrafusal spindle fibers. They are spinal motor neurons influenced by descending and local circuitry, not two direct commands emitted by the motor cortex. Gamma activity helps tune spindle encoding; overall reflex gain also depends on afferent transmission, motor-neuron excitability, background activation and mechanics.

An instructed-delay human experiment reported goal-dependent changes in spindle afferent activity and stretch responses during preparation [Papaioannou and Dimitriou, 2021]. This supports preparatory sensory tuning. It does not directly measure gamma motor-neuron firing or establish a fixed gamma schedule for golf. Avoid a one-to-one rule equating greater gamma drive with a known joint stiffness.

Example 32.2. A Restoring Sign Does Not Guarantee Passivity

For an imposed sinusoidal perturbation $e = A \cos(\omega t)$ and a delayed feedback torque $\tau_r = -k_r e(t - \Delta)$, mean power delivered to the moving coordinate over a cycle is

$$\bar{P}_r = \frac{1}{2} k_r A^2 \omega \sin(\omega \Delta).$$

It is positive when $0 < \omega \Delta < \pi$. Immediate viscous damping instead gives $\bar{P}_D = -DA^2\omega^2/2$. The controller can supply energy because the delayed torque is out of phase with current displacement. This manufactured calculation neither estimates human reflex gains nor proves instability of a complete delayed system; it demonstrates why a restoring sign alone is insufficient.

32.7 Coordination in the Coupled Chain

32.7.1 Interpreting Proximal and Distal Energy Flow

Geometry and inertia couple segment accelerations, while active and passive forces determine the complete trajectory. An impedance gradient alone does not prescribe the direction or amount of energy transfer. A centripetal acceleration also does not, by itself, establish positive work: force must be paired with the velocity of its point of application.

For a chosen segment, evaluate work rates from forces and moments, changes in its kinetic and gravitational energy, and any stored elastic energy. At a shared joint, equal-and-opposite forces can exchange energy between segments even when their contribution cancels in the total system balance. With the same conventions, a joint couple's net power on the two segments depends on their relative angular velocity. Check these balances before assigning a proximal deceleration to energy received by a distal segment.

32.7.2 The Whip Effect and Distal Acceleration

A distal speed peak can follow a proximal peak in a coupled mechanism. That timing is compatible with several combinations of active torque, passive torque and geometry. It does not show that proximal speed is nearly zero, that a wrist command is absent, or that energy necessarily passes through a single cascade.

Key Principle 32.3. Test the Mechanism Behind the Sequence

Compare a model's force, power and acceleration histories with perturbation responses and held-out motion. If reducing distal impedance changes release timing in that model, report the initial state, input and contact conditions. The result is a model prediction, not a universal instruction to relax the wrists. The flexible-shaft state and grip motion can change the club response even with identical joint-level parameters.

32.7.3 Impedance and Impact Transmission

Force is measured in newtons, not kilograms. A collision impulse is the time integral of force; peak force additionally depends on its time profile and contact compliance. Neither a force magnitude nor a short impact duration determines how much load reaches a particular tissue.

Stiffness stores and returns energy; it is not a dissipator. Increasing distal stiffness can change contact motion, resonances and transmitted loads, but it does not form a universal barrier that absorbs an impact. Damping, shaft waves, grip contact, geometry and the golfer's initial activation all matter. A response initiated by the collision arrives after a finite delay; anticipatory preparation and reactions after contact are distinct mechanisms.

No injury-prevention prescription follows from the simplified impedance model. Inertia and impact duration alone cannot identify a safe wrist stiffness, because the loading, allowable motion, damping, contacts and tissue outcomes are unspecified. The separate impact and damping chapters develop their own system boundaries; this chapter does not substitute a spring analogy for those operators or for measured clinical outcomes.

32.8 Stability Analysis: What the Model Establishes

32.8.1 Equilibrium Storage and Lyapunov Stability

Consider a time-invariant mechanical model with fixed reference q_0 :

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + \nabla U_g(q) + K(q - q_0) + D\dot{q} = \tau_{\text{ext}}.$$

Assume M is positive definite, K and D are symmetric, and $\dot{M} - 2C$ is skew-symmetric, as in a consistent Lagrangian representation. Write

$$U(q) = U_g(q) + \frac{1}{2}(q - q_0)^T K(q - q_0), \quad V = \frac{1}{2}\dot{q}^T M(q)\dot{q} + U(q) - U(q_*),$$

where q_* is a strict isolated local minimum of U . The equilibrium need not be q_0 : gravity can shift it. A positive-definite Hessian of the combined potential is sufficient for a strict local minimum; spring stiffness alone does not establish this condition. A semidefinite Hessian requires higher-order analysis.

Differentiating along the equations gives

$$\dot{V} = \dot{q}^T \tau_{\text{ext}} - \dot{q}^T D\dot{q}.$$

The kinetic derivative's $\dot{M}$ term cancels the corresponding Coriolis contribution; it must not simply be discarded as a small force. With no external torque and positive-definite damping, $\dot{V} \leq 0$. This is *negative semidefinite*, because it is zero whenever velocity is zero, including some displaced states.

Key Principle 32.4. A Bounded Equilibrium Result

In a compact invariant energy sublevel set contained in the neighborhood of an isolated strict minimum, the energy argument gives Lyapunov stability. With positive-definite damping and no other equilibria in that set, the largest invariant subset of $\dot{V} = 0$ is the equilibrium, so invariant-set reasoning gives asymptotic convergence. If damping is only semidefinite, undamped motions must be examined separately. A non-increasing energy does not imply that each coordinate error decreases monotonically.

Unknown external work cannot be removed by saying that damping is sufficiently large. A persistent load can shift equilibrium; an active environment can destabilize it. For example, $\ddot{e} + D\dot{e} - e = 0$ has one positive real pole for every finite $D \geq 0$. Damping does not turn a potential maximum into a minimum.

32.8.2 Tracking and Finite-Time Recovery

A golf swing is a moving, finite-duration task. Absolute kinetic energy is nonzero on its desired trajectory, so it is generally unsuitable as a positive-definite tracking-error measure. To make the distinction exact, use a constant-inertia teaching plant $I\ddot{q} = \tau + d$ with

$$\tau = I\ddot{q}_d - D\dot{e} - Ke, \quad e = q - q_d.$$

Then

$$I\ddot{e} + D\dot{e} + Ke = d, \quad V_e = \frac{1}{2}I\dot{e}^2 + \frac{1}{2}Ke^2, \quad \dot{V}_e = -D\dot{e}^2 + d\dot{e}.$$

For $d = 0$ and positive constants, error converges. Actual motion still requires the feedforward power and any reference work. This certificate is for the specified plant/controller pair; extending it to a variable-inertia humanoid requires consistent compensation or a different proof. Time-varying K adds $\dot{K}e^2/2$ to the error-storage derivative.

With $I = 1 \text{ kg m}^2$ and $K = 100 \text{ N m/rad}$, the poles solve $Is^2 + Ds + K = 0$. At $D = 5 \text{ N m s/rad}$ they are $-2.5 \pm 9.6825i \text{ s}^{-1}$; at $D = 20$ the repeated pole is -10 s^{-1} ; at $D = 50$ they

are approximately -2.0871 and -47.9129 s^{-1} . The middle case is critically damped, not a spiral. Excess damping produces a slower dominant decay. These values are manufactured examples, not wrist estimates.

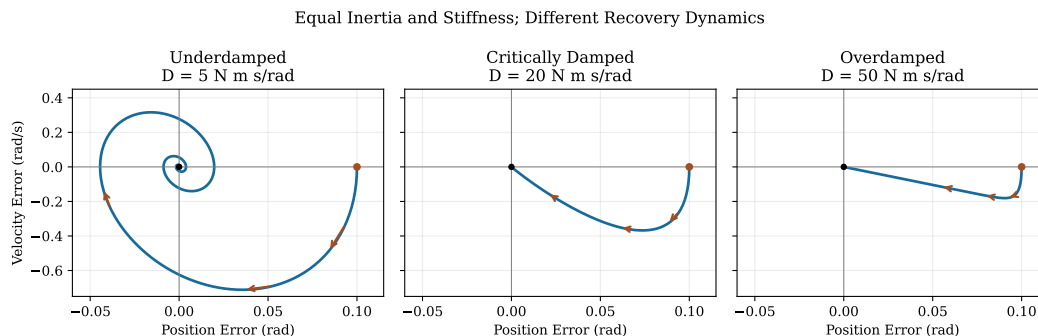


Figure 32.1: Manufactured Recovery With Equal Inertia and Stiffness: Only the Underdamped Case Spirals. All Curves Start at 0.1 rad With Zero Velocity; Orange Arrows Indicate Time Direction Over a Two-Second Interval.

For a linearized trajectory perturbation, $\delta\dot{x} = A(t)\delta x$ gives $\delta x(t) = \Phi(t, t_0)\delta x(t_0)$. The transition matrix Φ can amplify some perturbations over a short interval even when a related equilibrium is asymptotically stable. Use a declared norm and physically meaningful state scales, then examine clubface orientation, strike position or another output through its local output map. A recovery that happens after impact may not improve the shot. Contact switches, delays and uncertain parameters require additional treatment.

In Plain Language 32.4. What Apparent Ease Can Tell Us

Smooth motion and repeatable delivery are useful observations. They do not reveal how much muscle activation, elastic exchange or feedback was required. The explanatory target is a model that predicts both the motion and its response to relevant changes. A model that fits only an unperturbed swing can hide compensating errors in its controller and mechanics.

32.9 Learning and Identifying Impedance

32.9.1 The Identification Problem

Impedance must be estimated from input and response, not assigned from experience level. Specify the perturbation, operating point, measurement bandwidth, noise and model class. For the local equation $I\ddot{e} + D\dot{e} + Ke = \tau$, measured angle alone cannot identify all three coefficients without information about the applied torque.

Even known torque at a single sinusoidal frequency does not necessarily identify all parameters: because $\ddot{e} = -\omega^2 e$, inertia and stiffness enter through $K - I\omega^2$. Multiple informative frequencies or independent inertia information are needed to separate them. Neural delays and nonlinear mechanics can require richer models; a good fit of a spring-damper curve does not prove purely passive tissue behavior.

32.9.2 Practice and Task-Dependent Adaptation

Learning can change planned drive, sensory prediction, feedback policy and mechanical response. Burdet's directional adaptation is an example of measured task dependence [Burdet et al., 2001]; it is not evidence that practice freezes all impedance parameters permanently. Compare adaptation, retention and transfer separately, including changes in mean force and posture that could explain an apparent gain change.

32.9.3 Strength, Speed and Mechanical Capacity

Strength and speed training can change the forces and motions a golfer can produce, but those labels do not uniquely specify changes in stiffness, damping or coactivation. A larger capacity can alter the feasible control strategies; which strategy is used remains a measurement question. Mechanical power, metabolic cost and variability are separate outcomes and need not improve together.

Key Principle 32.5. Mechanical Conditions and Learning

Changing a task's constraints can help investigate coordination, while motion and outcome measurements can test the result. Treat a coaching interpretation as a hypothesis unless supported by an appropriate comparison. A mechanically plausible story is not evidence that a particular instructional method outperforms another.

32.10 Practical Interpretation of Measurements

32.10.1 Grip Pressure and Joint Response

Grip sensors measure pressure or force at a contact. They do not directly measure wrist impedance. The same contact force can accompany different coactivation, posture and tendon state. Evaluate grip observations alongside club motion and, where appropriate, controlled identification data. There is no safe stiffness range or universal impact-timing instruction available from the equations in this chapter.

32.10.2 Posture and the Mechanical Map

Posture changes moment arms, inertia and the Jacobian linking joint to hand motion. Virtual work gives $\tau = J^T F$. Differentiating this relation includes both $J^T \delta F$ and $(\delta J)^T F$; under preload, geometric terms can change tangent stiffness. A simple Jacobian transformation that omits them may be inadequate. A posture label alone does not identify these quantities or establish an efficient swing.

32.10.3 What a Waggle Can Reveal

A waggle provides sensory information during a small movement, potentially about contact, load and familiarity. Describing it as an impedance calibration is a hypothesis. Its frequency, amplitude and activation differ from a downswing, so a familiar sensation does not demonstrate correct stiffness for impact. Repeated observations can test associations without turning subjective feel into an identified mechanical parameter.

32.10.4 Combining Sensation and External Feedback

Proprioception, video, launch measurements and EMG answer different questions. A sensation of increased tension does not supply a stiffness matrix; video does not directly identify force; EMG is not a complete torque or impedance measurement. Combine information according to the hypothesis, report uncertainty and test predictions outside the data used to fit the model. Simulations and existing controlled datasets are appropriate teaching tools for perturbation analysis.

32.11 A Connected Model of Mechanics and Control

The muscle chapter explains how excitation, activation, geometry and tendon state produce force. The shaft chapter shows how those forces act through grip motion and flexible-club dynamics. Motor learning changes the policies and predictions used in repeated movement. Here, impedance describes a response around an operating condition, while stability analysis asks what happens to perturbations under explicitly stated dynamics.

Key Principle 32.6. Key Takeaways

Mechanical passivity is an energy property of a declared port. Impedance can include active muscle mechanics and delayed feedback. A forward counterfactual requires a specific input intervention and full initial state. Equilibrium convergence does not establish finite-time club-delivery accuracy. Measured force, motion, EMG and pressure constrain different parts of the model. Together, these distinctions make coordination hypotheses testable without assuming that the entire swing is either consciously commanded or mechanically autonomous.

Exercises 32.1. Chapter Exercises and Worked Answers

1. Phase-Dependent Hypotheses For address, backswing, transition, downswing, impact and follow-through, propose an observation that would constrain a phase-dependent model. Explain why the phase name alone does not prescribe stiffness.

Answer. Address constrains posture, mean load and initial activation; backswing constrains transport and changing geometry; transition constrains reversals and elastic state; downswing constrains delivery motion and available correction time; impact constrains contact impulse and initial mechanical state; follow-through constrains dissipation and deceleration. Perturbation response, when available from an appropriately controlled study, adds information beyond ordinary motion. Different combinations of drive, tissue response and feedback can produce similar unperturbed trajectories, so no universal stiffness schedule follows.

2. What a Decreasing Storage Function Means For the unforced spring-damper model, consider a displaced state with zero velocity. Is the energy derivative strictly negative? What further argument is needed for convergence?

Answer. The energy is positive but $\dot{V} = -D\dot{e}^2 = 0$ at that instant. The restoring acceleration subsequently creates motion. In a bounded invariant neighborhood with an isolated energy minimum and positive damping, the only trajectory that can

remain in the zero-derivative set is the equilibrium. That invariant-set argument establishes asymptotic convergence. It does not make every coordinate error decrease at every instant, nor does it establish tracking of a different moving reference.

3. Parameters, Samples and Information Count stiffness and damping parameters for 20 independent scalar joints and for two symmetric coupled matrices. Compare these counts with a 1 kHz record of 20 joint variables.

Answer. The diagonal approximation has 40 parameters; the symmetric coupled case has 420. The record contains 20,000 scalar samples per second. Samples per second and a one-time parameter count have different units. Neither specifies bits without a coding/precision model, and neither establishes a brain-wide processing limit or a guaranteed reduction in biological computation.

4. Distributed Feedback Distinguish an intrinsic force response, spindle afferent activity, a short-latency muscle response and a longer multi-joint response. Does lack of conscious correction imply mechanical passivity?

Answer. Intrinsic mechanics act through the existing tissue state. Spindle afferents encode signals influenced by motion and fusimotor state. Short-latency pathways can alter motor-neuron activity; longer pathways can integrate information across joints and include cortex. Their measured onsets and force consequences differ. An automatic neural controller can supply mechanical energy, so lack of consciousness does not establish passivity.

5. Net Torque and Coactivation Two idealized opposing muscles have constant moment arm $r = 0.03$ m and equal force. Initially each has local length stiffness 1000 N/m; after a change each has 2000 N/m. Compute net torque and restoring joint stiffness in both states. Can the result identify expertise?

Answer. Net torque is zero in both states because the forces cancel. Joint stiffness is $r^2(k_1 + k_2)$, giving 1.8 and 3.6 N m/rad. Changing equal activation can change resistance without changing net torque; muscle energy consumption is not measured by net joint work. These manufactured states do not identify an expert or a novice, and the calculation omits variable moment arms and tendon compliance.

6. Torque and Instantaneous Power For a fixed reference $q_0 = 0$, let $K = 50$ N m/rad, $D = 5$ N m s/rad, $q = 0.1$ rad and $\dot{q} = 1$ rad/s. Find $\tau = -K(q - q_0) - D\dot{q}$ and its power. Does the spring always remove kinetic energy?

Answer. The spring and damper each contribute -5 N m, so total torque is -10 N m and power on the coordinate is -10 W. Spring storage increases at 5 W and the damper dissipates 5 W. When the same displaced spring moves toward its reference, spring power can be positive; viscous dissipation remains nonnegative. Resisting displacement and resisting velocity are different statements.

7. Differentiate the Correct Storage Derive the energy identity for $I\ddot{q} = -K(q - q_0) - D\dot{q}$ with fixed positive constants. Then let $q_0(t)$ and $K(t)$ vary while retaining damping against absolute velocity.

Answer. For fixed reference, differentiation of $I\dot{q}^2/2 + K(q - q_0)^2/2$ and substitution cancel the spring work, leaving $-D\dot{q}^2$. For varying reference and stiffness the derivative becomes

$$\dot{V} = -D\dot{q}^2 - K(q - q_0)\dot{q}_0 + \frac{1}{2}\dot{K}(q - q_0)^2.$$

Reference motion and parameter changes can supply energy. Omitting these terms would turn an actively driven system into an incorrectly passive one on paper. Tracking requires a separate error-coordinate argument.

8. Damping Regimes Classify $\ddot{e} + D\dot{e} + 100e = 0$ for $D = 5, 20, 50$. For the critical case with $e(0) = 0.1$ and $\dot{e}(0) = 0$, obtain $e(t)$ and $e(0.1)$.

Answer. The cases are underdamped, critically damped and overdamped. Only the first has complex poles and spiralling phase trajectories. The critical solution is $e(t) = 0.1(1 + 10t)\exp(-10t)$, so $e(0.1) = 0.2/e \approx 0.07358$. The overdamped system's slow pole is approximately -2.0871 s^{-1} , slower than the critical decay scale. Increasing damping indefinitely does not guarantee faster recovery.

9. An Underdetermined Impact Estimate Given only rotational inertia $I = 0.01 \text{ kg m}^2$ and a time interval $T = 0.005 \text{ s}$, can one estimate a safe stiffness range? Interpret the dimensional scale I/T^2 .

Answer. One cannot identify a safe range. The scale is 400 N m/rad under the usual dimensionless-radian convention; choosing $K = I/T^2$ merely imposes $\sqrt{K/I}T = 1$. It supplies no loading, allowable excursion, damping, contact mechanics or tissue outcome. The interval is not automatically a forcing period, and resonance is not automatically instability. No injury or grip recommendation follows.

10. A Waggle Association Suppose a golfer records perceived waggle tension and subsequent delivery error over repeated sessions. What can this association establish, and how might it be tested without treating sensation as stiffness?

Answer. It can describe a within-golfer association under the recorded conditions. Separate sessions used to formulate the hypothesis from those used to test it; record task, equipment and relevant setup changes. Fatigue, expectation and preparation may affect both sensation and outcome. No mechanical stiffness is identified without a known perturbation and force/motion response, and association alone does not show that changing the sensation causes improved delivery.

11. What Identifies the Response? For a simulated joint, let $e = A\cos(\omega t)$ and measure the required torque. Show why an unknown inertia and stiffness are confounded at that frequency. Does adding grip pressure solve this automatically?

Answer. The torque is $(K - I\omega^2)A\cos(\omega t) - DA\omega\sin(\omega t)$. The in-phase term identifies a combination of I and K , not each separately. Known inertia or informative additional frequencies can resolve this within the stated model. Grip pressure does not automatically provide the missing rotational inertia, joint torque or operator; its relation to those quantities must be modeled and validated.

12. Simulated Perturbation Recovery Use a simulation of $\ddot{e} + 20\dot{e} + 100e = 0$. At $t = 0$, apply an initial velocity error of 0.1 rad/s with zero position error. Find the largest position error. How would delayed feedback change the inference?

Answer. The solution is $e(t) = 0.1t\exp(-10t)$. It peaks at $t = 0.1 \text{ s}$ with $e = 0.01/e \approx 0.003679 \text{ rad}$. Position error initially grows even though mechanical energy decreases. Adding delayed feedback requires a new response calculation and stability analysis; the undelayed result cannot certify it. Use a numerical perturbation or an existing controlled dataset for this exercise. A partner should not introduce an unexpected physical disturbance during a live golf swing.

Part X

Synthesis

Chapter 33

Where Disciplines Collide: An Interdisciplinary Perspective

33.1 Why Disciplines Matter

A golfer changes a cue, a shaft, or a training routine and the ball flies differently. What changed between the intention and the outcome? The answer may involve the chosen movement, muscle activation, contact with the ground, elastic deformation, club delivery, ball impact, or the measurement itself. Several of these can change together. A convincing explanation must follow the connections without treating a successful shot as proof of every proposed mechanism.

The central question of this book is how a golfer prepares and regulates a coupled mechanical system so that its evolving motion delivers a useful impact. Momentum, gravity, constraints and elasticity can help carry a movement forward. Muscles prepare that motion, continue to exchange energy with it, and alter its response to disturbances. The interesting comparison is therefore between specified movement strategies under specified conditions, with their benefits and costs measured at the appropriate level.

In Plain Language 33.1. Multiple Disciplines, One Framework

A mechanical model connects forces to motion. A controller connects information and goals to actions. A measurement model connects physical events to recorded data. An injury study connects exposure and other factors to health outcomes. These models can inform one another, but their conclusions are not interchangeable. A mathematically valid simulation can motivate an experiment without establishing what a person's nervous system does or which technique prevents injury.

This chapter builds those connections explicitly. Its numerical examples are declared teaching models, not measured golf populations or fitting recommendations. The worked answers also show how to challenge the framework: specify an alternative mechanism that predicts the same observation, then identify a measurement or intervention that could distinguish them.

33.2 Interdisciplinary Connections: Overview

The useful common structure is a chain of interacting models. The golfer's policy selects commands using goals, observations and internal state. The body and club transform commands and external forces into motion. Impact converts a particular delivery and contact configuration into club and ball changes. Flight, landing and subsequent motion deter-

mine the result. Sensors observe selected parts of this process, and their reports feed into subsequent decisions.

Materials and Environment Enter the Physical Models

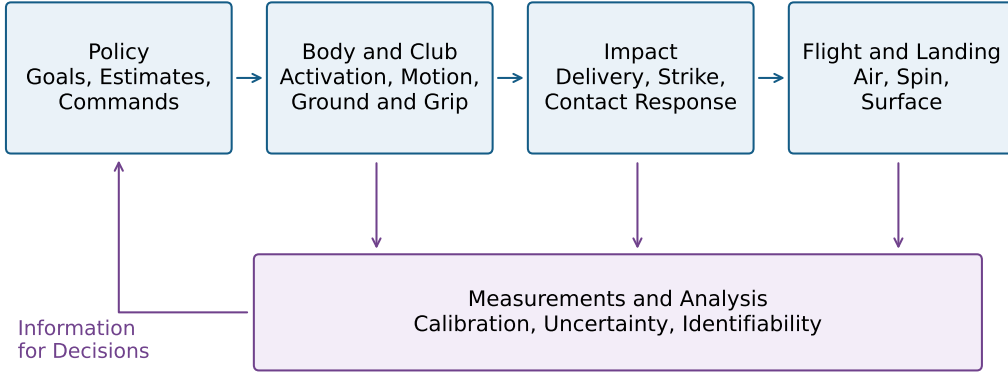


Figure 33.1: Connected Models Link Commands, Motion, Impact and Outcome. Measurements Observe Selected Stages and Inform Decisions; the Map Does Not Assign These Functions to Brain Regions.

Figure 33.1 is a map of questions, not a diagram of brain anatomy. Materials enter both the moving system and contact law. Ground and grip conditions constrain the motion. Neural and mechanical responses operate during the swing, while learning changes future policies. Uncertainty propagates through every stage; a precise final number does not imply that every upstream quantity was observed.

This chain also explains why one scalar cannot represent overall swing quality. Ball speed, directional dispersion, joint loading, effort, robustness and learning are different outcomes. Improving one may change another. A rigorous argument names the outcome, comparison and operating conditions before it calls a movement efficient, stable, safe or optimal.

33.3 Control Theory and Biomechanics: The Affine Bridge

In a local coordinate description, a constrained multibody model may be written

$$M(q)\ddot{q} + c(q, \dot{q}) + g(q) + d(q, \dot{q}) = B(q)\tau + J(q)^T \lambda + Q_{\text{ext}}.$$

Here q describes configuration, M is the inertia matrix, c collects velocity-dependent inertial terms, g is gravity, and d represents the explicitly modeled dissipative forces. Joint torques τ , contact reactions λ , and other external generalized forces must be assigned to the correct system boundary. Ideal constraints can be eliminated locally or retained with their acceleration equations. Contact creation, separation and slip may require different modes.

After choosing independent states and inputs, one possible model is

$$\dot{x} = f(x) + G(x)u.$$

This form says that the instantaneous state derivative is affine in the selected input. It does not say that motion is linear in initial conditions, that muscle force is globally linear in excitation, or that every biological model has this form. If the input is joint torque, muscle activation and tendon behavior may lie outside that simplified plant. If the input is neural excitation, activation and elastic states generally belong inside it. An input-dependent activation time constant can require a more general or piecewise description.

33.3.1 Input Coordinates and the Meaning of Zero

The drift f is the vector field obtained when the declared input is zero. It is not the set of effects that cannot be controlled. Control can change the state at which drift acts, and feedback can change the baseline used to describe the system. Under the input substitution

$$u = \alpha(x) + \beta(x)\nu, \quad \dot{x} = \underbrace{f(x) + G(x)\alpha(x)}_{\tilde{f}(x)} + \underbrace{G(x)\beta(x)}_{\tilde{G}(x)}\nu,$$

zero residual input ν retains the baseline action $\alpha(x)$. For an invertible β , this is an equivalent input description if the admissible input set is transformed as well. If β restricts input directions, it also restricts the controller.

Consider a rotor with inertia I and torque $u = -kq + \nu$. Zero ν retains the restoring torque $-kq$; zero physical torque removes it. At $I = 0.05 \text{ kg m}^2$, $k = 2 \text{ N m/rad}$ and $q = 0.1 \text{ rad}$, the former gives an angular acceleration of -4 rad/s^2 . The two zero-input experiments answer different questions. A zero-excitation muscle model likewise retains whatever activation and tendon strain already exist; it is not an instantaneous zero-force intervention.

Changing the state at the top of the backswing changes the trajectory through a fixed vector field. Changing equipment or tissue parameters changes the plant itself. Changing a feedback policy changes the resulting closed-loop vector field. These are three distinct ways to obtain a different swing, even though all can change the club's eventual delivery.

33.3.2 Control Authority Requires a Task and a Time

A large nominal drift term does not prove that additional input has little effect. For the constant-inertia rotor,

$$\ddot{q} = \tau/I, \quad \delta q(T) = \frac{\delta \tau}{2I}(T - t_0)^2$$

for a constant added torque starting at t_0 , with zero initial perturbation. The torque-to-acceleration gain is $1/I$ at every angular speed. With $I = 0.05 \text{ kg m}^2$, a 0.5 N m increment acting for 0.1 s changes angle by 0.05 rad and angular speed by 1 rad/s , whether the initial speed is zero or 100 rad/s .

A late intervention can nevertheless have little effect because little time remains, the command arrives after contact, activation develops slowly, torque is saturated, or available directions poorly affect the desired output. For a linearization about a specified trajectory, a fixed-time output perturbation has the form

$$\delta y(T) = C(T)\Phi(T, t_0)\delta x(t_0) + \int_{t_0}^T C(T)\Phi(T, s)G(x(s))\delta u(s) ds.$$

The state-transition matrix Φ propagates perturbations through the coupled dynamics; C maps state changes to the chosen output. A state-triggered impact also requires event-time sensitivity. Input limits, delays and units must be included before comparing authority across phases or joints. The computational-brain chapter (Chapter 31) develops a delayed-activation example that makes this distinction quantitative.

33.3.3 Minimum Effort Requires a Cost and a Horizon

Preparing a useful trajectory can reduce later correction, but proximity to the zero-input endpoint is not a universal minimum-effort criterion. For a controllable linear system with an unconstrained quadratic input cost, the minimum cost of an endpoint correction e is

$$J_{\min} = e^T W^{-1} e, \quad W = \int_0^T e^{A(T-s)} B R^{-1} B^T e^{A^T(T-s)} ds,$$

where the cost is $\int_0^T u^T R u dt$ and R is positive definite. A singular Gramian requires a reachable endpoint and the corresponding restricted solution. Saturation, nonlinear dynamics and path constraints can invalidate this unconstrained formula.

The result follows by minimizing the cost subject to the endpoint integral constraint. A Lagrange multiplier gives

$$u_*(s) = R^{-1} B^T e^{A^T(T-s)} W^{-1} e.$$

Substitution into the endpoint integral returns $W W^{-1} e = e$, and substitution into the cost returns $e^T W^{-1} e$. The weighting therefore comes from the dynamics and input cost, not an arbitrary distance between poses.

To see why the metric matters, nondimensionalize a double integrator using a one-unit horizon, a chosen position scale and its associated velocity scale. With unit input weight,

$$W = \begin{bmatrix} 1/3 & 1/2 \\ 1/2 & 1 \end{bmatrix}, \quad W^{-1} = \begin{bmatrix} 12 & -6 \\ -6 & 4 \end{bmatrix}.$$

The correction $(0.1, -0.1)^T$ is shorter in Euclidean distance than $(0.2, 0.3)^T$, yet costs 0.28 rather than 0.12. Position and velocity errors can reinforce or oppose what the available input can accomplish. Even this input cost is not metabolic expenditure or injury risk. Those require additional models and evidence.

33.4 Robotics: What Golf Can Learn From Machines

Robotics provides explicit ways to connect task objectives, constraints, actuator limits and feedback. A robot can be designed with rigid or compliant transmissions, delayed sensing, learned policies and adaptive parameter estimates. Human-versus-machine comparisons should specify the systems being compared rather than assign flexibility exclusively to people or fixed control exclusively to robots.

33.4.1 Principle 1: Hierarchical Control

A useful design can separate a task objective, a movement policy and an actuator controller. For example, a target distribution may constrain acceptable launch conditions; a trajectory

optimizer may search for feasible club delivery; a lower-level controller may regulate torque or interaction dynamics. Information can also pass between levels during execution.

This is an engineering decomposition, not proof that the motor cortex, cerebellum and spinal cord implement those stages in that order. Nor is a hierarchy mathematically necessary for every controller. A feedback policy can map estimated state directly to actuator commands. The important questions are whether its commands are feasible, whether it handles uncertainty, and whether it produces the intended response under disturbances.

For golf, a hierarchy is useful when it keeps the instruction at the right level. A target direction is an outcome goal; an intended club path is an intermediate task; a particular muscle activation is an implementation hypothesis. A coach who changes one level should measure the outcome at the others before attributing success to a specific mechanism.

33.4.2 Principle 2: Controlling Interaction Dynamics

Mechanical impedance describes a force–velocity relation. For a scalar linear mass–damper–spring perturbation with zero initial conditions, $M > 0$, $B \geq 0$ and $K \geq 0$,

$$F(s) = (Ms + B + K/s)V(s).$$

The force–displacement relation instead has dynamic stiffness $Ms^2 + Bs + K$. These quantities depend on frequency, direction, operating state and the chosen mechanical port. A compliant shaft, an activated muscle and a feedback controller need not have the same response merely because each can resist displacement.

For an elastic rotational controller with error $e = q - q_d$, nonnegative stiffness $K(t)$ and damping D ,

$$\tau = -Ke - D\dot{e}, \quad V_{\text{stored}} = \frac{1}{2}Ke^2,$$

and direct differentiation gives

$$\tau\dot{q} + \dot{V}_{\text{stored}} = -D\dot{e}^2 + \tau\dot{q}_d + \frac{1}{2}\dot{K}e^2.$$

With fixed reference and stiffness, damping removes energy at this port. Moving the reference or increasing stiffness while displaced can supply energy. At a clamped error of 0.1 rad, raising K from 10 to 30 N m/rad increases storage by 0.1 J. Calling all such behavior passive would hide an energy source.

Impedance control therefore suggests a research question: what response to a specified perturbation is useful at each phase? It does not prescribe maximal stiffness at impact or a universal early-soft/late-stiff schedule. Resistance in one direction may help face orientation while affecting another degree of freedom or increasing internal loading. The biology and computational-brain treatments discuss the mechanical and experimental distinctions behind this question.

33.4.3 Principle 3: Task-Space vs. Joint-Space Planning

Let a chosen club task coordinate be $y = h(q)$. Locally,

$$\dot{y} = J(q)\dot{q}, \quad \tau_{\text{task}} = J(q)^T F_{\text{task}},$$

where the task force and velocity must use compatible coordinates and units. The second relation follows from virtual work. Full orientation requires an appropriate local representation or a geometric treatment on the rotation group. A three-dimensional club pose cannot be reconstructed from face azimuth and path azimuth alone.

Redundancy means that different joint motions can satisfy the same instantaneous task velocity. Such alternatives may differ in posture, contact feasibility, energy, sensitivity and tissue loading. A kinematic null direction is not automatically dynamically harmless: moving in it can change inertia, future reachability or the next task Jacobian. This is why a good club-level result does not uniquely identify a whole-body technique.

33.5 Why Robots Teach Us About Muscles

The analogy becomes productive when actuator dynamics are made explicit. Torque is an input in one model and an output of activation, force–length–velocity behavior, tendon deformation and moment arms in another. A command can change immediately while muscle force develops over time; stored elastic energy can persist after a command changes.

Muscle recordings add evidence but need their own observation model. Pink, Jobe and Perry’s shoulder EMG study reports muscle- and phase-dependent activity; its abstract does not establish that a particular muscle simply turns off once drift takes over [Pink et al., 1990]. EMG is not a direct measurement of tendon force or joint torque. Electrode placement, normalization, activation dynamics and muscle geometry matter when connecting it to a mechanical model.

The useful robot experiment is correspondingly precise. Compare a torque-controlled rotor, a motor with series elasticity, and an actuator with delayed activation under the same output task. Match initial states and input limits, then perturb each system. Similar trajectories may coexist with different internal forces and correction capacities. Differences reveal which modeling elements matter, rather than which machine resembles a golfer in appearance.

33.6 Neuroscience: Motor Control and the Cerebellum

Control theory offers computational hypotheses about prediction, estimation and action selection. Neuroscience asks how those functions are implemented and how they interact. Mechanical equations alone cannot assign an algorithm to a brain region or rank one region as more important because a swing is fast.

33.6.1 The Cerebellum and Feedforward Control

A forward model predicts consequences of a state and command. An inverse model or policy selects commands intended to achieve an outcome. Inverse kinematics solves a geometric configuration problem; it is not synonymous with either a forward dynamics model or a complete motor controller. Planning a joint trajectory also leaves the force and activation problem unresolved.

Predictive control is useful when sensory information is delayed, but feedforward action need not be independent of feedback. A policy may use a state estimate that combines prior commands with new observations. The mechanical model then helps explain what information would be useful, while physiological experiments are needed to establish where and how such computations occur. Learning a useful prediction need not mean explicitly storing the functions named f and G in this book.

33.6.2 Motor Cortex and Feedback Control

The value of feedback depends on its information, noise, latency, the task, and the time remaining. The distant location of the ball's eventual landing is not an explanation for absent visual control: vision can also concern the target, body, club or environment. Proprioception and other sensory channels provide different information and can be combined with prediction.

Controlled reaching experiments demonstrate online visual corrections, and multijoint perturbation experiments show task-relevant responses involving motor cortex [Saunders and Knill, 2003, Pruszynski et al., 2011]. These results support studying feedback in context; they do not supply a universal golf-swing latency. A command triggered too late to affect impact cannot correct that impact, while forces and feedback processes initiated earlier can still be acting. Feedback after impact can also affect follow-through and subsequent learning.

33.6.3 Learning and Adaptation

A different shot after practice may reflect an explicit aiming strategy, implicit adaptation, altered coactivation, a changed state estimate, or a different movement policy. Mazzoni and Krakauer's adaptation experiment illustrates that explicit strategy and implicit adaptation can coexist and conflict [Mazzoni and Krakauer, 2006]. One successful trial therefore does not identify what was learned.

A discriminating golf experiment records the cue or perturbation, actual delivery and outcome, then examines retention, transfer and behavior when the perturbation is removed. If the claim concerns improved mechanical robustness, apply a controlled perturbation and measure the response. If it concerns implicit learning, use appropriate behavioral comparisons rather than infer it from a fitted trajectory. If it concerns a neural location, behavioral and mechanical evidence alone is insufficient.

33.7 Material Science: Shaft Design and Ball Physics

Equipment changes both the plant and, potentially, the golfer's policy. A comparison with prescribed identical grip motion asks how two shafts respond mechanically. A comparison in which a golfer freely adapts asks how each golfer-club combination performs. Both are useful, but they estimate different effects.

33.7.1 Shaft Materials and Design

An anisotropic laminate has direction-dependent stiffness. Axially aligned fibers can strongly support axial loading; off-axis plies can contribute importantly to shear and torsional response. A statement that angled fibers are more compliant is incomplete until the loading direction, constraints and response quantity are specified. Laminate analysis combines transformed ply constitutive laws with through-thickness force and moment balances [Roylance, 2000].

For a declared plane-stress orthotropic ply, use engineering shear strain and

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix} \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{bmatrix}.$$

Take illustrative constants $E_1 = 135$, $E_2 = 10$, $G_{12} = 5$ in GPa and $\nu_{12} = 0.3$, with $\nu_{21} = \nu_{12}E_2/E_1$. Inverting the compliance matrix gives $Q_{11} = 135.9060$, $Q_{22} = 10.0671$, $Q_{12} = 3.0201$ and $Q_{66} = 5$ in GPa.

For a 45° ply, rotation gives the constrained-strain coefficients

$$\begin{aligned}\bar{Q}_{11} &= \frac{Q_{11} + Q_{22} + 2Q_{12} + 4Q_{66}}{4} = 43.0034, \\ \bar{Q}_{66} &= \frac{Q_{11} + Q_{22} - 2Q_{12}}{4} = 34.9832\end{aligned}$$

in GPa. Axial response decreases while the shear coefficient increases. These are stiffness-matrix entries, not freely contracting Young's and shear moduli: extension-shear coupling can matter. A balanced layup can cancel some couplings, and ply positions determine bending response. A shaft's full bending and torsional behavior additionally depends on tube geometry, layup, boundary conditions, damping and frequency.

For an ideal uniform Euler-Bernoulli beam, a mode frequency scales as $\sqrt{EI/(\mu L^4)}$, with a boundary-dependent factor, bending rigidity EI , mass per length μ , and length L . Reducing mass does not logically require reducing stiffness. Nor does a lighter shaft automatically transfer a fixed saved-energy amount to the clubhead: the golfer may change work, speed and timing. A shaft stiffness label or an undefined shaft COR cannot settle that coupled comparison.

33.7.2 Ball Compression and the Coefficient of Restitution

For a one-dimensional normal collision, restitution is the ratio of separating to approaching relative speed,

$$e = \frac{v'_b - v'_c}{v_c - v_b},$$

under the declared positive direction and an initially approaching pair. Smash factor is outgoing ball speed divided by incoming club speed, a different quantity. Restitution also is not the fraction of initial club energy transferred to the ball.

Derive the distinction in a free two-mass model: positive mass M moves at speed $U > 0$ into stationary positive mass m , external impulse is negligible, and there is no rotation or tangential contact. Take $0 \leq e \leq 1$ for this dissipative collision example. Momentum conservation and the restitution law give

$$v'_b = \frac{(1+e)M}{M+m}U, \quad v'_c = \frac{M-em}{M+m}U.$$

The ball's fraction of initial kinetic energy and the total dissipated fraction are

$$\eta_b = \frac{mM(1+e)^2}{(M+m)^2}, \quad \eta_{\text{loss}} = (1-e^2)\frac{m}{M+m}.$$

Equivalently, lost energy is $(1-e^2)\mu_r U^2/2$, where the reduced mass is $\mu_r = Mm/(M+m)$. Restitution squared governs retention of relative translational kinetic energy in this model, not the ball's share of the initial laboratory-frame energy.

With illustrative $M = 0.200$ kg, $m = 0.04593$ kg and $e = 0.83$, the speed ratio is 1.48823, the ball fraction is 0.5086, and the dissipated fraction is 0.0581. The remaining approximately 43.33% stays in the moving first mass. These identities can be checked directly from both outgoing velocities.

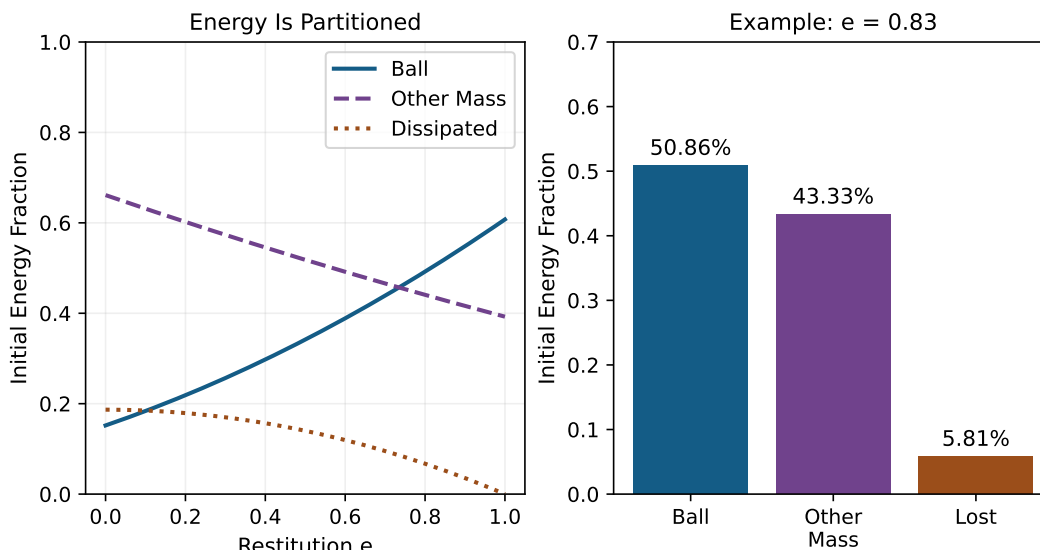


Figure 33.2: Restitution Does Not Equal Energy Transfer. The Free Two-Mass Teaching Model Uses a 0.200 kg Incoming Mass and a Stationary 0.04593 kg Ball; Fractions Sum to One.

A real club–ball impact can include rotation, off-center contact, tangential impulse, elastic waves and external coupling. Its effective normal inertia is not necessarily the head mass, much less the whole golfer’s mass. The impact chapter (Chapter 17) treats those distinctions. The number 0.83 here is a model input, not a statement of a current equipment-conformance rule.

Compression describes deformation under a specified test; it does not alone determine restitution, launch, spin or dispersion. Arakawa and colleagues’ normal impacts against a rigid steel target showed decreasing restitution with increasing inbound speed while normal compression increased [Arakawa et al., 2006]. That experiment contradicts a universal rule linking greater compression to greater restitution, but does not rank modern balls in a golfer’s driver or wedge. Fitting requires defined impact conditions and measured outcomes, including variation across strikes.

33.8 Data Science and Launch Monitor Analysis

Data science enters through an observation equation,

$$z_k = h(x(t_k), \theta, \kappa) + \epsilon_k,$$

where θ contains physical parameters, κ calibration parameters, and ϵ_k measurement error. A launch report can combine directly observed signals with estimates and model-derived quantities. A ball-flight sensor is not automatically a full-body motion-capture system or a measurement of muscle force. The [launch-monitor article](#) explains the geometry and observability of several measurement approaches.

33.8.1 Identifying Dynamics From Data

Even complete noiseless motion need not separate a plant parameter from an unobserved input. Consider

$$I\ddot{q} + kq = u(t).$$

If one stiffness k_1 and input $u_1(t)$ reproduce an observed motion, any alternative k_2 reproduces it with

$$u_2(t) = u_1(t) + (k_2 - k_1)q(t),$$

provided that alternative input is admissible. A hundred terminal launch reports do not remove this structural ambiguity. More samples of the same confounded experiment may improve numerical precision while leaving the underlying cause unidentified.

Identification can improve with independently measured input or force, calibrated geometry, known boundary conditions, purposeful excitation, repeated conditions and external parameter measurements. Its success must be checked for the particular parameterization and experiment. Strong regularization can select one plausible model without making it uniquely supported by the data. An optimizer's convergence is not an observability result.

33.8.2 Predicting Ball Outcome From Swing Data

Prediction is a legitimate goal even when the internal mechanism is not identified. Its evaluation must match the intended use. A model trained on one golfer, club and session may interpolate well within that setting but fail after a shaft change, injury, different sensor, or new movement strategy. Separate training and evaluation by the units over which generalization is claimed, such as golfer, session or equipment condition.

The complete prediction chain is body/club state, delivery, contact, flight and landing. Each stage needs its own assumptions and uncertainty. Face angle and path do not specify attack angle, dynamic loft, strike location, angular velocity or all contact properties. Ball speed, launch angle and spin magnitude do not uniquely specify wind, aerodynamic coefficients or the full spin vector. Carry estimates require an explicit flight model; total distance additionally depends on landing and surface interaction.

A model should also be compared with simpler alternatives. If a regression from measured delivery predicts ball flight as well as a personalized full-body model, the extra body parameters have not yet demonstrated predictive value for that output. They may still be useful for other questions, but that requires separate evaluation.

33.8.3 Personalizing the Model

Personalization means estimating quantities the available experiment can support, with uncertainty and a record of calibration. Segment dimensions, inertial properties, joint constraints, actuator capacity, elastic response and policy parameters have different measurement requirements. Their estimates can be correlated.

Keep an explicit inventory of measured, estimated, assumed and unidentifiable quantities. Report held-out errors and failure conditions alongside parameter intervals. When an equipment change modifies both mechanics and behavior, distinguish a fixed-policy simulation from observed adaptation. A credible personalized model may be deliberately modest: for example, predicting a distribution of launch conditions within a calibrated range without claiming to recover every muscle's action.

33.9 Sports Medicine: From Force Estimates to Health Outcomes

33.9.1 Net Torque Is Not Tissue Load

Inverse dynamics estimates net generalized forces under its model and measurements. Multiple muscles, passive tissues and contact reactions can contribute to the same result. A low net moment therefore does not establish low tendon force or low joint compression.

Consider a planar teaching model with two parallel muscle forces that both compress a joint but have opposite 0.03 m moment arms. Forces of 200 and 100 N produce a net moment of 3 Nm and a total compressive muscle contribution of 300 N. Forces of 500 and 400 N produce the same moment and a 900 N contribution. Additional external loads and joint reactions must still be included in a full balance. The example establishes non-uniqueness, not a clinical injury threshold.

Similarly, greater torque-to-acceleration gain does not imply that a larger torque is required for the same acceleration. In torque coordinates, strength limits primarily change the feasible torque set; in normalized activation coordinates, force capacity can also appear in the input map. Coordinate choice must be settled before interpreting a change in control authority as strength or tissue demand.

Injury risk concerns exposure, tissue capacity, recovery, prior injury and other factors over time. Robinson and colleagues followed 910 amateur golfers for five months and reported associations involving age, osteoarthritis and previous injury [Robinson et al., 2024]. They did not measure swing biomechanics. Self-report, population selection and incomplete questionnaire response further limit interpretation. Such surveillance helps define relevant outcomes; it does not validate a zero-torque technique as an injury-prevention intervention.

A mechanics-based hypothesis can propose that a change reduces a specified load under matched conditions. Testing prevention additionally requires suitable prospective health outcomes, exposure accounting and comparison groups. A decrease in one joint's estimated moment may shift demand elsewhere or accompany higher internal coactivation. The chapter consequently cannot derive a rehabilitation exercise, return-to-play decision or universally safer sequence from its equations. Those questions require individualized clinical assessment and relevant intervention evidence.

33.10 Coaching Science: Translating Physics Into Cues

Coaching connects a proposed mechanism to a learnable action and a measurable outcome. A cue is an intervention on behavior, not a boundary condition that guarantees a particular joint trajectory. Different golfers can interpret the same words differently, and a familiar cue can affect several mechanical variables at once.

33.10.1 Phase-Specific Coaching

An early change can alter the initial state for later motion, including momentum and elastic strain. A late change has less remaining time to propagate and may encounter activation or torque limits. Those observations motivate phase-specific experiments; they do not establish that speed problems are always drift problems or face errors are always late control problems.

For example, a changed setup might affect path, strike location and available torque directions together. An apparent improvement in face delivery might result from an earlier trajectory change rather than a conscious correction near impact. Record the relevant timing and delivery before deciding which mechanism explains the result.

33.10.2 Cue Design

Start with an observable difficulty, such as excessive variation in a specified face-to-path measure under a repeatable task. Propose one understandable cue, state its intended effect, and alternate or randomize comparable blocks with and without it. Record delivery, strike, ball outcome and perceived effort; include retention or transfer when the claim concerns learning.

Interpretation then has three levels. First, did the outcome improve beyond normal variability? Second, did the proposed intermediate variable change? Third, did that change plausibly account for the outcome under the experiment's controls? A positive answer to the first question does not settle the other two. Fatigue, order, familiarization and changed strike quality can explain part of an observed benefit.

Physics makes a cue more testable by identifying those intermediate variables and alternatives. It does not establish that a particular wording, more aggressive hip motion, grip change or release timing is universally effective. An unsuccessful cue can still teach something if the experiment reveals which assumed link failed.

33.11 Synthesis: The ZTCF Family Connects Testable Questions

A zero-torque counterfactual is most useful as a specified intervention in a specified model. Start from a declared state at a declared time, replace selected active torques with zero, retain or modify other mechanisms explicitly, and integrate to a defined horizon or event. Report what happens if balance, contact or joint constraints fail before the intended event.

Zero joint torque does not mean zero gravity, zero ground reaction, zero passive tissue force or zero previously stored energy. A branch with zero new excitation generally differs from one with zero activation or zero actuator force. These choices should be separate members of a counterfactual family, not silently conflated as the body's natural motion.

The contact mode matters for equipment comparisons too. Before any physical interaction with the ball, its compression law does not directly enter the golfer-club mechanical equations. A different ball can change the golfer's expectations, policy or prepared state, and its material response enters once contact begins. Assigning a direct pre-contact mechanical effect to ball compression would mix those pathways.

The comparison asks how the modeled future changes under that intervention. It does not uniquely apportion the actual club speed among muscles or prove that muscles did not create the motion. For a mechanically consistent model, an energy balance has the schematic form

$$\dot{E} = \dot{q}^T B \tau + P_{\text{ext}} - P_{\text{diss}},$$

with elastic storage included in E and appropriate work terms for moving constraints. Velocity-dependent inertial coupling can redistribute energy among coordinates without creating energy for the complete conservative system. Energy present at the branch point can reflect substantial earlier muscle work.

Different counterfactual interventions can interact nonlinearly. Removing two torques together need not produce the sum of their separately removed effects. A chosen attribution rule can summarize such interactions, but its result depends on the rule and baseline. This is particularly important when a zero-torque branch leaves the range in which the fitted model was validated.

The framework is therefore a common language for posing connected questions: what state was prepared, what forces remain, what alternatives are feasible, what can be observed, and what outcomes follow? Its strongest contribution is a reproducible comparison with explicit limits. Neural, coaching and clinical claims remain subject to their own experiments.

33.12 The Expanded Framework: From Rigid Bodies to Living Systems

Rigid segments are a modeling approximation, not a denial of deformation. Add degrees of freedom when they improve the questions the model can answer: spinal motion for segment-level loading, tendon or shaft states for elastic exchange, actuator states for delayed force development, or soft-tissue modes for measured relative motion. Each addition requires parameters and observations; a larger model can be less identifiable.

There is no universal anatomical degree-of-freedom count that follows by counting vertebrae or muscles. Define bodies, joints, constraints and contact modes, then determine the local independent coordinates. Redundant constraints or special configurations can change a simple counting argument. The spine and inverse-dynamics chapters (Chapters 24 and 20) develop these issues.

For a system containing the golfer and club, ground interaction and gravity are external, but so are aerodynamic loads and, during impact, the ball's action unless the ball is included in the system. A stationary ideal no-slip contact point can exert force and exchange momentum while doing zero point-contact mechanical work. That does not mean the ground force has no effect on center-of-mass kinetic energy: center-of-mass force times center-of-mass velocity is a different decomposition from the total body's external contact power. Real foot deformation, contact moments, slipping or moving supports require their own work accounting.

The same discipline applies to energy stored in tissue. Specify deformation relative to a reference, constitutive behavior, dissipation and active sources before describing recoil. Changes in mechanical work are not automatically changes in metabolic cost. A living system can maintain force while doing little external work, and the relation between those quantities is itself a physiological question.

33.13 The Future: Personalized Physics Models

A defensible development programme proceeds from calibrated measurements to identifiable models and then to tested interventions. Publish coordinate conventions, sensor timing, reference points, contact assumptions and parameter uncertainty. Verify the numerical implementation on problems with known solutions. Validate predictions against independent measurements in the intended operating range.

Next ask whether the model improves decisions. Compare a model-guided intervention with a suitable alternative, using outcomes that matter to the golfer and an evaluation long

enough for the claim. Better explanation, improved launch consistency, lower estimated load and reduced injury incidence are separate milestones. Existing sensing and modeling methods make parts of this programme possible; their availability does not mean that comprehensive personalized diagnosis has already been validated.

The most useful model may change with the task. A launch predictor, a shaft bench model, a full-body inverse-dynamics model and a clinical study answer different questions. Their integration should preserve their individual validation boundaries rather than hide them inside one impressive-looking simulation.

33.14 Worked Example: What a Launch Report Can Establish

33.14.1 The Scenario

Suppose a report gives ball speed 145 mph, launch angle 14° , spin magnitude 2500 RPM, club path 3° right and face azimuth 2° right of the target. Treat these as reported quantities whose measurement definitions and uncertainty must be checked. They do not specify the complete delivery, spin vector, atmosphere or body motion.

The golfer wants more carry while maintaining directional consistency. That is a useful objective, but the report alone does not establish present carry, an achievable distance increase, a unique spin optimum, or whether a particular muscle should act earlier.

33.14.2 The Physics

If the declared two-mass collision model applies with the illustrative parameters above, the 1.48823 speed ratio implies an incoming mass speed of 97.4313 mph. This is a conditional model calculation, not an independent club-speed measurement. A different effective mass, restitution or contact configuration changes the inferred value.

In an explicitly horizontal, positive-right angle convention, the face is 1° left of the path. A local illustrative direction model

$$\ell = wf + (1 - w)p$$

with $w = 0.76$, $f = 2^\circ$ and $p = 3^\circ$ gives launch azimuth $\ell = 2.24^\circ$ right. The selected weight is a local modeling assumption, not a universal law. Dynamic loft, attack angle, strike location and tangential contact remain unspecified. Face-to-path in this horizontal calculation does not by itself explain the reported backspin magnitude.

To predict carry, choose and validate a flight model with appropriate launch velocity, spin vector, aerodynamic behavior and atmosphere. To predict a zero-torque branch, additionally obtain an adequate pre-impact body/club state and a declared intervention. Neither calculation can be reconstructed uniquely from the five reported values.

33.14.3 The Feedback

A useful next session first establishes measurement reliability and repeatability, then tests a limited hypothesis. For example, compare strike-location and delivery distributions under matched conditions, determine which variation contributes most to the outcome uncertainty,

and evaluate a cue or equipment change against that baseline. If the hypothesis concerns shaft response, measure that response or use an independently calibrated shaft model.

Report what improved, its variability, and what mechanism remains unresolved. A measured club-speed change can be a result; a simulation can suggest a cause; a controlled comparison can test that suggestion. Keeping these steps distinct makes the feedback actionable without pretending that a launch report uniquely diagnoses the entire swing.

33.15 Key Takeaways

1. The affine description depends on state, input and baseline choices. Drift is not synonymous with uncontrollability or absence of prior muscle work.
2. Control authority concerns feasible changes in a specified output over a specified time. High speed alone does not imply a smaller input gain.
3. Mechanical impedance, internal tissue load, input effort and metabolic cost are different quantities, with different models and measurements.
4. Equipment affects material response, system dynamics, contact and behavior. Restitution, smash factor, compression and energy transfer must be distinguished.
5. More data do not resolve a structural ambiguity without informative observations or interventions. Prediction accuracy is not unique identification.
6. A counterfactual can connect disciplines by making assumptions testable. It does not replace neural evidence, coaching evaluation or prospective clinical research.

33.16 Chapter Exercises

1. For the two antagonist force pairs in this chapter, compute net moment and compressive muscle contribution. Explain what each result does and does not say about injury.
2. Compare a torque-controlled rotor with one described by $u = -kq + \nu$. Identify the two zero-input interventions and propose a perturbation experiment that could distinguish their responses.
3. For the 145 mph worked example, calculate the conditional incoming speed and launch azimuth. List the additional quantities needed for carry prediction and for a whole-body counterfactual.
4. A golfer improves after a cue. Give three explanations that do not require learning an explicit drift field, and design observations that help distinguish them.
5. Reproduce the 45° ply coefficients and the collision energy fractions. Explain why neither a shaft stiffness label nor ball compression alone determines distance.
6. Compute the two double-integrator endpoint costs. Explain why the closer endpoint can require more effort and why neither result gives a universally effective coaching cue.

7. A new command arrives after the impact event. Under what assumptions is its effect on that impact zero? Explain why this does not imply that feedback or muscle force was absent earlier.
8. Design a study that separates a change in estimated joint loading from a claim of injury prevention. Include exposure, comparison, uncertainty and the possibility of shifted internal loads.

33.17 Worked Answers

1. The moments are $0.03(200 - 100) = 3$ and $0.03(500 - 400) = 3$ N m. Compressive muscle contributions are 300 and 900 N. Net moment cannot identify the individual forces; neither force sum supplies a tissue-specific injury threshold, exposure history or clinical outcome.
2. Zero physical torque gives $\ddot{q} = 0$ in the ideal rotor. Zero residual command retains $\ddot{q} = -kq/I$, which is -4 rad/s^2 at the declared initial displacement. Release both from matched state and measure acceleration or apply a known displacement perturbation. Distinguish a real passive spring from active feedback using power, actuator and timing measurements; matching motion alone need not distinguish them.
3. Dividing 145 mph by the conditional speed ratio 1.48822836 gives 97.4313 mph. The illustrative horizontal launch is $0.76(2) + 0.24(3) = 2.24^\circ$ right. Carry additionally needs a sufficient three-dimensional launch state, aerodynamic model and atmosphere. A body counterfactual needs configuration, velocity, relevant activation/elastic states, physical parameters, external/contact conditions, an intervention and a stopping rule. None is uniquely supplied by the report.
4. Possible explanations include explicit re-aiming, a different movement policy, changed coactivation, altered strike location or ordinary trial variation. Measure aiming reports, delivery and strike, compare repeated conditions, and test retention, transfer or removal of the perturbation. A mechanical perturbation can probe impedance. These observations narrow hypotheses; neural localization requires additional evidence.
5. Substitution into the two rotated-ply expressions gives axial and shear numerators of 172.0134 and 139.9329 GPa. Dividing each by four gives 43.0034 and 34.9832 GPa, respectively. With the stated masses and restitution, the ball and lost-energy fractions are 0.5086 and 0.0581, leaving about 0.4333 in the other mass. Laminate/shaft boundary conditions, golfer adaptation and contact/flight conditions prevent a universal distance inference from either material label.
6. Multiplying by W^{-1} gives costs 0.28 and 0.12. The corresponding optimal dimensionless inputs are $0.8 - 1.8t$ and $0.6 - 0.6t$ for $0 \leq t \leq 1$; integrating each input and its remaining-time-weighted value recovers the requested velocity and position corrections. The first correction reverses part of its acceleration, making its smaller Euclidean endpoint error more expensive. These unconstrained, scaled calculations are not metabolic costs or verified human strategies.
7. If the command is causally generated after the event, initial perturbations are zero, and the model has no anticipatory dependence, it cannot alter the already completed event.

Earlier activation, stored energy, prior feedback and ongoing contact forces may still have affected that event. Post-impact feedback can affect later motion and future trials. A time constant is not automatically an additional pure delay.

8. First validate the load estimate against suitable independent measurements and vary uncertain muscle-force allocations. Compare interventions under matched tasks while recording strike, performance, exposure and relevant symptoms. A prospective prevention claim requires appropriately defined health outcomes, sufficient follow-up and a suitable comparison design, with prior injury and other confounders addressed. Randomization where feasible strengthens causal interpretation. Lower estimated net moment alone cannot substitute for that study.

Chapter 34

The Complete Golf Swing: Putting It All Together

In Plain Language 34.1. One Swing, Interacting Mechanisms

A golfer arrives at impact with more than a clubhead speed. Position, orientation, translational and angular velocity, shaft deformation, tissue state and contact conditions have developed together. Earlier muscle work changes energy and momentum; changing geometry changes how forces act; muscle activation affects both movement and response to disturbance. The collision then converts part of the delivered mechanical state into ball motion.

The purpose of a complete-swing model is to explain these connections and make comparisons testable. It should help answer why a particular change improves delivery, which assumptions that explanation needs, and what measurements could distinguish competing explanations. Calling a phase “drift-dominated” does not answer those questions by itself. This chapter connects phase descriptions to equations, worked examples and an investigation protocol.

34.1 The Full Model: Body, Arms, Club, Shaft, and Ground

Choose a system boundary before assigning forces. For the golfer and club together, hand–club interactions are internal. Gravity, ground contact, aerodynamic loads and, during impact, ball or turf contact are external. Ground reaction is therefore not the only external force besides gravity. For the club alone, the hand wrench crosses the boundary and can supply substantial power. Both descriptions are useful, but their force and energy accounts differ.

Let q contain the modeled body configuration and shaft deformation coordinates, and let v contain their corresponding generalized velocities. A floating-base orientation may require $\dot{q} = N(q)v$ rather than $\dot{q} = v$. Add internal states z when activation, tendon dynamics, viscoelastic history or other memory affects subsequent evolution. A suitable continuous state within a contact mode is

$$x = (q, v, z), \quad \dot{q} = N(q)v. \quad (34.1)$$

Modal positions without modal velocities do not determine shaft evolution. Likewise, identical joint angles and velocities with different muscle activation need not produce identical accelerations. Ideal contact reactions are normally algebraic unknowns, not additional in-

dependent state coordinates; compliant contact may introduce deformation and memory states.

For a declared torque-input model, write the mechanical equations as

$$M(q)\dot{v} = r(q, v, z) + B(q)\tau + J(q)^T\lambda, \quad Jv = 0. \quad (34.2)$$

Here M is positive definite on the chosen independent velocity coordinates. The vector r includes known passive, gravity, velocity-dependent and other declared loads with their signs; $B\tau$ is the modeled applied generalized load. The rows of J describe independent, stationary, active ideal constraints. A two-hand grip can close a kinematic loop; foot contact can constrain a floating base. Neither makes its reaction an independently commanded actuator.

For a fixed feasible contact mode with full-row-rank J , differentiate the velocity constraint and eliminate λ . Define

$$\begin{aligned} A_c &= JM^{-1}J^T, \\ D_c &= M^{-1} - M^{-1}J^T A_c^{-1}JM^{-1}, \\ a_0 &= D_cr - M^{-1}J^T A_c^{-1}\dot{J}v, \\ B_c &= D_cB, \quad \dot{v} = a_0 + B_c\tau. \end{aligned} \quad (34.3)$$

The reaction follows from

$$\lambda = -A_c^{-1}\{JM^{-1}(r + B\tau) + \dot{J}v\}. \quad (34.4)$$

It depends on input. Keeping the measured reaction fixed while changing torque generally violates the same constraint. The acceleration offset contains the moving-tangent term $\dot{J}v$ even for a stationary curved constraint. These equations are a derivation of the torque-affine mechanics, not an assertion that every physiological model is globally control-affine in neural excitation.

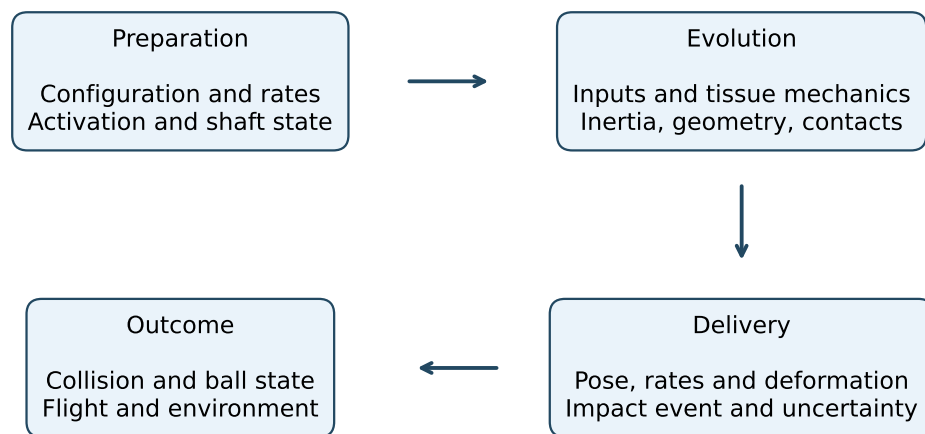
For muscles, an excitation input can first alter z , with force subsequently determined by activation, length, shortening velocity, tendon state and history. The attainable torque set can therefore depend on both state and recent inputs. A zero-torque actuator intervention differs from zero excitation, which need not immediately remove muscle force. Use $\dot{x} = f(x) + G(x)u$ only when the selected input and constitutive model actually support that representation.

Contact feasibility remains an additional condition: normal forces cannot pull a nonadhesive foot toward the ground, tangential forces must satisfy the chosen friction law, and a contact must be removed when its gap opens. Slip, lift-off, heel/toe changes and collision can change the governing mode. A smooth equation valid in one mode is not a complete hybrid swing model.

34.2 What a Drift Ratio Measures

A norm requires compatible units and an explicit metric. Comparing meters, radians, velocities and activation entries in an unscaled state vector can make a ratio depend mainly on unit choices. For one acceleration-based diagnostic, choose a positive-definite weighting W and an admissible torque set $\mathcal{U}(x)$. Define

$$c(x) = \sup_{\tau \in \mathcal{U}(x)} \|B_c(x)\tau\|_W, \quad D_{\text{avail}}(x) = \frac{\|a_0(x)\|_W}{c(x)}. \quad (34.5)$$



Measurement, assumptions, feasibility and validation throughout

Figure 34.1: Preparation, Coupled Evolution, Delivery, and Outcome. Measurements and Model Assumptions Are Needed at Every Stage; the Arrows Describe Dependencies, Not Validated Causal Attribution.

This definition uses available input, not the torque actually applied. Using the realized torque in the denominator defines a different estimand. If $c = 0$, the ratio is infinite for nonzero numerator and undefined if both vanish. Coupled muscle limits, unilateral contacts and asymmetric torque bounds cannot generally be summarized by substituting one vector called $u_{\max}$.

The ratio loses direction. In the dimensionless system $\dot{x}_1 = 1000$, $\dot{x}_2 = u$, $|u| \leq 1$, the drift-to-available-input norm ratio is 1000. Yet the entire evolution of the output $y = x_2$ is controlled by u ; drift contributes zero in that direction. A task-direction support function, or a finite-horizon reachable-output calculation, answers a different and often more relevant question.

For fixed q , fixed ideal contacts and the torque-input assumptions above, the instantaneous derivative is

$$\frac{\partial \dot{v}}{\partial \tau} = B_c(q). \quad (34.6)$$

Increasing speed can increase the velocity-dependent bias without reducing this derivative. Physiological torque limits may change with speed, and configuration can change M , J and B ; those are distinct mechanisms. “The inertia becomes large because the body is moving fast” is not a general property of rigid-body mass matrices.

For an independently checkable constrained example, take

$$M = \begin{bmatrix} 2 & 0.4 \\ 0.4 & 1 \end{bmatrix}, \quad J = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad B = I. \quad (34.7)$$

The constraint allows only opposite coordinate accelerations. Adding the generalized input $(10, 0)^T$ changes acceleration by $(10/2.2, -10/2.2)^T$, regardless of the existing velocity-dependent bias. Equal inputs $(1, 1)^T$ produce no acceleration change and are balanced by the reaction. Thus even a large available input norm says little about a particular useful direction unless constraints are included.

There is no calculated universal DCR curve in this chapter. Earlier numerical phase bands are removed because neither a declared computation nor the cited empirical study established them. A phase trend must be computed for a stated model, metric, input set, uncertainty and contact sequence. It is not a definition of expertise, conscious effort or metabolic efficiency.

34.3 Phase 1: Address and Initial State Preparation

At an ideal steady address, generalized velocity and acceleration vanish, but supporting muscle loads need not vanish. Equilibrium requires the applied, passive and contact loads to balance, together with feasible contact forces. The reduced zero-input acceleration need not point vertically downward: geometry and support constraints shape the permitted motion. A nearly stationary person may also have small postural motion or changing activation, so a video frame alone does not establish a full-state equilibrium.

Stance, ball position, grip and alignment change geometry and the reachable set of club poses. They can also change feasible support moments and muscle moment arms. Gravitational potential changes by $\sum_i m_i g \Delta h_i$ for the chosen segments; a wider stance or increased forward lean does not universally raise their centers of mass. The energy claim must follow the actual geometry.

Preparation is consequently more than selecting a photograph. The initial velocities, shaft state, activation and task intention influence what follows. Repeatable address geometry can reduce one source of variation, but movement variability, fatigue, equipment and subsequent control still matter. No claim of maximal control authority at address follows merely from knowing G or standing still.

34.4 Phase 2: Backswing and Distributed Loading

The backswing changes body and club geometry while muscles perform positive and negative work at different locations. Raising a segment can store gravitational energy. Stretching a loaded elastic element can store recoverable strain energy, while hysteresis and viscosity dissipate some work. A visible torso turn does not identify the strain energy of an individual tendon, fascia sheet or muscle fiber. Activation-dependent tissue behavior also makes “elastic” different from “metabolically passive.”

Angular momentum and kinetic energy develop as mass moves. They are not stored in a lag angle independently of velocity. Some segments begin decelerating or reversing while others are still moving backward. If the top is defined by reversal of a particular club-coordinate angle, that coordinate’s rate crosses zero at the event; other generalized rates need not do so. The top therefore does not imply simultaneous maximal speed, maximal tissue stretch or zero kinetic energy of the entire golfer.

Keep rotation references explicit. The McTeigue study’s publisher abstract reports upper-body and hip rotations relative to its measurement reference; those values are not a torso rotation of 90 degrees relative to the hips [McTeigue et al., 1994]. Subtracting two projected orientation angles also need not equal an anatomical spinal rotation in three dimensions. A useful analysis names the segments, axes, decomposition and event.

Backswing duration and amplitude are observations that vary with player, club and shot. A model can compare alternatives while constraining delivery, load and effort. The affine form alone does not establish an optimal duration, require a maximum stretch at the top, or show that the motor cortex has finished planning and the cerebellum has learned a specific drift field.

34.5 Phase 3: Transition and Coupled Sequencing

Transition is an interval of overlapping segment reversals, not necessarily one simultaneous reversal of all angular momentum. Define it with observable events, such as pelvic rotation reversal, club reversal and a specified change in hand motion. Compare the event times with measurement uncertainty; a universal 50–100ms transition is not needed to explain the mechanics.

The mass matrix couples accelerations, and velocity-dependent terms reflect the changing geometry of moving segments. Active torques, tissue loads and contact reactions act within that same system. A distal acceleration can result partly from proximal actuation, but a proximal-to-distal ordering of velocity peaks does not prove a uniquely passive cascade. Joint torques at several locations can generate similar motion, and the ordering can change with task, model, initial state and actuator limits.

For an explicit coupling example, a positive-definite two-coordinate inertia

$$M = \begin{bmatrix} a & b \\ b & c \end{bmatrix}, \quad ac - b^2 > 0, \quad (34.8)$$

gives a second-coordinate acceleration change $-b \delta \tau_1 / (ac - b^2)$ when only the first generalized load changes and the other loads are held fixed. Coupling can assist or oppose a chosen distal direction depending on the coordinate convention and configuration. This calculation predicts an instantaneous response, not an entire sequence of peak velocities or a causal energy-transfer story.

Ground forces and moments enter the momentum balance. For the combined golfer-club system, about its center of mass C in an inertial frame,

$$\begin{aligned} m \ddot{r}_C &= \sum F_{\text{ext}}, \\ \dot{H}_C &= \sum [(r_k - r_C) \times F_k + \mu_k]. \end{aligned} \quad (34.9)$$

Internal forces cancel from the total linear-momentum balance, but redistribute momentum among segments. A stationary foot can supply an external moment without doing contact work. For example, a vertical 800N force applied 0.2m horizontally from C has a 160N m moment magnitude even when its contact-point velocity is zero. These are illustrative values, not a prescription for a golfer's ground loading.

Ground reaction peaks need to be measured in a declared direction and reference interval. Total vertical force, each foot's vertical force, horizontal shear, center of pressure and free moment are different signals. They need not peak together or at transition. Muscles help create motion under support constraints; ground reaction does not directly identify which gluteal or other muscle was activated.

34.6 Phase 4: Early Downswing, Release Coordinates, and Shaft State

Define "lag" before measuring it. A projected arm-shaft angle describes one aspect of articulated geometry. Wrist flexion/extension, radial/ulnar deviation and forearm rotation are distinct anatomical motions. Flexible shaft lead/lag and toe-up/toe-down deflections describe deformation relative to a specified shaft reference. The shaft's angle to the horizontal is an absolute orientation, not a universal measure of wrist lag or elastic storage.

Maintaining a projected angle does not by itself prove active resistance, nor does its release prove a particular extensor contraction. Net joint moments require a dynamic model and external loads; individual muscle forces require additional information. Motion can result from active load, transmitted segment forces, inertia and passive tissue response together. Relative angles and angular velocities are informative observables, but they do not identify the control strategy on their own.

In an inertial frame, a clubhead following a curved path has an inward normal acceleration v_h^2/R_{path} , where R_{path} is the local curvature radius and v_h the speed. Tangential acceleration changes speed. Outward centrifugal terms arise in an appropriate rotating-coordinate description; Coriolis, Euler and frame-translation terms must then be included consistently. A large normal acceleration is not an additional source of outward power.

Shaft bending is driven by the coupled hand motion, grip loads, head inertia and offset geometry. Its deformation can change sign and can differ between bending directions. Earlier loading is retained through both deformation and modal velocity. High axial tension can modify transverse dynamics through geometric stiffness, but its effect depends on loading and the structural model; it does not imply a universally monotonic shaft stiffness or bend through the downswing.

The shaft-deflection simulations of MacKenzie and Sprigings show that both loading direction and previous bending matter, with radial loading capable of shifting the time of peak relative head velocity beyond a neutral shaft shape [MacKenzie and Sprigings, 2010]. This is a model result with specified boundary conditions. It does not justify assigning a fixed snapback time or the same deformation pattern to every player.

34.7 Phase 5: Mid-Downswing and the Remaining Correction Window

The time available before impact matters, but it is not interchangeable with instantaneous input gain. A fixed-inertia rotor demonstrates the distinction. With inertia $I = 0.1\text{kg m}^2$, an added constant torque of 1N m for 40ms changes angular velocity by 0.4rad/s and angle by 0.008rad, approximately 0.458 degrees:

$$\Delta\omega = \frac{\delta\tau}{I}\Delta t, \quad \Delta\theta = \frac{\delta\tau}{2I}(\Delta t)^2. \quad (34.10)$$

These increments are independent of the initial rotor speed. They are not measured golf responses: the model omits changing geometry, muscle dynamics, contacts and flexibility. They show why limited time reduces some corrections without proving that speed has removed input authority. Whether a fraction of a degree matters depends on the delivery variable and task tolerance.

New sensory information, a planned command already being executed, and the intrinsic mechanical response of activated tissue act on different timelines. Delays restrict responses to a newly detected disturbance; they do not erase ongoing force or preplanned changes. An arm-perturbation experiment found multi-joint responses involving primary motor cortex on a roughly 50ms timescale [Pruszynski et al., 2011]. That result does not supply a universal golf delay or demonstrate that voluntary control is absent in a named swing phase.

For a first-order illustrative activation model, $\dot{a} = (e - a)/T_a$, zero excitation gives $a(t) = a(t_0) \exp[-(t - t_0)/T_a]$, not immediate zero activation. Force also depends on muscle and tendon state. A late zero-excitation experiment therefore differs fundamentally from replacing all joint torques by zero. Anticipatory impedance and feedforward timing may remain important even when little time is available for a new feedback response.

34.8 Phase 6: Late Downswing and Flexible Delivery

Delivery is the outcome of continued coupled dynamics. A useful flexible-coordinate representation is $p_h = p_g + R_g r_h(\eta)$, giving

$$\dot{p}_h = \dot{p}_g + \omega_g \times (R_g r_h) + R_g \frac{\partial r_h}{\partial \eta} \dot{\eta}. \quad (34.11)$$

All terms refer to the same physical point and inertial frame. The expression separates contributions to velocity at one state. It does not show how grip motion would change if shaft flexibility were removed. That comparison requires another dynamically consistent trajectory, because the golfer and club react to one another.

The shaft-stiffness study makes this distinction concrete: a relative kick contribution of several meters per second did not equal the speed difference between optimized rigid and flexible model swings, and the tested nonrigid stiffness changes produced little speed difference in that model [MacKenzie and Sprigings, 2009b]. Head orientation changed as well. These findings motivate separate reporting of delivered orientation, relative deformation velocity and the outcome of a declared equipment intervention; they do not imply universal irrelevance of shaft properties or guarantee a fitting benefit.

34.8.1 Force, Velocity, Power, and Stored Energy Have Different Peaks

For one unforced, undamped mode of constant modal mass m_s and stiffness k_s ,

$$\eta = A \cos(\omega_s t), \quad \dot{\eta} = -A \omega_s \sin(\omega_s t), \quad \omega_s = \sqrt{k_s/m_s}. \quad (34.12)$$

Elastic force is $-k_s \eta$. At maximum $|\dot{\eta}|$, displacement and elastic force are zero. Maximum force occurs at a turning point, where velocity and instantaneous elastic power are zero. The elastic power delivered to modal motion is

$$P_e = -k_s \eta \dot{\eta} = -\frac{d}{dt} \left(\frac{1}{2} k_s \eta^2 \right). \quad (34.13)$$

Its positive maximum lies between those events. A driven, damped, coupled shaft shifts them further. Maximum modal velocity at impact therefore does not imply maximum restoring force, elastic power or head acceleration at impact.

For a linear spring, stored energy is $k\eta^2/2$ at prescribed deformation, but $F^2/(2k)$ under a prescribed static force. Reducing stiffness lowers the first and raises the second. Neither boundary condition describes the entire swing. If stiffness varies with an imposed preload, the derivative of stored energy also includes $k\eta^2/2$; the source of that parameter work must be included in the full system account.

The coupled equations and the impact objective determine useful timing. Energy transferred back toward the hands is not automatically wasted; it can alter subsequent movement and loads. A flexible mode can affect face orientation even when its contribution to speed along a chosen direction is small. Steel and graphite are material categories, not complete mechanical specifications: geometry, mass distribution, bending and torsional rigidity, damping and grip coupling also matter.

34.9 Phase 7: Impact and the Delivered State

Impact introduces contact forces on a much shorter timescale than the preceding swing. Specify the incoming ball and club states, contact location and orientation, material model, friction, and any external impulses being neglected. A rigid impulse model approximates a short collision; a compliant model resolves deformation and time. Their parameters and output meanings are different. A clubhead spring-effect test is not a universal ball COR limit, and its pendulum characteristic time is not the ball-club contact duration.

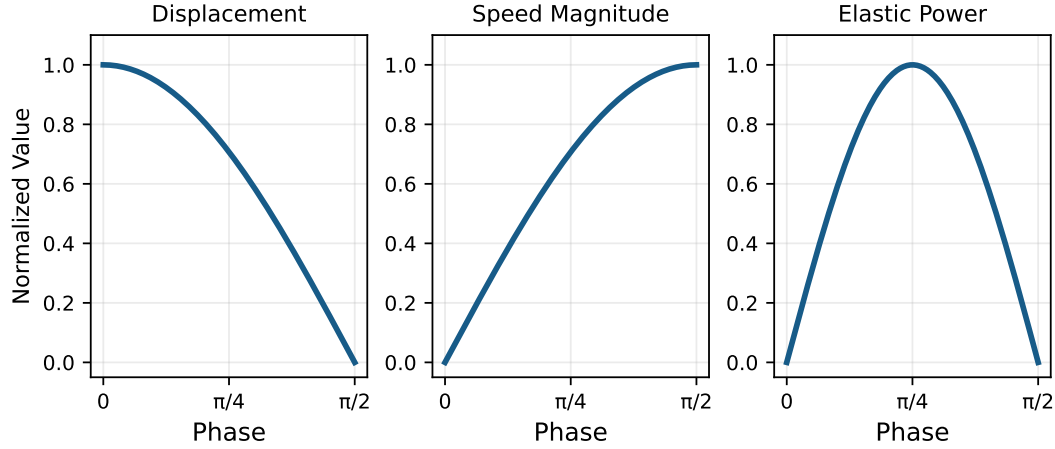


Figure 34.2: Displacement, Velocity, and Elastic Power Peak at Different Times. This Free Harmonic Mode Is an Illustrative Limit, Not a Measured Shaft Trajectory.

For a transparent reference calculation, consider only a central, one-dimensional collision: an initially stationary ball of mass m , a free translating club mass M at speed V , no spin and negligible external impulse during collision. Conservation of momentum and restitution give

$$MV = MV_c^+ + mv_b^+, \quad v_b^+ - V_c^+ = eV, \quad (34.14)$$

where e is the relative separation-speed to approach-speed ratio. Solving,

$$v_b^+ = \frac{(1+e)M}{M+m}V, \quad V_c^+ = \frac{M-em}{M+m}V. \quad (34.15)$$

Relative to the initial club kinetic energy, the ball fraction, retained club fraction and loss fraction are

$$\begin{aligned} \eta_b &= \frac{mM(1+e)^2}{(M+m)^2}, \\ \eta_c &= \left(\frac{M-em}{M+m} \right)^2, \\ \eta_D &= \frac{m}{M+m}(1-e^2), \quad \eta_b + \eta_c + \eta_D = 1. \end{aligned} \quad (34.16)$$

The relative-motion kinetic energy loses the fraction $1-e^2$; the total initial laboratory-frame energy loses η_D in this particular setup. Neither is $1-e$.

For illustrative $M = 0.200\text{kg}$, $m = 0.04593\text{kg}$, $e = 0.83$ and $V = 45\text{m/s}$, the ball speed is approximately 66.970m/s and the retained club speed 29.620m/s . The club retains 65.823% of its speed, or 43.326% of its initial kinetic energy. Ball translation receives 50.863% , and 5.810% is lost from the two translational energy stores. These numbers follow from the declared two-mass model, not a population estimate or a complete driver collision.

Off-center contact couples translation and rotation. Tangential contact, frictional sticking or slipping and deformation affect ball spin; the incoming face normal, local contact velocity and contact location matter. Grip and shaft impedance, clubface flexibility, ball

construction, turf contact and the collision timescale can change the effective model. A single COR and clubhead speed cannot identify every launch condition.

After separation, ball translation, spin and position initialize a flight model with atmospheric, wind and aerodynamic assumptions. The swing no longer exerts a contact force on the departed ball. A deterministic model gives a definite prediction only conditional on those complete inputs and parameters; uncertain launch and aerodynamic quantities produce uncertain flight. A ball curving right does not uniquely reveal which earlier joint or control phase was responsible.

34.10 Phase 8: Follow-Through and Energy Absorption

The ball removes only part of the preceding mechanical energy. Body and club motion continue, with active muscle work, passive losses, changing gravitational and elastic storage, and possible new contacts. Impact can abruptly alter clubhead motion while a finite-compliance shaft and grip transmit a time-dependent response to the hands. Muscles merely relaxing is not a complete explanation for the collision impulse or the subsequent deceleration.

Eccentric muscle action can absorb mechanical energy, but the sign of one net joint power does not identify all individual muscle contractions. Opposing muscles and biarticular pathways can produce different internal work distributions for the same observed net moment. Ground reactions can change total momentum without dissipating energy when the stationary contact is ideal. Slip, deformation and collision require their own power or impulse accounts.

For an ideal stationary constraint, $P_c = \lambda^T Jv = 0$ in the combined system. Internal joint and grip forces can perform opposite powers on adjacent subsystems, redistributing energy while their ideal total cancels. A prescribed moving boundary can do nonzero work. This distinction explains how support can redirect motion without functioning as a brake that converts all residual energy to heat.

There is no mechanics-based reason to label every follow-through high-DCR, uncontrolled, or devoid of cortical involvement. Its duration and load distribution vary with the task and player. Load estimates can motivate investigation of tissue demand, but an injury-risk claim requires appropriate evidence beyond a computed reaction force. The physical objective is feasible deceleration and task completion, with explicit assumptions about support and tissue behavior.

34.11 The Complete Energy Budget: Work, Storage, and Transfer

The swing connects a chemical energy source to mechanical work, changing body and club energy, and a collision with the ball. Those are different accounting boundaries. A ratio measured at one boundary cannot be interpreted as efficiency at another without the intervening balances. In particular, club work divided by reconstructed body joint work does not measure the fraction of metabolic energy converted to useful motion.

Define the Mechanical System and Interval

Before contact with the ball, consider the golfer and club together. Use disjoint body segments, a declared inertial frame, and the same start and end times for all quantities. Include gravitational potential energy and whatever recoverable elastic energy the chosen model resolves. Its total mechanical energy is

$$E = T_{\text{body}} + T_{\text{club}} + U_g + U_e.$$

For ideal internal constraints, a consistent balance is

$$E_f - E_i = W_{\text{act}} + W_{\text{ext}} - D.$$

Here W_{act} is signed mechanical actuator work delivered to the modeled coordinates; W_{ext} is signed external work not already represented by the included potentials; and $D \geq 0$ is the model's separately represented passive dissipation. Do not count the same loss in both negative actuator work and D . Gravity is already in U_g , so it must not also be included in W_{ext} . A moving prescribed boundary or an inverse-dynamics residual actuator contributes work and must be recorded rather than silently omitted.

This equation is not a metabolic balance. A net joint actuator can summarize unresolved muscles and tissues, but joint motion and net moment alone do not determine individual muscle work, activation cost or heat. Such quantities require additional measurements and a physiological model.

Follow Energy Without Assuming a Universal Phase Pattern

During the backswing, positive actuator work can increase kinetic, gravitational and elastic energy while losses and negative work occur elsewhere. Raising a center of mass changes its gravitational energy by $mg\Delta h$; the amount must be calculated for the actual system. A turn angle alone does not identify tissue elastic energy, which depends on deformation, loading history and the constitutive model.

The top of the backswing is an event defined from a chosen observable, not simultaneous rest of every segment. Body, club and elastic energies need not attain their maxima together. Through transition and downswing, active work continues alongside conversion of previously stored energy. Whether gravity or a particular elastic element supplies substantial power is a result to calculate, not a consequence of labeling the phase as drift-dominated.

At a joint, forces can redistribute energy between segments. For the club alone, power supplied by the hands includes both the dot product of force with its application-point velocity and the dot product of couple with angular velocity. That grip power is internal to the combined golfer–club system. Ideal no-slip contact with stationary ground can transmit force and angular momentum while doing zero contact work; ground reaction force is therefore not itself a dissipative mechanism. Deforming or slipping contact requires its actual power balance.

What the Reported Work Ratio Measures

Nesbit and Serrano analyzed four amateur golfers using reconstructed body and club models. Their reported swing efficiency is club work divided by total body joint work, with values of 20.2–26.8% in Table 3 [Nesbit and Serrano, 2005]. Neither denominator nor numerator

is metabolic energy or ball energy. The remaining fraction cannot be assigned to heat, and these four cases do not establish an elite-versus-amateur metabolic advantage.

When reporting a new work ratio, state whether its denominator sums signed work, positive work only, or the absolute values of work. For actuator powers P_j , define positive work and the magnitude of absorbed work by

$$W_+ = \sum_j \int \max(P_j, 0) dt, \quad W_- = \sum_j \int \max(-P_j, 0) dt.$$

Then $W_{\text{act}} = W_+ - W_-$. This separation does not resolve individual muscle contributions when P_j represents a net joint actuator.

For an illustrative mechanical budget, let $E_i = 100$ J, $W_+ = 500$ J, $W_- = 80$ J, $W_{\text{ext}} = 0$ and $D = 20$ J. The final energy is 500 J. If 250 J is club kinetic energy and 250 J remains in the other included energy stores, club energy divided by positive actuator work is 0.50; divided by signed actuator work it is approximately 0.595. These ratios use club energy, not the cited paper's club-work numerator. Both are compatible with the same balance, and neither determines metabolic efficiency. Initial storage also prevents treating an arbitrary output-to-new-work ratio as necessarily bounded by one.

Separate Impact and Follow-Through

To include impact, enlarge the system to contain the ball and resolve an interval spanning contact. Include initial and final ball, club and body energies, work across the external boundary, and dissipative terms. Energy left in the moving club is retained energy, not collision heat. Ball translation, spin, deformation and subsequent aerodynamic flight are distinct; ball kinetic energy alone does not determine carry distance.

During follow-through, actuator absorption, passive losses and external work alter the remaining mechanical energy. Recoverable deformation and changed gravitational potential may remain even when the golfer stops. Do not assert that every joule unaccounted for at impact has become heat. Check energy closure and identify the model's residual before interpreting a loss mechanism.

Connect the Budget to Drift and Control

The drift field describes evolution under a specified zero-input convention. It is not an energy source, and a large drift contribution does not establish low metabolic cost. Stored energy may have required earlier active work; zero net joint torque can coexist with muscle activity; and a different input convention changes what is called drift. A useful comparison holds the task and system boundary explicit, reports initial storage and signed work, and measures the resulting club state. This connects preparation, timing, geometry and active effort without assuming that reducing muscle work must increase ball energy.

34.12 From Phase Inputs to Impact Sensitivity

An endpoint sensitivity connects the whole trajectory, not just the norm of a vector field at one instant. For a nominal smooth-mode trajectory of $\dot{x} = F(x, u)$, let $A(t) = F_x$ and

$B_x(t) = F_u$ evaluated along that trajectory. The first variation satisfies

$$\delta\dot{x} = A(t)\delta x + B_x(t)\delta u, \quad \delta x(T) = \Phi(T, t_0)\delta x_0 + \int_{t_0}^T \Phi(T, s)B_x(s)\delta u(s) ds. \quad (34.17)$$

The transition matrix obeys $\partial_t \Phi(t, s) = A(t)\Phi(t, s)$ and $\Phi(s, s) = I$. For the torque-affine model, F_x includes derivatives of $G(x)u$ as well as f_x . For an output $y = h(x(T))$, left-multiply by h_x to obtain the delivered-output sensitivity. The kernel $h_x\Phi(T, s)B_x(s)$ includes later amplification, rotation, damping and task projection.

Equal input changes at different times need not have equal effects. For a torque-driven free rotor, a fixed torque impulse changes final velocity equally wherever applied, while earlier application has more time to change angle. In the stable scalar mode $\dot{x} = -10x + u$ over $0 \leq t \leq 1$, equal-duration unit pulses in the final versus initial 0.1s have endpoint-effect ratio e^9 , approximately 8103. This counterexample does not model golf; it disproves a universal rule that early inputs must always have greater endpoint influence.

Impact is usually an event rather than a prescribed clock time. Let the contact condition be $g(x, t) = 0$ at nominal T , with a transverse crossing $g_t + g_x F^- \neq 0$. The first-order event-time shift and pre-impact output change are

$$\begin{aligned} \delta T &= -\frac{g_x \delta x(T)}{g_t + g_x F^-}, \\ \delta y^- &= h_x \delta x(T) + (h_t + h_x F^-) \delta T. \end{aligned} \quad (34.18)$$

Parameter perturbations add their direct derivatives. A post-impact comparison must additionally differentiate the reset/contact solution. Near grazing contact the denominator can become small, and a perturbation may remove impact or change the contact mode entirely; one smooth derivative then ceases to describe the comparison.

For a simple crossing $p(t) = p_0 + vt = L$, perturbing initial position and speed gives $\delta T = -(\delta p_0 + T\delta v)/v$. Position at the event remains L even though position at the original clock time changes. Thus fixed-time and event-matched errors can lead to different conclusions about the same perturbation. Report which one is being used.

Observability is separate: do the chosen measurements distinguish nearby hidden states or parameters? One measured impact error cannot generally be uniquely decomposed into address, backswing and transition errors. Many input histories map to the same small set of outputs. A sensitivity matrix can predict effects of specified perturbations; inversion needs rank, conditioning, measurements over time and additional assumptions. A regularized inverse selects one explanation according to its penalty, not necessarily the unique physical cause.

34.13 What a Forward Zero-Torque Counterfactual Can Establish

Initialize the measured/model-consistent state at a declared event, set the selected applied torque channel to zero, and evolve the model forward with its retained gravity, tissue, shaft and contact dynamics. Declare what happens to activation, prescribed boundaries, feedback, grip constraints and contact transitions. Compare actual and zero-channel trajectories in a named output, metric and interval, including whether each reaches the same impact event.

The branch tells us what this specified model predicts under that intervention. It does not recover an observed passive trajectory, identify muscle silence or demonstrate that a player should relinquish control. Existing stored energy may have been created through earlier muscle work. Changing the control channel or retaining an active support law changes the meaning of “zero.” The state at which the intervention begins matters just as much as the torque being removed.

A pointwise ratio alone does not guarantee close trajectories. If the zero-input field is Lipschitz with constant L in a common valid region, and the removed input contribution has norm bounded by b there, equal initial states obey the sufficient bound

$$\|x_u(t) - x_0(t)\| \leq \begin{cases} b[e^{L(t-t_0)} - 1]/L, & L > 0, \\ b(t - t_0), & L = 0. \end{cases} \quad (34.19)$$

This follows by bounding the difference equation and integrating the resulting scalar inequality. It needs an absolute input bound, a growth bound and a time interval. A large ratio caused by an even larger unrelated drift component provides none of those guarantees. Contact changes require a mode-aware analysis rather than silently extending this smooth bound through a collision.

To investigate sequencing, compare the complete, zero-channel and carefully specified modified-input trajectories. Record whether a velocity-peak order persists, reverses or disappears, and examine the forces, work and state changes that accompany it. Persistence supports a statement about the chosen model and initial state. It does not prove that the measured player used the same intervention or that other muscle commands were unnecessary.

34.14 A Reproducible Complete-Swing Investigation

Begin with a concrete question: for example, whether changing a transition command improves face-orientation consistency at a specified speed, or whether a shaft change alters delivery after allowing the golfer model to respond. Define the target and tolerances before selecting the comparison. Speed, accuracy, load, robustness and effort can conflict; a claim of “better” needs an objective and feasible constraints.

1. **Specify the Model and Measurements.** Record segment geometry and inertias, joint definitions, body/club frames, shaft modes, muscle states and contact laws. Synchronize kinematics, ground wrenches and club/ball observations. Distinguish measured data, reconstructed quantities and fitted parameters.
2. **Define Events and Outputs.** State the top/transition definitions, impact detection, follow-through endpoint and whether comparisons use clock time or matched events. Report speed, face orientation, path, strike location and uncertainty without substituting one for the full delivery state.
3. **Close the Mechanical Accounts.** Check constraint residuals, contact feasibility, momentum balance and the energy budget. Identify residual actuators and moving boundaries. Reproducing motion used to drive an inverse model is not an independent forward-prediction test.
4. **Run Declared Interventions.** Change one stated parameter or input schedule from a common initial state, or explicitly reoptimize under the same objective and con-

straints. A frozen-kinematics shaft comparison and a dynamically responding golfer comparison estimate different effects.

5. **Test Sensitivity and Robustness.** Perturb plausible states, parameters and measurement errors; compare finite differences with the variational prediction where it applies. Resolve contact changes explicitly. A nominally accurate trajectory can be sensitive to timing or uncertainty, and a smaller net torque does not establish greater robustness.
6. **Separate Mechanism From Evidence.** Report whether a conclusion is an identity, an illustrative calculation, a model-specific prediction, an observational association or an independently tested intervention. Preserve failures and alternative explanations. Extending a golf model to humanoid tasks requires rechecking actuation, contacts and task constraints.

The work of Nesbit provides an instructive measurement/model example: motion reconstruction, flexible-club dynamics and work analysis reveal individual differences, while imposed motion and fitted ground support limit causal interpretations [Nesbit, 2005]. A new complete-swing investigation should state which observations were used for fitting and which were held out for validation. Evidence for one model's joint loads is not automatically evidence for individual muscle forces or a neural policy.

The resulting practical insight is specific: preparation changes the state from which later forces act; geometry determines their directions and coupling; activation and tissue dynamics constrain the available response; shaft state changes delivery; timing changes the contact event; and the collision and flight models translate delivery into ball outcome. These connections justify investigating the whole swing. They do not establish a universal passive sequence or a theorem that trying less must produce more speed.

Key Principle 34.1. Summary of Analysis

The swing is a coupled, constrained, sometimes hybrid evolution of a complete mechanical and physiological state. Drift/control labels depend on the declared input model. A drift norm does not replace directional gain, finite-time sensitivity, contact feasibility or energy accounting. Segment sequencing and shaft motion must be explained with forces and evolving state, while their causal interpretation needs explicit interventions and appropriate evidence. A useful reference connects preparation to delivery through calculations that can be checked, and leaves player-specific performance claims open to measurement.

Exercises 34.1. Chapter Exercises

1. Derive the reduced acceleration and reaction formulas from the constrained equations. For the stated two-coordinate example, calculate the response to $(10, 0)^T$ and $(1, 1)^T$. Explain why changing velocity-dependent bias alone does not change the input derivative.
2. Define an available-input DCR using a metric and admissible set. Distinguish it from an actual-input ratio. Use $\dot{x}_1 = 1000$, $\dot{x}_2 = u$ to explain why neither ratio determines authority over every output. Give conditions and measurements

needed to establish a rising phase trend.

3. For a smooth state-dependent club-velocity output $h(x)$, integrate $\dot{h} = h_x[f(x) + G(x)u]$ along the actual trajectory. Explain why these same-trajectory integrals are not the velocity difference between two independently evolved swings and are not energy fractions.
4. Derive the first variation from address through impact, including the event-time correction. Identify the extra initial states omitted if address is represented only by joint angles. Check the ballistic-crossing example.
5. Compare early and late equal-duration input pulses for a free rotor and for $\dot{x} = -10x + u$. State the output for each comparison and compute the damped-mode ratio. Explain why a desire to add 10mph does not identify the best swing phase without a model, constraints and objective.
6. Use a two-coordinate inertia matrix to show how proximal input can accelerate a distal coordinate in either sign. Specify a forward experiment that tests whether a particular peak-velocity sequence persists after an input channel is removed. Explain what that result would not establish about muscles.
7. A player has rightward ball curvature or a measured face-angle error. Form a local input-to-delivery sensitivity matrix and explain why one endpoint observation cannot uniquely allocate the error among address, backswing and transition. Identify additional measurements or interventions that would distinguish candidate explanations.
8. For the harmonic mode, find the times of maximum displacement magnitude, velocity magnitude and positive elastic power. Explain why simultaneous maximum velocity and restoring force is impossible in this example and why a driven shaft can depart from its timing.
9. Compare stored spring energy under prescribed displacement and prescribed force. Then distinguish a steel/graphite material change from a controlled change in bending stiffness. Specify fixed-input and reoptimized golfer-shaft comparisons and explain why their outcomes can differ.
10. Derive the three central-collision energy fractions and reproduce the illustrative speeds. Explain why $1 - e$ is not a loss fraction, and list the additional information required for off-center impact and spin.
11. Use total linear and angular momentum balance to explain how stationary ground contact can change motion while doing zero work. Distinguish an external support moment, muscle torque and a dissipative contact force. State how to test a claim about the phase of peak ground reaction.

12. Construct a follow-through energy account with actuator absorption, passive dissipation and changing storage. Explain why ideal grip/joint reactions can redistribute subsystem energy without absorbing total energy, and why net negative joint power does not identify every muscle contraction.
13. Compare high and low late muscle activation while requiring the same delivery and feasible support. Identify measurements needed to compare mechanical work, metabolic cost and robustness. Explain why neither control magnitude nor DCR alone proves one strategy faster, cheaper or more consistent.
14. Derive the stated equal-initial-state counterfactual error bound. Explain its dependence on absolute input magnitude, state-transition growth and duration, and identify what fails when one branch misses impact or changes contact mode.
15. Formulate an inverse design problem for a desired delivery state, with initial conditions, torque/activation constraints, contact feasibility and shaft parameters. Explain why multiple solutions can exist and what additional cost and validation would be needed to choose among them.

Glossary

Reading Across Definitions

The glossary connects three questions: **what moves, what supplies or transfers energy, and what the golfer can change**. These questions share a mechanical model, but they require different evidence.

For example, a changing forearm-club angle is kinematics. A bending shaft adds deformation and possible elastic storage. A joint moment combined with the matching angular velocity gives net joint power. None of those observations alone identifies individual muscle forces or the nervous system's command. Read **Lag**, **Wrist cock**, **Shaft lag**, **Power transfer** and **Inverse dynamics** together when investigating release.

The same distinction matters for ground contact. Foot forces affect whole-body momentum and the feasible motion of a connected system. In an ideal stationary no-slip model, a contact point has zero velocity, so its contact force supplies no power there even though it changes momentum. A force can redirect motion, transmit load or enable muscle work elsewhere without itself adding mechanical energy. Read **Ground reaction force**, **Constraint forces**, **Linear momentum** and **Work** together.

Finally, a zero-input calculation is a model intervention. It retains the declared state, constraints and loads; it does not erase earlier muscle work, stored energy or activation states. Read **Affine decomposition**, **Drift**, **Control authority**, **DCR** and the **ZTCF family** together. The useful research question is which measured outcomes a declared mechanism explains, and what observation would distinguish it from an alternative.

A Collision Check

COR illustrates why definitions must be used together. Assume two translating bodies collide along one normal, external impulse is negligible, the ball starts at rest, and rotation and tangential impulse are excluded. Let m_c and m_b be club and ball masses, $V > 0$ the incident club speed, and superscript $+$ denote the post-impact value. Momentum conservation and the COR definition give

$$m_c V = m_c v_c^+ + m_b v_b^+, \quad v_b^+ - v_c^+ = eV,$$

and therefore

$$v_b^+ = \frac{(1+e)m_c}{m_c + m_b} V, \quad \Delta T = -\frac{1}{2} \frac{m_c m_b}{m_c + m_b} (1-e^2) V^2.$$

Thus the ball can depart faster than the incident club even when $e < 1$. The loss ΔT is the change in the two bodies' total translational kinetic energy; it is not the fraction delivered to the ball. In an off-center three-dimensional impact, rotational inertia, contact direction and friction enter the calculation. This simple derivation explains the vocabulary, not a universal equipment test or launch prediction.

Evidence and Conventions

Unless stated otherwise, mechanical vectors use a declared ground-based frame approximated as inertial over the swing. Anatomical motion uses local segment axes. Lead and trail refer to the golfer's target-side and opposite-side limbs.

The wrist entries draw on a study that measured both wrists in local coordinates under different grips [Carson et al., 2019]. The tissue entries distinguish specimen-level mechanical evidence from whole-swing inference [Bonaldi et al., 2023]. The reaction-time example comes from an object-manipulation experiment, not a golf trial [Crevecoeur et al., 2016]. Pain terminology follows IASP's distinction between experience, nociception and tissue damage [International Association for the Study of Pain, 2020]. Equipment and stroke definitions refer to the governing USGA resources rather than treating a glossary as a complete rule book [United States Golf Association, 2026a,b].

A

Acceleration The time derivative of velocity in a declared reference frame. For a clubhead point, acceleration includes changes in both speed and direction; a large acceleration does not by itself imply large mechanical power.

Affine control system A model of the form $\dot{\mathbf{x}} = f(\mathbf{x}) + G(\mathbf{x})\mathbf{u}$. At a fixed state, changes in the declared input produce linear changes in the state derivative; the total map includes the offset f . Affinity in generalized torque does not establish affinity in neural excitation or a finite swing trajectory.

Affine decomposition The split into a zero-declared-input vector field and an input-dependent increment. It depends on the chosen actuator port and retained plant state. Elastic tension, damping, contact reactions and previously established activation may survive the intervention; the split does not identify biological passivity.

Angular acceleration The time derivative of angular velocity in a declared frame. For two points on a rigid body, $\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$. Here $\mathbf{r}$ points from A to B . The angular-acceleration term and the angular-speed-squared term are distinct.

Angular velocity The axial vector describing instantaneous rigid-body rotation. A body's points share angular velocity but generally have different linear velocities. For three-dimensional orientations it is not generally the vector of Euler-angle derivatives.

B

Backswing The movement taking the club away from the ball toward the transition. It establishes configuration, velocities and tissue/shaft states. Raising mass can increase gravitational potential energy, but the amount and location of elastic storage require force-deformation evidence.

Ball compression A rating derived from deformation under a specified compression test. It is not a universal material modulus, cover hardness or impact COR. Dynamic response also depends on loading rate, temperature, construction and the test protocol.

Beam theory Models relating distributed loading, deformation and internal forces in slender structures. The appropriate shaft model depends on bending, torsion, shear, rotary inertia, geometric nonlinearity and boundary motion; Euler-Bernoulli theory is one approximation.

C

Catapult effect An informal description of elastic shaft recoil contributing to clubhead motion. Recoverable energy was stored through earlier work. Its contribution depends on loading and timing; a straight shaft at impact is neither a general optimum nor proof that stored energy has all become ball energy.

Center of mass The mass-weighted position $\mathbf{r}_C = \sum_i m_i \mathbf{r}_i / \sum_i m_i$. It differs from the center of pressure under the feet and from the relative vertical loads on two force plates.

Centrifugal acceleration An apparent acceleration $-\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$ introduced in a rotating frame. For fixed-axis circular motion, inertial acceleration is inward, of magnitude $\omega^2 r_\perp$. Real shaft forces must be found from the complete force balance; the frame term is not an additional energy source.

Centrifugal stiffening A change in a rotating structure's transverse dynamics caused by rotation-induced axial tension. The resulting geometric stiffness depends on the configuration, loading and supports. This is not a change in the shaft material's elastic modulus, nor a universal scalar stiffness increase for every mode.

Cerebellum A brain structure involved in coordination, timing and sensorimotor adaptation. Internal-model descriptions are hypotheses about computation; they do not establish that the cerebellum explicitly learns AffineDrift's chosen mathematical drift field.

Clubhead The club's striking body. Its velocity, orientation, angular velocity, mass distribution, impact location and deformation interact with the ball and contact conditions to determine launch.

Clubhead speed The magnitude of velocity of a declared clubhead point at a declared time, commonly just before impact. Specify the point, sensor and measurement convention. Distance also depends on strike, ball launch and flight conditions; population averages are not individual targets.

Coefficient of restitution (COR) For a one-dimensional normal collision, $e = (v_b^+ - v_c^+) / (v_c^- - v_b^-)$, with signed velocities along the same contact normal and a positive approach denominator. Here c and b denote club and ball; superscripts $-$ and $+$ mean before and after impact. It compares relative separation and approach speeds, not ball speed to incident club speed. Neither a universal golf-ball COR of 0.83 nor an 83 percent energy-transfer fraction follows. Club spring-effect conformity uses governing equipment test procedures [United States Golf Association, 2026a].

Collagen A family of structural proteins important in the extracellular matrix. Tissue behavior depends on collagen type, arrangement, cross-linking and interaction with

other constituents; collagen content alone does not determine stiffness or recoverable energy.

Composite material A material combining distinct constituents, such as carbon fibers in a polymer matrix. Fiber orientation, layup, resin and geometry affect a shaft's directional bending and torsional response.

Connective tissue A diverse family of tissues containing cells and extracellular matrix; examples include bone, cartilage, tendon, ligament and fascia. Their constituents and mechanical functions differ, so they cannot be assigned one stiffness, strength or elastic-storage capacity.

Constraint A modeled restriction on configuration, velocity or contact state. Joint geometry, two-hand grip closure and foot contact impose different restrictions. A real contact may slip, separate or deform, invalidating a permanently rigid constraint assumption.

Constraint forces Reactions enforcing a declared constraint model. For ideal stationary bilateral constraints they do no total work on admissible motion, although they can transfer power between bodies. Moving supports, friction and deformable contacts require their own power balances.

Constraint-force method A dynamics formulation solving accelerations and reaction multipliers together. With J the constraint Jacobian, $M\ddot{\mathbf{q}} = \mathbf{r} + B\mathbf{u} + J^T\boldsymbol{\lambda}$ is supplemented by the differentiated constraints; $\mathbf{r}$ collects retained loads and velocity-dependent terms. Reactions generally change when applied input changes.

Control affine See affine control system.

Control authority The effect attainable with the admissible input set. Instantaneous acceleration authority is a set such as $a_0 + HU$; finite-time authority additionally depends on dynamics, time and constraints. It may vary by direction and posture. Faster motion does not itself increase the mass matrix or prove loss of authority.

Control input The declared external command or effort in a model, such as joint torque, motor voltage or neural excitation. These are different ports with different internal dynamics. A net joint torque is not an individual muscle force.

Coriolis force In a rotating frame, the apparent force $-2m\boldsymbol{\omega} \times \mathbf{v}_{\text{rel}}$. The relative velocity and frame rotation must be specified. Velocity-coupling terms in generalized-coordinate equations need not correspond individually to a torso-frame force.

Cortical bone The relatively dense outer bone tissue, distinguished from the porous trabecular architecture. Material modulus, apparent structural stiffness and whole-bone strength are different quantities and depend on direction and geometry.

D

DCR Abbreviation for Drift-Control Ratio. DCR compares modeled drift with bounded modeled control in the same acceleration or task-projected space. A reproducible value declares the weighting matrix and norm, admissible control set, regularizer, effective plant, and state. A ratio using realized input rather than available control capacity is a different, policy-dependent diagnostic.

- Degrees of freedom** The number of independent local configuration coordinates after accounting for constraints. A free spatial rigid body has six. Counts for an arm or whole golfer depend on modeled joints, flexible modes and contacts; redundant orientation coordinates are not additional physical freedoms.
- Dorsiflexion** At the ankle, movement bringing the dorsum of the foot toward the shin. The term is also used for wrist extension. Neither use defines the forearm-shaft angle called wrist cock; wrist deviation is a separate motion.
- Downswing** The movement from a declared transition event to ball contact. Segment reversals need not occur simultaneously. Duration is measured for the particular swing, club and event definition rather than fixed at 0.2 seconds.
- Drift** The state derivative when the declared control input is zero, with the effective plant, internal state and load protocol retained. Gravity and velocity-dependent, elastic or dissipative effects may contribute. Late-downswing dominance is a model-dependent quantitative claim requiring an explicit comparison.
- Drift field** The vector field $f(\mathbf{x})$ in a declared affine model. Its value depends on the state and retained loads. A sequence of pointwise zero-input evaluations along an actuated swing is generally not a trajectory generated by that field.
- Drift-Control Ratio (DCR)** See DCR.
- Dynamics** The study of how systems evolve over time under the influence of forces. The dynamics of the golf swing are governed by Newton's laws and the constraints of the body.

E

- Eccentric contraction** Active force production while the specified muscle or fascicle lengthens. It can absorb mechanical work, but fascicle, tendon and whole muscle-tendon length changes may differ. It is not confined to follow-through.
- Effective mass** Directional inertia seen at a chosen task point. For an unconstrained positive-definite M , task Jacobian J and unit direction $\mathbf{n}$, $m_{\text{eff}}^{-1} = \mathbf{n}^T J M^{-1} J^T \mathbf{n}$. Constraints change the mobility operator; a zero directional mobility has no finite effective mass in this sense.
- Elastic energy** Recoverable energy associated with deformation. A linear spring stores $U = \frac{1}{2} k \delta^2$; a bending beam stores $\frac{1}{2} \int EI \kappa^2 ds$ in the corresponding small-strain model. Viscoelastic hysteresis and continuing external work affect how much reaches another body.
- Elastic modulus** A constitutive measure relating stress and strain under specified conditions. Young's modulus E describes uniaxial elastic response; anisotropic and nonlinear tissues require directional or tangent measures. It differs from a whole structure's force-displacement stiffness.
- Elasticity** Reversible deformation on unloading within a material's elastic regime. A material can exhibit both elastic response and irreversible plastic deformation; viscoelastic recovery may also depend on time.

Elastin An extracellular-matrix protein contributing to reversible deformation. Its abundance varies by tissue and sampling method; a single percentage cannot characterize all fascia.

Ellipsoid A surface defined by a positive-definite quadratic form, with a filled ellipsoid including its interior. Higher-dimensional ellipsoids can approximate uncertainty or reachable sets; general nonlinear reachable sets need not be ellipsoidal.

Endurance The ability to sustain effort over time. Golf endurance depends on cardiovascular fitness, muscle endurance, and mental focus.

Epimysium The connective-tissue sheath around a whole skeletal muscle, continuous with internal connective-tissue networks and attachment structures. Force can be transmitted through several paths; the sheath is not an isolated spring with a universal stiffness.

Equations of motion Mathematical equations describing how a system evolves over time, typically derived from Newton's laws or Lagrangian mechanics. The equations of motion for the golf swing are nonlinear and high-dimensional.

Equilibrium A state at which the declared state derivative vanishes for specified inputs and loads. Static force and moment balance is one mechanical case. Ongoing rotation or translation can instead be a relative equilibrium in an appropriate frame.

Euler-Bernoulli beam theory A slender-beam approximation that neglects transverse shear deformation and normally rotary inertia. Bending moment and curvature satisfy $M_b = EI\kappa$ with a consistent sign convention. Shaft taper, composite anisotropy, torsion and large motion may require extensions.

Extension An anatomical joint motion defined relative to local segment axes. At the wrist, the hand moves dorsally relative to the forearm. It is distinct from radial/ulnar deviation and cannot alone define cock or release.

F

Facet joint A synovial articulation between adjacent vertebral articular processes, also called a zygapophysial joint. It contributes to spinal guidance and load sharing.

Fascia Connective-tissue sheets and networks with region-specific architecture. Mechanical tests of fascia lata demonstrate direction- and time-dependent response; those results do not establish a universal fascia-versus-tendon ranking or whole-swing energy contribution [Bonaldi et al., 2023].

Fatigue An activity-related reduction in the capacity to produce force or power. Its measurement depends on the task and protocol; altered swing performance can have muscular, neural and other contributors.

Feedback Information about the current state or outcome that is fed back to the system for control or learning. Proprioceptive feedback informs the nervous system of body position; visual feedback informs the golfer of ball position.

Feedback control Control that uses information about measured or estimated state or outcome. The useful correction depends on sensory delay, processing, actuation and time remaining. A late swing can still be influenced by feedback to earlier events.

Feedforward control Control based on a reference or prediction without waiting for the resulting error. It can coexist with feedback throughout a movement; feedforward does not mean biologically passive.

First mode The lowest-frequency mode within the declared class and boundary conditions, such as the first elastic bending mode after excluding rigid-body modes. Whether it dominates a shaft response depends on forcing, damping, supports and the measured output.

Flexibility In a scalar linear structural model, compliance is displacement per applied force and is the reciprocal of stiffness. Anatomical flexibility instead describes range of motion under a protocol; it is not generally one material constant.

Flexion An anatomical bending motion defined for a joint. Wrist palmar flexion moves the palm toward the anterior forearm; it is a different axis from radial/ulnar deviation.

Flight Ball motion through air after separation from the club. Initial position, velocity and spin combine with gravity, aerodynamic coefficients, air properties and wind to determine the trajectory.

Follow-through The phase after impact where the golfer decelerates the club and body, coming to rest. The follow-through is important for balance and controlled deceleration.

Force A vector interaction, measured in newtons (N). For a fixed-mass body in an inertial frame, resultant external force equals mass times center-of-mass acceleration. Forces at different points can also create different moments.

Force plate An instrument measuring resultant contact forces and moments in its calibrated frame. Center of pressure is derived under declared contact-plane assumptions and becomes ill-conditioned at low normal force. It is not a direct measurement of individual muscle forces.

Forward dynamics Solving for instantaneous accelerations or the state derivative from state, input and loads, including constraint reactions when needed. Numerical integration repeatedly uses that result to predict a future state.

Forward model A model predicting motion or sensory consequences from state and command. In neuroscience it is a proposed internal computation; a useful engineering predictor is not proof of a particular neural representation.

Frame A reference origin and orientation for describing quantities. A ground-fixed frame is approximately inertial over a golf swing; Earth's rotation means that it is not exactly inertial.

Frequency Cycles per unit time, in hertz (Hz). Angular frequency is $\omega = 2\pi f$. A fixture rating of 240–280 cycles per minute corresponds to 4.00–4.67 Hz, but does not specify the modes of a moving, gripped club.

Friction Contact interaction opposing relative slip or supporting a no-slip state. Static friction can be essential at the shoes and grip even when there is little relative motion. Its magnitude cannot be dismissed by comparison with other forces without a contact analysis.

G

Generalized coordinate An independent local variable describing configuration, such as a joint angle or flexible-mode amplitude. A chosen coordinate chart and constraint reduction determine its meaning.

Generalized force A quantity dual to a generalized displacement, defined through virtual work $\delta W = \mathbf{Q}^T \delta \mathbf{q}$. A rotational coordinate has a torque-like conjugate, but its axis and scaling must be declared.

Generalized momentum The conjugate quantity $p_i = \partial L / \partial \dot{q}_i$. Euler-Lagrange dynamics gives $\dot{p}_i = \partial L / \partial q_i + Q_i^{\text{nc}}$. Conservation follows for a cyclic coordinate with zero nonconservative generalized force; zero applied force alone is insufficient.

Geometry of motion The study of how systems move in space, considering constraints and symmetries. This is the subject of a companion series to this textbook.

Glute maximus The gluteus maximus, contributing to hip extension and external rotation. Its swing contribution depends on activation, posture, loading and interactions with other muscles.

Glute medius The gluteus medius, a hip abductor with region- and posture-dependent rotational actions. Its anatomical role does not specify its activation timing in an individual swing.

Golgi organ A Golgi tendon organ, a receptor supplying sensory information related to muscle-tendon tension. Its signal is part of a neural circuit, not a direct conscious readout of net joint torque.

Gravitational potential energy For a nearly uniform field, $U_g = \sum_i m_i g h_i$ relative to chosen zero heights. Raising one segment increases its contribution, but the whole golfer-club change requires all segment heights.

Gravity The interaction between masses. Near Earth's surface, gravitational acceleration is approximately 9.81 m/s^2 downward; weight is the associated force, not the acceleration itself.

Green The smooth, manicured grass surface on which the hole is cut. Golfers aim to land the ball on the green, from which they putt.

Grip The hand-club contact arrangement or the club's handle covering. A rigid two-hand grip creates loop constraints in a model; real grip compliance and slip may matter for force transmission.

Ground contact Interaction between shoes and the ground. Normal reactions, friction and moments support and accelerate the body subject to unilateral contact and slip limits. They do not invariably propel the golfer forward.

Ground reaction force The resultant ground force on the feet. Together with other external forces it governs whole-system momentum change. Its components, peaks and timing must be measured; weight alone does not determine them.

H

Hamstring A member of the posterior-thigh muscle group. Most cross the hip and knee and contribute to hip extension and knee flexion; the short head of biceps femoris does not cross the hip.

Hang time The interval between ball launch and first landing. It depends jointly on launch speed and direction, spin, aerodynamic properties, wind and relative landing height.

Heading An orientation or travel direction in a declared horizontal reference. Body facing direction, target direction and ball velocity direction are distinct and should not share one unlabeled angle.

Heel strike The onset of heel-ground contact, commonly used in gait analysis. A golf heel-contact event must be identified from the particular foot and swing rather than assigned to a universal phase.

Hip rotation Often used in coaching for pelvis turning, but anatomically refers to femur rotation relative to the pelvis. Pelvis orientation in the laboratory frame is a different measurement.

Holonomic constraint A restriction expressible as $\phi(\mathbf{q}, t) = 0$. It also restricts velocity through $J\dot{\mathbf{q}} + \phi_t = 0$ and acceleration through a further derivative. A joint or contact is holonomic only under the declared model.

Hook Pronounced flight curvature toward the golfer's lead side: left for a right-handed golfer. For centered contact, face-to-path geometry helps determine spin-axis tilt; off-center gear effect and aerodynamic conditions also matter.

Horizontal plane A plane perpendicular to local gravity. A sloping ground surface is not horizontal, and a golf swing generally cannot be represented by one fixed plane.

Horsepower A family of power units. Mechanical horsepower is approximately 745.7 watts; metric horsepower is approximately 735.5 watts. Use watts for an unambiguous mechanical power balance.

Hydration Water content or fluid status, with the tissue and measurement basis specified. It can influence tissue mechanics, but there is no single water fraction that characterizes all fascia.

Hypertrophy An increase in tissue or cell size; muscle hypertrophy increases muscle cross-sectional area. Strength also depends on neural drive, architecture, leverage and the testing task.

I

Inelastic collision A collision in which the bodies' modeled macroscopic kinetic energy is not conserved. Energy is redistributed into deformation, vibration, heat and other modes; it is not destroyed.

Inertia The mass-distribution properties relating momentum to velocity and force/moment to acceleration. In a rigid-body model, configuration changes can alter generalized inertia; speed alone does not increase mass.

Inertial frame A frame in which free particles move at constant velocity and Newton's equations need no frame-inertia terms. A ground-fixed frame is a useful approximation over the time and distance scales of a swing.

Infraspinatus A rotator cuff muscle contributing to external rotation and glenohumeral stabilization. Its activation must be measured rather than inferred from a named swing phase.

Injury prevention Actions intended to reduce injury risk. Mechanical analysis may identify loads and candidate mechanisms, but a claim of reduced injuries requires appropriate outcome evidence.

Input-output behavior The relationship between a declared input history and measured output, including initial state and disturbances. Torque-to-club motion and neural-excitation-to-ball outcome are different systems.

Intermuscular coordination The organization of activity across muscles. Coactivation can affect both net moments and stiffness; a small net moment does not imply small individual muscle forces.

Intervertebral disc A structure between adjacent vertebral bodies consisting of an annulus fibrosus and nucleus pulposus with associated endplates. It transmits load and permits motion; pain cannot be inferred from a rotation measurement alone.

Intramuscular coordination The organization of motor-unit recruitment and discharge within a muscle. Force also depends on muscle state and contraction history.

Inverse dynamics Estimating resultant generalized forces from measured motion, inertial properties and external loads. It does not uniquely identify individual muscles, establish neural commands or guarantee that a desired motion is dynamically feasible.

Inverse kinematics Finding configurations consistent with task positions and orientations. Solutions may be multiple, absent or sensitive to constraints. A mathematical inverse-kinematics solution does not prove that the nervous system uses that algorithm.

Isometric contraction Active force production while the specified length or joint position is held fixed. A fixed joint angle does not require every fascicle and tendon to remain the same length. Zero external mechanical work does not imply zero metabolic expenditure [OpenStax, 2022c].

J

Jacobian A derivative matrix connecting coordinate increments or velocities to a declared task representation. For point position, $\dot{\mathbf{r}} = J\dot{\mathbf{q}}$; the dual force mapping is $\mathbf{Q} = J^T\mathbf{F}$. Orientation representations may require additional rate mappings.

Joint An articulation between bones, permitting or restricting relative motion according to its structure. Biomechanical joints may be modeled with fewer freedoms than the real anatomy.

Joint angle A relative segment-orientation measure under declared anatomical axes and rotation conventions. In three dimensions it is not generally the angle between two bone centerlines.

Joint complex A group of articulations contributing to a motion, such as glenohumeral, acromioclavicular and sternoclavicular joints with scapulothoracic motion. A modeling label does not make the complex a single hinge.

K

Kinematic chain Connected bodies and joints defining possible relative motion. A two-hand grip and ground contacts can close loops. Kinematics alone does not establish the direction or amount of energy transfer.

Kinematics The study of motion without considering forces. Kinematics describes where the body is and how fast it is moving.

Kinetic chain A description of interacting segments and the forces and moments transmitted between them. Demonstrating a particular energy-transfer pathway requires segment power balances, not merely an order of peak angular velocities.

Kinetic energy For a rigid body, $T = \frac{1}{2}m\|\mathbf{v}_C\|^2 + \frac{1}{2}\boldsymbol{\omega}^T I_C \boldsymbol{\omega}$, with inertia and angular velocity in the same frame. Clubhead kinetic energy is not all transferred to the ball.

Knee flexion Bending at the knee. Its effect on body height and loading depends on the simultaneous ankle, hip and trunk configuration and the contact forces.

L

Lag A coaching or geometric descriptor, often the angle between a declared forearm direction and shaft direction. Specify the lines and angle convention. The same angle is possible with a rigid shaft; it does not measure shaft strain energy.

Lagrange multiplier A scalar or vector variable enforcing a constraint. With constraint Jacobian J , the generalized reaction is $J^T\boldsymbol{\lambda}$. Multiplier magnitude and units depend on constraint normalization; a multiplier is not automatically a physical contact-force component.

Lagrangian mechanics A variational formulation using the Lagrangian, commonly $L = T - U$, together with nonconservative forces and constraints. It includes forces through generalized work and is equivalent to Newtonian mechanics under the same model assumptions.

Launch angle The vertical angle at which the ball leaves the clubface at impact, measured from horizontal. Launch angle significantly affects carry distance and ball flight.

Launch monitor An instrument estimating launch and sometimes club-delivery variables using radar, cameras or other sensors. Specify which outputs are directly measured, inferred or simulated and the calibration and uncertainty of each.

Lie angle The club's shaft-to-ground geometry under a specified sole/reference condition; dynamic lie describes delivery at impact. Lie can affect the orientation of a lofted face, but it does not by itself determine club path.

Ligament Connective tissue linking bones and contributing to joint restraint and sensory function. Its response varies with site, fiber direction, rate and loading history; it has no universal elasticity ranking relative to tendon or fascia.

Limit of motion The range reached under a specified active or passive measurement protocol. Anatomy, load, pain and muscle state influence that range; an observed endpoint is not a universal tissue-failure threshold.

Limiter Something that limits or constrains the system. In the golf swing, constraints (like the kinematic structure) act as limiters on motion.

Line of pull The local direction of muscle-tendon force at an attachment or wrapping point. It changes with posture and determines moment arms together with the joint axis.

Linear momentum For a fixed collection of bodies, $\mathbf{P} = \sum_i m_i \mathbf{v}_{Ci}$ and $\dot{\mathbf{P}} = \sum \mathbf{F}_{\text{ext}}$ in an inertial frame. Internal forces redistribute momentum but cancel in the total.

Loop closure constraint A relation ensuring that paths through a closed kinematic chain meet consistently. Two hands connected through the arms and torso while holding the same club form a loop in a rigid-grip model.

Lordosis An inward curvature of the spine viewed from the side, normally present in the cervical and lumbar regions. Its magnitude alone neither diagnoses pathology nor determines a golfer's injury risk.

Low back pain Pain in the lower-back region, with potentially multiple biological, psychological and social contributors. It is not synonymous with a particular injured structure or proof that swing rotation caused tissue damage [[International Association for the Study of Pain, 2020](#)].

Lumbar spine The lower spinal region, usually containing five vertebrae. Its motion and loading depend on posture, external forces and muscle activity; net joint moments do not uniquely identify local tissue loads.

M

Magnus effect A transverse aerodynamic force associated with rotation of a body moving through a fluid. For a golf ball, lift and side force depend on spin, relative air velocity and flow conditions; spin-axis tilt helps determine curvature.

Manipulator A mechanical device with joints and segments that move. A robot arm is a manipulator; the human arm is also a manipulator.

Mass A measure of the amount of matter in an object, measured in kilograms (kg). The mass of the clubhead and shaft affects their inertial properties.

Mass matrix The generalized inertia in $T = \frac{1}{2}\dot{\mathbf{q}}^T M(\mathbf{q})\dot{\mathbf{q}}$. It is symmetric positive definite for independent coordinates with positive kinetic energy in every nonzero velocity direction. Dynamics also includes bias forces and constraints; redundant or massless coordinates can invalidate invertibility.

Measurement A quantitative observation with a declared method, units, reference and uncertainty. Distinguish measured signals from quantities inferred through a model.

Mechanical advantage A force or moment ratio for a specified mechanism and configuration. Ideal power balance couples force advantage to a reciprocal velocity tradeoff. Long limbs and high rotation speed do not provide unrestricted force and speed amplification.

Mechanical coupling Interaction through connections, constraints, inertia or material deformation. Coupling can transmit forces and redistribute energy, but its sign and usefulness depend on state and input.

Mechanoreceptor A sensory receptor that detects mechanical stimuli (pressure, stretch, vibration). Mechanoreceptors in fascia and tendons provide proprioceptive feedback.

Medial rotation Rotation toward the body's midline under the relevant anatomical convention, also called internal rotation. Joint-relative rotation must be distinguished from pelvis or trunk rotation in the laboratory frame.

Meridian A term with several meanings. A proposed myofascial anatomical chain and a traditional medicine energy meridian are different concepts. Anatomical continuity does not establish a special energy channel; see myofascial meridian.

Modal analysis Analysis using deformation modes of a declared linearized structure and boundary condition. For undamped vibration, $K\phi = \omega^2 M\phi$. Moving supports, rotation, damping and nonlinear deformation may require a time-varying or expanded model.

Modal coordinate The amplitude multiplying a chosen mode shape. In $w(s, t) = \sum_j \phi_j(s) \eta_j(t)$, rescaling ϕ_j changes η_j and the modal parameters; normalization must be stated.

Modal damping Dissipation represented in modal coordinates. A linear viscous model uses terms proportional to modal velocity; the modal damping matrix is not necessarily diagonal.

Modal decomposition Expansion of deformation in selected modes. Truncation requires checking whether omitted modes affect the forces, motion or energy relevant to the question.

Modal frequency See natural frequency.

Modal stiffness The stiffness associated with a normalized mode or modal basis. For a single mode, $k_j = \phi_j^T K \phi_j$ and $\omega_j^2 = k_j/m_j$ with the matching modal mass.

Mode A characteristic deformation pattern of a specified linearized system. The lowest elastic frequency need not dominate every forced response.

Model A deliberately simplified representation with declared variables, parameters, inputs, constraints and validity limits. A rigid-body tree may need closed-loop contacts, muscle states and flexible shaft modes for a particular golf question.

Moment See torque.

Moment arm The perpendicular distance from a force's line of action to a specified axis in the corresponding planar geometry. More generally, torque is obtained from $\mathbf{r} \times \mathbf{F}$ and projected onto the axis.

Momentum See linear momentum.

Motion Change in position over time. The golf swing involves complex 3D motion of the entire body and club.

Motor control The process by which the nervous system activates muscles to produce desired motion. Motor control involves feedforward and feedback mechanisms.

Motor cortex Cortical regions contributing to movement preparation, execution and feedback within distributed neural circuits. Movement is not generated by an isolated cortical command directly specifying every muscle force.

Motor learning Changes in movement capability through practice and experience. Improved performance can reflect several neural and behavioral processes; it does not alone identify the internal dynamics model learned.

Motor unit A motor neuron and all the muscle fibers it innervates. Motor units are the basic units of muscle activation and force control.

Movement variability Variation across repetitions. Its consequences depend on task sensitivity and coordination: different joint motions can produce similar club delivery. Reducing every component of variability is not a universal objective.

Muscle Biological tissue generating active force. Skeletal muscle can shorten, hold length or lengthen while active; force depends on activation, length, velocity and history.

Muscle fiber A skeletal-muscle cell containing contractile structures. Fibers are organized into fascicles and connected through extracellular matrix; activation is coordinated through motor units.

Muscle spindle A sensory organ whose afferents provide information related to muscle length and length change, modulated by its own neural drive. Its output is not a direct measurement of whole-joint angle.

Myofascial Relating to muscle and its connective-tissue networks. Mechanical continuity permits interaction but does not quantify whole-body force transmission or recoverable energy.

Myofascial meridian A proposed chain of anatomically connected muscles and fascia. Evidence of continuity or force transmission along a particular path does not establish a dominant whole-swing energy pathway; that requires loads, deformation and power measurements.

Myofascial release A label for manual or self-applied soft-tissue techniques. A change in symptoms or range of motion does not by itself demonstrate rupture of adhesions or a specific fascial release mechanism.

Myth An informal label for a disputed belief. Technical criticism should instead identify the precise claim, its evidence and a possible falsifying observation.

N

Natural frequency An eigenfrequency of a specified linearized system. An undamped scalar oscillator has $\omega_n = \sqrt{k/m}$; its frequency in hertz is $\omega_n/(2\pi)$. Damping, shaft supports, grip and attached clubhead alter the measured response, so a fixture rating is not a universal swing frequency.

Natural motion The motion a declared effective plant exhibits after the declared control input is removed. A forward ZTCF trajectory is one precisely specified construction of this motion.

Neuroscience The scientific study of the nervous system, including the brain, spinal cord, and peripheral nerves. Neuroscience provides insight into how the golf swing is controlled.

Newton The SI unit of force, $1\text{ N} = 1\text{ kg m/s}^2$. A one-kilogram mass weighs approximately 9.81 N near Earth's surface.

Newton's laws In classical mechanics, free motion is uniform in an inertial frame, net force changes momentum, and interacting bodies exert equal and opposite forces. For fixed mass, $\mathbf{F}_{\text{ext}} = m\mathbf{a}_C$.

Non-holonomic constraint A velocity restriction that cannot be integrated into a configuration-only restriction on the relevant domain. Some rolling-without-slip models provide examples. Sliding contact does not automatically impose a non-holonomic constraint.

Non-muscular power A source-accounting label that must specify the boundary. Elastic recoil can deliver power while the tissue or shaft loses stored energy, but that energy may have originated in earlier muscle work. The label does not establish energy independent of muscular activity.

Nonlinear A description of a map or dynamical model that does not satisfy the relevant linear superposition properties. An affine input map can coexist with nonlinear state dynamics; a constant offset also makes a map non-linear in the strict algebraic sense.

Norm A nonnegative measure of vector magnitude satisfying definiteness, homogeneity and the triangle inequality. Comparing mixed positions, angles and velocities requires explicit scaling or weighting; a norm is not automatically an energy.

Normalized drift fraction A separate quantity written as drift / (drift + control), ranging from 0 (pure control) to 1 (pure drift) when these labels denote nonnegative, comparable magnitudes with a positive sum. Both zero makes the ratio undefined. It is not a vector sum, cancellation measure, work fraction or percentage of muscle activity.

Numerical integration Computing the future state of a system by discretizing time and integrating the equations of motion step-by-step. Numerical integration is used to simulate swings.

O

Objective A declared criterion, such as expected score, dispersion or task error. Distance and consistency can trade off; an optimization claim must state their weighting and constraints.

Oblique muscles The internal and external abdominal obliques, contributing to trunk rotation, lateral flexion and abdominal function. The resulting action depends on side, posture and coordination with other muscles.

Observation The act of measuring or perceiving something. Launch monitors provide quantitative observations of swing parameters.

Optimality Being best relative to a specified objective and admissible set. Local, global, nominal and robust optimality are different claims; no single swing is optimal for every criterion.

Optimization The process of finding the best solution to a problem given constraints. Swing optimization involves finding the control strategy that achieves the goal.

Oscillation Repeated variation around a reference or equilibrium. A moving shaft may oscillate about a changing loaded shape rather than its unloaded straight shape.

Outcome A task result such as ball position or score. Impact sets launch conditions; aerodynamics, wind, terrain, bounce and roll also affect the final result.

Overload principle Training adaptation in response to demands beyond the accustomed level. This is not an instruction to exceed tissue capacity; useful progression depends on dose, recovery and the individual.

Overstretching An informal term for stretching beyond the intended or tolerated range. Range alone does not determine tissue injury; load, rate, duration and individual state matter.

Oxidative metabolism Processes using oxygen to support ATP production. They contribute to sustained activity and recovery, while brief high-power actions also draw on other energy pathways.

P

Palmaris longus A variable forearm muscle, absent in some people, that can flex the wrist and tension the palmar aponeurosis. It is not a required late-downswing driver.

Parallel mechanism A structure in which multiple linkage paths connect bodies. A golfer holding one club with two hands can form such a closed mechanism; simply having several limbs does not make every connection parallel.

Parameter A quantity characterizing a model, such as mass or elastic modulus. A parameter is distinguished from the evolving state; uncertain or time-varying parameters require explicit treatment.

Passive stiffness Resistance to small deformation under a protocol intended to exclude active muscle contribution. It depends on posture, loading history and tissue response. Zero net joint torque does not establish zero activation or purely passive stiffness.

Performance How well a goal is achieved, measured quantitatively. Golf performance is measured by score (total strokes), distance, and consistency.

Perimysium Connective tissue surrounding and linking muscle fascicles, contributing to organization and force transmission. Its role is not described completely by a single series spring.

Periosteum A connective-tissue covering on most external bone surfaces, absent from articular cartilage-covered regions. It participates in attachment, growth and repair.

Perturbation A change or disturbance relative to a declared reference. Linear sensitivity describes sufficiently small perturbations; finite disturbances and contact changes require additional analysis.

Phase A stage defined by events, such as backswing or follow-through, or an angular position within an oscillation. Event phase and oscillator phase should not be interchanged.

Phase portrait A representation of state-space trajectories and vector fields. A low-dimensional projection can hide state variables and cannot by itself establish stability or a limit cycle.

Phase space The space of dynamical states. For an unconstrained planar point mass it has two position and two velocity dimensions; canonical mechanics often uses position and momentum instead.

Phenomenological Describing observed behavior without resolving all underlying mechanisms. A phenomenological relation may be useful within its calibration range without identifying a causal neural or tissue process.

Planar motion Motion confined to a plane under a declared model. A planar golf approximation must specify the phase, fitted plane and tolerated out-of-plane error; the full swing is three-dimensional.

Plane of motion A plane used to describe or approximate motion. An inclined fitted plane can summarize part of a club trajectory but does not impose a physical constraint or describe every segment.

Plasticity Irreversible material deformation after unloading; in neuroscience, a change in neural organization or function. These are different meanings. Permanent tissue deformation should not be equated with beneficial mobility.

Position The location of a point in a declared frame. Joint angles describe configuration; they are not Cartesian point positions.

Potential energy A scalar energy function for conservative interactions, defined up to an additive constant. Gravity and elastic deformation can contribute. Nonconservative forces do not generally have a configuration-only potential.

Power The rate of energy transfer by work. A force supplies $\mathbf{F} \cdot \mathbf{v}$ at its application point, and a moment supplies $\boldsymbol{\tau} \cdot \boldsymbol{\omega}$ in matching frames. Large force with little velocity can mean little mechanical power.

Power transfer Energy flow per unit time across a declared boundary. Interface forces can transfer power between segments while cancelling in the total balance. Net joint power, individual muscle power and shaft storage are different quantities; a proximal-to-distal peak sequence alone proves none of them.

Precision Closeness of repeated measurements or outcomes to one another. It differs from accuracy relative to a reference and from a causal explanation of the observed consistency.

Prediction A forecast conditional on a model, state estimate, input and disturbances. A successful prediction does not uniquely identify the mechanism producing the motion.

Proprioception Sensory information and perception related to body configuration, motion and force. It draws on multiple receptors and neural processing; it is not a noiseless instantaneous state measurement.

Proprioceptive feedback Sensory information about body position and movement from mechanoreceptors in muscles and connective tissue.

Proteoglycan A core protein with attached glycosaminoglycan chains, forming part of extracellular matrix. These molecules interact with water and other constituents and contribute to tissue behavior.

Putter A club designed mainly for putting, typically with low loft. A putt can initially launch and skid before rolling; it is not necessarily pure rolling from impact.

Q

Quadriceps The four anterior-thigh muscles contributing to knee extension. Rectus femoris also crosses the hip and can contribute to hip flexion; swing timing is an empirical question.

Qualitative Describing properties using words rather than numbers. Qualitative observations like "smooth" or "powerful" are important in coaching.

Quantitative Describing properties using numbers and measurements. Quantitative data from launch monitors enables objective analysis.

R

Radial deviation Movement of the wrist toward the thumb (radial) side. Radial deviation of the wrist is part of the wrist motion during the swing.

Range of motion (ROM) The angular or displacement range reached under a stated active or passive protocol. It depends on joint geometry, tissue state, loading and measurement convention.

Rate of force development The slope of a force-time curve over a declared interval. Measurement depends on the task, onset definition and filtering; a group difference does not establish that training caused it.

Reachability The set of states attainable from a specified initial state within a specified time using admissible controls and disturbances. It is model- and constraint-dependent, not a direct statement of an individual golfer's capability.

Reaction time The interval from a defined stimulus to a defined response onset. It differs from reflex latency, EMG onset and electromechanical delay. An object-manipulation experiment found goal-directed muscle responses around 60 ms; that is evidence against one universal delay, not a measured golf correction time [Crevecoeur et al., 2016].

Recoil Recovery motion of a deformed structure as loading changes. Returned elastic energy, damping and continuing work jointly determine the response; recovery need not occur only after complete unloading.

Recovery Return to normal function after fatigue or injury. Recovery time between rounds allows muscles and connective tissue to repair.

Recruitment Activation of additional motor units. Muscle force is also adjusted through motor-neuron discharge rate and depends on contraction state; fibers are not independently recruited as arbitrary continuous actuators [OpenStax, 2022c].

Reference frame See frame.

Relaxation A reduction in active muscle tension or, in material testing, stress decay under held deformation. A reduced command does not instantaneously remove all muscle-tendon force.

Repeatability Closeness of repeated results under specified conditions. It is an observable property distinct from accuracy or the mechanism producing it.

Resistance A context-dependent opposition to a specified change. Inertia opposes acceleration rather than consuming energy simply because motion occurs; friction and drag can dissipate energy.

Restitution Recovery of relative separation motion during impact, characterized in a normal collision by COR. COR is a relative speed ratio; energy loss also requires masses and the full collision model.

Restoring force A force that acts to return a deformed system to equilibrium. The spring force in Hooke's law is a restoring force.

Resultant A sum of compatible vector quantities in a common frame. Forces can be summed, but moments also depend on their application points. Clubhead velocity follows kinematic mappings, not an unqualified sum of segment speeds.

Retraction Posterior movement of a structure, such as the scapula moving toward the spine. It is an anatomical description, not a universal prescription for address posture.

Reverse engineering Inferring candidate mechanisms from observations. Swing motion alone generally does not uniquely separate drift, applied input and constraint reactions; identification needs additional measurements or assumptions and uncertainty analysis.

Rhomboid One of the rhomboid muscles, contributing to scapular retraction and downward rotation. Its activation pattern is not determined by the name of a swing phase.

Rigid body An idealized object that does not deform, maintaining its shape and size. Rigid-body models simplify golf swing analysis.

Robotics The field of engineering and computer science concerned with the design and control of robots. Robotics provides insight into human motion control.

Rotational kinetic energy For a rigid body about its center of mass, $T_{\text{rot}} = \frac{1}{2}\boldsymbol{\omega}^T I_C \boldsymbol{\omega}$. The frame and reference point matter. A peak in angular speed need not coincide with a peak in energy when effective inertia changes.

Rotator cuff A group of four muscles (supraspinatus, infraspinatus, teres minor, subscapularis) surrounding the shoulder joint, responsible for shoulder stability and motion. The rotator cuff experiences significant loading during the golf swing.

Rough Longer grass outside closely mown playing areas. Grass or moisture between face and ball can change friction, launch and spin; spin is not invariably higher than from a clean lie.

Round Play over the holes assigned for a round, commonly 18 or 9. Duration depends on the course, format and pace of play rather than a fixed biomechanical timescale.

Run Distance traveled after first landing, including bounce and roll under the chosen convention. Landing velocity, spin, ground properties and slope determine it.

S

Saddle joint A joint with reciprocally concave and convex articular surfaces, exemplified by the thumb carpometacarpal joint. Its motion is more complex than two independent ideal hinges.

Safety Management of the chance and consequences of harm. No single swing technique or equipment choice guarantees freedom from injury.

Sagittal plane An anatomical plane separating left and right portions; the midsagittal plane passes through the midline. Its orientation follows the declared anatomical reference, not every segment throughout a swing.

Sarcomere The contractile unit between Z-discs in striated muscle. Active sarcomeres may shorten, remain approximately constant in length or lengthen; contraction means force-generating activity, not necessarily shortening [OpenStax, 2022c].

Scaphoid A carpal bone on the thumb side of the wrist, participating in wrist articulation. A diagnosis cannot be inferred from swing appearance.

Scapula The shoulder blade, a triangular bone on the back of the ribcage. The scapula moves during shoulder rotation and contributes to arm reach.

Scapular dyskinesia An observed alteration in scapular movement or position. Association with symptoms does not establish that it caused injury or that correcting the appearance will prevent injury.

Scapulohumeral rhythm Coordination of scapular and humeral motion during arm movement. It varies with task, elevation and individual; a fixed ratio is not a universal health criterion.

Second moment of inertia In beam bending, the area second moment $I = \int_A y^2 dA$, measured in m^4 about a declared section axis. It is distinct from mass moment of inertia in kg m^2 ; bending rigidity is EI .

Segmentation Division of the system into parts. The golf swing can be segmented into phases or into body segments.

Sensing Detecting information about the system or environment. Proprioceptive sensing provides feedback about body position.

Sensitivity Change in an output with respect to a specified input, parameter or initial state. Derivative, finite-difference and statistical sensitivity are different measures and require units or scaling.

Sensorimotor integration The combination of sensory information with motor commands to produce coordinated movement. Sensorimotor integration is critical for movement control.

Series elastic element A compliant element in series with an actuator in a muscle-tendon model, often representing tendon and aponeurosis. Fascia includes several geometries and load paths and is not universally a series element.

- Shaft** The structure connecting grip and clubhead. Its bending, torsion, mass and damping interact with grip motion and head loading. Elastic energy is stored only to the extent that it deforms.
- Shaft flex** An industry label or measured flexibility under a stated test. Labels such as regular and stiff are not universal stiffness values across manufacturers; compare force-deflection behavior, geometry and boundary conditions.
- Shaft lag** A shaft-deformation descriptor requiring a reference axis and sign convention, such as head displacement relative to the grip-end tangent. It differs from the forearm-shaft angle called lag. Strain energy depends on curvature and torsion along the shaft.
- Shaft plane** A plane defined using the shaft and an additional independent direction or point. A shaft line alone belongs to infinitely many planes. Address geometry does not constrain the subsequent club trajectory to that plane.
- Shaft release** An informal term for a phase of changing shaft deformation. It should be distinguished from wrist release: one describes flexible-body motion, the other joint/club geometry.
- Shank** A shot struck with the club's hosel rather than the intended face region. Ball direction follows the actual contact geometry; the shot alone does not identify an arm-activation or sequencing cause.
- Shape** In ball-flight discussion, the trajectory's curvature, such as draw or fade. Launch velocity, spin-axis orientation and aerodynamic conditions determine the path.
- Shear force** A force component tangent to a declared surface or section. Its tissue effect depends on the area, direction, distribution and simultaneous loading.
- Shear strain** A measure of distortion involving a change of angle between material directions. In small-strain notation, engineering shear strain is twice the corresponding tensor shear component.
- Simulation** Running a mathematical model forward in time to predict system behavior. Swing simulation uses the equations of motion to predict ball flight.
- Sinus** A cavity or passage in bone or other tissue. The frontal sinus is a cavity in the forehead bone.
- Sinus tarsi syndrome** A clinical label for pain and related symptoms around the sinus tarsi, a space between talus and calcaneus. The label alone does not identify one lesion or establish a golf-specific cause.
- Skeletal system** The framework of bones and connective tissue supporting the body. The skeletal system provides the structure for motion and is involved in force transmission.
- Slice** Pronounced flight curvature toward the golfer's trail side: right for a right-handed golfer. Centered-strike face-to-path geometry is one contributor to spin-axis tilt; contact location and gear effect also matter.
- Slop** Informal mechanical terminology for play, clearance or backlash. Physiological joint motion and soft-tissue compliance should be described by their actual geometry and load response.

Slow twitch See type I fiber.

Smooth A qualitative description of motion. A technical smoothness measure may use acceleration or jerk and must specify filtering and time scaling; visual smoothness alone does not prove efficiency.

Snap An informal description of rapid motion or recoil. Replace it in a mechanical argument with the measured joint velocity, shaft deformation rate or contact event intended.

Soft tissue A broad anatomical category including muscle, tendon, ligament and other non-bony tissues. These tissues have different mechanical and physiological properties.

Sole The lower clubhead surface that interacts with turf. Bounce geometry, delivery and ground properties jointly determine the contact response.

Space The 3D region in which motion occurs. The golf swing occurs in the 3D space of the golfer's body and the area around the ball.

Span The distance between two points, like the arm span (fingertip to fingertip with arms extended).

Speed The nonnegative magnitude of velocity. Speed omits direction; equal clubhead speeds can yield different impact conditions.

Splay To spread or open apart. Foot splay at address affects balance and weight transfer.

Sprain A ligament injury, ranging from stretching or microscopic damage to partial or complete tearing. It differs from a muscle or tendon strain.

Spread The spatial distribution of a quantity. Force transmission through connective tissue can be heterogeneous; continuity does not imply uniform loading.

Spring An idealized elastic element with a force-deformation relation. A shaft can behave as a distributed elastic structure; reducing it to one spring requires a declared coordinate and loading model.

Spring constant The scalar force-displacement stiffness k in $F = k\delta$. For a uniform small-deflection cantilever with a transverse tip force, $k = 3EI/L^3$. Other supports, taper, load positions and deformation modes give different relations.

Square In a declared target reference, a square face has its horizontal normal directed along the target line. A conventional square stance has the foot-alignment line approximately parallel to that line, not perpendicular to it.

Squat A movement involving hip and knee flexion and extension with coordinated ankle and trunk motion. Its contribution to swing loading requires force and motion measurements.

Stability A property defined relative to a reference and disturbance class. Lyapunov stability keeps sufficiently small state perturbations small; attraction additionally brings them toward the reference. Balance at address and stability of a moving trajectory are different questions.

- Stabilizer** A muscle that provides stability to a joint or body part. The rotator cuff stabilizes the shoulder.
- Stance** The position of the feet at address. Stance width and position affect balance, weight transfer, and rotation.
- State** Variables sufficient to determine model evolution given future input and disturbances. Positions and velocities may need muscle activation, tendon or shaft states, and contact mode; measured kinematics alone may be incomplete.
- Static** Not moving or changing with time. Static stretching holds a stretch without movement.
- Stiffness** A local relation between load change and deformation change, such as $k = dF/d\delta$. It may be nonlinear, directional or history-dependent. Structural stiffness, material modulus and active joint impedance are distinct.
- Stimulus** A physical or sensory input that elicits a response. Proprioceptive stimuli inform the nervous system of body state.
- Straightness** A geometric description relative to a line or target. A straight ball flight and a planar or smooth club path are different properties.
- Strain** A dimensionless deformation measure relative to a reference configuration. The relation $\epsilon = \sigma/E$ applies to a uniaxial linear-elastic model, not to every tissue, loading direction or strain magnitude.
- Strain rate** The time derivative of strain. Material response may depend on this rate and loading history; the direction and size of a rate effect require a specified tissue and protocol.
- Strength** A maximum load or force capacity under a specified test. Muscle strength, structural failure load and material failure stress are different quantities.
- Stress** Internal force per area represented generally by a tensor. Uniform uniaxial stress is F/A ; local multiaxial tissue stress cannot usually be inferred from a net joint moment alone.
- Stretch** An increase in a specified tissue or muscle-tendon length, or a mobility exercise. Changes in range of motion do not uniquely identify a material mechanism or establish injury prevention.
- Stretch-shorten cycle** Active lengthening followed by shortening. Performance effects depend on activation, tendon storage, timing and comparison conditions. A kinematic stretch alone does not quantify stored energy or guarantee enhanced power.
- Striations** Regular patterns of light and dark bands visible in skeletal muscle under a microscope. Striations result from the organized arrangement of contractile proteins.
- Stroke** In golf rules, an intentional forward movement of the club to strike the ball, subject to the governing definition and exceptions. It need not be a full swing or make contact; scoring also includes applicable penalties [[United States Golf Association, 2026b](#)].

- Structural** Relating to component arrangement and physical properties. These constrain possible motion but do not alone determine the motion actually selected.
- Structure** The arrangement of components and connections. Modeling choices about joint freedom, contact and deformation determine the mathematical representation.
- Subscapularis** A rotator cuff muscle contributing to internal rotation and stabilization of the humeral head. Its activity depends on the task and individual.
- Substrate** The material or base on which something rests. The turf is the substrate for the golfer's swing.
- Supination** Forearm rotation at the radioulnar joints that brings the radius and ulna toward their uncrossed anatomical relationship. Palm-up describes one arm posture, not a frame-independent definition.
- Supinator** A forearm muscle contributing to supination. Its mechanical contribution depends on posture and coactivation; it is not assigned a universal late-downswing activation.
- Supraspinatus** A rotator cuff muscle contributing to arm elevation and glenohumeral stabilization. Anatomical function alone does not determine pathology or swing-phase activation.
- Surface drag** The skin-friction part of fluid drag, distinct from pressure drag. Total aerodynamic loading depends on relative air velocity, shape and flow regime; neglect requires an estimate appropriate to the body or club studied.
- Sway** A coaching description of lateral translation, often of the pelvis or trunk. Define the segment, reference direction and time interval before evaluating its effect.
- Swing** The complete motion of the golfer with the club, from address through follow-through.
- Swing plane** A plane fitted to or defined from selected swing geometry. A full club trajectory can change plane and deviate from planarity; an on-plane label needs an explicit reference and tolerance.
- Swing sequence** The order of declared events, such as motion onset, peak angular speed, peak power or EMG onset. These are not interchangeable. A pelvis-to-trunk-to-arm peak sequence does not by itself prove optimality or causal energy transfer.
- Swing speed** See clubhead speed.
- Symmetry** In mechanics, a transformation leaving the relevant model or action unchanged. Continuous symmetries can imply conservation laws under suitable conditions; visual bilateral similarity alone is not such a proof.
- Symptom** An experience reported by a person, such as pain. A clinical sign is an observed finding. Pain may accompany tissue injury but does not establish its presence, severity or mechanism [[International Association for the Study of Pain, 2020](#)].
- Synapse** A junction through which a neuron communicates with another cell, chemically or electrically. Changes in synaptic function are among the processes involved in learning.

Synchronization Coordination in time. Distinguish simultaneous events from a useful relative timing pattern; optimal timing depends on the task and constraints.

Synergy Coordinated action, or a low-dimensional pattern used to describe muscle activity. A fitted synergy can summarize data without proving that the nervous system commands exactly those basis patterns.

Synthesis Combining findings into a coherent explanation while preserving their assumptions and evidential limits.

System An organized collection of components working together. The golf swing is a biomechanical system.

Systemic Affecting the system broadly. A whole-body effect or training benefit requires evidence rather than following from the adjective.

T

Tangent A local first-order direction along a smooth curve. A tangent line can cross the curve, as at an inflection point. At nonzero speed, the velocity vector lies in the trajectory's tangent direction.

Tangential acceleration For a smooth path at nonzero speed, the component along its unit tangent, equal to the time derivative of speed. It can increase or decrease speed. The normal component changes direction.

Task A specified goal and success criterion. Golf tasks can prioritize scoring, launch conditions or dispersion; minimizing immediate distance to the hole is not always the best scoring strategy.

Task-space The space of selected outputs, such as clubhead position, orientation or launch variables. Its dimension, units and metric depend on the task; it need not include every end-effector variable.

Telemetry Transmission of measured data from a remote source. Radar sensing is a measurement method, not by itself telemetry; an instrument may separately transmit its results.

Tendon Connective tissue transmitting force from muscle toward its attachment. Its recoverable energy and stiffness depend on structure, loading and history. Different tendons serve different mechanical roles.

Tensile strength The maximum tensile stress sustained under a specified test. Specimen region, direction, strain rate and area definition matter; a universal tendon-versus-fascia ranking is unsupported.

Tension A pulling force applied to a material. Tension in muscles and tendons transmits force.

Teres minor A rotator cuff muscle contributing to external rotation and glenohumeral stabilization.

- Test** An experiment or trial to measure performance or understand a system. Swing tests using launch monitors provide data for analysis.
- Thigh** The portion of the leg between the hip and knee. The thigh muscles (quadriceps and hamstrings) are large and powerful.
- Thoracic spine** The spinal region usually containing 12 vertebrae and articulating with the ribs. Trunk rotation is distributed across this region and other joints.
- Thoracolumbar fascia** A connective-tissue complex in the lower trunk, linked to several muscles and skeletal attachments. Its geometry can transmit forces; its quantitative contribution to a swing requires a model or measurement.
- Threading** Passing one object through another. Threading a shot between trees requires precision.
- Threshold** A specified boundary for an event or response. A pain threshold concerns the onset of a painful experience under a test; it is not a tissue-damage threshold [[International Association for the Study of Pain, 2020](#)].
- Thrust** A pushing force along a declared direction. Ground reaction has several components and should not be reduced to universal forward propulsion.
- Tibia** The larger, medial lower-leg bone, transmitting most axial load between knee and ankle. The fibula also contributes to the lower leg's structure and load sharing.
- Timing** The relative time of declared events. Wrist geometry, joint moment, shaft recoil and peak clubhead speed are separate events whose relationships can be investigated.
- Tissue** A collection of similar cells and their extracellular matrix. Muscle tissue, bone tissue, and connective tissue all make up the body.
- Toe** The clubhead region opposite the heel/hosel. An off-center strike changes impact mechanics, including head rotation and potentially gear effect; its outcome depends on the design and contact location.
- Topline** The upper clubhead edge visible at address. Its appearance alone does not determine launch angle.
- Torque** A moment, measured in newton-metres (N m). A point force contributes $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ about a declared origin; a force couple can create a pure moment. Torque is not power.
- Torso** The trunk, including chest, abdomen and back. Its mechanical contribution must be evaluated with the pelvis, arms, contacts and club rather than designated the universal primary driver.
- Total energy** The sum of the energy forms included in a declared system. Mechanical $T+U$ is conserved only when the net power crossing the boundary and nonconservative conversion vanish and the model has no unaccounted explicit time dependence.
- Total momentum** See linear momentum.

- Trajectory** A time-parameterized evolution of position or state. A geometric path alone omits timing; ball trajectories also depend on launch speed, environment and aerodynamic behavior.
- Transition** The interval around the change from backswing to downswing. Different segments can reverse at different times, so a single event definition must identify the measured quantity.
- Translation** Rigid-body motion with unchanged orientation. Any rigid-body motion can also be decomposed into a chosen point's translation plus rotation; center-of-mass translation can coexist with rotation.
- Transmitter** Something that sends or conveys something else. Tendons are transmitters of muscle force to bone.
- Transverse plane** An anatomical plane dividing superior and inferior portions. It is horizontal in the standard upright anatomical position; a moving segment's local plane need not remain horizontal.
- Trapezius** A muscle with upper, middle and lower regions contributing differently to scapular elevation, retraction, depression and rotation. Activity depends on the movement and load.
- Trial** A single repetition of a test or swing. Learning occurs over many trials.
- Triangular ligament** A shape-based description that needs an anatomical name to be specific. The wrist's triangular fibrocartilage complex contains several structures and should not be treated as one generic triangular ligament.
- Triceps** The triceps brachii, contributing to elbow extension; its long head also crosses the shoulder. Its mechanical role is not confined to follow-through.
- Triggering** Initiation of an event. The first observed downswing movement does not alone identify the neural command or mechanical cause initiating the other segments.
- Trunk** See torso.
- Trust** Confidence in one's ability to execute a task. It is a psychological concept rather than a mechanical measure of reliability.
- Turf** Grass and soil surface on the golf course. Turf conditions affect lie and ball contact.
- Turnover** An informal description of changing forearm, hand or clubface orientation. Specify lead/trail side and joint axes; it is not universally forearm pronation.
- Twist** Rotation or deformation about a structure's longitudinal direction. Rigid rotation and material torsional strain are distinct.
- Twisting moment** Torque about the long axis of a structure, causing torsion. Twisting moments on the shaft cause torsional deformation.
- Two-plane motion** A coaching term requiring an explicit definition, often comparing planes of different segments or phases. It does not mean a two-dimensional pendulum. Spatial motion can involve multiple changing planes.

Type I fiber A relatively slow-contracting skeletal-muscle fiber type with substantial oxidative capacity and fatigue resistance. Force depends on fiber size and state; slow does not mean incapable of high tension.

Type II fiber A relatively fast-contracting skeletal-muscle fiber category with subtypes differing in metabolism and fatigue resistance. Its contribution depends on recruitment, size and contraction conditions.

U

Ulna The forearm bone on the little-finger side, prominent proximally at the elbow. It should not be described simply as the larger forearm bone; the radius is broader distally.

Ulnar deviation Wrist motion toward the little-finger side, distinct from palmar flexion. Its relationship to club release depends on hand, grip and three-dimensional motion.

Ultrasound Sound waves with frequency higher than human hearing, used to image tissues. Ultrasound can assess muscle and tendon status.

Undulation Wavelike motion or surface variation. Green undulation affects the path and speed of putts.

Unit A defined measurement standard. SI unit names include newton and watt, with symbols N and W; consistent units are required in power and energy comparisons.

Upper body The arms, shoulders, and torso, as opposed to the lower body (legs).

Upswing See backswing.

Urethane In golf equipment, commonly a polyurethane cover material. Performance depends on the full ball construction and contact conditions rather than the cover name alone.

Usage pattern The distribution of use across tasks, equipment or time. A claim about a golfer's pattern requires recorded activity.

V

Velocity The vector time derivative of position in a reference frame. Its magnitude is speed; both magnitude and direction matter at impact.

Vertebra One of the bones of the spinal column. The usual counts are seven cervical, twelve thoracic and five lumbar vertebrae, with sacral and coccygeal vertebrae below.

Vertical The local direction along gravity, with upward opposite to gravitational acceleration. It is not generally perpendicular to sloping ground.

Vibration Oscillatory motion about an equilibrium position. Shaft vibration can occur even after impact.

Vigorous An informal description of energetic effort. It does not specify maximal effort or a measurable mechanical intensity.

Viscoelastic Exhibiting time-dependent mechanical response with both recoverable storage and dissipation. Loading history, rate and relaxation matter when estimating tissue or shaft energy return.

Viscosity A constitutive relation between fluid stress and deformation rate; a Newtonian fluid uses a constant dynamic viscosity. Soft-tissue viscoelasticity is broader than a single fluid viscosity.

Visible Observable by sight. Forces and muscle activity usually require inference or instrumentation; visual appearance alone does not identify them.

Vision The sensory process of sight, distinct from a mental image or goal. Visual information can guide preparation and later responses even when a current event cannot be corrected before impact.

Visual feedback Information obtained through sight and used to update action or learning. Seeing an event and having enough sensory-motor time to change the current strike are different questions.

Vital Essential or critical. The downswing is vital to swing success.

Vitality An informal description of energy or wellbeing, not a defined mechanical or physiological measurement.

Volume The amount of space occupied by an object, measured in cubic meters or liters.

Voluntary movement Goal-directed movement involving voluntary control, often with substantial automatic and feedback processes. A late command's origin can precede the final downswing; short remaining time does not make the motion biologically passive.

Vortex A swirling motion or region of rotational flow. Air vortices around the ball affect its flight.

Vulnerable Susceptible to harm under specified conditions. Quantifying injury risk requires population, exposure and outcome evidence.

W

Walk A gait usually characterized by alternating support with periods of double support. Its energy cost and pace depend on the person, terrain and carried load.

Warmth A thermal sensation or elevated temperature. Tissue temperature can influence behavior, but feeling warm does not establish protection from injury.

Water The compound H_2O , a major tissue constituent. Tissue water fraction varies by structure and measurement basis; a universal 70 percent value is inappropriate.

Wave An oscillation or disturbance propagating through a medium. Waves in the shaft propagate along its length.

- Weakness** Reduced force capacity relative to a reference and task. It may affect function, but a single strength measurement does not determine injury risk or swing quality.
- Weather** Atmospheric conditions including wind, rain, and temperature. Weather affects ball flight and playing conditions.
- Weight** Gravitational force, approximately mg near Earth's surface. It differs from mass and from the time-varying ground reaction during acceleration.
- Weight transfer** A coaching term that may refer to changing foot loads, center-of-pressure motion or center-of-mass motion. These are distinct measurements; declare which is meant.
- Weighted** Biased or emphasized. A weighted average gives more importance to certain values.
- Wellness** Overall health and wellbeing. Golf can contribute to wellness through exercise and outdoor activity.
- Whip** An analogy for rapid motion in a linked or flexible system. A flexible shaft's phase and energy exchange must be measured or simulated rather than assumed to straighten at the ideal instant.
- Wobble** An informal description of oscillation or orientation variation. Oscillation can be stable; the word alone does not diagnose instability or poor performance.
- Work** Energy transfer through force acting along displacement, $W = \int \mathbf{F} \cdot d\mathbf{r}$, with the corresponding moment-rotation contribution for rigid bodies. Mechanical work and metabolic energy expenditure are different.
- Work-energy theorem** The change in kinetic energy equals the total mechanical work on the modeled particles or bodies. Clubhead energy can reflect work by interface forces, gravity and other interactions, not only an isolated muscle-work term.
- Workload** An exposure or effort measure defined by its protocol, such as repetition count, duration or mechanical work. It is not generally one quantity measured in joules.
- Workshop** A place for learning and practice. A swing workshop is where golfers improve with instruction.
- World record** A best verified performance under a category's rules and measurement conditions. It does not establish an absolute physiological limit.
- Worst-case scenario** The most adverse case within a declared uncertainty set and objective. A worst case is meaningful only relative to those bounds.
- Wound-up** A coaching metaphor for the configuration at the top of the backswing. Rotation or muscular tension alone does not quantify elastic energy available for recovery.
- Wrap-around** Something that wraps around another object. The fascia wraps around muscles.

Wrapping In composite manufacturing, laying reinforcement around a mandrel or form. In anatomy, a muscle-tendon path can wrap around a structure, changing its force direction and moment arm.

Wrench The combined moment and force acting on a rigid body about a declared origin. With moment-first ordering, $\mathcal{W} = [\boldsymbol{\tau}; \mathbf{F}]$ is dual to $\mathcal{V} = [\boldsymbol{\omega}; \mathbf{v}]$, and power is $\mathcal{W}^T \mathcal{V}$. Origin, frame and ordering must match.

Wrist A joint complex permitting flexion/extension and radial/ulnar deviation with coupled carpal motion. Forearm pronation/supination is primarily radioulnar rotation, not a third interchangeable wrist hinge.

Wrist cock An informal description of forearm-club geometry, often associated with radial deviation of the lead wrist. It is not synonymous with wrist flexion or extension. Grip and three-dimensional conventions matter; the angle alone does not measure tendon or shaft energy [Carson et al., 2019].

Wrist release An informal description of changing forearm-club geometry during the downswing. It may involve wrist deviation, flexion/extension, forearm rotation and club motion. A measured angular change does not alone identify its muscular cause or elastic energy source [Carson et al., 2019].

X

X-factor A measure of relative pelvis and upper-trunk orientation using declared axes, projection and swing event. Different two- and three-dimensional definitions can yield different values. It measures geometry, not torso torque or stored energy, and a larger value is not universally better.

X-rays Electromagnetic radiation used to image bones. X-rays can reveal fractures or structural damage.

Y

Yard A unit of length exactly equal to 3 feet or 0.9144 metres.

Yardage Distance measured in yards. A golfer needs to know the yardage to the green to select the right club.

Yell A loud shout, sometimes used to psyche up for a swing. Some golfers use vocal cues to trigger execution.

Yield To give way or deform under load. Materials yield plastically (permanently deform) above the yield strength.

Young's modulus See elastic modulus.

Z

Zenith The highest point in a declared spatial direction. A segment's highest position need not coincide with its reversal event or with zero velocity of every other segment.

Zero A value defined for a particular quantity and convention. Zero declared control selects the model's drift value, which may itself be zero; it does not establish drift dominance or absence of muscle activity.

Zero-Torque Counterfactual (ZTCF) family Model interventions that set the declared applied generalized-control channel to zero while retaining the declared effective plant and protocol-specified loads. The family contains pointwise samples, stitched pointwise traces, forward trajectories, and achieved-state branched trajectories. Zero declared control does not imply zero muscle activation.

Zone A region defined by explicit boundaries. A comfort zone is subjective and should not be confused with a mechanically verified stability region.

ZVCF The instantaneous Zero-Velocity Counterfactual acceleration evaluated at the current configuration and declared internal state with both velocity and declared control set to zero. It is neither a state nor a released trajectory.

Zygapophysial joint See facet joint.

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